DevOps RAG Knowledge Base

Cleaned, term-corrected and topic-sectioned DevOps transcript dataset for retrieval-augmented generation (RAG) ingestion

Source: 10 DevOps videos from the Train With Shubham YouTube channel (auto-generated Hindi-to-English transcripts). 443,580 raw words reduced to 257,518 DevOps-only words in 659 sections.

#	Topic / video	Raw words	Kept words	Sections
1	Docker	45,081	33,928	87
2	Kubernetes (Concepts, Architecture, Workloads, Networking, Storage, Scaling)	112,536	82,697	211
3	Kubernetes on AWS EKS: Production GitOps Project (Terraform, Helm, Argo CD)	24,338	10,817	28
4	Jenkins and CI/CD Pipelines	53,823	35,417	90
5	GitLab CI/CD	20,308	15,383	39
6	Argo CD and GitOps	72,259	30,627	79
7	Linux for DevOps	44,723	22,914	59
8	Shell Scripting for DevOps	30,251	16,414	42
9	Computer Networking and System Design	30,878	6,953	18
10	AI Agents for DevOps	9,383	2,368	6

Not included: the DevOps resume-building video (career advice, not technical content).

How this dataset was cleaned

Noise removed: greetings and filler (hello, bhai, brother, okay, dude, friends...), audience interaction (can you hear me, comment boom, like and subscribe, chat questions), channel promotion, course and community pitches, resume and career talk, jokes and unrelated anecdotes, and Hindi leftover words.

Term correction: speech-recognition errors for DevOps terms were normalised, e.g. Kuber Nits / Cubinities -> Kubernetes, Junkin -> Jenkins, Gut Lab -> GitLab, Argos CD -> Argo CD, Cube Ctl -> kubectl, My Sequel -> MySQL, Enix -> Nginx, Diploid -> deployment, Seech Mod -> chmod, Pad -> pod, Cider -> CIDR, Sink -> sync, Persis -> persistent, Amble -> Ansible, Owas -> OWASP.

DevOps-only filter: every sentence was scored against a DevOps/Linux/networking/cloud vocabulary and command patterns; sentences with no technical signal in their neighbourhood were dropped. Near-duplicate sentences (the speaker repeats himself often) were removed.

Structure: each video is split into topic sections of about 380 words with a topic heading, so a chunker can split on headings and every chunk carries its own topic and source video. The companion file devops_rag_chunks.jsonl contains the same sections with metadata (id, video, topic, text).

Known limits: the source is a machine translation of spoken Hindi, so some sentences are still awkward or contain mis-heard words that no dictionary could safely correct. Treat commands shown here as hints and verify them against official documentation before running them.

Docker

Source file: transcript_9bSbNNH4Nqw.txt | 33,928 words kept of 45,081 | 87 sections

[Docker] Docker introduction and architecture / Docker images and registry

What happens, where did Docker come from? When did Docker come into existence and how is it being used now? Architecture are going to see that Docker when you will go to the interview if someone asks you I asked you how Docker works, so you Can you tell me how to install Docker setup? Will also do it on AWS instance We will also do it at your local level so that you get no There will be no bill cost and you can do it well.

Learn what Docker images are What are containers, what is networking? There are advanced concepts like Docker Volume Storage Docker Compose Docker The registry will look at all these things. Tell us in the comments if you are ready for this. Is deployed in which you have deployed that website You made it very well, you made it perfectly you made that website but client When asked for delivery of that website and. I saw that she did not even leave and then you say this is working on my machine but it is Not working on clients machine so this The IT industry remains a classic example.

That it is running on my computer I don't know why your computer isn't working. Docker is a tool that lets you It basically creates containers and Applications run inside containers Docker creates all those environments. In which you did your development from the letters You were on a Windows machine and your The client was a Mac, he was on Mac. The libraries on Apple's machine There are requirements that will not be available on Mac. He makes it and gives it wherever whatever required libraries are there The required environment is that of that box inside or reaches inside that container and you ship this container via image to the client and this image you.

Can be run on basically any computer This image will work great as will this container image was running in the computer I didn't understand this at all Docker let me tell you what happens when. You talk about Docker The tool is an open source tool where You can develop cool stuff anywhere. Things developed anywhere exactly That's what happens, now first of all understand that this Docker Where did it come from?

[Docker] Docker introduction and architecture

Cloud and this problem in a company named Dot Cloud I know it is working on my machine, not working on your machine so they did that thing Containerized Docker was born there. Made available at a conference whose The name was Paikan, in which it was publicly It became available, this is about 2013 and 2017 CN CF Cloud Native Computing in Foundation, this is a foundation there. Cloud Native Tools Are Basically Open Source Through contributions are very cool This thing has reached there and now it is very Most people use it every The organization uses Docker only, which I.

Before Docker containers are used All the organizations I have worked in If you use Docker then read Docker If it is very important for you then History Civics Geography so Jaan. Lee Docker's that came in 2013 in 2017 Reached CNCF and then now everyone Now let's understand that Docker is using Why do people use you again virtually? Sometimes we have to create such an environment It happens that whatever runs on your computer She is walking because she There are some libraries in the computer.

There are versions, there are some tools and because of that thing is running on another computer. Maybe that thing is not there and that is why you need a container Or you would have to create a virtual environment. Because of that virtual environment That thing works on other computers as well. Is this Docker your virtual machine? VM is like a virtual machine What happens is you were given a piece of hardware On that hardware you can create a virtual machine You set it up and there you go.

You can run anything, so can Docker. You can run anything but VM. There's a huge difference between VM vs Docker And there is a huge difference between Docker, let's understand What happens is that it creates a new concept. In that your virtual machines or VMs can be used. There is containerization in which your Containers are used, which are also part of Docker. There may be another Docker-like tool available. Whose name is Pod Man Pod Man There is another tool called Container D Container D can also be his so Both containers and virtual machines What difference does it make?

[Docker] Docker introduction and architecture (part 2)

I tell you Let me tell you about virtual machines. They use what is called Hypervisor: So what does a hypervisor do? Is the hypervisor from your operating system Space resources allocate everything and run your operating system on top of it Is your operating system mean Windows or Yours or yours So consider that I have this viz machine. It is above me Docker Engine What does Docker Engine do itself? It does not have an operating system of the host. Lets use the operating system If this is my Windows and I So what happens to this one that creates its own container?

It means that if you have If you have 8GB of RAM, you can run one or two virtual machines. It starts happening but if you have 8GB of RAM So you can run multiple containers. Let us see diagrammatically that in virtualization and containerization What are the differences diagrammatically? Which may be from let's Runs a hypervisor on top runs the hypervisor and Talk directly to your operating system and allocates its resources Provides resources What are resources Yes, is it your LAN port or yours?

Whatever the hard disk is, it is complete Containers have to be allocated on the other side How does your hand operate with your lats? The system is Now all these sources are used These are the resources of this operating system. The RAM that is nearby is the hard disk and this If it shares it with the operating system So you can run a lot of lightweight containers. In case of virtual machine you can Allocate just enough to make it big There is a difference, a much bigger difference than this.

The modules are made and kept comfortably. Let's say, now let's go to Docker. Architecture Docker Kaise Kare So look at the Docker architecture if You talk about Docker. There are three very important aspects of architecture and texture. One of the important things is your Docker Engine then it's yours Docker D Your mind and Docker CLI: Three very important things, what is Docker Engine, Docker D what is mind docker CLI. Understand it like this, that the Docker engine should also be It is called Docker Application Container Engine, the Docker application container engine All containers are running in the background.

[Docker] Docker introduction and architecture / Docker installation and setup

Doctor D Mann or Dr D is such a The service that manages your containers that container that I just gave you before Told that there are light weight containers which Inside you can run shared applications. With memory they run which Docker D It happens to all those applications is managing the containers but this There is another thing inside Docker DI which is called Let's say. Container D Container D Ye Container D I'll tell you what happens, so container D This is basically a tool of CNCF.

Tool that allows you to make such robotic portable containers You can now build Docker in the background Container D uses Container D basically A little one written in Golang. Project is where you can create great containers If you can make it, you can manage it. Container de same use in Docker background does now this will help your containers They are managed but these will become containers. You will enter the command so you will get CLI. It means it goes command line Interface so you write Docker commands docker login docker ps docker build docker Runs all of that through Docker CI and Those commands arrive at your Docker database.

Which is a background process and that Container D says, this container Make it, run it, run it like this and all this Things are running in your Docker engine There is another thing called Docker. What is the client the Docker client where But basically all the things you need for Docker You can see that,. Connects to API What is API yes Application programmable Interfaces are things that are on your back Docker client Or Docker Desktop, whichever you use. Docker talks to the engine and your A simple example of this will show you the containers.

I show you architecture through architecture I'd better tell you how I Installs Docker directly. Let me show you, so if you like this Docker This is architecture, if you want to see it live. You'll need to do a few things first What you need to do is install Docker If we have to do this then we will have to install Docker. He is going to do it locally also and We are going to do it on AW also, so let's go first.

[Docker] Docker volumes and storage

First, let me tell you about the local The first thing you need to do is go to a new. Write down Download Docker Desktop, download Docker Desktop and install it on Click here to docker desktop Pay Aapko The Number One Cutline Software For all these things now Docker Desktop wants. The login will also be done and with this you will be ready. You have signed in to your Docker account. If you have created an account on Docker So what you have to do is first go What is Get Docker Get Docker in Docker You can download Docker Desktop from Docker.

If you go directly to the Docker website If you go to Docker's website, So there you will get the download link. You can download it directly on your desktop if you I will install it and basically install it After I got a Docker Desktop here Mine will show up, so this is your Docker. What Docker Desktop does now Sees how many containers you have How many images are running, how many volumes build scout extension ye saari Shows you things, what is this I'll tell you everything slowly.

For now, understand that Docker Desktop You have a client that supports Docker behind you. Allows you to do cool things with the engine Understand this much now, we have understood that What is Docker Desktop and why? I took it on local but AW How to install Pay, let me tell you So the first thing you have to do is go to AW If you want to log in then it is yours. AWS Go EC2 Pay S2 In the instance we do our entire The full meaning of the game of Docker As complete as the end to end video is.

The entire AW of will be on e2 instance only You can use Docker to practice this. Let me just launch the instance. Instance Now this instance that I will take is I would choose an Ubuntu instance because it's easy. Too much access remains, Ubuntu is quite good now I what will I do I will take this image again Ubuntu lei Pay close attention to the instance type here. If the free tier is eligible then you mean In this, if your 750 hours of the entire month If you have a free tier account, you can use it.

[Docker] Docker images and registry

What is the problem in this, do a CP And a live memory is created which is yours All the projects are Project One. No Titu Medium free tier I have to pay its charge per hour so this is one if you want I can also use t2 micro but Again I have to make many projects on this. Maybe it gets stuck somewhere Best is to play on the safe side and take t2 medium you can buy t2 micro to save money You can take a pair and create it.

You can do it if you do it for PP. I press the key and as I create I will pay and download it here, it is done. It's great that's great what should I do now I want to enable SSH in network settings also. I will keep this SSH of mine active. I like him very much Now I will keep whatever I have to do next. What to do when you have a lot of storage space You build a lot of Docker images You create projects and your storage is also less.

If it starts happening then the dub that happens to you Provides up to 30GB of free storage per month. I'll make it 15GB, you can make it 8GB if you want. You can keep it, but I'm keeping it to be on the safe side. I will make it 15GB and after doing just that I Let me launch the instance, if it's I created an instance of Docker locally. The desktop will be installed and AWS Hum Log Docker One thing comes and installs it.

Will do successfully initiative Shot instance is running then instance running If so, I can select this instance. I will go to the instance state and Start and stop if I have to If you practice with let's then You can also stop and practice again. I have to start again, can I will click on connect I will go to SSH and I will use this SSH command I will copy it, I've done it. Please read the instructions first Open N SSH client so let me open the SSH client so this is my terminal where I'm happy to open my SSH client.

[Docker] Docker introduction and architecture

SSH client nothing happens You have a terminal that has SSH in it. It is installed so if you look carefully I If I write SSH here, then this message comes So you have SSH installed. Docker in one shaw key.pem then ls docker in one Shaw's yes it is in this download folder now Run This Command to Ensure Your Public Key Is Not viewable publicly, which is what happens. If not, read from Seek AOD 400 Only permission is granted, now I have got the entry Connect to your instance then enter this command I'll take it and connect it from here.

I will say yes to the instance and my Instance is done, now it's connected. My instance is an Ubuntu instance. I mean this is basically Ubuntu. Your AW VM is two instances I will update this system so sudo Aye Pati Gate then I would have updated the system I am updated so that I have the latest security patches. Once updated, What you have to do is come and grab this package. If you want to install it then type sudo at get Install Dakar D Aao What do about Dakar Dao?

You will get Docker Engine and Docker D Mind I will get Docker Seal I will get Docker D Docker sorry container d get everything that All of that resides in Docker's architecture. You will find the item, then click install and enter I have done it, he is saying I will take 289, take it, there is no scene, it went well. I am driving,, So he installed all the things, if you If you want to see it, you can see Container D.

Installed Docker service Given that Docker socket is installed, as many There were also important things, install them If you gave it, then your Docker service is Docker D is Docker K is Docker Engine. If you see that it is not

running then do sudo system ctl status docker enter so it's saying that Docker Application Container Engine is Active and Running So Docker Container Application Engine. What happens is the same as what happens with the Docker service. Docker Engine Full Form Docker Application Container Engine So Docker Architecture is right before our eyes It's starting to look good, let's do this too.

[Docker] Docker containers and commands / Docker introduction and architecture

Thing it now has Docker D running behind it. Docker D which is Docker D mind now What's going on inside Docker D mind good container D is running inside which is Inside that too, it is basically Docker's It's architecture, quit it. What you told me about this Docker Sia What does that mean if you write Docker? These are all the commands that you can use with Docker. What commands can you run with it? Docker run exect ps build pull push You can login and run all the things but, when also i do docker ps docker ps w command that makes containers visible whenever.

I do docker ps so I find this strange I am getting this error, how to solve this error So understand the error, what is it saying that Docker D Man Docker D Usko Connect. Unable to do so because permission is denied. That thing does not have permission, so my Who am I, my current user, camel that Ubuntu doesn't have this permission for Docker so this Ubuntu needs to add me to the Docker group. It will have to be done and only then will he get permission how to do sudo user mode user mode Meaning, modify the user and add him.

In a group whose group name is Docker now Add which user, whatever my current login is. These are the users, this is the dollar user, what does it mean There is a variable name which is current There is a user, please enter his name like this. Given if you do docker ps now you will still Why is this error coming, because your The Docker group is updated with the user. It has been done but it has not been refreshed due to which that if he sees things on this terminal then he Refresh sudo or new grep Docker will generate your group bit with this command.

It will be refreshed, refreshed now you can Two Docker ps Now you will see error not occurred Container ID means if a container is running. His ID will be visible and the image will be visible. The command will appear, everything will appear. This is docker ps, with this you What did Docker SEAL I change from Docker D to a Sent a command that, show me the container.

[Docker] Docker introduction and architecture / Docker images and registry

I hope you understand your Docker architecture. I must have come across the same thing you can do with Docker You can do this on desktop as well using Docker. Opened the desktop, now here in the corner Terminal is given, click on it. Terminal opens, what can you do now? If you want, you can enlarge it and write it down. Docker ps and any number of containers. How much will you see them in your local area? Mast so what did you basically do Have you also understood the Docker Architect lecture?

It means taken Module Two I hope you understand And you must have also understood that How to install Docker in Module 3 I hope you understand, come on. Three done, now let's move on to Docker images Containers talk about both of these What are these, let's talk. First of all, understand Docker Images here if I run this command But if you have any image then it should be visible It goes away but, what is this image? You might remember that you were in college or Did you ever write letters in school?

What are letters, such a paper There are small things in that paper Something is written and in that small writing What do you make of what happened? Your answer You make a script, you make an answer script If it is correct then you will have just this much blog. If it happens then you will pass like this only. I just happened to be joking I was a topper, so understand this blog like this. There is an image and the answer from this image You create the script and understand that the container, that's another container from this image.

If it is made then you can post that image anywhere. With two lattes you gave to your neighbor that, take this letter and pass that image. Put it somewhere, you make a container for that thing. You can right, how is that image created? It is written using a Docker File Docker File Bha But Dockerfile To Image Banao will go to create container from image but image is the image such a blueprint? In which all those steps are written that you use to run an application What is needed, what kind of environment is needed what should happen and how you can create that image.

[Docker] Docker introduction and architecture / Docker images and registry (part 2)

You can directly create and run the container. It's like a blueprint, so understand it like this Write some instructions in the Dockerfile There will be an image based on that instruction. It is created and you can replicate that image. You can create any number of containers. If you have these nut shells, then you If I tell you, first of all you will write Docker The file contains all the instructions. What will you be for Build Image then Instructions to Create Image and what a Dockerfile makes An image is created and what will be in this image.

Everything will be there, there will be an environment thing so that in a container Application should run even if the source of the application is The code is also inside this, you can see this image. That a container If the container is created, then there is a mantra that you If you want, you can memorize it, it is made from Dockerfile. Is a Docker image and is built from a Docker image Docker Container But Agar Mujhse Koi Image If I want to fetch it, I always have to write the Dockerfile.

Will we no longer need pre-built images? You can also bridge already built bridges now. Images can also be automatically saved to your computer. You can bring it and create a container from there. Let me show you how to make pre-made How do you get the images first? What can you do, this is your system, do it here is Docker login, what is Docker login? What will happen to you with Docker login? The Docker ID is as I just showed. I didn't have a Docker ID here.

I logged in to Docker here. You can, but if you give your personal access If you use this token, you will never forget it. Please tell me the token that you have to log in to Docker. Docker Hub is this one thing where Store your images across all repositories If you live there, go to Docker Hub and log in there. Sign in Log in to Docker Hub The same ID as your Docker ID. Your username will be A B C D on it Go to your account settings, go to account settings here.

[Docker] Docker images and registry / Docker introduction and architecture

If yes, you can reset it here but here There is also a personal access token, click on it Do it and generate a new token here. Delete gave these three accesses and I will generate it as soon as it is generated. I saw the token here, now you remember it Because y is a token You can generate it anew every time. What I'll do is I'll go back to Docker here. I'll go and beat him up here Enter login suxi ded so now you can use Docker Hub Docker Hub is a place where you can log in to But all the Docker images you have.

Any private images are here what will I do now I want to run an image, how do I pull it? If I do it, I will do it A Docker pull is an image named world to world to There is an image called World, I am using it as a bridge. I am about to enter, what is a bridge? Pull means you pull an image from Docker Hub. You download it like you would git clone any repository locally by doing You can do the same by doing a docker pull on any You can download the image from the repository.

You're done with docker pull world. If you look at Docker images you will get The image of the world has come to you which is 18 Someone must have created it a month ago, but. If it has come to you then it has become your image. What do I have to do with this image? Container is made from so what can I do what is the name of the docker run image World Bus Enter and Can You See This From Docker this message shows that your.

Installation Appears to Be Working Correctly what does it mean We installed Docker correctly. Local AW local chaho So you can test docker pull always You don't have to docker run if you use If you do this then they will first pull the image. Create a container from that image Enter saw made me docker pull val He had no need to do I ran it directly, from Docker and these and those things, I got it.

[Docker] Docker introduction and architecture / Docker images and registry

What happened to the Docker client? Contacted Deem Docker Deem said Pulled the image of the world then did the docker created the container and saved its output to your When I showed it at the terminal, I understood that. How the Docker architecture works Docker D man Docker CI all these people It is connected, now, you have seen the image I have taught you how to pull, but which image can I pull? I will show you MySQL address. I docker pull MySQL If I do To install MySQL or The image that made MySQL run.

If it happens, he will download it too, it is so correct. Let's also look at this beautiful image I downloaded it now from this image what can I make, I can make a container so From the Docker Images I can see that my Pass MySQL image 602. What is it now, I will die in a second. SQL run sakta sata run main Next, I'll show you what Docker containers are. I will show you in it, but still, if you Want to know how a container is made?

With images, remember that the Dockerfile To The image is created by building it and running the image. Container is created so this is the image I have If I do docker run image do this then This image will be created but the mycl No, that's my SQL is a server, login to the server. If it happens then you have to have one Putting a password in an environment variable It is called MySQL Route Enter your desired password and value. You can enter the root, so by doing this you have a once I gave MySQL root password and The name of the image was given by MySQL, just you Press enter, watch MySQL.

Chalne Lag IDB Inish Ei Jashan Has Ended IDB Initialization Ended Temporary Server Started and finished will come to you after some time that MySQL has started can you see This is MySQL community server. Everything is going well, MySQL. Inet Process Done Ready for Startup Simple your MySQL started running, how much That's sorted, now, the question comes. That's MySQL that you played. This container has been started MySQL is running This is going on in front of me, now I am So whenever you run a container how do I exit If you do, your terminal or your.

[Docker] Docker introduction and architecture / Docker containers and commands

He blocks the screen, so that's why You should always return the terminal or your container to the What does it take to run it in detached mode? I will tell you what happens too, but first understand. What to do in a situation where Is our container not closing? It's a system, it's a VM, log it back in. I connect and in a new I write fine on another terminal What CD downloads means this is from my local I am connecting back to your AWS Pay and I'll do it here.

Can docker ps and I can see that MySQL container is running so image The container created from this is running Containers also have a life cycle. The container is running now and will stop again. It will restart again if it gets killed This is the life cycle and it is the life cycle of the container. If it puts it in, look at your container here. Showing that status up about a minute means it's been running for 1 minute if I do docker stop that stop and the container ID I will put the ID docker stop the container.

Hit enter and see what it does to the container. Will stop it means if I If I go to the screen, I'll stay here for a while. After which my screen got blocked Unblocked due to a running container So great, I'm back on my screen. I've understood that if you have Docker You have run the run command by mistake. If it is not run in the background then your The screen will be blocked and you can switch to someone else. By going to the screen or some other meaning Unblock it by logging back into the server.

You can get it done, you can stop it, but if you If you want to run it in the background, The command is docker run -d what does this do It detaches and runs in the background. If it remains then I enter it and now I will make it taste good. It is not happening and if I go to another place Go to another on the same instance On another terminal, I will open Docker P.S. Your container must be running fine.

[Docker] Dockerfile and image builds / Docker introduction and architecture

How do I deploy a repository from Docker Hub or and I can pull the image from I can create a container by running it, but The question arises that the image you have pulled. What I have done is a publicly available image But I have a project,? I have a project from Lets to Java. Make me a container for that Java project. How do I make it, how do I do it, so You will need to write a Dockerfile for this.

Let me How do you write a Dockerfile? I will teach you Java project. You will need many methods of Java. To get the project, you either write a simple Java git up and create a simple A Java application will come up which will have a SRC. There must be some code, it would be and that A simple app will come, simple or you will be mine You can visit my repositories on GitHub There are many projects, notes and. Spring Boot Bank Pop and many more There are a lot of projects here Write two Java and you will get a simple Java application.

You will get a simple, absolutely plain Java. Application is in this thing and simple date current Prints the date as a developer Why not learn to write Java programs? Because what will you do with the work of a developer? What you need to do as a DevOps engineer Docker to run this Java project If you want to learn how to write a file then let's do this. This is a simple Java Docker project, we call it People clone their local or Let's bring it, so MKDI R Projects so I'm doing a project called Projects.

I create a folder in which I put all I will keep giving you projects, then CD Projects ls git clone now I have my which java docker I clone the repository that is clone it. I took LS and it came so simple Java Docker this which My project is this Dockerfile is dummy And what is mine in SRC The source code of the application is kept. The

Dockerfile is already written, no problem. RM -v - if writing from scratch v means verbose so it appears to be deleted If it does, then Docker removes the file.

[Docker] Dockerfile and image builds / Docker introduction and architecture (part 2)

Enter then the removed Dockerfile was visible that It has been deleted, let's create it now. Dockerfile create Dockerfile Bhaya Docker what would a file be Let's understand that some set is in the Dockerfile There are some instructions There are instructions and those instructions Because of this your application runs and. Give it inside the Docker images. Your application code should be There must be libraries that are compatible with the operating system Information about how to use it which are basically repositories or Any framework has been done or any.

There are tools, there are packages, whatever you need All that should be inside it, so this The image is a box type image which Everything is inside and it grows up. A container is made, now how will that box be made? You need to create a Dockerfile for the thing wich are instructions to build image so let's If I give you instructions now, you will never You must have made Maggi sometime,. Instructions have to build a recipe for Maggi so this If you give the recipe to someone, he will have it.

The image will appear and he will be able to make Maggi from it. If he can make Maggi from it then it will be fine The Dockerfile is made from Image is created from container let's just e Docker writes the file. First understand that Which application do I want to run? I want to run a Java application. The first thing I need is a Environment will require a pot and pan You will need a computer with Java installed. I'll use an image for that too.

Whose name is base image then pull images writes to from so I will write here What is the name of the base image now from Java If you want to know the base image of JD. How do you take it, it is simple You can use this image, open it. JD now has this image name next to it. There is a tag for the image, what is a tag? This image has a version of the image. From 17 open JD's 17 I want a Joe JD has an image called Alpine, so this is a This makes the image very light weight.

[Docker] Docker introduction and architecture

What happens is you get that environment which already has the Java Development Kit installed so you can get it back You don't have to install it, it's done now. What to do now you have a working DIR Working Directory Entry If you want to walk, you will need a container in this container. Create a folder in which this application is placed. If you keep the code then let's make it work. In the directory / a folder named app I want to go ahead and create it inside the container.

Which is going to be the container now I What will I do in this, I will copy everything thing that is stored in my host or source to my container so i dot likha. I will give you copy dot, meaning copy from here and My container's working directory entry I want to know whether I am on this host or whatever I want to do. The source code stored in the computer that goes into my container and that Once the container is running, you have to copy it.

Isn't it the source code, so he's writing the copy command I copied it exactly to the Word directory. I will make it simple, Java S Main Java, what is this Java C, whenever I need to run a command, I use I run all these people from work. Syntax of dir copy run this Dockerfile, so run a little bit of Java. The speciality is that the Java code is It is first compiled and then executed. If so, to compile Java into C it command is required and your file name is what is the file name so what is the file name see inside ARC med j so med j.

If your file name is then it is what I said so what am I doing source code name Copy the main.java file from the folder I am doing this and putting it here so my. When you're copying, you're just ARC only Copy your main Java code. Where is the / a s main.js So what does this mean in your source code? From the one that is placed on the host, you Copied the made java and installed it inside your container put it in the app path and compile that thing.

[Docker] Docker introduction and architecture / Dockerfile and image builds

I did it, it got compiled, then at last I enter the command CMD whose name I will enter Java and comma class name so my class. The name is main, so the main class name is me. I will put it here, done, what happened to me? Let me explain it once again, first what is your goal my goal is to have a I want to create an image that will connect Java A to my Java A. To run it in a container, you need a Writing Java file, sorry, writing Dockerfile Padi first pull a base image which Gives All Required Tools and libraries so like my java p so.

OpenJDK requires Java to run So for Open Jedi, I pulled it. I have pulled the base image and now that base I will create a folder inside images. I'll write here, create a Folder where the page will be stored to page The code which will be stored is P code. The command for that is work dir now what am I'll copy the source code from your host machine to your container so which The container is a folder named PSLTapp which. Inside the container is its source I will copy the code which is on the host machine is kept so copy command src to My container is done, now what will I do?

Compile the Application Code So I'm going to use my application code I will compile it to compiled Java s I am done now, then I Run the application I will run the application. This is also a command, this is also a command You have written run here, CMD here It is written that why do I explain such discrimination? Whenever you write a line in the Dockerfile Each image created from that Dockerfile Line is a layer from which the image is built.

If this is happening then all these lines are from work ddir copy run this all Lines for creating intermediate containers It happens but the CMD is such a There is a special command that once your. To run the container once it is created The command you enter from above lets say this Your container is created, here is the complete code inside it.

[Docker] Dockerfile and image builds / Docker introduction and architecture

Everything is fine but you have to drink the water inside So you will like this straw,. You can drink whatever you want So you can run this command if you don't Even if you write it, it will become a container but if you. If you are writing this command then after the container is created After how much this command will be executed It's cool, isn't it, whatever is written till here The image will be built using these instructions. The image created by a particular line The container that will be made from it will have a command will be injected simple now this You can also overwrite the command when you.

Docker run is the command that you use when you can also overwrite this command in whatever you write. If you can, then CMD can be written over this. There is another elder named Entri You can write point in it like this too. I can write this but I can write this when you All commands when you enter the entry point You write that the entry point should not be overwritten. CMD may overwrite CMD and Entry points are both syntaxes that Not used to create intermediate containers The container is created, followed by the commands It is used to inject is done escape colon dk enter my docker.

The file has been created cat Dockerfile my docker become a file Now the Docker image is created from the Dockerfile and Docker container is created from Docker image. What is the command to build an image from a file Docker Naming the build image is also very If necessary, you can do the naming ceremony with high tea. As I named this image. Java App I named this image Java App and after giving the image name Java app I will just put it here dot What does dot mean context?

What is context means the Dockerfile that Docker This image can be created by taking the file into such a context Now where is the Dockerfile kept? DirectoryEntry Step one out of five means what is that? The first step is running the where clause base image He is bringing the Step to Work Directory to P is making, you are watching removing.

[Docker] Docker introduction and architecture / Dockerfile and image builds

Intermediate container that creates a container every time removes it and creates a container Removes this thing called layers so step one is a layer step two is. A layer step three is A layer step four is A layer again, all these layers are yours now. You can see the image named Java App has been created. Docker Images You Can See Java App Name Your image was created 32 seconds ago How right Now the image is created from the Dockerfile. If the container is created then docker run this command If I put it and put Java app then my ai jaaye enter docker current date is Wednesday November 6th Itna Kitna Mast To.

Our source code was made in Java Docker Current Date Is so it went great Now I want to make some changes in the app. In the source code I will go and replace the made java with lettuce. I don't want to write docker, what should I write? Whenever you run application code that is hosted on your If it is kept on the machine, it changes You will also need to update the image and. The container created from the image will be the updated image Updating the image if the container is Very easy docker build high t image The name of the Java app and the image of the Dockerfile.

If it is kept then dot enter then see The previous layers took it from the cash Liya using cash so layer by layer He stores it and takes it from the cash but. Where he felt that SRC should be copied running there he ran I did not take the cash there, now I have it. The latest updated image of that image has arrived. How correct is that you have understood how to make any code work smoothly. Be it PDF, Python PDF, any PDF, you can use its Docker Can you make it, do you understand Docker Java?

Let's explain about Python as well Let me explain a Python project. Would have picked up Which one is this flask, let's get it. It runs its Dockerfile is also written And look, there's entry points and all that. The thing is written, now I have to do it myself.

[Docker] Dockerfile and image builds

I'll make it let's make it so the code Clone it and it will come back to you. In projects you have used a simple Java Now I did python-tk app a Flask app I picked up Flask is a framework micro framework for rm new Dockerfile or removed Dockerfile. Let us understand what is in this project. What all things are kept in the dm is app.py First of all understand the requirement and run Take what app.py is, that's the code Where your flash cup says Run Pawa.

If you are going to run the application then read it. I have the application code in Run Pawa The code to run the application is written What are the requirements, let's open it and see. Cut requirement.txt so that flask contains 2.2 is written, meaning the particular version It is written about Flask which you need to install. If you want to do it, let's do it first. What I need to do is to create an EM Dockerfile. In this Dockerfile What will this Dockerfile contain?

First of all it will be from because we You have to pick up the base image and bring the right base image We have to bring it and from this base image we can do things. The base image of this application will be I use Python for this I can now use Python 3.7 as well. I can use 3.9, 3.8, whatever yours The developer said that for this project this Write the version of Python you need here. Work DIR made What will I do now with the name of the app?

I will take the code from my local or my Done inside the container from the host machine now I'll run whatever libraries I have The command to install is pip install -R. Requirement dated Ekti, this Pip How to install it and how to do it When I found out, I did a simple search. How to Install Python Requirements Ekti, in whatever answer he got, he said this Told me to write pip install here r path to Your Requirement TT My Requirement If it is kept here then pip install han.

Come Requirement is simple so either in your read It will be written in the file or you can ask the developer Ask or do some research and find out.

[Docker] Dockerfile and image builds (part 2)

DevOps engineers can write any Dockerfile Now after doing this either I can enter I can make a good point that which is the file that python runs. His name is Rund D Pwa or so I can put that Or I can enter it using CMD Now let us understand that the example I have given Shown here is this Python file in it. Python is written in the entry point CMD Why is it written run in it, then the entry point I'm written in Python, so it will always run.

If run.py is written in CMD then it will run If you did not enter any command externally then run.py will run but this run.py It can also be overwritten because c. AD is a command that can be overwritten The entry point cannot be overwritten. Lets you get Python Nut JS or whatever I want to create a working base image. Took the directory in which all your sources are located The code will be that source code on your host machine copied to the working directory from If you need any libraries, install them.

Finally run the application given escape colon double k enter Dockerfile create I've gone from the Docker build to the latest I made this one called Flask P. Let me turn it down a little bit, then the flask Pop done and here the dot bus He will play enter cool, so what is this? This is what I'm doing, this is how I'm portraying myself. From a Dockerfile Look, he downloaded Python 3.7 first. Did Great, download it, very The big image is 211, so right here.

Is there any thing I will also go ahead and teach you how to How to Multi-Stage Large Images You can compress through Docker builds I'll teach you this too, let's go now. We are into Docker images and containers. Playing what is that bo warning There is no problem with the warning given, I am successfully tagged now. What will docker do Run image name Flask app now this Flask pop is on this particular port It works because it is a web application.

It runs on a particular port, so you in your code or ask your developer Look, on which port it runs, it is saying If it runs on port number 80 then it is fine.

[Docker] Docker introduction and architecture / Docker containers and commands

If port number 80p is running then you can Set the port number 80 of the container to that of your host You can map port number 80 to let's port Whatever is written first is your. Host's port colon port number 8 he wrote which is the port of your container, I did not understand the host port. What happens to the container Like I told you, it's Your operating system is your host It has resources that are available to Docker Engine Docker Engine shares and manages its containers runs now inside this container which.

The application is running on port number 80 running and your operating system Its port number is now mapped to these two ports. Or else Publishing the port for Bindies So you have to write in the command it means -p Port number 80 on your host machine Of The binder is simple enough, let's do it. If you see this page in the background If you want to run it, write down the background. Docker run -d -p 8080 flash cap done If the entry starts, what will I do now?

I'll do a docker ps and see if my flush cap is working. My flus cap is running so I'll go to my eToo instance. Instance or its public IPv6 Address is there an HTTP port on this The application must be running Enter and is taking a lot of time. For some reason it is taking a lot of time, let's see if you feel like it, man. Your application is taking time and is not working. If so, you go to this and this container ID is Jack Docker Logos Docker on this ID You can view the logs inside the container.

What's running inside the Docker container It is running, enter and he is saying what is running On All Addresses Port Number 80 but it is not working on port number 80. For some reason, it doesn't matter if you And drive it to the port, what can you do? Docker ps this container which port I should have gone for number 80, but why not? It is not working because your port number 80, it is not available to the whole world So what do you have to do with your You will have to go to the instance.

[Docker] Docker introduction and architecture / Docker containers and commands (part 2)

You will have to go to its security You will have to go to security groups and By going to the group you will see that these bound or Then there is nothing in the incoming traffic. I'll add a rule and tell it to That you can set port number 80 to Anywhere Meaning anyone can access port number 80 and save it, now you have saved it. What has been done to port number 80? This is your application welcome to Deb 0 So whatever your application is, it will run.

There is another path in this application, slash Health if you put slash health. You will write that the server is up and running and you can see it. Write that the server is up and running. If it is running then chch means your application is running Fine, it's going great, so whatever. You can docker log in and see whatever your container ID is. You can see it in there, so I saw the slash Health hit this time slash this now if I will do / profile now profile name so something.

If it is not there then it is showing as not found Now I checked in the docker logs. Every time I have to give the command with logs. Is there any way to run it that your container gets attached on this screen So that I can see all the logs in real time. Got exactly the right Docker logs as you did It was written that instead of logs, what should you attach? What happens by attaching it is that whatever you The container terminal and your terminal, these two will be attached and You will keep seeing it live here so if I Here I do refresh.

Slash something like that so look here real I can see how cool the time is again. This Is How Docker Attach Works Again, the screen of your container is here. If it has arrived then your screen will be blocked. Go and unblock that screen, Docker. Stop container ID then now container stop It will be, the container will stop. Your screen is there too Get unblocked here will go you will see vidn unblocked Now just like you did docker stop You can also start Docker.

[Docker] Docker introduction and architecture / Docker containers and commands (part 3)

You will have to stop the container docker stop Do it and this is your container that will stop. Will go stops after a while He will stop after some time in stopping It takes some time, yes it's done, stop now You will see that by refreshing it, side cant be. Riched but if you have done docker start So the container has started, now you can run it Will you refresh? This way you can do a lot with containers. Can you play something let's go docker ps So your flashcap is running right now And MySQL is also going on which is my Is the SQL going on?

MySQL Joe Is there a database? Can I go inside that database? I can enter and after entering the database which is running Can I access it at all? If you can do it, he also has command over it. The command is Docker what can you do by doing docker ps whatever You have to get inside the container or You have to insert yourself inside the container. To see what is going on inside, pick up the container's ID then Docker Ezek means Execute me an interact terminal If you want to make it, then - it, it means interact.

I will enter the ID of the terminal and container and Like my shell is bash so I will put bash and press enter then docker ekak -. ID and bash of IT container now this bash 5.1 What is this, this is your SQL Now you are immersed in MySQL. Meaning if I write MySQL here The username is root and the password I entered is He had also put a root, so he Root Welcome to MySQL you great my Show data has come in the SQL.

Database DevOps You can create a database of great DevOps Maybe you misspelled O Create. Create database devs so great query show data Once the bases database is created, you can use Docker Running MySQL inside a container now. In this way you can exit and go to the container. You can come out from inside one more time now. After exiting, you will be back to your instance. If you can, I've shown you how to build Docker images.

[Docker] Docker introduction and architecture / Docker containers and commands (part 4)

Taught you how to use the flash cap of Java if you want. I can also make nuts from GS but I use wood Highly recommended that you also practice that docker ps what does this mean when running Containers look like Docker images. It happens that the number of images you have It looks good but, you made Java P Its image was created and why its container It is not running because its container job. All I had to do was run the app and close it.

To run kiya world Whatever it was, it was printed and his work was done. If it happens then it does not have to run all the time But if you always have to run something then To you ITD this command has to be run then for Example me a. From Ubuntu Lates I am now on Mac O fine This is my Mac, this is Docker Desktop. My Mac is O, give me here I want to run a container that is always running.

Docker run nutu I want to run nutu in mac os But keep showing me his interact terminal. If I want, I will do it D - ITD Enter Has the container left? Ubuntu is running but this is what I said If IT had not been installed, it would not have worked. If it had come and closed Docker stop if I do this one If I stop the container, I stopped it, I stopped this container. If I do docker run ubuntu then that It won't work because what is his job?

Turn off the UTU and start it, so if I will docker run Ubuntu So let's close if I do Docker P.S. Nothing is coming because his work what was it, it stopped, but if you If you want to keep running always then high itd like this You have to write D which means in detach mode. IT means interactive in the background Terminal is fine so I hope you have containers You will understand a lot about I understand a lot about images.

He must have gone, if he has come then chat with me. I hope about Docker Images and Docker Containers in understanding about this module This must have come, this was our module four and If five has arrived then you can visit my repository.

[Docker] Docker introduction and architecture / Docker networking

It doesn't take much effort as If you used Python there, then here Use node and then copy the work directory code whatever you want to install and then run the command docker write the file docker. Creating an image Creating a Docker container It's easier if you get it up to here. Fifth Everything seems easy to me till the module. It's been over an hour since I read this far. If you are going to do it then Docker networking now Let's get started with Docker Networking.

It happens, understand that this is your computer is this is your cutter A Docker container is running on this computer. Running Docker container and you have another Docker container running Here's this one from Let's,

which is Docker's container that's a flask container This flask is the container of water and the other The container is my. SQL is the container of the database Now this flash cap container is a Isolated Container this is the container of MySQL this is a. It is an isolated container, which means it is can't talk to this can't talk to this Maybe but if you have to choose between these two If you want to get things done, you need a network.

That's why the concept of Docker comes in. The Docker network is basically seven There are seven types of butts If you want to read the official one, I will tell you Let me tell you from which second docker network Let's get started. Joe Docker There are seven types of networks. The first is the host that your Docker The network the container is on and your host If the network of the machine remains the same then Your port number is running in 80 host.

You have added the host to the Docker network then the port number of the Docker container is also 80 It will continue like that, one thing happens and then another thing happens. Is The default which is There is a bridge, it means like this of yours This is your container, this is your operating system. Host is your from this host to this container Building a bridge to network you If needed, they provide pull by default. Those Docker ones are, by default a bridge.

[Docker] Docker networking / Docker introduction and architecture

If it happens then there is a bridge network,. This is your The default network remains the default host. The bridge is gone, then it happens that the user Defined Bridge that you can create a network of your own You can also enter the name of the Custom Bridge. It is also said that you should build your own bridge. Enter name User Defined Bridge There is another one which has the meaning of nun in it. No network to or from Docker container Can anyone talk, can he talk to anyone?

Could be isolated completely isolated There is no internet or anything else going on inside. If there is nothing then this is a non-network Why do you need a network if you have one? Running a container whose job is just to If you want to run things or need internet then you Give me a non-network, there is another network Which is called Mac Wallan. What is Mac Vallan Mac Vallan Basically on two MAC addresses It can communicate generally Docker swam when your a cluster related If so, it uses it in the same way.

There is also an IP address and an overlay. These last three are generally Docker With Svam you read Docker Svam What It happens, like you just bought this laptop You have created Docker containers like this. Build Docker containers on 50 laptops and send them to everyone Two clusters run together, which nowadays say Kubernetes uses it like before Docker used to use the form for that. There are three uses pert which are important for you is reading and is this four host bridge user defined and your nun so this four for you.

It is important how you study Docker network so what do you do docker network ls If you write this then see this network ID Haybridge is this, the host is this, the nun is. In front of you if you want to have one of your own What can you do to build a network? Docker network create docker network create let's You have created a network called My Network. Made and - de - de means driver So this is what I have given you a cha pa 6 seven These are the drivers that told the type of network So let's give me a.

[Docker] Docker networking / Docker introduction and architecture (part 2)

If you want to create a network named bridge, then enter If I do a docker network ls I can see it. Is my network a bridge type network? This bridge has been built, what is this by The default one is made that host one is this Nun so you can see bridge is a default. Driver Host is again a driver my network is a custom bridge and none is no then The four you wanted are right in front of you. MAC VLAN is not covered by IP VLAN overlay Do not use Docker SM because it is outdated That's why it is good, what is its use It will be used in a project.

I will get you a two-tier project done. Tier project means how is the project The project is so beautiful that it is yours. The project is running a Flask API. There's a flask running and a The MySQL container is running fine. MySQL container is running Now this flash cap is a Docker container. Is this a docker Container, this is MySQL, this is also a Docker Container Now both of these are Docker containers. Someone is doing well independently I am not facing any problem but now I have to Getting Connected How can this be connected to MySQL?

I will get you to use me for this thing. Docker network will be required, let's docker network this one. The project is currently using Docker Network connection of a two tier I will go to my repository for applications. Here on repositories I find I will do two tear flask p so be it gone to tear flash this is my complete application It is a very cool application and it is working great. I will clone this, why should I do this? Just to show you the concept of

Docker Networking is fine, how many projects are there now?

My two projects are gone, the third one is done. Take the project to Git Clone this enter to tear flask on your The clone is done, now I will make a CD and I will enter inside the tear flask. It's good, now understand this post, there is post in this also. If you have Pawa then cut app.py and read this a little.

[Docker] Dockerfile and image builds

No one needs to understand why because It's the developer's job to write what your job is You just want to know if this application Yes if it connects to a database. It is written in it, Confil MySQL from environment variable so this is something There are environment variables that you need to pass You will need to operate this flask pop. Whose name is MySQL Host MySQL User MySQL Password MySQL database so these are the four variables that you pass If you remember, you will have to do Docker.

Run han e this is what happens e because of this you pass the environment variable,? How is the Dockerfile required for this created? If we go to the Dockerfile then this Dockerfile If you look carefully, this is from python3.8 Slim means so this code is Python use Uses version 3.9 and smaller images. Small Image set the working directory in the container then working directory entry What he's doing is some packages. If you are installing this application It requires some packages to run.

He requisitioned and to Copied a MySQL client named package that it also installed and Installed the requirements, the rest Copied the code and ran it from CMD but, this is almost similar to how we. People made Flask apps and Java apps Writing exactly the Dockerfile would have been much easier. First get the base image working DirectoryEntry read the code And then copy the entire code and run it. If you do it, it's very easy to write a Dockerfile. It's easy, let's run Docker now.

The file then Before doing docker run we need to docker build I will have to do this in which I will use the image of this page. Name I will write Flask back end P then flash back and P or should I call it. Instead of flash back and p, I write two tear back end two tier back end this is the name of it I put it in and after doing a dot, I put it in. The back end of the tear-floss cap is where But the code is written in Python, build it I do it so I put it on build Let's get this right.

This will be a great build this image I have taught you how to create an image.

[Docker] Docker introduction and architecture

It's very cool, now this is going to be the container. Let me show you the diagram, so this is what I made. I've been in it, what have I done in it What do I need to get as environment variables? My SQL host or whatever its environment This is the variable I need to pass to this container. This container is placed in some different environment If you want to give variables then it is fine like this What's gone is my already MySQL.

I am using the container that is running. What will I do with this MySQL? I inside What is a database called? I need to connect to this container. What is needed for this, then I will study here. I can see in the wind what variables I have I need a host, I need a user I need a password and I need a I need a database, let's say I need this and which one? The port is running on port number 5000.

P is running Let me try this Docker images. I want to share images that I have a two tier back end This is the image, docker run to tear back. I want to run this image,? My what-ifs for Butt Run I want to run it in the background I will add this command to my If you want to publish port number 5000 then -. Fifth note of musical scale 5000 colones 5000 so my system's 5000 and The container costs Rs 5000 and I have to run that too.

Do I need to give it an environment variable? SQL Host is one way in Docker that Whatever is the name of your container, you can call it that. If you can enter the host name, let's say mine. MySQL has a container, its name is MySQL let's say so I directly my I can put MySQL in the SQL host. What other environment variable is host Done MySQL User So MySQL My user name is root,? Root user now what is the third well yes when also you are putting environment variables every Han e ye for an environment variable If you have to put it separately then Han e ma sicle.

Host Han e MySQL User then Han e What MySQL Password then Han e ma SQL deb good MySQL password was mine Root and MySQL DB What Was My DevOps this is done so basically I.

[Docker] Docker introduction and architecture / Docker containers and commands

I have run the docker run command which is on my port Number 5000 mysicle host This will run this environment variable and My two-tier back end, this is the image. You will use the tag of the image. If you have to give it also then the tag of this image is latest so I'll colon latest here I will give you this cool latest image with Will work. If I hit enter, my flask is a container of water. If it's running, docker will work.

The container is not visible So if your container is not visible This simply means that the container is an exit. Has gone or has been killed, if he is dead If you are taking In Active or Acetate If you want to see the containers, go to docker ps. Add a highlight, this will show all containers enter to can you see re baap re two tear The back end container died with a status exit If you can check its logs then 582 DD you.

Mast if you can check the logs then docker logs whatever your container ID was 582 DD I checked its logs and now it says. There is an operational error unknown server host MySQL what does this mean this means MySQL to Is a Flash Cap is Unknown He doesn't even know who MySQL is. Because they are both on the same network. What am I doing now I will do this which is my container of MySQL Funny touring name so I'm instead of MySQL as a host funny touring dal.

I see that my y is the docker run command Funny touring in this my because container You have to enter the name of the host name,. There probably won't be a container named MySQL It will be called Funny Touring, try putting it in. Docker ps It still hasn't worked, what does this mean? This means check the logs again. Do it with Docker Checked the logs and found unknown server host funny Touring, in the network of this flask, this I can't find MySQL container, so let's go.

Create a network and call it that network Let me make it, so what do I do? I will do it, I will do docker ps and this MySQL is going on, I will delete it.

[Docker] Docker introduction and architecture / Docker containers and commands (part 2)

I'll do it for now so Docker Stop, I have given the ID of MySQL guy And docker rm this stop and remove that which. MySQL is going on, stop it and remove it, remove it, not just the container, remove it and see how I enjoy it, now what will I do, I am a mother. I will create a SQL container which I will name I will put the host name MySQL than that First Docker Create a Network Create a Network I will create a Docker network named.

Two-tier network called a two-tier network I will make it the driver who will use it The bridge that the bridge driver will use, then its what does that mean I'm a network. I will create a name that is Two Both these containers are in the tier and the network. Let's get this network done Two Tier Now I want to call both these containers as How will I give you the network,? Tell me how to deliver the container. The container of Docker run -d MySQL first run There is a container in my hand There will be a SQL named My SQL in the Hyfe - Network I will give you.

Tier network two tier now han e Environment variable so Joe MySQL Joe Specific to the container it contains Environment variables have to be given like In SQL Route Pass. You can put it but for this MySQL you will need You have to enter specific ones like route I entered the root password now yes my SQL database so you can use the value of the database. You can put anything from let's to dips put it put the DevOps now DevOps also put what can you do now network put the name put this is now MySQL put the image of finish then docker run -d.

Container's MySQL Network Are Two Tier that I have created and this environment Let the variable remain and I will go away, now it is gone. Same network I will give my flask So look at the container of the flask What is there in it, it is also a two tier back end The name of the database is DevOps, MySQL's name is. Password is root, correct username is root That's right, your host will come in which If your container name is my sequelize The name of the container is MySQL so its name.

[Docker] Docker introduction and architecture / Docker containers and commands (part 3)

MySQL will come in the host and which The environment variable is preceded by a -. What is the two-tier network, what will happen now? MySQL is also two tier. It is on the network and the database is also two-tier. Meaning the back end of Flask is also too You are on a tiered network, now you will enjoy it. Enter Now if you look at it by doing docker ps Your two-tier back end is running Flash cap MySQL is all going on and if I checked its Docker logs.

I am not facing any problem, everything is going well. I am here, how did this happen, this is why Because Docker Network LS when you do two tier This is my two-tier network. Created if you inspect it If you do, then Docker

Inspect to tier if you are using this with Docker network to the Inspect Network Do you inspect the network? Two tier Then you understand that there are two things inside it. Containers are a magical whatever and the other My Sickle both from the same network If we are inside, we can talk to each other.

Let's run this application and see. If yes, then this application is running on port number 5000. If it works then I will activate its 5000 port. I will go to the D rule and port number 5000 Activated it, saved the IP anyway. I have ruled it well, now I am in two I'll go to instances running and I'll go to my I'll pick up the IP address, this 35 p, this and here. Go to colon 5000 in http re http I will go and check Enter and your application is running smoothly.

Let's make a totter application Flak and Masal Submit It's printing now let's see two. Did This is MySQL container now what should I do Karunga Docker Ekk - IT Main MySQL inside the container of Ghusuk -u root -p root whatever mine I gave the root password and used the database. Which database name should you use? O Show data If the name is Bases or Devices then use DevOps use database DevOps so only use DevOps is used DevOps database Changed Select Star from Messages to This table name is messages.

[Docker] Docker introduction and architecture / Docker containers and commands (part 4)

Create containers and manage those containers How to communicate using Docker I've taught you this network. Your Docker network is also cleared. Let me know if you understand things up to this point. I can proceed further to the Docker volume. I'm going to go so if in Docker Talking about volumes and storage This is a very important concept and it I would like to explain to you by N. He is coming late from this application. Understand that if I had done this docker ps this to the container containing the database The container containing the database is called MySQL Container What will happen if I restart the Lets.

I did it from Docker restart this container id done i Restarted the Docker container Restart means that container stops. It will start again and everything will be complete What happens in that case if it gets refreshed? Just look at that case, if I put this thing I refresh So here I can see my strength. If this is closed then here I can see my data but if I put the item back IT HAPPENED MY AL -Y ROOT -P Root, this is what I did and used DevOps.

Show Select Star From Messages, so this is the data I see. Instead of restarting it, if I delete it for example I did docker ps and this is my database. EC iso I Docker Stop this End docker rm what will happen if I do this Tell me what is my data that needs to be restarted? Pay was coming in this situation when this data What if I remove the base? If you tell me in the comments, I will Deleted it, now it's in the Docker container I run back bean container docker run Name MySQL It Runs Back in the Same Container.

When I refresh it my All data is gone and can't connect to Server MySQL Docker PS should be running Docker restart this That's why it was showing can't connect to server Because I deleted that server. It should work now, when I restarts the two-tier back end. Ideally it should work, work But can you see the data judge gone data gone Now this is a problem with Docker volumes. End storage what is this it is such There is a way you can do this through Docker.

[Docker] Docker introduction and architecture / Docker volumes and storage

Persist data for all containers so that the container is accidentally If it crashes, restart after stopping it. If you restart it again then the data is stored For example, like this is your MySQL. This container has all the data inside it. If this container is deleted then all The data will also be deleted, so we What do we do with this container that holds the data? His/her of the binder Along with our host, who Hard disk has its own file system Let's make a path in it Create a folder and map that thing.

With our container data Whatever happens, we call that thing a concept. From docker volume ls docker volume ls whatever Aegistify SQL Data So you have a volume called MySQL Data. If you have made it, then docker volume create MySQL Data Enter MySQL named Data The volume has been created, now this volume is by Which location will your host be in by default? You can see that in Docker You can see from Inspect MySQL Data If yes, it says that the mount point is / ver Lip Docker Volumes Ma SQL Data Under.

If you want to know how much volume you have created, then MySQL Data This is a path that you It is placed on the host. Host means your In one to one system, this is the path, if you If you CD into it, you will see it now. It is saying permission denied because he is a The root user can access that thing. I have to launch my container with this path If there is a binder, how will I do it, let's show you.

If you give it to me, I will do a docker ps. This is my The SQL's container, for now, I'm keeping it Docker stop my SQL's Containers and Docker are my Removed the container ID of the SQL. I want to do the next thing I know

what I will do, what I have done I created a volume called MySQL Data. I'm with my MySQL container by Bindy I'll tell you how I'll do that So the command that remains is docker run my The one with the SQL container that was the run command so yes.

[Docker] Docker volumes and storage / Docker introduction and architecture

I will do it, I will add it anywhere where next to the network I do it, - and - and means volume I want to attach it to the volume to whom. MySQL data, this is my volume. With whom my container slid inside it Ver/Lib/My AL This Is For You How did /var lib/MySQL What is it that your container lives in? The data of MySQL that resides inside is stored in this of my host if it is stored in the path MySQL Data Volume Container's version lip MySQL with binder now i do.

I press enter, now this container is running. The funny thing is, now I'm going to restart it. Docker restart your ya which flush I have a cap, I will restart it. Let me show you the complete practical, so here it is. Test Dummy data something deleted dummy data anything I put it now what I'll do is docker PS I will do it and this is MySQL. I will docker run M& Docker should basically stop and rm to docker stop and docker rm docker stop container id end docker run My so I have my MySQL data.

You blew up the container, did you blow up my container? Data also It will fly, let's see how it will happen. What I'm doing is using my Docker I blew up the data for the SQL, but now If I drive AI, he will say that my SQL is saying MySQL It will try to connect but it won't. If I do Docker Volume Inspect MySQL data in this particular location I'll go and see it, so sudo AU first. CD that particular directory with super user permission I get into the path and do an ls I see that all my data is secure.

All my data is secured so now I What will I do with the same volume? I will rerun the container with a new tensor I will make Docker run MySQL's butt which Volume is I use MySQL data I will give you the volume and press enter. I will put the container By doing docker ps which is my back end I will restart Docker so that He tries to connect back to the database. This is how you can add Docker volumes You can persist your data and more Is there any way to store data in a Docker volume?

[Docker] Docker volumes and storage / Docker containers and commands

There is another way to get the money, what are you doing? MKD R Volumes A volumes named Created a folder and inside it a KDI Create a folder called Are MySQL A path where all the data of MySQL is stored. If it remains stored then PWD By doing this you got a path home Ubuntu to Volumes MySQL What Can You Do Now. I know that by going to your projects go into the tear floss cap and whatever you Used to build my own Docker container.

If the container had been there then this SQL one MySQL so docker stop my SQL and Docker are MySQL this which is your container if this you Delete it, if you delete it and its path. Give the path to the volume that is there. Even if you just created your current folder Docker volumes can be run in two ways. It works, one is the name volume Docker volumes created by creating volumes and the second one is path so if I This is a simple run command, my docker run where is this one.

Docker run, yes this is the command Where I wrote this MySQL data What has happened is this, this is MySQL data. Ned volume and if instead I put Do I have a home Ubuntu volume? MySQL is also My work is done, enter it and create it. If I do CD volumes And inside that I'll go into MySQL So I have all my data backed up. How cool is it that whatever data you get You should hide it and mount the volume.

You can do it, high fun and volume You can mount it comfortably and this is How Docker Volumes and Storages Work Docker so if you're studying Docker Volumes allow you to save your data. This is a very good concept so I hope Your module seven in which we Data is saved through Docker volumes I hope that is clear to you, so this Docker Volumes now understand Docker Compose. What happens is very interesting The topic is why do Docker Compose,? Because you may have noticed that I am very I'm doing manual work, what am I doing?

I am again and again I go into a two-tier floss cap then I do a docker ps then I get someone from Lets If you want to delete the container, docker stop and.

[Docker] Docker Compose / Docker introduction and architecture

I'm doing all the things in the world right Like this, like this, I am completing it I am doing everything manually and docker rm If I am doing it manually then as a DevOps. You should automate as much as possible What is Docker Compose Docker Compose is a tool that lets you automate things I help him a lot, how so his who. If I explain the concept then Docker Compose Basically there is a file, a confession file. Which is your wa ml file or email or else YAML yet another mark up language its full If you need the form then you can remember it but one markup language lives where your.

You pass things by value Why is it named Docker Compose? It can store one or more containers helps to make it so whatever we People just did that docker build again Docker run then docker volume so all that Things Docker Compose does in a single file. This is the best part of it, so let's go. What a Docker Compose file looks like. This repository is a two-tier flash cap inside this I placed the Docker Compose file There are two ways now, either I use this Docker I can explain the compose file to you.

This is running on the latest version of Compose. Docker Compose Version two this is the latest one now here You will enter services, services means that How many containers are to be made and how many people were there? How many containers were there in those services? Container Banana is MySQL and the second container to be created is Flasks of two containers to be made of me so you Here you can use the command of Docker build. In which if I have a Dockerfile like It is made from a build contact but if I don't have the doc file so that It is made directly from the image, so I.

I can write the image directly here. I'm going to use the image from MySQL Now in this docker run command you what happens to the environment Is Environment Now since which second in the environment variables were my sequelize database. A variable named MySQL database so you can put anything from let's You put a database named DevOps then A MySQL Route A variable named password in which you.

[Docker] Docker introduction and architecture / Docker volumes and storage

Rooted it with let's and what all can you do You can do as much as you want in the docker run command You write it, will you put it here like you can write volumes. Which volume are you using in Volumes? Were MySQL data you were using And that's what you were mapping with / ver / Lib/My AL Correct, this is what you were doing And then what else were you doing? Network networks were using correct What was the name of the network?

Tier was the name of that network, it was this, now it is And the port of MySQL is A particular port is port number 3306 If it works then you can add ports in it. 3306 mapped with 3306 So you can do this too, so be it. Went Your flask sorry MySQL's container Whatever information is there in Docker Compose Your build command is whatever the run command is I would have written the same in your value pairs. How was your flask just now?

This container also had a name, so you can also name it. If you can put it then for example it container name was what myas so its container was also called MySQL, so this became one. The flask that was there now is not from the image. It gets built, it gets built, it gets built and there seems to be a context for it. Context means where the Dockerfile is placed. If so, this directory entry Satay kuch to for example to Tear back end whichever you like You can enter the container name, then the ports.

Which port does it run on so this one The application runs on port number 5000 and 5000 has been done, now this application is something So The environment uses the environment now. Where do you find these environment variables? You can get it from here for Example MySQL Host MySQL host is an environment variable that is set by the container will be called MySQL, then yours and What are the environment variables? SQL's user is MySQL's There is a user named root and then there is your my The SQL will also have a password like the user is So what is the password, that too root.

[Docker] Docker introduction and architecture / Docker Compose

If so, the password is also root, then it is my AL DB is the name of the variable in the environment. The names of the variables in Flask. Here the name of the DB is DevOps DBs were renamed DevOps and whatnot. Your networks because you If you want to put it in the same network then two tier The name network also seems to be correct now. The number of volumes you Write down the names of those created like my S K Data volumes you have created are my SQL Data This is the volumes you created.

Created as many networks as put his name too What networks have been created? If you create these two-tier networks, Just put its name and it's done and its The value is done, so basically you have the same I have

confirmed it in the file that Ss means. How many containers are there in total, so MySQL is one is a container and a flask is a container so total Two containers now MySQL agar bat karoon So what will be the name of the container?

Which container image will be used? What will be the environment of the container that will be used? Database name DevOps root password root What will the ports be 3306 volume what This will remain and so will the flask ones. For the application or for the container that The build will be done through a Dockerfile. The name of the context dot container Two-tier back end or whatever you want You can use ports, this port environment. I put the environment variable network and this How did you make a great confine?

Given that this is the entire process for running containers Confine is Escape Colon Duplicate Enter Now you Docker Compose is also installed on this system. You will have to do this with a separate tool. The installation method is very simple: sudo AT Gate Install Docker Compose v2 so version two you have to install Because it is the latest, I entered it and It says Docker Compose is installing. How to unpack setting up Docker Compose v2 If you install it then basically you have EM Docker I wrote the compose file myself from scratch and Now you're going to run it, let's run Docker.

[Docker] Docker introduction and architecture / Docker Compose (part 2)

Compose up this is the command to run the Docker Compose file whatever you have written enter now It says additional property version is not available. Aloud, I probably misspelled something made a mistake Is And I do this every time, the version is done, now just Docker Compose you Try it now, you shouldn't have any problems. Enter See building means which is the docker build command It also executed the command that created the image. The docker run command that follows will be With which the container is created and the command is also run.

If he gives it to us again and again, we will give four different Running the commands docker build docker run docker build docker run has been replaced with One Single Command Called Docker Compose. How great very great very much Correct Superb but look what the problem is here What's the problem with MySQL yet? It is not built and is saying 0 tier back end that can't connect to MySQL server Basically a two-tier back-end container It was made before the MySQL container.

It has happened, which is a mess, this should not happen. It goes unlike that Docker, well yes Docker. Even if you press Control Q in Attach If yes then it gets quit so this is done now What I will do is use Docker Compose. I will write that this one with my flask container that Container Depends on which MySQL so unless Until MySQL is made, this is my Don't become a flask guy, but do this. If we try this, we can Docker Compose up Enter is saying that he is still saying that Let him start MySQL first.

The entry point was created as soon as He became the one who had two tier back ends I started it but where does the problem arise? The container for MySQL would have been created first. The container is formed but MySQL's which Container accepts not ready for He makes connections later, meaning until Our MySQL is not completely ready. If you can say so, then you will get one thing for that thing. It seems that what is called a health check In Docker Compose you will find the MySQL ones Health checks have to be included with the container What are health checks?

[Docker] Docker introduction and architecture / Docker Compose (part 3)

It happens that you can enter the command to test that For example, if I am in MySQL I pin the local host to myself with words as long as it's localhost until it pings. Retrying every 10 seconds Retrying does not setup retransmit [music] Pushing is happening MySQL admin ya which is mine MySQL is to try to ping him. Doing routes and routes on localhost with password until it passes the health check MySQL's container up Similarly, you will not be able to use the Flask application.

You can also enter health check in Health You can deposit checks as long as You don't do that MySQL admin because Right now in the flask, there is a little bit of MySQL. There is an admin but you can do this if you are a local Host 5000 Pay until status code is 200 Don't light it until it comes or it will happen. Some till curl local hos 5000 health right Keep exiting until it comes from when it comes right If it goes then run it, it will basically be healthy.

Just to make sure your container is running is or is not running if the container is not running So what can you do with that container? You can restart it, restart it also by one key. If you do it forever, then that What happens in

case your health check Did not pass This health check did not pass Your container will restart. You should put the same thing in MySQL also. You can do the same thing in MySQL also. You can enter that if you pass your health check.

If it did not happen restart always restart How true is it that you have completely Confession is set with it that the first container Make MySQL's container name MySQL Pick up the image of MySQL. Database name: debs; put root password Root Dalo This will store the entire Dalo volume. Give him network also and every time Restart until the health check passes and repeated health checks of 10 seconds Keep retrying five times in the interval. If you try retrying, Docker Compose up enter now it's saying property Health checks are not allowed at all.

Instead of health checks, it's a health check to This extra feeling hit me so it health check escape colon double k done Now I will return Docker Compose up and I.

[Docker] Docker introduction and architecture / Docker Compose (part 4)

Will See That If This Works Enter, you are looking at two-tier applications. Restarting here It was not found but that operation was after error It didn't stop, it kept restarting. Finally, this is now working if I now I'll go to my server and restart it. I should also stop the container. Let's say I stopped I removed it, I destroyed everything I saw this and I am stopping the container. Do how easy it would be to be just a single person? The command that Docker Compose up ts it just an if I'll write it down now like I stopped it.

This won't work, this side can't be reached right now. If I just write the command Docker Compose up and this is next to Docker Compose up i - Should I put D? Enter container MySQL in the background Running container two tier back and running end He will also do all the health checks and everything. Docker will do everything, try both. Restarting the containers is running See if your data is also safe then you can keep the container Destroy it, do anything, your volume Perseid is and with the help of Docker Compose, we.

With the help of IM Docker Compose you What did the container do? Creation Image Building Health Check dependency check restart check network Volume: Everything in one file put in the command And This Is How Docker Works Pose works so right that you A two-tier approach using Docker Compose. Created the application and Deployed Three Tier So Project Is A Three Tier Application No, then there will be another project in it or It will be different, so I hope you like Docker. I also understood the compose module.

It seems it's quite crowded here If you have understood everything till here then Tell me quickly in chat that you understand everything. We saw in Introduction to Docker that Why do we study Docker and how do we study it? Have you seen the complete architecture of Docker? Seen how to install Docker locally How to install in AW Also see how to install Docker Images Docker Containers Docker Networking Docker Volumes, Docker Compose, all these things We saw almost two and a half hours.

Let's understand what a Docker Registry is.

[Docker] Docker images and registry / Docker introduction and architecture

Consider these Docker images as two-tier. Floss cap to tear back and floss cap and Java app all these images are in your local If you need to put it on a server or to store in a repository such as. Docker Hub then gives you this image properly tagged You have to do it and push for it You may also want to use the registries Docker Registries or Repositories are needed and before that. I'll show you what's useless Delete any images and containers as well.

How can one do this with these single commands? If I tell you then docker system prune What happens with this, whatever your There are containers in the system that are not running All those who are stuck like this. There are images that are not working as well as There are volumes that are not being used. Docker will prune the system if it blows up Warning this will remove all the stopped Containers All The Networks Not Used All The Dangling Images All the Dangling Build how do you want to know.

I want to continue and look he I removed all the build caches. How to remove the Docker images I will do this flash cap and it is very simple Docker RMI means remove image Now whatever image ID you have in the image,. All your images have image IDs. You can pick him up and make him disappear Yes, but, if I have 500 images like this If it is there then it will take that much time, right that much. It will take time, what should I do in that case?

In that case there is a command, I do the same thing as Docker in general. Remove all Images are like a command in it Docker has images and why images and why Meaning all the images that have IDs bring those

images so if I just Write Docker Images. I will get all the IDs, now what will I do? Docker run mi docker run mi remove Images Forcefully -F Forcefully and I've put the command in parentheses inside the dollar. First, this command will execute Docker images with a The output of this will be Docker RMI.

I will go in and bomb all your images.

[Docker] Docker Compose / Docker introduction and architecture

Docker All images will be deleted yes this The image is still running because of its The container is a stopped container so It is kept there, even the LCA could have blown it up. So docker ps -a This is a two stop container, good if you If you want to remove stop containers, remove them. There is also a Docker Compose command of Docker Compose a Removal is fine no Docker Compose is down Perhaps down Docker Compose down yes correct then docker What happens to compost down as many.

There are containers, it stops them and deletes so one command up one Command down and then stop and remove it. It gives so easy that if once You understand Docker basics and then Understood Docker Compose, then you can use Docker Compose How much can you get all the work done through this? If you have docker rmi If you run the command, all your files will be deleted. If you look at the Docker images So you won't get any image, I understand.

Coming to how to delete images how to update images Docker Compose Up via Docker Compose And if you write hype ha fun next to it Build Docker images forcefully builds back pulls back. Docker commands are many when it builds. It is very interesting, there are many commands. There are many interesting commands. That it will be easy for you to learn Docker initially. You will have to work hard later in Docker Compose You can automate things by putting another The thing I would like to show you is Docker P.S.

We did it and this is my two Tier Flask is Cup I did this with Docker Attach this, now its screen is attached here since if I do it here then here. Pay this attached if I press control q or p If I had given it, it should have been removed, but never mind. Generally control p attach detach from control q But I think it's in my terminal. It is not happening but anyways you can docker ps by doing or by doing the same DirectoryEntry Control-Q doesn't do it, but it seems it does.

Not working but it's fine no matter what You have a second backup that someone else Access the same system by going to the terminal You can, that's it, stop now.

[Docker] Docker images and registry / Docker introduction and architecture

Docker Registry as I told you Docker Images This is the image of the two-tier flask pop. It's in your local right now I want Push it to a repository that is Docker Hub's repository so what I will do is First, make sure that Docker login works for you. Making sure you login to Docker You have already mentioned Docker earlier. Logging in with your username and personal access You have to enter the token here, then what do you All you have to do is tag your image.

How to tag an image tag What does the Docker image tag mean? It means your old image which is a two-tier flask a flask that is old This is the latest image, you can rename it. You can rename it and if you want to keep the image locally then save it like this It's fine to write a normal name, but if. I mean I need to use the Docker image tag All you have to do is rename the old image and Enter the name of the new image as Train With.

What does it mean if you have to go to your local computer or from your server I want to push some image to Docker Hub. If you want to post, you use Docker Push. If you need to pull an image from Docker Hub If so, you use Docker pull now. This push has already happened here, I'll go Ch Is This Docker Hub on your Docker Hub and here But I refresh I'll see that my two tier back end is this image.

The last push less than has reached here A Minute Ago How True Now Once Image Do what I can do if I get there? I can simply say in my Docker Compose I am sure, you build the image again and again Don't build what you were doing Repeatedly fetching images from the build context Were building, don't build it, instead. Escape colon double k enter Docker Compose down and Docker Compose Up The image is that it is not built locally anymore. What is happening is that it is fetching from Docker Hub like This has brought zero in your image right now.

Downloads were refreshed I will do it, I just downloaded it myself That's why he is not coming but if you are also two tier Back end will run this thing two tier back end.

[Docker] Docker images and registry / Docker introduction and architecture (part 2)

I'm going to give this image a two-tier back end. I pull it to my Docker desktop. Username and password oh my docker here If you are not logged in, try Docker login. Can Ho Username TrainVShubham Password Personal Where was my access token somewhere around here? You can authenticate well and do local I will authenticate myself, I am done logging in. I am going to pull the train from Dockerpool.com. I will do this to tear back and latest Once your image is created, tag it.

Do push that to Docker Hub and now that You can now use my image anywhere. I have images from all over the world but can You See Docker Train Wish Bun to Tear Back Key The image was created 7 minutes ago and it I've got it, how cool is this Docker? How cool that you can Docker Compose docker Images Docker Hub Docker Push Pull You can automate things, it's so cool. I gave an example, right, that Docker what happens what does docker say it is, it is running on my machine, but it is not Running on client machine now tell the client.

That, you just have to do docker run and this The name of the image is just over and its application is there. That is the beauty of Docker now. You must be thinking that, I have Windows is in Windows Docker Desktop is what happens in Wiz. You have one This is the Docker Public Registry if you want. Please explain it to me in simple language, I will understand. Let me try using Docker images. At your image from Let's which was the image of its Java app Which one was not Java?

No, if you remember we had a Do which P it was that was made, the flask P ECS Remember this one Flask ECS This one, whenever I build it Docker Build Haiti flask. P latest dot so whenever I built it I did this one which was my python Picks up the base image of 3.7 Now Python 3.7 The base image is a very large base image. If it happens then whatever comes out as my final image Isn't that a very big image of the application?

Ban aata who is a very concerned thing that there are so many big things in your system Images are there now, so this application is very The blank application is a very simple application.

[Docker] Dockerfile and image builds / Docker introduction and architecture

Tomorrow when this application becomes very big If it goes then its image size will also be very large. If it happens then let us see what our Flask app was what we just had. What the size of its image is. How long it takes to download If it is taking time then where the size of the image It will take some time to download if it is large. It will take time to upload and create the container. If it takes time, it takes a lot of time,?

The image is yes Let's see, I think something like 600, 700 If the image is of MB then Docker images so my flash cap is around No. The pack is 1.01 GB, can you understand that this much Big Image So What Multi Stage Docker Build Divides your image into stages Lets see the image is created from the Dockerfile. What makes a Dockerfile a Dockerfile It becomes yours The image is the image you have now. 1.01 GB has such a large image that If a container is made, it will also be very big.

What happens inside it that from It happens that from that, he picked up a base image. The image is of Python which is 994 so your base image is the same 994 Now it is your code then Snip then your requirements files and All these libraries together have so many GB. If I were to measure the size of this thing If you want to reduce then the biggest one is I can reduce the base image to that. Sometimes what happens is that this base image is v im Dockerfile ye base image 3.7 Sometimes it is required To install libraries, every Sometimes we cannot use small images.

Requires library installed But only the library to Required to install stall to run libraries or Then to run the application, Why do you need a larger image? Every time you pick a base image that is a stage this is the stage so like you From Wrote python3.7 then in the whole code. If from is not written anywhere then this from python3.7 this is a stage As much as your base image can be in one stage The size is that much plus the total size in front.

[Docker] Dockerfile and image builds

If it happens then your total size right now The base image of stage one is If it is of 994 then the total size will be Around Ruffle 1.04 GB will be fine. This base image is called I'll give you Edge, meaning nickname Alias Edge. I would have written the back end or the builder. I named this stage Builder Stage So what is Builder Stage This is just the stage in which you Libraries and packages will be built and Then what I'll do is I'll go to this builder stage.

I'll stop it here and start a new stage Let me create stage two in which I will create the base image. I will use the base I will use that small base image. From Python I will use a 3.7 slim small base image. I will use the slim one

which will be the size Around 400 me B size choti le What will I do if I take it So far I have everything installed. The one that gets installed is the site If the package name is installed then I will create a workaround named / app.

I copy I will take my upper stage, right stage One copy - no From The builder is at the upper stage. DirectoryEntry What else is there and I have requirements which is already installed once the requirements are installed, it will What is the need for a large base image? If installed in packages then this Particular location remains Have us r local lip python site packages this particular The place remains So I'll copy this from here. I wanted an image but once the requirement It's installed, then why do we need a larger image?

I'm going to look carefully at the Dockerfile. The base image for creating the image Base image this from python 3.7 If its size is too large If it is large, so will the size of your final image. Agri will be big Now why is this big image needed? To install packages then Get image Large image Install package Do it, then there is no need for a big image. You don't need to reinstall it Copied it, run the application and do something.

Will go show run Of Docker build -t Flask now this one image name a little bit Keep the flask small Mini Dot So Its Image Name Flash App Keep the mini.

[Docker] Dockerfile and image builds / Docker containers and commands

What the stages are doing in it. How many stages have been completed. This is non as multi stage build one stage With your big image and a stage for you Now you people will see it very well with the small image. You will be surprised if I do Docker Images to Flask P Mini is just 142 MB flash cap big which was 1.1gb so Basically you have created a file from a 1.1GB image. Multi-select your images with up to 140 images Compressed it with stage build, but How is it possible that from 1GB you can go up to 140GB?

That is the power of cashing and Multi stage build because there is only one Your stage is not being used. Divided, used a new stage Yes and the slim image is by The default size is something like 140 GB. I might have written 400 here, but That's not 400 He lives very little, how little he lives Let's see, it remains 125, Can You See 3.7 Slim 125 So the base image of stage two is that. He came and he will come out, what about 140 Isn't it cool how you can get 100 GB from Day 1?

Because we have just finished Our Multi-Stage Docker Builds How much are multi-stage Docker builds? Interesting was just monitoring and logging. In Docker so like you did Docker P.S. Your running any container from lts If so, Docker runs in the same way as it is now. Flask A Mini Latest Flask A Mini should be run on port number 80 Run to port number 80, this should run Welcome to Touro is going on now let's call it. If you want to debug then docker ps you and go to this ID and open Docker You can check the logs, whatever your logs are.

The logs are there, you can see them in front of you if you want. Another thing you can do is to add as many logs as you can. You can redirect to a file The command remains no hoop no hoop no hoop if you would know no What does hup do? Any command Runs in the background and outputs It also transfers it in one file. From the let's I did Docker Attach and ID name, oh my god, this is too much has grown up So let's docker ps I did this My ID is no Hoop Docker.

[Docker] Docker introduction and architecture / Docker containers and commands

Attach ID name and em percent at the end I put end and pressed enter, now look no Hoop Ignore Input and Append Output to no pod enter now my pm is running now i. Did / Health so here comes the server is up end running and I did Slash Err or Slash Thanos Whatever I did, all this data was hooped out. Is being stored in a file named can you C HTTP Health Error Thanos I enjoyed that you were able to collect logs.

How can you collect logs from Docker? No hoops ahead for containers to attach Docker Command M percent now whatever I do here Let's say I followed this up with Thanos is bad like this I have written something, even that is for you. This is being collected in the logs here In this way you can very easily monitor and You can log your Docker containers Look, are you enjoying reading Docker? Introduction to Docker 3-minute course of one and a half hours You are seeing a great video for hours.

Things are falling into place the way Docker does. What is Kubernetes, what is Kubernetes Nits so right now in Docker you read Docker You must be having a lot of fun reading it. Docker this is Docker that is Docker in containers You can also analyze the logs and Docker Compose is so cool docker multi stage so Destruction was a lot of fun, but one of Docker's. Docker We Go Directly into Production In production We do not use only Docker. Containers are not used in production.

Why do because like I just did docker ps Simple if I docker Two characters of the container ID for Stop Fur B I will also write. If it is not gone then it will end and this Docker If after stop I write docker rm If you remove Docker, then whoever moves around will use Docker. If it is a container, its data will be deleted. This will be exited, so Docker Chances of container crashes are high There are more crashes in production Docker containers are therefore We do not operate containers individually.

In production this is why we use Docker containers what do you do orchestrating, what does orchestra mean?

[Docker] Docker introduction and architecture

Orchestra stays right complete great one He is playing a song like this on a mic. There is a drum playing just like that Kubernetes is one such orchestration tool. Which manages the Docker containers that a Docker container killed if it crashes The second Docker container is up immediately Go get a container with plenty of storage space. If it is there then allocate that storage and room. Two memories so that it can be used within certain limits. Containers run sometimes you will see us It's so cool to open amazon.in [music] amazonline.in comes in handy if you If containers are crashing then those containers auto-heal and if traffic increases.

Container runs are being run for 100 people Traffic has come, five has become 50 based on Some criteria to Kubernetes Docker containers are running in the background They just keep going this way that they have auto. Scaling and Auto Healing capability, auto scaling and Has auto healing capability so that if Docker container crashes even in production I want it to automatically get up and running. If you want to read grave policies then grave Kubernetes have some things like pods So what are pods?

Pods are such a unit resides inside which your Docker containers run They are living there but now this pod of One or more docker inside Containers can run now I get a lot of. In case you want to make pods, you need to deploy had to make is a collection of pods Reps Deployed You Have to Make It, now this is deployed, you can back it up. You make the front end separately and deploy the end. Do you make it separate or do you make the database separate?

To get those deployment to talk to each other You can get the network you need for this. You connect us with the service Deploy to users so if you have an Accessing those deployed to the outside world If yes, then you feel the service. There is routing in that too, so you have ingress routing will take place So all these concepts, if you If you want, I can give you Cobra Nits Cats in One. Let me know in the comments, this is it I gave myself a small intro that How Docker works in Docker Kubernetes It is Kubernetes, that is the entire framework.

[Docker] Docker introduction and architecture (part 2)

The entire architecture is what your Docker Limits prevent containers from crashing Lets the scale heal and we cannot run Docker alone Therefore, Kubernetes has to be run in a cluster. Industry standard tool for reverse Proxy MySQL is a database that Production grade database remains very All companies use it and the web Applications are generally available in many companies today. They are asked in interviews Requirements are also product based Web applications are generally used in a company. We are going to create a web application using Nginx.

Pairing up with And MySQL First of all, understand what a web application is. What is it and what is Nginx and ma What is a SQL to a project? If you understand architecture, then the project is That's basically your web app that runs in Django. It is written now that such programming is being implemented Language is a framework that The language it uses now is Django's from MySQL database and this complete The application will be served from engine to engine. This is also kind of using three containers.

If you have an application then your first container The first container to be built will be that of Nginx container the second container that will be created will be created your Docker's sorry Django container and which. The third container that will be created will be that of MySQL. Container Now this is the container of MySQL, this jungle will talk through the container of, in this also we People may have to set up a network Can he have to install a network?

The most important thing is that and that's what Nginx If it happens, I will tell you its work. Like whatever applications there are for Django Or any web applications that are Let's move on to the port number. 8000 now we will run on port number 8000 8 Maybe someone else in the 1000s Routing is going on but that routing It should be done internally, not externally, port number. Have to give so that internal routing can be done A reverse proxy server is a reverse proxy what does it mean that you give them a proxy pass so that if.

Your URL came up as localhost 8000 or If your URL is slpup or slushhome If it comes, please redirect it here internally.

[Docker] Docker introduction and architecture (part 3)

From IP LETS 3.28 7.1 Whatever it is and beyond I If you put a slash, the routing to the engine server will be If there is a slash, then it will be written like this. Redirect to Django if it is / db So please redirect me to MySQL. I understand how much If you are interested then this is the engine setup. Django setup ma cycle setup So I have created this application in my Git. It is placed on the hoop, what do you have to do?

Here Expense Tracker Notes Page 2 English Two Tier Floss Cup Spring Boot Bank up 90 days every type of project here You will get the notes, click on this. This is a powerful app, we use this app end to end. If you are going to build then code this Take it and copy its URL here. Going to our instance, how many of our Projects have become Falas Cap a Java pop two tear floss cap third and this git Clone toother flash cup no junko notes.

This will be my fourth project today. I have done this, the notes have arrived, now let's look at this Understand the app Django Notes Understand the app what are the things in it Where are these projects from? All this is from the build that is there html4 If all these things are given then you have a file named Dr. You can see this test db root this You will also need to create a file named dotty env from where to connect to the database which Enter the required environment variables here.

If you have kept it then there are two ways either you can also pass environment variables. Directly or as an environment variable Can you also pass the Dr. I will show you both things, let's do this right now. If we have cloned then you understand gone that nuts a is basically your back and My Notes is your front end and the name Nginx is the folder where the confine of Nginx is located. If it is kept then what is the confine of Nginx The confine of the engine says that if somewhere on this server on port number 80 also http Django container Django container this.

[Docker] Dockerfile and image builds / Docker introduction and architecture

The container will be named after our back end if Its port number is 8000 this URL is somewhere. If it also comes then it is reversed through slash proxy kareega means what does lets do from this is my instance this is my instance is if I do this my which instance. This is the IP address of the slash colon 80 port number 80 If I put a slash, where does it go? This will redirect This will redirect This works on our internal Django container If it happens to Nginx then people will do all this too.

I gave an overview once that Which folders are there, API that The folder where all your code is stored, now the code is done, clone it. If you see the Dockerfile, click Dockerfile. A v Dockerfile so what is its Dockerfile Speaks from python3.7 work directory / back end copy requirement.txt to ai back end so what is this Firstly, he is doing the requirement and TT. Where is this person copying his The working directory is created and then What are you doing?

Updating your container He will upgrade what we have done to It's the same as it was in Tier P, Li My will install the sequelize client then pip. Install MySQL client and so on will go and install the requirements then Sorry for the remaining code back end. The port will be copied to the working directory. Number 8000 will be exposed so the docker If we had created the file earlier Let me show you what we did with this The first app we created was our two-tier app.

Flask app This was our two tier If you open the Dockerfile of Flask You will see that you will find almost similar things. Python3.8 took a slightly slim image All these are required in the work directory. Gone escape colon db enter so our Docker The file is in front of you, if you want I can return it. I can also create it but I have created this file If it is made from the same then it is almost Similar is fine So what I'm going to do now is to create this Dockerfile.

This is my Dockerfile, I will build it How to build docker build when t notes p and dot notes p dot and enter this By doing this I can use the Dockerfile of my Notes app.

[Docker] Docker introduction and architecture / Docker Compose

I will build it, let's take notes. Dockerfile is being built very Good Come on, let's keep building this work. Directory p all this that stuff done Now you understand what I have to do Three containers to be created One container to be created What to create a container of? Ajax We have to make a container of wars and another container I want to make MySQL, it has been built. If you have an image then you understand the image Dockerfile. If you must have gone, I can tell you again.

That's all this Dockerfile does. Copies the application to its container Whatever required libraries are there in Install it and that's it She does, what's most important is This is the Docker Compose file where the entire engine is located. All this is written but, this one which Docker Compose file isn't it what we want You write, let me write it, then remove the new thread. We will compose this and remove it. That this Docker Compose file has been removed, we People are going to build, let's build.

Em Docker Compose dwa ml docker We're going to build Compose. Whenever you are building with Docker Compose First remember to put the version two version Let's put this 3.8, now whatever you want If you want to create a container, add its services. Services let me get the first container which I want to make that banana of Nginx which is the second I have to make a container for the wars. Or if you write anything in my notes, then Django and the third container that needs to be created I want to make MySQL so as many.

You also have to make containers for all three of them. Write the name that these three containers are to be made for me. The container containing Nginx is From which it will be made or it will be made directly from the image Can be built or can be built from a Dockerfile So save it here double enter save. CD do CD Nginx This Nginx folder has two The thing is Docker The default conf is or whatever the default conf is How do you write, please simplify it.

Nginx default conf file just search it ajax Default conf file on the bus Beginner's Guide Go to Nginx.org here you can find that How to write a confession file about all of them.

[Docker] Docker containers and commands / Docker introduction and architecture

I have to write this part in small words, sorry. Server that tells about the server What should be the server's confine location? Meaning URL if slash is in your URL So redirect the proxy pass What do these people say about this particular local Host 8000 Beyond whatever URL you enter redirect so it does that if. If you want to add any images, do that. Two more and with the help of this I made two tier Create an encx file in Flask app not ter sorry Django nuts p me a I have made Nginx's default conf in it.

80 Listen 80 means whatever your engine The server will listen on port number 80. First of all create an AJAX server Let me show you, look at the Dockerfile. From Nginx 1.3.3 Alpine Copy Default Copy to T nxk dd.com so this is the path this is the path in your container and this dot default.xml would also work, so would yours. It will work, you will see this IP address of yours You will go to here you will go to the IP address If HTTP will show you Welcome to Nginx It is so easy to use Nginx.

As much as this Docker run command is required, I definitely need that engine image plus You also need to copy your fashion file into it. That's why we have written the Dockerfile so it's stuck again so we have a This is the screen for backup then docker ps And docker stop this and docker rm this ends then. This will stop that container which is running its There is an engine container that will just stop let docker stop f2 f3 done stopped.

This engine of yours is a site that cannot be read. If it is right then your engine will be boosted in this way. What is extra in the Dockerfile? This is from Nginx, meaning that base image pull It will do it and then copy it as default. If Confine has so much work to do at this place then We need to build this thing through Docker Compose. If you want to do it then I Docker Compose now what do I have to do to Engine I want to say that, what do you do, you build it.

[Docker] Docker introduction and architecture / Docker volumes and storage

You build and whatever you build context which is the build context and Pick from a folder whose name is dot / Nginx dot / means this corresponding directory entry in the folder. Pick up the file and build it now from Engine Enter the name of the container name and name You can put anything, put Nginx. This has happened, now who is that Nginx? It runs on port number 80. If it works then which ports do I need 80 Colon 80 These are my ports now and What I need, all this has been built.

What I need next is that The Annex container restarts. If it shuts down and restarts, it If the container is restarted as well Restart Always to Restart Always I restart my container now that's my job Nginx done, finished, I'm done now, I'm leaving. My Django container depends on my On the container of the SQL, first of all I will I'll sit down to write the SQL's container,? I will write Django later, now MySQL. Let's sit down to write a container of The container of SQL has no Docker If the file is not there, it will be built from the image.

From which image is MySQL titled image. The build that will use those ports will be Which ports will it use? Will use 3306 and is mapped to 3306 will remain and which The environment will use it like we did. Last time I did MySQL you want So you can put the environment like this, my AQL Route password and let's put you root and. Let's move to a MySQL database and you put test a dib ye MySQL put the root password and database of you can also add volumes if you want.

Volumes What put the volumes to MySQL data colon / ver/lib/my sql ye We last saw Docker in it. How do you give volumes in volumes? You can move on to the most important part and If you want to enter the network then networks you can put this Please put some name in it Notes P so Notes P name. You should create a network in which this thing You will put it, now whatever volume and Input the network you created below.

[Docker] Docker introduction and architecture / Docker volumes and storage (part 2)

Two so what is MySQL in volumes Data This is your volume and what you Networks which one is this what is it Thay Notes Pa Apke Network Ho Gaye Toy. Your work on MySQL is also finished. Docker's Cha is done, now your work is done Sorry Django's I always know Django's Docker No, why should I tell you, then let's go now The work of Django Again will be the image that. Even if it is built, where will it be built from?

If you do a dot, what will happen to the current? The Dockerfile is kept at the same place. He will take it from there container name now container name here very It is important to know why because we People have said in the engine that the Confine Usman Django Underscore Kant. This war is under the container, so if you want to pick it up from there Your container name should be JangoCont or Then whatever the container name should be, it should be like this.

Should be updated in confession then here Pay Jaake Django Kant Your Container Name Ho It's done, what else do you need, environment file? The environment file is new. Under file what does this one say that from You can now import the environment See what the name of the file is. If you take then Django notes in P which If the environment file is dot invi then you dot here You can name this file as invi, it's done now. This container also Restart Always Up Always Up Similarly yours is Whatever MySQL will be, that too restart will always be there all these All three containers should always be restarted.

If by mistake someone has failed well then Now the structure of our three containers has been created. That's it, yes, this container too. If it remains in our network, it will also get a If the network provides NetWords is a notes app, so is this one. If it has come into the network then your Nginx Your container has been created for MySQL. The container has been created and so has your Django. Container bun b some things in the container of the jungle Missing is like ports not written we People put ports and Ports seen due to environment file How much easier your work has become by hosting us.

[Docker] Docker introduction and architecture

The database name and password are not all of these. I had to put it, I would have taken it from the environment file So whatever our port is, 8000 is mapped to colon 8000 now if you If you remember, we fought these wars. We were creating containers through the Dockerfile there. This Expose did not write anything after 8000 There was no run command so you could do whatever You want to give commands to that container. Write this in your Docker Compose file If you can, what you have to do is write simply Is The command is fine, simply write the command and the.

You can write any command you want with let's You want to give a command in a shell script Yes, first of all you want to give a command that Yes Unicorn what is a Unicorn. G Unicorn is a tool that helps you Serves applications in production So Jia Kan is a W SGI There is a file like your Notes app. Drove on 0.0.0 8000 so whatever runs, it will run locally Inside the host container you have now installed ajax It is said that If the user enters a slash in the URL,.

If it redirects to this application then If you just put a slash in the URL If yes, you need to enter the port number. There will be no need to write it all down. Your application will have to be running internally This is the power of

Inex, just for demo I'll show you through that's done now one The other thing remains your Django application. This happens whenever it connects to the database. If it happens then that migration has to be done Some of its commands are python manage pwa migrate python.py migrate this one There is a command which can be used to access any tables.

There are tables and columns which are databases All those who are going to become I become - - no input what does this do It is that you are saying that after migrating. If this command is run directly then it The command to run my Django app is Now my Django app is also ready. The container for the war has also been made and I think everything is ready, let's go and see.

[Docker] Docker introduction and architecture / Docker Compose

What I will do first is simple I will Docker Compose up now than doing up I will understand that it will appear before me what is going on and what is not going on. It could also be that my Python Container of Sorry Django Container of MySQL It pays but I put it in here If it is not there then it is good that I reminded you, let's go quickly. Let's put VIM Docker Compose now mine It depends on the container of the war If it is on MySQL then I will write it here.

Depends on what depends on MySQL pay so my container of Django Depends on MySQL And what is mine It is the container of the engine and the engine also has a mother. If you want a SQL to your wars then Ajax Depends on what my not my own Django engine Depends on the war and your war may Depends on the SQL, it's simple Ajax. It depends on the wars and this is over now. I do Docker Compose up Done Let's See First That MySQL I will pull it, I have pulled MySQL.

Bridge Complete Let's see what happens Next looks quite interesting because In a single command you can do the entire Are you applying up? Please give me the full application up here. DevOps is fun Come on, he's building Django right now, oh ho. If you If you pay attention, what happens Tell me, something is happening, so basically Your MySQL Is Ready Your Angel Nicks saying host not found o Django app Django. App not found and what not He got that No, never mind, the Docker entry point is here.

Can you see unknown server host db then The host name is the DB, is it like this? We will correct all the errors Resolve First enter a health check Let's give you that container of my MySQL Put this first condition that it should run properly Let me tell you that MySQL should work. Because it is a database, it should run Let's go let's go Now such interesting scenarios come up that You write everything and then something else happens. If not, then first of all look at v m ad evi Go to the environment and see what If yes then the database name is test db is it correct Username is correct DB password is root.

[Docker] Docker introduction and architecture

DB Port Yes DB Host Yes What do you need to put in the host that my The name of the SQL's container is that pouring. If you have to, the name of the container of MySQL will be I have already put MySQL so this is one This was the first mistake that we rectified. If you have done it, then the container of MySQL name is my s no my note no docker Which is the name of my sequelize container in compose He has to put his own container name.

I have not even put it, no problem, so let's go my Enter the name of the SQL's container. Container underscore name what have I entered my AL so this is the name of the container now I want I am not ready until MySQL is ready If the container does not move by then, I. What will I do, health check, I will put it And I'll put one to the test As such command which Do MySQL Admin Pin it For all this comma separated pin.

Kare for my host lo local host P means that MySQL will run in this container. It's going to ping the same local Host with the username root and its password also the main root should be this Test condition should run Interval: It runs every 10 seconds. Try running the MySQL container first. If that is the case then we will proceed further. Will do it so it's done escape colon d k Enter this is done now yes the second one which was There was also an engine issue, so VIM which in Ajax.

There was a default conf, there was some issue in it Yes, look, he's holding Django app. Is not a Django app, it is a Django Container it was catching it done let's do it The work is done, now return it. At Docker Compose up now, ideally everything should work. Because we also did MySQL Fixed Docker too, sorry Django. Has also been fixed and the engine also Mistakes have been fixed, please provide Docker support. Enter Come on, let's see us C Host not found in upstream O Basically host not found in upstream This means that Jangpo is not yet ready.

[Docker] Docker introduction and architecture / Docker Compose

Why is this Django app searching for me? Let me tell you why you are looking for this Django Have you ever been to your Docker sorry yes Do you change Docker's confession? By default Docker Compose doesn't load its cache If he picks up things from you then you have to. I have to specifically tell you that build also do build also then you will get a command. I have to write Docker Compose up - - build but how do you write Docker Compose up - - build but now I have this thing Notice that when I do this, every moment of my life.

Things will start building back up, let me do it. Can you see it starts building back up I went back and everything started working, great. Running can you see docker Entry point confession complete now he says Host Not Found Confession Completely ready for start now until my MySQL is not set up properly then. Till then the engine will keep getting this error that Host Not Found Django Container Not Found As soon as my MySQL setup is done then. It will go to the Django container and say what container Acetate Unknown Server Host Let's see, let's see, let's Continue seeing that my MySQL when Things will work only if you are prepared properly.

If MySQL is ready or not. First of all, if you are a beginner then this much Don't be afraid after seeing all the errors and think what is this? What is all this happening unless your It won't work until things get sorted out. Unknown Server host don't talk let him do it tra main mai The time when the SQL container should run, my The sequelize container will run the Django container on its own. Things are going smoothly Or we can do something else.

Are our Django containers restarted until then This continues until a particular test case If it doesn't pass, let's do that. I Why is this Unknown Server Host coming up? Hitting an error I do it, I do it, fix it, stop it Go stop, so in such a situation videos generally where the errors are coming. I have it because of the same errors you people also learn c so vim Docker Compose now i What I want is the health check I did in my.

[Docker] Docker introduction and architecture / Docker Compose (part 2)

I did the same health check for the SQL. Make your own containers for your wares I will do it for you and tell the wars that, as long as it's yours CMD Shell In CMD Shell MySQL I will not ping the admin. I will make curl that localhost a until your up and running Till then keep trying, give it a curl. Give the command and tell me what happens in curl means like we are sorry on browser Go and write this IP address 34d like this.

You should have written something like this, this curl, this curl also happens curl -f now here you Please provide your URL http colon localhost colon 8 000 beyond which your The application is running as / admin. If the URL is If it comes out well then it is correct. This became my health check condition that As long as this test case or this health test return the container until it passes keep going it's done and that's it until This won't work until your.

A, which one is above, Nginx depends If he does, then Ange Knicks won't even be there. Let's get started by naming it something else Let's keep it, let's also give it a D name. Just because I feel like this I am confused with the host name, let's see Let's escape colon double k enter that's it. Docker Compose up - ain build so I hope You can see this build command in my photo. The build is visible Enter what is saying service janggo Depends on Anfa So I changed the name of this DB.

If you want to do it then it depends on the database, this is done. I will do Docker Compose up halfway through Build and back up the photo what do you say The Container Name MySQL E already in use or to Docker Compose down to. Your container was already running, now what should I do? I'll back it up docker ps I will check if the container is running. Docker Compose up not running - ain build Enter Let's Go Name E already in use why are you saying this If you are facing this then docker ps.

Try a container and see if you can see it.

[Docker] Docker introduction and architecture / Docker volumes and storage

This is MySQL if I exit docker rm remove the first container now look at your MySQL container. Is the reason that your MySQL container is Not running and I'm looking at Docker Compose Is there any problem with MySQL? Went image MySQL container name ma Sequels are Ports are Environment ma SQL root password database all Something feels right to me, volumes my SQL Data Volumes Yes Whenever you. If you make it, then definitely put a dot here, dot MySQL Data to isi jhaj par banaye waise Well, there is no need for a dot, just look at it once.

I take it that the old volume we made how was that made yes can you see dot SQL data is inserted in /, maybe that Issues are coming, it's done, networks. I have put Notes on and I have put Restart on it. Needed Docker Compose up - of build enter let Tried to run MySQL something is going on This had MySQL getting exited first I

was having a lot of trouble with this right now. The SQL is going well Is, let's play our MySQL one more time.

Lag Jaye Yes MySQL Ready For Connection Now he'll run his Django container Why is the engine container failing? The one did not run, let's see how you can see the unknown. Server host my SQL Server named this MySQL Why can't he find the host? Host Not Found Django Container Ye Django Container It is unable to connect to the database Why is this happening for some reason? Still Argo CD container acetate Coming with code Is, let's try it out. I should have, but, I think for some reason, one Let's take a look at our There is a main compose file, what have we done in it Did I make some mistake while building the engine?

I have done everything, I have built the context Did This is also bindi environment file also please give I had given all this also I have given the image container NameDB Underscore Container ODB. Is there something like an underscore container? DB underscore container in environment It is written what could be that the DB Its something to do with the underscore container and DB Relationship Yes, let's see.

[Docker] Docker introduction and architecture / Docker Compose

DB Underscore should not be like that container let's give it a try Let's give it a try, by the way our This entire Docker Compose file is the same Whatever we made is the same but. Let's just give it a try, this could also happen that it's some kind of particular thing Let's try that and see if it works. It's well and good, if it doesn't work then he will D bug will do Come on, we should not go to an unknown server Host MySQL whenever you have such an unknown.

Server host see what your notes are up to Actually, this work is done by the developer if some It may or may not work but you can change your settings in your power bank. You can go and see what he's doing when Trying to connect to the database This is the host and the DB Underscore Host in db underscore host whatever Your V.I. Whatever will happen in M.invi, it will be this It has MySQL right walk Let's see here I did DB I have done the root test db everything looks correct It doesn't seem like anything is wrong.

Here let's just see what we have here container is kept name container what is put here [music] The container name is Django Container DB Let's try a container. What difference will it make to the container name? I have put it in double quotes, by the way double There is not much problem with single coat but Let's give it a try let's use the DB container. What is the name of the container of its engine? Inserted in the Nginx container let Ang Nix container inserted And what is my other one, of the wars?

Container name Django container Django container inserted Again, let's give it a try, it may be something There may be an issue with the container name in Docker Compose. Remove as many containers as you can from the bottom. All the containers from Docker Compose Up turn it up let's C Let's Give It a Try Confession complete, but the engine container should not be created. Until Our Docker's sorry not Django's container Until it is formed, the engine should not be formed but Let's C Let's Give It a Try to See What's Happening Its with Unknown Server Host.

[Docker] Docker Compose / Docker volumes and storage

He's messing me up, I don't understand him If there is a network then everything is fine. That's right, networks in all networks I Think all put right in Is We are interesting, some are very interesting. Interesting work is going on, let's see Network Django Notes P o I understand If I do Docker network create docker network create high which Notes app - de bridge if I a network. I specifically create the bridge There is a type of network called Notes P So this network has been created, now I The Docker Compose file contains a network given that now let's see what.

I have created a network and I which is giving network in Docker Compose file If he can connect with her Docker Compose Up Lets See Let Us C well, another thing about this was this. ApnaDB which was the name of the environment file DB is the host in the environment file And if you put something of your own in front of DB then e. The DB in the embed env was the underscore container, that was one thing that I suddenly remembered.

Went to Docker Compose up - - build Let's see what he's saying O MySQL data permission denied so here But someone named MySQL Data was made Does the OM remove it or. You can also give it permission to This is MySQL data, just delete it now. Sudo rm - rmy for SQL data why is this happening this is such a folder has been created that shouldn't have been created. Go to compose, this is MySQL data and There are networks, right, remove it from here,?

We would like that whatever volume is there, we Create externally Docker volume if Create my sql data to my sql data I create a volume named MySQL Sorry Your Notes is named. I have created the network, now I will go and

see am Docker Compose up - - Build now according to me this thing should work I want to know why, what is that engineer saying? Undefined Network Notes P Invoke Compose project I had created the network,. Docker Network LSA Notes PVC Docker Compose o did you understand what I said I messed up what belongs to the wars was not even connected to the network small mistakes im pretty shy.

[Docker] Docker introduction and architecture / Docker Compose

People must have written in the comments In the network we have containers of Notes P Argo CD was not even put into the network, small Issues keep coming up, so your wars. Container is MySQL of this whole one I should have him in the network. It was not even put in the network, it messed up So what is the name of the network notes P like here note p Done Let's go Docker Compose up - ain build Let's See. Fingers crossed let's see how it is Goes Let See How It Goes, there's still something going on.

I understood that real life problems are a how many problems can arise What was our problem in the wars, everything was fine It was running from there, network was not provided. It will take a lot to run this, let's run it now. Plus will you put this is and you put this is how You Trouble Shoot In Docker dicker was done docker save this Saved it, the notes arrived, on which date. Made it, that too has come, it is so cool,.

All your logbooks should also be collected and this through the entire engine I have not entered my deployed number. The port number is not listed anywhere. I created a network of notes. That notes app network called Django I passed the container in the argument I didn't do it, that's why it can't connect. The rest of it is up to him to find that small network. If we talk about this simple thing then it is that these big people Such small things keep happening on the channels.

All the containers that were created will be removed and Docker Compose up -d karunga so that It will start running in the background, cool I can operate the entire computer now. If your app is running in the background and If you look at this container Despite removing it, this data came back. It simply means that your data can also be parsed. Almost quite next level engine my This project was a SQL, it was a flop cap. Simple Java Tutor P and Project I want it now, meaning my heart is not satisfied.

Mange More We need more projects and you Let's give some and see the project but I Hope you will understand this much How to run Docker is clear.

[Docker] Docker introduction and architecture

Everything will come with practice, so we did this I also did a project with Let's see this three tier application with Docker Compose is what we will do in Java. Let's try the application, let's go. Let me try the application and see it now. There are so many applications on Notes. Lots of these expense trackers Let's see, I have affection, this is our batch I have done this for the learners till now. By running If This Works or Not, let's run Docker Let's see the file of this application.

This is a project of Mavin And it is made of multi stage That's great, let's give it a go Let us first see about this project. Let me tell you a little bit, so this I will tell you how three ter A is. If your browser is fine then The UI is made on a thyme leaf that will shine on your back. It is built on Spring Boot. Spring Boot is a Java framework Yes, it's, go, go, die, disappear Java lat java that one java no java bahut It is the most popular programming language The framework is Spring Boot, so your.

Spring Boot is your back end and which The database is Again sorry take database on MySQL Let's take it, so there's a database on my site. Your MySQL DB is this three. Tier application is yours now this spring Boot is the back end, it is built into a from the particular tool whose name is Mavin now what is mavin. Mavin is a build tool that works well If it is made by the organization then it is good Maven is a build tool that allows you to I will build a great application and it's done.

This is what you have to do in today's project. The first thing you should do is to install Docker We need a composition in which we can use these three applications. People will be building these three containers we People must be making what is the best part now This is about thyme leaf and spring boot Spring Boot and Thyme Leaf is this you a. You can combine it in the application just Like our previous application Build the front end and back end into one I gave it to you in a thyme leaf spring boot.

[Docker] Docker introduction and architecture (part 2)

Only you can combine that through Mevin Once built, our base image will The base will remain Whatever image remains, it will remain Mavin's again. In the base image of Mavin, we have Java Build the Spring Boot app and Externally take a MySQL database and try to connect it This is our simple project, let's do it. Let's move on to the project. Let me tell you a little about this What do you say about this entire project? You don't need to do anything for Maven.

This is any application that is running on Java. Application and Java Application Run Time There is a maven jar installed by default. It comes with Maven for download and this Maven has some properties which are by default You don't need to write anything down SRC is no longer useful to you developer It is useful because on development. He writes code, so here's some test cases these are some test cases and What's in your main resources In resources you can see it in static asset css these are all these things.

Important part because it properties, understand that this These properties are stored in environment variables. What is the name of the application Expenses App? These are the expenses of your application. The meaning of data source is database Connectivity JDB (Java Database) Connectivity is the thing he uses and my Now pay attention to what connects the SQL Read from this URL and I got this much. I understand that your MySQL The container name should be MySQL The port of the container should be 3306 Database name must be EXPENSE Name of the tracker and your application Expenses must be the username.

Confession that your Spring Boot app has It means the configuration of Spring Boot P Confine here is that something variable and its Correspondingly Meaning this username is in MySQL. The username will go and the password is this my The password of the SQL will also go to MySQL The name of the container will be this, the port will be this. The name of the database will be this, I understood it. Your application's properties are now View the source code of the application.

I won't go into the source code. I'm not a Java developer, aren't you?

[Docker] Dockerfile and image builds / Docker Compose

If you are there then you can definitely see it but this much If you are happy with me then this is your Java. This is the application, now let's see the MV ad. Again this file is also some of Maven commands that are by default In Maven project you need to write No, what is important to you is pom.xml It tells you that your application. Like my application dependency use Spring Framework Boot and Spring Framework Thyme Leaf Creating a UI in the Spring Framework itself There is a thyme leaf for then for validation There's something this that my MySQL connector use.

On so all such mayvin The Spring Boot plugin uses this, so this All the pom.xml you need is finished, that's it for now. Let's take a look at the Dockerfile and understand it. How do we write this Dockerfile? First this is a multi stage Dockerfile so First we have to pick up the image of Maven, name it Have to give builder then mvn clean install your app through Maven from. We have to create a stage where only this which is copied from the builder You have to take it and copy the jar and just run it That's it, let's create the Dockerfile.

Sit down and take references in between There is nothing to fear if you take this source Git clones the code Clone It's Done Now CD Expenses Tracker Well one more thing whenever. You are picking up any such application from The Internet to its original author Please give me credit, I did this for you. There is so much love for the learners of our batch. Let's build you got enough shout outs come on What will we do now? First of all, this Dockerfile And we can remove the Docker Compose file.

We give because we are from scratch About to build remove Dockerfile remove Docker Compose file both I have removed it, I will make it from scratch I Am Dockerfile now java application. Creating the Dockerfile for the Spring Boot application How will we make the base first? Image will pick which maven's mavin's We will pick the base image because we need a An environment is needed which is also open to the environment.

[Docker] Dockerfile and image builds / Docker security and best practices

JDK should also be Open JDK, it is fine but Which exact version do we want? You should be told to look at these expenses There is a tracker app, there is a Dockerfile inside it. No, but Sneha who is our learner If they created that Dockerfile then Maven's We need a particular version, that's what we need. 3.8.3.8.3 and Open JDK's which version want that want 17 this is our first This is the stage, with the help of this stage we This is the first stage.

The one where we built the Jar, meaning Java Application Run Time So basically the jar file is an executable file. I will become using maven is done now I work DirectoryEntry main remains maven mavin install and when you Do Ho build jar when you do so how can I. Create a runnable jar using maven so this pam Excel you have to write mv n clean I saw this like this with such a start. I learned by watching, but maven installed Before doing this you need the code.

Copy everything from your host machine to your where here from the host machine Now copy it to all the containers and run Maven clean install now if you noticed So this is our code, this expense. This particular work remains then mv a clean install from maven Your application will be installed and finished. If we copy then we will reach stage two where we Only will execute the jar file from the Abab The stage where the jar file will be created here.

This is run maven, create jar file or jar The file will be created and here we will have a small Now we will take the image and run that thing. I don't need to run the file as a maven. We should simply open the JD's 17 Alpine, this small image can also be useful to us. Maybe this is a small image of Open JD 17 Alpine Always Not Small Image If it remains, this will also get our work done, let it be.

We will copy high fun from the equal to Now there is a naming of the upper stage. I will copy it and put it on the current In the file, there is also a working Create a directory Work DIR I paste it in the current directory.

[Docker] Dockerfile and image builds / Docker introduction and architecture

Its name should be the same as the application I show you what I was doing in properties. Because your SRC is for him who is the main one Resources are application properties In that its name is Expenses app, so which The name will remain the same as the file name you want. App.js escape colon double k this will cause my application My application will be built. The build will be done, my Dockerfile is created If you want, you can build it and see.

Try building it with docker build Nutty Expenses A dot what does this mean I am keeping it here I want to build the Dockerfile. Try it out and see if you can find out Is there any issue with the Dockerfile anyway? It would have been known in Docker Compose but already Check the issue of the old project. We will be very careful not to make the mistake Will run every single thing Let's see Mavin The installation started and everything was downloaded.

Build Success What Matters So Your application has been built. The application is built, now this build The work that was done is over, now he just has to run. Can you see running in this your The application is running, well done Just look at the Docker images you have Expenses was just an image of P 379 Now imagine such a big project. Spring Boot is only 400 lbs. I will create compose d y now it's done Understand what I need here.

What do I need first Version 3.8 Let's write this down in advance Write that, this is done now services What services do I need? First of all I need a java P and then I need a ma al db this I need two things, one of Java Need application and database of MySQL. If you want, then the Java build will be Where is that context kept? Directory and this is what Java This is what the container name should be. There should be expenses to be the container name should be Don't forget the network now.

Expense A AWS, add some such name Expenses Pay AWS, enter this name, Now the environment This is the most important part here. Is most needed Your environment But the environment of Java Variable is already application property then whatever is set in your Java Its environment variables will remain here It is already set, if you overwrite it.

[Docker] General DevOps discussion

If you want to do it, you will have to understand a lot. I will explain it in very simple language Look, this SS is your Java app. Go to its main folder, its resources Go to its application properties Now go and see the variables that you see in front of you. Spring Data Source Username This username Which is written in front, Spring Data Source YAML as it is copy Do it, take it here and paste it, but when you paste it.

You will do it, everything should be in upper case. Meaning it has to be the name of the environment Needed Spring Data Source Han URL because when If you write this then only it will map to this thing. It happens that this is a property of a Maven. Whenever you build through He has to write it, now copy this URL Here JDBC, copy this URL and This goes and throws Two this slammed URL now simple spring Data Source Username Now Spring Here Data.

Source Username What is the username here If you have a route, then put the route here too. Spring Data Source what's next password So password what is test @ 123 i Hope Test @ 13 yes it's done this way You have set the environment variable of your java A. If you have confirmed this, you will not have to remember this. All you have to understand is that whatever Variables in Application Properties The same is written in upper case with a dot.

I understand that I have to put an underscore in its place and run it. It has come, it is easy, it will be done now, it is done now This is my application Depends On My MySQL depends on My Sql Depends on the DB let's tell MySQL db. Let's restart it too Always restart it as well Yes, on which port does this application run? This application is not internally an accurate tom cut It runs on the server which has port number 18.

If it is 8080 then you should also check it out once. You will have to know on which port this application is running. I learned Java earlier but I am a full time I am a Python Developer Platform Engineer.

[Docker] Docker introduction and architecture

Where I had work on Django Python I understand Python things quickly. I read long ago which ports Is 80 colon 80 done running on 8080 is ours The application is done, now it has a health Also check that 8080 P application. Your Java app has been created. To the SQL app, so MySQL The image that will be used will be the image of MySQL He will use what he The container name it will use is the same container name. The container name should be what is written here.

If it has happened then that is why MySQL should be here. The container name and the networks needed here The same network is being used here as well. It should not be a last minute mistake We will do it, this one is done, now this one too. Environment Will there be a variable environment in this? The environment will remain my SQL Root what is my password Root now MySQL database Now the database is what you have to put not root. Password is not root then test@123 root password will be adopted test@1223 What will now remain is MySQL database Our database will remain the same as the one we have uploaded here.

Sorry So MySQL Database Expenses under tracker so both of these The variable will do its job. Enter your work in password and database name It will be done now I need a restart Always have to restart it again And what else is needed in it. Ports, ports, we need it in 3306 colon 3306 this is our ports should Ports are also done, now we need a health check. What commands are inside square brackets? Will put Which is what MySQL admin will ping will on the host local will ping localhost host why I'm doing this because I know MySQL takes time.

It is being made so long as it is thoroughly tested. Until that happens, I will keep the container above I will not run any Java container that the user Its name is Route and the password is its test @ 123 one. Another thing happens by filing secret files that you can also get the password from the secret file. It became one Now I will put it in Interval of 10 seconds and then And what do I put in Start?

[Docker] Docker introduction and architecture / Docker volumes and storage

You could have also put a time out, a small time out. If nothing happens for 5 seconds, So time out done restart always it's done If you create a network what is the network What is the name of the network Expenses Pay. Network Expenses would have created a network And the volume that will remain, those volumes We can use the same volume as The previous one was used dot / MySQL You can use a volume named data. Ver / Lib / My Sql I think now ours is completely made It is done, yes one thing is left that our Java is not working unless If our MySQL DB doesn't become Put Depends On here.

What my SQL Underscore DB Enjoying reading Docker Compose that the container commands you used to write were written by Docker Build, Docker, Run, all in one place. If you are driving cool then version 3.8 This is your version of Docker Compose The file version is services two containers become There are Java apps in which Build sorry context dot means current. The Dockerfile is stored in the folder Build the container name Expenses P My container's name will be Java. Network Expenses P AWS Network Me I create something in the environment variable passed in Entering JDBC data source URL too This is necessary because it allows us to interact with the database.

Connectivity will be Spring Data Source User My username is root, the database user The name is root and the password is testA3 until I get my MySQL DB My Java app won't restart. Always a health check here if you want can put Yes, put a health check here as well. There should be no need to put it in there but If there is no one, what can you do? You can tell the CMD shell that, MySQL admin no you curl you curl Do you want to curl a simple

curl?

On port number 8080 on your local host Your application is running or not running curl -f http colon Your application is running on localhost 8080 Is it running or not? If it is not running, it will be Then exit with status code one. Finished and that's it, done, restart Always put now looking at ma SQL DB takes the image will use MySQL's Container name will remain MySQL this network It will have the same network environment.

[Docker] Docker Compose

I've put the variable restartalways He will keep showing his stomach and in health check Health check is also fine, volume is also good. I gave the network, I gave everything is all set escape colon double k enter this time No Mistakes Let's See Docker What happens with compose up, press enter and finish. Depends on already defined at line 14 Probably Turned Depends On Twice Depends on It already depends on what I have already done. Be it Gone is the escape colon double k enter now.

You can run Docker Compose and see what happens. I am validating home no to ec docker Compose networks must be a mapping to we I'm Docker Compose so networks that mapping It should be there, have you put it or not? The mapping of the networks is Mappings should never contain dashes If it comes, you will have to remove the dash from here. This mapping is mapped with colon If it happens then press escape colon. Enter Docker Compose up enter Let's end with that being said your build The stage has started.

You have your Java app. Build Let's Run MV N Clean We have already tried installing it and it works. This Will Work oh what are you o is saying binder 3306 port is already allocated o docker ps Your Notes app is already running. He is walking That's why we can't run it on the same port. If you take down the notes, Docker Compose Down so the old P had The port was already allocated, hence the problems. If it was coming then let's take it down and this Let's start with that, that's when I started the project.

This instance is called Tad Medium at the start I know that multiple I'll teach you how to apply if you want. I will be teaching all day long Let's see if the DB container is now stopping. It takes a little time to start and stop But fine single click down single click in How much and what do you want in life? Go back to Expense Tracker [music] Docker Compose Down first by downing as many I have removed the running content now.

I will do Docker Compose up is not showing on my photo's screen If it happens then Docker Compose up this is inserted into the command enter lo Docker Compose up MySQL inila o can you see.

[Docker] Docker introduction and architecture / Docker Compose

Spring Boot Pu it is interesting it is interesting Let's see it until it restarts It will continue to do so as long as our database container. The container of Spring Boot is both health If the checks don't pass, drive it, drive it. Run it is exiting now but as soon as your start the database It will connect to the database but Your database should start that too Why is the database not starting? Database Taking Some Time its five retries five tries Let's see what happens after [music] MySQL cannot connect This issue always comes up with you too.

Is that MySQL can't connect? The solution to this is that he Retrying that was MySQL. Why is the container not running? Ambulatory Let's go Wet We do one thing, first we Let's try the SQL and see it. Let it go and then we will go to our Expansions will run Java A understand I came first to the one with MySQL. Just to see that Somewhere MySQL when it's set well go MySQL is set to be tested. What's my Docker Compose Up MySQL now play you first my Play the SQL and then you do something Expenses P to Exit Ho But MySQL Why is the SQL not working?

If you look at the error log, you will see a yellow color. You will get MySQL oh no I don't know where MySQL was Getting treated right are there any particular region [music] stop and look Let's see what issue Atlee is coming up with. In this he is saying that Authentication is very common in Java. Crazy errors happen, you will go to see So this will continue, but I understand this. I can't seem to find my MySQL container. Why is it not running That's the main thing TIM Let's see what I've done with this Everything is given correctly The network is also provided and the volume is also given.

Given is VIM Docker Compose first just And just by running the container of MySQL Let's see Depends on my DB all this is correct all is correct. If this happens then it will take a lot of time, so one thing Do Only and only for the one with MySQL picking up MySQL DB this raised such a person.

[Docker] Docker Compose / Docker introduction and architecture

Why it is Crushing a second then Docker Compose Run Specific containers see that Docker Compose What is just for and container only Docker Compose Client, so let's just And just us running MySQL DB. Just to check that our ma SQL D is also not running properly so docker Compose put this photo back up Are Now which mother Let's run SQL Underscore DB So MySQL is starting the server. Remained [music] Has run MySQL d in the path e accessible Consider choosing ready for connection MySQL is running, then what is the issue?

By Address Port [music] 33060 Have we done something wrong? If people don't do it right It is MySQL, so ours is going on. I'm trying to get the data source username. Data Source URL Source Password Everything I have kept it almost the same Allow Public Key Retrieve True What is this Is bonfire public key retrieve true o this I have not lit the fire of the public. Retrieve is saying something This let If I haven't put this thing in jdbc MySQL asus Expenses Tracker It's inserted but I have not put a particular key on that side in the program.

Let's try putting That JDBC MySQL expands a. Tracker Allow Public Key Retrieve Equal To True Use SSL False I think everything is almost Similar Database Name Expenses Tracker Database Name Expenses Tracker Password It's also the same, I think everything looks good too. I should have worked, ideally I should have Worked Let's See Docker Compose up half of what is the build saying This is MySQL data. Data sudo rm han r MySQL Data remove Docker Compose Come on up, let's see again, this time we That URL, give me the correct one again.

Is based on the repository let's see how this Bar If It Should work run mv en if it won't run this time What We Can Do Is You Got It. I can try it, let's C o He is doing something doing is no active profile finish route Second is doing something o and that was the. Equal to true is exactly what it is, let's See Allow Public Key Retrieve True My what is SQL dockery atli let's see let's C can only connect to a MySQL running.

[Docker] Docker introduction and architecture

In a Docker container one allow public key true We will know in depth that, the issue is It has been solved but how has it been solved? Allow public to know this particular setting that allows you to create a JDBC Java database have to be put in the query. There is an official one here I did some are nd and found the following solution that when you go here and give the public the truth. Ho this query help me to connect with the database so if you connect your Java A to the database trying to connect and If it is not working then might be allowed to the public.

Retrieve equal to true you are missing How cool is it, I also like it very much I have learnt something, come on let's try it. To access the application the application is Running on port number 8080 which might not B which might not be publicly accessible So in security will go And we will edit these bound rules, in. We will edit the bound rules and here Will add it to the rule 8080 Anyway save rules if you want this thing You can also run it through AJAX now.

If you run it through Ingenious, it will Your 8080 is here Enter and Your Expenses The tracker application is up and running which Expanse built into Spring Boot Tracker is a free to use MT license. Dear Mohammad Tariq Alsaa also error If there is an error in your application, please check it. You don't have to resolve it, it's your job You have deployed it now the error is Code errors that will resolve your DevOps Developer not engineer so its next job Let's log out of the developer.

I am logged out, I am looking for a log out. It's, Expenses Tracker in this repository The repository is here pull request You can create that done with engine s Well, whoever creates the pull request. You have to create a pull request. The directory of my Django Notes page I will put the engine folder inside it here I will copy it, I have an engine. When the folder arrived, I am in Nginx. I'll go inside Ajax default confine on that I'll go and I which is the name of my container, I just put that name Shall I put my name here?

[Docker] Docker Compose / Docker introduction and architecture

P and this is what runs on the port number Just put it on 8080 and I'm here on the server. I will put the upstream server 80 its Well there is no need but if you want you Can Expenses App Ye Ho Escape Colon Double K Enter That's it anyway it's running on tomcat server. It is still there but if you want to do it then it is there and Go here and pick up the engine code. Lok kan The code is not the same as the Agen code Agen I taught you Agen Kisam then No problem, yes I have sent you the notes.

Ang has read it correctly So here I'm going to go and from Docker Compose I'll pick up this engine code, finish it and Depends on Khatam VIM Docker Compose and just go to an end here. I'll put the engine in, just a little bit of

it. You'll need to look at this to see the indentation. Will need container name engine container restart And it depends on what it depends on Your Expense Which Java A Java P Name Station what was its name java p ha so java.

It depends on Java and its network That network will also be in the same network. To Networks Expenses on Network That's it, now we're running this great engine through it. Escape colon double k enter everything looks correct Docker Compose Docker removed all containers compose up - - build because we have to I also have to build the engine, so what? Docker o Docker Compose up - afn build let's just engine ecus We will build this if permission is denied. It is MySQL data, that's why Because your volume is CreatedDocumentsFragment [music] Fifth note of musical scale data over escape colon double k now that's you Docker Compose up won't kill permission error.

You run the command that was there, now what do you want? He's still speaking to MySQL data. Sudo rm -r my sequelize data removed Watch my videos, I will remove all the errors as I put it is so that if you get this error. I have already told you that, such and such walking like this should let's see tom cut server Started not 808 internally and also doing routing Simply means if I Go to particular IP http this.

[Docker] Docker Compose

Should I go to a particular IP address? Go to port number 80 and press slash So my expenses are running without any The port number your Expense Manager is running on, please tell me its SSL also. Let's take this IP address and anyone can take what you say Gaddi Gad Co Code Audi ID Whoever it is, take it, what can you do now? Will come 0.01 means how much is 0.01 into 1 Take the rap, what is it, and then go and have a You have purchased the domain, then use that domain No DNS mapping like mine is.

DNS mapping is done, now DNS I'll tell you what I'll do in mapping. I will tell you here that, this is the IP address on this If you want to map it, I'm here on the domain. I'll go and click on Manage DNS. In Manage DNS and Manage DNS I will go and tell him that, make a record. Make a record, add a new record whose name is put Two Docker, put its name Docker and in the option Choose a record, what is a record?

It will take time because it gets updated I take a little time Let's try HTTP s a little bit. It will take some time to update but it should Have work going on here What Archnet Colon 88 should have work no problem STT p Although this is not ATP S. It's very unfortunate I have learned a few things, it's quite over. [music] Velminik If so, what should you do if you feel like this The application is running smoothly on this IP address But this is happening because this is Nginx.

Who knows if it is on localhost then vm The default configuration of Nginx is local. To consider particular as server name then This is your domain name, save it. You have saved it and now back to your application. If you build, this is your sudo rm han R Java A data, you have to do it separately. You should have made it, no problem, you can return it. You do it with Docker Compose up - ain build iske Run it in detach mode, -d means This will keep playing nicely in the background.

[Docker] Docker security and best practices / Docker introduction and architecture

What I'm going to do now is whatever our Docker dot I told the engine that the train was starting. It was not known that the engine had yet to host the local host. Knew Server Name Now Server Name What has been made? Train.com Our container has just been started docker ps container all started is engine Everything is fine now what should I do? Here I am a great I will re-add the Docker record. Pay name your Docker will come and the value that will come That's your IP address again, so copy it.

Do it and go here and throw it and this HTTP Again you have to remove it and here you have to customize it. Put a custom TTL of 600 seconds Time to live so refreshing time for it will be done now your server also knows About Duckertrain.com Your Godaddy Joe. DNS is also known to Docker Train Vi what can you do now about such a Incognito now boys looking at incognito Don't get so excited, I do this because old cash sometimes.

Trains.com design supports a secure connection Meaning, HTTP A is not working. Still if you are not secure If you want to continue, then continue with the site. Do End as soon as you click on Continue with site One click will get your Expense Tracker running. Introduction to Docker Architecture Volumes Multi Stage Two Project Too You got it done, what are you waiting for? There is also content where we have Docker Scout and we're going to talk about Docker It Let's quickly understand what Docker Scout is.

What happens whenever you build Docker images Now that you have created a Docker image, let's This is your Docker image created with Lets. Two tear flask pop image made so two Tear Flask was an image of whatever app it was made of Now this image is created from a base image which is the base. The image can be of Python or anyone. Maybe right, it is made from the base image And from this base image your new image is created. It is also possible that this base image something in Security related threats If there are related issues or vulnerable keys That's why we keep the image for security.

[Docker] Docker security and best practices / Docker images and registry

Scanning takes place and the image is scanned. This scanning is done with a tool called Docker Scout says this is the work TV also does it, of course it does. Trevi's job is to scan images and files. Similar to system scanning Docker also has an image scanning tool of its own. I've created a tool called Docker Scout. You can easily install it in Docker Desktop. When you get to Docker Containers you will find images that we Viewed Docker Images Viewed Docker Volumes Then you see Docker Scout, you have to click here.

If you minimize it a little then it will be advanced. Image Analysis with Docker Scout using Docker Scout helps you with image scanning How does it help me, how can I How do I run Docker Scout? First of all Understand that by having Docker Scout do it for you. As many images as you have Is it on your local or on your Docker Hub Docker stores all those images in the repository Scout can scan and view that somewhere in its base image or its final.

There are no security threats in the image so let's take it from this one of ours is two tier back end latest. I have picked up the application and now look at it. He immediately did a vulnerable scan to see if he had Critical Issues Zero Is There Any Critical There are no six high severity issues. There are issues of medium severity but, this How will he know what the issues are? You will need to run some commands for this commands you go to terminal this Your terminal has opened, now here you will see.

All you have to do is create Docker images and from here you can If you want an image, you can get it from us. Which image do you want right now? Our two-tier back and so two-tier back Copy the name of the end image from here. Took a copy and all you have to do is docker Scout Quick View So from Quick View this which I saw this 66 52 and image in front of you. It will first authenticate with your Docker Hub.

You also need to login to Docker Hub here.

[Docker] Dockerfile and image builds / Docker security and best practices

There were six high, six medium and 52 low Savarin means anyone who is very critical The base image you used is no longer there. If you do research then you have to research that You can use a better image than this one. Docker Scout tells you all these things. If you want to know more in depth then One thing remains common, vulnerable end Exposure CV ES to Docker Scout CV es and your image name from let's your. You can do it and then the team will decide that This is due to the use of a particular package.

Fix it and so on so Docker Scout very This is a very interesting tool An interesting tool that lets you customize your images. Can you analyze it, I enjoyed it interesting Now let us understand your second bonus. What is Docker in it, understand it like this that you Have to build Dockerfile Have to build Docker Compose And I have to make a dummy and make three things. How much time will it take to first edit the Dockerfile? Tell me what happens, first write from Yes, we then write the work after that.

D I R then what do you write then copy Open the source code and then run some commands. Run CMD and complete your application. If you want to give labels in this then that too You can write and so on but this is the work. You have to do this manually in the Dockerfile. Docker has to be built manually What it does is it's such a readymade template makes you for I need an example of a Python application. A Dockerfile for Docker Compose and a readme So I just have to write docker in it and in that Select Python from the options.

Here is my Dockerfile Docker Compose and dm me this All three will be made, show me this, see, I'll go here now and I'll create a folder here. I make M KDI R Docker it test a docker called it test I create a folder then cd docker it test inside this folder I went now I'll just write docker in it now. Docker Scout and Docker It are both Docker Additional features come with the desktop if You will need to do this separately Docker Ignore makes it of Docker Ignore what does docker ignore dot docker mean.

[Docker] Dockerfile and image builds

Ignore is a file in which you Whatever you write, that will be your final decision. In container creation or image of p Sometimes there is no copying in creation, is there? What happens is we use node modules or environment Folders keep it all in their code base But we don't want node modulus and environment variables and all these things. If the folders are also copied in your image then you If you can put that in Docker Ignore, then Docker Ignore that thing while reading the file.

If it gives then this Docker it CI that file also It also creates a dcker file. Compose also creates a readme.md. If he creates it then come on let's get it Started what application platform does Your Project Use So Your Project Go is made in. I will give 1.1.3 no 1.13 enter whats the Relative Directory with Lending.org of Russia Main Package Now if you have a program is main do go or whatever your The package should be in this directory. Or whatever directory it should be in I want it, please give me the path, don't go to me now.

Dot what port does your server listen on How is the application port sometimes If the number runs on 8000 5000 then my let This application runs on Rs 8000 per month. This is a math application, it just asks you questions. On the basis of these questions, this one is asking Docker will create the file for you to view now Warning: No good mod file was found but It is required to build a CreateToken. By running Docker Compose up it became The application has become everything, if I look at your We created this folder if I ls I will do the Docker file dummy and make Kapol all these.

If I go into the cabin, you will see Maybe these are some comments about it but If you look carefully Services Server Build Context Target ports everything database if. If you want to use PostgreSQL then he has everything I have written it and have also put the health check. What's the matter volume secret put everything I have given it, I had a lot of fun escape colon double k Enter cat or else VIM ha cat le cat Dockerfile your Dockerfile also he.

[Docker] Dockerfile and image builds (part 2)

I have made it completely, it means it is great and Well done for creating this user, what a great thing. Is from alpine latest s final what The thing is, multi-stage has also been made, well done. This way your docker in it within a few minutes, not even minutes within In a few seconds your Docker Compose Dockerfile and also creates a cat readme Docker What is a dummy? All those docker in the dummy. The commands are written as let's docker build what is the command to run docker what is the command and then to push to docker What is the command and so on so interesting.

Stuff Related to Docker I Hope You Like This Short You must have liked this bonus if you got it.

Kubernetes (Concepts, Architecture, Workloads, Networking, Storage, Scaling)

Source file: transcript_W04brGNgxN4.txt | 82,697 words kept of 112,536 | 211 sections

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / Persistent volumes and storage

The architecture of Kubernetes is different How does it look like hands-on setup? Do Local Pay Eat Pay kubectl What are pods and what are namespaces? What happens is everything that is basic He is the king of the nineties and he is also an advanced man if Talk to Deployed StatefulSet DaemonSet ReplicaSet set cron jobs everything that. You are asked in the interview if and If you take it in depth then it is like networking persistent for Services Ingress Storage ment volume storage classes persistent Volume A Clemm if talk like in production.

You can customize the architecture to fit your applications. Get your notes, your laptop, and everything ready sit because the ending in the i e case cluster Pay start hum log Minikube and local And we'll do it here and there, but we'll end up with ES. Running a three-tier application on a cluster Doing CI/CD with Jenkins vs. GitOps with Argo CD There will also be a deployed MongoDB so that You should also know a little about MongoDB. Many people say, three tier Make the application, make it with the net, then the net python3 application will remain last but not the.

I'll get you live on trains.com which I am going to start Deb 0200 Passion Batch Nine which is going to start from 25th January Its registration is already going to happen. Started like today I am Kubernetes I am going to teach, similarly I am completely I am going to teach DevOps from January 2025 and I want to see you all there link. Crashes if there is too much traffic If autoscale were done then in 2014 they The initial name and rank of Kubernetes was this He started and then this thing.

Tell Kubernetes, it's been 10 years. Better late than never, let's understand that this is such a powerful Kubernetes application. If it does, you will have to understand it first. What is Kubernetes, so Kubernetes one by one Kubernetes is basically an orchestration tool orchestration tool, you saw that movie Which will be Hera Pheri, then Hera Not in the ferry, Raju lives here. There is a Dhoom Dhaada orchestra, only that So your Docker containers live in Your Docker containers live from Let's The front-end Docker container is back.

[Kubernetes] Kubernetes Services and networking

The end is done, this is the database behind. How are all these Docker containers? When should the scale be automatic to whom if late? Kubernetes gets it done in this manner and if you If you look at the total letters of Kubernetes k followed by 1 2 3 4 5 6 7 8 and then s. That's why it is also called K8s,? This project is now a CNCF graduate CNCF Cloud Native Computing The Foundation is an organization that Strong projects like Kubernetes happened Argo CD Done Helmets are given a She gives them resources and everything.

If there is going to be a tutorial, please like it. Let us understand what Kubernetes is. Why do we use it, why do we read it? Nits first of all what is DevOps engineering There are roles in it of Kubernetes which The demand is very high because Many people who used to make applications Earlier monolith applications were made, now they. All the micro services have been done, these two What is the difference between Imagine that there is a very big shop.

There is a very big shop from Let's D-Mar I don't know what you are doing D-Mar It must have happened, my, I mean I live in Pune, this is behind me. Monolith means a large software A large application that contains a single You have uploaded everything to the repository. Meaning, if you talk about amazon.com then log in Pages Signup Page Cards & Products Sub Some are uploaded to the same repository. This is the biggest application that is running now. The entire application is done here.

Lets you get a little If you want to update something, you will have to Why do I need to update the application? You get a small store where your daily When you receive items, you have micro Created services, created microservices Given that a large application can be divided into smaller ones Divided into services, now what is easy? You will get the service if the service is running If it is not working well then it is on service only. There will be an impact, other services will continue to run.

If it will not affect her then Managing microservices architecture It becomes easy, plus it becomes independent.

[Kubernetes] Kubernetes Services and networking (part 2)

Such services exist, hence its cost also which is shorter because it is a smaller size than the Lets C service is authentication service for that. You will not find a big server in it, but in small server or If you need to manage services If you want to orchestrate then your The tool that remains is Kubernetes, so nowadays. Microservices in the industry The demand for architecture has increased significantly. That's why your demand for Kubernetes If it has increased too much then you should know Why are you reading Kubernetes?

Why so many people don't know Kubernetes what does Kubernetes mean to them I don't know what Kubernetes is. If you want to read them then it should be clear to you. Should because your monolith and Microservices within microservices Architecture is being used a lot. That's why Kubernetes are in trend and very Everybody is adopting it now. What is there in Kubernetes, that everyone wants it and everyone wants it too Hum Kubernetes One Short Video Everybody wants a CKA Joe Certified Kubernetes Administrator Everyone wants to clear that too.

Why because the demand in companies is not that much right now It has happened because if you talk to freshers If the basics of Kubernetes then do it. Even if it is there, one gets a job very quickly. That's why I'm making a video, so quick attention Listen to how the architecture of Kubernetes What happens, what is it that happens in Kubernetes? It works so well, let me tell you, this is it, this is the architecture of our Kubernetes I understand what this is, so let's go.

I understand that your Kubernetes is Kubernetes. I said that this is your kit which you have given to us. Why is it that you have seen the Docker logo? If you have seen the Docker logo, then you What is the logo like? Which is in the shape of a ship and from the top The captain of the ship is now carrying containers Who is it that is steering the ship when. He who drives with the steering wheel The steering wheel is the logo of Kubernetes is that your Docker containers who drives is me, this Kubernetes now that How will it work?

[Kubernetes] Kubernetes architecture (control plane, nodes)

I would like to tell you that whenever you are in a company somewhere You go from LETS to TCS infosys.com I used to sit next to him and write the same code and in the evening. When we used to go to have tea, we There are startups but due to some reason MNCs grow so much because of this If it is so scalable then understand it like this. Take that startup means your Docker container Startup is fine because there is only one container Poor thing, it can't scale big.

Runs with difficulty, sometimes stalls Sometimes it works but those are Kubernetes policies Understand that it is a multinational company MNC It is a multinational company, now there is nothing left for MNC. Architecture due to which MNCs Such a big MNC has been formed, so let's understand First of all understand how that MNC came to be. The architecture of Kubernetes There is such a server in it or there is such a server Nod remains whenever I say Nod whenever I will say node, it means server.

There remains a node that is different from the rest. Manages the servers and the rest Understand the meaning of servers in such a way that a node means Single node One server Multi node Multi node means more than one servers and this thing Let's say. Cluster, this is what is called a cluster. At Kubernetes, we are a cluster level But you don't need a server Managing your entire cluster which can have multiple nodes like Node one node two node such multiple nodes.

Consider that such a node which is your master node is a node master node and other nodes like these Done your node one let's do this gone node one this is your worker node worker. Node Now look from MNC's point of view where do you work where do you work In Cognizant, in Wipro, in TCS, here You work here, now its head The quarter is somewhere far away but. Are you in any one of the R&D Work at one location within the branch Let's go to Pune location or Bangalore location Headquarter in Mumbai or Cognizant's If you talk, you work in Pune.

Head, you work in Noida in Bangalore Quarter there California or else New York is somewhere over there in the USA Similarly, the master node that remains is yours.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Kubernetes Services and networking

There is no work there, all the managers The CEO and CTO sit there. Only work is given that in India Pune becomes a node Pune node becomes a node Bangalore node became a node Noida node People actually go and work on these nodes. Whatever you might be saying, I am your I'm from a startup, I don't know, but it's If you have architecture then Kubernetes is also in the cities One such architecture is a master node. Lives where there are some services that Just sitting there working on the worker node If you get the work done from the worker node then your The actual address of the worker node.

What is actually in Docker? Containers are running in it Docker containers running on worker nodes Press and your Docker containers would be running. This is what the Docker containers are running He is sitting on the master node, so if you Kubernetes interview asked where Do these containers and applications work in a. Kubernetes Cluster so they work on the worker Node None of the Containers Work on the Master Nobody works on the master node. It works on worker, now what is it like?

Those people do it who deal with the master's worker What is it that is getting the work done? There is a master node, it provides some services. Like an API server that What things are running in the entire worker node? What is happening in the entire master node He keeps a record of everything that is going on. Just like your team lead would be your team lead Isn't it your team lead, what does your team do? If you keep telling me the work is done then the API server There is such a unit, there is such a server, there is such a Service that acts as a communication gateway.

API server with each component What remains is to communicate, that's why It is very important that you understand the API. What is an API application? If you talk to Kubernetes Cluster You can't talk directly You have to talk via API server So API server is a very important job. If you see the arrow also, then API Remove all arrows from the server and API server If you are there then it is the most important unit which is used for communication with in The Cluster.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Pods

Your team lead said, our requirement is There is a requirement for a man, a man who DevOps is an engineer from Let's, so Joe DevOps hiring of an engineer or. It is called HR, so it is such a service. Whose name is schedule what Schedule is one such service that works which will run your containers on worker nodes schedules that you get one on one from the lats. Annex is to make a container of Engine Aces. Reverse proxy server or my SQL, which is a database, gives you a Scheduler to create a database container He will tell the API server that, I am just I'll schedule it and create a container.

The scheduler's job remains to Schedule containers and pods so pods are The unit resides in cubic knots inside which When your containers are running, the container If we talk about. Or if multiple of them run simultaneously then If the pod is then schedule Responsible to run this pod to run The pod and inside the pod are basically your containers. The box is your pod inside that spot is your The Docker container is running,? HR has announced an opening and I got you a job, that's right, but your The data that will remain is when you created that pod.

When did that opening come and when did you get the job? Everything lives in one database and Database is called Kubernetes etcd to etcd is a key value pair The data store that stores your data All the Kubera running in the NiTS cluster. Data of all those things is stored in etcd. Let's store data in etcd Your butt is gone, it remains the same. There is a big head manager who looks after, all my servers are working. If not, it is called your project.

Manager Project Manager to Project The manager is in charge of everything. Down it's going on no it's going on No, he remains the controller manager anyway. Checks if your nodes are running Whether your pods are running on your cluster Are all these things of yours running properly or not? Whether the services are running properly or not The controlling manager is very important. This control manager is now the master node.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Kubernetes Services and networking

You will get the services of the headquarter. I understood that there is an API server in it. Which happens with every complete cluster The scheduler does the communication The pods inside which the container is running I will schedule it, it is etcd. Where all the data is stored as a database is the controller manager node of every cluster Your complete fate will take revenge on someone. He controls it, now it has become the master node but where actualy worker node Work is going on, if there are any services there too Like you are in Pune location or Bangalore.

Be it on location, in Gurgaon or anywhere else Yes, you have a manager there too. Which insures that you arrive on time It's not working, have you logged in correctly? No, you submitted the task correctly or No, are you even working properly? Let's call that thing kubelet can be deployed at the entire cluster level. Pay works but kubelet your worker It works at the node level and observes that If the pot is not running properly then it should be running properly.

If it is then it will call the Kube Late API server. The API server then the controller Manager and etcd each have their own The cube will store information while watching Is it true that your pot is running,? On the worker node, but this worker node is what you People from outside must have visited FOCIS TCS sometime. The pods You remain isolated,. To change the matter from outside to inside There is a person whom we can also call a proxy. They say, have you installed my proxy?

He also does the same thing here and there, if You have to access these pods as a Outside User Seeks You a Service Whose name is service proxy or kube-proxy is also called this service. Because of this you can access the API server as a user. Pods inside the from 's Cube Latte so whenever we create applications and provide services for people. I am sitting in Pune Bangalore This is how we communicate here. The two servers are server one or node one.

Or node to node 3 these are all nodes this one How to communicate with others In the same way, Tung Teams Music that increases your anxiety.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Pods

Or is that notification from Slack the same way? These servers, these nodes are a Container network interface with another Let's talk on the network which is called CNI It is also called network, it has many Examples are Vanette Calico, which. Because of the network, all the workers nodes and the master node, it is connected to each other He is able to communicate but, your HR I told you and you told me the team lead. Project manager, tell me, tell me manager, tell me You also told me about this pod of ours.

I also told you about the service that destroys Lanka, but You didn't mention the director. Every company has a CEO who He directs that, do this and that. Didn't talk about it so if you Instructables to your Kubernetes cluster I have to give it to you, show me how many pods are running. Show how many nodes are active. If you create this, you need a controller. It is called cube whistle or cube Controller so cube whistle is one such tool which talks to your API server to complete the task.

You can order the entire cluster of, this is the architecture of Kubernetes. You can do this thing yourself in just 10 minutes. Show me what you have to do and complete it your kube is a cluster of nits it Your cool cluster is now like this What's inside of you stays inside it. Yours remains a master node is precisely a master node and Multiple worker nodes is fine and multiple worker nodes now this is the master node There is an API server inside it which Communicating with everyone.

Helps correct API server and API server if it needs to run things tells the scheduler then if the entire Controller to manage the entire cluster There is a manager and then you have an etcd. Which also stores your data, etcd is done then worker node per worker What happens when a kubelet on a node kubelet where he sees that all. The pods are not running and the pods that would have Those pods are basically something that Inside which your Docker containers are running There will be docks and containers running.

You can access this pod directly.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Kubernetes Services and networking

If you can't, you'll need a service for that. If you need a proxy then now this is the complete This is the network which will be the entire communication That will be communication Via a network interface, also known as CNI. If you say so, then basically you have a Kubernetes I learned to draw an architecture diagram of how is this what is this tell me what is this is one API Server. API Server this is the API Server then This is your schedule this is your schedule then what is it This is your controller manager then what is this is your kubelet.

Proxy, which is for outside users To the outside user, enjoy all these pods will give access via service proxy and this The service proxy will talk to the API server. This API will talk to the server schedule and Will talk to etcd and Controller Manager and also talks to your kubelet will do it and it is your CNI or else. Container network interface and this API server kubectl talk will kubectl end like this you Architecture diagram of Kubernetes You will come back with a great performance in the interview.

It's done, if you had done a little hands-on, it would have been fun. If it comes, get it set up Get our AWTP done on local. Pay instagram2 assisting account do let's go and Using route email like my train starts Let me zoom it a little, to 150 Let's next this on Zoom and Let's save this MFA my password. We have come to our Now what I did on the AWS console Your region is selected and kept. This region lives in the North,?

Virginia Ohio California Oregon Mumbai If we live in Canada, then, we Indians Mumbai guy picks it up, Mumbai guy We picked up the Mumbai region, an Eetu. We have to create an instance and we can create this to instance. People practicing their Kubernetes That's, so we clicked on two instances. Go to Nut now, we'll get Nut Server. People 244 means in April 2024 this A version of T had come, select this one. Take the x86 architecture because it is common The processor also remains then in Docker If you face any problems then take x86 set It remains to be confirmed, if you want to confirm then do it.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / Kubernetes Services and networking

True Instance Type T2 Micro Free Tier remains eligible but You are running such a big architecture. If it is there then you can run well on T2 medium Your work is fine, do the job and delete the cluster. Do it again, I have to do something new again Create it and then delete it again, then t2 You can also create clusters on Medium,. There are many ways to create a cluster. That the best way to create a cluster is The first one who was Kubernetes There are ways to create a NITS cluster.

There are different types of brothers like one The way your cube lives Edm cube Edm I played the cube Edm I have made my first video in kubeadm How do you handle two instances? Be two or more and on those instances Install kubeadm and If you combine all three into one cluster, This remains one way, but now it costs increases because one two three you. To create a cluster, three nodes will have to be created. Which we call Minikube gives you local pay or a single You can make Minikube servers on bricks.

Live Minikube clusters that you make You can have a third way, kind cluster which we know today Learners are Kind Clusters, So Kind what happens in like kind means. Kind is not like this, kind means Kubernetes Nits in Docker means your Docker container There must be some Kubernetes going on inside. Very crazy so kind today learner I will also tell you about the Minikube Let me tell you both that there is no scene, so kind We do this in clusters and if you.

If you go to the cloud then EKS AS EKS Meaning Elastic Cover Kubernetes Service You AWS Cloud Kubernetes Ho Ekas Ajar Kabar Nitas Service You Ajar You can do cloud computing and GKE. Nits services remain, meaning you get a cluster I don't have to manage AIDS or Azar Va We will do it for you, not kubeadm. How does one install the whole but I want the full tutorial to be You are not experts in Kubernetes first. Go to local pay kind cluster, we people things to do then I'll tell you in local I will do it in AW also.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

There are different companies which are in their cluster. I have made it myself, I will not take the names of the companies Because they gave me the If I haven't done it then why should I take your name for fun? I am doing so Sivo Vulture and if you If you want to do it at the production level then an RKE Rancher Kubernetes Engine R Kei Yes here it is Enterprise Level P If you need Kubernetes. If we want to do it then there is a rancher and then our sew, the company that CO is Cloud computing services such as AW.

Is it through kind of cluster or mini Learn to play local games first through Cube. Let's learn now, take Titu Medium Because it won't work for less than that. I will create it using RSA and PM key I will create a key pair and this is my K8s This shot has been downloaded here. Work done, now I go into network settings I turn on SSH traffic and HTTPS ATTP All this traffic is also on I will do it, the storage is big.

It is important because you have a very Create a cluster and place applications inside it. There is a lot of work to be done to make it, so this If 8GB of storage doesn't work, or So take up to 20GB or even 30GB because 30GB. All you have to do is create a server. T2 medium bus latch instance of done Lunch is great, absolutely now successfully initiate lunch This is done, now open it. Through the SSH client, this is my This is the SSH command, I will copy it.

I'll take it and open it here in my terminal. This is my terminal opened up a little bit. Terminal zooms zoom also happened with this Well, I would have maximized something completely. Otherwise I will go for CD download. The SSH command I copied I will paste it like I paste it I will get an error that your If private key is not protected then It is your private part,. AOD 400 has to be done because right now it Too many accesses to the file are open Permission is to open the box which gives you all in one.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) (part 2)

Run the command back with SSH this time If your server is connected then connect So the server is done What do I have to do with this server now? I have to understand that I have to take a stand on this. Creating a Kubernetes cluster on the same server How can I make you do this for me? What is Kube and Docker? Cluster like this then If you want to do something on Google's Minikube Install it and you can Minikube You can also install Mast Wiz P now.

Vij has nothing to do with it, so I will do There are one and a half thousand forks here and all The concept is of Kubernetes and if you. If you want to create a Kind cluster then you can create a Kind cluster. Click on the cluster if you want a mini If you want to make a cluster of cubes then you can use mini Click on the cube and let's see what happens. Let's create a kind cluster first and see.

Let's make a small kind of cluster and see So first of all you have to install Kind If you have to do this, this is a shell script which Kind downloads it from here. Whatever is written here, this is here downloads and one more thing remains kubectl which I mentioned which There is a director who gives all the orders. You will have to download that too, so this is one There is a small script that will tell you everything. If it allows you to download then this script I took it and let me bring it over here just a second.

Here we are installing Kind I created a script called DSA, enter and That script I have lying there. I pasted it here, it's done, so now that what are you saying without bash means shell The script is a y name hynam this whatever Its architecture is curl allo kind a this. By downloading a kind from a particular URL Bring version 20, we have the latest version 25 How did this can also do it? 25 It is the latest because of his kind installation vahan d v 0.25 If it was written then d v 0.25 0 is written, so we've done that, now we.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) (part 3)

People chmod kind means kind which these file will make it executable and it A command named Kind will become yours then Basically, these three commands can be used to control your kind. Kind Kubernetes and Docker will be installed There is a tool that allows you to If it allows you to create a cluster, it will be done now. You can also download Cube City in the same way. Kind and Cub Seattle Whether the installation takes place or not escape code w k enter c ao 777 install k a iske little The script which gives us permission has downloaded / install.sh.

Enter Kind and kubectl Installation complete means that our This is a particular shell script With this you can play Mast Kind and kubectl can you install kubectl What controls the entire culture? The Cube Controller now lets you drive kind You will also need Docker for this, so let's do Docker also. Installing Docker sudo is very easy AT Get Update First Update the system Install all the packages you have. Download and get it, then all the updates will be available. Get it all downloaded let's go, now you can install Docker also.

Sudo at get install dkr d come this big It is necessary because if there is no Docker then Kind Kubernetes in Docker How to run Kubernetes if Docker work is done Yours is fine. It's great that it happened Now docker ps to see that your Docker ps installed correctly or not Permission nut tell docker install so good What permission denied came, let's fix it. How to fix sudo User Mode? Your current user is Ubuntu Hu Mi Kaun I am a current user,

I am your Ubuntu The user does not have access to Docker.

If it says permission denied then sudo user Give mode access to the Docker Append to the group with Docker Do it to whoever is my current user From dollar user through environmental variables. Can also be accessed Append and newgrp Docker newgrp Docker Meaning you refresh your Docker group Once the user is updated, it will be updated with Docker. Add to the group, refresh, enter Now if you do docker ps it will run smoothly If it is installed, then Docker is installed.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) (part 4)

Kind is installed kubectl It's installed, try running it. You will tell me what to do, kubectl What is low, it's just a version of this, maybe yes If it is just a version then you write kubectl. He told me the version of Docker High Fun version. This is the kind of fun version that I told Given that these three things are with you I don't know if you want to install it. I mean, if I have a Mac with a normal processor, I Directly for Mac, the Mac command Copy that and put it in your terminal.

I will also teach you Blue Green Blue Green You have done this, now what should I do? I have downloaded Kind, he needs permission. Give it to me now I'm cool if I kind of nuffin I will do the version, it has become kind among us. Docker Nuffin has also been installed The version docker is also installed in my kube CTL There is no version in this, there is no half in this If there is only version then cube seattle. It is installed so in my local also I Kind Docker and kubectl done I even went beyond what I had on O2.

kubectl Kind and Docker Y have been Done, now how do I create the cluster? I want to create a cluster, so what should it be like? There will be a difference of just one small thing between us. Like I need three, I mean the number of The number of nodes is what we get I want three, three nodes, three nodes Instead of two nodes we get three nodes. If you want three nodes and one master node then If we need total four nodes then we need four nodes.

How to create a cluster? The cube starter has a conf file, there's a confession file that you have to write. After doing this I will hit enter and A DevOps Live Course Near Me from Lets which is the second one I have an AWS I have a course and then the third one is I have a. I have a Python course, so what do I have? There is a Generative AI course and it's all Courses DevOps Row to Zero Batch will help you If it is found then basically what it is it's a key and it.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / Kubernetes architecture (control plane, nodes)

If you have to give any value then Cub Nits I have filed everything in it, so those who Whether he came to DevOps or DevOps There is no coding at all, this is very wrong. There is a lot of stuff in the app, what can I say? Kubernetes tus i l about YAML code ye YAML Whatever code you are writing in the files on line remains the same so escape call double k this. If you save it, it will remain the same as it was before.

What we have to do is confine the cluster What cluster do you want to build? Confession how then became an MKD IR A Kind of Low Called Cl Cluster Create your own CD named Cluster. In this cluster there is a There will be a file that contains your cluster information. The conf.fig file will now be Confine Kind Meaning Ek To Kubernetes and Docker that one kind but That Kind It's a Tool would have been another Kind what type of file is it kind this.

The cluster creation file is fine. The file that creates the cluster is now this file This is how you can use the API in the Kubernetes cluster. If you give it to the server, then an API version of it will also be given. The API version also remains, so the one The version is a particular version. You have to create a cluster of Kubernetes. If you need nodes inside the cluster, write nodes How many notes do you need? You need notes, you need multiple nodes,.

How they did it, nodes. Start making a list of what I What do I have to make Do you want to make a roll, this is our main There is a master node, it is also called. It is also called the control plane. It is also called a control plane, what do you say? Control plane so the control plane If you want to make it, then role control plane now this control plane What I told you will happen, Kubernetes End Docker will be built through Docker images.

What kind of image will you need to use? Est this kind name image is in it You also have to create the node which is now the latest version Off Cub Nits is going on, he's going on right now.

[Kubernetes] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Kubernetes architecture (control plane, nodes)

I don't want to take too much risk Buy the latest version because it is long If you have a video then take the stable version. Let's take version 1.31.0 Think 1.31 the most This is the latest one, I picked it up, now this My control plane became the same way you Make another one what do you make you a worker. You create a node and which image to use in it If you want to make a node then this is the image.

Copy and paste it, now how many do you have? You have to create a total of four nodes. It has three workers and one control plane. If there is a master then this is how to copy and Taking it like this, it became paste and paste Now it'S3 workers and one control pen. Your work is done, great, but your The cluster is a Docker container Your EC2 will be running inside If outside, the port number of the EC2 and Its port number will be different, so it.

Therefore, it does not have to be mapped Extra Port Mapping Extra Port Mapping Extra port number mapping if needed What you have to pay in the container. I have seen both, if you see them then quickly Tell me in the comments because you are absolutely right. Direction to the container port The port number of the container is 80 and it is mapped to that port. Is the host's port number 80 is and which Transmission protocol to be used Control protocol used in TCP And similarly I am going to set port number 80 And what does port number 443 mean?

Is ours What is HTTPS and 80? If you remember when we were here We were building a brick instance, so we No, SSH is enabled on port number 22. Port number 80 and port number 443 We enabled these two ports. Similarly, you can also add two ports to this cluster. That's it, our cluster confine escape is done. Colon double k enter I hope you understand It may have happened that what did we do? The value of Created object and created list of nodes also made it fun escape colon dull k enter so Look, you are also slowly learning email.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

What do you have to do Kind Create Cluster Kind Create Cluster So You Kind Who are you telling to make a cluster, Kubernetes? The TWA cluster appears to be back. Go now the D cluster is done but this The cluster that will be formed will be made from the confession file. Han An Confine no confine file which is mine A TW cluster using a conf file I am creating a cluster named Kind me so enter so creating cluster TWA Cluster Assuring the Node Image.

Kind node version 1.31.0 great your cool new image that I am bringing the preparing notes, look at one Doti is creating four nodes so you can How many were needed? One doti four nodes were needed. What is a plane control plane? Scheduler to API Server to Controller to the manager and Running etcd Starting So you see, these were the services that you He is managing the entire cluster. Starting services Installing Storage classes mean you also need to store some You will have to do it for that, your disk If there is a storage class, you can do that too.

Joining worker nodes so any number of Worker was nuts now he's with your cluster Joined Joining Worker Noted And you can now use your cluster. You can use your cluster. kubectl cluster info hyf fun Context Keyne TWA Cluster what is this kubectl cluster info So what remains is the kubectl cube CTL remains the tool that manages your cluster. If you can handle the whole thing then what are you doing here? You will get the name of the cluster. I will have to explain a little about the context.

You will have to give what the context says That's my Kind Teads cluster Give info about it, enter it is saying Kubernetes control plane is running at 127.0 1440 389 so localhost on local. One is running and where is the DNS running? If someone goes into this then it is basically yours Please tell me if the server cluster is running fine. How many nodes are there in the cluster? Nodes boom so he'll tell TWA The cluster control plane has a node named which is ready whose job is the control plane this The control plane is and the API version is.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

TWA cluster worker is also ready And it has been going on for so long and this too This is the version and there are basically one, two, three workers. We have one more master and three workers. Is worker forest 3 And there is another master, this is the complete one which we have We had made the diagram with our hands. I turned it on and checked, so our cluster is working. I am left with no option but to tell you what to do.

I have told you through kind, now I will tell you the same If I tell you through the Minikube, then an MK Create a folder named DIR Minikube Low CD Minikube. You should not feel like this, you are kind Perhaps they have partnered with the people like this There is nothing you can do about Minikube here. You can use cube starter You can also go to the repository and download Minikube You can go to the installation, there I have It is said that by updating the system sm.

I needed this thing, put it in, so this What is tomorrow W Gate which I have to install All the tools that are required for this are there. Kind of pre-requisite this put Docker You have already installed it now. Meaning of enable What is it, enable means system ctl enable docker means your docker your Even if the computer restarts late Docker will restart automatically. That is why it remains enabled, enabled means on Start up the service will restart This is done for you, you did this too now.

You install Minikube then Mini The latest one from Cube is MD64. Bring the Linux one, then this is that URL is c url means curl you can download Le Ho are doing a Minikube called In a folder or in a file you can. Bring this particular thing, this particular We can get mini cubes from the URL. Here you go 99 now 99 Done try it here Minikube Now that Minikube is executable. If you had seen it, your Minikube would also have been installed.

The Minikube is also great. Installed, now you can have fun on Minikube You have now installed Cube Seedle. Start the Minikube start driver docker v m true then type this command Minikube start nfn driver because you should pay the Minikube on.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / kubectl commands

I want to run it on Docker, I want to run it on Docker too Docker works e tr v a e true You then have to write When you are running on an EC2 instance. Because it is a virtual machine, so enter Minikube version 1.34 it is, the latest one is Minikube. Pulling image 0 this is that thing, he is pulling all these things and Your Minikube is also forming clusters Meaning, today you should destroy our instance. You can also have a cluster of Mini Cubes running.

A cluster of Kind is also running Then teach me how to create a cluster of ES as well. It is happening so it takes a little time but You can do the same thing at your local place as well. All you have to do is go to Minikube Start and Installation of Minikube Or whatever you need for Mac OS, just Copy it and install it quickly. This is the complete method for this, it is great Hit install and everything will work fine Will work This creating Docker container requires two CPUs and Oh my god, with so much memory.

Father, what will happen, we have two clusters running. Preparing Kubernetes version 1.31.0 on Docker 27 Look at the network we're running on Kind Yes, it is 1.312, so what should we do? kubectl is now comfortable to use Mini Kube Cluster and Default namespace Meaning if I do kubectl get Nodes was just telling me about three nodes. This is just one nut telling what this is We got scammed so when you mini If you install the cube then your cube seat It was pointing to the Minikube by default.

You've seen Kind Cluster before. If it was installed then do it by default Now it has become over-ad, so if you Get notes to do so you high han Context Kind TW S Cluster. If you do this then you will get the blessings of your kind caller. In the command kubectl get nodes this I have told you in the command that, how you Context scans to different contacts If possible, your Minikube should be on the cluster. Did you understand that Minikube Cluster?

If it is still running then don't make me run anymore, I don't want a Minikube, so you can also delete the Minikube if you want to delete the Minikube, you.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / Kubernetes architecture (control plane, nodes)

Simple Minikube Delete Write Do Boom Delete The Minikube cluster will be finished now If you cube seattle gate notes then that It will say Can't find the server error is coming Can't find the server because by. Default that set on which one is this mini Cube Butt Yehi If You Minikube Gate Nuts Use context, whatever your context is. What am I kubectl Confine Pect Which Kind TWA I'll use the cluster enter switch Two Kind Tead Clutcher Now if I do kubectl get nodes So now that everything is set for me, I understand.

I explained how you can use different clusters. You can set contacts for different texts You can create clusters and run things on them. Can you, this is our KB NITS cluster? In the journey of making, the Minikube was made.

Kind cluster also became EKS AS GK You must get this done in our mega project. Doge End Me kubeadm, that too Thank you, if you make me do all this then that too If you get it done, let's go and see Cube Stadium right now.

Let's take kubeadm like I told you Bare Metal gives you a Kubuntu cluster. If you want to create it, you will have two servers. You can make one of them your master. You can make the other a worker and both of them You can join by becoming a master. You can also get multiple workers to join And this is your cuberium, so its I will tell you the diagram, this is the diagram I would have made it a little more beautiful.

Let's just say it like it is Your master node is done. Node Done Cube on your Master Node Adium will install it and it will mean Inish is where ever kubeadm inish It happens, he becomes your master. Node and then later you can do 1 2 3 4 as many as you want You can get worker nodes to join it. Let's read this quickly, first of all you What you need is a masternode and a worker node so what I'll do here But this creates a new instance.

Let's put the name of this one, Master. Master Kates, you could have added one more name. Worker Cats, this is not Windows, I regret it.

[Kubernetes] Kubernetes architecture (control plane, nodes)

Tauba nuts take it master K8s To make it, we need nuts which are t2 Micro won't work, we need T2. Medium is fine, you will need t2 medium so that we are completely on T2 medium Get the pair that the cluster can handle. We also have to keep in mind that this Our workers join the node,. Port number If he joins at 6443 then he also considers himself You may have to open it later,? Take the storage, otherwise we are just for demo.

I am looking for 10GB storage, will it be good? After doing this you can create two such instances. If you want to make it then here number of instances to put it why because we have a I need a master and a worker, you want. Instance two launch tense done great So this is your master and you are a worker. If it goes then it becomes your master and it becomes your Worker node then the name of the master is master only Let's keep it and this is the name of this one too.

If it becomes master then change its name. Workers do it so if I do this and put ktest in the filter here If I give it to you, you will get it, where did it go? I have been I will put in the state running for whomever it is. Whoever is going to be my runner I would put him I'm here running for All State. If I put it then master cat and worker Kate, we've got this, great, now I If I make these two SSH then Master I will go to CAT, I will go to Connect and this.

We are not copying and creating a new one here. Let's create a new terminal and We will do SSH to the new terminal,? Make it a little bigger in the download Yes, paste the SSH command. This is our master cat, so its By changing the name Add master cat's and some color too. Put it in, what will be its color, it is purple Master Cats Done Master, we have done Master Kate's best. Will give If you connect your worker CATs also Worker Cats took As and here's a new Opened the and here's what I did.

[Kubernetes] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Kubernetes Services and networking

A little bit of this in downloads where our The pay file is kept there, go there and do SSH. To our worker's test, worker's To the Bannet guy, this is what we do. Worker this is worker and this is our Master done easy, this is the master, its worker also give it some color or make it yellow done So whether he is a master or a worker, both. We need to create a Kubra Nits cluster. Let's just make it now what do we have to do now We have to go to our cube starter In the repository where we have this All the steps to install Kind Cluster.

All the steps to install Adium is here So according to this you just have to keep going. This Kubernetes knots cluster and cube This is one such tool that is going to be created through Adium. Which runs Docker container disks in the background Container Making Services or Container Using Management Services Kubernetes will make you Kubernetes clutches Let me tell you what Container D is. Container D is an industry standard CNCF is a container management tool. A graduate and has a background in Docker This is what I use, so this is how it is If I give a background then container d So we are going to make it using.

You must ensure that all instances In the same security group, we If two instances were created from the same form Both have got the same security group. 6443 Go to security group and expose If you want to do it then let's expose it. Where did it go, so this is our instance of this, we People will go to the security group, this one Go to the security group and edit in bounds. After reading the rules, we will make an ad rule.

If we need a port, we open it. There are some commands that you can use Both master and worker have to be executed There will be some commands which will only You just have to pay Master and something. This is a

command that needs to be executed only on workers. Let's start one by one are now executing both master and Worker, first we disable swap We will give it swap memory first.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

Turn it off and I will go to Master first. Swap off I go to worker and swap off We have to do this command at both places. Execute and both are necessary Kernel Module for Cable Networking Some times the policies are about networking. Overlay Net Filter Us Master and Worker You have to enable it on both Done and Ye Worker Don System CTL Parameters that IP tables contain are for networking because your needs to connect multiple instances. How was the kubeadm and Minikube one?

Everything was running on the same instance here. If you have different instances then for this You have to do all these things so Copy it, run it on the master first and then Go ahead and do it, worker, now what should I do? Have to install container d look Container D like we have Docker installed Container D would also have been installed if I had done so. Sudo apt-get update which is important Things need those rings its a docker key When a key ring comes, you will add it to the system.

Install container D by updating Then restart container D. You can check the status to see if it is running or If it's not working, I'll copy it. First I will throw it on my master Enter, then I'll throw it at my worker. Enter now containerd is the tool that Kubernetes The control plane of the Kubernetes is like etcd API Server Scheduler Controller Manager on that side worker note on kubelet Service proxy, all these tools. Container D will run so container D will see Let's check whether it is installed or not master install ho on packing Container De Done Ho gone container d installed and active.

Running on Master worker jaake dekhta The container is being installed, after some time it will install ho jayega and it is active and Running on worker queue suppress out of here. You will come and why press on the master also When you come out, container D will be placed so that your Containers for all components Yes, it is installed now you will get the actual. Installing Kubernetes components in Now you have to do this, the version is v 1.29 If you want, you can use the latest version as well.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) (part 2)

You can also do 1.32, you just need to go here This thing has to be changed and what else can be done He is updating the system Also important are packages like curl. Download the system update and cube lat and kubeadm and kubectl It will install these three things and that's it. It will hold that if you stick to the same version I'll copy this and paste it. I will give you first Master Pay Enter and then I will paste it on worker and then enter cube.

Kubeadm is needed on both the place, the master and worker have been done, just see All these things that we installed Is it installed or not? Cube CTL Done Install Here kubeadm Ho If the installation is done then your master and worker Both note the common things that had to be installed And now it is executed only on the master. Node now has to do things that just If you want to install only master node then first What I said in my diagram Whichever server you are using kubeadm on.

If you do it, he will become a master, so right here So if I do sudo kubeadm in it What should I enter on Master, now what has this become? He is becoming a master, meaning in this Installing Kubernetes The control plane of the Niti will become the API server. Scheduler Controller Manager etcd SARI things will be installed in it because This is the master, this is the control plane. All things will be done detected that the Sandbox Image Registry and Cuten RAM Inconsistent with use but it is not recommended No, there is no scene, it will continue.

There was a warning that it was not a controlled scene. Plain Using Manifest Folder Done Control Plain Creating Static Pot for API Server Controller Manager Scheduler Can You See It has become ours, this wonderful thing has become ours. The control plane has now been created, which we kubeadm it did that from sudo means root user but you can always use root user You will not be able to run things on your own. You will also have to run your own user from the user of Meaning I am an Ubuntu user so I am using Ubuntu.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / kubectl commands

Which is the current working directory She's a home stew, so what should you do? A new DirectoryEntry You have to create it in that directory which you The admin has conf files that are stored in the ETC grave. You can save the files in Kubernetes Admin Con. Once you have done this in the folder, then you what to do with that dot kubeconfig There is a folder inside which the confine is kept If its ownership is changed then what will it do?

This will make your user who is yours Current user is who am I who am I Ubuntu this Nutu can also run the cluster as a root user as a main. If the user is able to run it, then this is a small step. What I told you to do Whenever a cluster is connected to another master And this is what your worker nut communicates So a container does it from the network interface. If the CN network is Calico in this case What do you want to do, apply.

Have Project Calico on your cluster? From kubectl apply which calico name network of all its manifest files All the important things that happen to him Kali will install them to run Enter and put, everything happens. What is a cluster role? What is all this, you deployed video? The difference between our master and worker The connection is set up correctly. The worker doesn't even know. Let's put some commands on the worker. By the way if you have any If you want to make the worker node join the master All you need to do is create this kubeadm token print join command this is it just here.

You have to go and run it and it creates the token. By joining kubeadm, this port This token is passed through and anyone can use it. Can join the server now if I would do kubectl get nodes like We had seen the Minikube cases,. I did kubectl get notes so Notes were also visible in their Kind cluster I had taken notes, how is the current situation with him now? Gate nuts have only one which controls There is only one plane and it is ready for me now.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / Kubernetes architecture (control plane, nodes)

This worker, this worker also has to be added to this I want to add it to the master node, let's add it. I will give it to you, now what will I do with my workers? Nut p boonga kubeadm reset by mistake We don't know what happens then. Kabhi Worker Node Pa Jake kubeadm It If done, that worker node becomes the master. Kubeadm reset preflight check From you to whatever things are in kubeadm If you remove those things then changes are made to This by kubeadm Inate and kubeadm Join will be reverted.

Our worker, now I join this worker How will I get it done, how will I get it joined, this is mine There was a join command, Master P, I copied it. Tax I will take it and go and write it on this worker note. Starting the cube late cube late Where does it live? It lives on the worker node. Worker node on that kubelet start Given and this node has joined the cluster then This cluster of yours, beyond which right now There was only ready control plane, now if you do kubectl get.

Nodes can you see a new guy has joined He is not ready yet because right now He will get ready, comb his hair, he will be very and sharp. If you do watch, it means every two seconds To see the output of the command kubectl Get nuts so you'll see 40 seconds 42 seconds Come on, is ready, so this. The way you did it is so wonderful Cluster of cubes, you have made it very nicely. If you want, kubectl run any pod Can you run it with Let's, what is a pod?

You will find out in the next video, but You can run whatever you want on it like kubectl run Nginx me engine run which is To run Image Engine from the latest image, enter. Pod Engine cool pods and whatever in createdted That cube can also be created through ADM. I hope you enjoyed it, I have taught you how to build a cluster. Minikube kubeadm Kind In the mega project or end of the from you If you will see EKS too then it is great, let's move ahead.

[Kubernetes] Kubernetes Services and networking / Pods

Microsoft told me that it means monolith. Told the architecture, told the setup also I got the kubectl done and told you everything. Move things, make things happen, what's the point in that? In this he first understands the whole me I will explain the whole story, so first understand it. The story of how you run a Dogger container It happens that Kubernetes is fine A Docker container has to be run but this The container you drive is an individual You don't run a single container.

You run a container in a pod of sorts. In a pod, one or more in a pod More containers can be run but now Kubernetes What is the work of Nits Auto Scaling Auto-healing is what you need. Deployment deploy you need Kubernetes If you want to deploy then after your study You have to learn deployment now this is Deployment is how you expose yourself to the outside world. Accessible Doge Who Was The Outside World If you had service then you also got the service of Kubernetes.

You will have to learn, there is no service, so take this Look, it's, what will you do? You will also have to learn service, Kubernetes Making cheese is not that straight forward. You would have just run a Docker container The work is not done. Let's run that Docker container. One of our engines is a Docker container from Ecus Nginx is a Docker container of Nginx The first pod of that Docker container Then a copy of that pod will be made. Deployment will become a service And only then is your User You are a user who can access it You will get this user and you will be able to access it.

If you are like this for Angel Nicks then it means Like you mentioned about the engine, probably It must be correct for other things too Just like Engen's, your MySQL is the same. It would also be for It would also be applicable for MySQL If I pot it in containers If you learn to make all this service then I will be fine. You just have to change the container, then everything else. The configuration needs to be changed a bit, but The story is that P Deployed Service or At the basic level, you have to learn it.

[Kubernetes] Namespaces / Pods

You can modify the containers you have. Running and MySQL on the same cluster If it is still going on then there would be confusion,. The group lives in a happy family in which the house People living in an abandoned family home People live in the same way in Kubernetes cities There are also groups, what are they? There are groups in which members of that group All the related resources are in this pod. Deployed service contains all these resources If there is, then that group is not called a group.

The group is called the namespace it's called namespace or Then What does NS say, namespace say So what is a namespace? A namespace is a There is such a unit that particular name All the resources available in space These groups will remain for the group and other groups. If this engine will not disturb the Do not disturb mice cycles with Pod is This pod will do it, this one will not do it. This service will not deploy it not isolated so that you can manage If it becomes easy then first of all we If you read about namespaces, I What will I say to my Mini Cub?

I will come out and I will be kind to everyone. Some Kind Clusters So Kind You have created a cluster, you have made everything, how much It's cool,. You've made a mini cluster. In this particular session I Whatever commands are there in the particular video I create a repository for everything. Whose name is M KD I R Kubernetes I will create a folder for these one shots. First I want to start with namespaces. From so kubectl get name Space kubectl get namespace or else You can also do this as Cube Seat Gate NS yes ns means namespace so you get the same.

Default means that which has no If his God does not exist, he will exist in the same way You did not create any namespaces And if resources are created in it then by. Default goes to the default namespace If you haven't created any namespaces So it goes into default mode. What happens when you have a Kube Node Lease? The nodes of the Kubernetes cluster are formed Where does his information go so cube Node lease public as much as outside services What about the publicly accessible If it goes to the public system level.

[Kubernetes] Namespaces / Pods (part 2)

Where are the services you are running? If you are looking at the kube-system, you will need to have storage If you want to get it done then there is also a path in the local storage. The pods are running so if I do I am telling you that, I have a cluster on the server. Which pods are running in So Kube CTL Gate Pods S kubectl get Pods enter No Resources Found in default namespace Meaning kubectl get Joe The namespace is the default, with nothing inside it.

Running but the system is our kube-system So kubectl get pods in the namespace - An in the namespace kube-system Enter Now Look, all our components were core DNS etcd and some APIs of the kind. Server Controller Manager Queue Proxy and The scheduler keeps track of where everything is running. Rha kubectl get Ns Cube in the system so that's why in Kubernetes you can run multiple A's with different names. In space, just now, for example, I Am I going to teach you Nginx?

What am I going to do kubectl create ns means create A namespace called Engine EKS Ek Cube Seattle Create Engine Cube Seattle Create an engine to me namespace enks Get an excellent engine name. The space has been created, now I pod in that namespace I will create it, I will deploy it, I will create a service And if I get it accessed by my user then this. What will I do, all these things are based on I will make a simple cube, ctrl run cube CTL Run Cube What is CTL Run.

Like your docker run and image name so cube seattle run engine so this is the name engine - - image is equal to What does it mean to be alike. Docker Run Engine and Engine create containers It starts moving, look, I'll show it to you. If you do docker run engine then the engine will start running kubectl run Nginx and image which engine image lifted I will take it, so this is mine. Photo also came cube seattle run ajax Image engine enter pod ajax CreatedDocumentsFragment Meaning it had to be run in this particular namespace If I Cube Seattle Get Puddled in the namespace.

[Kubernetes] Namespaces / Pods (part 3)

By mistake your pod is default. Has been created in the namespace, so when something like this You can also delete your pod if you make a mistake. If you can, kubectl delete pod and pod Name of Nginx enter pod deleted then your pod Can be created, can be paused, can be deleted It is possible. If your image is not found in LETS then Image Pull back off error crash loop back off error all this Errors may also occur, so upload the image properly.

Where do you get this image from picks up from Docker Hub now I have beans I will run the command kubectl run agnes image Annex in the namespace within a namespace. Here, this time, I will define these The namespace Ajax Enter Pod Ajax Create Now if I do kubectl get pods o No Resources Found But kubectl get. Puts in the namespace Inks Boom Your Engine of the namespace This pod has been formed inside and in this way you Creating Pod namespaces from the Command Line Now what I want is the command.

Don't do too much on the line because the command You will forget which command you typed. Drove That's why this pod namespace is deployed Service, all these things, we are will be created through ML files which will be Manifest file is also called manifest File sote power of manifestation. If you want it, it will happen, then whatever you want will happen Write it inside a file and the same thing will happen. That's its system, so let's manifest Let's learn to create files, delete all these.

kubectl Delete pod named Nginx which namespace Inside Nginx is the deleted kubectl Delete namespace Nginx so namespace and nix too Deleted sorted, how very correct thing Yes, but what do we have to do now? Create a folder named Nginx, ajx Now cd Nginx if I want to make an Nginx, what should I do for it? I need something, I need namespace, pod Needed If I create deployed files then EM name Space dot wa ml e m namespace dava What this is, I'm basically creating a file.

I'm living in a place called namespace and its What am I inside I will make a manifest inside it I will write a file I will write a file like this.

[Kubernetes] Namespaces / Pods (part 4)

That the namespace is created, so like To create a cluster, we The kind of conf file that was created was It was a cluster, so it would also have a kind of name. Space now like its API version was kind of Dot x Ked Come and So On will be its version Why v1 because version one of Kubar Kubernetes Currently running KBS version 1.31.0 on which we. People are running API servers, that's why Its version v1, write this much, this is it Now the information that needs to be given is gone.

It contains multiple information will go yes so what is this kind namespace is the key value pair is API version v1 is again API version v1 your key value. The pair is the name in the meta data is an object Inside which meta data is an object name Please enter the name of your namespace. From lats Nginx is over, that was all there was to escape. Colon entered Now what will I do now, I will say it now. Cube C to kubectl Apply it whenever you create a confession file.

Either you create it or If you apply then create means only once Create happens, apply means create also If it happens and gets updated then I always I apply kubectl Apply this file of mine. The name is namespace ddl. This is called your API server. Navigate to and apply to your cluster two enter namespace Engine CreateDrawerNavigator If this is done, then this namespace is yours. Manifest File Everything in Kubernetes is a manifest file and this manifest file A Kubernetes when you apply Resource cube becomes the object of

Kubernetes If you do your kubectl get ns then You are here engine.

Pod makes me Padwa, I am going to make pod now, pod now, pod What will remain in me is simple, your kind that will remain and What will be your API version? What should be the name of the data that will be there? If you want, I can give you the name of the pod. Ajax pod Now the meta data or the angle spot is Entering a namespace within a group If you want, what will be my namespace? Ajax because I am inside the engine namespace I want to create this pot, so Your pod's metadata is now complete.

[Kubernetes] Pods / kubectl commands

I need specifications, I need specs I need specs of pod, bh now what should I do I don't know how to write spec now. It's very I will tell you a simple trick, pay close attention You have to see where you have to go G P and write Kubernetes Pod Bus and you will get Pods Kubernetes here. Cuban can go next time in a pod now I still said it in English. There is a spec which is the template of the pod, you can also see from here, I will explain it to you.

You can also see from here that the You have to create containers in the spec Meaning there can be more than one container. Let's make it so there is space in it for you. You can create containers one by one or more. More and the containers that will remain You will have to name it like every single A container contains an image and If the name remains then the name of the Container Engine. Which Nix image to use Nginx, you have to use the latest image.

You need the latest everything and everything that your ports Which ports will remain? The ports you have are multiple. Whenever there is multiple, you can list Remember that by dashing you can make a list If you can put it, whatever you have The container port will remain your engine. Generally it runs on port number 80. That's all you need to do is use a Docker Compose command like You write that you write the name of the image in it. Do you give me the port from which second I have to insert it?

If you write any environment variable If you have it, you can put all that too. If you can put the specifications here, it Your pod manifest file has been created Kind pod API version v1: pod name in meta data Nginx its name in the engine pod namespace. Space has become spec or specification What containers did I get called Angry Birds? Nix and uses the latest image editing software. In port the port number is 80 you have installed it. I put confine and the manifest said that I want to manifest this escape colon w k Enter and you simply wrote kubectl Apply -f pod YAML and enter pod If you do kubectl get pods something.

[Kubernetes] Pods / Namespaces

You will not get it because of the problem of the pod In the meta data of the file, you namespace what did you put engine to cube Seattle Get Pot In The namespace Engine. Will you do your pod running great, your engine, it's moving inside the engine pod. How to get inside a Docker container Docker used to enter inside it one by one. How simple is kubectl Ekak - it interact terminal at ghus Go inside whom do you want to get into, inside the engine pit If you want to enter then please.

What is the name of the Nginx pod namespace? Space Nginx Now when you enter, which shell I will use nfan bash enter this what happened han what happened to bash kubectl Ekak - IT engine pod in the namespace Eugene -. Bash is an object of what is happening. There was a particular command to do this when If you are also using bash then in between Make sure to hit space otherwise he will think that What is he doing? Try LS inside and see, it is running fine and One thing lives in Nginx Welcome to Nginx If you come after writing this then curl it Curl 12701 Can You See Welcome To.

Nginx, your agency is running but this So it is not going well, if you tell us If you can show me how to run it on the browser then we will enjoy it. We get happiness, it's not fun But learn to have fun because cube CTL Get Pods in the namespace Ang Nix you do then you can your pods You see, but that pod is not working properly. How is it going, you need to know The debugger is very important, so I This is also mentioned here in the syllabus of this course This is how you learn debugging.

Debugging and troubleshooting here How do you debug in kubectl? I will also teach you how to check the logs. kubectl describe pod Nginx pod so you your Can you describe the Nginx pod? Using the namespace Nginx enter man I kubectl Describe Pod Nginx Now Look At Your He would tell you everything that happened with the pod. That your pod name was this namespace this No priority was set by default service account node this pod node number one The start has been directed at the waste worker.

[Kubernetes] Pods / Kubernetes architecture (control plane, nodes)

Time started at this time, it This is the IP address, this is its container ID. Its image is state is running and volume Mount by default this volume is mounted now What happened when that happened? First assigned to Nginx Pot Cluster who assigns then the default scheduler who The schedule he assigns to the pod remains that this pod is on this cluster will go to this node then queue bullet. He pulled up the image of the engine, then why? He pulled over and Then I ran the engine container and it All the data is passed to the API server.

The API server stores it in etcd Given so that you can read all this and understand You are thinking that this architecture diagram is just And just reading is one thing, then command. It is also very important to understand by digging through It is important that the API server is configured with Kube CTL said create pod engine then he Pod Schedule said near the schedule. I decided that I will go to the first node. I'm going to schedule this pod again.

The cube that was lying there, picking up the image ran the image and then your pod finally I started understanding and that's why kubectl Describe is an important command and Types of kubectl units inside it. Engine Pod in the namespace Engine Hyph Nuffn Also to get inside the bash pod If it is an important command then give two commands to your best Prepare this pod, then Docker. You create pods just like you create containers. If it is made then see what is special about it.

It is not a specialty to make this pod scalable Making is a specialty and for that you Looks like deployment What looks like deployment When you make deployment, just like you made pod. Made by the way v m deployment dwa you I will wonder what you are making, you are kind of making it. Are yes what Deployed what will be the API version v1 Then you will say, when we made a mistake As you said, we are here We went and wrote Kubernetes Deployed.

Ne and deployed here opened end like The API version opened and was written here.

[Kubernetes] Deployments, ReplicaSets and rollouts / Pods

My name will be engine deployed. The speck that remains is what is deployment? By the time people read it, they understood that Meaning the container is inside the pod It works, why should I deploy, understand this. Whatever you have to say or whatever you have to say This is a pod you can make another one, make another one You will take it, but tomorrow you are making this pot. For more on amazon.com amazon2 three non-working your entire India is coming there and buying shampoo and soap.

If the traffic is too much then you don't need one, two, three pods, you need a group of need pod reps Such was the ReplicaSet This is what ReplicaSet would have been. How is this, how to put it inside, this is This is the result of rapt, by making you rapt. Will give three Cheese ReplicaSet Of Set Stay Full set deployment, if you maintain the rep If you want to do it then as the name is ReplicaSet then ReplicaSet. Let the set do it, so this StatefulSet What is deployment and what is deployment?

There are three different First let's understand why this thing is deployment but Before you understand deployment, you need to understand This word will be called Rep Controller Rapcase. You cannot assign all the traffic to a single pod. You can give it because the poor pod will die Then you make rapes of it, so rape is rape. Of the controller pod Reps manage Rep work. Meaning if you said one pot then one pot If you said two pot then its ident tomorrow.

Two pods will become clones, aren't they twins? Does So what is a ReplicaSet if you said I have this template of a pod, this is a template of the pod in which I want to use the image. Angry Nicks should be more ReplicaSet Four will become your one, two, three, four pods. This is ReplicaSet The set is then a StatefulSet that each one Pods can be created and deleted at any time. The StatefulSet is what each pod says I will give you the numbering, number one number three, number four, and so on in each pod.

Key state maintained in a by-number sequence If it remains then that state is fully set and The third is deployment and ReplicaSet The set remains almost the same, in that too you.

[Kubernetes] Pods / Deployments, ReplicaSets and rollouts

You can only make reps but deploy you a Rolling works great Updates Rolling updates mean if your Four pods were formed, four ReplicaSet were formed, you have your. If you change the image of the pod then the ReplicaSet set In this, everything will change completely and What is the problem that your application The blue application is running. You have to click update so that it becomes yellow in colour. The whole application converts from blue to yellow. It is happening in between, there will be down time in between If the application stops and starts again, it There is a downtime issue What is deployed?

This pod will continue running until the first one This one is getting updated, this one is part chat This one is being updated while it lasts. When you look at it hands-on, you make it deployed in it Reps, please tell me how many reps I have. I need two reps from let's Who needs it, who needs rapt, so this I need someone in whom I can recognize that This thing is mine, for example, I like this. This is my pod, the engine in this pod Ek is going on, two reps have to be made of it.

Deployment can recognize these pods. Should therefore label this pod You have labeled me DevOps,. Similarly, this pod was also given a label. That Nginx pod or p ejax And to select that pod, press a Do you need a selector as well? Do you need a selector as well? If you want, what are labels and selectors? Please give a label to each pod. Give him a naming slap that this, this P is the P of Ajax and say it in deployed. That Selector Select something that has a name is ajax so if this pop name and this pop If the name matches, then copy that item.

If you apply your reps to that thing Let me tell you, I need two reps. I'll go into the selector and I will say, match the labels. Label it Any label that has a page Nix will get him matched but his So I need the engine too, so I'll make a template, template and In the template I will say that meta data.

[Kubernetes] Pods / Deployments, ReplicaSets and rollouts (part 2)

What Labels meta data in the labels template What happens is your pod's template remains engine, what did I do? General information about the pod in the template means what will be specified in it and what remains in the speck of the pod Containers live and what's in containers Remains name ta engine container whatever you. Naming the image remains engine Latest and what are your stays and ports live. If the container port is exactly container port 0 then what did you do here you created a deployment what is deployment control the repen To do your one state particular.

File a file to go to the desired state You write that you want it deployed. Make two ReplicaSet, make two twin replicas and Any Pod That Matches the Label Pop Engine Ecs Select him, meaning I have the power. There is no way to duplicate anyone's Whose butt should be made to make a ReplicaSet? I want to make this whose app is Nginx now He is my selector whose app Annex Now I have to find the label of whose puppy is this?

Nginx is good this is a pod its label is Ang Nix is pick him up and make him rep This is what I'm trying to do template. What happens is that the puddle you have is the same here. If you have to write, then even if you don't paddle If you write it, it will work too, then you Write metadata in the template, meaning whatever You want to enter the name from let's enter it. Nginx Write something like this in the pod of deployment engine.

You can name it anything like this It's yours, you can do anything. Labels Engine It is very important that the What is the label of the pod you are making? Angelina is her label what is the spec Specification of your Pod which is your container spec so what are the containers in it Name Nginx Image Engine Latest Port 80 Done Escape Colon Del Key Enter Done. kubectl Get Pd In The namespace Nginx It's going on now what did I say Deployed will also create pod and give pod also If it is created then it is of no use.

What can I do kubectl I should delete my own post.

[Kubernetes] Deployments, ReplicaSets and rollouts / Pods

I enter the engine deleted Gun now what can I do I simple I can tell my cube Seattle that Apply the manifest I created. Which I created as the deploy manifest Apply that then cube seattle apply Han f deployment YAML enter Getting error one creating deployment version v1 cannot be handled Deployment Strictly Decoding Error Spec. ReplicaSet so your version of deployment It remains there, there is no rep in it. If rape happens then you will have to do all these things.

Is escape calculus w k enter cube seattle apply h f Deployment Enter Deployment Agent DeploymentCreatedDocumentsFragment In the namespace Engine Two pods are ready meaning if I do kubectl

Get Pod in the namespace and Genetics. A Jinix became two pods, how cool is that Cube seattle get pot became two pods and right here. If I do two kubectl Scale A lot of traffic has arrived Scale deployment named Nginx Deployed whose namespace is engine uske rapt kas kitne do five Two Enter Deployed Playment Scale Cube.

If needed I can use kubectl scale Deployment reps can do one and boom If I make cube seattle gate pods one I am running it, I will do kubectl. I'll run a small pod, a pod, understand. That ReplicaSet also control rep How many reps are running stateful Sets also maintain reps deployment also maintain reps, just dill T updates your rolling What would the update be. Do something after asking for his command. Teach me rolling updates, I am not Let's know why we do rolling updates I scaled my deployment from And if I did five reps, then my five reps Became Get Pot Five Became Now About It.

If I write to you, I will More information is being received that this study worker two is on it's on worker one it's on Worker is three P, this is two P, this is one P. Less information is available, the question is this All the pods are the latest version of Nginx. I am using it, use the latest version I want to update you from Lets a little older version so what should I do I will find it here Nginx Docker.

Version So here I have different ages You will get versions as there is a version 1.7.3 and one is 1.27 there are two versions 1.7.3 and d Let's make 1.7.3 one and 1.6.2.

[Kubernetes] Deployments, ReplicaSets and rollouts / Pods (part 2)

There are two versions of me, in between these two I have to update it, how will I do it then cube CTL set What is the image CTL or roll Out kubectl Which engine is deployment? Deployment enzymes that are deployment Space Kisam Engine Me Set Image is equal to what Anagen colon 1.7.3 Enter Set Image So the command for its rollout is Let's take a look at the kubectl roll out. Command again you get everything directly here will go set image kubectl Set Image is a command which is very easy to use.

Cube seattle set image whatever your image is Whatever is your deployment, take what is your Take the image, now this Nginx Equ. 1 has been paid what is it so what This is the command kubectl set image kubectl set image whose mine [music] deployment Set the image of the deployed organ In which namespace should I put that? Of your Nginx container in the namespace What is the name of my engine container? What image do you have to set 1.7.3 Enter deployed image Updated if you get pods can you see running error e pull means your You updated the image, there must be some error in it.

Image pull back of error is coming, which means you are bringing the image of Nginx, she is messing around Is 1.7.3 The image is messed up so here it is. It should go, no problem, so he is saying The image is messed up, no problem. People come with another image kubectl set image Let's take a look at Engine 1.7.1 and see the updated image. Get pot now Running Image bridge let's de back this that this Why is this happening? We kept it close If it is right then you should go I am getting an image pull error, why is it happening?

Set this image to latest What problem are you facing with your engine? Next to the container name Nginx Key 1.7.3 What I was giving was the image I had to give. 1.2 7.3 which is not an image at all The container name of the engine is equal to the engine key image colon whatever your tag is enter Image updated now you see get pods now.

[Kubernetes] Deployments, ReplicaSets and rollouts / Pods (part 3)

Container creating but when this container Was it being created or was an error coming from LETS? Still we had containers running that It is known as rolling updates, this means a lot. It is a good example that rolling updates You are doing it and while doing it, an error also came. Some pods are getting sequential updates and if you do get pods now all How cool it is Similarly if I will set the image from let's 126dt I have updated the old image.

If done then image updated get pods did something The pods are still running and the rest The containers are being made and the rest are ready. I am not going to run, I will get the pods Now some are terminating and becoming containers. Still some are moving and now all are moving If they are there then they will never be able to access all your containers. I don't switch them off all at once and keep some on. Interesting, back to the latest image.

I am Kubernetes how you deploy You can create multiple pods You can make reps and keep rolling those reps. Setting the image by updating and setting the name You can set anything you want to do, let's see. Let us tell you where we are in the syllabus. Where have we reached, if we talk about the core concept then We have made monolith microservices I have done the architecture of Kubb Nits I've set it up, I've seen the cube seattle. I have

seen the pot, I have seen the namespaces See also Labels and Selector which is Anosh Deployment in workloads ReplicaSet Greedy About Reka Sets talk do so kubectl delete -f Deployed, now delete your plan.

It remains set if you do CP copy Diploma in Computer Science Email is yours Deployed Ban Toi M Reka Set Same as it is, just what instead of deployed putting you ReplicaSet Of. What do you need to do instead of using the set and engine deployment? Jana Ajax ReplicaSet Sets namespace Age X ReplicaSet Give Me Should match label Angel Depp replace pod? You have two containers, so whatever you have Deployment remains the same as your ReplicaSet side The only difference remains You can use the Rolling Updates role in deployments.

[Kubernetes] Deployments, ReplicaSets and rollouts / Pods (part 4)

You can do all this back roll out replica If I can't do it then escape column double enter kubectl - Apply -F ReplicaSet Set of Set Enter ReplicaSet. If there are two created tokens and two ready ones, Any too many in the deployment ReplicaSet set There is no difference, the only difference is that You can roll out rolling updates to the deployed. You can use Deploy R for updates. More Flexible End Thatchi which deployment use that ReplicaSet But deploy is a little more advanced.

How to manage that rep If your system is updating If the application is being updated then all reps No, updates should be done little by little. That rolling update in the rolling update here You don't get it, I understood it simply, it's cool. ReplicaSet's set is clear now, ReplicaSet set is It's done for us, you Tell me StatefulSet, tell me stateful We will set it a little further when we. If you read storage then it will be with storage Will you read what D mind sets are, D?

Someone gets food, it's by the way D The idea is to set these pods and cubes. CTL Gate Pods in the namespace Nginx Ye These are pods, who got them, who got them? Meaning you have three workers, worker one worker two worker three or worker two Worker Three to Cube Seattle Get Pod Hyn O Wide this pods are assigned to which worker nodes. If so, this pod is worker one. It was assigned and this rep is a worker The poor worker assigned to Two is assigned to Three.

D Man Sets insure that as many If you have a node then at least one pot should be run. Let's create the D mind set so kubectl Delete the High Frequency ReplicaSet Set are and a d dae m o n set d wa m l create a d mind set dmn set. The method of preparation is the same as your Its method remains Kika ReplicaSet because Everything remains the same, so even if I For example, I might have said this Daala ReplicaSet Set StatefulSet Deploy.

Have DAEMOON sets all the same work Your rap controller does everything Let's see what happens with But D Mind Set.

[Kubernetes] Pods / Deployments, ReplicaSets and rollouts

It is important that one ReplicaSet is running on each node. The name of the set is what work is, the more you Say Rep, make that much Rep, State Full what happens to the set of work ReplicaSet ho w a state maintain with himself. Let's try what is the work of deployment Rolling updates as much as possible Scaling, auto scaling, auto healing, all that. If you keep doing it, it will continue to do its job,? How to make this Simply copy it Low ReplicaSet sets were the same as D Man sets It just happens that the kind changes.

If a ReplicaSet moves on the pod, you will find the ReplicaSet in it. There is no need to insert selector match Labels are needed because they are needed. Metadata in template The Energy D Mind Pod Label is on the Energy Pack If you understand once it is over Let's say the name key and value are in the yal file. Kind Joe Kind Joe Resource You Have to Create Its name is API version which API version is used You have to enter its name in the meta data which is your.

Namespace is what you want to give your name to. The specifications that he works in If you create a pod in the specification template The template of the pod remains and that's it. I have to press Escape Colon Dull Key Enter Cube CTL Apply - FD Man Sets Enter D Man Sets Cutted kubectl get pods in the namespace Engine, guess now, we have three nodes. Three pods were also made anywhere I didn't say how many reps I wanted. He said that D Man Set has to be made, so D Man Set made One worker node per worker node But one pod worker to pay one pod worker three one.

Pod O Saheb, this is such a correct concept that With D Mind Set you insure that all One pot on the notes should be made at least one. You make him do more reps and make him do more You can, it's cool how sorted the cube is. There's a lot to do Dem set became ReplicaSet set Let's talk about Ban Gaye Jobs and Cron Jobs Let's talk about jobs and cron What are jobs?

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / Namespaces

What is a job, MNC or startup? I'm not talking about the job I jobs mean you a container To perform a single task from a container To get a single task done means not you want that thing to run as a server. There are plenty of courses going on, you want that there should be only one job, it should run and be finished Correct so container has a task that Task is a job, it runs and. Be something like taking a backup and patching You want your system updated?

Have a job then you create yes one job let's create jobs right now So creating jobs is not very difficult. What you have to do is v job dot job write dod y now that You are making it, it will either run single or Then it will run in parallel batches from let's. You have a task to take a backup, either you Run it in single thread and take backup Finish or you can do this task in parallel Let two or three parties break down the same task.

You can do this also, that is why Joe Kind What will be the kind of job he will have? The job will remain and its API version will remain. What will remain will remain batch because batch Processing runs in parallel processing If the batch is v1 then this batch will be v1. In the same meta data meta data you name You can write Nginx job or whatever you want. Have to do a job, let's print some of your You have to get it done and that's all you have to do Engine Job, please write any demo job of yours.

You have written the demo job, now you will get it from Lets. If you want to run it in namespace agency then you can also run it outside in the namespace engine. If you want to drive it, you can drive it outside, it is important. This is the specification of how much you are interested in this job. If you want to get this thing completed then complete I want to run this job only once So one completion is you single running this job Do you want to run it parallelly?

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / Pods

What would I put in parallel parallelism I want to play single if I do two here If I give it to you, they will become two pods and do the same work. I want to run only one then which The template I want to use has the same What is the same meta data as my name? Demo job pod template where came meaning You are making a pod, you should understand it like this and move ahead with it. Labels All you need is whatever your I have to put it like P batch job batch job Something like this or write batch task, batch The task is done, now use your template.

Metadata is done, write its spec that which What are the containers to be made and what are their names? Whatever will remain yours put the batch container nach re batch means either one will do the work or one There will be a lot of people working in batches. Many pods will work now this image Whatever you use, it is an image that is busy. A box is a container that serves as a Only for running commands like how an engine works To surf websites MySQL work remains busy with the database.

The box's job is just to run commands. This is its image if you want it Take the latest now I have the command in it If you want to run it, busybox needs a command. If so, what command do I have to run in the shell? I want to run a command in the shell. Whatever will happen, it will echo saying, echo, that's all and run this command after And sleep just this much I will tell you and Sleep after 10 seconds, that's it.

I just have to do the work and that's one more thing. Restart policy remains restart Policy Restart Policy means if this work expires or this container fails You don't want to restart it. Policy Never Restart It what I want is that one of my Work is to print the echo, this is my only small work. If you do it then escape colon double k enter kubectl get no apply -f job dal Enter what is saying Rita policy ayo ale If I spoke stammeringly then I would call Ri.

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / Pods (part 2)

I have written that now Lita should not be taken away. If you give me leave then I will go to work and this Rita I'll write restart escape colon double k Enter and here I apply cube seeder Job Enter Job Demo Job createdted. Jobs in the namespace Engine Ecus can you see Why is the demo job running after 10 seconds? There is a timer in it, so cube CTL Get Pod in the namespace Engine Demo Job Status Completed Meaning, he ran the 10-second echo.

I had to print and print If you did it and it got over then it means that the study will also end. State remains in the container Creating then running then terminating Then after completing it, you will get an interview like this. Telling Completed Running Terminated Image Pull back all the errors that came up Tell me what is in it, it is completed. Cube CTL logs Pod / Demo Job Something In The namespace I have also given the engine log. Who Whatever logs he had, whatever logs came in Echo You can see all of that from the kubectl log.

I can run that job again. I applied for the job and went back and locked it. Started walking Getting it, but now that was one job, that he The work is done, now you have to delete the kubectl delete -f delete job You will have to recreate it and then you will see. Container Creation Now Get Pods Running Meaning that 10 second gate is running right now. Logs get logs man r kt He has become 10 Second so job it's very easy What is your container in the job some task You should do that task, it is your job.

Once done, your container will start running. Your Pod will not stay running What will happen after your pod is running? Pod completed will be Easy kubectl Delete - The job is fine, what did you teach me, namespace P D Man Set ReplicaSet Deployment Job and Similarly, you want this task to be any particular schedule. This is called a cron job What is cron? Cron means following a particular schedule So cron job means a task that you can Create a schedule that will keep him studying.

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / Pods (part 3)

It will keep happening, it will keep going on, but it will be complete only then He would have told me when that schedule was completed. Let's teach it to you, so for that What you need to do first is a V Im Kr You will have to make the job dull in which again your Kind will be that crone. That minute backup so I take backup every minute I want something like this, I'm crazy I am going to create a job, its namespace Engine namespace Engine will now be important.

What is spec, which spec is important What do you need to put in posting schedule I will have to make a schedule, this is a schedule How to enter schedule. You have to eat cron and for cron You will need to learn which task you will be assigned. When to do it, let's say my task is backup and I have to do this backup every day every minute every hour every day of the. I will give you my schedule is done, this is my Task every other minute Will work and what is the task that job template You will have to give me the specifications as well.

You will have to provide that template in that job. The name of the pod in the metadata of the pod that will run anything will remain You can add whatever you want to add in minute backup. Labels Don't forget the labels and put anything Enter the backup or enter the same name. Minutes Backup anything you want to do You can also put production, you can put anything Yes, this is done, now your container Look at the specifications, this is the spec above.

Isn't that the spec of your job template? One single task job What is the template for that single task? Template and What is your cron job? You can add that job template to Spec. What I'll do in spec is this my pod If it remains, what does it matter? I will remain mine, my name will remain minute Backup containers The container that will use the image is the same as busybox. Because I need to run the command busybox will use a container named busybox Now if you want to write the command in a single line.

[Kubernetes] Pods

If you want to give it then you can do so in square brackets. If you want to give it again, you can use it like this First command h then -c then you what to do, what is this, this The arrow that remains after this arrow, you want. Keep entering cool commands line by line. You don't need to put hype It'll be kind of a multi-line thing, so what I want to do first is I want to get an echo backup started back.

I want to get this eco started up Then once this is over, I will do it again. I want to be a I want to create a directory Ansh you who Whether you can do that, let's say I'm a I want to create a directory named will be Backups DirectoryEntry. D I R -P Data One Data Name Another I want to create a directory, and this I want to backup whatever data I have. I am telling you that you can write the task inside and What do you want to do and get it over with?

You want to write echo backup Completed and your backup is complete If it happens then this is what you want, now this is what you will need data to do this. When you stack a container or If you give it then no volume is required for it You will see the volume mounts. I will also explain in depth but what is volume There is basically a unit which You can create a folder to store it temporarily. If it gives then name data Volume and path or whatever you want The mount path is where you put a slash.

A Volume A folder will be created named Demo Data. A folder will be created called Backup. What will be the volume whose mount path Slash backups done and what did you do with it With this you basically did this demo. The store will remain in volume, this is done and this After doing this, you enter its restart policy Two Restart What will I do on failure of the policy? If this pod of mine fails, restart it or else don't do it and this You have to mount the

volumes,.

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / Pods

You have to give volumes too. First explain to me what volumes are. Give it to me then you will understand, understand me in such a way that This is your pod There's a container running inside it called this pod. There is a container inside and this is your host device this container and this device this between I want to parse the data So that there is no data loss, that is why you. You create volume and that volume If you mount it on Y then it just Why do you have to do this because I told you If you give this Make a Directory demo.

I just did backup and just echo And if printed demos had been given then this would also have happened There is no need to do it, it is done and here. You put in the volume whatever your volume is Name of the data Enter the volume and your path. Host Path What will be in the host path that The path will remain yours slash demo data and whatever you type what will it be I will also do it for directory backup.

This is an actually working example backup The volume is backups that's done this actually A working example is that you have two folders Backup and demo data will be created and whatever demo. There will be things in the data if e then that Backups will be backed up and backups If it comes complete and written like this then it will be your every. It will run every minute, let's try it and see how it works every minute. Minutely Backup Started Backup Completed Backup started Backup is completing No Well, I've made it a minute every second.

Every minute so that you give me He doesn't have to wait for a minute, he has to wait for a second or a minute. Removed and done every minute so now My cron job is to copy files from one folder to another. To backup a folder, run the cube. Ctl apply -f cron job enter error what are you saying There is error from server creating cron job v1 Cannot Be Handled Strictly Decoding Error spec unknown in job template spec. Meaning within the field spec d job template To your spec Job Template to your spec inside template Labels it didn't get him so VIM Labels that come in the template in a cron job.

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs

It remains inside the meta data, so put it inside Escape Colon Double Enter Done, now I will make a cube here. Seattle apply -f cron job enter then He is giving me trouble, what is he giving me? Unknown Field Spec Job Template Spec speck job telet scud speck to mean vi i Spec job template in M cron job Speck dot speck it went wrong speck Meaning your job template container so this. If it comes inside the template then this template right, this will come inside it and all this will also come inside If it starts coming then what generally happens.

You can directly access the official website of Kubernetes. That remains the best part, another thing that I would like to tell you that your certified Kubernetes NITS Administrator exam is held In that too you need Kubernetes. If you are allowed to read the documentation, now What I have done is cron in spec The job specifications include a The job template is that job template Everything should be included in the specifications. I am working on the Spec Job Template in Spec. Volumes of containers then containers Volumes should not be I am cron Jobs that mount volumes to containers and the volume is yours independently It should be so this will come out for you.

I'll give you all this code on GitHub by the way. I will give it to you so that I can correct any errors that I get. The required value is valid values on If there is failure or on failure then what else did I put? I'm Cron Job Containers restart policy to restart The policy is for containers that are restarted Policy for that template on failure has happened. CubeSeatleApplyFinally your cron job, you have created this whole thing. Wrote a cron job and wrote so much in it How much is it, you are just a small part of it.

Google2 cron job schedule you put the schedule job template you job You put the template in the spec Put the template in the spec container the same But now I will tell you what I wrote. A little backup and all that was teaching So I also put the volume down. I know you should not have put it but learn it.

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / kubectl commands

If you don't want to back up directly I want to copy paste things, If you want to put it then even a simple small one will do. Will go it's done this volume you It will be very useful in the future, so I have just given you If you have told me, then your cron job is created. kubectl Get Cron Job in the namespace Nginx so can you see minute backup Schedule Time Zone Fall Suspend Falls Now The last schedule has not been suspended till 38 seconds ago, that means schedule.

Yours is over Backup kubectl Get Pods in the Name Space Nginx can you see the minute backup is for today It has gone, meaning it is the same as it was for a minute. It's gone, now whatever is coming next minute should also happen can you see 66 sec 6 Now you get a backup every minute. The message of complete comes, how cool it will be Isn't it true that the first one is complete? The second one went to check its logs now.

Copy this kubectl logs pod / A In The namespace ajax to backup started backup Completed so how sorted it is now every minute If you scheduled it, as soon as a minute. It will end with the creation of another container. 110 is 110, which means 60 seconds and 50 seconds. Seconds means another one in the next 10 seconds The container will be created and it will be periodically backed up. Will take Boom, another container was completed and he backed up how crazy that you If you are giving any thing as a task then it is one way.

If you have to give it only once then make a job if Whatever you do once, you have to do it daily. To schedule is to schedule again and again So create a cron job just to give you the complex What is a part, the structure of the ball You have to remember, but, what. Questions asked in the interview and its structure Definition and use in interview The case asks you to remember this or that Do not ask to You won't have to remember this will ever be asked.

You should also see this, it was, I enjoyed it.

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / Pods

How to create a cron job in Kubernetes You have also learned something from some website. I have to bring it again and again every minute on cron Create jobs and bring them in a scalable way kubectl Delete, look, I will read the gate and make it five times. It must have been done, not made good every next minute will make whatever its next minute is whenever. When will the next minute be over whenever it's over then it will be his next minute so cube saty delete h f can Also create a cron job with job.in backup.

I understand that in Kubernetes you I am giving all the examples one by one. That you can get jobs like cron jobs I would like to know all this by the way I am using cron jobs I have a little bit about storage and volumes. If you have already talked about it then please Talking about storage volume So let's talk about storage. Let's do networking, we'll see after that I would drink water Buddy Let's go back to when we created the cron job.

When I created the cron job, these volumes were And volume mount, I did something. Understand what this is, that whenever You create a pod Whenever you create a pod, that pod becomes a server. Would have been running on Yes correct, that pod would be running on a server. Or the right one would be running on one node. This pod is now running on the node This pod can also be deleted by you at any time. Anything can be destroyed at any time It may have been deleted or destroyed.

Anything can happen inside this pod. The data that The data inside this spot will be will be What about the data that is inside the spot? What will you do with it if it gets deleted? For example, you are here deployment P kubectl apply hif f deployed and you kubectl get Pod in the namespace Engine. Four seconds ago there was a pod called P v7 It happened automatically because reps control The controller says that he has said so many rapes. That's how many reps you need for an auto healing container It crashed and he rebuilt it but this What happens to the data of the container that is deleted?

[Kubernetes] Persistent volumes and storage / Pods

The data is also gone so you will need to restore your containers. Data has to be parsed to retrieve your pods data and to process that data Your Let's say with the host that your host is the host means your The server shares that data with your host. It's bound up with the pod, be it darned? If possible, you will need to host this data here. With persist so that the data If you go then you have to get it done how will you do that to that thing For this you need persistent Volume and Percent.

Volume claim requires both these things Both these things are required, please understand. I am having both these things, is this percentage Volume and Persistent Volume Claims Is it so like this Understand this is your host machine this is your host machine is e2 docker whichever is your host machine. It has some storage inside it from the let's 30GB of storage is taken, right 30GB I have taken storage out of these 30GB I want to allocate some sections. Of these 30gb from let's 1gb means gigabyte so allocate in 1gb If I want to do this then what should I do with 1GB?

Created persistent volume created a volume I made it and told him that, my host Take the volume of one tongue from the machine, come on and If you make it then I said I am Went. Persistent went to the volume dwa ml and I

One Kind Perseid Volume B created in which API version is yours v1 API version v1 I made it now its spec sorry Meta Data. I told you that the name is mine local persistent volume local pv its which Labels The labels reside inside the meta date.

I took a local name, that one Now I put a speck in it Specification that this persistent This is the volume, which is the persistent volume. What should be its capacity, so I put it in What should be the capacity, so whatever capacity They Depends on storage so I want If 1GB is equal to 1000 then one. Jeeb I want what should be the access mode access Mode access nodes not modes that I read I want to get it done, I want to get it right or.

[Kubernetes] Persistent volumes and storage

What I want to do is I read write once, I want to get it read right once. This is what I want, this is what I Want Now I Want persistent volume I should retain that if my I also deleted this persistent volume. This is my volume here. If I retain it on the host machine write it here persistent Volume Reclaim Policy persistent Volume Reclaim Policy Written so I want my Joe persistent Volume that I allocated a jib storage What should be retained?

Class Storage What is Class Storage Class names remain e.g. LocalStorage One remains local storage local storage Meaning the local of your host machine Storage Similarly, you can also use cloud storage You can also use network storage. You can do different ways of storing data Just like we save data on hard disk. Let's save it in the pen drive Storage is the same as saving to DVD class name means local storage you can use your local If you are storing data on the host machine then this.

Public is the storage class name and that is why you will also have to provide the path of the host. I will tell you in one path that my sm If there is a path named AT SData then this I have a path where I live allocat h toy / a at /. I am allocating 1 GB in the name of data. Kubernetes is a pre-requisite of Docker contains Docker volumes Pre-requisite and I m persistent volume host path in p Capital remains The P of path remains capital P.

If so, by doing this I apply persistent VolumeCreatedDocumentsFragment Get Pv Get persistent Volume To I will note that I have a local PV whose The capacity is 1 GB and its access mode is. Has read right once reclaim policy is written that it stops status available Meaning, this is a jib of the host machine, it is a tongue Of persistent Volume remains it. It is available, it is made, it is available now. What do you need to do to determine the available volume?

If it is mine then I will claim it only then It could be mine if I put any here. He will come and take you away similarly to your Persistent one GB from the host machine volume has been made just made and that It is available, now you have to claim it also.

[Kubernetes] Persistent volumes and storage / kubectl commands

If I have to, I am percent Volume claim Dawa A now you claim that If you will, let's claim it now. How will you do it again Kind persistent volume Claim Now the API version v1 API version v1 Meta What is the data name? Persistent Volume Claim Local PVC Now this claim is just a claim that I am claiming now this claim which He will also do it so that it becomes a tongue. How much have I been in one life, how much have I been If I want to take it then I can take 1GB of storage.

I want to lift the entire tongue and whatever is your storage class name What should he be? The same local Storage, what do you understand about local storage? Escape call WQ is not coming yet In kubectl get ns namespaces Go and see, at the end, local path storage Know the local path storage in your Docker Binds together with containers to form a storage. If it is there then the storage class name is local Storage is fine Is persistent about volume claims again?

Kind is my persistent Volume claim means this is my percentage I am preparing a claim for the volume. I am preparing the claim, that claim What does it say that I request 1 GB? I am requesting 1 GB Request from whom To whom will I make my request? One GB request from persistent Volume I am doing So right now I have the kubectl get PV how much is available with me 1 GB available Listen again very carefully, very carefully. From the host machine you have 1GB out of 3GB I have got it removed, I am asking you to get it removed.

If you have persistent volume then 1gb niklaav available Now if I have to claim it then what should I do? Cube CTL Apply high f persistent volume claim which 1GB is available, I requested it. Trying to claim 1gb Enter Now what is he saying about resources? Persistent volume if misspelled I'll go into claims and resources SO y R K went The resource is fine Resources, I'll apply it now. Persistent Volume Claim CreatedDocumentsFragment Done The one GV that was available is now available He is no longer bound, why?

[Kubernetes] Persistent volumes and storage / Pods

Because persistent volume Claims gave him claimed it So if I do the gate PVC then my Near Local PVC Cha Iz Bound to local storage got a jib He needs to have my data first, right now. This is my pod, this poor little pod Its data was getting deleted. What I want is this my pod in the host machine's mnt/data save your head Keep your data safe, so what have I done? Created a 1 GB persistent volume One of that 1 GB persistent volumes I made a claim, now this claim I'll give it to Spot.

How will I give it, what I will do what I was deployed Which cube was my deployment? CTL Gate Deployed in the namespace nks this It is deployed, its data keeps getting deleted So I'm going to delete this cube. CTL Delete between deployment and deployment It got deleted, now I have such a deployed Banani How can I create one whose data cannot be deleted? Sorry, how will I make it in the container that is there What should I call the container?

I will tell you Volume Mount the volume now inside this container The HTML file of the engine which The HTML file resides and resides on the mount path whose path is slash ver Dad, Dad. Html so I want r dad The path of the HTML should be mounted by whom? With My claim with my local PVC a GB of mount with him It's now what's mine persistent and that's my volume If the volumes remain outside my container Volume Ay, that's my volume name.

What is the name Local TV persistent this is my volume Whatever you put in BVC, now its persistent Volume Claim What is the name of the claim, local PVC? This local PVC is the name of its claim. You can name the volume anything. I feel like you can put PVC in me good I think I put it, it means What is this container that is the container of the engine? Its path, also dubbed HTML, is a The volume will remain mounted with its name.

You can put anything like my volume Two my Add the volume you like and Create a volume named My Volume and that's my volume of that claim Name it the one that has 1 GB.

[Kubernetes] Persistent volumes and storage / Namespaces

I mean, these lines of it are not If it gets too much, reduce it a little. I created a 1 GB persistent volume What was available was what I did as a persistent Volume claim which was named local PVC I requested this claim from I have bound it and this is PVC. How many GB does PVC have? One GB is one Where is the tongue named MNT/Data I'll tell pod if there's a jibe on the folder Create a volume and name it My The volume and the data it contains.

Path de do let / ddd So what I did was change the container volume. This path is persic at volume in mount Binder with claim given escape card kubectl apply -f What deployment Enter deployment Created in the namespace Engine pods have arrived, pending now. It is pending If it is pending, it means that it is There must be some problem with the path and all this. If something like this has happened to you, then whenever it happens to you, Try describing kubectl Describe Pod / Whatever the Pod's Name Is In Space Age And what is he saying, zero out of four nodes?

Are available for persistent volume local PVC Not Found Why was the local PVC not found? Your persistent volume Its namespace was the default name. If you have space, you can use it for Angie Nicks. If you have to run it together then you have to enter namespace will have agency And similarly your Joe persistent Volume Claim You will have to enter the namespace for that also. Engine Ex and then this error of yours is local PVC Not Found error will be resolved kubectl Apply or what to do first Get PV End Get both PVCs, get PVs and this.

Delete the local persistent volume Do it kubectl delete pv c / local PVC whatever my persistent volume and which was my persistent At volume claim was the default namespace I had it, I removed the default name. I removed the space one, now what should I make? kubectl Apply -f which is in the Nginx namespace Volume Data Email Engine namespace If created, kubectl get persistent Volume in the Engine Ecce Name Space now it's available now I can use it persistent volume claim de.

[Kubernetes] Pods / Kubernetes architecture (control plane, nodes)

If I do PV then I am bound Now what I will do is kubectl get Pods in the namespace what engine exes Running understood that you I every I am also seeing him introducing error on stage. As if tomorrow you will get volume creation If there is some error in it or there is an error in the pod, you can resolve it. If you want to de-buck it and learn to use it now This is my pods running means cube CTL describe this one which is pod this in the namespace.

Engine so he didn't get PVC at first Then he rescheduled it again PVC got Engine Local PVC got And he rescheduled it. In the ver data inside the app The ATML that I have is my MNT S How to get it in the data, let me show you the kubectl get pods in the namespace Nginx do. Han o wid now see that your So which worker node is this one running on? This is running on a Teads cluster worker and This worker is not a node but a Docker container so if I do cube.

CTL Gate node so I have this y which is my pod this and it is running on a worker node with A worker node is basically a Docker container. If there was no Docker container, you would Directly CD SM AT / data And you would have got it, but he is your host, isn't he? Your e2 instance is the two instances on that The worker node is not inside the Docker container If so, you. Have to do docker ps and find that worker where node this is my worker node now this is that The ID of the container is now I want to create a container.

I'll go into the Docker unit within this container Id bash I container these now I worker node I've been inside and got the LS done and MNT CD M AT Data Mila CD Data End Data. I didn't get anything because then I Nothing was created, perhaps the path if something goes wrong, you feel like that. Why because you have calculated the persistent volume Created a folder through and bound it. This is how it works, when I will take you towards a mega project The end, so there we will see that my SQL data and how you persist it.

[Kubernetes] Kubernetes Services and networking / Pods

You get it done and how do you run it, but Before that, this pod is kubectl Get Pa in the namespace These engine pods are working, but look. If you are not there then how will you show it then a small So I want to tell you something that this It is a pod, well you have its data. I took the binder here, but this pod is for this It is deployed, it protects you from the outside world. To make it accessible to external users It happens and that happens via service proxy then you need to create a service resource I will have to wait, would there be any service?

Whatever your resources are, let's Deployed that deploy if you get external If you want to access the world, you need a It seems like service, understand service in such a way that one. There are too many pods that redirect You grouped them in deployed There are so many deployment, where should I go? At the service, you can see the diagram here. Service many deployed pay you Let me show you how it redirects how am I going to go over here I will serve you Dwa made in a service.

I will give it to you, now it is very easy to create a service. What you have to do is just a Seconds you just Kubernetes Nits Service Just write this much Kubernetes Write down the service and you will get one service. The file will open, now what is the service? If you need to expose your pods To the outside world, you seem to be a service. This means you need labels and selectors It will be installed here also and on which port will you open it?

The API version of your service will be Will remain v1 And your meta data will be the name. Of the engine Service Engine's Service will be your name engine service namespace what is your Agnes now that spec You will have to give a selector there. Selector because you get this service You will have to find out for whom to expose. Whatever the label is, it's engine. Whoever has the label P Angle, I will expose you, simple, where should I expose you?

[Kubernetes] Kubernetes Services and networking / Pods (part 2)

I will expose at the port which at the port I will do protocol tcp protocol. I will do it now, there are two things in this, one port and there is a target port and to port so Port means The outside world, which is outside your service. Which port do you want to connect it to? I want to open the outside world on port number 80 I want to get it connected and the target The port is which of your pod Which is the port you want to connect to?

If yes, then the port number of Nginx is 80. I'm from the outside world If I hit port number 8, it will go in. The pod running on pod 0 within the cluster It will redirect to that, that's all written is now Your type of service is different. Types such as node port cluster IP Load Balancer External IP headless Service what is all this, if I cluster If I change the IP then the value of your cluster. IP address is that one that comes out and A port is assigned within it and You can map with that if you I If I write node port instead of this then your entire.

You will need to remove one port of the node. Lets port between 300 and 32000 I have to find someone and that's you. Here we have to write this by doing node port Anything from 3000 to 32000 can be fine and load balancers for the cloud in general use because you put a load in it. You also attach the balancer and cloud If you use some services If you don't write it then by default the star IP will be Types of service are created and which remain

separate.

You can set any other IP in that external IP. Enter the static IP address and headless Service General Use with State Full Sets What we haven't seen yet So type cluster IP this I will put this I typed cluster ip done and that's it. 80 in the world and inland Port number of the cluster Port number of the pod 80 will map to For Anything That Has P Nginx So if your deployed label has.

[Kubernetes] Kubernetes Services and networking / Namespaces

If it is Nginx then it will be exposed escape Call Duplicate Enter kubectl Apply -f what service dull enter what is coming service catch p v service.ts Because if you want to provide multiple ports. Port s Ports This is done Apply Service CreatedDocumentsFragment Now is the time to access this service on. Your browser So if I go to my browser here is my ip address 65 so its port number 80 so my p is running Why should I enter because it is not working?

Your cluster is a Docker container. Port forwarding of the Docker container Does matter IP address needs to be exposed There is a separate command for that, the command is like this Type is Cube CTL Port forward kubectl port forward now. Which service's service Engine Port forwarding of the service is required. I am kept in namespace Nginx Change the port number 80 to 80 of your system. My IP address If you want to expose me to his address kubectl Port Forward Service Engine Service via an engine access port 8080 Address E to this now when you do this.

It tells you that you need to log in to port number 80. If you don't get permission, you can bypass it. You may not have this issue in your local area. If you are facing TP issue then you What to do, use it and use it High means it will run within an environment He then type this command sudo e kubectl Port Service. Enter what is he saying, address already in Use this address, it is already in use. Please block port number 81 what is in it Open port number 81p, here it is, it started working.

Forwarding from 00 281 So I mean, for my brick instance If you have to expose port number 81 I'll go to my instance | Security I will go to IP Address I will go to Edit. I will make a rule to whomever I tell port number 81, please expose it and save it. Expose as soon as I expose and here But I wrote port number 81 and boom Welcome to Nginx so Creating an Engine container from your namespace. Deploy from Pod from Container to service and finally reach you.

[Kubernetes] Kubernetes Services and networking

I hope this is what the journey looks like for you. Take any application, that application You can make this service by taking any application. You will make a service out of that application. You will make it deployment by making its pot. You will make a small namespace and run it. Pick up any application, let's take an Example of B M take notes P So Notes P has a branch DevOps By doing this, this is a development branch. One I have a P, I like its Kubernetes If I want to make it, what will I do?

I will copy Treat It As A Small Mini The project is fine, I'll copy it. This will turn off port forwarding. I'll do a git clone of my I cloned the Notes app. In cd Django notes p I went and git check Out DevOps I came to DevOps Branch P where I have kept all these things, now understand. How does this thing flow, flow like this suggests that you first So a container has to be created and that container's The image is there, it has to come from Docker Hub if you want.

I'll make you a fresh flow First you need to create a Docker image. Let's create the image, so this Docker If you have a file, create an image from it using Docker Build Han t notes A K S dot so I create an image of my. Let me make a cool image of the notes. Become gone This Python 3.9 is a bit older, so That's why there is such a scene, I did this one created the image Make your notes Now this image is created, Docker Images.

Notes This will not work locally, in Kubernetes it If you need to deliver it to Docker Hub, then do it on Docker Hub What do you need to deliver first? You need Docker Hub login required Docker Hub login I need to log in to Docker Hub first. I logged in to Docker Hub. I took it and as soon as I logged in, I got a I want to create a token. I can create a token here. Give access I'll give it to you, I've generated it.

If it is asking for password then it took this token.

[Kubernetes] Deployments, ReplicaSets and rollouts / Namespaces

Is logged in Succedence P Cats if you have to tag me in this Docker image tag which is the name of the old image He and his version and the new image is new. Where should I put the image? Kubernetes will pull the image only if it will be publicly available and can also be done privately. You can, but again you have to give him access. If so, it will go to your Docker Hub Pay & From Here Then to Your Pod, I understand now, from here it is If it goes into the pod, your image will be delivered.

What am I going to do here I am on MKD I will create a folder named IRK8S So CD's A now what I'm going to do here I want to make it and I am. Deployment mel a deployment How is it made again, I will come here You said deploy, here it is mine It's deployed, I'll just copy it. I have already learned how to write a deployment. Are It is so simple, just change how it will happen That I have a deployment, this is an engine of deployment Place Notes P is deployed in labels I.

I will put it on my notes, I will only rape one of them. I want to keep the match in level P, I will put it there Notes P This is my Notes P and match Labels also have notes on the back of the page. I will put it in and name it after the pod Notes and the image that will be used, which image I will tell you that image will be in your Docker. The name of the image you pushed to the hub what is train vishu bam notes p kets so this I will paste the image in my deploy.

This application is on port number 8000 runs done finish escape colon double k Enter finished Done Now I need a namespace You will also have to make notes by typing them. Namespace DWA and its kind here The name will remain Space And the API version will be what v1 and meta What will the data be like? Name Notes A These Notes A Name Key namespace So, Cat deployment in which namespace o namespace did not enter what and I In M Deployed I will put that this which.

[Kubernetes] Kubernetes Services and networking / Namespaces

I have a P that goes into a namespace whose Name will be the namespace note p so in namespace If it works, I have a namespace. I have become deployed, now I will make Service D y a so how do I want in my service That my kind service became Whose API version should be v1 whose API. The version should be v1 and its metadata should be named Notes Fifth note of musical scale Service then what else is there in it notes After the service, your name should be there.

Space Notes App Notes app namespace Yes, and in its specifications Selector what will be anything with pop notes app is a selector now the ports in it what will be the protocol they will use. Your TCP port will be The port number will be 8000 and your The target port of the container The target port will remain 8000 is done now if you want you can type it. You can also provide cluster IP if you don't provide it. By default cluster IP will be saved colon dob enter your deployed name Pay service has been created, so what we have done I thought you had posted an image here first.

Pushed the created image to Docker Hub The pod created is enveloped within the pod deployed. My pod is covered inside the deployment. I made it deployed directly. This will be deployed on service so I have Service was also made So here it is yours Service Done Now let's build one by one By the way, it also has a namespace in the middle. This thing is in a namespace. All these resources will be in one namespace If I am there then what will I do now kubectl apply -f namespace name Space Created Draw Navigator That there is no error while creating the container.

The container is now running, what's the matter? This port number 8000 is for me If you want to expose the kubectl port Forward Service Notes Service of P which is in namespace notes P. Its port number is 8000, map it to 8000 with and his IP Expose the address You need to spell enter service correctly. If you have to write Service Done Forwarding Port 8000 Now you Go to your security group and edit it.

[Kubernetes] Kubernetes Services and networking / Pods

In these bound rules and here the rule tax You have to give port number 8000, add it and Save it with Anyway IPV F. You If you go to your IP address and port If you go to number 8000, you will get one Notes on Snapping Bhadakta Cha is Running on Kubernetes so I hope you understand that. How is the journey, it was said that it is mine Diagram a journey to learn how to In different namespaces you can have different Showing the journey of the application and the user.

You are getting access to and it's the Beauty of Kubernetes So Your Kubernetes I also see Kubernetes's projects side by side. I am going while getting covered so services is also covered Now imagine if I drink / so these p open if I will / Nginx So if I open Nginx, I will do this like this. I want it to open like this, it should open like this I redirect things I should also redirect the services, meaning like this Understand that my A service nearby.

I have a service from Let's is an engine There is another service of I have a service from Nginx and another one Service is let's say notes of. One is for service engine and one is for service notes of P I want / whenever I say Nginx then go to this service I will say when. I would say / Notes, so on this service. How will this happen, then for this you need one thing You have to use what it says Ingress so what is Ingress routing your traffic Reroute traffic to your cluster.

He does it and this is what we call it Ingress so if you have a high level of it If you want to see the diagram, this is your cluster. This is your cluster within this cluster Your pods are inside this cluster. There are pods, which are many, right here. There are a lot of pods, this is your pods. Lots and this Pods are being managed by a deploy This pod is being managed by a deploy from And to access this deploy If you need a service, you.

A service has also been created, and with that So you have another deployment and It has more pods And that's how it's managed for him too.

[Kubernetes] Kubernetes Services and networking / Namespaces

You have created a service, now this service So access this It will take this service and it will deploy it will access but if you have multiple Microservices architecture for services What happens to such microservices? If you want to manage it, how will you route it there? You will need an ingress. Ingress is a service Basically it is a resource that allows you to Manage Traffic and Routes So if I Bola / The engine will go this route if I spoke /.

I feel like you will go on this route I want this, how will it happen? All this will have to run in the same namespace Is I have all this in the same namespace If I want to go, let's go, what will I do? I'll make a small edit here and I'm Diplo that this is namespace notes p place I am in the ajax namespace for now run I will give the service an Nginx name as well. I will launch it into space so that my beans Both my engine and notes page in the namespace If I can run the cube then CTL Gate Deployed in the namespace notes p so this.

This is deployment, please delete it for me. If you have to, kubectl delete Deployed Notes App in the namespace A Note So it got deleted Similarly, Notes P's service I deleted that too, why should I delete it? I am doing this because namespace is common to me. Keep If so, I will delete the service as well. Similarlly kubectl delete namespace I also want to change the namespace of Notes Page to Notes Page. I'll delete it, If it goes away then my space on Notes P will also be deleted.

What do I do with kubectl? Apply -F You can deploy the service Deployed kubectl Get Pd In The Name Space Nginx so I have engine key Deployment is also Notes P's Deployment is also kubectl get Service I also have engine service notes. If there is service also then make the diagram which It became mine so it Nginx There is also a service and my notes are The service has also been made, let's start it now. For this you need a The ingress controller holds the ingress Looks like a controller that works at your cluster level But it helps in routing.

[Kubernetes] Ingress and ingress controllers / Kubernetes Services and networking

The ingress controller remains Nginx's and also But Nginx's remains ingress Where can I get this control, if you Search for Nginx Ingress controller Kubernetes so. It would do if an open source community created a An Ingres named - Nginx Created a controller that redirects you End to end load balancing in reverse proxy It helps me, the only difference is that you. This which I want to create an ingress controller for Engine. If you want to make it for Kind Cluster then you will get it What should you do if you need to take care of Kind will depend on the cluster documentation.

Go then click on ingress now ingress Expose Your HTTP and HTTPS Routes from outside the cluster to your Services within the cluster that we have here You are trying to do that for. What will have to be done to create a cluster You will have to do port and map again. You e The files of the controller are the email files. If you have to apply that then kubectl Apply -f Kind of Ingress Deployed If I have to, I'll come right now.

LS Ingress Controller's Boom Phat All done, now I will do kubectl Get AS namespace I'll see you'll see that a namespace called Ingress and Nix. It will be created and you will do kubectl in it. Get pods in the namespace Ingress Engine then The services with ingress controller are one is in a different namespace then it is yours It is complete and running, everything is fine What does completed mean? That's done and this is the pod that's

running. Running Done similarly to how you got the pods Similarly, if you want to get the service also The ingress controller has a service loaded This service is running under the name Balancer.

I will go to my engine and create a new file. Ingress Dawa I have a file named MA Ingress YAML I will make a small work in which I will that I will write an API version, this Where can I get the API version? You should go to your documentation and find your Ingress version will be available, Chach is known as Networking Catered i1 so this comes under networking so This is the kind of it that is.

[Kubernetes] Kubernetes Services and networking / Ingress and ingress controllers

Ingress meta data is the name Ingres Agnes Notes of Nginx and Notes you This is the ingress of this namespace. That's the engine namespace, this is intentional. I did it so that all the services are mine Created within the same namespace are Enosh this notation because its It is also kept in the documentation but if you. My paths are gonna be path / Nginx so whenever slash enz Nginx come whenever If you come across / and Nicks, enter a path type as well.

Two Prefix to Path Angle Nix path type prefix whenever this path occurs So which service is on the back end? Redirect to the service on the back end The name of the service you want to redirect to what is the name of that service Engine service what is the port of that service. Service name engine Service to name engine The service and the port of the service are port number 80 is the service on port number 80 Now she is similarly giving me another.

Path type will have to be created pre Fix whose path I will put / app and The service in the back end is called Notes on service whose port is 8000 whose port is 8000. If it gets redirected there then whenever I where did you say My port number is the number after the good port It happens Port number of port 8000 After number 8, this is whenever my engine. The port number of the service should be reachable only when I put the engine like this, if you see the diagram But look, if I put an engine in, that engine.

It will go to the service which is port number 80 here. Whenever he slashes he'll Note: The service will be on port number 8000. I want something like this, it's mapped on me If I want, let's do what I want. Show me then Ingress becomes escape colon w k enter kubectl apply -f ingress enter Error validating ingress ml7 If you choose to ignore this error, turn No validation off, we will write it correctly. How to make kubectl Get All in the namespace Annex So what's going on with me I've got my nuts going on Deploy is running Deploy of Ingenious service is running.

[Kubernetes] Ingress and ingress controllers / Kubernetes Services and networking

My ingress is also running here now. A little tricky part comes This is an ingress. Running through a controller that There is a controller called Ingress Nginx which Kind cluster hum log ne install You have to expose him Let's do it how do you do it so kubectl. GetService in the namespace IngressEngine Now this ingress controller You have to expose your. kubectl Port Forward Ingress controller, which Service / Ingress Engine Controller service that names space ingress Its port number 80 comes to me in the engine.

Mapped to 80K port already used is its port number is 80 we People have to be mapped with 80 and then what have to do Yes, then I His. Address 0.0 I have a binder, I have to do this now. It says permission denied so sudo han e Let's see how I can improve my Ingress Forwarding the controller's port He is saying that, he is already in the address. If you use it, you will have to use some other port. Let's change the port number to 808 I'll map the number to 808, let's go.

If installed then your ingress on port number 808 The controller is working, where did you go? Go here and go to the security group. I will go to Edit in Bound Rule and change the rule. I will give you port number 8080 What is going on in my ingress controller It's going on Meaning this port The number is 8080, what's going on beyond that? Engine Ingress Controller not found running what's going on with his slash Nginx now have you ever found / a pay what's going on can you see this on the Slash app?

Notes page found, can you see page not Found something like this, something came, how cool This is why this page is not found. Because this is what it is Was Slash these, as many people as Django They will know that it is your app only. I want this not found, it's coming, let's do one right now working I'll make a small change to it Its ingress file is this one in it I will make a slight change that my. Slash A which is in the jungle when slash When you insert it, the port opens automatically.

[Kubernetes] Ingress and ingress controllers

If the above commands are not visible then There is one small thing I can do that I can write the command in this line or Then I typed all the commands I had written so far. I will give you all the commands from there. Copy paste and write by referring from here If we can, then this is our one. I haven't changed my service yet. Did I change the pass ingress so If I apply Ingress, then kubectl apply h f ingress conferred Now I'm going to run my ingress back.

This is my command kubectl port Forward Ingress Enter to Port Forwarding It's done, now I'll go and see what my pass it is running no my ingress Now can you see my port 8080 when I. I'm doing slash So this is opening my pop so my ingress The controlled pop is opening now what do I care I will / ajx then Nginx should open but ajax not found is coming. Nginx is also a There is something that runs on slash, meaning How can I tell you the concept of slash?

I am going to explain to you how It happens whenever there is an application,. That application is running from let's port Number 80 is your IP address. 13.4 from lats 4.57 something so if here port number 80 If it happens then the port number is 80 Also called generally invisible port. If you have HTTP here, you have HTTP like this and put port number 8 so port number 80 Disappearing, only your IP and a slash remains so if your. If Slash comes then Nginx will run and same story is also with your app so if I slash When I give it, my app is opening my jungle.

The pop is opening but right now I need such a / If I do an X, my engine will open and then its What do I need to do to add a note? It will have to be What happens in Entenken gives you your You will have to go to Ingress and your Ingress Slightly changing the meta data of. If you need it, I will tell you one thing exactly. Information such as Enosh now I'm going to nick my angel I am telling the AJAX controller to I am saying that your ingress is.

[Kubernetes] Ingress and ingress controllers / ConfigMaps and Secrets

He is the one who is Kubernetes Da Aao, who is in that Your slash is what I need to raise. Edge ingress Kubernetes.io now what should i do Rise target only Now which target has to be raised whenever My slash comes whenever I slash it should basically have port number 80 or whatever. Redirect so if this happens at my end If there is slash Nginx then I like this thing Rise Nixus redirects to port number 80 Go to the port number 80 mentioned here.

I want this so escape calculator double k Enter or you can also read more about Anosh What can happen, you can directly write here You can Kubernetes If you can write note ingres well then basically. What happens to the ingress specifications? You can customize it however you want that whenever I write images, something like this happens So here you can find a guide for images. You can write different annotations for different images. Can you get this time out from Ingress? Rise target to slash wherever He will come and write it down, so up and last I will remove the slash end, now I will write this I will do kubectl apply nuff ingress now.

I have updated Ingres and now I am What will I do with my from Ingress? The command was port forward, I ran that. I will give it and it will be running port forwarding is running now I'm going to go into this slash if I Likha Nginx so welcome to Nginx crazy now. If my slash is not there then my app will run. Used to be what a good thing means it was fun that I 8080 did this I put the part slash so my page is opening this I When I inserted the part, my Nginx is opening.

You can avail as many services as you want. I have all that, keep adding and fat This will continue, Ingress is such a crazy thing. Tell me why Ingress is burning Generally no one explains with examples. Everyone tells me there will be traffic ingress Root ella bala real error occurs when ingress occurs I explode, then you understand. That's it, great, now let's move on to Ingress. People confine secret read confine The Secret to How You Can Store Whatever You Want in Your Pod variable or whatever your data is how You can deliver it in a completely secure manner.

[Kubernetes] Namespaces / StatefulSets, DaemonSets and Jobs/CronJobs

I think state full sets left It won't matter, it won't matter, one To read the Confine Map and Secret Let's try a new example now I've played enough, let's turn it off. I've done a lot of crazy things in Engen is like cron job ingress persistent Volume ReplicaSet Set Service Pod Job Name Space Diplo Cube CTL Delete -f Han f namespace So Joe Nginx. There is a namespace, I will delete it. What will happen with this, whatever happens with Enks The resources were all deleted and How crazy it has become that there is only one If you blew up the namespace in the click.

If you go, your ingress will not survive. I'm going to teach you state full set and With StatefulSet I'll teach you And, would this state have been a full set then? I will tell you everything, I will tell you everything comfortably Let's get started on this right away Let's understand first, now look at the kubectl get pods in the namespace Nginx. If there is no engine namespace then there is no pod. I will meet you, everyone is gone now I am here as a new person I will create a folder MKD I R My AL, what do you want to do with MySQL?

Ho Humse MySQL Aisa Lega To, understand that when we are in apps Whenever we work in production So how do I make apps? I will explain it to you very well, when We also have some applications like Django application done or a flask Application done, Flask done or else I'll make it full screen, yes. Django is done, Flask is done and your Spring boot is done Spring boot is done All these applications are gone, so all these How are the applications if your state is maintained?

Even if it doesn't happen, it still works because UI is user can use anytime He stopped anything could happen so this basically StatelessWidget DB Done my SQL done, MongoDB done, all these. Those databases are stateful applications are stateful Meaning the data in MySQL and your There should always be Perseid timely in it Transactions means read write, all this. Can't even mess with When It Comes To databases Case and if you have normal applications You have to maintain its state it's fine.

[Kubernetes] Pods / StatefulSets, DaemonSets and Jobs/CronJobs

You can do it, even if you don't want to do it, there is no If there is no problem then it is a small difference So whenever your state less. There are applications that you generally use. Deployment ReplicaSet set ReplicaSet sets dep DaemonSet extra but when your stateful There are applications like MySQL Gone is MongoDB, in that case you can use Do you do StatefulSet so what is a StatefulSet that. The beans belong to their fraternity that in this Reps are also maintained, in this also rep The system remains the same, but what is the benefit in it?

It remains that if you make five reps If five reptiles are made then these are the pods that are formed pod le MySQL aapne five pod I made five reps, so this pod. This pod is made to carry a state with you. Like this one pod is done this pod So this pod's number is up One of this pod Two of this pod Three of this pod Four, whenever a pod is deleted, its. The number of the new pod that comes in its place will also be It will be four or whatever has been deleted There will be a number and its state will be as it.

It gets transferred to another pod So this is what StatefulSet do that the pods in it are sequence I am numbered, I will show you. I'll give it to you, so what will I do now? My SQL so I'm with MySQL I'm going to explain StatefulSet to you very well. It is not very difficult, it is the same calculation. The book is like yours was in Deployed so V I am a MySQL StatefulSet stateful setd gml this is what I do I will give it now, what kind will remain, my kind is.

StatefulSet kind is my StatefulSet Again, this comes down to workload, like Deployed ReplicaSet set to its API version what will remain v1 sorry apps s v1 a1 its The API version will now be its metadata. I will do something new in this, whose name is It will remain, it will remain basically MySQL State full set and its namespace That will remain MySQL, so a new name.

[Kubernetes] Namespaces / ConfigMaps and Secrets

Space should be MySQL first One before this, if I don't forget, one Create a namespace that is named after the null Provides kind namespace and kind API version v1 and meta data name ma sqlservr.exe. I mean this namespace I'll give a spec to the StatefulSet like this I will tell you the name of its service in the spec. Kas so I want three of its pods formed three pods formed now I It has labels and selectors. I will use the concept selectors selector what do I put In this, I will put match levels in the selector what p my al so you can put anything in this ho demo MySQL pa my.

SQL, it's your label, you can put anything You can tell me the important part This is where templates come in. Falls which is Labels, labels, so these are the labels You have put p MySQL that as it is You will put it here also so that this template. I could give you three reps, now what else is there in you have to do these things which are templates. Labels These If it comes in meta data then meta data labels MySQL it's done now your spec in it will come because you have to create the container right.

You have to make the containers now I have to make one Creating a container named MySQL my A container named SQL is to be created containing the image. Which will be used and which will be used my Which SQL version should I take? Take 80 and use the ports in it. The container port will be there in He lives, he lives, the container Port 80 a sorry not 80 it remains 18 engine Its value is 3306 done so your. Your container is created

and everything is done.

The container of MySQL remains Always asks you for environment variables So in environment variables it says that, do anything to get me MySQL route Password one, so environment variable two. This is the environment of MySQL. How to enter the variables and that too Again, I'll tell you, all you have to do is Kubernetes A NetSat Full Set Environment Search the variable like this, directly. Search like this and here StatefulSet Now you will see the full beauty of it in this Now what to do, well There are a lot of things put into it.

[Kubernetes] Persistent volumes and storage / StatefulSets, DaemonSets and Jobs/CronJobs

People have kept the engine running and the engine It does not require environment variables. What do we have to do with that environment? Is of the environment in which MySQL Route Password is MySQL root password and What will be its value? Take the value now this is MySQL route password environment variable and more If you want to add something, add another one. Give you MySQL database if you want can enter database yes Create a database called Value DevOps. Container created name image port whatever Had he put the environment mass root I entered the password and finished.

Made for StatefulSet but what is this says that if you want a SQL If you want to make a state full set then read this Storage required Volume required Storage Volume I told you persistent Volume. The volume will be available in a StatefulSet only. If you have to put it then you have to make it complete there only. I'm going to create a volume of this, like You must have seen the set in the example also. I saw that the volume mount is taken and Name Something is entered Volume Mount Path It is inserted, this is its volume mount.

These volume mounts I picked up from here I'll take it and I'll go over here and put it in. Volume mounts what comes its name comes let's say my I mount a volume named SQL Data I will give it to you and give you the path to it. Mount What will come in Path Mount Path / Ver Sliv / Were Live Ma / Were Live MySQL this will be its path now Volume mounted Its volume mounted Path Done Now Volume Claim Key Templates.

I also need persistent Volume Persistent volume and its persistent volume claim waise Its claim template is also required and this It's just a template, you don't need anything extra. I don't want to make it from here, it's within spec. If it comes then it will be in the line inside the spec The Volume Claim Template will now appear here. A MySQL data, this is what you put in. Whatever your volume name is, that's what you put it in the meta data now its You have to enter the spec.

[Kubernetes] Kubernetes Services and networking / StatefulSets, DaemonSets and Jobs/CronJobs

I'll copy this spec as it is and I need the storage class, otherwise I will I will remove the storage class from here. Because ours is the local storage one. Here's what my volume claim is for. The name is MySQL data which reads and writes once Access to resources that are a single gig then it's my The work is finished, my work is done. I created a StatefulSet that is a ReplicaSet It should also be a service of its pod So vm service I will put dull now its One of the specialities of the service is its service.

The specialty of this is that the headless service is headless Service means headless Service It means that everything will remain the same kind. It will also have a service and its API version It will also be v1 only, there is no scene, its meta data as well The name will be MySQL Service only and the. There is namespace, that too must be MySQL. Yours Joe The cluster will be the cluster IP, that will be none. This means it is a headless service.

The service will not be exposed going forward This is for internal use only, if you have any pod MySQL data then usko service You can connect through the service name. You can connect through but it is external Should not be accessible to the world and thatchi -, the name of this is MySQL. In this you will also get the name of the port for the service. The ports that are to be used port that is 3306 more Target port 3306 and yours Protocol TCP so you have done it now.

What is the purpose of this, that you service is MySQL service this service Only one particular StatefulSet binder with so escape colon w k Enter now I'm going to my state full set. I will tell him that, whatever you have The service name is for this StatefulSet. Service Name: The service name you use Is that what is going to happen, MySQL? Service to MySQL service name The service will be associated with your item. Notice that this is my StatefulSet. Each pod has its own unique structure for each reptile.

I have added my service and kept my purse at Volume claim template also added If so, everything in a StatefulSet is defined in The YAML Manifest Only then the service name as well.

[Kubernetes] Kubernetes Services and networking / kubectl commands

I also added the environment variables. Volume claim templates are also provided. After doing this double click escape column Enter now I am State Full Set I will run it, I will run the service first Because the service name is referenced there. None of them, you can choose any one of the two. If you can run the thing then kubectl apply half f a service you created service one The service whose cluster IP is non headless Service also speaks now Cube Ctl Apply -f State Full Settle Enter.

He is saying decoding error templates O templates do not exist, templates do exist So we're in a state of full set and I'm going Let me see where I put the templates, so this. CTL Apply Online Service Account No Service account no re StatefulSet YAML Enter what is he saying now again Why do I know the volume claim templates? In fact, I even checked here that No other templates, yes here it is. Do you ever remember YAML? You always pick YAML up from here but yaal here the templates only Then why was he making a mistake oh I know why he was making a mistake because this.

Inside and its indentation outside I will understand what I have done I also did that volume claim template. Has reached the indentation level where Have your own template that will clean the volume Template: Yes, now it is correct, now run it. kubectl apply -f state full set enter state full set Created Pods in the namespace My AL So can you see state full set zero is We can see the container creating but name. When people made pot in deployment, its what would the name be engine deployment x y Root P K Something Look at the name here Zero.

Get pods One by One Now Create things are getting attached to him what is crazy what is crazy this is very MySQL state full set zero. Ready State Full Set One Creating Now State full set which third also creating you C Why is the third pending coming up? I have shown it by making a state full set. The advantage is when you do kubectl GetSVC in the namespace MySQL then You will get the service of MySQL.

[Kubernetes] Pods / Kubernetes Services and networking

There will be no external IP. The cluster IP will not remain and you can use it whenever you want. Access this service in your applications How can MySQL 0 dat You can access the service name by doing this So the domain name increases very easily. Creating DNS is also called that I will explain it to you later but your service It has also been made, so if you want, you can avail this service. You can drive it anywhere, but what about me right now?

I will do kubectl get pods in the name I will do space MySQL now you again You will say that, it is because of your photo. That command went out, look, I am here with the photo. Let's take a look at this and I'm state full I opened the set, now this part is You will not see the cube Kubernetes If it is the majority part then it is so and so. I must have overlooked it Not at all, so here comes my pod now.

If I get inside this pod, the cube CTL unit IT what is the name of the pod ma SQL state full set 1 in the namespace my. Nfan Bash Enter I Entered My / Han u root han p what was my root password right now Route i inside MySQL How crazy that I had a state full MySQL's deployment through the set. Di MySQL is going on and that too scalable so now what I will do is show data Bases you can see named DevOps database I added a variable to the environment variable Environment variable was inserted MySQL.

The Underscore database and the values in it He had put that DevOps, so the name DevOps Once the database is created, it becomes so easy. Brilliantly cube CTL eject pod did this pod unit I did this, let's delete it kubectl delete Pod MySQL state full set zero in the namespace MySQL That Last One Bash. Remove it otherwise he will cry, if you remove it I'll delete that pod, pod my SQL State Full Set Zero Deleted Done now kubectl get pods in the namespace MySQL can you see what I.

Deleted the same name Pod of the same name It's back 8 seconds ago and that's it How Kubernetes Maintains the State Pod Always Whenever we create something else,.

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / Deployments, ReplicaSets and rollouts

I made four pods from lettuce and named them They will be random, delete any one and get a new one. It will become a name but it is not so in this case In this case, the pod with the same name that you Deleted becomes the same as that, that's the specialty of StatefulSet and its service. Volume template and its variables The environments all remain tightly coupled together That is why whenever a new pod is created it always He takes his state with him, that is.

Why it is known as a StatefulSet to come Hope you understand StatefulSet well I must have come across how a StatefulSet is Different from deployment in deployment which pods are made with different names every time. This is one of the biggest challenges in the ReplicaSet. Rapes are different and they also have different names but Rolling update roll out, all this cannot happen If that also happens in deployment then every Each thing has its own speciality, let's do this. We have done everything till now State full set means your core concept All the workloads are done, all the workloads are done Network policies remain in networking I will get it done for you in the project.

Almost everything is in storage too. Only the cuffing map and secret remain, let's go Now let's read Conf Maps and Secrets. First let's read what is Cuffing Maps Understand that this is the StatefulSet you have created a StatefulSet in which my The value of the SQL database that Debes inserted. Nor if you have to change this value tomorrow If you need it, you will have to come to the StatefulSet and do this. I want that as much as I Environment Variables and Variable End The values should be kept in a separate file.

For my environment variable and normal If any variable becomes a variable then I will create a file and name it Kong. Map Dwal and this Confine is and is The confine is v1 and your meta data That is my name. There is space, that is your name, space is yours will remain my The SQL is just yours, whatever you put in it. You enter the data, you enter the variable And when you enter the value, just like the data you Now enter the name of the variable here.

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / kubectl commands

You can, like my You can see the value of AL database here You can put it here like there was DevOps. What is the benefit whenever you Changing the database value You will need a stateful file which is so It was a large file with a template inside. The name of the service is also attached, everything is attached. You won't have to bother with it once it's made. Put it down you will configure this and this Any number of key value pairs in Confine You can use it, I'll show you how.

You can use escape colon and it's done Confine gone then kubectl apply han F Confine Map CreatedDocumentsFragment in the name Space my AL so what can you see Confide means there is a variable now this. If you use it in your StatefulSet then I'm going to go into the I'm StatefulSet and In the StatefulSet I will go and say that which is my environment variable MySQL. The database is its value, that is the value I will not take that value directly.

I will take value from somewhere Where will I get that value from? Name me SQL whatever you can put MySQL what was Confine from MySQL is Confine Map This gives you an edge Confered will have to confine ref meaning What does Confide mean? The confine you have entered On reference so your confine is MySQL Confine is the key that is inside it Let's see what that was, call Escape. Cat Confide base so you have to put my The SQL database stores the key information about its state full set.

I am like this so what does it mean to you The environment variable is my sequelize database From whom will he take its value? Value from whom from confine now from confine's key If it takes a value, what is its reference? Confine is MySQL Confine Database Escape Colon Double Enter I will do it now I will do kubectl Apply State Full Set Enter kubectl What good is it whenever you apply? RequestInvalid Patch with Invalid Value Oh, is he talking about map metadata map?

Does kubectl delete Ha Ye MySQL State Full Set In The Name Space MySQL deleted this [music] a one second state full set / MySQL StatefulSet deleted Now that the StatefulSet.

[Kubernetes] ConfigMaps and Secrets / Namespaces

We deleted it and kept it deleted and whatever A new state full set has been created in which That's our value, let's try it. It is still giving the problem, error from Server Bad Request Error Setful Set Version v1 Cannot Be Handled Edge Setful Set Strictly Decoding Error Spec Template Spec Container Environment Value from cuffing map key ref key. Should go to state full set which is of ref It is R, capitalize it is fun Kubernetes Cube Seattle Apply St Full Set Where was he getting it from?

How is the usage, see the environment. An object containing a list inside it that contains The variable containing your environment name is his name again Confide this value then within that you. The name of the config map and then its key is like this You have to remember, but if you don't remember If you want to keep it too, then anyway you have a way. It's to refer to a documentation So this Create a database named Confides. Will go Datchi Get Pudd In The namespace MySQL Running means that confine add and value there You are not getting any error after picking up from Like you have normal plane in confine.

Data in text key-value pairs Similarly, if you encode it You want to enter generally encode passwords If you want to do it, it comes in secrets. V I am Secret that email secret what remains in it This also has the same API version as yours v1 Your kind will remain, secrets will remain,? Secrets don't just remain secrets. Remains fine and whatever meta data you have What will be the name of the secret in it? My AL Secret C RE T MySQL Secret namespace will be yours, will it be a mystery?

Namespace MySQL is done now What will your data come in this again? MySQL's Route Password or any password you need to enter So you can enter it but you can enter a normal password like this. You cannot root it like this and put it on the base 64 encoding is required, so this route you will take base64 encode escape colon d base64 encode Maybe it's something like this, echo this password. If it is taken out in 2 minutes then its base is 64 Creating encryption only or You get very high level security No one wants to provide security for it.

[Kubernetes] ConfigMaps and Secrets / StatefulSets, DaemonSets and Jobs/CronJobs

What is its intention whenever this secret If it is created then a binary file is created. Whenever we do not understand the file This will create a binary file of the secret, from which Then you will not be able to retrieve the password. To understand, its work is security and It is not its job to maintain secrets. Secure secret files in binary format If you want to use it then you have encoded it You can add a comment here if you want, base64.

Encoded for root so this password is for root name The base64 value of is the escape colon double It's done, it's done, now I'm on my I'll call it a StatefulSet like I did in my. The value was taken from the SQL database. With the same confidence as the root password If there is value, who will take the value from? Secret Key From the ref whose name is Secret's My SQL The secret, and the key, is MySQL. Route The password is MySQL secret.

Where will MySQL root password come from The SQL secret which is inside the file If the root password is written, it will come from there. Escape colon double q enter now kubectl by -F Secret CreateDrawerNavigator Enter State Full Set If the Confidant Full Set is completed then kubectl Get Parts in the namespace. My so you can see 3 seconds 7 seconds 9 seconds when you completely change the secret I made your pods again now Your StatefulSet is a StatefulSet You cannot enter your password and your.

The database name is not exposed I Have Not Exposed the Root and the DevOps Whatever I put into this thing, I haven't exposed it, but still, if I kubectl unit - it. Hafna IT State Full Set Someone Else Take One Take in the namespace my SQL Han on Bash I get into it my ackl -y root -p password. If I put R OO boom I'm entered I've made my Password as root as it is not entered state In the full set I added via secret file.

That too was encoded in a secret file and then And still I am able to access Even the name of our database was DevOps. If you had not entered it then show databases here You will see how cool DevOps has become.

[Kubernetes] Persistent volumes and storage / Namespaces

This is your only shot Great in MySQL Joe production level But there is a database of the state of that thing. Full set service of that thing its secret Learn its namespace, its confine map, all How much time did it take you Whatever it takes to learn, half an hour is enough, so now you have learnt it. Do it because our storage unit and completed storage unit We have seen persistent volume Persistent Volume Claims Reviewed Checked storage classes, checked conflict map I've seen the secrets, when I.

I was reading here, someone here I was reading one, some just wandered around like this. I had reached there wandering around, what did I read in it that the data of my Nginx The mount path should be / yes R Share Nginx ATML and Hum Log Data. The path you were entering was probably wrong, so try it once. Just for curiosity, enter this path Let's see your persistent volume Just for Curiosity in Just for You If it works then it is well and good, if it does not work then it is.

My work on MySQL is done, so kubectl delete -f namespace then I delete the namespace MySQL. All the state full sets all the secrets All confine all services related to MySQL will be gone to thatchi of Having a namespace cd dot so What I mean to say is that our. It is deployment in what we have mounted The path is laid out, I say this mount The path is probably wrong, so I can change it. I'll check back persistent volume. Will be revised again escape colon w K Enter Now what I will do is first I kubectl apply -f namespacel namespace ajax that it has become cube what will I do now.

CTL Apply Han persistent Volume First I will make persistent volume how much If I make it 1GB then persistent Volume I made it unchanged, it remains. kubectl Apply persistent Volume K After what I will make a person volume I will make a claim, I will make a person volume Claim made right now persistent This claim has now become a volume claim. I will use it in my deploy. If I go to VIM Deployed, here persistent Volume Claim Local PVC Bun It's gone let's escape column dk become kubectl apply hif deploy done Deployed has also been created kubectl Get pods in the namespace.

[Kubernetes] Persistent volumes and storage / Pods

Both pods are currently pending. You will also learn kubectl Gate Describe it in the namespace engine By describing See Describe Pod so what is he saying default Schedule zero out of four Available Unbounce Persistent Volume Claim o persistent Volume Claim Unbounced. What is the problem cube CTL Get PV in the namespace [music] Engine O It has been released Is your purse volume available first? Only kubectl Delete Hyn F Han Any Delete Local PV From the namespace engine first then delete Can I delete a PV persistent volume?

Let's give Persistent Volume Deleted now kubectl apply -f I will create the persistent volume persistent volume ctl get pv in the namespace ang Nix bound right now she's bound so Cube if bound with claim. CTL Get Pods in the namespace Ang I'll nix running saw that if pod run If it is not happening then try describing it. What is the issue and then run it we what was our objective that the path we had given a mount I think that was probably wrong.

There is a bond with the path so the data is not shown. If I am doing it then let me tell you for contact that when We were doing persistent volume, so Inside your Docker container where the mount path / mnt/data usm data show. Maybe we were not doing anything I had entered it wrong, now let's see kubectl Get pods high o wide now that I understand We'll figure out which pod we need to enter. If yes, then go to the cluster worker and check it.

Cluster worker's which is a Docker container How to get its ID from docker ps What did I get? Was a cluster worker and his ID was also found. If it went then kubectl not docker unit contains the ID of the container. Enter worker nut th have come inside Let's see what we have in Worker Nut 3 CD into MNT. Data folder in MNT Let's go to the data folder. No I'm guessing persistent Volume if you want to do it then you have to do it locally.

Have tried that which of persistent volume The data should be visible in the Docker container He is not visible even for a second in this demo.

[Kubernetes] Pods / Autoscaling (HPA, VPA, Cluster Autoscaler)

What is data o can you see demo data w The data was what we had created in the cron job. If you created a folder while backing up, Demo data this folder so well let's become If this happens then the work of persistent volume is What happens is that your data is inside the pod. A reflection of the same data on your server Run on the host or Docker container, whichever If it is there then it should be on MNT data name.

A folder has also been created and demo data also It is made, perhaps that is why it does not come inside Because I am not entering the correct path But we'll figure that out as well. No scene, no scene, try again take what is in it kubectl delete -N Nginx namespace Nginx then delete - end No delete -f ajax key namespace deleted. Let's get back to ours, what are you up to? Persistent volume claim has been achieved Done class done conf map secret ho Culture networking is gone, everything is over.

That's it, let's go scaling now. Schedule scaling end What does scheduling mean? HPA VPA What is it all about HPA VPA Node Affinity Tut Staller Resource Quota Limits probes, what are all these, I teach you. Whatever pod you have Let's say where did my pod go new make a pod This is your pod, this is your pod The pod is running on

your host machine. From lettuce this is your host machine this is Your host this is your host machine now this host There will be some confine of the machine like this is my t2 Dot Medium.

How much is the TDD medium generally 4GB There is rum and its CPU is A two-core CPU has two virtual CPUs. It has a dual-core CPU and Your 4GB of RAM remains, let's take your Joe Pod Now that's the pod of Nginx Pod Nginx's pod now pod p traffic. You managed the pod of that pod and that pod It is increasing, it is increasing, bear it And this 4GB rum pod alone is enough. If it goes then there will be problem, 4GB If RAM alone eats the pod then it is wrong.

If you have another such pod Lets Whose MySQL So what if this consumes the entire 4GB resource?

[Kubernetes] Pods / Namespaces

If you do not get it, you will have to pay whenever you want to get your pods. If you decide the request and limit That you have to read now is When Ever We Will Read Request Quotas So Now Kissam Banana If you want, request quota is created in this. If it is made in the engine then deployment whenever You are created whenever your containers are created So what do you give to that container? You give the request end to the container Limits So you directly say that this is deployed file this is mine There is a deployed file, what do you find in that file?

You have to do it where you put it Always look at the port container specific pod If you have to give request and limit then You mean what your containers are then you. I have to go to the containers and tell them. What do you want to me, come out Gone, I need a request. I want to request my CPU I need a CPU, how many 100 What do I need and I need memory. The request is so much that this pot is going to run I need this much CPU and this much memory.

If there is dal roti then I will eat this much dal roti Max, Mummy, I don't want more than two rotis. Limits again to you CPU limits will have to be set if Your CPU can pull up to 200 GB. Not more than that and similarly You can also get up to 256 MB of memory of which you can pull, if not more than that What happens is that your pod is as. If you want further then you can read Make reps, then they use their resources.

Distributor to Load Distributor will take it, what happens to your node It happens that it does not come to a complete halt. Because as much as you have proportion If you are eating in the same then this is a waste of resources. There are requests and limits that you If you want to insert then press escape colon double key enter ctl apply -f namespace a engine Key namespace created kubectl Apply Now my deployment is. Persistent volume and all these things.

[Kubernetes] Resource requests, limits, probes, taints / Pods

Believe it or not, high persistent volume Make will have cube CTL get PV in the namespace engine to This is released, it needs to be deleted. kubectl delete PV Local persistent Volume which is the namespace It is in Ajax, I will have to delete it. I Persistence by creating Persistent Volume I will create the volume claim and then I will apply for my deployment now. What are you saying, I always mess up I will do whatever you request. These are not requests, they are requests.

As there are resources, so there are requests and If there are limits then what happens to requests? How much do you need to run the pod? What are the resource requests limits? It happens that you run the pod so much To avoid over-accessing memory Limits so I always that something so in I mess it up, what happened again? Cannot be handled as a deployment strike Decoding error unknown field spec template Speck Volume Mounts Volume Mounts We did not change the volume mounts.

What happened to the volume mounts man this container volume mounts containers I accidentally took it out You should take great care with indentation. The indentation is because you are using the Vim editor I am doing what is generally used in the industry What happens is that your VS Code is used. This error in VS Code is due to the linter. That's one thing, so kubectl get Pods - and Engen These pods have arrived now you fano wid then you will show that What is running, showing everything Now you can do kubectl.

Describe pod this one in the namespace Nginx and if you check it then you will meet Key Limits for this pod is CPU 200 The memory limit can go up to this point. The request can go to the CPU so much If I can take requests then limits end Probs you have no limits and probes Limits and Requests You Have Deployed Your. You have put the beans nicely, just like this You can also add your own StatefulSet. What you have to do is directly here I have to go Yours is called MySQL or My Sql You have to go to CD My SQL and your The StatefulSet is in the im

StatefulSet.

[Kubernetes] Pods / Resource requests, limits, probes, taints

In that too, you can manage resources and You can configure the limits well and you can do this In any pod, in any pod, you Resources as much as it needs RAM and CPU. If you can get that for him then it's yours. Concept about resources and limits Your has arrived, so resources and limits. What are the probes, understand it like this that whatever There are three types of probes. Lives Probe Redness Probe End Startup Probe Livens Redness and Startup Probe means this is a The request remains Probe means there is a request that your pod it your.

Whether the pod is working properly or not is working r not simple so this pod is this Any port number from pod pay lets say 3000 Or something is going on at 4000 or 8000,. There is something going on at 8000 with Lets That app of yours was this Django app. Let's say that one is running at Rs 8000 but You have to make sure that it runs at 8000. If it is still there then you will use probes when a pod is created, it will be an internal The request will be sent to port number 8000 like this.

He will make a request and after making the request if your pot is alive or dead live Otherwise, there is a liveness probe test. If your pod is being created then ready It's done, it's ready, it's ready The probe kills your m when your prop pod If it is starting then at the time of starting. Start up properly with the Startup Probe You will need a different method to determine whether it is over or not. If there are different types of requests then this Let us try implementing it's going to be interesting so what are we will do whatever we want, we will do it...

In the kits we were watching, We put everything on deployed names If you had given it then you will go to VIM Deployed now. This Note P is its container port. 8000 is this beyond this my AI is running or I am not going to see him LiveSoCERTV Livesya is our direct slash,. Port 8000 on that, so that's a slash on 8000.

[Kubernetes] Pods / kubectl commands

Will make a request will make HTTP Gate that if that request comes correctly then well And good, if it doesn't come correctly then there An error will come that is still missing his pord. Your pod is not live, similarly as Live video whether your pod is ready or Otherwise my notes are a test which we He had created an app and shown it, where did his. P A, you could see us here, the engine I was trying it, I was trying it here Well, I just turned off the ingress controller.

It is kept that way, if it is not opened then 8000 Our walk to the port To see this, we used two probes. Mare prob means request initial Request Escape Column Dull Enter Lets Give it a try to see if it works or else kubectl Apply High F Service CreateDrawAnaNavigator Is container creating so right now container is being created. The container is still being built. He has been there, that's why it takes time Because it's a large container I Installed on Docker Hub, will build the following He will make him run slowly and now he will start walking.

If it is ready then what will I do now? I will watch it in this pod, this is my pod. No gate Describe Pod this one in the namespace Nginx enter so speaking what redinus probe feld gate http this one pay 8000 dial tcp connection refused why connection Refused because it's your app. He doesn't have this 10.24.53 But this will not work on this port, so if you If you run it on your own local then it will be yours. It will run on this what the locals say If the host runs it then that will work but this is But it is failing but I am watching it.

I wonder why it didn't liven up the probe. It is fine because this is our IP. We are on this IP address. People are not allowed but their own redemption Where did the probe go? Line probe Here it is LiveSoCERTv vala pod is running kubectl get pods No kubectl Describe this pod called Notes P deployment then Pods / note p deployed in the namespace Annex, so your lives probe is.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Pods

Hit it and it was correct, some problem If not, that was live, your pod is Correct means it is running so if you Will you also try port forwarding for your What is your service called, Notes? App's Service Notes app's service which is in namespace Nginx and which If it is running on port number 8000 If you open it, it should ideally be 8000. Pay working so probes just checks It's a Way of Checking If Your App is Working Or not, like we just put this IP I checked by entering the port number, it was.

The prop also internally checks the redis prop. It failed because that IP address The cluster was picking up the IP if it was local If the host or any particular IP is picked then it. If it had also passed then the work of the probes would have been this much Just to Check If Your Pods Are Running successfully is not just about work So, we

have started this project of ours. I put the live probe in the deployment. If you want to add an interval, you can add that too.

Ho late i want initial delay Or put some periods in it, that too You can put it, so there you go. Probes where did yours go so Linus probe You also requested with example See redness probe it was failing Because cluster IP if you again any e. If it is on KS cluster then that will work fine There will be no problem when we complete the project If you do, I'll show you that too. Probes Limits Tents End tolerance Interesting Tents and Tolerances So what are the tenets and tolerance?

With all ten lines and all but if you remove these lines then yes good diagram We made a diagram, it's. It's Good Diagram Now whenever I tell my cluster that cube seattle apply deployed to ai The server tells the scheduler its arrow Let me shorten it a bit, the API server says. It is the scheduler that can schedule the pod. That worker node one has to send the pot to worker Whether to send pot to node two or worker node But if you want to send the pod correctly then the schedule is correct.

[Kubernetes] Pods / Kubernetes architecture (control plane, nodes)

One who decides on which node our pod will be The schedule will be correct now as we speak. The concept called tolerance and tolerance Tent as the name suggests, tent tant humlog If you tease me then let me tell you have a relative who is taunting you. What is it, you will not go to his house, correct? Because he hits the ton, why is he a useless man? You will go the same way if you use the scheduler You want the scheduler to read this node number one Pay should not go to node number one or node number Pod Schedule on One.

If not, what can you do with node number one? Tent Is a Way of Telling Your Kubernetes You can create a pod on this particular node of the cluster You won't schedule that is tent if it is tented but whether or not our something. First of all, you come and set up your tent. Get it done c kubectl get en all namespaces, we could have taken out what all we had made. Father, you have made a lot of things, so in the engine I go to the CD engine Went, now in the engine, this is a pod.

This is a paddle, it's a great engine Now this is a beautiful, good-looking pod. Pod This is the pod, if I run it, where will it go? kubectl Get Pod In The Name Space Gate Pod in the namespace Engine, so all this stuff is going on here. If you want to delete it, press kubectl delete name I will make a space engine Everything will be deleted now, I will tell you What am I demonstrating? I'm trying to do this to you I am demonstrating that I now I will create a pod and when I create that pod Yes, it should be scheduled at some node.

Will probably be on worker node one It will happen on two, it will happen on th, but if I tented worker node one so it Pod not scheduled on that node That Is Basically Tents and Tolerance. I'll bring the nodes and I'll say it now. kubectl taint kill taint node who to node this which is my cluster worker tent it, tent it, something You can also add any kind of thing to it.

[Kubernetes] Pods / Namespaces

You can tell me that this is what it is that is production prod equal to true Because there is a production node, then above it. If you don't want to run it then put product two and this Say no, no, schedule, no, with a colon. Schedule means that on this node Do not schedule pod enter no teads worker It's been a tent. What will I do now, worker? I will also tent two workers Three I will also make a tent, what does this mean?

This means if I take my cube CTL apply -f pod first to name I will have to create a space, the namespace is now created. If cube seattle apply -f pod I will do Pod will be created but kubectl Get pod when I do in the namespace angel. Nix this pod is pending this is pending why is it because one two three this three node All three of our tents are done. Beyond whatever becomes tent If the schedule is not met then if I cube CTL I will describe the engine pod.

I will describe it in the namespace and Janix is then saying failed scheduling row Out of four nodes are available one node is Untolerant Ted these control planes and three. Node Head Un Tolerant Train Production equal to true so that your control plane It also happens to be tainted and that is why Pods on it can never be scheduled And you tented the remaining three. CTL Gate Pods This is pending Poor guy, if we give him a hand full of notes, we People Cluster TWA cluster as we People had pitched a tent, worker one, this one.

Let's make a tent for Worker Two. Did you not tent node TS cluster worker To tent it and no schedule Now put a minus in front of no schedule. I will give minus, which means whatever the tent is remove minus means remove then enter no Teads clustered and two untented which tent got removed now will you do get pods. Because he

got the pod, now if this If you describe it, he will definitely get something. Scheduled whom to tide worker to Because the worker was too untenanted, he was sent away.

[Kubernetes] Pods / Autoscaling (HPA, VPA, Cluster Autoscaler)

The container started well and went well. I completely understand how you can now pitch tents with lettuce. I put the tent back to Worker Two Now my pod is still running, why? It won't make any difference, after this you can buy any pod. kubectl Delete high a pod so delete the pod now I Let me tell you that now I have all the nodes tented all nodes view kubectl I also told this worker to Tant Nod. Schedule and tented it now it's tented How will I run it on the pod also, I am Go into the pod and listen to the confession of the pod tell the container to confine that container.

Meaning not the conf of the container but of the pod Go to Spec and say it Tolerance, tolerance and in tolerance, say that Anything whose product is whatever you had put Prod what is the operator. Equal to hum log ne product equal to I had put true, now the product is equal and value true then any thing that has. Your tent is lying there, which is the product end value is true what to do with it Effect No Schedule So if you have any node that has prod Equal to true and no schedule entered.

Tolerate it and move beyond it Escape Colon Db Enter Now there is tolerance in my pod. Given that any node any node that has There has been a tent in which product is equal to true and effect no schedule is already. It is tented, how will it work on it kubectl apply -f padl enter pod Created pods in the namespace Angle Nix Running It worked, it didn't work before, why? Because there was a tent, but not because there was a tent.

I used to walk, now I have gained tolerance, so I am walking. I hope you find peace and tolerance A lovely concept, I understood it very well. Went It would be great if you also understand tensions and tolerance. I must have come and next we will go Horizontal Pod Autoscaling and Vertical Pod Autoscaling on HPA and VPA This is a very important concept when you go to study Kubernetes because Kubernetes is from NITS. The main function is auto scaling auto Healing is all its work, so HPA VPA Vertical Pod Autoscaler so this is for auto scaling It is a very important concept Let's read this in a little depth.

[Kubernetes] Pods / Autoscaling (HPA, VPA, Cluster Autoscaler) (part 2)

First of all let us understand this auto scaling what is auto scaling Because everyone is using microservices now. Architecture has come and everyone has come Optimize your applications based on traffic Now we have to scale it like A means horizontal pod auto scaling then comes VPA Vertical Pod Auto Scaling Then Another. He comes whose name is Keda I had never heard of this Kubernetes event. Driven autoscaling is generally too much. It is not used but I can also tell its meaning.

Let me first understand HPA VPA Horizontal pod autoscaling means if your The pod is that pod is mast uspe trough trick waf Crazy amount of traffic has arrived This pod will scale automatically. The scale will automatically increase and such reps will become his habit. I will go and do all the reps of this pod. Your deployment is like it is not like We put five reps in Deployed And in Horizontal P Autoscaler you have said I told him that, if this much percentage is on CPU.

If the load becomes too heavy, three more rep Make it so that if you say so, then he will be a Great under horizontal auto scaling will make your pod reps now this happens. HPA in which the pods form raphes then Here comes VPA means vertical pod Autoscaling that earlier your pod was like this and The maximum limit that the pod used was. He uses 100 from resources let Yes, but there is a lot of traffic on it. It can handle a maximum of 200 mph of traffic.

Can If you want to auto scale it then The pod will become Hulk means this pod There will be so much power inside, this is a big one. The scale will be changed and the first 100 And there are 200 resources, now they are 500 MB can pull up to 1000 so you can have a Horizontal increases the strength of the pod. In a pod, you increase the number of pods. You increase vertical pod autoscaling You don't increase the rep now only the pod Resources and Limits Have Requests and You increase the limit to that person and the scale to that person.

If you do it then generally VPA or vertical Pod scaling for stateful applications It is used for if you have a mother SQL is a StatefulSet.

[Kubernetes] Autoscaling (HPA, VPA, Cluster Autoscaler) / Pods

If the state is full set then you can put VPA in it You can, but if someone sends your normal application Let's take a look at the Django Kubernetes The application is normal, if you like it. If you want to scale the application So you can put HPA beyond that and You can increase the reps so that you do five. 50 on this But if the traffic increases, the traffic will increase by 50%. If it goes then it becomes five then it becomes five, you will get the rapt, so this is the Beauty of HPA but, I understand HPA I came in Replica increases VPA in Pdf.

It becomes bigger, what is hard, hard Kubernetes Event Driven Autoscaling He sees how your events are. Re-routing is a high traffic situation If you are going to put it on the Air HPA then it will There is too much overload and stress. If he is living then he will put VPA then based on The Nature of Yours HPA of metrics and events or Selects any of the VPAs But, here you have a very important. The word spoken is whose name What is a matrix?

Matrix means quantifying force resources How much CPU is being used? How much RAM is being used and all these Matrix is based on this Horizontal pod autoscaling or else Vertical Pod Autoscaling Actual Work So first of all this cluster of yours. It has come, where has my cluster gone, here it is This cluster of yours is kubectl Gate nodes, these nodes, how many Traffic is coming, how much is happening? Matrix address is using command called kubectl top node or kubectl top pod if you do kubectl top node Error Matrix A AI Knot Available you do kubectl top pod.

Matrix API not available to Cube kubectl get namespace in system So in the kube-system you have a Matrix Server should be there and that Metric Server on your node and on your pod. What kind of CPU is this, what scene is it, all this is here Tells you have the Matrix API If you don't know how to do it, then what do you have to do? Confession load balancer is not one of them.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / Autoscaling (HPA, VPA, Cluster Autoscaler)

If the Matrix server remains, then that Matrix You have to install the server no matter what. No, I who The notes I will give will include all the In the history command that will remain, I will do this. I will give you the full command if you make a Minikube ho Minikube agar You are a Minikube, so Minikube pay a lot It is easy to use the Minikube directly. Can you do Minikube well I a little bit If you come down, will you bring chill again, Mini?

Cube enable matrix server so these Minikube The add-ons that are there, add-ons You can go there and add mini cubes. Enable matrix server then you will enter So now I just installed the Minikube server It has not started, that is why it is coming like this. If you have your cluster on Minikube If created, enable Minikube add-ons Matrix Server works fine after doing this You can, but if you are on a Kind cluster You have a particular URL, that I. I will give you the URL something like this type is Kubernetes Nits Kind S cluster Matrix Server such that running Matrix Server on Katas Client Kind Cluster and then You have a particular.

This is from this location, this is the location HTTPS.com CATUS 6 metric servers and releases Latest download from this type of URL You have to take all the URLs. Where to find Cube a Cube Starter Go to the repository, there I have done this HPA VPA kept in folders All the things and in this I put that whole cube Kubernetes Sys Matrix Server Releases Latest. All download components are provided in it Copy it, you've copied it, now you go. On your cluster This is your kind of cluster so go to this and paste that kubectl apply h f GitHub.com The manifest file of Kubernetes also remains You can also apply from GitHub.com like this.

You can, there is no scene, it goes very well. Enter fast fat, all things now Created a Matrix Server named Then we gave him a roll and went back and forth. Given a role and then created a service I made a deploy and it started running, so this is us What people have done with the Matrix server I installed all the things that were there but.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Pods

Sometimes that thing doesn't work because we Running on a brick instance pay HTTP HTTPS us If the entire server is not signed then we If it is not signed then a TLS this security. If you have to bypass then for that What to do on Edit the Matrix Server You have made the deployment plan,. Edit this for the deployed metrics server So kubectl in and kube-system then inside the kube-system namespace Your deploy that has the Matrix server Edit it, then take it here and put it here.

Two kubectl - & kube-system Edit Deployed Matrix Server Enter Now This Your deploy has opened which you downloaded. Had to take tension for Matrix Server You don't need to write that much You can download it now. What you have to do is put containers in it. If you want to find containers, then these are the containers You just have to add two small lines in it. Add the Security Bypass to Deploy Under container d Argo CD so container k Argo CD Just two lines of security bypass.

I want to add that kubelet insecure tea ls that your cube lays which TLS is run on your worker node If it doesn't have HTTPS security Even then it will work, no problem, kubelet. Preferred Address Type Internal IP Host Copy the name, external IP address and What you have to do is go here and throw it away. Went and threw it here, now what to do about it This is what the indentation is messed up Fix it, so I've done it Fixed the indentation.

Colon Delay Enter Deployed Matrix Server Edited Now Deployed Edited Deploy the file, now you have to restart it. You will have to restart it is very simple kubectl roll out restart deploy Matrix Server What will happen if In your namespace kube-system which The deploy you have entered should restart. It will be rolled out, meaning slowly every one Pot by pot it will restart Enter Deployed the Matrix server and restarted it. Once your Matrix server is restarted Then what do you have to do, then I will see you That kubectl get pods in the name Space kube-system in kube-system.

If you have kayak pods then can you see matrix The server is still pending and is still running.

[Kubernetes] Pods / Namespaces

Pending till For some reason it is still pending Matrix Server Kako Pending [music] I installed Hare Matrix Server. If I have given it then there will be no problem Should Cullet Insecure T LS Restart the [music] This too should have been deployed Why is kubectl get pending? Sorry Top Node Matrix API Not Available Matrix Why is the server pending? The cube pending coming CTL Describe Pod Matrix Server This one in the name Space Cube In the namespace of the system kube-system Let's describe it and see what man.

All my nodes are tented so my The pods are not working oh my god so right now one by one I will untick all the notes, sorry. For that because while teaching I make a mistake It was said that all our notes are tainted, so we As many nodes as there were, we had worker one. Let's make it no schedule minus It has become tent, worker two has no The schedule is reduced to zero and the attendance is over. Worker three should also be given no schedule.

Let me give you the attention, now as soon as I Attended now I will do kubectl Gate Pods in the namespace Cube System Done running saw Kubernetes in so many. All these are concepts but the fun is in this. The fun is in this if you like all these concepts. Your metric server is now installed. CTL Top node so you'll know which control plane It has 4 CPUs and 669 GB of memory. The worker node that is sitting there is 238 Worker Note 2 284 Worker Note 3 299 Ye now I want to see who is the heaviest What's the heavy pod that I'm driving?

Cube Seattle Top There's no pod if I want to see that Which is the top pod in the engine namespace? There is an engine pod which has two 2 memory If he is sitting with it then you can see it like this Can you tell me which node is the most. Which one is taking up memory and which one is taking up CPU? The pod consuming the most CP memory. Ctrl delete pod Dottle delete -a puddle as pat deleted now If you cube seattle top pod engine karega you won't get anyone even if you do MySQL You will get none and the default is none.

[Kubernetes] Namespaces / Deployments, ReplicaSets and rollouts

You don't have any pods running. Engine taught state full set persistent MySQL to Volume Taught H Notes An application A new Deployee Pod Auto Scaler Let's see the thing whose name is Apache What is Apache, this mkdir. Apache What is Apache Nginx There is a server like a web server which You can also download the application to If you allow it to be deployed then the CD becomes real. Complete next level em name spaced I Create an Apache namespace. Kind Name Space API version v1 meta data name what is my Apache is just that, Apache uses the Apache namespace Apache itself keeps the namespace escape colon dk enter kubectl apply -.

Apache namespace and my Apache namespace The space has been created, now what is my job? This good has to be deployed and that Deployed to HPA Want to scale automatically so v I am Look at the deployed email, I will get you revised. I mean, writing deployed means this much too It won't be difficult, I'll write it quickly. Our deployment will remain Deployed then our meta not API version What will be our app sww then What will remain of us, the name will remain intact and Our name is Apache of.

Deployment is fine and what will be our name? Space will remain our Apache, then what will remain Our specification will be that we need ReplicaSet We need only one by default. Will remain P Apache selector done then This is our selector's criteria. What will match labels match labels be like this Match the label that ends in Apache, now again the selector and the rape one. The work is done, now we need to create the template. Is template templates not template meta Its data and name will remain Apache is done, now what can you do about it?

You can put containers in containers that name Off the container Apache then Image of the container Apache Generali httpd is httpd is such a tool remains whereby You can create servers and all that stuff. It is very crazy, you should have bought version 2.4 It is a good latest one if you want it. Take the latest one, put the latest one like this, Let's get the latest version.

[Kubernetes] Kubernetes Services and networking / Pods

What will we do with the ports? Container port of the good ports what remains 8 remains so I am here Container Port A Container Port AT Now here is the most important thing for you. The task is to ensure that the container's request end A limit will have to be set because that is one It's a best practice This is practice, you can modify it later. You can do it both as HPA and VPA So what should my resources be? Request CPU Kitna Chahiye Again 100A How much more memory do I need again?

Max my CPU that pulls up and 200 It goes up to M and the memory that we have Let's go and pull my memory up to 256 A. I want it to be mine, Apache. How much would it take to prepare the deployment? 1 minute It must have taken 2 minutes but it became an escape route. My deploy is ready so kubectl apply -f deployed low entered What about rapt again, you have done it so many times Still not able to write reps to ReplicaSet keep practicing it will happen.

Selector does not match template label then You have read the meta data of your reader and meta data Labels are not included in the data name. Labels are not killed P, this should be your label. In the template and deploy as soon as you do this Created pods in the namespace well You will need a pod running Apache Deploy. If I want to make it accessible, I need a Service Dull made the service how kind API version before service metadata, this meta data API version is up and down.

Manifest only scan will kill API version v1 Now Meta Data P No Name Ach Service namespace Apache and your spec should be will go Selector what you want to select Ho Anything To Have Labels P. Whoever has Apache, we will support them. We will select it and after selecting it What we'll do is just a little ports whose protocol Will be done TCP whose port number will be 80 and Whose. The target port will be 80 and so on. What is the difference between target port and port I mean, I have already read all this of yours, but this expose The port in the cluster and the target port.

[Kubernetes] Kubernetes Services and networking / Pods (part 2)

Meaning [music] container port so jisko container i hope you Now this must be great for revision. Put the cluster IP, cool IP Capitalized service and cube by doing this CTL Apply Han Service Dull entered the field, what are you saying? Selector of type string cannot Unmarshal, selector labels are good for you. Where will the selector labels be located What doesn't fit into your labels remains yours. Normal selector and you will do it. Apply and service create great now you Will do kubectl get all in the namespace So what did you get from your rep?

I wanted a set and I got it, your deployed I got your service, I got your everything has been received What happened to it, it's, it's done for you. I've got everything, just a little thing I'll give you I would like to tell you a very small thing that What you might find very useful is that when Whenever you access a pod, you. HTTP whatever your service name is As - Service name then dot Default Dot svc dot clr dot local this when you do Yes, this has become Apache, it should be Apache.

Apache and if you do this with curl HTTPS CURL HTTP Apache Service default local this if you are from outside If you do it then he will not get it but if you. Doing it inside the pod is not possible without any namespace. If you do it inside the pod then it becomes your domain Name: This becomes your domain name for that pod. What is the best service to reach It is running on port number 80. Whatever cluster local is inside is now the default Service is now your default location.

Apache will come Apache means namespace so It's not happening right now because you If you are doing this from outside the cluster If you did it from inside the cluster, it would work. Just keep this in mind that if You need

to access a pod to access a service. If you want to do this, you can also access it through DNS. kubectl Describe the service Good service in the namespace Well, when you do this, here it is DNS is not showing but this is its The name is this dot namespace dot your cluster.

[Kubernetes] Pods / Autoscaling (HPA, VPA, Cluster Autoscaler)

This is how it comes out, I'll do this Next, when you do HPA, you will be able to access it. Deployed it all createdted forward kubectl port forward to whom my service or service slip good I will expose the service which name Space is good, run port number 80 82 Let's map 82 to 80. I will give it to you and I will give you the address 0.0 enter what is he saying permission Denied it, no problem, by doing this We will run the same command again now.

I started forwarding it, now I will go to my I'll go to the security group on the instance. A security group in bound rule I will add it, which will be named port number. Change the port number to 82 as you did to 81. I'll give you this port number: 82 port range I added it and now I am changing my IP. I will get the address where did my IP go This was the address,. If you go here If I run port number 82 then it works.

Meaning my good deploy is working great. It works fine so here's my Till now the work is fine but the problem that is occurring now is This is a good server, we call this good server Leave it in public, let's train with. If everyone comes and accesses my cluster I want the cube CTL Gate Pods In the namespace apache when there is a pot right now. Moving on then I want the cube CTL scale deployment Apache Key deployment Hyphae Fun [music] Reps three become in the name If space becomes Apache, then scale is now me.

If I do cube saddle get pods then it becomes three, this container creating container. Creating and running three became but, he You have seen the traffic that was about to come. Visualize and scale pods manually What happened was scaling, this auto For auto scaling if scaling is not done Now I have to do Horizontal Pod Auto scalar to i m h. Padwa now here I will make my Horizontal pod Autoscaler So its There is a little API version,. What stands out is the auto scaling system.

V2 Auto Scaling s v2 then Auto Scaling v2 Meaning when your manifest containing auto scaling Their A version is like this now Whatever kind it is, that kind is what it is.

[Kubernetes] Autoscaling (HPA, VPA, Cluster Autoscaler) / Deployments, ReplicaSets and rollouts

Horizontal Pod Auto Scaler Auto ng Even though it happens but its particular The name of the resource type Autoscaler Now put some meta data to it Key Name Apache HPA Apache HPA which namespace. It will remain in Apache, now whatever it is The spec will be very useful which Scale Target Ref Scale Target Ref does not mean you What you need to scale is the pod. Have to scale Deployed have to scale Scaling ReplicaSet Sets is StatefulSet what you have to scale If yes, then tell me that scale in target ref.

If you want to scale it, I need to scale this thing is kind deployment whose name is good deployment whose name is good deployment and whose API version This is SS v1, so this is the identification. API Version / v1 Name Deployed Kind Deploy all these things, your one deploy Now we have the details of what we made. Keep going and Joe Max Reps want that five pods should run in these cases. Resources Whoever has good capital and good resources then Replicate this on the resource and Which resource will be my CPU My CPU will stay if my CPU targets If my CPU's target type Utilization is.

Let's say utilization means that I have so much CPU I am utilizing my target Average My CPU utilization should reach 5. Please make me a rep if my average Once your utilization reaches five, you can become a rep Give it to me, I want it, so what did I say? Is in the matrix type resource Name CPU If my CPU average Utilization of your capital will remain in this if My average CPU utilization is five. I'll tell my autoscaler to escape colon double k enter now our three reps we People would first scale up and build a reputation.

If so kubectl get put in the name Space You are a running man, now, this thing that you have just done URL di at works iske to. It's not working because I stopped it a little bit. It will continue and it will continue on this, what is the method of making load? A Breaking Point Load Test to Find It Before Your User to Start Testing for, let's free our account.

[Kubernetes] Pods / kubectl commands

Email Here is the password yes god what a joke It doesn't matter, told me to do it. If you run it, this is its code Yes, you should confirm, this is what Kind of, this is it, that's all Just let's start testing. Testing what type of script should we Start with Browser Script Protocol Script Create Protocol Come on, get me a script. Script Step D HTTP Step This is not HTTPS localhost. Gave He does one thing, all this nonsense We don't go into things ourselves.

We will make our own load tester How will we make a tester, so one thing will be done Let's create a pod that We will name it load generator, load. Create a pod called generator from it First, let's look at the kubectl get. HPA in the name Space Apache My get hp in the name of Ach C AE Ach get HPA because I have used my HPA kubectl applied -F HPA Horizontal Pod Auto Scaler CreateDrawAnaNavigator. Which is my CPU utilization that is just 1 But get HPA is still % it a little If I make it smaller, if I zoom in, will look cool.

You should get HP min pod max pod now Currently a rep is running, right now a Rep is running and minimum pod so much now I How do I do stress testing in this? I will do it then kubectl run so what do you get from kubectl run You can do anything with kubectl Run If you can make a pod, I'm going to make one now. I want what I'll call a load generator If the name is then the interact terminal is kubectl run interact terminal Load generator name interact generator name.

There will be a terminal that will use the - - image Please write to me in chat and tell me who is this? This is the image that I told you about, which is generally It is used to write commands, its name is If there is a busy box then I will use the busy box. In the Apache namespace and its open in the - ain bash I will do it. Enter so what he's doing right now is the load generator will make a busy box and give it to me will open it and give it to you kubectl run it load What happened to the generator?

[Kubernetes] Namespaces / Pods

It usually takes a little time It's, we can do one thing. kubectl Get Pod in the namespace Apache then load generator clc crash loop back Why is the cube showing an error? Try CTL Gate Describe pod load generator in the namespace what are you saying, there's no bash executable in it. If there is no bash then it doesn't matter, basically The command we entered was - bash you can put something else in its place / Can you put bin sls so that it works.

He doesn't have a bash. The busybox one didn't have bash so We delete that, delete pod, which load generator in the namespace Apache remove it now We will make a small pod. kubectl Run load generator A load generator named make a pod whose image will be what busy box and which Load generator will be created in namespace Created Pods in the namespace. Try it and the load generator is complete. Crash loop back off now watch this Why is the crash loop backing off repeatedly?

Because our busy box has what does it do just enter the command If you do not put it then this problem is occurring then one It works, remove it, delete it do it. I keep it on, I keep it on kubectl run -i interact terminal - - TT wa time to live in it Keep the load generator on And we will use the image of busy box. Waiting for you and namespace Apache I will make it and give it to him.

/ s will give sh because bash will find it I was not able to open it now It's done, I got it, it's open now. Run a command What to do W Get W Gate W Gate means you can access any URL Bring something from there and give it to me. All the options are there, you can even remove it if you want. Can you get what http I What did you say if you have Apache? Service Apache Service If you want to go inside then its namespace is Apache dot SVC Dot cluster dot local when you put this and Do what will happen if you do it?

Apache services that are part of the Apache namespace He's in and hits on it now.

[Kubernetes] Pods / Namespaces

Enter what is happening that file assist file Exist we can't open ix html5 file Agist, so let's change it a little bit. Are han q And if it works then your load is generated You can see that I have used the same name Which of the following URLs did you hit with the space? It's still going on, so you're seeing how much Late that it works it works it verses it Vers it work it verse means what is happening I created a pod which is in the same namespace.

I am here and told that pod, come on, run this. I am generating the load. I'll go into my downloads and I am saying that my server is down right now. Meaning there is some load generation going on in it. I can send SS to my server from other place I will try to do it from the side Try to SSS your server I will do it and call it kubectl. Get HPA

in the namespace aché Now you can see the CPU utilization.

It's been 18 and if you and I do it kubectl get Pods in the namespace Oh, one two four, oh my god It made a lot of good load Our generator is already running. Running how crazy it is and here If I stop this I stopped him and he finished. It's done, I stopped it, this is mine load generator This was the load generator whose job was only and I stopped it just from generating load. At this Cube CTL Get Pods in the namespace What Good Running is kubectl get Horizontal Pod Autoscaler in the namespace See now CPU Utilization has come down to 2, CP.

Utila my % has arrived so it means this much now If there is not much CP utilization then that You will have a great time, what can I do now? Let me show you the cube in front of you. CTL Scale deployment A deployment Nafan Reps equal two in the namespace Apache Deployment The scale is done, now see what I will do. kubectl get pod karunga gate p and I'll just put a witch in front of it, witch I will apply the paste, now you can see it slowly.

[Kubernetes] Pods / kubectl commands

My pods are getting smaller, can they? UC my pods are slowly running out First one, two, three, yes, see it will gradually reduce. It's been a while, two seconds, this, this, this It's going slowly, what should I do now? I will check my HPA CP Utilization One Now If Low CP If there is utilization then keep only four pods from it. I'm using kubectl delete -f Why are you deleting Deployed Just to give you an idea that in these cases for the first time I create So there will be a pod and then a cube.

CTL apply when deployed now I created it, so there is a pod, there is a pod Apache Deployed's Cube Seattle Gateway Apache what should I do now Doonga Cube CTL Apply -F HPA Horizontal. If you also install pod Auto Scaler then horizontal pod Auto Scaler is also running fine on it horizontal p auto scaler is also running Now I'm going to watch my kubectl. Get Poords in the namespace great can you see yourself he Why did you make it five because I loaded a lot.

I had generated too much of it, that's why He automatically created running in it. Running Created automatically How many four have you made, now see one being made on its own. Can you see one was CreatedDocumentsFragment Because the CPU was being used too much. Application so this Django Naats App Is This is Django's nuts pop if I go to this nodes I should put some good resources on it. Resources such as Request CPU 100 MB Memory 128 MB and similar Limits so 200 MB End Memory 256 if I put it like this then this is my Deployed is also currently consuming resources and.

Have to copy In writing to myself, I have given you It is shown in writing but I copied it from I get out of this I get out of this Exit LOAR Generator finished load. When the generator is finished, the kubectl get Pods in the namespace to load The generator is still running, delete it. Two kubectl Delete a Pot load generator in the namespace Apache so The load generator has just finished working. Load generator is done now what do I do I am a simple one, I gave you that bus load Why did you make a generator?

[Kubernetes] Autoscaling (HPA, VPA, Cluster Autoscaler) / Pods

To show that if you are in another pod You can add a pod within the same namespace. If you want to hit service dot namespace dot I hope it is also called DNS. I also need to create its HPA here. If you want to make a dottle, then Apple itself is not I'll pick it up from here so I'll make it and cat HPA and this I as it is I use this. If I pick it up, what am I doing now?

That your Not well deployed on your notes I'm going to scale the deployed. Who is going to scale the notes app deployed will name it Notes The HPA and its namespace may Inside Nginx was Notes P Deployed this All right, clear escape colon d. I don't understand what I am doing Am I am here Powering your Notes app So right now Apache His story is over, so Cube CTL Apply High F Name Space or kubectl apply n f deployment dll. Deployment CreateDrawerNavigator No matter how much the resource utilization is kubectl Get pods in the namespace Nginx So you can see that Nut AI has become deployment.

Is similarly kubectl apply -f Your service has also been excellent. What do I have to do next to If you want to apply then apply kubectl h f HP's app has also become a bean like us. People made it for Apache anyway. I'll have fun when I karoonga kubectl get HPA for The namespace engine that my Notes app is running on So right now the CPU utilization is at 12. I made the first pod 69 seconds ago and this was 23 seconds ago, it just got a little Made it a while ago, like we did with HP.

I'll do it again, it's only 38 seconds, it's only three. I know why it is stuck on the platform because which is also the CPU utilization of the HPA If you are a distributor then divide your money in three pods. Get it, can you see it 56

seconds ago Created just 10 seconds ago 11 seconds ago Two more became That Is The Power Of Autoscaler Because your application is Notes Pa. It was taking more CPU than it should have.

[Kubernetes] Autoscaling (HPA, VPA, Cluster Autoscaler) / Pods (part 2)

He made more pot than what you have now Notes P has become highly scalable Auto scaling is ready, cool now I'll put everything on GitHub for you. Father, it's been six hours, there is no scene left yet. I will read it, so right now my work is on this. If it's done, then the kubectl delete namespace Nginx ang remove nicks and similarly i paach I will also remove it. I will not remove Apache. Why because in Apache right now I HPA So I told you I have just told you about the vertical pod.

I will also tell you about autoscaler, I will not leave you. Vertical Pod Autoscaler View Vertical Pod For what is autoscaler, you The files that reside in the Vertical Pod Autoscaler If so, you have to download them. The files that reside in the Vertical Pod Autoscaler You have to download them from the group. I will give you the link to its git repository. If I give it to you, what do you have to do with it? Who to clone Official Kubernetes Auto from repository scaler here's this one so Kubernetes Auto Scaler from the official repository Look, it's still running an hour ago.

What is Qnet Auto Scaler in this Repository Content Auto Scaling Related Component for Kubes Cluster Auto Scaler Vertical Autoscaler Addon Reiser Charts End What is the cluster autoscaler? Cuboidal or those that form clusters scales them to the cluster as well You can also increase the notes but this kind of I never did it, so that's for another day. If we need this, it keeps the vertical I'll code this and clone it into my I'll go here and put this in git Clone.

I will give you the code for the autoscaler. That's great, that's a very big code, that grave nits If people have written it then how big will be the meaning So it's coming along great, it's happened at 50 so far. Are you Went Done 100% done so what do I have now The code for the autoscaler is here now. What I'll do is go into this autoscaler and go to CD. Autoscaler and then vertical pod in it How to create a vertical pod in Autoscaler I found out because I was in autoscaler I'll go here, here, vertical pod Autoscaler is written and what is in it.

[Kubernetes] Autoscaling (HPA, VPA, Cluster Autoscaler) / Pods (part 3)

Everything is there if I install it If you want to do it, here is the installation guide. I can see it like this, it's written in front hgt clone otr gt then cd autoscaler Vertical Pod Autoscaler Then Check It Out Whichever particular release you want to go to. Let's do it right now This is something hacked, something someone No problem, we will do it, what is there in this, dot There is a folder named / hack inside which. There is a script called VPA update s which You have to drive yourself Enter boom everything went away Deployed CreateDrawerNavigator The work of the scaler is over, which means.

The repository was auto scaler finished its work Gone is the difference between vertical pod auto scaling Pre-requisites are required for this You have installed it, now what will you do? You can have a great Vim Apache app. PA Dull Bana Dooge Vertical Pod Autoscaler Again, its type will be vertical. Pod Auto Scaler Vertical Paw Auto Scaler Its Function whose API version will be and will remain again the same Autoscaling S v2 autoscaling v2 but a little bit in it. The catch is that K8s also come, this How long will we remember this?

Documentation is provided for you No need to remember v1 is by the way So autoscaling kv1 is the API version of autoscaling Now its meta data will be in, please put its name. Please enter Apache VPA and its namespace. In the Apache namespace, your auto Scalar Vertical pod T Scaler Name You gave me a spec and put it in again like You provided a target reference in the HPA. Scale Target Reference in this bus You will need to provide a target reference.

Will you give the same reference as you gave that had done the time what was your name Apache Key Deployed good and what else is there in it What API version was entered? The target ref was the only one in this target ref is now the one to scale What you are thinking of is Apache Deploy now. A little about the update policy If you have to write then what will you do in the updated policy You can do it, you can put the update mode Update mode means you can use vertical pod.

[Kubernetes] Autoscaling (HPA, VPA, Cluster Autoscaler) / Pods (part 4)

What updates do you make in autoscaling, e.g. I told you in the explanation that The updates you make to vertical pod autoscaling The thing you do is called your pod. Based on your use case what can you do You can put it on auto, auto, that you Increase it automatically and you can make as much as you want. Meaning, as much as you gave in its deployment It was the same but I want it to be auto scaled So if there is auto scale then I put auto then.

There are three options for this, you can choose update mode off You can keep it if you don't want to scale. You can keep it auto so it will auto scale or Then you can put initials so you get a You can also give a good number, I have done auto Escape Colon Double K Enter kubectl get. Pod in the namespace Apache is running a pod kube CTL Get HPA in the namespace CPU utilization is at best 1 so first which is a lot of pods CreateDrawerNavigator Now what do I do?

I will delete the kubectl -f HPA then HPA is removed I will give because horizontal pod auto scaling is if I load this on I will hit you hard if I give you a load. Horizontal Pod Autoscaling Pod's Reps It will make me not want to get raped by the pod. What I need is my vertical pod auto If scaling goes this way, what will I do? Cube CTL Appla Apply - & PA Vertical Patler I will bomb it, the spelling of spec is wrong has written it in your IM VPA.

I'll go and correct the spelling of spec S Fifth note of musical scale S Y Done Apply H Apache Vertical pod Auto Scaler Apache name not currently in the default namespace It's in space. Apache VPA is in auto mode CPU 25 GB is using this much memory Using in KB is provided true If it is yours then what will I do on one side. I keep my watch on one side I will kubectl get va pa for namespace good I will watch that its good How much resources is the VPA using?

[Kubernetes] Kubernetes Services and networking / Namespaces

You could have given it some other name instead of local. For now, local means your The server name itself is a little different and here I What to do I will give it to you, my Apache service is broken. kubectl Get Svc In The namespace apache my apache service is running My pod is also running from Get Pods in the Name Is Get VPA My VPA is also working fine. Cause some havoc let's cause havoc They do the same, they run their load generator So load generator come kubectl run How do I load a generator?

The load is the generator, nothing, poor little thing. Image which takes the image of the busybox and runs its namespace in Apache I enter the command/bin/s and After some time he would come to my busy box. I will tell the busy box what are you doing True Do W Get While is a loop while true meaning on fine it loop keep going. Go while true do w get command What was - Why are you telling me this right now? Tell me how internally a pod with the same name How will the service hit the space?

Write the name of the service and then the second part After this your good namespace will come again Service will come, then cluster will come, then local so this is your DNS your. End of service DNS for internal use What will I do after I do this? I will tell him to just do this You did it, then you After you've done this Yours Done, just do it [music] what happened why is it not working I need to restart my service.

Is it no matter I did not think I would get this service back. Restarting will have cube CTL port Forward which service Apache service which is in the namespace Apache Its port number is 82 mapped to 80. Tax and address at 0 put it and what k sudo -e this The port has started working, now I will see that This good itworks of mine will continue. Where is he not running my service? We will get this started now, we will get this started They will say, let's go to Apache service, you Apache Let's start running Apache in the namespace gone it works it works now I.

[Kubernetes] Pods / kubectl commands

I will tell this kubectl Get VPA For Name Space Apache is now using 25 MB of CPU. It is 25 M now, he will increase it now. My pod will increase the number of pods. It won't increase his CPU The thrush hold of increased slightly It will take some time to see him I will increase the butt, he So Tworks is coming. Once upon a time, kubectl got Pudd in the name If we try space apache then Apache Deploy is still running.

At this, why is it from 25 not increasing will continue to grow, will continue to grow It will take some time to grow This is because it may be the current CPU usage. I want to see how much memory I have on my pod. kubectl get pod in the name Space Apache not pod get pod instead of top Pod so Apache is taking 83A and memory 9. I am taking it, just a little more left, 85 happened, it is understood that his He has some limits, he has some limits.

There is a limit now and I am pushing it. You can also have your pod auto scale. I have done 109 and this is how you have Enlarge Your pod You have enlarged your pod Now if you wait for a minute, then that too will happen and. It will scale further and have more CPUs Take He will take it, it's cool when he says crazy I remember having time He has gone crazy, he speaks very coolly. Sounds pretty good to me, that's an 84A CPU.

That is using the core Gate VPA 109 Oh my God, understand I am coming how 25 app was running you Great just scaling right now so this is Vertical pod out scaling where. kubectl Get pod in the namespace If you do it, there is only one person and he will keep worrying. Even a little load was applied, pat very much All the trees were being made, now that is growing Try describing this with kubectl down l kubectl Describe Pod Ye in the namespace.

Apache right now Look how cool it is I don't see anything like that right now, but it's. kubectl get VPA now only at 109, I will load as much as our load The generator is giving decent load in that too Giving kubectl Top Pod in the Name Space Engines Not Excess.

[Kubernetes] Resource requests, limits, probes, taints / Pods

Apache x86 now top node look node get top node So your control plane is at 20 But your worker three is on 31 CPU is sitting with 6 638 A cores. I was given a CPU at 31 in it. Can you guess who is on worker three? Pod is running Let's help you cube CTL get pods - in the namespace -o Wide which is our work load generator He is running on the worker's th is getting killed it works it works.

It works, it works, so load is generated. Going right now by doing get pod get VPA Look 109 it's been 10 minutes we VPA vape is playing I hope you get vertical Pod Auto Scaler must have been understood. Horizontal Pod Auto Scaler is also understandable. There is also a cluster in the workload All services have been completed in networking also Ingress Network Policies in EKS I will teach you, everything is in storage too. Pas Volume Claim Storage Class Conf Map The secret is all in scaling a schedule HPA VPA has become tall resource Done Limits Done Probes Done Node What is Affinity or Node Affinity?

Watch Affinity Meaning of being attracted to someone What is called affinity is For example, if I like it too much If I feel thirsty, this water is enough for me. It becomes more precious, so I will be more Affinity to I will have more affinity towards This is if but how do I find this thing I am using its characteristic that I. If you need to install your pod on a node If you want to schedule it then about that node.

Please tell me a little bit that the name of this node is equal to this or type e equal to this key e equal to this resource is equal to this so if so. I will show you pod's confession. Like you write node affinity Kubernetes If so, it has a simple, little old node in it. At the 24th July and 26th July 2024 Go to node affinity then if you get node Affinity is the concept similar to node selector where you can choose which nodes to place your.

Want to run a pod? You can put it like you would for your pod.

[Kubernetes] Resource requests, limits, probes, taints / Kubernetes Services and networking

Put that in spec, Affinity node affinity expression that This might be matching with the letters that your The pod is that zone, in this zone is yours. Node is in this zone so if it If there is a match, then your pod will be in the same If the schedule is done by going to a particular node then. It is also called node selector. It is also called affinity, so this is Unit Node Affinity Tense and Tolerance Ye Both of these work kind of similar But what is the use of tension and tolerance?

If you do not need to deploy the pod on any node Any node needs to be scheduled If you don't want to get the schedule done then you can tent it. You do it, no definite, it says the opposite, I want to get the schedule done on this only. If I want it, it becomes your node. Affinity to HPA VPA node affinity tense By prop tolerance resource quota limit we People also finished scaling and scheduling. If we have done this then 1 2 3 4 pa units are ours The next unit is over now, yours What remains is cluster administration.

Control whenever you read R back or Role-based access control is simply this: that the resources of Kubernetes Such as Pods Deployed State Full Sets. Services and so on are yours to access whether or not there is access to a role is formed, a role is formed like ours The role that comes with vegetables is the same. I still have Kubernetes Just like you had a house in the cluster. Just as this was your house, so too was Kubernetes. There will be a cluster and within that cluster, as many There will also be users, what can those users do?

Goes to Role Base Access Control You need to read two things. Service accounts service accounts End is the second one, user remains the second one Your User Service Account means your remains of temporary user type and that In a service account, you are basically a user. What do you do with let's say I am a user? If I am a manager then a service called manager account or I can have a If I am an administrator, then the administrator name can have a service account as well.

[Kubernetes] Namespaces / RBAC and security in Kubernetes

Whatever name you enter, it's a service account. Maybe treat it as a user only user This happens at your entire cluster level. This is your service account. Namespace can also be at the namespace level It can also happen at the level, but your users occur generally at the cluster level Now at the entire cluster level in the same way. This increases your namespace level. Roles that are deployed in this namespace can I get it? If I can do these roles and that service To assign this role to an account, you need Roll Binding roll binding and similar There are no roles at the cluster level.

Cluster role cluster role and that Cluster rolls for you binder Cluster Roll Binding Cluster Roll binding so when you read r back In simple language it means that Your User or Service Account. If you can control access, You can do it on two levels, either namespace level that resources within a namespace Put it in and you will access that level. Or you can do it at the entire cluster level You can cluster it if you namespace If you are doing it at the level then you should read it.

Roles and role binding if you cluster If you are doing it at level then cluster roll and Cluster role binding so this is your simple If there is a concept then now we are giving examples. Role base access in that Create a service account in the control namespace will create a role will create a role I will also make the binding and then I will tell you. kubectl get Nodes Mast Near You All worker nodes are ready. All things kubectl get ns to pass Apache namespace is MySQL.

Namespace is kubectl get pods in the namespace: What do you see in the good namespace? The Apache deployment is running the same way. Which we were looking at things kubectl delete -f dot to - What happens to the F dot? All the risos All the courses are in it, burst If deleted, delete the namespace Apache It is done and the deployed has been deleted properly. Whenever you do Role Base Access Control it Do you understand as a concept this is also very important to you?

[Kubernetes] Namespaces / Pods

It is important to understand that kubectl Auth Who Am I that who am I kubectl What happens if Information about the current user If I get it, my username right now is Kubernetes. Kubernetes admin and I am in the group cube Adium Cluster Admin System Authenticated So my username is Kubernetes. Admin now at Being a Kubernetes Admin I can see what I can do I'm using kubectl auth so whenever you If you do R back then you will get Auth related information.

I have to run a lot of commands like kubectl Auth can I cube whistle th can I mean what I can do this can I get pods. Can I get pods, yes I can get pot what do i do now cube CTL Apply -f namespace so I'm going to create an Apache key namespace. I will create space, now I will see am kubectl can auth sorry auth can I Get Pods in the namespace, so like I'm an admin and a. I am very upset because of the meaning of admin user.

I can do anything by entering all the namespaces I can because I am an admin user Cube whistle at can I get pods yes now Do what the problem is in this? It's like you can do anything URN Administrator, you are a free bull, do whatever you want. Sometimes you need to uninstall different applications. Has to be made and you don't want every A user can access the application For example, think that the name of the good I am deployed, look I am deployed.

Deployment is so strong, deployment is very Next Level and Ye Production Key Deployed think so if I do cube CTL apply -f deployment YAML enter Deployment The Can I Gate Deployed in the namespace. Apache yes so being an admin I am everything I am able to drive but tomorrow if someone else of mine The user came and that too it was our production. The intern has come so if I asked can I Delete deployed yes ch it so risky so I wouldn't want anyone to delete something like that.

If I give permission, what will I get for this? Let's zoom into the meta data Name Apache Role Now enter Apache Role in it You can do whatever you like with Let's I want to make a good manager.

[Kubernetes] kubectl commands / Kubernetes Services and networking

I'll put that here, namespace, so This role operates at the namespace level. Has Apps S v1 R Back Authorization K Come on, this is the starting part. There is a group and what remains It would have y ye above take ye group There are many such groups,? This black group is the one in which Only v1 is written, that blank group A S v1 means the workloads For the deployed deployment that seems to be. There is also a batch group which is cron If you feel like doing a job and all those things then it is like this.

API groups are available now, you can also resources you can provide resources that you What access would you like to give to this role? I All this about seeing deployed and deleting I would like to grant access to the pod I would like to give, I would like to give service access to the service so much access. I would like to give so many resources now comes Verbs Now Verbs Very Important It happens because whenever we cube We write CTL commands,.

What do you write kubectl get pods kubectl apply kubectl delete kubectl Watch Understanding kubectl Create kubectl apply kubectl patch, these are all verbs we use So basically what I did is I took a. What the roll does is that whoever has this roll That will be the deployed pod service. Can apply Can delete can watch can create If you can patch it, any user who has good This particular role will be called Manager. If you give me a role, I will make it for you now.

If the role is ready then I will apply for it. kubectl Apply Now f roll now I want to get this roll which is my can delete the post can delete the post Deployed can update all these things. If I apply this then I will get a good manager. Roll Created role in the namespace Apache to Right now I have a role called Apache Manager. If anyone needs a binder, I will use this I'll give you the binder, but I need it first.

What will be a service account? It happens that he is basically a user, so v I am service account dwa a what should i do now I will create a service account.

[Kubernetes] RBAC and security in Kubernetes / Kubernetes Services and networking

Your name will be Kind, Service Account service account will be its kind and Its API version will be v1 Kind Service Account AI Version v1 Meta data will now be my service account. What do I want to name it good user I created a service account named User I will give you what is active on my namespace level. It will be, now I will put it in the namespace. I also put it in the right namespace. Service Account Escape Colon Dk Enter Cube CTL Apply -f Service Account Service Accounts Achi User There is one created fault user and one good one If there is a user, then I have two users created.

There are two users, the default user, so our Remains the default user and is a good user Now what about me? The user can get this particular deployment Let's see if we can kubectl Auth Can I kubectl Auth Can I Get Pot in the namespace yes so my Kubernetes admin user. That is, get the pods inside the namespace I can do it, but if I do it here High as it is equal to good User Hyph Han Edge means kubectl what.

I can get the pod namespace in good shape As a good user no I can't because My service account is just CreateDrawAnaNavigator Roll Binding Dull and here I create a binding. I will give it here, I will say kind role Binding role binding that it is such an emul file in which I will take my user I. I will take my roll and let both of them go by Bindi Now they make the API version which It will remain simple and backed.

Authorization k iv1 this r back authorization ai The version remains, its meta data will remain Name Roll Binding Roll Binding Now role binding whose manager's because Apache what is this role my role Apache. If it is a manager then the role of a good manager Binding is fine, the role of a good manager The binding is done, now its namespace which It will remain, what a good application. Get this particular cool namespace for subjects Meaning, what kind of people are evolved in this?

The first one is my Y Evolved What kind, which kind is mine, evolved?

[Kubernetes] RBAC and security in Kubernetes / Pods

Service account to but usme here it's a user I'm a user here If I create a kind user then who is mine what is the user what is his name. The user name is apache user apache User and API Group What is API Group Sorry API Group, this is your R Back's The first part remains that it is known as API. Do escape colon null key enter kubectl Apply Online Roll Binding and Enter Now What is he saying error passing role Binding Converting Line 10 Did Not Find Expected Indicator I AM Line.

Number 10 for role binding what have you got subject to this Oh Or this list which is there, generally it is inside like this It remains like this, it remains like this and your role. Ref will also be inside it finish like this If it remains in the subject, kind of this role reference In this you just have to keep the indentation inside Escape Coil Duck Enter Apply Roll Binding. R Back Authorization Apache Manager Role Binding Get role binding in the namespace Well, now look at the Apache Manager role The binding is based on the role of the Apache manager.

With I gave you a binder to bring vegetables or whatever. We have just named it Apache Manager Apache Manager Now this good manager this is my apache user With this Ba has come with the ending So now if I very carefully cube. Given a role called Manager and What is the access gate to that manager? Can delete the pod can delete the pod Can update so on now tell me Will it be able to get the pod no cube whistle I can get paid in the name of what I did.

kubectl Apply High F Service account achi user to kiya cube seat The Can I Get Pa namespace Age Users do something that This is ours, what is its name, this is the role, this role. Let's take a look at YAML and see how it works. The pod is right, let's try one thing. Gives one Atric and here we kill a hash. Just to check if my user judge getting some Kind of access and not then after that If you filter it out, double escape colon.

[Kubernetes] RBAC and security in Kubernetes / Kubernetes Services and networking

What I will do is kubectl auth can I Gate Pods As a user Apache user No user dot Well, what do I feel because this. If you are an Edge user, you will get a little something in it. I have to give a namespace as well Let's give it try kubectl Auth can i service account so check it once If you take it, what do you have to pay for it? kubectl Auth Can I Get Resource S System.DataServiceAccountNamespaceService Account name and then namespace let's follow this Let's try the System Service Account Let's try this: kubectl the can I got paid.

Can I get it read this complete statement The System column service account is currently in service. What is the name of the account good user now Next Apache User then this Staging What is staging, by the way, let's look at it in this What is staging in its context? [music] Staging I feel it's not a namespace Yes, you are watching Staging Is a namespace. There is a namespace after the service account. That's why they have given staging, so our The name of the namespace is Apache then what will come In this, the service account name is Service Account Staging service account name is not the same as before.

Service account will come then Apache then service Account name then namespace service Account name then namespace Apache is still no cub whistle th can i Get Pods As System Service Account Apache good. User System Service Account Staging [music] API group added to Tiller rules Put resources into API groups happened I have put all the namespaces in. The Can I List as System Service Account namespace: Username is correct. The Cube Seat: The Edge Case Mistake to Avoid Being aware of basically user can be name Similar Sytex Accounts and Can Trick It is me A System Te default.

What I'm trying to do I am looking at its user as apache user am I my Apache Why is CanI not giving user access? Gate pod Gate pod Gate pod Vase pod In The namespace good No, if I remove this, can I get it? Pods as User in the namespace Apache Come, I have just found its particular I saw the syntax, even in G, there are many here.

[Kubernetes] Pods / Namespaces

There was something there, but finally got the perfect one. Syntax it is kubectl auth can i Get pods as a user then your name Space will come, what were we doing before? Cube Seat Loth Can I Get Pods As a User Well then no one was coming, I have to say then we people System google2 auth can i get pods as a user. In the namespace Apache yes why Because he has a role Now Cube goes a little deeper. CTL Gay Auth Can I Get Deployed in the form of a user when User in the namespace Yes I Can Get it, but now look at the small trick in it.

I did a great job in the role I played. Sorry, my role was YAML, in that I did everything I made a fire, now see what I will do I know I'm just telling him a few things. I will allow it as I will allow it You can only get the pods Only the pods could be gated and The API group of the pod remains blank. If you live there, then look at this good manager now. What is the role this good manager has a role that he can gate pods Datchi L Auth can i get pods as a user ach user In the namespace.

I can get pods but can I get deployment Can I Get Deployed as Apache user in Apache namespace This Deployed yes why is it coming we I M Roll. Why tell me because you have given the role Edited but kubectl apply -f I did not apply for the role, how is that possible? It will work, so whenever you edit the role After that apply it, it is confirmed now you kubectl can i get pods yes i Can get pods because in my role.

Getting the pod is allowed but can I get it? No, deployed because it is not allowed. I will go and give my a deployment too. I will give it to you, you can deploy it here and Services both take and deploy which That API group comes in the API group Generally if there are apps then I have this API group. I unlock all and purse resources I unlock deployed services.

[Kubernetes] Autoscaling (HPA, VPA, Cluster Autoscaler) / Pods

Verbs in just and only unlock the gate I have done escape column double enter kubectl I forgot to auth again, please apply it. First apply role dal to role base access Control is configured, now do kubectl Auth can I get deployed as an Achi user yes Can I get it? Can you get a Service as an user yes but can I get HPA Horizontal pod Autoscaler am I HPA no, because it is not allowed So if I roll in the dump.

Go and click on Deployed Services here. A small HPA on the side or If I also add the horizontal pod auto scaler I should also put HPA and press Escape and enter. Apply to and can I get HPA now Why would the name of the BHP be something else? That's why it may be possible, but what do you do in the role? Go ahead, but yes, let's try a little more test. kubectl Auth can I delete pods as a user Apache user the namespace Apache can I Can delete pods Cannot Why because the one with delete in verbs I haven't put it yet, I'll put it now.

You would have just entered both apply and delete. Escape colon, double k, enter, apply. Role to Role Base Access Control Manager Confirmed now can I delete pods yes then This was your role base access control that you Give that roll a serve Binder with Account Two and One Bar. Binder given everyone applied then you kubectl auth can i access it from You can check the control now like I cube CTL Auth Can I List pods As an Apache user in the namespace Apache.

I can't I can't because I did not add the list here but like I just add it to the list and I apply for the role and I get it instantly. I can do my list pod that is your Enjoyed the concept of role-based access control That's how you learn Role Based Access Control and enter your service account namespace. You can make rolls at the level now that you monitoring logging doing things at that level Then you need to specify cluster roles and cluster We have to learn all that role binding, so let's go.

[Kubernetes] Monitoring with Prometheus and Grafana / RBAC and security in Kubernetes

How does work happen at the cluster level in R-Back? First of all understand that cluster The work we have to do at this level is Why do I have to do this with the whole cluster? Looking at the metrics of the entire cluster Monitoring the entire cluster's logs I see I would have wanted it at that level. I understand that whatever our R back It happens if you create a service account in it. You are like an user, MySQL is a user.

This gives them access only up to the namespace level. Should get pod delete pod and so on but when If you create a user at the cluster level, Cluster Role Cluster Binding This is all you need This happens because you need to run the entire cluster. To fetch logs of pods across the cluster Have to raise the entire cluster's deploy So what should I do now? I can give it a I will take it to the cluster level,? Cluster level I will work for role base Access control in an Apache namespace We will not work, MK DI R Dashboard Create a dashboard What will it mean for your entire cluster?

Give it to me, what have I done to you? The cube that is the cube starter repository In the starter repository I kind A great full of clusters how do you can setup the dashboard, I have also done that. If it is given then what should you do first? You have to manifest your dashboard If there are files, then they have to be applied kubectl apply -f ye Particular files are recommended What does this do to your dashboard? Once the manifest file is created, I'll go to the CD dashboard and everything is right here.

Work fine so kubectl Apply -f, this is the name of Recommendable Board. How did you become this GitHub user content? Kubernetes Dashboard of these Kubernetes Official web in the official GitHub repository It is bringing the dashboard from, I am picking up his email files. Service account created in Kubernetes dashboard Created a Service Kubernetes dashboard named Secret made some then conflict map made one Created a role called Kubernetes Dashboard Roll made that Kubernetes Dashboard Roll. What access should the cluster role have?

[Kubernetes] RBAC and security in Kubernetes / Kubernetes Services and networking

Created a Kubernetes dashboard Also created a cluster role so that it could be used for the entire The cluster must also have access to the role Binding the cluster to this role. Role binding done for this cluster role Then I created a deployed service and If we make this then basically if we Check the syllabus so I have given you the namespace. Taught how to create Taught how to create a service account I have to learn to make a service and make it a secret.

Taught Confine is to teach role base access I have taught you control, I have taught you how to make a role. Binding is taught by deployed service I have taught everything except deploy cluster roll so let's just cluster roll now I'll teach you how to make it, so let's go now. We are a dash Board admin user dashboard admin user If you make a Dwa then I will be a grave nits I want to create a dashboard and Create a user to access.

If I want, what would be the name of that user? I'll kind of create a service account. If the API version remains v1 then it will also Fine Service Account User You Can Use Any I will create a service account only for this. In the meta data I will say the name admin user so an admin user named one i am creating a service account named Space, it is very important for you to escape. Colon dot which is the namespace, you need this namespace daani namespace Kubernetes Dashboard So Cube Kubernetes Dashboard Name If you want to enter the namespace of the Vim dashboard.

There will be a namespace in the admin user. Your Kubernetes Admin Dash is fine. You have an admin user named Service account has been created, now if you want, you can do it Enter this sign three times in the file and. You can also write another email file in this file. Just one of many projects There is an email file and as many as you Deployed service ingress all in one He puts it in, that too works, it Complete fine if you get it in one go.

I want to make it so it's completely fine now What am I going to make here?

[Kubernetes] RBAC and security in Kubernetes / Monitoring with Prometheus and Grafana

Cluster Roll Binding, why did you roll the cluster? You haven't made it, why are you making role binding? Because cluster rolls so Kubernetes is already ready for dashboards Cluster Role Already for Kubernetes Dashboard It has become what I want for myself The dashboard that was available for the user. Admin User This user with just that role Binder is there so I need cluster. Role binding now has the same API version stays r back dot authorization case iv1 this is the API version meta data of this What will happen, you can name it anything.

Let me put the name in Admin User binding is for admin user Let's get the binding done, give me the namespace This has a grave of Nits dashboard take namespace now its role ref. What will be the role reference and what will be the subject and role ref so What will be your subject? Kind of the one above, this is a service account or Such a name to copy and put here What is the name of the cluster admin? Your Kind Name and Yes API Version API Your version comes in Sorry API Version API Group comes in API Group This will be yours only after R Bug authorization.

Enter, this role is the cluster role. I have to attach that cluster role with me. What is the name of that cluster role? How to get cluster admin kubectl Gate Cluster role Cluster role in the namespace Which Kubernetes Tus Dashboard So Kubernetes what is the name of the crystal roll in the dashboard. There are a lot of names, a lot of roles. Find out now what is the name of this role There is too much roll, its cube is subtle Get Cluster Roll In The Name Qnet Dashboard QNET Dashboard has lots of The tap is yes.

It is written here that, which one If you want a role, the name of the role is Cluster Role. The name is cluster admin, so this is from here. You can also copy the roll ref and get the shortcut here. API Group Role Based Access Control Authorization Keyed Kind Cluster Role Because I need to enter the cluster role name Cluster Admin So a person named Cluster Admin.

[Kubernetes] Monitoring with Prometheus and Grafana / RBAC and security in Kubernetes

If I have all the access then I will put it in Escape key Del key Enter key Ctl Apply Now Dashboard Admin User Enter Now what is he saying The Cluster Role Binding Admin User. Bidding invalidates the Subject Zero API Group Unsupported Lr Back Authorization Case Ao Supported Values Block So let's see what the dashboard admin has API group is in Kisam Subjects R what did you put back In this. Take the subject, what did you put in the subject? If there is none, please put a namespace.

We people So here we are Enter only one namespace on it. Let's stick to the syntax name Space Kubernetes Dashboard It's Done I became a dashboard admin user and now I am using Cube. CTL Apply -f Dashboard Admin User Service Account Admin User Unchanged Cluster role binding back to admin user Binding I have got the role, now what will I do? I will apply this thing, apply this Now I look at my cluster I still have time to drive it. For this you need a kubectl named proxy.

A command has to be run but access to it kubectl then find the token to In the namespace Cube Tudge Dashboard Create a token admin user so what do I do My new admin user. I will create a token for that. When I got entry, I took such a big token. Copy this token here by doing this Let's take it, copy this token somewhere. kubectl proxy kubectl proxy nfn Address E 0.0.0 enter Now starting to serve on 8001 whose The port is open, so now my dashboard Where I will monitor all this.

I will receive it on port number 8000 so let's go Brothers start looking for it, so this is ours. Instance Security Groups of this instance Let us go and make a new rule in these bound rules. I will put a rule that Give me access to my 8001 anywhere else Save it, 8001 has been granted access. The it instance that I have running has a Get the IP address, here is the IP address End Port We need to see what's going on at 8001.

Can't provide a secure connection This is gone, now what exactly should I do?

[Kubernetes] Monitoring with Prometheus and Grafana / Kubernetes Services and networking

If you want to go to the location then I have done it here. Exact location dala / ai v1 namespace Kubernetes Dashboard Services Kubernetes Dashboard Proxy allows you to Particular Location Go will have to Enter Side Can't Provide Secure. The connection is fine if it is a secure connection is unable to give, it means that what The port number is also 8001 as 8001. I will need a binder, let's see what's going on. To serve on 8001 now let's refresh this Yes, he is not giving it here, why?

If he is not giving it here then he is giving it here kubectl proxy 8001 address zero The one and only kubectl Proxy Try kubectl proxy what is he doing now starting. To all on 127.0 is serving on localhost Not serving on localhost Papa, I don't want to get it done on local host. Starting to serve local host then let's Looking at this, Cube Seattle Proxy Expose kubectl Proxy Expose Service Expose IP No How to Use kubectl Proxy To Access QNET API cube whistle something like this is.

Let's try this: port, address, Ch Let's Give It Try Cube Seal Proxy Port Number 8000 Address Equal To 0.0.0 Kube Seal Proxy Port Run on it And get exposed at this address, let's go. It is still showing the same starting to Serve on so much showing but not very sharp of if this. Will work 81 OO for Biden tch means 8001 is working now what should I do with this I have to go to a particular location and here slash daal boom ho for.

For Biden Cub Proxy for Biden Cube GTL Proxy for Biden sees If yes, then it is saying that it is able to access Dashboard by Cube Seedle Proxy Absolute We are also getting the same error. UI read cube seedle proxy session Regarding using Cube Seedle proxy Port Accept Host A Accept host dot astic aise karke kuch to bole It's coming, 21 people have given it a thumbs up. Let's get started with the Kubernetes Tus dashboard. Let's put the token, we have put a big token It is beautifully saved here.

Here is our token, let's copy the token. Go here and throw the boom token.

[Kubernetes] kubectl commands

What happened to him why sign in No Insecure Access Detected Sign In Will not be available access dashboard over HTTPS and Using Local What should I host, HTTPS or Access only via localhost. I have to do it, we will go to the local now. Local Pay kubectl get Nodes Only No, the cluster is not created yet. Let's put ours on git, now run it locally What will happen, whatever happens, get the gate It's great here, we have our own MT repository.

I will keep playing videos for as many hours as I want. How do I push the code to GitHub? Remote Add I will do origin ad origin and this is my This is the URL, I will add it but this URL Why will my access be denied? I'm going to take an access token here. Go to my GitHub and go to GitHub's settings. I'll go into the settings and then I'll go into a developer settings and Then I will create a personal access token and I will call this personal access token generated.

I will expire access to the repository Why should I generate this token? I am generating this token because Why am I getting this from my local? Pay this token to gab.com from the machine at gab.com But I want to access the

remote control. Git is added status, add all these things, get d dot all What did you say about adding Git Embedded? Repository o Git This is the status which is not in the middle A repository named Auto Scaler So, I don't want that CD, this is the one If the autoscaler itself is also a repository.

If you don't put this then RM Hyner Auto Scaler Remove auto scaler git status be Went and we made MySQL similarly. Our Django Notes app is also a song. Repository LS CD or Joe Django Notes If you have CD then you can write notes on it. This is in this RM -R dot git dot git If you delete that folder, its git If everything in the repository is also removed, then Git Status is fine, ach auto scaler hum delete. The gate Go L S CD Apache Auto no scaler ach auto scaler now If not, then git commit hynum aided Case in One Short Manifest is in the manifest file.

[Kubernetes] Monitoring with Prometheus and Grafana / kubectl commands

I am doing git commit and git Push Origin Main Ho Now git push origin master because local If the branch name is master then push to master. Because it is a different part of a different thing. Dashboard Admin User Let's get your work done That's all I had to do for this file. Full of git let's clone this now What I'll do is I'll go to my local and I'll go to the CD document and work. What was I supposed to do, yes, first of all me You had to create your own cluster, so how could you create a cluster?

It becomes a stop Go to our confine in kind cluster Fig this Joe Conference So I am [music] Confine is how I create my cluster I want to Kind Create. The cluster end will remain the name within that cluster TWA Cluster End [music] own Confine y this will be it Assuring node image Preparing nodes Writing Confession Starting Control Plane. What will I do with this cluster of mine? I will remain in this with my Kubernetes I will make the dashboard really cool for you.

I will give it to you, it's done kubectl get notes so these are my It is not ready now, it will be ready after some time. I have to put all the files, now they are all ready for me. Go to the dashboard folder and I created a dashboard admin user and he If you want to apply then kubectl apply naf Dashboard Admin User Service Account Admin. User Created Drawer Navigator / v1 / where are you getting this from here I have given the complete path here.

Is / ai v1 namespace proxy so this I'll copy it and go over here. I will put it on my enter, it is going great no error no nothing now i use it I will pick up the token Here is the token, copied the token and. I slammed here and sign in boom you have Mast der in nothing u display because If you haven't made anything yet, let's make it quickly. What will I do if you give me this cube By sending CTL proxy to NAM I'll keep running it in the back end and make something what will you make.

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs / Monitoring with Prometheus and Grafana

What will you make a CD of? If you want to make engine things, would you make everything? Your set of reps It is running and deployed is also getting ready. How many containers are being created, so every thing is happening now these things that are happening All the running that is happening right now is this thing You will see it in the dashboard. Refreshed it now nothing to display Hair is speaking but if it above namespace If you change it and put the engine here then You can see your deployment is also visible.

The cron job is also showing pods two Succedence since which second is it going on Confinet and what is your ingress What is in the engine ex note ingress. There are things, everything you can see, how much This is so true, it is next level. Workloads if you want to see them The status of this is something like this that D Man Your sets are running, we have seen everything. What's covered in the sidebar Let us look at the workloads In 2018, we used cron jobs in workloads.

Set up the daemon, deploy the job Pod ReplicaSet Controller State Full Set Service Me Ingress Service of Ingress Classes Conf mapped to Key Config persistent Volume claimed secrets Storage Class Cluster in Cluster Roll Binding Roll Cluster. Events not yet done namespaces Network Policy is not yet announced Nodes persistent Volume Role Binding Role Service Account Custom Resort Definition I haven't done it yet, but if you pay attention Look, you've covered all the concepts. I have covered all the concepts, almost 99 But all the concepts are yours, enjoy it.

I am telling you how you are one of the best Kubernetes I mean, it's been seven to eight hours in the same video. It's a different matter but you have done a great job in just one video. Matrix Monitoring Logging All Viewed CRD cool I'm going to tell you about But how do you feel about it here? Till now you have done a great job here with

your cluster. Monitoring Kubernetes Dashboard Cluster Monitoring Logging in, see Li, obviously we are the people project.

Monitoring Prometheus and Grafana in We will do it together too, but for now, this thing Enjoy the workloads Enjoy yourself, you will get everything.

[Kubernetes] Pods / Custom resources and operators

If you want to check the logs, you will need to check the logs in each pod. Let's say this is a pod, which one is your Nginx? Take this pod of Nginx. All the things that happened in this pod You will find all the events logs. We have seen it very well that how do you install the dashboard of your Kubernetes niches You can make it, it looks very beautiful The dashboard is also a luxurious pendant pod. Can you make a dashboard of Cube Kubernetes?

Let's move on to the next concept which is Let us see some of its advanced features which There are custom resource definitions, well, these too It comes in advance only, let's see that too. If we talk about custom resources First of all you have to understand the definition If you have to find out what this resource is, then Kubernetes has pods of services it occurs. All these things are deployment Isn't this what you're playing the whole game on? All resources are pod is a resource Deployed ingress is a resource now all these Resources are something that helps governments I already know if I do it, Cube.

Ctl create pod or kubectl run then He knows how to make a pod cube Seattle create namespace so he knows this I want to create a namespace, but what is this namespace? If it happens then you need to create such resources which is not in cubic Kubernetes by Default but you have to make your own From custom resources let me create a database Where I am no longer. I am feeling that I am my own Kubernetes I am creating a resource as if someone had code must have done and created a pod that You need to write containers in the spec for this.

If you have to put images in containers then This is a structure, a similar structure. If you want to make it for yourself, custom Resource definitions seem to be a bit more Tell me in detail, I will tell you like your pod It happens. Like you have a pod, what do you do in that pod You can create an API version of that pod. Which is v1 but if at the same place But what happens when you are deployment?

[Kubernetes] Custom resources and operators / Pods

Deployment When deployed, you write the API version. AS v1 correct so what are these apps here Why this app is not basically your group The name remains and the API version remains yours. The API version of this resource Pot remains a resource as it remains Its group remains blank, meaning here There is nothing, only v1 and v1 what. This is a version and inside it you what do you do inside it You write spec directly, spec means Specifications of the containers you will be making Then your names will appear in the containers, then your.

The image of the containers will come and so on you can see this You make all the things but this structure Now you have to define it,. I created my own resource from Let's Where the Pods What Kind Happens The right kind is pod, not even s pod If it happens, what kind of pod is ours? Let's make a kind of pod. I have to create my own resource, let's Batch Me for Myself Kubernetes has to create its own resource I don't even know what a DevOps batch is but.

Still I will create my own DevOps batch where How will the pay spec be that the batch name again Duration of the batch, then which batch Walking on the platform What is the platform then batch mode. I can't create myself out of thin air like this specification should be shared with this resource. The resource needs to be defined and this is Custom Resource to Custom Resource Definition There is a structure that you can use in Kubernetes You can create your own resources, let's do it.

If there is then M KD I R CRD a CRD Let's create a folder named cd CRD Now here's the Pei im a I DevOps CRD a DevOps' own custom resources. I will create my own custom resources for DevOps now. Whenever you create a custom resource definition You can create custom resources from that definition only. This is the group name, what else can you add? You must have noticed that whenever you create a pod When you make it, you can make pod PD pod but when you read you write cube CTL Gateway Get pods s plural same as in names you.

[Kubernetes] Pods / Kubernetes Services and networking

You can write plural or this particular which What will be the plural name of a resource? Will DevOps Children What will happen in Singular as you saw The pod's pods will be deployed. If it gets deployed then what is singular DevOps Batch Singular and Kind What Just like pod has a capital P, so do I. Write the DevOps batch of the DevOps batch like this in capital letters I can write capital b this is me What I'm doing is creating my own resources.

You must have also noticed that whenever you do Yes, service and whenever you create a service, but When you get Kubernetes service so kubectl get service but now cube Do you also do CTL Gate SVC? If these are short names then here you can enter short names. You can also enter names like Passion Children And WS whichever you like short name and you can put now you want this whatever Custom resources are those of a namespace If there is scope inside too.

Scope Name What will be the name of the scope Spaced namespace that you this DevOps Baches name Use the resource in any namespace can you do versions version can you version. You have given the first Now new versions are made from time to time. I can also store it with storage true so I can basically make anything myself. I can also create database units of You can also store your own data in Kubernetes. You placed the object, now that object Some properties There will be properties there will be properties Like there is a pod, there is a speck in the pod What are the properties of Spec?

Your containers names of containers and so Non-deployment what remains Spec Reps Labels and selectors and accordingly His logic is written in the same way as I am I write about the properties Spec Type object. What else do I have to do, type this The object will come in spec, then what I want to put it in this I want to enter properties and see Open API. Description This is the name of the DevOps batch Name of the DevOps batch now in the same way me and what do I want to add name duration.

[Kubernetes] Custom resources and operators / Namespaces

Platform mode so I want to enter the duration let's type String No duration No duration Description This is the. Duration of the DevOps batch so this DevOps batch duration ho mode that I this DevOps How am I going to teach the batch? From top to bottom I did a custom Created the resource definition that Kubernetes had previously I didn't even know, but will he know now? CTL Apply Han Adeps Sdi Enter Now What is the error saying when retrieving? Current Confession of Resources AI Extension k1 Resource Custom Resource Definition Group Version App Extension Kind Resource Name namespace from Server Dev Resource may not be mt resource name may.

Not be MT so the resources we have given Custom Resource Definition I may have entered the name wrong. Custom Resource Definition (DEFI) Definition, wait, we'll look at this. Kubernetes Simple Kubernetes Custom The resource definition now contains Go ahead and try it yourself. This, may be the name is not correct Let's Cube Seal Apply Again Error When retrieving current confine resource API extensions for correct resources Custom Group Versions on Kind API. Extension Kind Custom Resource Definition name blank namespace blank from Server Dep CRD Resource May Not Be Mounted Let us once again see that this What is the capital D of metadata?

Gave I have entered meta data escape colon double k enter End Cube Seedle Apply Custom Resource Definition API Extension DevOps Batches Due to a mistake created by your A lot of time can pass, it's over now. It becomes a resource, however, you should also have one of your own. Custom resource has been created, now its benefits What is it, the advantage of this is that you can create a DevOps custom resource using be like what kind of it would be like you.

You make pods, you make groups and ingress You make all this, you make a DevOps batch You can keep its API version,. What should be the name from let's I entered Passion, I don't know how to sell it now, I have no namespace If you put it in then you can put it in the namespace. DevOps Name Batch Name API Version Use I entered his name and then I put everything and escaped the call.

[Kubernetes] Custom resources and operators / Kubernetes Services and networking

DC Enter kubectl Apply High F Dep What is Sarmal CR means Custom Resource CRD means custom resource Definition Enter batch as CreateDrawerNavigator Gate Now I can put it here. DevOps Batches Enter Passion Batches Now I What I can make another batch cps2 now I let Made a new batch of cess c2l which I named Put passion batch lets to 10. I don't call it simply passion, instead I call it passion. These operators have also become our CRD. Used for programming as in Golang Python has operators and these What are the operators like now we Created a custom resource definition in Debs batch Now to program it, schedule.

For that there are operators like Argo CD operators are Prometheus operators, then all these operators so you can use this custom resource Definitions that create custom resources You can program it and. The Kubernetes API is also very important for It happens because like all these pods and Your deployed ReplicaSet Sets service ingress All these things are available to you via API You can also access Python's. API pata OPF so If you write OPF Python then it will Python Framework Two Right Queue Operator Python Framework So there is a Python framework that you can use to build Kubernetes.

Examples of how you can write Nits code I'll go and look at the examples. Tasks you imported to Cope F D Man I made it so basically it's too much In Open Source and Kubernetes Generally, you have to make a contribution This is not asked in interviews. If you don't ask what, then CRD is not generaly You are asked, API is not asked generally Operators are not asked for you Yes Helm will be asked Service mesh asked You will be sent to security.

Security Standard Image Scanning Image You have to do the scanning, you can do it via tv you can do it by Docker Scout if you can see here you go to your Docker. It's fine on desktop, your Docker on local. The desktop will be on that Docker Desktop. All the containers that are running late clicking beyond the Minikube or whatever it is You can maximize it here and you can do it here Docker Scout Likhit Sakaata Ho And You Can Do.

[Kubernetes] ConfigMaps and Secrets / Pods

Image Analysis with Docker Scout You can do any Please upload the image, I have mine. Grafana is Grafana from Let's or someone else take it kind of note on which we can deposit our cube Running a Kind Cluster of Nits Vulnerable, you can see that there are six Critical. Severe is 15, Very is 15, Medium is 29, Low is 15 Savarin is specified and by doing this you can have fun You can also scan your images using. Docker Scout is all you need in Image Local And you can pick it up from here or you Also go to your local and Docker Scout by the way.

It does not come on this, there are plugins for this. It runs only through plugins, so if you now You can study it from your Docker Desktop Docker in Docker Desktop Scout is enabled and this allows you to track your. You can also scan images over the network. Policies Secret Encryption We Have I saw that base64 is the code and base64 D-code: These two types of encryption exist You can run any network CI/CD integration in policies which will be a complete project with GitOps.

I will get cloud managed services done in it. In this we are going to cover EKS AS And GK will keep this for some other session. Monitoring will be done in the project, it will be yours Deb Trouble Shooting you saw How do we debug in kubectl? How can kubectl provide logs There are resource usage analysis top nodes like We did kubectl here Top node, so here we have resources I get the utilization that I have control over. The plane is the top most node then the top Which pod is taking up the most space?

Let's say in the namespace to I have this pod which has a lot of space If he is taking all these things then we have If covered under resource utilization If you talk broadly, we have. Matrix Server Logging Monitoring In Operators, we saw an overview. Home Service Mash Connect API CRD These All We have covered all the things, now what is left is are that your projects and some and Advanced Tools, till now you have seen so many.

[Kubernetes] Helm and charts / Kubernetes Services and networking

We have created all the manifest files. Has made so many things, I know a lot Over Writing a Velminik file Writing its kind Making its template and many other things It remains but if I tell you that there is such a. There is a tool because of which you can do all this manifest You can get rid of the hassle of writing files. For each environment you need a separate You don't have to create a manifest file every time. No separate manifest file for the tool Everything will have to be packaged, so how much?

If you enjoy it, the name of this tool is Helm. Helms manager is Tus package What is a manager like you do at your place? You write edo at gate Install Unzip like you would at get install unzip or at get install docker you at

which that there is a package manager for your System, you do that, what is happening? As you do with your package manager Unzip and install Adicate to your system And all these things get installed like this Similarly, you can see things through the Helm.

You install it using your Kubernetes The manifest file is too much. I did not understand clearly, please tell me a little more. Tell me a little more in depth about Helm happens to be a package manager. For example, for cube Kubernetes you Like a lot of Kubernetes manifests There is a file, you have to make an engine, you have to make my. Do you want to make a SQL or do you want to make your own? You have to create an application from Lets you Create your own notes page or any other page So what do you write about?

Manifest file as you would in your deployed You write that it has become very big, like you In the manifest file you can add your deployed writes. You write your service, you write your you write Ingress you write HPA and As many more compacts as you can have As You can write, you can write secrets. Will you write for MySQL too? You will do it for your own benefit also and if you Deployed service ingress HP of all these Look at the templates, they all have the same things.

[Kubernetes] Helm and charts / StatefulSets, DaemonSets and Jobs/CronJobs

There is a kind and then there is its kind API. Version its API version meta data its Metadata namespace Its namespace End So on then when everything has to be the same then why No, we package it with a package manager. Why not use a package manager and that The package manager installs all of these things. Helm is a package manager that will You can manage your Kubernetes workloads Manage Kubernetes applications can you can deploy and so on today. Let's understand how to actually use Helm.

If you want to do it then let's go to our server and say this is what I'm going to do, so let's go to our server. I will create a folder named Helm. Cd Helm and inside Helm I type install Gonna do Helm charts Helm charts What happens, for example like you Here's the engine, here's the engine, and the engine What's inside is your cron job. Ingress is Persistent Volume is ReplicaSet sets is DaemonSet is service deployed end So on, you made this yourself and that's it.

Thing if you need a package in an automated way If you had to make it with the manager, you would have made it with Helm's You can make it through but for that thing you. Helm you have to go If you have Google 3 then go to Helm's official website Go The Package Manager for Kubernetes This is The Best Way to Find, Share, and Use. Software built for Qubus to Qubus in All the applications that are made are all yours. Can you make it through what should I do now If you want to install Helm then install Helm You download a script to do You can do it and with this script also you can Helm.

Curl f s gumd a to gaye naam a The script will be made for you and that script The content inside will give you this If I get it from the URL then I will come here. I'll go and copy this to my Helm. I will paste it in the folder and press enter And I'll do it then get ahold of this shell I have received the script, cut the message. If I try it, it will All means all functions and all shells The script is written so you can play Helm You can install it so you don't have to do it yourself.

[Kubernetes] Helm and charts

All you need to do is install a small I have written it in the file, now that file You will need to make it executable. From the Seech MO 700 you get its executable Now you will have to give permission for that cool green color Ho L S Han L Executable Too Done, now execute it dogg. A Enter Helm version 3.16 already latest mast aapka Helm is installed if you are using the Helm version If you do it then you will not be welcome Helm.

If you do version then which version of Helm do you think is right for you? If you get all the versions of Go, then help There is also a version of Go written in Go lang. With you can create new help Chart you can create environment you lint You can do it, meaning if you accidentally do something If you have made a mistake then something in the template. You can list, you can push, you can pull. If you can do it with Helm then let's go first what will we do, we will make a Helm If we make a chart then how do we make it?

Help A chart of I from Create Let's If I want to make it, then good I'm creating a chart named Helm Apache Helm am Helm create apache Helm enter then Creating Apache. I made a helmet inside which I am Apache I will play it for you so can you see A folder named Apache Helm has been created for you. Cd apache Helm now if you ls If you do, you will get Charl Charts templates and values and what is it Helm's own creation How did Help create all these things?

If you do tree in it, right now the tree is in it. Otherwise what is a tree, that whole tree Structure tells you where your folders are Let's install what's inside sudo att install tree small command What will make you so beautiful? This will show what Helm has done. I've made things for you and here I am I will give you tree enter so you have Helm Created Charts Fill Charts Then Templates and inside templates Deployment hp.in survey service. Account done so what you were thinking here that If I make an engine, then its You will have to write deployed, you will have to write HP.

[Kubernetes] Kubernetes Services and networking / Deployments, ReplicaSets and rollouts

He already made it so that everything would have to be written down. Thatchi of him that you will get your Kubernetes Complete return of nits manifest files I don't want to write that it is your template, meaning if. You will go to the CD template and click on the LS button. If you do this then you will get all these deployments. If you will do this Deployment so you are not fully deployed. Likhni is the Zoa version of deployment Whatever kind is deployed, it is all automatic.

It is written and its values are Inside such curly brackets include well Held Full Name Labels Include Held Labels Recca Include Values Di Replica Count each selector labels so what is this thing. This is a templating engine and this template In the engine you just get the values provided and those values have to be transferred directly here. I will get injected, then Literally nothing to do if you are not in service If you go to the service department, You literally don't have to do anything with your port.

No, why have these people done this, but You will also need to enter values in the target port here. The target port will have to be written like this and there If you have to specify the target port then that If I do it for you, it's a small. All you have to do here is like you Service D Port is written as Target D Write the port sorry service dot target port Target Port Now What's the Best Thing About It.

You can find the file with the port values. Go and edit it and whatever that port is. Target port your values will reach here I go and add the target post here. Whatever you do will reach the protocol. Just put the protocol that is running and it will be added. Service completed, fully deployed ingress No writing it all down, templates for you will get those templates which have been created automatically. I have to use it, but, these are the values Where does it come from, so whatever values it has Open Values now in this these are the.

Default value use for ach Helm this is a YAML formatted file declare variable to Be passed into values let me say Its deployment replica count should be two.

[Kubernetes] Helm and charts / Kubernetes Services and networking

I wanted to do two reps, so I did two. I wrote down the image I want now. I don't want to have a repository should be used httpd because I have a good server The image for that is available on httpd Pull Policy If Not Present. Meaning, if my image is not presented then pull it or else don't pull it tag Let's tell you about the particular httpd If you want any version from lates to 2.4 then you. You can also add tags here and all this The values will go there and get injected now.

If you need anything else, you can add that too. If you want to create an example service account put create true and for everything like For security port security you can put any service you want. If you want a cluster IP, you can use the cluster IP If you want note port then you can put note port Put it here, for example, the port number is 80. People there named Target Port If I added a field, then here I will put a target Add a field called Port and enter 80.

Whatever you want to add in the template, you Add to the values you get in the values If you want to add something, add it to the template. This will keep getting injected This is how Helm If you want to enable Works Ingress Set true here, any class of Ingress If you want to enter a name, then enter it, any resource. If you want to add CPU limits, you can add those as well. Let's say I want my These run on 100A CPUs, you can go up to 200A.

You can also enter that, basically order scaling. You can configure it for your Helm chart. Those values to your manifest files This is how cool Helm is now What are these four dots,? If I do v 4l inside this v 4l Whatever chart you are making inside this Are you making a good Helm chart or engine? You are making MySQL, you are making it Details remain about what its version is what is his name what is his description what is its type, what is its version and.

API version whatever, P version whatever to enter You can put that here, so double escape colon Now I have my Helm chart.

[Kubernetes] Helm and charts / Kubernetes Services and networking (part 2)

Going into his values, all those values I put what I need for an Apache It is for example apple is good for me what do I need for Apache Where did my httpd go? 2.4 tags are needed and this service is needed If it happens then I have given the port number in the service Enter the target port and port Done Now what I have to do is my job. What do I need to do to package this chart? Helmet has now become a helmet in this package You can import and export You can do anything with this package.

Generally I put the package in the wrong place. I have done this, let's remove it Generally what you have to do is get out of it. This is a good thing, you should take it to If you want to package then lm Package Helm Done Successfully Package Helm so here is your good Helm package. This command was also correct dot means here Package whatever you have. Two but that location should be package inside it. Which is not right to get you out of it what am I going to do now Helm install Helm install my package is done Now I need to install that chart.

What I do is I change my chart. Let's say Lets do this with Apache instead of Nginx So instead of an engine, this is my Apache. Apache engine both are same kind of this Apache which is the httpd server that I am going to give you an example, that is it now. I need to create a dev environment for this. Making it a production environment and create a staging environment If so, I can name it directly. Help install dev and then my Chart Name Helm Enter Can You See DevOps Deployed But namespace Is Its Default If that's wrong, what's the problem with that?

If you do kubectl Get Pods then you will see it in the default namespace 's pods are also running, how accurate is that Meaning you have to enter hardly two or three commands. Padi and your running is good if you get If you provide services then you will get good services. The service is also running if you deploy it.

[Kubernetes] Namespaces / Pods

You also have to run the deploy of De It's so cool, what should I do now? If I want to delete it, I can write it directly. Helm uninstall DevOps apache help uninstall DevOps apache and All the pods I had, get pods all Deleted Gate Deploy All Deleted Gates All the services are deleted which are running of Kubernetes. She is just walking Now why did I delete this? Because whenever you install You install any Helm charts DevOps from Let's as well as his name Space Create namespace This is very important to create a name.

Space is very important, let's say DevOps I am creating a namespace with a good name. Am Enter create namespace not you directly put a space in the middle of the name DevOps and before Obviously there is a command where You create a namespace and the namespace. You also give the name of it, I am forgetting the command But let me just give it a try, so The command is create namespace and Han A DevOps to DevOps name namespace ye use. Will end at the same time install it Time Hyfe Fun Create namespace will also Enter done so what does this do you whatever You are creating your own chart.

Name your chart and then you Enter the name of the namespace you want to create. Add two more flags on the side that - ain create namespace to let's say You've built good servers for DevOps. That's going to go into the DevOps namespace. Similarly, if you are building for production yes then Helm Good for install production Which servers and charts to use There is going to be some good help with the Helm chart. Templates already help in this I just made a list of charts that are tucked away in a folder.

Changed the name of the installation put production apache and namespace Production Apache High Free Fun Create namespace boom so now my If it is deployed in production Apache also If I do kubectl get pod in the. Namespace DevOps So My DevOps's Pods is also running and my production pods I want to run from the let's If there are three pods instead of two in production What can I do directly with VIM?

[Kubernetes] Helm and charts / Pods

I can go into my Helm apache Helm and Go to Helm and enter my values. If you had put two in the beginning then two key In place of three, I want to put three reps. I want to put it, I put it, put it now What will I do Helm upgrade Helm upgrade i just and only I want to upgrade my production. Prod and name your chart which is Help Enter Production Has No Deployed If you are a religious person then you should also upgrade your account first.

You will have to go here and get its package. Whatever is my Ach Helm chart, go to it I'm 16 this is the P version, change it a little bit. I will do it I'll change it a little bit, one escape column. Dub k enter so I have made my chart I

have added a value to it, now I am going to package it. I will do it at the Helm package apache Helm such enter save it to ye So I just got this package now I can do it directly upgrade then my app name which is.

Apache of production then here is my Apache Helms Production Enter Dekhge My Production Apache Has Been Upgraded Happy helming production Apache has been Upgrade This is the second revision now my dev. How many pods were there in the Get Pods in the DevOps Disability DevOps had two of my but because of my Recent upgrade to the production Apache I have three pods, see how easy it has become. Managing multi-environment so right now I your dev to production and just two minutes It will take me more time to create the staging Helmets are possible now.

The question arises that what should I do for my revision? Number one was the one with only two pods instead of three. If I had to roll back beyond that, Helms is a fan of my Joe production. That was my revision number one, revision number One in which my particular two pods are running I want that one, I want three. If you don't like it, Helm roll back production. If you understand that through Helm You can do anything like I did just now.

[Kubernetes] Helm and charts / Kubernetes Services and networking

Created Apache Helm as well as I can Cube Seat No Cube Seat No Helm Create any Take node JS P is a page of Nana Jas which I want to run. If I want to help create a new character, please enter it. You have made your nut like a package. Do it Helm package now before packaging it first You will have to upgrade the inside of it. I will have to update the values so I am then run in the cd node Jas P Go and put your same tree inside it.

It is made that all these things are chart dull Now in the charts You can go ahead and name it Nut GS P What is its type Application Its version. You have a small change in your templates Go and look at your service port a little. It's a manual you have to change Thing so what will come to you here like this The port at the top remains the external port. Target port is the port of your container So here is one Please enter the target port so that the container He should also know what the port of So escape colon double k entered now.

I will change it and you can now go to Docker Hub. Here's Docker Hub, so Docker Hub if you remember it. We must have read a lot of notes and If you had put things from all over the world then what would I do? I can go into my Docker Hub and Any image, anyone, anything, I want it I can run mongo deb if needed I can run mongo deb which is good. Has Docker Hub, then by signing in to it Now I'm going to run my own things,?

Notes app Afzal's Portfolio is another notes app This is Node's app, there are so many Tutor apps. In the image, instead of repository, use Agen. Engine by default has such a place holder I will put the engine on the train wish node Pull policy if not present tag i take I will take The latest one is the node pop. Give it to me, this is for the secrets And all this is fine, service account is fine Just tell me what to do here.

[Kubernetes] Namespaces / Kubernetes Services and networking

That this one I have is on port number 8000 It runs and its target port is also 8000 I understand that this container is worth Rs 8000. It works and I get exposed whenever I want to. 8000 pahi boom scape colon double k Enter my values have been upgraded so I have received a letter from Nat Jas, some somewhere. I also did not write any email file. I did nothing, I just went and took the image. I've put in the values now I'll Helm Package Nut Jas P package do package It's been done perfectly Install Do I Need Node for DevOps Now If you want to make a dev node.

Nuts for JS app DevOps JS P Banana In The namespace DevOps Node is named DevOps Node. I want to create an app in namespace and I will also give it a create namespace. DevOps Nut Jas P of my chart My name is P, my release name is this What is the name of the chart that will be used? JSP This is the chart you will use Han DevOps Nut Create namespace Boom and your DevOps Nut Jas P deployment L Gate Paw in the namespace dev node your p running The gate.

Service in the namespace DevOps node so your Its port number is 8000. Port Forward which one for the service DevOps this one Copy paste the name of the service. I need to map port number 8000. What else should i do with 8000 Port in the space dev node namespace Dev node namespace after forwarding I am its What to do with the address 0.0 Enter forwarding from 8000 Now I. I'll go to my brick instance and see that Is my port number 8000 open?

It will be open because my notes It is P, that too used to run on port number 8000. I'll take the public IP and go here. An HTTP request on port number 8000 by hitting batch write your plan on Learning Jenkins whatever it is I want it I can change whatever it is but mine too DevOps test. It is made in what f minutes 10 minutes two minutes

pata No, and this is how you can help through something. App whatever you like from Let's someone Our [music] Helpoversies CD can also be made, let's go to Helm Let's see through Argo CD help through Argo CD or what else can I do?

[Kubernetes] Helm and charts / Monitoring with Prometheus and Grafana

Helm's through Argo CD yes Argo CD you make through Helm let's Now let me tell you about the cool Argo CD Argo CD What is Argo CD? I will tell you in depth as well but if You can make an Argo CD in two minutes. Is one second So you have to install it here. In Helm is not here, you can directly go here will find it somewhere Helm show Argo CD Helm write like this Two Helm Argo CDs wrote this and here You will get everything, how to add it.

If you want to do it, it will also be somewhere in it. [music] Man Argo Sea when I take you end to end I will tell you when I teach you, but if you If you really want to know how to do it then. There are some things that you like Repository needs to be added like a I will give you a small example Helm repo d Helm repo add but which repo. I want to add, if you don't know then search him Let's search for Prometheus Helm Search is Helm Repo Search is Yes, Helm Repo Search engine to search Engine atli that command is jh click.

Helm repo is not happening Search Here is Helm Search Repo what are you doing Helm search repo now Nginx you have to search So he will tell that Prometheus Community. This exporter is something, Nginx such a Helm Repo Argo CD No Result No matter how many you have installed You will get these things now, a small one. Install Helm Which tax to install? A Helm Repo stable named Stable There is a repository which you can also search on Google. Helm repo if you can There is one called Stable Charts.

If so, what happens to the stable chart? Generally just to check out things to report them Repository of stable and stable charts put the link and as soon as you do it like this Helm. Repo Ad Table and pick up this link and this one here go and wander Enter now in your computer your basically This repository should be added to the system Now this stable repository is being used. I'll tell you what it is, so here you are.

[Kubernetes] Helm and charts / Monitoring with Prometheus and Grafana (part 2)

Go and you can just check stable karta That is basically you have some topics which are like this It is for quick start,. There is not much work left for it Even more if you want Maybe he removed it after seeing artificial things. Nginx here it is now let's say me engine Install Via Helm To Isam Me I will click and if I get it via Ajax. If you want to install him then you need to directly what to do look this is what I was looking for Atlee was not available, similarly Argo CD If I had to do Argo CD, I would.

Directly Argo CD's Helm Chart Use from here and lets me If you want to install Prometheus I can install Prometheus directly from here. Helm Ripo The Prometheus Ripo Update done how cool it is let me show you I'll show you how to install Annex. Click here directly on the engine TL;DR Meaning of Too Long Didn't Read You don't have to read all this, you just do it directly. Helm install whatever your release name is OCI Registry Docker Vsami These Charts So That's all you have to do, that's all you have to do yes and that's it let's try it so Helm.

I would have put the install my release Am Enjan Helm Enjan Helm Enter This Ho got Helm install engine Helm can you. What happened, yes your engine Helm has been deployed in the default namespace and this All the things you will do if you kubectl get pod Get pods so you can see Helm is now. Our production Apache also comes with names. Similarly, I have my DevOps Also, uninstall DevOps from here. I can do it, it has become so simple for you Add your Kubernetes Manifest to the Kubernetes Applications I will give you more examples of how to manage Let's tell you Mango Dib.

I want to install MongoDB so I will go directly to Mango and start MongoDB Container Image MongoDB Go Helm Install OCI Registry Versus MongoDB copy this Take it here. I will go and slam the Helm install tanami MongoDB and here I will put it Using MongoDB Creating a namespace for the name Half Create Name.

[Kubernetes] Pods / Helm and charts

Add a space so that the fraction pod don't become a word Enter Helm install MongoDB Helm pulled Done so if I will do clear and kubectl get pod in. The namespace mongo is currently running an init container It will end after some time and that watch My seal get pot for mango and that's a little After some time, yellow running also started for some time. After zero out of sorry one out of one running This will also happen and your Mongo The DB will run perfectly.

Running is so cool now you Directly kubectl Enter the name of the unit in the IT pod. Enter namespace mongo half b put two enter lo pod k Got in, LS done, the whole pod is made mongo shul enter le MongoDB Connecting to MongoDB Local started working Host Direct Connection MongoDB. What you are saying is going on for two minutes. I am downloading everything and giving it to you Kubernetes is creating its own manifest file He is managing himself and in this way You can see things through the Helm.

Deployee has added Helm a You can see from the list what you have done. Things added in the namespace Mango to In mongo your MongoDB is Helm and In Nginx you own nothing by default. There is nothing, so in this way you can see what you have installed. DB Helm Co In The namespace Mango Boom To Helm has some life cycle. Helm you Do you create or install? Upgrade, roll back, and then If you delete the history Look, I told you to install Helm.

Helm uninstall told me help upgrade also I told you and also told you to help roll back. Container vs Side Car Container, these two concepts are its If the parent topic is removed, the pods It comes in that we are inside the pods What we generally run are containers. Run it from that also there is init container and there is a side car container, by the way Normally you can only have containers in pods. What is the difference between this and the init container?

In the container of the side in the container like this Understand that you have inet in both conditions Both container and side container conditions Has a main pod or a main container.

[Kubernetes] Pods / Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

The pod will always be the same, the same pod. Whatever you make, it will be a pod. What do you create a manifest file for? You create a correct manifest file Whose kai kand remains pod right kind Pod so whenever you unit container You make it, you make your side container Kind will always be a pod, just what's inside it. The containers that will remain in the space will be There will be two containers in it, two or more There can be more containers but one The container remains your main container in which.

Your main application should be running on the rest of the net container side container these two containers To help your main container It happens like you create a database with let's. You are going to create that database. Took the main container and put the image in it Let's look at PostgreSQL as a PostgreSQL tool. If you upload the image then the main image of this PostgreSQL Before the container is made, it needs to be made from let's A folder is required or a path is present.

Must have or need some connections If there is one thing to initialize it container Guess what is an it container Things to do before running the main container Should he give an init container lets. I started a project somewhere. I am here to run that project A folder in that particular location You should be present, just like we do. If we make a directory, then one The folder is created and then automatically A unit container before creating a deploy You can run it wherever that folder is created.

First the unit container will run, the folder will be created or Whatever his task is, that task will be completed. When the main container runs, this is what happens. Init container side container It happens like Batman,. Batman used to fight alone at first, but then He found a side kick Batman and He became Robin, so now both of them fought together. You have a simple manifest. A single manifest file will be created. Whose kind will be The pod and the speck in it will have two Containers will run, one container is your main page It will work and the second container will be from let's.

[Kubernetes] Pods

That could be for logs. On the left side, both things are moving in parallel. Side Car Container as a Helper Container Saw Just like Robin is Batman's helper. This side car container is for your new container. When you show the example hands on Let's get hands-on, so let's get a CD. MKDR do MKD I R it A folder named containers or pods makes CD cards now here I am what will i do em it container d y here now. Here I am going to create my init container's manifest I will start writing the file, now I will always remember it.

Keep the manifest file of the it container The kind that lives in him is always in the pod This will remain because you can run the containers inside the pod itself. If you create it, you can also create it inside Deploy. You can, but generally for this example, pod There is a unit inside which you can choose one or You can run more than containers which The API version will remain v1 and its. What am I doing now about the unit Next I make its spec Meaning, I will give its specifications now.

Inside the pod I launch the init containers init containers inet Containers Innate Containers Your Main Containers If they are separate from your main container, your main container will also If you remain, containers are your main containers. I understand what containers are. Your main containers are in containers Your main place will come where your actual Running applications in init containers I am supporting him or lying to him. Whatever container is needed for this, now it is mine What is the idea, listen, my idea is that I will make such a name such a thing.

Box man or put its name Main container I will create the main container and This is the only line where you will enter the name of the container. I will put the name in it container So I have two, I have one unit container and One main container, now this is the unit container. That image will use the latest Busy Box will use the image and run a command will run the command which will be a shell command and.

[Kubernetes] Pods / kubectl commands

In that he will say that, there is a shell like this command and speak Echo Inish Ization started dot like this and a little. Will pause for a long time like sleep for 10 seconds then it will echo again in which it will speak Ligation Completed Done, I realized this was the previous pod. He just started a project called Inila Ization will hit print then for a while will stop and then come again inila completed done now. This is a 10 second wait, for that 10 seconds what should I do if the main container should run only after I will put it in the image.

Busy Box I will get the latest Busy Box here. I'll put a colon and enter a command. That, whenever this busy box starts, a shell run the command what main Container Started main container started means it What am I proving by writing a container? I'm proving that my main The container will start only after initialization and Initiation complete, all these things should be How crazy is it that I should check it quickly? I can do that, by running my containers First of all these checks, all those things can done escape cold drink enter this I saved the cube and applied CTL.

Han f it container YAML enter in test Created by Can You See in It Taste Pod Inish easing and if you do kubectl Describe or log this pod Look, man, kubectl logs this pod. What is the default container main saying? Container out of main container it container not Found why this is known because inet test The name of your pod is the container it is inside. You will have to lock it if it is running. kubectl logs in test inside that The container which is the main container, its logs Show me what he is saying, I am not found o v i m it container me men do not.

Why is the found speaking ah echo forgot If the main container is started then it basically I made a slight mistake in my command. He is saying that I am not found, it doesn't matter. This is done, now what should I do to save it? I will do it and apply again cube CTL Apply Applied Ho Can You See This it test is invalid fobd n pod Updates may not change the file spec container Only addition to assisting a kubectl.

[Kubernetes] Pods / kubectl commands (part 2)

Delete a pod it test so which unit test Why do we remove the pod first? Because what is already initialized is a container You are trying to edit it which is a wrong thing to do with politics If you stop it, it is good that you will also get errors. It is understood that if in it containers You can stop a moving iron container and If you make edits in it then cube units will Not allow it so now kubectl apply.

High Quality Item Container Emulsifier in It Taste Created Logos or Logos Inet Test I need high-fancy logs from the init container are also needed and Logs from the fancy main container should also be entered. Can you see the main container? Started and before that I had to have it container I need logs of inila ejection started Sleep 10 inila completed i think those who There was a sleep command, that too with Echo. Once printed, put it in the it container. Whenever you want to end any command Nor do you need a further This colon has to be put like this, semi colon Now you have to apply the cube.

Whistle Gate ports in the nameless space Can you see it test completed to which initial The initiation that was there has also been completed which is the main. There was a container, that too is complete, now I can do it. kubectl delete pod it test Done deleted now look I will cube CTL apply -f now you my photo S cube may not be visible because of this CTL apply -f in it contain createdted. Pods kubectl Get pods Now Viewing Ho in it taste zero out of one in it initial is happening means kubectl logs it Test K in the Container It Container Initialization Started Initialization.

Completed It container work is finished If yes, then now this status will change and if I'll get pods Completed I understand that it How Containers Work Until It The container was running kube CTL. As long as your it container

was running What is his status in the gate pod there? It because it's going cube Seattle Apply within the container I apply did it ctor now watch kubectl Get pods to can you see now initialization running And as soon as 10 seconds are over,.

[Kubernetes] Pods

There was a sleep of seconds, after that the pod initialized It's so crazy how it contains and in 18 seconds Completed understood how does your innate Containers work interestingly that first you. Inila is do then your main container Drive your sidecar containers this way There are also side car container DAA now in What example should I give in a side car container? I'll run two things again always your it Containers and your side containers If only the containers are running then just What a difference there is here too.

The pod will be there and your API version will be there. Will also become v1 and your meta data you can ho naam side car test side test You are creating a pod named where In spec you are giving what in spec you Why are you giving containers? You are giving containers in this, you have side Why didn't you give the container because if you give an init container then first it initializes. It will happen, there will be a run, then it will end, but if You need to run containers in parallel, side by side.

If you want to run a side then you will need to use containers named You will have to put it inside that spec, so here I create two containers, one in my. The main container will remain the main container and the second He will remain mine will remain on the side car. Container: One main container and one side car container Now what do I do in the main container I will put it in the main container, I will put it again.

Image Koi Le Sakta Hu Busy Box Let's take the busy box and I'll put it in what command should I run in the main container What is the use of the side container? Are any of your containers running with let's? Collecting his logs his logs or He has to copy something side by side Its job is to help, so let's I have a main container whose command is what do I want to do here I here. Pay an infighting I want to run a loop and then end after a while.

Whatever I have written in it No, I want to finish it with let's I'm here for an infighting The shell command is h where i is c.

[Kubernetes] StatefulSets, DaemonSets and Jobs/CronJobs

I will do what else I will do while true while True means while true means infa I will play loop inf loop inf loop and here I will just do what I want to do. What does that mean that I, print again and again I will get it printed and then after that here If I leave it on the wire log, it will be mine. The main container is the one that repeatedly, the var log will keep printing But what is this doing, this is an application What is One Place Store.

Do my side car container is my What is the function of a side car container? As soon as my main p is leaving there All the locks I put this to all the locks I will print the busy box on the screen. Should I give him the command that, you Run the shell command where you will Wherever you go, what will you do, simple Which is CAT, CAT means display the content. I understand what I am doing I am just showing its contents and Let's show me only 10 pieces of content.

If you want to show me the last 10 then I will tell you what happens to the tail - tail hif Show me the last 10 items in this file. This is the main container whose job is What is to Produce logs this side container what is it To display logs, I will comment here. Let me explain what this container does. Display logs and what this container does Produce Logs and in the same way you can use the engine here Run the container and its logs are produced.

I will stay and run the container from one side. Who will keep watching his logs just a little The thing is that if you store the data somewhere What will you need if you want to get it done? Mounts will be needed, volumes will be needed You will also need to provide volume mounts. That you can keep data sharing then name I put shared data shared logs and its path I put the mount Path is this / ver/log so this / Ver / lock rahega mount path.

Same mount for the top container Path mine for the container below as well The same thing will remain in the lower container.

[Kubernetes] kubectl commands / StatefulSets, DaemonSets and Jobs/CronJobs

I will give it to you, what will happen to me if I The container which is the main container is produced will keep logging, will keep logging, will keep logging. The container on the other side is Will continue to display simple escape colon double enter kubectl apply -f What is the side car container enter saying Spec Container Volume Mounts Name Not. Found Shared Lock then you need a volume I'll have to create a mount as well, so let's do one. V also creates a volume mount.

Im side car container so you first spec Also create a volume mount by going to will be called Volume One You will need to create volumes and name them. Give two names shared Logos, this is the volume mount name,. You have to create yourself and its MTER by an MT DirectoryEntry Created Parts Can You See It Taste Well, he failed now because I told him Deleted side test is running. What's crazy is the side test is running Meaning if I take kubectl logs Whose side I will take the car test logs and from that Main Container.

If I take out his logs then I will get nothing. Why will you get it because the work of the main container What was the main container's job? Produce Logs But Is A Side Car Container whose job is to display the logs if I will check the log changes of that container. Side I will check the logs of the car side car container Which is this enter can you see so a container main container The side container producing the logs. Displaying his logs and this Like two brothers, both destruction, both great walks And in this way you can init container and can implement side car container Yes, now its a little bit of real world.

Please tell me the implementations as well so that now The concept has become clear, it container means If you want to run the main container before running something. If you want to confine, put that unit in the container. Two more if the side card containers both have to be run parallelly, helping Like container, put it in the side container put it done a little bit of it real.

[Kubernetes] Kubernetes Services and networking / Pods

Tell me the world use case so it container You can see the use case of Engine in it container So you will get the official documentation of Neti. Give some example use case, I see a lot of injuries or something here. Do not use it container with engine not shown It doesn't matter, even these people have their It hasn't been shown that much, it doesn't matter. We will make it ourselves in our project. We will run containers, what will we do with our The project will remain, that project will continue.

Some things will have to be initiated, those things we People take it from an it container and I show What I do in the evenings is I If I had to introduce something in the future. Like I have a Spring Boot Bank app So I am into his Kubernetes. I run the container generally like my The state of the SQL is full set in this can you C is not here, where did I put it? This is its real world use case Like I have a deployment in that deployment What do I have to do?

Is it that MySQL has been made or It is not made that much, I just want to see it. What I do is I use my back end Deployed is he his MySQL connected If he remains connected to MySQL then MySQL won't last until it does. If I am not deployed then my main The container is the main container which is the banking app The main container is what it needs, a mother Connection to the SQL So What I Do. I tell the init container to wait for Taking the image of a busy box of MySQL I run a command that until my MySQL is not up, until then you Keep waiting and as soon as MySQL.

Once it is up, the work of the IT container is finished. The main container runs so you have a real I have also seen all these world use cases. Hope you understand that Kubernetes What is init container and side car container in It happens, let's move on to the next topic. Our next topic is service. Mesh Service Mesh Baap Re Crazy Topic Service mesh so the topic of service mesh Let us try it on the local train, done.

[Kubernetes] Kubernetes Services and networking / Pods (part 2)

First they should understand what a service mesh is. My god, it seems there is so much Kubernetes above too. What is Service Mesh Service Mesh Look up Service Mesh like names It is like that, his work is also service. It's a mess, there's a pod from here. The service is running, there is a pod there. The service is going on when I tell you this about our Three dear application, how does it work? I know I'll show you three dear There is an application and a service is running in it.

Voting service is running, other service The result service is running again A worker's service is running which If the votes are calculated then these three The service is running now. This is the service port number. Moving

from Lets to 5000 This service runs on port number 5001. This worker node runs on a separate port. What is all this, these are all services I know you don't get this completely in your mind Let me take another example Because this is the project that In a little while, we're going to do a.

Let's take another example, that you have There is a great application called Zomato Mast is your application which has a The service being run is named menu so I pick up my phone. I go to the server service whose name is The order then goes to a third service. A system like Zomato is enough All complex services are running now. How are these services all these services? You need one for gateway and through that gateway all Services Imagine that you are standing, now you don't know People, where are you from?

Which traffic should be routed where Which traffic to redirect where and to redirect this traffic We Need a Tool Called Service Mesh so what is service mesh service Mesh gives you a gateway through which you. You can route your services You can basically redirect A very popular complexion There is a tool called Stee So this is a tool which is a very popular service. Mesh tool so o read way too much It's not difficult, but I'll give you a trick. I will tell you how to read AO, so you should go to AO.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / kubectl commands

Write g p just o demo and just there I have to enter it, I am getting started What is Steo is an open source service Mesh that lets you see transparently Layering the Distributor Application. There is service to service communication goes in a cluster with encryption your Load balancing is automatic. Gain control becomes automatic Matrix Lock Traces Everything is visible There is something very cool about your service. Separate open for separate Minikube for K8s Different for shift, different for cilium Different for someone, similarly find yourself here What to look for is to look for kind because We are working on Kind Cluster.

To Kind Create Cluster Stio Testing: First of all, we take a stereo A new cluster named testing will be created. A cluster is created on which a ST test is done. We give MKDI R I have done this O AO, now I will go to CD AO. What can I do here, I am kind of I will create a cluster. Kind of create a cluster. Name Steroid Testing A Steroid Testing I'll create a cluster called. This will continue to happen comfortably, what can I do?

Find a Log Line That Matches Your Target Multiuser system detected no problem So basically you already have a cluster here. It's going on, let's go to our local We travel on local trains, what about us on local trains? I will do what is there in it, stereo practice Now I already have a cluster running locally. kubectl get nodes so I have a TWA clusters like ours run on the net kubectl get Nuts Same I also have a kubectl in local Sorry, a kind of cluster is running.

It would have been a big deal if you had set it up locally. The download will take some time because SO The package that comes with it also contains some of the AO's. If there is something interesting then put it first We would have downloaded the entire package. Let's download it and take some time takes y, until then, let's see what we have. From which second in local is the namespace local You didn't expose this much of mine, did you? My MacBook Pro is so kubectl Get namespace so this is some of my local There is namespace, there is already something of AO in it If so kubectl delete ns so system.

[Kubernetes] Kubernetes Services and networking / Ingress and ingress controllers

I will delete the one which is already my AO. I generally create a P namespace. That yes let's by the way Our release has been downloaded to I will do this directory in ls. 1.4.1 The latest AO is 1 24.1 that is your If the pass is downloaded then CD S 1.4.1 now it has license in it is binary Manifest files are tools are dummy. There is a manifest, there are samples whenever you need something If you encounter such a situation, always read readme.md.

If you want to take it then read the cut readme.md So what does the dummy say? What am I saying, our school Title is an open source service mesh There are some things in this dummy. Community Introduction St is composed of This component to stand in a side car There are proxies, but so are microservices. As many microservices as you have It would also have a side car container running. What is the function of the side container? The main container's logs are all Keep giving things to him, keep moving you The work of the toyo also remains the same, side car Containerized ingress and egress traffic.

Things to consider for service to server communication keep giving Keeping Enviro safe if you have a history of osteoarthritis If you get a chance to give a good interview You can directly say that. Styo Yus What Styo Yus An

what is a side car Container proxy that allows what that what It provides as many server services as you like. Of all the services that remain, whatever There is traffic, it keeps injecting logs. Microservices on the home side's proxies to Handle ingress and egress traffic during the service Mesh is not an overlay, it is a simplification.

Enhanced, it also has good AOD What is OD, it is a control of AO Plane stays; it provides a service Discovery means as many of your applications You will see all the services I have. Service discovery gives you that Inside you have the pilot, the sidel. The gallery houses the pilot who lives Configure your proxy server Sidel Certificates Issue If you need access to Gallery validation if needed Remains to ingest and aggregate logs. It also has an operator who will help you can operate with service mesh.

[Kubernetes] Kubernetes Services and networking

It became clear that one of the There is also a service called Studio D, so you All this remains a theoretical part, we call it theory. I don't enjoy it much, we are practical I enjoy what we do We will go to practical here. Move to the studio package directory Yes, the installation has come in the AO package. Directory Contains Samples and St ctrl o A shell named controller a executable located inside the bin folder If it is there then it is by making CD samples where your.

I have all the samples, oh my god, so many. There are samples, there is also a sample of open telemetry. Lots of it and yours in the bin here Ste CTL is also kept so if I Karunga dot s et ctl Enter so cool he is telling io confine Command Line Utility for Service Operators. To debug their Ste Mesh use Ste Ctl Admin Analyze Auth and so on to stay We execute all the step commands with CTL. Like kubectl to Kubernetes Runs the command in the same way as from Seattle runs the command now this binary is its If the location is here then it means I am all over.

If I can't operate a computer, I'll have to If you want to run the computer all over then copy me. I will have to do it or I will have to move this sti ctl kithar/usr local / What happens to my computer without this? Which are local binaries like MKDI R PWD these are the commands, those commands will also include Ste CTL Enter rey yeay what are you doing overwrite yes Rename ST permission denied no problem. Do sudo whenever you get hooked like this If you have to do it, use sudo.

What you have to do is your shell Script or whatever executable file it is It has to be moved to user binaries This is your job, what can I do now? Ctl install Book Info Demo Profile No Gateway YAML What does this mean? Project which comes by default in AO Let's install it, so let's step CTL Install Han F Samples Book Info Make sure that the directory entry you For AOD What is AOD is a component.

[Kubernetes] Namespaces / Custom resources and operators

Have AOD Installed Installation of AO Completely done means I have AO is installed AO runs from CTL AO is not just a I have the command installed. I will run it, how to run it What do I need to do to get the namespace? Label to instruct step to automatic Inject if into any namespace I want to play it, what is it, service mesh? Which handles the rest of your services Yes, but it is a third party tool. Allow inside your Kubernetes cluster If I have to do it, what will I do?

I will give the default namespace a label that Sterio injection enabled if Sterio If injection is enabled then this steo s a Third party tools can also help you manage your Kubernetes. If you can do some things in the cluster then by doing this I want to label my namespace with Cube CT namespace Default St Injection Enabled. Enter so namespace default not labeled not why labeled not labeled why should it be labeled why namespace label to instruct stu Automatic injection nozzle Why is the Deploy Not Enabled label appearing?

No, we will search namespace. The default not labeled st will search like this [music] namespace Not injected, that's a very interesting error. Why is the namespace not injected? End kubectl label namespace default st injection enabled - - overwrite by See, the same thing is coming, get namespace Let's see get namespace with label yes oh this is already enabled interesting. I was able to do it somewhere before can you see your stereo system and default is your default namespace Sterio injection enabled which is, we were just watching it for no reason.

Your stereo injection has been enabled There is no need to worry, so let this label be I have already given you what is your bio injection? Enabled should come like this, it should not come like this earlier I was waiting, I am coming now, good Interesting Install The Kubernetes Gateway API CRD I used to like. It was explained that custom resource definition is such Things that are in Cube Kubernetes by default It is not present now, if you make it so then install it, copy this and here.

Go and paste the cube Seattle Gate CRD Gateways Enter ja deployed application now i Deploy your sample application Deployed Application Book Info Sample There is an application like this B Book Info application.

[Kubernetes] Kubernetes Services and networking

Which will be a product page of Bali A Python from where it will redirect to your java All services are your microservices All these are on product page reviews. Will redirect to Details page Ratings page will redirect so let's go now What do people do now? Have kubectl apply -f raw git Whatever of Hub User Content Sample Book Info This is We will run it smoothly, I copied the file. Ml file and what we did I went here and threw it here, apply cube seattle Han F Raw GitHub User Content AO Enter my service details created.

What is the service, I will tell you this, which is my Book Info application is the details in it The service is built on Ruby on Rails. If there is a service, then it becomes your service account. It has become deployed, the service has become Rating has also been made, so this rating is a note Jas's rating has also become a service. Review one review two this reviews name There is service, that has also been made and what else There is a service called Product Page.

There is a Python service, that has also been created. Your entire application deploy apply if i do cube CTL Get All to my deployed product Page Rating Reviews All Things have become if I have to be individual. If I had to open them all one by one, I would Let me show you how it is If you want to open the product page then click on cube CTL Gate. I take SVC service product page How do I open the service? kubectl port forward svc / Product Page Product Page Which Ports Running on port number 9080 if so 9080 Colon 900 product page end After doing this I will run it on port number 980.

I will go to the new local Host I will open 9080 and you can see it. My product page is opening This is a Sample Book Store Application Consisting of Three services as shown below: On, all these things are going on for me, like this. You should go, interesting but, right now Provide me with a separate product page service. Run a separate service for rating reviews. I will have 50 microservices like this So how many will you manage?

[Kubernetes] Kubernetes Services and networking (part 2)

Will manage the URL so you feel So let's move on to our project now. As we move forward, all these things have become The application will start getting all the services Get all the pods, all of them are running now, validate. That the app is running inside the cluster by Checking the page title in the response We have to validate ourselves that whatever we have The application runs within the cluster. Title: We have got this in the output, so Which was their output Simple Book Store app I have got this title, I mean it They wanted it, it's coming to us.

Open the application to outside Traffic means this to the outside world now Open the application, what did I say? If you need to close the application to the outside world If you have to open it for me, what will you think? It will take a gateway, so let's build a gateway for it. kubectl Apply Nuff Samples Book Info Gateway API Book If Gateway Enter Here Pay Enter Book Info Gateway The Created Way has arrived, which will solve all these problems. Redirecting services just like a Traffic Inspector coming from here Service redirected from here to there A has been redirected here and so on The gateway has been created, now what will I do to change it?

Service Type to Cluster IP by Annot the Gateway will then annotate the gateway to the cluster If you create IP then cube seattle nutted gateway If you create a cluster IP, then the Bay Gateway will now. The cluster will run on an IP so that the cluster Now he can manage the traffic at the level Try the Cube Seattle Gateway to see if you get one. Did I get the gateway or not and I have got the book Info Gateway from Stiio, this is its There is a particular location, if you want, click on it.

If you can go to this will you go http curl http is particular What will you do if you find the location? You will get it now, so basically you will get a gateway. To which you can redirect it now I'll show you how to drive it, drive it outside, done, now I'm going to call this gateway.

[Kubernetes] Kubernetes Services and networking / Monitoring with Prometheus and Grafana

I was trying to run Gateway but it wasn't working. This gateway of ours is not working. I was staying because now I call this gateway I'm going to open it, let's open it. If there is a kubectl Port Forward Service Book Info Gateway St Port Number 8080 To I would open it on port number 8080. I am entering my port number 8080 and it should be open. If I go to the local host 8080 enter localhost page Can't be found because there is nothing on 8080 Yes, but it is given like this in simple B The application consists of three services and.

Your book reviews, everything here, can you see which service to route where? I did not come, that is why the service mesh It's important, now let me show you how Which service is routed where this page is opened? That is the power of O Can You See Now Cube Seattle apply samples add ons so I'm here I will add some add ons, I have added them now. kubectl Rollout Status Deployed Killy in the stereo system now what I will kill you in my stereo system Which is my dashboard that's a service mesh shows what the actual service mesh is.

Page I requested reviews from here Went to ratings, went to ratings and brought them back I went into details about how and when this entire flow happened. Nothing shown namespace traffic nothing showed In the Vision Version graph, you can now see some It is not the last minute What can you show on the display? It's not coming yet If yes, I got it then this is your service mesh that you have an AOD that is your data plane Data collection and Grafana Prometheus Wherever you want that data.

Stee this way it's going great now if you want to see its services then you can You can see the services, if you want to see the conflict then you. Is If your yes is this then generally your triangle The service remains and the details you can see here If it is square then it remains your deployed. Came from the details so this book Details and these book reviews that we see Where are they coming from, they all are Coming from different services and o gives you a visual of this whole thing.

[Kubernetes] Kubernetes Services and networking

The service mesh reveals what else What is happening with your services? Interesting tomorrow you have 50 micro There will be a large architecture of services You can easily run SO in that too. You have to do one thing, you have to do this cool SO I have told you with examples that What is a service mesh? A service mesh is your How your data is shared across different services How to redirect traffic flow He is showing Now all the names of the services have come, all at once.

I enjoyed how you used to service Where are your services headed? There are more projects then one project Your mast pura usne jankan ci sdi GitOps Argo CD will all remain one project will remain on MongoDB because as of now we. People made as many pods as they could Deployed made all those MySQL related. A little bit of MongoDB too It will have micro services, meaning your The front end will be separate and one of a kind. The service will be running on the back end of your network.

The database running on the JSP will be your Mongo. If it is running on DB then this will be a chat There will be an application that I am going to provide you. You will have a 3D application. In which python.kv is gone chat application So let's start with the chat application. If they give it then what a great thing it is, oh my god I have read a lot, just a little blank Give me space, it's good, this is ours The project is called Chat.

Application on Kubernetes knots which your basically There is a three tier application where your The front end is running on a writ jas and the back And your nut is running as it should and then your. Mango is running so here I am A web page something from Lets for the front end So let's make this, let's take it, let's react. Taking and let's take our back end Let's take node.js and get our database. Let's take MongoDB, this is our Database is great, now what do we need, we need these What everyone wants first of all is like the rules Accordingly, we will need Docker containers.

[Kubernetes] Pods / Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

Docker will come for this too, Docker will come for this too will go and its docker will also come so What do we have after all this Docker containers? If we want to make pods then make pods Nange it creates pod from Docker container then Its pod will be made from here and its too Once the pod is ready, we will prepare its pods too. Once the pod is created then We also have to make its deployment. It will be fine, its deployment will also be done.

Its deployment will also be made. Deployment will be made once again. I have made it deployed, then what will you do with it? We serve cheese We will make it, the service for this thing will be made front. End service and back end service and database service once our If services are created then these services If we want to talk to each other What do we need? If it happens then we will do all these things now Who will do all the things, see, I will tell you I will explain it completely, I have a few.

If three projects are added after careful consideration, then The first project will remain yours,. EKS Cluster EKS Cluster But the second and third, this second one The project will remain, it will remain your Minikube The third project on the Minikube It will remain on your kind cluster. Kind Cluster Done and Joe Monitoring with Promise You can get the graph done on the client cluster. If you can get it done by anyone then this The way I completed all the projects divided into clusters of types Very crazy, let's start right now.

This is our project chat. Application on K8s 3 and this will be your mini, if I do it on the Minikube If you are staying then pay locally so that we can Also try it out a little bit. This is my terminal, here is my local. We will come on local a cd dot So this is my local, look, this is my local. It has opened well and we will If we open it well, this is our We are here at the terminal on Local and Local.

If you open it, here you can click on Terminal New The terminal will open and here we will see the mini If you start the cube, then the mini.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / Kubernetes architecture (control plane, nodes)

If no cluster of cubes is formed then we How to make a cluster of people mini very Keeps it simple Minikube Start Nuff Driver is equal to Docker so Docker Cluster of a Mini Cub with driver. What is this thing, so VS code is yours What to do if the code editor remains VS Code install then type this VS Code A very powerful code editor. You can download it from anywhere Can do it for ws2 Can do it for mac If you can do it for someone then it is our wish.

VS Code here and Terminal here Done so your Minikube Hum Log Start Karenga VS So Minikube inside a terminal of code star driver docker enter so Minikube Version 1.33.0 is great on our common 64. Driver based running has started Creating and Preparing Docker Containers Kubernetes Generating Certificates Keys Booting up control plane conflicts are back Hope you have understood all these concepts now. If I do Minikube sorry kubectl get notes if i do this You should have a cluster of mini cubes ready.

The control plane is ready on the local platform. He had installed the chat app CI/CD. By doing pipelines and all that. I had put it in but now what will we do in the future? It is written in the plans that we plan to Implement CI/CD Pipelines and Kubernetes Manifest for Content Orchestration So basically he has deployed a cool full. Stack Real Time Chat Application Built what are we going to do with this application If you are going to do cube Kubernetes then future The plan that Kubernetes was going to do.

If so, check out the project on GitHub. If we fork this on chat then Click on the fork and as soon as you fork If you click on it, then this name will appear on Full Stack Chat. This is an application of the main branch. Fork means you can cut your You can access the repository in your GitHub profile I clone this thing and fork this I don't say that repository to repository Full Stack Chat is done now. Get the code and this link comes up through which You can git clone this link Now it comes, what will you do with it?

[Kubernetes] Pods

Git clone how to do git clone First of all, you should create a folder where But if you run it then file open folder open Click here now I want to. There is a folder inside it, I am saying something I was running things, now I'm a new I create a folder called projects And let me create it here and this. In the project name I opened Done Once it is open, here is my A folder named Projects opens. Here I clone a git repository I'll give it a try.

I can do this from here in git I can also go from here and open repository Now what will I do? Clone I'll create a Git repository and click Clone From Gab I will put now the repository's Name here Full Stack Chat page which recent page It happens that he has started appearing in me. I will click and as I click I will give it to him again and he will ask where is? If you want to insert it then it should be inside the same document Work inside train vi shvam vo apne me very well, local Look how much everything is lying around, why not?

Current workspace d to workspace so I am using this If I add it to the workspace, it Remove everything or dismiss, never remove, it's done. Close the source and go to your files. What all has evolved, if you If you want to make

any Kubb Nits locally then I'm telling you what things look like So the back end is the SRC inside the back end. Meaning SRC means its code will be Docker So this is its Dockerfile. Docker so here Its Dockerfile is of back end then its The Dockerfile of the front end is front The end Dockerfile is here, a great front end.

Docker build and then create a pod And then we have to make a deploy of that thing as well and If you want to make a service of that thing then all this. A pod is made only when it Docker has an image, it goes on Docker Hub. Include yourself in this project via Docker Hub as well. If you need it, go to Docker Hub and install it.

[Kubernetes] General DevOps discussion

Go and log in so this is my logged in and I'm here at Docker login So I'm going to do it here with my Docker Hub is my credential create. If I do that, go to Docker Hub and create an account. I will create a personal access token. I give it read, write and delete permission. Is generate done so this is the command from docker Copy and enter the terminal here You can open Full Stack Chat anywhere. You will build the Docker image and push it to you.

You will have to build the front end Docker image. If you do it, you will have to push, so let's go quickly. Build and push Docker images First of all let's go to the back end so CD I went to the back end and ended up here. This build is happening, can you see its name? Docker Hub Similarily what you would do is cd You will do the front end, you will do the front end, you will do the CD now You can also see the Dockerfile here by doing ls.

Docker built the front end and the back end Created a Docker and pushed both of these to you If I had to do it, what would I do here? Which is my Docker images in it again I have to go through so many images. Will push so docker push oh yes if your Let's find out who has the name tag If you want to change it, I would have told you its command also. Its command is very easy, you just have to write it.

Dock Image tag and the tag of your old image That will happen and then the tag of your new image You can take that and push it,? With Docker Image Push you can push your image If you push it, here is your front end key. Image and back end image here I am pushed, now what should I do? Where did this MongoDB image come from? I will take this directly from Docker Hub So in Docker Hub if I go directly to And write it here Mango then you will get that of MongoDB Latest Image Directly from Docker Pull Mongo.

If I get it then basically I will have my own Chat should be back-end pushed to the repository and my chat app front end is also pushed It's done, in this way I enjoyed myself.

[Kubernetes] Namespaces / Deployments, ReplicaSets and rollouts

Both are done, this is done and this is also exit. My work is done with this, so both of my The images have come to me on the front end and back end now what should I do with these two. I want to use images and create my Pods using these two images If I want to make a pod, what should I do first? Inside the Full Stack Chat app is a folder called K8s done first give me a namespace If I want to get the right one then I will be new in CAT.

Create a namespace called Chat P Chat -P I will give it to you, now see what habit I have. If it was IM then the escape code would be something like this There is no such thing, you can directly write well in it. You are done, now you will go to CD Cats. Go to the folder now kubectl create -f namespace then do The command remains kubectl create end kubectl apply so if you have Something already cries if you confiscate it.

If you want to do it, you do kubectl Apply if you are creating something for the first time So cube seattle create both moves cube seattle create namespace half aml namespace Enter the Namecheap chat page. You can see that the namespace of the chat page 6 seconds ago Done It's done now What I'll do is it's my namespace Now the next thing I will make is Make it directly deployed. Deployed whose back end deployment dva back end deployment made directly If it gives, then understand it like this that if you have back end What do you need to do to create a deployment?

Things will look like API version apps / v1 then yours will come what kind Deployed then what will your meta data come The name of the deployed will appear in the metadata. Back end key from let's Deployed then what else will your name come in it Back end deployed and your namespace front sorry front ad no chat p to Your namespace is done on chat now its. Your spec will come inside in which your ReplicaSet I always forget the rapes. There will be one, there will be only one ReplicaSet then You will find yours here You will have to give the selector, you will give the selector.

[Kubernetes] Pods

You will have to write a lot of match labels. Here's what you can do Go to extensions, click on three dots I go to Kubernetes Find such a nits and cuts which is this Install it, if you're done then Kubernetes Develop. Deployed and you get all these things So it becomes very interesting Kubernetes done Minikube upgrade Available to 1.34 currently later I have to do it later, not now. God that's it match labels then now Look, it immediately started giving error.

Because you downloaded the extension Didn't that the Kubernetes one will tell you everything? These suggestions will give all this for example here Pay app Chat, this chat is called, I will tell him that I will tell you that this will be my label, chat. Let's put the back end label that back end The back end label for Deployed will be You can also put it like this, then you can put it Tem. You can insert the let template and in the template What you need are containers.

Raat To Container Me But Container Yours If it comes within the spec, then here the spec and what remains in the template is meta Data remains as metadata or all memory. How will it be again, you have a cut Read from the excellent documentation and enter the name Something that back And pod and its namespace take two chat pages. All that is important in this is labels this is important Does not match label labels match labels why Labels are happening again and again and you will find them here.

Pop back end use the same label here too will have to do done then the meta data is finished What do you need in Containers Image What should be the image that we want? People had posted it on Docker Hub,? Similarly, there are containers here. Pass list must be array item image Then what is your name, put it in the name You can chat on back end so this is chat p back end ho How do you ignore this,? That's why I like Vim so much, it's so cool Look how it happens here, so much is coming so name chat back end Ports Now go to your Dockerfile and.

[Kubernetes] Deployments, ReplicaSets and rollouts / ConfigMaps and Secrets

Port: Let's go, we need a port. Number 5001 P he is saying expose so let's go What do we need to put in the pay ports here? If you want to enter the port of 5001 container then click here Our container port is good, he gave suggestions. Given 5001 container port 5001 and What all do you want to put in it and yours and Is there any environment variable so your Look in the Dockerfile and find an element variable. Is it yes node environment production so There is a node environment variable named whose value is production so What will be your environmental impact in this?

There will also be production which he wrote here What has happened, what is written in the production, this is it Put the value in the value, simple now With this you will be done with your deployment. Gone means obviously it is not over Because in this deploy you will get your mongo The DB also needs to be attached to the front end as well. I have to attach all the things that will remain for now. Much of our back end should be deployed Now copy this thing like this The front end will be yours Deployed y so front end Deployment will remain the same, only changes will be made.

What will happen like back end deployment By the way, the name of the front end will be deployed Space will match labels in front end The back end will be replaced by the front end. Front end done, this back in templates The end pod will be replaced by the front end pod. Labels will replace the back end with the front end. Front end, here are our labels now. Repository Chat app Front End So this chat app Copy our front end, that's it.

Our front end repository is now on the front end. Go to the end Dockerfile and see that Which port to expose then Docker went to file expose port 80 If he's speaking, so the front end. Deployed Containers to Port Number 80 now in front end deployment What are the environment variables? Find in Dockerfile then in Dockerfile I exposes an engine copy of the engine Confine also in this node environment Projection is given, let's also find its node would have put environmental production.

[Kubernetes] Persistent volumes and storage / Deployments, ReplicaSets and rollouts

Here's what I'll do: Production as it is, I will do it to me The front end has also become deployed e.g. I needed the back end as well as the front end It has become mine I am now deployment Now I will use my MongoDB If I create deployment then this is me as it is. I will copy it and create a new file here. I will give mango DB Deployed y a so of MongoDB as well This will become deployed, what will change in it?

This will replace the front end deployment. MongoDB deployment will be namespaced The match will remain the same and will be done in the app. MongoDB will do this so MongoDB will be meta The data in the MongoDB pod

will be Mongo DB pod will be namespace same chat app And in AI MongoDB it will be containers. The image will be used whichever port is of MongoDB It happens and generally 2 or 1 or something like this So the port remains, we can find it online.

I can find something like this lo MongoDB port 27017 This is the default port of the environment. There is no need but in MongoDB you If you have to provide your password, please provide it here. If you can lift it, you can do this also, so be it. Your deployed is gone, but, ask for this. The deployed DB will be MongoDB so its Your persistent volume with Deploy And if there is a persistent Volume claim then persistent volume will also come and PVC Persistent volume claims will also come.

I will definitely come and make it, so this is done. Your deployed persistent volume and Let's create a claim so you get MongoDB That all this has been deployed, now what do I care? Persistent Volume Directly And Whatever my persistent volume is I'll pick up that cool one, so percent here. How about the volume Here it is, so this pod has become persistent. Volume Yes Here it is Yes got it So this is our persistent volume so I took this as my pleasure I will take.

If there is a space, it will remain on the chat page, so the chat page name persistent volume in space these and Storage Class Name Storage Class Name Us What should we keep as storage class name?

[Kubernetes] Persistent volumes and storage / Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

Keep local storage so storage Class name, we are doing this on mini cubes. Storage Class Name for Minikube Let's see the storage in the Minikube what is the class name Is o So it's saying that if you get a Minikube If you want to pass in / data then you. Make something like this so that I can do the same. If we do, we will copy this and here If you paste this then the access mode will be read write Once we did the capacity 5GB Just enter this much data on the host.

Host data then persistent volume Also, the persistent volume has now been created in the same way. We will also make a claim, it is, persistent volume A claim can also be made, let's make a claim. Persistent Volume Claims Again Claim Persistent Volume Claim let's go here, they must have made it I was just looking at it, there was a If the persistent volume claim is this, then API. Version Kind persistent Volume Claim and From here we can go to our Persistent Volume Once you have raised the claim, copy it again.

Is access modes which are read write once read Right once its access mode will be now this Claim persistent Volume by How Much If you request it The resources it is requesting. 5gb will remain in storage, so my MongoDB PVC 5GB of storage Requesting now this persistent Volume Claim and persistent Volume Hum If you install it inside the container of MongoDB then. We will go into the deployment of MongoDB. Here directly where we people He has made a container and we are inside it.

Will put own Volumes Now we are in volume I want to put the name of my volume, that's you what can I put mongo data mongo data Now he has made a volume named. Persistent volume claim whaye Oh my God, it will come, I have written so much about it. Persistent There is a volume claim but there is a tent volume claim. What is his name what is his Claim name is that MongoDB PV C then MongoDB PVC Its a persistent Volume Claim named If so, we have attached the volume to it.

I just want to see one thing once that this Containers and these volumes are all one Done, the containers will remain in line.

[Kubernetes] kubectl commands / Persistent volumes and storage

Volumes are in the same line I'm done, let's run MongoDB now. Let's check if MongoDB is running or not. The namespace of our chat page has been created. What I will do is kubectl apply - First MongoDB's persistent volume I will create an enter persistent volume kubectl get persistent Volume In the namespace chat a dun so pass mango. DB Pavee has 5gb of driv only retain It is available, let's claim it now. kubectl create not apply -f mongo DB persistent Volume Claim Dwa Enter To MongoDB PVC CreatedDocumentsFragment MongoDB with Vala now with PVC What should I do if my chat is on MongoDB?

The one that picks up the image of MongoDB Deployed to kubectl Apply Nuf MongoDB Deployment Dull Enter MongoDB Deployed Created pods in the namespace Chat page So can you see container is creating for. Creating a MongoDB container If you want to see how to create a container If it happens then watch kubectl get

pod in the namespace Chat app Enter To MongoDB Deployed is now running so mast aapka mango. DB deploy started running, done now You can change its password if you want.

As you like, let's say generally the user The name remains, it remains. Mango remains admin and the password remains They can put anything you want in your secret. You can put whatever you like If you can, ask for admin and secret, it's yours. Entered the password and this vim SK You put the colon with double K and After doing this, if you apply it again to your If you do it, it will be confirmed and your Perts. If you run the back end and the front end I think it's very important to run the front end yourself.

There is no need for much front end so let's go only the front end will go Deployment l apply -f front end Deployment Enter Front End Deployment Created in the namespace Chat P and mine. Front End Deployed Error is coming Q let's see what is the error kubectl get sorry Describe Pod Front End Deployed and Describe the front end deployment. Crushing the kubectl g Describe pod this one in the namespace Chat what are you Went kubectl Describe pod this one front end Deployed Chat P Why It cannot find the pulling image.

[Kubernetes] Pods / kubectl commands

What is host nut found key Docker host nut found So what this means is that the front end Upstream will require you to run the back end also need. I'll come back to drive the front end I want an end, so what are we doing now? Will kubectl apply hyp f back Let's try applying end dit and press Enter Back End Deployment created in the name Entered Space Chat on Container Creation Let's see by the way apply back end. Before doing this, this front end If you delete it, delete it again.

The pods of the back end, back end If our running is there then our back end is running. If so, kubectl apply front End Apply Front End Deploy Enter Deployed created pods container is Creating Error Again Error I Think It Has Some Kind of Error Again Crash Loop Back. kubectl Describe pod ya in the name namespace chat Why is the app being successfully pulled image size now you can do one thing docker logs sorry Docker logs not kubectl. Logs is an example of this front end deployment.

You can see what he is saying on the namespace chat. Getting Host Not Found in Upstream Back And in engine conf then koi confin need something for Front End What will be needed to host the front end Not Found ETC Engine, let's see what this and this. Needed O I understood, I understood, so what is this of yours Engine is your code, this is the front end. If there is code of the front end then the code of the front end which is Docker It's a file somewhere in an Nginx file.

Confession is using the name of Enked Confession is using engine conf and that Speaking of which, the / API has a back end. The back end is generally the name of your pod. If the name remains then I will get back end service. You will have to make it and then what does it say naraz maashuka humari crash loop It is hitting back off error only then it will run How much do you call it tightly coupled? What is your problem or which is of the front end?

Pod it connects via engine which There is a reverse proxy server now that the back up The service at the end is not this service If it gets created then this pod will have a Location.

[Kubernetes] Kubernetes Services and networking / Namespaces

The location named API is inserted and that The service is inserted in the API location name ch is back end so this back end name We will have to create a service called back end. If you create a service, you should close it now. This is our full stack right now, all this A new service will be created in case Will you give me back end service? Creating a Service Again Simple API Version or Then what to do, only Write service, write service, press enter It would have been completely done on its own, what a sorted.

If the thing is fine then write the service meta data on Now the name of its service should be Should back end back end otherwise our Masuka route has been selected and its name. Space will also be a chat pop up or else it won't work. Looking for the back end then kubectl Delete - Let's deploy the front end,? Until the 's service is completed, he will If you are not agreeing then please chat on namespace. Which is the back end selector?

If yes, then go to the back end and see the deployed one. That its p is back end so back end The back end of the service will remain fine now. Its port will be 5001 and the target port that the container will have 5001 will remain, that is enough, this is done Your back end service and similarly you will create your front end service. Dottle Again Ready for the Cube Nights Extension Magic Service Likho And But The entire service becomes so sorted I'm for the love of your life and your volume.

I was searching here and there, if I am here If persistent had written a volume, keep it complete. What was I doing in vim 10 12 Just for the hour you have made such a big video Kidding, you should use Vim and learn how. I write all these things now, this is his name. The name of the back end will be the front end. There will be a service called front end. Its namespace will be Chat app. This will be its service named Chat P.

The selector front end is the front end which The front end port is that through the engine If it is open, it means the port number will be 80.

[Kubernetes] Kubernetes Services and networking / Namespaces (part 2)

You created and pushed a Docker container. If you make a pod deploy, then this deploy also The service was made and the back end was also made pot Deployed and service is created, let's do this Let's apply kubectl apply. Back End Service Back End Service CreateDrawAnaNavigator Done Front End Service Done MongoDB persistent volume of back end service Done Mongo Deployed Done namespace front end deployment should have been done right now. Han kubectl Apply -f Front End Deployed Enter Front End Deployed Created by Potts in the namespace chat Papa, now our Masuka is ready.

I am ready to run away, I saw that As an engineer, you also need to be able to read code. Otherwise you will get trapped and we will do it on our own. I couldn't figure out why our pod wasn't moving. If it was running then why was it not running because Nginx had a dependency that you She tells the service that, the name of the service. I can run it simply so I cube when I play CTL Get Service in the namespace Chat P enter so my back end is running my The front end is running what's up so cube.

CTL right now my back end and front end are running If it is, I will port forward it. Port Forward first I ran the service first back end Let's take it because our front end is A little mean, irritable Lee error comes. First we run the back end, then the back end and back end which port p is running on 5001 p to 5001 colon 5001 and this is the back end of. Namespace is also Chat A 5001 Enter Forwarding from 5001 means it works like this If it is there then escape it and put m part and we press enter, then this.

Similary Hum will continue to run in the background People will have to do kubectl port Forward whom of my service whose name Is front end and the port. It runs on port number 80 and the namespace will remain Your Chat A 8080 Enter Why able to listen because permission Denied so let's tell sudo your own local. It is to have the machine in your own house without permission.

[Kubernetes] Kubernetes Services and networking / kubectl commands

Never mind sudo -e kubectl Port Forwarding will do this. Enter Password, put it in, forwarding has started working now. After ending this we will put M Part and press Enter. Go to Network and Create Account. It says Sign Up 404 Page Not Found Saying localhost API auth sign up 404 Not Found so on local host it will do something I am not getting it on / AI, why? Because its back end is not working properly Maybe so kubectl get pods in the Namecheap Chat App Backend is now live The back end logs show that the back end why is kubectl logs not running.

This back end is deployed which is the namespace chat app. If he is inside then what is he saying oh father, he is making an error. Mongoose error the you rr to Open tell back end back end your back end The deploy worked but how did it connect to the database? Will connect to back end then you will get MongoDB I will also have to provide the URL, so let's go here. Create an environment in which MongoDB puts the environment value What is your name MongoDB?

URI whose value you should have entered something What is the URL of the value MongoDB? Generally the URL of MongoDB is something like This way your MongoDB will look like this Its starting protocol remains then Whatever your MongoDB's MongoDB if password and username remain What is the password for. My username is Mango Admin and this What should I do if I have a secret password? I'll call back end deployment Mango DB Username and Secret This is my password This will connect to MongoDB and what is this MongoDB this MongoDB If MongoDB service remains, then MongoDB.

I will also have to create a service for I just forgot to install the MongoDB service. We forgot to create a MongoDB service. I went to the service again, I will write boom puri The service is done, I am enjoying it a lot. Name MongoDB selector app MongoDB take a look at its selector what is MongoDB is selector MongoDB Done Ports 2717 and our target port is the same. 27017 bus name here in this Space will also be added which will become chat pop So this my namespace also got added so Mang If the service is created then apply kubectl.

[Kubernetes] ConfigMaps and Secrets / Kubernetes Services and networking

Applying the service of Heinz F MongoDB Two enter MongoDB service creation Now this service has been deployed to me. I have to use these backends for my back end And this has become a deployed red wall. What should I write in the url of MongoDB? What is the name of this developer of yours belongs to the developer who maintains the repository Check that repository once. Put something in a dummy somewhere in the repository as if it is inserted in it that you A great little user named MongoDB or RE The name is to be entered as an environment variable and it.

Its URL should remain so MongoDB is the Protocol MongoDB as admin is the user Name secret is the Password @MongoDB is the Service Name should be the name of your service. If you want MongoDB then your service name What is the service name of MongoDB? MongoDB this is correct then your admin The user will be named MongoDB Admin. In MongoDB Deployed the user is his name is MongoDB admin root If the password is secret then this secret is also correct. This is so correct right now DB name is admin so Copy this line and paste it here in your back And go to deploy and hit it with a slam.

This is done, so this is how you Confirmed your MongoDB URL What else do you want JWT? If you also need a variable named secret What name do we need here? JW What is the secret name and its value? If it is secret then secret file will be created. Not a Little Your JW Secret Just put this in, what does it matter? The secret key of JWT is different. Plus you have not added anything in your deployed If you want to keep it, what should you do?

Make it and name it Secrets Again the same auto in Dy A and secrets Complete secret likhe boom secret ban Chat secrets in the bus meta data. In the namespace chat app this is the data JW token instead of password in This is our complete completion. Put it directly then chat a secrets name There is a secret file which we have to use. In our back end so JW We cannot directly value the secret.

[Kubernetes] ConfigMaps and Secrets / Deployments, ReplicaSets and rollouts

Value from aayi from wat Secret Key Ref will give us the Secret Key The name of the Secret Key chat app was Secret Key. If it is named JW, then it is JW's Where will it come from? From our strata data So let's go back to our back end. Even JW Secret in deployment Dal MongoDB You RIA and what not If it is there then tell me and where did it go? This port 5001 is a port named If you want to add environment variable all then that too If you put it then Enter a name called name port whose, whatever value remains, it will remain.

I have 5001 dans and this It is inside double quotes because these People also put normal in it, so We also did normal work, no scenes were done. I understand how to work somewhere If you still feel there is an error, check your The environment variables of the containers If there is something wrong somewhere then this If it happens, our secret has become service. kubectl apply -f Secret Done, what did Secret say? V1 Can't Be Handled A Secret Illegal Base64 data, oh, I forgot this.

I thought that the secret I want to put is this base If you want to insert it in 64 encoding then copy this write there online base64 Encode and go here and put your item and. Base64 value now if I go to this and I will apply secret then create secret Yes, it is good that you have done everything Now the concept has been done in one project. kubectl Apply to the deployment of the back end I have to take him inside, what is he saying?

There is an error creating the back end of wine. Deployment Deployed Verge v1 Can Be Handle deployment why unmarshal number into Go Struck Environment Variable Spec Template Spec Container Element Data Value of type string where we put string People have deployed this value. If that string is not there then he is crying. Never mind we removed it I will apply, still having problem what are you saying is the environment variable dot spec template Spec containers do values of type String O It Must Be String Lyrics.

[Kubernetes] Ingress and ingress controllers / Namespaces

No matter what the given string is, it must be, and if I put this in, hit enter now. Back End Deployment Created Potts in the namespace Chat P back end is running front end is running MongoDB is running on three tiers. Application is running kubectl port Forward Kiska Service Vis / Back and which is in the namespace chat p I will open port number 5001 with 5001 End Enter If Running Then If Running. It will continue, now go to your local Host 5001 check lo not found means error Nothing is coming to him, so he got it.

Its URL will have something else like / auth or There must be something else but it's working, something is going on It's been there, done. The same way you did the back end Port forwarded the front end as well Also do port forwarding what about the front end. Did port forward on the port number 80 and this 0 and run this why

permission din sudo no e this is the same command I started it, entered it and also gave the password, oh already.

No, you can do it through ingress. This is through ingress now. Let's run it through Ingress now. How to run this through ingress If you have to create a new ingress here Ingress Dwa M L Now I'm just here Magic Ingress I will Qnet Ingress And, our Ingress is so right. Networking is so much fun. Kind Ingress Meta Data Chat P [music] Ingress Chat P's ingress is absolutely solid and Label its namespace Chat P chat P namespace What label will you put, let someone put this.

He is not in this, he is putting labels like this Do My Ingress or Chat a Ingress Two chats and ingress, put it like this, speck Rules Host Now what is your host should be. Chat ts.com is the same one we were running just now. Where has our chat gone, this chats.com of ours Host must be ATTP PA path prefix slash if I give slash So what if our back end service works? Which one is the name of the service, what is the front?

End because our front end slash Pay should come and the port number of the front end It is 80, this is the name of the service, this is back and back.

[Kubernetes] Ingress and ingress controllers / Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

Which service should I go to? Make another one of the same which has the same prefix put and if you write / API in it If yes, then back end service back the port number that will go to. Running on 5001 Just a little you have to add a note as well, otherwise it will be a little Metadata gives problem A simple notation of yours in Enosh and Enosh ajax dot. Ingres de Cannes its re right target It should be that it stays slash what it Your application runs on that slash.

Just put this much, this is enough, and if there is more Give you some SSL redirects If so, you can do that Ridden Target like Nginx Ingress had inserted You can do the same with SSL. If you don't put in HTTPS, a warning will appear. CTL Apply -F Ingress Enter Chatap Ingres CreateDrawerNavigator [music] Done running but not on y slash login You should just walk on it, then I am getting fucked up, I am not in love. If you are running Ingress on a Minikube If so, you might have to change it a little bit.

What to do kubectl get NS You get a new namespace called Ingress. The engine I told you about when I was teaching you Ingress controller of the Minikube It is very easy to install the ingress controller. Mini Cube Aid Once enabled, Minikube will be added. There are ons in which you have to enable Look at the ingress, right now that ingress is enabled What is happening is what I would say chats.com On writing that page, it should open on the slide.

Why is Ingress automatically logging in? If it was not working then you had to use Minikube. You will also need to create an ingress It takes a little time, it's a little light This verification process will take some time. May have been revised that a code whose back And the front end is lying with me. What did we learn about Kubernetes? Made it into cube nits means deployed Created Service Created Ingress Persistent Volume Pass Volume Claim Name Space Secret made all these things which You were a variable, you were a confident person.

No problem, this is a verifying ad It takes a little time in the ons,?

[Kubernetes] Ingress and ingress controllers / Kubernetes Services and networking

As long as he is taking his time, we will watch. Let's take what is left in our syllabus to chat application on case three tier to Minikube Mast Deploy Ban He will go now persistent volume also became Everything has become all sorted now. Next up is a CI/CD. Integration of Kubernetes with Jenkins CID GitOps with Argo CD in this one too I will create a three-tier application using Argo CD. If you have it, tell me and you can drive it very well.

After the d on is enabled please run Minikube Tunnel and Your Ingress Would Be Available at 12701 fine ingress add on is Enabled Now my ingress is enabled. As soon as my ingress is enabled Q CTL An increase engine in the gate ns namespace Automatic addition can you see in two minutes kubectl get is added first Ingress Gate Service because the ingress. There is a service in the name Space Ingress Engine has a service whose name is your ingress controller So that ingress controller needs to be exposed to you.

Which will run on port number 80 Now kubectl get Service is available Our port number is 80, there is already something in it. Running so l so f -i port what happens to the number 80 l so whatever The port number is any port in your machine Get the ID of the thing running on the number. If you go then nothing is going on right now uncle

Is it a good thing to have an ingress controller engine? What I did cube CTL Port Forward kubectl Port Forward service Which ingress engine controller which is namespace type ingress Agnes has port number 80 as port number Map it to 80 and this is the command.

You must run the command with sudo permissions because If it gives problem in port number 80 then sudo -e give this password to what Address already in use,. What to do in the kubectl get service namespace Chat p which is the service port number of the front end If the cube is running at 80 CTL Delete -F Front End Key Service. Back up service Two back now go kubectl apply - Apply F ingress also and you Now port forward your ingress Controller port number 80 Why is Enter Address Already in Use appearing?

[Kubernetes] Ingress and ingress controllers / Kubernetes Services and networking (part 2)

Port number 80 why is he giving this HLS OFFI port number 80 port Number 80 P Is something going on? Port number 80 p local If this is what the host is doing, how can I stop it? kubectl delete service done and if you will do this, but it will still continue. Is handling connections for A How to Stop a Looking at the port are these lsof -i colon port number But lsof fun i port Number I get 80 but he is not giving anything.

It's very strange, it's now Whatever port we forward for this They will do it on port number 81, what about that? 81 per tax I gave it to you, but whatever you take, that is the fun. It comes only on port number 80. What should be done what should be done kubectl delete -f dot boom delete everything Tax Delete everything now, how will you do it? You will go to port number 80, you will not go now,. You won't walk anymore, what should I do now, Cube?

Ctrl apply -f to namespace apply it Then create the persistent volume of MongoDB Then make a persistent Volume Claim and Once these people are formed then you can. Let the things go, burst, it's gone now See Port Forwarding of kubectl How much hard work do you have to do now? If you do it, it will work, see that kubectl Port Forward Service Ingress Controller Running I think I'm having a service ingress controller We destroyed the entire system to run it. Did you see now I enjoyed it now if I do localhost then it is running fine.

If I do 127.0.1 take 80 http1 7.0.0 1 Calculus 80 Colon 80 Re so can you see the page of Nginx open which This means it is the ingress controller of Nginx. If I will slash and localhost enter now it opens great If I will / ai So look, it's returning a 404 API if I. I will do slash so this and if I do chat ts.com which is ATTP Your Pay has opened, now in this / API Pay Your not found can you see that / API pay.

What is not found, this is your not found then comes when its back end runs then you even got Ingress running, it's great.

[Kubernetes] Monitoring with Prometheus and Grafana / Kubernetes Services and networking

I saw how you completed the whole Automated my app to check git status Oh my God, you people have created so many things. How long did it take, one and a half hour, good, now What I'll do is git add everyone. If you check git status then all these things are done. Ours is done, now I will do it git commit num aided Cats Git added the manifest of Kubernetes Push Origin Main and I will do this thing in my branch.

If I push it, then remove all this right now. In the folder you will find all your manifest files. Is dead Well done and this was your entire project that User logs in then goes to User's Dashboard User can see the list and so On security layer it does this and that Complete your application. Deployee B is also completed on Mini Why let's move on to the next one, which is a Net python3 tar application that needs monitoring We're going to do Promise and Grafana to again aapko vi.

Deployment of database in specification Deployment of the database service Redis Deploying the service result of Redis Diploma voting of service voting of results Service working so on so this now you make. Let's start monitoring it now. Observatories Waybill three large It is based on interesting principles like Your first principal is Matrix II is yours logas and then comes your third With the traces matrix you do you monitor the metrics? From log from you do You do the logging and tracing What is the difference between these three?

The matrix shows that what is happening in your server what Meaning what is your RAM, your CPU usage? What is your network like so what is Happening in your server, this will give you the Matrix End monitoring lets if you I want to know why that happened. Lines, all these things are very important to have logs logging error line then yours remains How is that, if it happens then reach this How it happened, that's yours for How. If you talk about

monitoring then The most important tool we use are and is From Prometheus Prom to you a matrix You get metrics i.e.

CP usage and You get your rum and whatever else you want.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Pods

Goes from Prometheus then you get the same If the matrix of Prometheus If you want to do many shows, you Grafalco you need to show this somewhere else Another one from Let's live some Grafana. There are many tools available to help you with data mining. You can also show it to the power BE You can also take it to the dashboard where it was made You can also take it to Kibana, so wherever. You have tools like Loki Prom Tail All these are for capturing logs.

If you need to talk to tracing So the tools remain like Yeager Open Telemetry has all these tools which will help you a lot for your Observative What remains is that you have to monitor that. Whatever Kubernetes cluster you created Whatever application you are running in it How will you monitor that should be your Primary M Now I will show you diagrammatically Let me tell you what to do with the let's. You have a cluster of Kubernetes. This is a cluster of Kubernetes knots now.

Any information about the components of the cluster Can you tell me how many let me see People remember that in Kubernetes What are the components that I need from it? You will understand that, yes Kubernetes I have taught it correctly, well done, so V We have a master node and we have workers nodes we have master note we have. Worker Notes Then Master Node Pay We Have Something very cool, uncle is none other than If master note pay we have something ch is known as an API Server remains the correct API server.

Yours and the API server are basically the whole Keeps talking to the cluster and If you need to schedule a pod then you have something known as schedule We have to keep data. Controller manager so we have many such components that make up your cluster Now this API server is completely ready. Who will your worker node talk to and Worker nodes also have some or the other It is a strong thing which is non-age kubelet cube lay and that cube lay sees that.

Your pods are running here or not here Your pods are running or not The pod is not running, this is your pod running here happens to be Correct this P and this is basically your entire.

[Kubernetes] Kubernetes architecture (control plane, nodes) / Kubernetes Services and networking

There is a whole cluster of Kubernetes which is like this The API server can talk to these people. Can talk and so on and vice versa if you need to talk to the API server If you have to use something like this then you have to use something like this. Cube CTL this is a totally cool thing of yours Architecture lives on, but now here's this If we have to monitor the architecture then If we talk about this worker then he.

What is a worker and what is a worker node? This node is one and this node to If all the things in this node are for Example CPU or whatever it is If we put all this into its matrix. All these metrics will have to be exported Whatever data we have about this, we will give it to the people. If you want to send it then we have to use a service known Edge node Exporter we have to use a service known as node exporter so what is node exporter.

Does the node exporter of your nodes Takes information from nodes takes information and a particular port Number remains f is generally 9100 is exported to that country and then Exports Tax How Many of You Got. This is what Node Exporter does for you. It takes things from all the nodes and Export it as the CPU will take the network He will take the rum and whatever else he has from here. Take it just exports it to ever I want to but this is about the node.

The matter is over, right, this is about node exporter. It has been discussed, meaning if my worker Node is whatever workload is running on these nodes All those things will be exported. My API server is completely Is the cluster being monitored fully? The API that is talking to the cluster What should the server have so that once the data is exported, the master node Has a particular namespace How many people know the kube-system? The system kube-system is a namespace The namespace in which your API server is located etcd Controller Manager All these services Even these services are running You have to monitor what you do.

360 degree monitoring for Kubernetes niches cluster so for this thing you need one more thing I want what is called cube state Matrix Cube State Matrix So just the cube exporter or.

[Kubernetes] Monitoring with Prometheus and Grafana / Kubernetes architecture (control plane, nodes)

Node exporter will not work for you. Need Cube State Matrix Cube What are your state metrics,? The state of the cluster is your Kubernetes The controller in the condition of the cluster Scheduler Manager is not running correctly It is not running properly or at all Components are in the control. The cube is not moving properly How is your service proxy doing? How are ingress controllers running? How many of you have just Realike Porter or Docker Container because Docker container logs took your CPU Took it all, took it all away, Kubernetes is a much bigger robot architecture.

In which you have many components which Can go down at any time, that is how you have To do an overall 360 degree Kubernetes Monitoring Kubernetes Monitoring is like this. Let's understand this thing that Something in a cluster of Kubernetes There are components that can give out some Data is fine, but where will this data go? This data will go to your Prometheus This data will go to your Prometheus has one Architecture That Allows You to Scrap DataT allows you to scrape a lot of data.

The one who stands firm in this will be only and only The data will not come from the node exporter. Lots of sources There will be many data sources in it Data will also be available from your API server Data will also come from your schedule. With all the components you have How many data will you send to Prometheus? Of you are now getting something crazy What is happening here is that all the components All these components give our Prometheus will send data it was not till yesterday it.

Was just a node exporter that is one of your Sends out data from a small brick instance We have a lot of things right now that Sending you data for example your node. Your cube state matrix everything so this Data about everything will come to your Prometheus So what is Prometheus? Prometheus is a time Series Database It's a Time Series Database That Has Abilities to Scrap data he can scrape data he data It contains a query server also resides and what does the query server do.

Prometheus using Prom Ky L Using query language, he can access your data.

[Kubernetes] Monitoring with Prometheus and Grafana

You can also query and even graphs gives to create and all this data is available to you in S.A. Stubborn Source You can also export all this You can export data from this data source. Ho in your You can find all this in Prometheus Correct Prometheus. You can now export this data. Our visualization will be done by exporting I will go into visualizations such as graphs that Your charts are created, graphs are created When this data is created, it will go here.

Visualization tools now in visualization Which one is it, it is a visualization tool Yours Graphador visualization tool general Grafana in interviews I am pretty sure you are asked for it. You must have heard a lot about Grafana. That, creating dashboards on Grafana is very easy. Tell me what graphna is that I allow you to Give visualizations that visualizations that data to you gives sources To visualize like here data source This is where the data is stored. There is a data store and this data store Data can become the data source of Grafana Store means where your data is stored.

The data source means where it is stored. Your data is coming from there Graphador Complete Graph Waf Ye Aapko Banay This type of graph will give you a good meaning This is a very simple graph, we. They are going to make very strong graphs, so all this Things are going to happen to you so this is how a Stack for your monitoring of a grave nits Cluster Looks Like Kubernetes Monitoring. It looks something like this if you If you want to understand it diagrammatically then this is how What Your Kubernetes Monitoring Would Look Like Now the question is that I have here kubectl get.

Notes I have my cluster ready here It has three great nodes, how many of them You are now comfortable on creating a How many people will make up a Kubernetes cluster? Kubernetes Cluster very comfortably that You can make mini cubes through kind of through bana sakte ho der r many ways to Create a Kubernetes Cluster But now that. We have a lot of things going on in the cluster. What we have to do is we have to do that I have to monitor what things It's running, I have to monitor it now.

[Kubernetes] Helm and charts

Promethazine also has to be installed stat We have to do the matrix and connect these people also. There are a lot of things we have to do learned to create a manifest file and We can do many things by making a manifest People saw the question, how did we quickly. In a single click all the things at once I can setup the entire manifest in You can use all the files at once. The answer comes to him Helm I will go and click on install them.

Installing Helm is here Installing Helm now What do I have to do here? How to install it on your system I will also do Helm from one of your scripts If you can install it, install it. It is very easy to do, we had seen it. Curl f ssl l whatever and these get Helm th this is where its script lives If your Helm moves then MKDI Do I have anything about R monitoring? Mkdir here Monitoring creates a folder called monitoring and here we are I will do the curl command that was raised,.

Get Helm, we will get the Helm here. Curl f ssl - get Helm enter if you please get held This is the shell script that installs Helm If she has come to you, then Question is Shall Script L S Han L Is this executable? Does it have the x permission which is execute no does I don't have to do anything to help myself 700 get Helm file for user I will give executable permission and if I Lace. My get Helm is now a table with x permission executable permissions has come close Now I will install it.

Held from here install your Helm Will gethead a enter verifying check sum Helm install into user ban Helm to Helm nafan. Try this version, I think it's just version then version build info version v 3.16 3 Latest Helm You're Done Install Well done Now whenever I buy anything in Kubernetes. If I do it, I do it in an isolated group. I am for example if I have good engine If I want to run a SQL or anything, I do that group in an isolated group what do you say if you can tell me.

[Kubernetes] Monitoring with Prometheus and Grafana / Helm and charts

kubectl get namespace so my Have lots of namespaces Good ingress All these are the same as me and another namespace I will create kubectl create name Space Monitoring then created a namespace. I will give you monitoring in which everything All the monitoring work that will be required I will do it now what is my next job? Me the most At the first Helm was the Prometheus Community's Joe There is a Helm chart, so you will have to add it. If I tell you what I'm going to do I am trying to then Prometheus Community where are you writing Am I Prometheus Community Helm Chart So when I also do Prometheus Community.

Helm chart so here Prometheus Community Helm Chart Help Here Charts can be found in Prometheus's All those things are written on it, all those things It is written that you have a Prometheus. A whole server of a Grafana can be made so everything is already done So if you want to explore, you can You can read the charts by clicking on them. Straight away once Helm is your setup If it is there then just add it to the repository. Helm Repo The Prometheus Community Prometheus Community Get Hope Helm Chart To What will this do?

This will cause the Helm repo list to fail. I will understand that my Whatever number of helms the Prometheus community has Charts are all the manifests. This is its URL where the hand I have the charts, I have the latest ones. I need a repository where all the updates are available. If things happen, I will do it Helm Repo Update What will happen with this I have as many There are also charts, they will be updated. Successfully got an update from the Prometheus Community Chart Repository Update Complete Happy Helming and Kubernetes Look at the symbols of the nineties, they are also crazy.

Add whatever you want to the Helm repo. What you have to do is Helm through What to install is a Prometheus Installing Stack Prometheus If so, install Prometheus Stack now. Where to install Prom Stalk You to your Prometheus community Prometheus Community Prometheus What is a community? Prometheus Community My Helming Is What I'm Looking For Cube Prometheus Stack is what I'm looking for now it has some properties.

[Kubernetes] Monitoring with Prometheus and Grafana / Kubernetes Services and networking

Like the namespace you will need to use So that hyphenated namespace that we use. What is it that has to be done Monitoring, like in the case of Docker, we What would you have done in Docker's case? We need to add namespace monitoring. Which is your grief and your Prometheus So can you tell me if I Nearby is a Prometheus pod made from lettuce. Is It became necessary to give him access to the outside world. What do I need for I'll have made a pod and given it to me Or has that deployment been made What is needed if you want to expose the port of.

Gotta give me absolute node port ch e a Service is what I need here. Setting here is setting the null that Prometheus's service Which one will it be, that node port will be and You can give whatever port number you want. Port number should be less than 30000. You can give anything up to Rs 32,000, I will give you Rs 30,000. To set up your graph If you have to do it Grafana.com Service.node.port Grafana's nut port You can enter something else, enter 31000.

Grafana's Notepad 31 is now available The default Prometheus stack comes with Your cluster IP address will be displayed to us. If we want to build a port, what do we have to do with it? Have to set up have to set up Grafana.service.type how many people I understand that whatever comes by default This is what comes by default in Grafana. That's your cluster IP in Grafana Or in Prometheus or in whoever he is The cluster IP remains and here when we If people change it, it will be called service type.

What node port will we have to do? The spelling of nut port is nut port correct And similarly we will have to set it up. Nafan Set which service is it Prometheus.service Dot type e to node port till here everyone It is clear that Helm will have to install Prometheus Take Prometheus Community Q Prometheus Tac now namespace I will use. Monitoring if you see anything set in the Helm like set anything in values null If you want to do it then you can use - - Set Now I have a note of Prometheus' service.

[Kubernetes] Monitoring with Prometheus and Grafana / Kubernetes Services and networking (part 2)

I have to set the pot for Rs 3000, so I did it. I want to set the notepad of the graph service 31000 so I did the type of service I have to set the note pot, so I did this. Enter Changing the Note Pot of the Service of Datchi Us So I did it and with that being Shade Boom Prometheus Stack Deployed namespace One revision deployed in monitoring has been done. Check its status by Running Cube Seattle namespace monitoring now this to me Tell me how difficult it was yesterday when we Looking into Docker Compose We are going to graph the set by set by set.

Prometheus was all this and now it is Just one command and with help you if Doge Cube CTL Get Pods in the namespace Monitoring Can You See Your Prometheus Stack node Exporter is running node Exporter is. Running Node Exporter Is Running Node The exporter is running now tell me this Why should a Doti four node exporter be run? Are the reason being you if your diagram If you look at it, we have one or two There were three notes, but here we have a cube.

CTL Gate nodes is four nodes. Four nodes There are four components for the lake, not four. Node exporter is used for nodes We Have Cube State for Components Matrix, so one master and three worker nodes. There are exporters and your master node Whatever matrix is there in the control plane That is your cube state Matrix Crazy how many people seem to understand that We have cube state matrix which is. This we have nods given by given by nod Exporter And everything is Running Everything Is Up and Running now the question is access it How to access it, how to access it.

One Look First Understand kubectl get Service in the namespace Monitoring, so we are in monitoring If you want to get the service then do it for your own work. What is it, look here what is useful for you If you want to see only that much then Prometheus is useful among the stacks and from the Prometheus stack Grafador is everything going on. Prometheus does his work and Graf does his work If it is right then what can I do Doonga kubectl correct kubectl port forward Whose service is mine, my service is mine whose name Prometheus Stack Cube Prom.

[Kubernetes] Monitoring with Prometheus and Grafana / Helm and charts

Prometheus this Prometheus port running on port number 990 But it is going on, I want to expose it. It is running on port number 990 I want to map that to my from port number 1990 which is port number 190 I am running it on port number 1990 The namespace I want to map. Running in Monitoring and At the Same Time The address of this That's what I want to expose 0 I want to expose its address. I'm Prometheus' 1990 exposé enter so can you see Forwarding from 0.0.0 port number 90 is mine, it is up and running on pot number 1990 So I'll jump into my instance.

I will go and see what happened to it in my instance. I'll go to the end instance and what do I do? Karunga Any Guess Main Security Group I'll go to these security group I will go in bound rules I will edit the rules and I will edit this. I will tell him that my, I just have one D rule. Do it ch e Give it to 990 and any where and just save the Rules 90 is now up on my instance.

I will go My IP address uthaoonga and if i go on a new and HTTP always don't make this mistake its 990 But if I Gone boom I have Prometheus Server running for. Me a Prometheus is running for me how cool Is this what I did Helm It's installed, that's all I did. I'm all Set but how many people know that Prometheus We have to maintain a file in Whose name is Press YAML in which you. Confines of scraping have to be inserted correct how many of you remember that Need to enter the configuration for scraping are you using now that in Helm Configuration of scraping.

All are already placed, all are already placed Something has happened, but it is not coming back. It's fine, but you can see your alerts. All the metrics we have Are set up Can you see how cool this is That you should have your Confidence You Have Cube State Matrix. You don't even need a Sea Advisor because the cube state matrix is your cube Whatever resources there are in the cities, whatever There are ponds, she is giving information about all of them.

[Kubernetes] Monitoring with Prometheus and Grafana / Pods

Your node exporter is as many as you want in your node There are also nodes giving things of all kinds End You can update it by going to Datachi. Interesting how Prometheus explodes Sara The work has been completed using Helm charts Now I'll run this thing in the background. Run it in the background Enter, so it's running in the background. Similarly kubectl get Service In the namespace monitoring what will I do now I my Grafador Fana here is my Grafana correct this is it My I'm going to expose So kubectl port Forward kubectl Port.

Forward Service Grafana Prometheus Tack Grafana its port number which is running in The one who lives outside is 3000 Generally I want to use port number 3000 for port number 80. This namespace comes into monitoring and after that I What should I do to address this as well? To Asus Address 0.0 p I will open enter exporting from 3000 so at 3000 I will go to my Instance security. P Generally you need to know the port number 3990 This whole thing, it became life, became much Easier now I can add rule here and make it 3000.

I will add for anyway and after doing this I will save it and as soon as I save it This server of yours is the same as Prometheus. What will you have running on the same server? Pay your Grafana is also running means what Helm is brilliant, you have achieved it in one stroke Kubernetes also means Prometheus in Cubart I have also made a tank and the Helm is also dead in it. How sorted is it now, the question is what is its password admin and admin but if I enter admin Then my login fails.

If you have login then it is failing What should I do now, my login failed. If it happens then it may also happen that just to Improve security of your cube nodes The number of pods running in the cluster Make it M% and run it in the background. Just because of the security you have Also, if any pod is running, its password is something else. It is possible that his password may be something else. If possible, you can enter the password from your pod.

[Kubernetes] ConfigMaps and Secrets / Monitoring with Prometheus and Grafana

You can take it out or you can use your deployed one. CTL Gate Secret kubectl get Secrets Where to Get Secrets Prometheus Graf Prometheus Stack Grafana to which Prometheus Stack The graph that you were running was a reflection of this thing We have a secret password,. This Prometheus Stack has to be removed. Grafana is it running in namespace monitoring So this service is what it is If you want a password, then this name out of what is going on in space monitoring What will I do Han O.

Json Path then Json Path in Json based on JavaScript object notation requires its path and You will put it in double codes, what will you say about Atlee? If you want then those secrets are there Secrets he generally keeps in dot data and also those who The admin password is yours. If we do it like this, kubectl get When you enter the secret admin password, it Your password arrives How many people understand your password? L Get Secret Whatever Your Secret Generally Forget the passwords that remain secret Have you all gone if we change If we have to put it, then where do we put it?

In confine and if we give people the password If we put it then we put it in whichever way it is That Kubernetes resource in which we people How to create etcd using password or secret. Maybe my has secrets Enter the secret password in secret and The variable you put in the confine Yes, now there is another speciality that This secret is Base 64. If I keep this secret I will pick it up from here and if I am in it here If I enter it and try to log in, the login fails.

What will we do with this, we will pipe it will decode the - ain in base64 and Decoding will reveal the Boom Prom operator The password was named Its hacker man is understanding that some. Anyone's password by going to

any service how to steal Here you take the secret and base64 decode it Enjoyed tomorrow you can anytime any of your Project Banaoge Let's Say MySQL. If I pick up this prom operator and I if This time I will put it smooth operator enjoyed how you in Grafana I have to login as admin, admin is not working.

[Kubernetes] Monitoring with Prometheus and Grafana / Namespaces

Going to run fine how many of you got this Crazy now, we are new in Grafana. We'll go here and connect Go and add new connections, how many of them You think I'll go over here and take Prometheus I will search and this Prometheus ad How many people remember this thing? I'll go to The New Connection Prometheus I will go Why not because there is a Helm chart where Stack of Kubernetes Prometheus on If you go to Connections, you will reach Home.

You will go and see that your Prometheus is already set Up Prometheus is already set up why because you Helmet up, Prometheus is also set. Turned up and your cube state matrix Prometheus and all those who are Grafador [music] Add Any New Sources All You Have to Do is just build a. Dashboard Click on Build a Dashboard Your which is this Prometheus it's already added If you want to add, this is the URL http.com PR Monitoring How it was discovered Whenever you go to any particular university.

If you want to reach the service then use the service. Name dot namespace if you put then you get there I have now added the data source. I have to make a dashboard, now what should I do? I will do ad visualization here I'll go to ad visualization, this is my data source. Prometheus is that I take everything from Prometheus I am now here in Matriculation, you who You can put whatever you want with let's Network Container Network Receive Byte These containers receive bytes from the network.

Advisor used to give which is related to your Docker Monitors the kube state of containers Matrix is also giving the same thing, so container Network Receive Byte in which now here you. Can you put a namespace here, how cool? It came, if you put the namespace then the namespace I put now whatever namespace I Let's say me in the monitoring namespace If you want to run all the things from Anything in the monitoring namespace. I can see everything that is happening in front of me, let's So tomorrow I have not done any other new application.

Created the Three Dear application on this Deploy the application to Raising the namespace of the deployed application I have to take it late, I did it well tomorrow.

[Kubernetes] Monitoring with Prometheus and Grafana / Namespaces (part 2)

Run query Apache's namespace is visible You have made me a kube-system, a kube-system I ran a query of the kube-system Whatever requests you have, I will give you all the things. Looking how cool this is that now I have to do it only according to the namespace Dashboards need to be created with just the namespace Create an accordion dashboard are you having fun. Came interesting how many of you thoroughly enjoyed The Power of Technology The Power of DevOps Using Helm, Helm is a package manager.

Which we first do component by component Components used to be things that are now a package. Gave you a command help install Cubase Stack Prometheus Cubase Tack and Vid That everything is done, another best thing told It has the Kubernetes Prometheus tank which They know from which second they maintain it. Dashboard is to be used, so if you want to use dashboard If you go to, all the dashboards are for you. Already made Prometheus's node for you of the exporter Kubernetes Scheduler persistent Volumes Pod namespace All dashboards of the cluster are already created Yes, everything is already done by you all dashboards built.

Still, if you create your own dashboard Want to create your own dashboard? Go to the Write Queue Grafana dashboard Grafana Dashboard s Write Next Grafana Dashboard Write and Grafana Dashboard Key Grafana dashboard will come on the official website Grafana Labs Click on it and like. You will click and you will see many dashboards You will find a lot of things, just search. Kubernetes Prometheus just not Look, it did not come, search on the internet and yours came. Loki Stack Prometheus Case for Storage Prometheus Dashboard that suits you, that suits you feel like you can take whatever you want.

Kubernetes Views is a GraphQL dashboard. Namespaces are all there, you can take them all. Are you looking for Cube Deploy Full Set? D mind set matrix so I'm going to click this If I take it, look here, deployed memory Use deploy on available this all thing to you Here I am showing you what all is yours. You can use whatever you like Whatever you like, here you can find it I just typed Prometheus in the search late.

[Kubernetes] Monitoring with Prometheus and Grafana / Pods

Search Prometheus Now here you go Which one is needed from a Qubes cluster If you want Prometheus, click here. Two this is cluster CPU cluster disc all these things now this one Putting the dashboard into my Grafana So what should I do? I will go to Grafana, this is my Grafana URL and new jaanga new jaake I will click on import. I clicked on this gf.com URL I have an ID, I will go there and throw it away. I will load it, as soon as it was loaded he said Now you have a Prometheus data source If I want, I'll go to the data source and it I'll click Prometheus.

I clicked on it and now I can import it. As soon as I imported it, can you C is not just the last 30 minutes, but the last 5. Things will start coming later Can You See a Pod You have 31 pending pods and you have 31 running pods. You have two pods in your cluster. 35 containers have already been moved and a waiting tax is being charged. Two containers have been terminated now. Let's terminate two containers now I can edit this to make the container look risky.

I will give and go here and color Mode Background Solid and Its color red If I give it and save it, what will it do? If any of my containers terminates If it is happening then I will quickly cover it with red color. Identification help you too Do storytelling that two containers are dead There are so many running containers running from lettuce If you want to show it in green then edit Do it, add color to it. Green Because Running Green Looks Good Too 35 Running Waiting Blue Contain Terminated Red how cool is This is how cool this is Now, this is very crazy.

I'm going to do something And we do crazy things one level higher I will take it with me to my GitHub Whatever project you take up, The project is ours,. That everything is given by God, if someone But put your hand on it, no one will refuse, so one Kubernetes could have taken up some project of Tus or some would also take up the project.

[Kubernetes] Kubernetes Services and networking / Namespaces

It's a very solid Python with dot net in it. It has nat jus PostgreSQL all Yes, we will clone him. How to git clone your voting page We will take cool CDs, voting is over now. In this voting paper I have Ketus Kubernetes I have some manifest files like It is deployed, it is a service, it is Redis, everything I have things, what can I do with them? I will check this once to see if its namespace What is Result Deployed its namespace once If I look at it, it's not in any namespace.

When any manifest file is with any name Which namespace does she not live in? If she lives in then that is the default name So if I do cube in space CTL Apply -F Dot Bhat Fat Karke Deployed created service created Deployed Redis created whatever my. There was a stack here, this one [music] python2 l gate pods so my database is running redis is Running result is running so what is this Basically this is a voting application where. You can vote for the dog or cat here.

This is our beautiful voting application. Tire application is running kubectl Gate service Gate service so you can see Voting service should run on your port number 5000. Voting Service There Is An Application Voting service whose voting is running at Rs 5000 If so, cube CTL Port Forward whose service is our voting Service Its port number is 5000, that is me. I'll map it to my 5000 and Address For my Public and its service name is Vote 5000 and keep it in the background I will run Enter forwarding from 5000 now in my security group.

Open your custom IP save rules and like I Voting: Now when you go to vote, you Do you wait for the result as well? I will also open the result service. The service is the same as the voting service. The services have arrived, we will add this also. Giving We are 5001 in our security group. This is my IP address Pay Apne Paas 5000 Port Pay Dog Cat Voting is going on, take it you can Participate in that voting age well and.

You like dog more or do you like cats more, look at me So I like dogs more, so I will vote for dogs.

[Kubernetes] Monitoring with Prometheus and Grafana / Kubernetes Services and networking

I will do it and vote for that What will I do on this, simple So I'll do a small thing on that vote. I want to use my Kubernetes cluster if Possible a don't hit it because again your If there is too much load on the server then close it. Do it don't hit it now i can directly I will go to my dashboard I'll go to my dashboard and I Directly to your Kubernetes networking workloads and can. I see all the networking here at UC.

In any namespace now I will select the namespace which one Default namespace can you see that website My average rate was not opening and everything increased. The Can You See The Network Here It has increased a

lot, why because everyone Tried hitting on that and as many as I pods were it started lets say. I got the result which was your voting service. If you want to see it, you can go to the voting service. The Voting Service Got a Lot of It Current Rate of bite received lot so whichever I Workloads should see the results service.

I can check the service to see how my work is going. Has been the last how in f minutes How cool is this, how crazy is this now I can see the entire cluster that my What is going on in the entire cluster? It's fun Now let me go directly to Prometheus. Your cube Kubernetes let me get a new one I want a dashboard now from this one. Need for My Kubernetes Nits, which looks very sharp Dash is perfect gonna take this dashboard Here's this dashboard, we've taken this dashboard.

The thing is visible and I can directly see it in my In Dashboard, go to New, go to Import I went and pasted the ID and it was loaded. All my data is scraped in Prometheus It is happening, I have imported it and Just Like That I've got everything in its place What is going on in it, how much CP is being used? Which of my nodes is having the most More like cluster worker one on most Too much load, can you see this at my Cluster worker one group the maximum load now.

[Kubernetes] Monitoring with Prometheus and Grafana / Kubernetes Services and networking (part 2)

Being a DevOps engineer, you can see that Your kubectl get pods this voting Pod kubectl get Pod -O Wide Vote Worker Cluster One then you Directly identifies which nodes. There is too much load and you go there You can do scaling, you can see the workers There was a forest and the load on it was too much. Because our voting service was that was on worker one and because of this Dashboard You Can Tell Your Team You Can Tell Your managers that we need to work on This particular node Scaling this node.

Will have to live in IPL Cinemas Hotstar How All This People Scale They used to have these dashboards all the time They kept monitoring where they The instant you notice the scale is rising. Would have liked helmet monitoring where We have developed a three-tier application. Monitoring also seen on Grafador and Every Matrix Available In Kubernetes, I enjoyed it, now you can enjoy it too It is going to be the time when we will do mega projects. We are going to be in this mega project People who are going to use the cluster will There's going to be a cluster of KS.

KS Elastic Kubernetes Service AW is a managed Kubernetes service What is the benefit of ES? The advantage is that you have control There is a plane where all the services are running. Your API server and your etcd Controller, Manager, Scheduler, all these things AW manages so you get the load You don't need to, you just need to setup the cluster. You have to do it and deploy your application. What is the application, your application is one three Tier application that runs on ES Whose CI/CD is going to be of Deaf Seps through where we are going to use Jenkins.

Have evi ova saner kube docker email notifications and this thing that will be GitOps That would be through Argo CD using our git repository then we Helm your cluster We are also going to monitor through where. Prometheus and Grafana use both these things By doing this we are going to monitor so this A two-and-dime mega project is on Kubernetes But Its Small Requirements You should know how to create an eCase cluster.

[Kubernetes] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Kubernetes Services and networking

Let's quickly make EKS Cluster right now If I do kubectl get notes then These notes that you see are of our kind. The nodes of the cluster are now here I want to create a KS cluster, how do I create one? Understand this thing which is of EKSAW If the service remains, it is a managed service. If I search for EKS here Elastic Kubernetes Service Here one of your clusters whatever your clusters I can see all of them, so I am here Create cluster is also great from here.

I can also create the cluster from the UI but this Banana is not the best approach I will make Using AW Sal I, first of all we Installing AW Sal I to People. It is very simple, you just need to go to googleclassroom.com You can install AWS Ciao here. The command line install is provided in curl HTTPS AWS CI this that complete this Copy it, it's just a small thing here If the unzip command is being used then your system. If I need this unzip tool then you can install it.

Take sudo at get Install azip install this unzip tool Now what do you have to do to complete this? This is the complete command, you just have to copy it. AWACLI what is it There is a command line interface through which you Clusters ECS containers or bricks instances S3 bucket anything you create from here This is done, what

should I do now? I will confine my AW, meaning I will To the local machine, to this machine of mine, Connect to the server with your AW I will do it then add Now you will have to confine AWS here.

Your access ID seems to be Joe AW This is S's account, this is one of them. I'm doing this because I will create the user AWM my in the account and the credentials of that user I will give them access to my server. I'll give you all this, this TWA User next is now granted permission to this user. You will need to run ES, so you can do it here. Pay either by searching ES and give full permission Or you can give administrator access.

You can get administrator access directly So that he has access to all the services If you get it then VPC for ES and ES and your EC2 instance You need access to all these things.

[Kubernetes] ConfigMaps and Secrets

This will be achieved by this, after doing this we If people click next then we will update the user's Details put di ES TWA user usko granted administrator access and Create user with this and your ES. Twd user this has been created once this The user has been created, you have to click on it. You need its security credentials Go to Security Credentials If you want to go and create an access key Go to the Create Access key and this is access key, it is such that.

It is very secure and through that What can you do with the command line? If two secret access keys are created then I copied the access key and here I went and threw it away, done, my access key is gone. I have given this secret access to my instance I took it and I went over here and knocked it down. I lost my secret access key. Default region name can be whatever you want. It is in Mumbai region which is AP South One If you want, you can live here in AP South One.

You can enter whatever your default region is. Even if it remains a nun, there is no problem. Portfolio Terraform All Buckets My what does it mean it means This is my AW this is my S2 instance my The figure from the AW account is now. What I'm going to do is I'm going to cluster here. If I create it, creating a cluster is very simple. This is our project, a mega project. By going here you will find it in the repositories You have to go to the repositories and search Wunderlist and Wunderlust Mega I put all the commands in the project.

How is this project keeping all A to Z? What's going to happen, everything is going to happen It has been put away, what should I do now? Just go over here and tell me a few things. You have to install it like we did I have setup AW CI cube CTL so we already have EKS CTL means Elastic Cover Kubernetes Service It also has a controller, so that too. Enter boom and y In a short while your e CTL will be installed then e CTL what is your version 0.19 that's right this is the version of your.

[Kubernetes] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Terraform and infrastructure as code

Pass required now once you have your ES CTL The setup is done, now how will you create the cluster? Create cluster like We the People of Kind Create Clusters Similarly, CTL create cluster now What else will be there in it, the name of your cluster Let's say I want to enter the name of the cluster. I want TWA or ES TWA ES to obvius si baat If there is an ES cluster, then there is a TWA cluster. This is the name of my cluster and what I want to give it, I want to give the reason here So what is my reason, I want.

That this is my cluster, it is South One. Live in this region which is Mumbai region And what I want is your version of it. Whatever version of Kubernetes you can find here You can put it in your version from Let's 1.31 or 1.30 or whatever it is, put it here. You can use the latest version 1.31 that is running. You can do it, it is 1.32 but it is elastic Cube Netservice for that ES 1.31 running now you have to do this without node To group without node group without.

What will happen without node group? Your cluster will be just that and Only one cluster will be created and its control will be maintained. A plane will be formed with no nodes attached to it. I will add the course to EKS, can you see it? I will give it to you, great and this will become your cluster. If you go then at least bring something in it, it is recommended Policies and Found for VPC CNI Add On no scene if you like this.

Add ons in the cluster means some more add ons add an addon like Prometheus Age Can you Open OIDC means Open ID Connect? He keeps doing this, you can add that too I have created the add-on Successfully createdDocumentsFragment Then we have to add nodes to it. Adding nodes is also very easy for you. What to do is go to the same repository What I told you and how I told you told me about the advance,. EK CTL Utilities Associate IAM OIDC provider so by doing this you OIDC Add the provider, what happens?

[Kubernetes] AWS (EC2, VPC, IAM, S3, ECR, EKS)

Your master machine is the source of all the other machines. Once the machines are connected, all you have to do is You have to do it and once you do it, What you need to do is create a node group. Node group means node group is such a thing which contains one or more nodes are from let's that I made I want three notes or two notes If you want it, I need at least two more. You can say this maximum two times and let's So you wrote five here and you wrote Nodes minimum two and maximum five What does it mean, minimum two notes always.

Should keep going but in case of scaling Like we scaled the nuts The case can hold up to five nuts, so it The thing seems to be that you need to look at the type of node. You will need to decide which type of nodes you will need. If you need something too large, you Run the application if t2 large if small If you want to drive it, it can run on T2 medium also. Keep t2 medium then at least t2 micro don't Keep it because it fills up quickly.

The node volume size is what you need You can keep it as it is limited to 30GB. Free tier but still free tier so something is left No, it's a paid service, so there's no free tier. Forget it now, after that we will do such a thing. Access will also be required, that is why SSH Public Even if it is imposed then where will the SSH public come from? Where can I get it, it is a little below. You can create a key pair here.

Take it and whatever pair is formed, you can see it here. You can attach and SSH by putting Enter access and enter node volume size Total by putting minimum and maximum nodes Specify your note type by entering the number of nodes. You can enter the name of your cluster. Enter the region and create your node group. If you want to give then my EKS cluster TWA Cluster in AP South One Region It has become a case for me, what should I do?

[Kubernetes] Terraform and infrastructure as code

I have to do CTL create node group So ek ctrl create node for cluster Group Cluster Name TWA Cluster This happened but before that I opened the ID. If this happens then this EKS CTL Utils Associate I am OIDC Provider Do this You will have to and the reason for this will be the same. The region where your cluster is located I mean, what will remain of South One then? The name of your cluster will be which is Your TWA Cluster and Your Hyfe Fun Approve this will be and e ctl of your IM.

Turn on an Open ID Connect provider allows you to differentiate between the rest of the instances Now you can connect after doing this. Create a node group for your cluster So let's create a node group for the cluster TW s cluster that cluster What is the region of AP, my region is AP South One. What is the name of the cluster, tell me both of you. This name is not the cluster name, it is yours. The node group is named with lets TWA cluster ng means node group this is your The node group name is then your node type.

If it becomes too much then give T2 medium also. If you want it from your cluster, you can put it here. Lets you set the node volume size to 29 GB. If you don't want only 20 tongues then you can come here. 20 Put Han Afn SSH Access Also Needed You will need to enter your SSH access. It is not equal to this, it will be higher here and Here you will have to enter the SSH key. What will happen to SSH Public?

If it is made then you can use it if you want. You can create a new one too, but it's fine this I have used enter, now what is happening? Creating node groups in version 1.31 For Your ES Cluster and Deplat TWA cluster is so perfect Not with in I think 20 minutes you a. If you can create a cluster of ES then a little and if we talk about the project If we do this, then this project is quite next.

Level project is where your Argo If CD is also being used then this ES cluster But we will also make Argo CD.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / kubectl commands

Monitoring will also be done with Prometheus DevSecOps is also going to do it with Grafana Jenkins' along and one of his three dear to the application Deployed again like I said. This is good yes by the way How does eCTL create its ES Pay cluster? It creates cloud formations from clouds Formation One If you look at it, EK CTL TWA Cluster creation complete, you have a TWA cluster made an eCase cluster using cloud formation beyond that if. What's built into our servers?

If you want to see, you can control your server. The plane became the security of the control plane Create a group Ingress Note Group Internet Gateway VPC Route Table All gateways, oh my god, so many things Built in your EKS TWA cluster. Total 32 things made, what if you Go to Stack Action and select View. It happens by viewing

and viewing and you complete it. What the entire cluster looks like You can see it completely, not this, here comes one. You can see the entire cluster here.

TWA Cluster NG has two notes You have two notes created. Group in TWA cluster so our If the node group has been created in the DBS cluster If I do kubectl get Notes kubectl get Notes to Can You See IP 192 168 21 100. AP South One Computer Internal is Ready i 192 This is ready so the two notes that you had It is ready but, there is a control plane in it. Why is it not showing because the control Your plane remains managed in the AW Side To amazonbusiness.in nodes - - use context is equal to whatever your Kind of like the TWA cluster, whatever that was.

I have to do it, which one was ours, take it, I forget the commands by the way, whatever I have put all the commands in my git. All the things we did are uploaded if You would be better off with another cube kubeconfig Use of Confine Dub Cluster Can you do this right now from Let's Cube Seattle. Use Context Kind Clustered Use Do enter context this is done Now I did the get nodes, so this is my kind of The cluster's gate nodes have now returned to me.

[Kubernetes] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / AWS (EC2, VPC, IAM, S3, ECR, EKS)

What can I do if I have to shift the case? Am AWS EKS The document is fine, it is of an AWS. This is the official document of EKS Cluster. There is a command for that, what is that command? It tells you to tell your AWS You can update the cube Confine is for that you AWSE case Update kubeconfig AP South One and the Name or whatever your cluster is. The cluster was named after Let's TWA. Cluster was inserted then you can see your cube You can update the config now if you If you do GATE notes then you will see ES people.

I will give you a great project complete. CI/CD integration, all these things, oh my god! Father with this Kubernetes Tuzz in one We are about to have the final end of the shot video. People Core Concept Workloads Networking Storage Scaling Cluster Monitoring Advanced Features in Cloud Native We told me in the comments Tell me if you need AS and GKE also. CICT integration with Cube Nits We saw microservices on ES. Leave a like and go away if it's 16 hours for If you liked teaching then this is Only 16 hours are left for Deb 0 to Hero batch.

Which is starting from 25th January I was live in it for more than 100 hours.

Kubernetes on AWS EKS: Production GitOps Project (Terraform, Helm, Argo CD)

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[Kubernetes on AWS EKS: Production GitOps Project] Helm with Argo CD and production GitOps project / Argo CD projects, App of Apps and multi-cluster

You have come here because today's video I'm going to talk about Kubernetes. I am going to do a multi-tier project. Multiple Databases Multiple Services Use It does. UI It's built in Java. The catalog is built in Google, The checkout page is built in Node.js, orders Made in Java, the card is made in Java. It's I am going to do the entire deployment. On EKS cluster via Terraform Using Argo CD Helm and Nginx Ingress controller. You heard the Kubernetes I am teaching today.

The project is absolutely zero. I will teach from But what is Kubernetes? If you If you do this then you will deploy this project It will be very easy to do. This I have divided the entire video into three phases. Where in Phase One The architecture of the project and We are the ones who set up the infrastructure. In Phase Two, Kubernetes and To understand the application We deploy it on EKS auto mode, we are going to do it completely end to end.

The third phase is GitOps Deployment. Where we are Argo CD Git Actions How to see all this stuff When the developer commits the code, automatically upload it to GitHub Actions build that thing on ECR Deploy it on private images and. Helm Argo CD's through your cluster I'll be up. Let's start the project Fuzz with Architecture and Infrastructure. An AWS Containers that are available on GH on the official repository Those people are doing such great projects. One of those retail stores Sample app project that can be downloaded from your DVOS The has done a great fork and It is good that you add more chilli and spices to it.

The orders are written in Java. Here you have Java, it is connected The service that is checked out from PostgreSQL It is connected to Redis with Cart The service is connected to DynamoDB. The service with the catalog is connected my from the SQL. This whole multi-tier This is a microservices application. Monolith Applications Why didn't you take it? Tomorrow The service that you provide for your orders includes some There was a glitch. Did the rest of the service impact. Everything of yours will continue to run smoothly and you will get everything One item not in the entire repository Microservices This is the biggest advantage and Kubernetes for Microservices.

[Kubernetes on AWS EKS: Production GitOps Project] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Kubernetes Services and networking

It is considered best if you have a micro Services need to scale. Product of the project Introduction I hope you got that It's a great project. It is an e-commerce project. You have a whole monolith, a huge one. You don't have to maintain a repository. Small services you can also decouple him You can. It happens, micro services and monolith Architecture. Why Use Amazon EKS Auto Mode in iOS? First of all, this whole To do the project you will need a You'll need to create an account on AWS.

You can enjoy this project. You can run it on AWS. If it's Compare production Kubernetes to EKS Let me make it a little bigger. Zoom in I give EKS Elastic Cubinetti Service provided by Amazon There is a production ready Kubernetes service. Where you can create your own clusters and The control plane part is Amazon He looks at it. It would be fun if you could explain it through a diagram. If you need to pay for production If you want to go, you have to use cloud services.

From Amazon, Kubernetes clusters you can create Production ready is the EKS. Let's understand what EKS is. If EKS Talking about simple Cuban is cluster. If you have asked how a cluster is formed? This is what is called the control plane. The control plane is your entire cubeset. APIs to it Server, Scheduler Your Controller Manager All these things reside in it. You only need to pay these worker notes. The applications you deploy are your You'll manage. The heavy lifting that happens The control plane is managed by Amazon.

All you need to do is apply. First of all You need compute, meaning your EC2 instances. You need storage Meaning your EBS volumes would need to be reinstalled Is the network mean what your CNI container Drivers of network interfaces then VPC It contains IPV4 addresses and IPV6 addresses. If you If you want to provide outside traffic then load balancers You also need an LB. This, this heavy lifting with EKS ye You also have to

manage EC2, Then manage EBS, manage VPC, Manage ALB.

I'll also manage your load balancer. Even if you talk about security We'll also manage security. The images used in the programs that will be used Machine AMI will be Amazon Machine Image which is used It will be immutable.

[Kubernetes on AWS EKS: Production GitOps Project] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Argo CD projects, App of Apps and multi-cluster

This will also eliminate security risks. Data will be in the plane via Amazon EKS Auto mode. Carpenter for scaling you Let's use your lot load has arrived. Automatically your nads Are there EC2 instances that can scale? If the load is too low, it scales down. The cluster is so easy. Super Simply click on Create a Cluster. Quick Configuration with EKS AutoMod Tax Your cluster name will automatically appear. For security, you can create recommended If you click on the role then Joj will be able to see the roles.

You can select any field for the node. Automatically You will need your own VPC. How to create an Amazon EKS cluster Auto mode enabled. Whenever this cluster is formed, it takes less than 10 minutes to make It is the fastest. Meaning if you are production ready Talking about clusters, it is less than 10 minutes. When you go to its computer, first how I had to manage the nodes myself,. Add this node Add EC2 Add forget It's not like that in the background.

Carpenter is being used which is EC2 Will automatically scale instances. 10 Wait a minute, within 10 minutes this cluster It will be ready in a great way. You must have understood what EKS is. What is the auto mode of the EKS? Let's say that our Infrastructure is what we have today. The infrastructure to be built in the project What is it and which one do we need? First, you Creating an EKS cluster on AWS. You will need the AWS CLI. I'll show you what the command line interface is.

Secret Access Key is coming, default region The name is coming, the default output is coming. This is coming because you have AWS You must have the CLI installed. This Installation of whatever the project requires I put everything in Read Me Meaning if you want to make this project locally What do you need to do to drive? CLI Terraform kubectl Docker Hell How will you take it? The thing has been told here that downloading AWSLI If so, run this command. Hum log connect on AWS CLI How will you do it, how will you do it, first of all What you have to do is nothing much, hard work.

[Kubernetes on AWS EKS: Production GitOps Project] Helm with Argo CD and production GitOps project / Helm and charts

Go to the AWS console and search for IAM You have to take it and you will click here. Whatever you need access to For example, you need access to EKS. Full cluster admin access if needed Select that. Go to the terminal and run a command. You can do this only if you have AWS CLI version two is installed. You have to run a simple command Take a look. AWS S3 ls that's it. This will reveal that AWS has as many S3 buckets as it can you see what?

With the AWS CLI We're done. That's what you can do with Terraform in one short video. You will understand that Terra is an infrastructure Edge Code is a tool that will help you with this. To automate the entire infrastructure of via code which would have helped a lot If there is an error and it does not happen then Terraform will try to. If you want pass installed then come on tf Let's install Do I have Terraform? Version I have terfm version 1.57 Terraform version is out of date with the latest Version is one to have low let's go br Update Terraform Done low blue upgrade Terraform.

We will do as you say, what should we do? The thing with your cluster of Kubernetes To do it. kubectl version Client version 1.33.2 so kubectl also We have it installed. If you have If not, it's written in Read Me. Helm Let's also take a look at the Helm version. We'll go here and Look here it is written Helm, so this Just copy it, it's cool I copied this line and went here. Boom so I have the Helm now The Helm file is downloaded.

By doing a little chmod 700 If you make it executable then ls huffn I have to look at the gate Helm, I have permission Yes, read, write, execute permission I have given it because of Rs 700. Rest of the users There are no permissions. Downloading Helm what's the matter, give me the password So you all know our password, test it. @123 Just Kidding Helm Done Install So Hell Huff Version, no -, just write the version. Why do you do this, it's a disaster If it is installed then as many pre-request visits It was there, we installed it.

[Kubernetes on AWS EKS: Production GitOps Project] Terraform and infrastructure as code / AWS (EC2, VPC, IAM, S3, ECR, EKS)

Initialize Terraform and Review infrastructure code structure. Meaning Our project structure is We need to review it. From Terraform What do you want to make? We have to create it using Terraform. Here's your VPC Virtual Private Cloud. Is our EKS Elastic Cubinity Service. In this we People will enable auto mode so that this Let the pat become our own cluster. This Terraform will also install the Argo CD Inside our EKS cluster. Helm too will install our EKS cluster In. Ingress of Nginx controller so that outside traffic inside and outside, all this is one If it can be connected, this ingress controller will also.

The rest of the retail stores will be as good as the app. Manifest files are like files that are used to create pids. Deployment has to be created, service has to be created Yes, Ingress has to be made. All these things via Helm charts will be taught to you. First, I'll go to my repository. Because everything is in our repository. I'll move this to the main branch. CD documents in a good way Go to the folder. Retail Enter store in retail store He is gone, now what should I do?

I found the Git repository here. If you want to clone it then this is the command Just copy it and go here and do it. Let's git Clone If I slammed it, my repository is retail. Store Sample App will be cloned Clone will be fast? One The folder is Argo CD. You will find all these files on Terraform. AIDSf Argo CD Locals Main Output Read Me Security Variables Version You'll get all these things. Infrastructure VPC And your EKS cluster creation. Both We use prebuilt modules for this.

You will create a block called module. For more details, see Terraform. Your The EKS cluster and your VPC are small. Search for extensions and Terraform And this is Hasik and Terraform, right, 574. Keep showing love like this, this victory is ours This is a victory for the entire DevOps community. Click here and if you want to use Terraform Now you will open the files, ah yes, colorful, ah yes Watch now talk about module VPC If you need to create a VPC.

If you want to create a VPC here, What is a VPC?

[Kubernetes on AWS EKS: Production GitOps Project] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Terraform and infrastructure as code

A private There is a cloud inside which all your The resources will be there. If you have a If you create VPC then this has to be created Name CIDR Range Tenancy Availability Zone CIDR subnets have to provide all of this. Source Terraform AWS Module VPC This One Line All the files your VPC contains Security Availability Zone gives you all that No need to pay attention. Low Registry AWS Terra Form VPC and that's all you have to do. Click on the Terraform AWS VPC module Two and that's it, I got this code from here.

I have taken it, you just use this, I have copied it like this I did this and went here and threw him down, that's all. I have done the work, that is so easy In Name you have entered the cluster name in the variable Created a VPC of Variable Cluster Where did the name come from? Here I The variable tf is created. Where Cluster Name Default Value Retail Store Environment Dev. The Kubernetes version is yours. VPCT's VPC's Take whatever CIDR block you want.

Argocide What will be the namespace of the same? You can configure the variables from here and You can use it in main.f. Availability Zones Public Subnets Net Gateway that provides the outside network address Translations for your public and private I have set it to true for subnets. Internet Gateway External Internet Access I've made it true. It has been made here and by doing this your The VPC module has been created. I also configured the EKS cluster. Source Terraform AWS Modules EKS AWS Get the latest version.

Cluster name Cluster version Cubitz version. Joe End of Cluster Points have to be given public access What? EKS Auto Mode I mean, I told you about compute nodes. You will do EKS Auto Mode of your cluster, and in that Enter the VPC ID and create a key so that Encryption is also required. If the cluster login If you want to keep it enabled then make it true. If I'll go to Terraform here cd Terraform I'll go and write it here, Terraform In it.

Helm aago cd everything through this I installed it and my Terraform init What has happened to the Terraform command?

[Kubernetes on AWS EKS: Production GitOps Project] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Terraform and infrastructure as code (part 2)

Initialized in the folder so here My Terraform is done initializing. What do I need now for my architecture? I look very carefully at my What do I need in architecture? My architecture requires an EKS cluster VPC Argo CD Hale and Injection Ingress Controller Until my VPC is created, my EKS cluster will not be created. Until my Until the EKS cluster is created, my Argo CD will not be made. Until Argo CD How will the internal application not be created? Wise if you need to install anything Like this is my Terraform folder If you turn this off, it's Argo CD.tf In which I have configured Argo CD Everything about the Argo CD is written down.

If I What if you applied Terraform? This is Argo CD Main VPC EKS will become everything and connect to each other. I'm going to start with I will create a VPC. Target is equal to module which is mine The module was VPC, make it enter. My first VPC It will be created. Will find among all the files that our Where is the VPC? It's in Maine and in What does a VPC need? Only that much will do the job. This will take some time because VPC It has to be completed.

If you do this, VPC creation will start. Module Creating a VPC is great. As long as VPC is created, let me tell you that How do I automate this whole thing in Terraform? How to do this in Terraform using Argo CD I'll tell you what's going on. If you If you use an AWS provider You can also read the related components of AWS. Resources Mast Aisa Chutki Me Modules You can install it through. I also got a provider for Helm. I also took a provider of Kubernetes.

Time random string I used for everything I have taken the provider. I am telling you that, whoever helps me The provider is on my EKS cluster. Take all the commands you want to run. My help should run on EKS cluster. Kubernetes is my end of the cluster Points whatever token is configure everything So I've put everything in versions.f. Variables TF I explained it to you. You If you talk about production then your load Which route should be in the balancer?

[Kubernetes on AWS EKS: Production GitOps Project] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Terraform and infrastructure as code (part 3)

What should be HTTPS traffic? Security Groups I need some security from myself. If the outputs I need to show some output, like cluster What is the name of the cluster? What is the suffix of the cluster? What does it mean at the end of the cluster? What is the VPC ID? Your You must have become a VPC. You have become a VPC. In the output you can see that VPC Sider Block and which VPC ID these It's created. Copy this VPC ID and Let's see what actually happens in your account.

Whether it is made or not, because I don't know. Your VPC is in US West Two Search the ID, I didn't find it, no problem. Here's the Retail Store VPC And this becomes your VPC ID paste 090? Once the VPC is created, you can now What can you do? You can even create a cluster. I run Terraform apply Target is equal to module dot now which My module was in Main, what was its name? VPC's name was Retail App. EKS copy that and go and throw it here module dot rip EKS my enter That's all I have to do and my EKS The cluster will obviously form after 10 minutes which.

Created a manual cluster named cluster We did. Cluster I need access to. EKS Make me an admin. EKS I gave the admin cluster access. Here it is AWS where did it go? AWS EKS Update kubeconfig just copy this and Go to any of your favorite command lines Drop it. Added CubeContext Done.kubectl Get Ns Can't list resource nodes. These APIs group cluster scopes. Whenever you have such access issue Come by the way ours which AWS EKS which The cluster was being created using Terraform.

How many resources does it have to add 40 to it? I created an EKS cluster t WS EKS Cluster In US East Sorry West to end my user Is not able to list Nads just you Amazon I told Q that, this is not my user. He himself describes the cluster. Your cluster is active and running. That will list the node groups and will add a policy. I only have access to the cluster. I just paid AWS Go to Access and give access to the cluster Nodes were not given.

[Kubernetes on AWS EKS: Production GitOps Project] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Terraform and infrastructure as code (part 4)

Nodes are auto Managed by EKS Auto Mode. Yours The cluster has access to everything. If I do kubectl get nodes boom. I have my node Ready. I have my namespaces ready. We are a cluster of retail stores. We are making it through and it will take time. Have a great day learning DevOps, Git enjoy it cool I asked you where you joined from. If you have seen it, please tell me that too, like Maximum The votes that I have received are for Linux.

Kubernetes is then Docker then Jenkins then Gittip good Everyone got votes. We've initialized Terraform has been given. The structure of the infrastructure's code. Deploy Core Infrastructure VPC with Proper Networking and EKS cluster. EKS It takes around 10 minutes for the cluster to be created. EKS cluster takes around 10 minutes to be created. Secure Access to the Newly Created cluster. The command was entered AWS EKS update kubeconfig will create your new cluster. Just put its name here then kubeconfig Your update will be done, this is also very easy.

Shubham153 this is my GitHub where it is You will find all the projects now. Go here and search for Terraform. Terraform for DevOps Search it cool. Here if you find a If I had to create a simple EC2 instance Look at how much effort it takes. Is it a keep pair security group, egress, Ingress instances everything. Many Instead of writing things down, you can use modules. That's what I'm doing with this project of mine. Less code but also reusable code There may be production deployments.

If someone is telling you like this, 500 lines so we have a Terraform Make it. Our Terraform will be created. The EKS cluster that will be created using Terraform How will things go inside? What did Terraform just do? An EKS cluster, we need to build an EKS cluster. This The applications inside, this is the Argo There's the CD, this Helm, this engineer Yes, there is ingress control, how will all this come? Here you will find all the micro services. Like UI, which is UI micro The service is written in Java and Spring boot html css all It is combined.

To run it, The Dockerfile is written here.

[Kubernetes on AWS EKS: Production GitOps Project] Helm with Argo CD and production GitOps project / Helm and charts

Maven It all builds through. If you need to install this on your local computer, please do not forget to install it. If you want to run it, then it can also be run through Docker Compose. What's in it The important part is that, this is mine Retail Store Applications Pod Ye How will everything be made? This will be made of charts. Helm charts will help you understand the whole process. Not writing the manifesto of teracubanities You just need to dot the values in the charts.

Y Just insert your values in it The definition of what is paid is what is paid That's already a Yammel template. For example, if you want to see deployment then deployment The template always remains the same. Similarly, the micro- There is a service and there is a chart inside it too. No, where will I find my Kubernetes's manifest files. Because It's all happening through Helm. I'm back I'll tell you what to Helm if you want. Helm says that the should be kept separately, Separate deployment, separate service, separate Do not make ingress from.

I have given those values here in different micro Enter the values in the chart in Services I have given it. If you want to change the image then Yes, change this line. The values of Hell, Hell charts, these will reach my cluster How? It will arrive via Argo CD. CD stands for Continuous Delivery. Argo CD I have kept complete documentation of this also. Argo CD's You a Project You don't have to make it. I have already written retail store Projects that source repos have this repository He will pick it up.

Fork the repository and this is your The repository will be done. In What will the applications be? UI The application of this is in Argo CD Helm Charts How will he pick it up? SRC in PATH The values dot eml written in the UI chart It will pick up whatever has happened and deploy it. I understand that Argo CD is a repository He will go to this branch. He will go to this branch. This is the folder in the branch SRC UI The chart shows the values written in it.

He will use the values and deploy them.

[Kubernetes on AWS EKS: Production GitOps Project] Git and version control / Namespaces

I give it to you on this particular repository Jayega Retail Store App SRC I will go. Go to UI Go to Charts. Charts will go into the values that are in it and which The image's repository value is written He will deploy it. Ingress So, the template remains the same for everyone. It should be fun no matter what happens because this If you want to add this repository to the project You just clone and just images You change it here and there.

Deploy the application from this repository You can. This is a plug and play Repository. If you go back to the code Go to Terraform Go to Argo CD then Here it is written that all the things How will Argosy do Cube Citadel? Path in the Apply Argo CD namespace Modules What Argo CD / Projects means Argo CD / project in this path The project was created in this path: Argo CD / applications became applications They. Your application It'll be up. Our EKS cluster is now created.

Go to AWS console Let's refresh and see. Our old one got deleted which We created it manually and the ones that were created automatically It was created using Terraform and is up Retail Store 71H9 It's done. Because who added all these things to the CLI user? What I had to do before A cluster was formed in which I found many Things were done manually by AI, not me. If I run this simple command AWS EKS Update kubeconfig Region US West to name retail store to Will my cluster connect?

O By the way, nodes will not be created for the first time. Because there's an empty cluster. Cube CTL Create NS Lets Demo Nginx Try creating a demo engineics namespace Let's take it. kubectl Run Injectix Look at the autocomplete, it's all coming up. This image is Nginx, namespace me. kubectl GATE Notes If I do it then I will automatically The note will be ready. It will look like this because the kubectl get pods in the namespace Demo Nginx A Little Time It will take time.

Watch kubectl get pods Low Watch is not the same Tell me, install watch, see what a watch is. There is a small command which you have to No need to run a command repeatedly It will have to be done as a complete process.

[Kubernetes on AWS EKS: Production GitOps Project] Helm with Argo CD and production GitOps project / Argo CD projects, App of Apps and multi-cluster

Come Watch Cube Seal Gate Ports If you start running then you have come to running. kubectl get You have also created a node in NUDs. It has so much strength Security is there. Every 14 days, every 21 days These notes keep refreshing For security purposes. Whatever is in it OS AMI is used for Amazon Machine Image It is not immutable. Top Notch in If you like it If you want to read more then Amazon EKS Auto Mode Search like this and this mast will give you its complete Full documentation will be available.

My Kubernetes cluster is created. Configure Cube CDl Access to the newly created cluster Gone. Deploy the Platform Services.NGX Ingress Controller Certificate Manager and Argo CD. This Platform services are what you call yourself Deploy it. Terraform Apply Target not this time I will. Because now if I If I apply it completely then how much Those which are already created will be refreshed. Terraform that right now face me by face. EKS I've done the clustering. Argo CD and Ingress Controller and my I will take up whatever Helm is.

I know it's been an hour since our project was seen I am not coming because it is architecture if so you get the framework If you go, you can run any number of projects. Because your EKS cluster It's empty now. Argo CD will go into it. Node The values of will change. My What extra do I get in the cluster? I need it if my cluster I need a certificate that my application SSL Use a certificate. Use HTTPS So I just set it to true.

If you also want to add monitoring Yes, you just have to uncomment it. The entire stack of Prometheus and Grafana is yours Cluster Pay will be enabled. I told you that you are told that This is Jenkins. My Let's focus on our Qubes cluster. Let's release Helm Argo CD now. Argo CD is being created. Meaning Argo CD projects are being created. This file is Projects Argo CD. Retail store Create a project named Sample Application The code is available in this repository.

In this namespace is putting in. The applications that are These are all the applications of ROCD. The chart of the retail store with the cart It's being made.

[Kubernetes on AWS EKS: Production GitOps Project] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Namespaces

Deployment is taking some time That. You will find a snapping application most Public Secret Shop. You can say I worked on an e-commerce Application for a client which includes Front end built on springboot java Using Maven back end on Node.js golog Database with MySQL PostgreSQL DynamoDB Caching Using Reddish. Infrastructure built using what teraphone Application Deployed on EKS Continuous Delivery using Argo CD income fantastic Developing Argo CD Project And Application Baat Argo CD Project Created. Argo The admin password of the CD is sensitive.

Enter the namespace of the RGCD If you need to buy Argo CD locally It will work only on cluster but if you want local I want to access using port forward. Its command has also been given. Give me the end point of the cluster. Cluster name suffix OIDC if you Connect to this cluster with Open ID Connect If you want to get a third party application done then that too. I have given it, I have given it to the security group I gave everything, I did not leave anything behind, Ingress Nginx Load Balancer is a of load balancers.

If you got the URL, you could have copied it. How useful commands are If you want to see CD apps, check this out. If you want to see the cluster then see this. If you want to see the notes, check them out. Get Potts If you want to do it then let's do this Get pods. Cube CTL Gate pods All the namespaces All the namespaces I have some too, bring them all. Our certificate manager has been created So that we can do SSL, HTTPS We've got to demo this, engineers.

Engineering He has also become a controller. Two because here I'm going to kubectl which is the command I pasted get service ingress and give me the url 2. This URL Pay my load balancer to my application Deploying. You Have Deployed the Most Public Secret Shop on an EKS cluster. Meaning, if tomorrow I You can also clone the repository here to any I'll play it on my laptop. Prerequisite EKS and AWS CLI If configured, if Terraformed, it Your application will be deployed.

You should change its public images. Just put in the image you want.

[Kubernetes on AWS EKS: Production GitOps Project] Argo CD projects, App of Apps and multi-cluster / Terraform and infrastructure as code

That application Will be deployed. My God Mission Authorized Order ID Placed Order Place Your Order Arrival Your application is working Perfectly Using Terraform Argo CD help Argo CD has done this I don't know how I deployed myself. kubectl Get pods in the namespace Argo CD then Argo CD running Everything is running now I have an Argo CD UI then Terraform output I will write this down and whatever output I get. Argo CD See the port forward command for Here. This kubectl port forward Our port is 8080 on localhost's 80-80 I want my Kubernetes cluster connected.

I don't know yet, but absolutely I will tell you, I will show you the password Yes. More Terraform output in the Terraform folder Write it down. I haven't entered a command. Secret Can you see the secret? I must have put it in the Read Me somewhere. Find a secret somewhere Yes here is the port forward to arg cd UI kubectl Argo CD Get Secret Copied the initial admin password I threw this one at my terminal. I copied this And go to your UI and enter the password.

I went and hit sign in Hacker Man, so this is your Argo CD. All Sync All Healthy how this application How did all these things come to be? What is chat me Aago CD in one Write a shot at Argo CD's UI CLI Using YAML Files Completely Explained I will go into it in depth, for now I will tell you this much. Applications and projects so these The project is a retail store project. Whatever things are written, name namespace, You can also create everything manually.

Go to Settings and go here Go to Projects | New Project Go as you enter project name here My new something to describe my new Engineics put from LETS Description put something create. If you have done that, you can project through UI. If you need to add it to GitHub If you want to add a source to the repository, you can edit and add ad source here. On the same thing I did through the code Name namespace Description Source Installed Repos Gate's rip repository.

Destination: Enter the full address of your server. Everything in Kubernetes is a YAML file and a Manifest.

[Kubernetes on AWS EKS: Production GitOps Project] Helm with Argo CD and production GitOps project / Kubernetes Services and networking

Should that if I if I Here is my main app, this is its source I should change it to. Like in the UI I go to And go to UI, then Source and then Main. Let me commit this to you. Given in the main branch It should not be done directly. Because right now The current images are in it using Argo CD What image is that? Which image is it now? If you click on it You will do that, the public uses the image It uses images.

All these things are manifest, right, these are public If he uses the image, the public image is public. Our own images You will have to create it. It will cost Build, deploy, all that. Deploy Platform service ingress controller Engineering Certificate Manager ROCD done. We need every single micro Business purpose and technology of the service Understanding the stack is essential. As a DevOps Engineer only and only on ROCD Deployed. Quickly picking up code via Terraform Deployment done. That is not how a DevOps Engineer Works.

One that is real In essence, a DevOps engineer His mindset is DevOps. DevOps means What happens to Development and Operations? I got the CD folder, and They have mistaken it for a folder, which we have not. I understand that in docs and images. Cart's Read the Dockerfile to see what it uses. 2, Amazon Linux, Redat all these public You can use the images. These mavens It is using java 21. V means your If you read the source files of the cart files, you will find Java files somewhere.

If you have Java files It Simply Means the Docker Should be doing maven build mvn install MVN Clean will do all of these things. Multistage Docker builds used here Yes. Something User added for security Purposes and basically made a jar file So if you want to learn Java applications You have to build it. That you understand your cart, Cart uses Java. If you look carefully at it Look, go dot mud men dot go men dot go Meaning, like in Python files, main.py It happens that way, there's main.go here.

If you have any Docker in the golang If you want to write a file, how is it created?

[Kubernetes on AWS EKS: Production GitOps Project] Helm with Argo CD and production GitOps project / Kubernetes Services and networking (part 2)

First you should install Amazon Linux. Will pick a base image. After that you Download all the Go packages in Go. They will download your main from go build The entire application will be built. Once that It has been built, you can access it from app.main So generally, creating Docker images, Creating Docker files is easy. If you Understand who wrote the code in the background. The question arises that, you tell us Got that cart functionality It's written in Java from Let's. Which is behind my The SQL is DynamoDB, which is behind Redis, how do we configure it?

Went to the charts and Get into the values. That the image that is being used, you build it By doing this, the image is created from the Dockerfile and you can use it. If you go any lower, you will It would make sense that there was a The public image used is of this version. The service is running on this port. Deployment strategies, help, Ingress, Role Based Access Control, Kind I have brought all these things here. Many people don't even know that What are they doing,.

Business of each module and each microservice Purpose Usa Technology Stag Review Dockerfile you made with Helm chart There are templates, they have values, they are very It is more important, if Let's come to me. My card service is the card service which Helm chart is what values does he use In the background it uses DynamoDB So I also did the public image of DynamoDB. It is kept there, if you don't want to use it If you want to do it, what can you do directly?

Can you provide DynamoDB endpoints? It is important for you to understand that Helm Charts How are they being used in this? How do you understand service? EKS Auto Mode lets you gives the data plane, which is your Node, security, and your storage. Joe Third party products like Estio You can also configure it yourself. However, if you want to see service to service then you Let me tell you a simple thing. Microservice Overview Catalog Service Uses Go and MySQL and Provides an API for product catalogs.

That the product information is shown on your UI You can.

[Kubernetes on AWS EKS: Production GitOps Project] Kubernetes Services and networking / Helm with Argo CD and production GitOps project

Cart Service Java Dynamic DB Uses customer data in the cart Stores. Check Out Service Temporary Customer data is provided for Redis caching Order Services manages my Java with sequelize data to keep its order and The UI is your front end that lets you access the API. Calls are handled by the app service that connects to each other. This is a typical E-commerce microservice architecture with each Service handling special. Service to service communication is now complete. Review Helm charted the applications of Argo CDs.

You've checked the definition and sync policies. How the service is being exposed is very It's important. All the services that are there if you go into it In service, they are exposed on port Number 80 if you look at all these port numbers exposed at 80. All these HTTP Port number 80 is exposed and ingress The controller is just redirecting them all. Check Take out the service. This also exposed on port number 80 Now its ingress is affecting everything He manages it.

Go to the UI ingress,? In Ingress you have to do everything The thing you will find is that this is the UI of the retail store. GitOps, it's been an hour and a half. GitOps what happens is that I put something here I made some changes to the application's UI. What is your GitHub, SCM? Source code management is fine yours which ECR is ECR what is that elastic container Registry means whatever images of containers He will be in ECR. Argo CD What does EKS Cluster Deploy do?

The question arises regarding this ECR. How will this Docker image be delivered? I have written the ECR. Fully Managed Elastic Container Registry You can create the repository here. Here UI Image from UI Lets Push Me If you want to get it done then demo private UI image tag mute mutability mutable means change Maybe. No, it is up to you and that is all you create. What I did was I created a private repository Made this private now I'm just on this I will click, it's done, click, now see what It will say View Push Commands, click on View.

Push Commands and This Is How You Push An Image to Your Repository AWS ECR Get Login Password Username AWS Ye If so, copy this to anyone of your Go to the terminal, they go to their Kiro.

[Kubernetes on AWS EKS: Production GitOps Project] CI/CD concepts / AWS (EC2, VPC, IAM, S3, ECR, EKS)

I have to build the UI using let's. I'll just do it Docker. AWS ECR Login Password Copied it and posted it here. The Dockerfile should be here. It will take that much time to build the image. On that much time to build And he will build the whole thing and so on. They are masters of DevOps and now they I want a project. It will come to you in your ECR and it will be The name of the image is just what you will find in the help file.

I'll tag and I'll push here. Go and see that Helm. The Argo CD latest image So manual. I don't want to I wait for hours to see when my will come It will build. When will he Not deployed. Vi is no age CI/CD Continuous Integration You've seen the Continuous Deployment CD. I took that Argo CD will say that whenever your Source deposits changed, I've seen those things. CI How will you do it? CI Continuous Integration means that anything in the code Detect the changes that have occurred.

CI Continuous on CD Inti Integration. What would your Docker do first? Check out the code Then if your code is latest then its Detect Changes Then What Will Happen docker build then what will happen docker tag then What if Docker push to ECR then what? Which will be the tag of my latest image Update tag in Helm charts. This is what All the steps are there, if it happens automatically If I go, I will have the latest image that will be created.

Its value will be shown in the Helm chart. From here to here In Continuous Deployment I used Argo You have already set up the CD. This becomes my complete CI/CD pipeline. The catch is if I did not do this CI and I Directly updated in Helm Charts Go here manually. Can my Argo CD deploy that thing? Meaning, this entire pipeline of CI/CD Isn't it from GitHub? This will be built on Docker, this will happen again This will be pushed into ECR Then this will be deployed on EKS.

DevOps lives Code Build Test Deploy CI/CD now there is problem in this Do what it is? Changed it and this Argo CD Once deployed, your application must It'll do.

[Kubernetes on AWS EKS: Production GitOps Project] GitOps principles / Helm with Argo CD and production GitOps project

GitHub Does it match the code on the Is it? A small error in DevOps There are chances that if you Engineer DevOps Engineer Manually Updated the Helm charts and added Argo If the CD picks it up then your code. The code will be checked out on Git. That and the production in deployment. That's why the concept of GitOps came about. What is GitOps and GitOps and DevOps what does GitOps say GitOps Bolta Your Git and Your Version The control system is the single source of Truth.

Meaning whatever is in your Git pay code The same thing will be written in your deployment. If your git commit is a If you want a replica, then in your EKS clutch There must be a replica. If it has 10 If it is written then it should be 10 here. How do you do something like change the code in GitHub? That is why it is known as GitOps. That's why fuzz three of this Project Is The Mostant Fuzz Till Now. That here you have to understand GitOps a lot.

Review GitOps Principles and Benefit Single Source of Truth. Meaning I was quietly building this No, this is such an unpacking of Docker, this quietly I built it and I quickly got here. Docker tagged And did the docker push. Because I am doing this docker push my Helm chart will pick it up. Can I upload my source code to my GitHub? Whatever changes you make It will be on GitHub. New Folder Whose name will be GitHub Work Floss And you'll need to create workflows.

This only that changes in the code were detected in the code If changes are detected, docker build will be done. Gets triggered, Docker pushes, gets updated Helm Charts Pay and Argo CD that thing Deploy it automatically That Can Be Achieved Using GH Actions. Let's start the GitHub Actions right now Talking about. By the way you The image is in the private repository. If you look at this image, Where do I put the value after picking it? Helm Chart Put it on.

Let's understand GitOps right now. I'll take you to Git. Git Pay and here I'll take you to the actions In. You can find all these workflows on GitHub.

[Kubernetes on AWS EKS: Production GitOps Project] Helm with Argo CD and production GitOps project / AWS (EC2, VPC, IAM, S3, ECR, EKS)

Defaults to static HTML Deploy the pages. Deploy static HTML pages, you just configure this Click on the button. Docker Want to publish a container? The GitHub Actions is called Docker. This work Flow uses actions that are not certified by GH. You can also schedule this Run it every day at 8:00 PM on the 23rd minute If I push anything to the main branch Or should I pull over the main branch? If I create a request, then on this registry The build of this image should be ready.

If it is built, it will not be locally produced. GH already has They have installed machines on which Ubuntu is installed. Docker will then build it. Docker's metadata is a docker build endpoint. Joe We entered the same commands as ECR. If I had to work on my own, If you have to write from then I will tell you. Kiro, I will tell Kiro, Kiro of Do me a favor, enter your idea I want to generate dot GitHub work Flow for This Application Which does three simple things?

What Detect Any New Changes Build and Push Docker Images to ECR And update the values in Helm charts boom enter now what will kiro do i mean ai. He will make a plan, he will make a plan As many files as you have, as many There are things in this whole system, all of them He will read it, he will read it all. That latest cloud There are as many 4.0 LLMs these days. After understanding your source code well, one can AI can also create workflows for things.

All this data is local cache. Spec will contain your There is a requirement, let's say you require Do you do Git Actions workflow for retail? Store Application CI/CD Pipeline which includes I have a requirement CI/CD The pipeline has to be automated. This happens whenever the main This will happen when the code is pushed to the branch. This will happen if there is any micro service change. If multiple microservices change then this What is the second requirement? DevOps Engineer, I Want Docker Image to B.

Automatically build and push to ECR. This is a project that I have created. If you like this If the requirement seems right then you can apply for it.

[Kubernetes on AWS EKS: Production GitOps Project] CI/CD concepts / Kubernetes Services and networking

DevOps and Gates Practices Or if you have a requirement then write like this Give it here and he will understand that too and for you He will make a design. I have written down the things here and told you Joj I wrote down the things here that Detect Changes, Docker Bill, Tag, Push all of these I was just showing you the design. If he makes it himself, it will be a design. Like this Main CI/CD Work Flow CI/CD.L Trigger on push build then build service like so It will also create a complete email like this.

Create that task that the first Tasks create a folder dot GitHub work Flows. Task 3: Send an email if my build fails so on. Can you see Created Task If you click this task will you do it First create main CI/CD as a DevOps The engineer thought that first git work A folder with Flos will be created, then path based. The card catalogue will be filled and so on again Reusable work flow will be created build service again AWS Authentication will be ECR access, how will the authentication happen, you will have to tell your.

Secret access keys will have to be given to GitHub They'll give it. Docker build multi Platform, all these things, service Specific configuration, configure go Service, Configure Note J. A DevOps Engineer in Day to Day Work It does. Create Main CICCD Workflow File. Create dot GitHub work flow. Here Your GitHub wfls folder is created. Cycle If you want to make it then GitHub in CI/CD he I made it myself. Wrote it and that too was told by the from DevOps It was.

Create me The CI/CD workflow file has been created. If you go to CI/CD, whatever your Docker push and build is action, setup build exe Yes, need AWS account ID It must be, you need a reason. All these things, all these things here, update the chat Path. By the way Using this I have already created A Gotab Actions workflow that shows what you can do You can get the status, all this is made up. Git stay do it what does stay do All these things are temporarily hidden and You will use it later, now.

[Kubernetes on AWS EKS: Production GitOps Project] ConfigMaps and Secrets / Git and version control

I'm going to a new branch called git checkout get os so I created a new The branch has been created. This new branch I already I already I have written deploy. There is a simple pipeline called deploy which what does it do? If you push my In branch GitOps Path SRC means if in githubs branch this The src file in the GitHub branch If I look at my developer, the developer is right here. Will push the developer to change the UI So that will change in the UI, change the catalog.

If I do it, I will make changes in the catalogue. A DevOps engineer will do this. If there is any change in the repository, run it. Here you have to apply your mind that I The entire repository does not need to be changed. Read If you change me, my pipeline will not work. If there is any change in SRC then all these I'm going to run my pipeline. What do I need in the environment? Generate credentials, you did it. Go To Your GitHub Repo Settings Secrets End Variables Actions New Repository Secret.

The user who has access to your ES Do not use the ID of the same user who created the cluster. I'll name it GitOps admin gops admin this is a new user Who should have what access now? I don't have to give it, I have to give it to ECR Where did ECR go? ECR Me Re, give me admin for now, but you What do you basically need? Should i give to my GitHub Actions user In I AM So That It Can Build Push Images to ECR Use the best security practices What's going to happen now?

I will see what permissions I will get. What's the point of create access, man, that's the Power of AI I don't say this that AI will take your job and you will be in your job Using AI to improve your productivity. Here in the middle are agents who use tools are using use AWS. Access Keys of GitHub user. Settings After going into Secrets and Variables I'll go to Actions. My secrets are stored in that repository. I configure the repository secret. How is it written here, go to your Get Repos Settings Secrets Variables Actions New Repository Access Key ID Secret got its name and Secret Access What happened to the ID of that user?

[Kubernetes on AWS EKS: Production GitOps Project] ConfigMaps and Secrets / AWS (EC2, VPC, IAM, S3, ECR, EKS)

The access key is Create Add a Secret Access The ID has arrived, what else do you want? Need secret access one more Create a repository secret and enter its name. What is its value for secret access? AWS Region Check the AWS region in Read Me or No? What AWS region is the region? This is how you have access to GitHub. I have given you the details of my AWS account. Not the entire AWS account Given. On means when will this job run?

Changes to this source It will run on AWS in an environment You have the secret in your region. There are three jobs in Jobs. Meaning Detect Changes, Deploy and Give Summary. Meaning it runs on GH's runner. You don't need to run it on your local GitHub will automatically run on its runner. Output Change Service Steps Changes Output Change service. Changes and changes if detected So what should happen So my detect changes will run so change Service will help you understand who C-Which service has changed?

Parallel jobs means this Multiple jobs can run at the same time. Here Pay credentials are configured. You have Access ID, secret access key, all this So I've kept it under observation. Log in to ECR Do it. You have access to ECR. Everything was pushed manually, all the same The commands are written here, the name of the repo How did you find out the account ID region? ID region because here if you ECR Go to So whenever there is a push on ECR, if you go to your private registry whenever If it is pushed, what is the name of that image?

This is your account ID then DKR ECR US West Region Amazon AWS your One way is to enter the name of the repository and then the tag. ID DKR ECR Region Amazon AWS then Retail Store Your Image Name Then Service Name of UI Cart And So On Then If the repo is not there then create it. Create will build and push docker build with the same command docker build docker push The same command will be pushed. Once the image is If pushed, the values of the Helm chart So updating Helm values For this, first you have to see that help Where is the chart placed?

[Kubernetes on AWS EKS: Production GitOps Project] Git and version control / Helm with Argo CD and production GitOps project

Just doing this and that successfully Updated image and that's it. Once you've updated the images. Commit those images to GH too. Because when you commit on GitHub What happens? If you commit to GitHub If you do, the work will start. This The jobs that were not detect changes were deployed. Summary This is your CI This was your CI Continuous Integration Once you've committed Done the work of git and then Argo CD It will start there, what did I do? It has been said in the applications that if Any commits to the GitOps branch of this repository If changes occur, these values are deployed.

Tax two values Deploy CI Jobs Pay Values Here changed and Argo CD I deployed it. I mean, whatever I Gateway will go to deploy that Argo CD. He'll give it to my cluster. Git Status Now ER policy is ECR policy. Test Commit This is a small version named Test Commit. That my All images Build it. You So I said that you will change it on read me and Not good if the pipeline gets triggered He's just showing you a demo. Orders are read on the UI to me Test commit done so now I do if git Status So I have five files changed.

My status became five ready to commit. If I commit this.Get Commit hfan.m added minor changes to src Microservices I'll put something like this for testing. Committed Should my pipeline run? Because what the pipeline says is when to come Pipeline Speaks On Push To The Branch Gate Offs so if I do git push Origin GitOps Also If You Do This. You can fork this branch of this repository. Try pushing in yours and you will find out my I am pushing and saying that Not working git Push Origin Gate Offs Enter.

The push is done, as soon as the push is done, now I Where will I go to my I'll go to the repository and go to GitHub's head Recent Push? Pipeline is running added minor changes to SRC Microservice Testing.Click Do that. What's in the Deploy Cart Getting an error? Build and Push Cart Pay Came. User is not authorized to perform ECR Create Repository I gave him access to push but If you are not given access to create repository Now what I will do is create that user.

[Kubernetes on AWS EKS: Production GitOps Project] AWS (EC2, VPC, IAM, S3, ECR, EKS) / StatefulSets, DaemonSets and Jobs/CronJobs

I'll give you access to the repository. Amazon Q So if you write Amazon Q ECR Create Repository X Access Was Not given to GitHub. We give you ad permissions here. No problem, let someone do it Get to the User Get Actions Using the Policy Dot JS ECR Create Repository Access Was Not Granted Given to the GitHub Actions user, what is that? The ECR that will be stored in this repository If you have a policy, someone will assess it. What he is saying is that the ECR policy is missing is That's so cool.

List Create Read Update End All ECR Related permissions on AWS Ask this Does this user have all the permissions? He will tell which permissions and Needed? The job is not to take decisions on your own. The pipeline itself is messed up. If you tell the AI, "!" Build this end to end makes garbage. You might not know that ECR How to add a Create Repository policy Yes. The permission issue you encountered should now be Resolved Your CCD Pipeline Should Work Walk Smoothly Click on Renew Jobs so that If you missed something, you will get it back.

Reinfeld doesn't have to pretend to be committed Click on Jobs and all these things fail. It's done, click on rerun jobs and let's go. Just back then, he created the Jobs Matrix. Check Cart Catalog Now Out Orders Deploy UI All this is happening Yes. If you click on this now Look, push it, push it, be happy. Build and push, everything started happening. Come This is the best part of DevOps. You will say that, GitHub Actions in Make one short video where you are yourself Write things down line by line and I will do it.

Errors appeared in the project. Such strong people Let me take my projects to the next level Let's give you a deployment summary. It happened on this branch. Built End push Docker images done. Helm charts The update has been made. The commit has also been made. Argo The CD will sync it. He said that Argo The CD will sync automatically. It What is the name of the branch? This branch The name is GitOps. Not from the Gate Offs branch. Because when we Demonstrate the project in the beginning When we were doing it, we used public images.

[Kubernetes on AWS EKS: Production GitOps Project] GitOps principles / AWS (EC2, VPC, IAM, S3, ECR, EKS)

The private images are on GitPop I have kept all this in the branch. Explain in the branching strategy file That's my branching How is the architecture designed so that developers can change the code? If it's in the main branch, There will be manual deployment. Public ECR Images will be used and the umbrella chart will be complete. If the developer Changed the codes in the production branch If in GitOps branch then GitHub Actions trigger The bull will build and push to hell Will upgrade the charts and Argo CD.

Private ECR images will be synced and updated. It means that my Argo CD is GitOps This is the configuration of Argo CD. I want to apply in my cluster. Because right now what I What is going on in the cluster? The Argo CD running in my cluster right now It's running on the main branch. kubectl Apply -f What are the applications of Argo CD? All the applications, whatever they are in these namespace Argo CD Enter, now what will happen Now that I have updated my Argo CD configured which My updated target revision is which is.

You can Argo CD if you look at the applications Look, everything is being processed. If you look at the target, you'll see Branch and all are healthy. You posted it on GitHub. GitHub Actions fixed that thing Build Test Currently in testing stage I haven't added it. I built it and Pushed to ECR. Helm Charts Updated and synced with Argo CD And Argo CD is syncing right now. He will do it slowly because the entire public will Move from images to private images Neither will it complete.

Just run this command echo Me. Here is my application running. Because I have added that UI to the GitOps branch. Go to the GitOps branch containing the images Go to the GitOps branch now see the UI I'm going to change the source of the UI I'm going to assume I'm a developer. The developer changed the UI for that I went to his resources, templates. Went and went to the home page and said What did you say? Secret Shop by Train Everything a Secret Agent Needs Minus the paperwork.

It will process it and only do it if it happens.

[Kubernetes on AWS EKS: Production GitOps Project] Monitoring with Prometheus and Grafana / Terraform and infrastructure as code

Otherwise don't do it, let me commit. I'm in a Git branch and want to commit directly to git branch as I make fewer commits. What will my Git actions say? The push to the branch has happened. If I'll go to my sample app githubs branch So the push has happened. Why weren't the others deployed? Because I have written that when If the service changes, change it accordingly. Services architecture with gates drop a. Build End Push is pushing the UI now. Helm Values It will update.

Build and push UI is happening. By the time the Build and Push UI is running we Let's drink some water. You must have Argo CD skills. EKS Auto You should also know how to mode. Tick So where were we, you saw how Access to Argo CD as an admin You have entered the port with the credentials You also saw the forwarding of GitPos The principal should check your application. Observe Sync processes of RBOCD and Just did the health monitor. Declarative Configuration Management Gate Ops gotb work flow seen.

Load You got the balancer URL. Monitor application health and Performance metrics have been done. Practice Trouble Shooting Common Deployment Let's practice the issues. There are some common deployment issues that you A changed image, changed secrets. Learn to identify Roll Sync Issues in Argo CD. These Sync Issues Occurs when your configuration is incorrect I made a video on how to roll back. Tell me what infrastructure you have. Code Terra Perform Kubernetes Cluster You managed EKS on auto mode. Container Orchestration done on GitOps Methodology Viewed AWS Cloud How do they integrate services?

Saw Production Trouble Shooting End Monitoring View Declarative Infrastructure Management (See DevOps) You got the idea of the practice version How is monitoring and control done? You may also have seen Observability Practices I have not seen you, what can you do to me I don't have much to do here Go and have your Where in Terraform Where did we go in Maine? Find Prometheus Do you see it somewhere Here's the Terraform of Prometheus, so advanced. All you have to do is go in and, please uncomment the part and so much Apply Terraform to uncommit the part To give.

On your cluster Monitoring and observability should also be enabled It will go.

[Kubernetes on AWS EKS: Production GitOps Project] Argo CD notifications and integrations / Terraform and infrastructure as code

Of having infrastructure s code. I am able to tagged Kubernetes Prometheus and named it Put it in space. I have of this thing package modules were available so he I turned it on. Modern Microservices You get the idea about architecture patterns. Deployment Strategies Video I'll link it to you. You understand the deployment methodology How does production go? You have seen end to end EKS management of You've seen the DevOps workflow. That we can put it in a cool recycle and, I have brought it for you.

If you like this If you want to destroy the infrastructure It's a simple command. Prometheus I want to enable monitoring and I need Lindin Let me know by tagging me. Let's say you ever deleted an EKS cluster. I entered the destroy command. Run Terraform will see what needs to be done. Whatever he creates, he will destroy it. I need to upgrade my cluster. If you want to upgrade the cluster, it is in your hands. You can back up this cluster and create a new one.

You can migrate to a cluster. Multiple Waze if you want a video on cluster upgrade So let me know in chat. Still devastating Argo CD Devastating. By issuing a certificate to your cluster Shit, man, I destroyed it. If you go to SRC, in UI Go to the charts and go to the values So you will get that your ingress which Shows your GUI, is ingress Engineering is being used in that. Which is SSL redirect false. The email in the manager is not my email.

Whatever your email is, it's your email.

Jenkins and CI/CD Pipelines

Source file: transcript_XaSdKR2fOU4.txt | 35,417 words kept of 53,823 | 90 sections

[Jenkins and CI/CD Pipelines] Monitoring with Prometheus and Grafana / Notifications and email

How to Setup Kinect on Virtual Machines Where we are at A W EC2 Jenkins is going to set up Jenkins. How the UI Dashboard looks like on Uspe Jobs How to make a freestyle project How to write decelerator pipelines Like in production, multi Agent Multi Node Jenkins Agent That Also View. How to write shared libraries And if you want to become a DevOps engineer If you want, you can have role based access. Authorization: All these things are required by the user.

There are Kubernetes on Argo CD and its DevOps Seck will monitor Prometheus. Ops which is an aspect of security Jenkins' that's also going to be covered in this where we are going to meet TV SonarQube All the things you'll do if your pipeline fails. If it happens, you will also receive an email notification. Two, because there are so many DevOps I'm going to bring it in the videos and if you If you look carefully, I have written a lot of Debs has also created tutorials.

Do it too, let's get started What is Jenkins? Understanding Jenkins First you should understand what CI/CD is. There is a project in the organization, let's create a project on GitHub. It is a tool where You can store your projects It is now used for code management. The project that remains on GitHub If the developer writes the project, then who Developer writes Now developer this project Code push Lets build this project now That happens on Docker for builds. We use Docker, so this is on Docker.

The build takes place and then an ECT Instance Pay or any AW Pay We are deploying this Kubernetes cluster. People can use this on EC2. Deployment Do we monitor Applause? Let's use Prometheus monitors now next Time whenever the developer pushed the code again It will be built again If it is deployed then this thing will not be continuous. What should be what should be Continuous means one after the other This thing should happen again and again and this thing What happens if your build from let's.

If he fails, he will not be able to get further deployed. There will be no problem but if your build passes Done, you can immediately take it to production.

[Jenkins and CI/CD Pipelines] CI/CD concepts

If you want to deliver it, whatever is the time to market time to Market, it has to reduce its market has to be reduced and who reduces it? The DevOps Engineer You Want to Be What is the work of a DevOps engineer? To complete the process in minimum time All these things like code build, deploy etc. You can automate this using a tool called Jens, what is Jenkins job? Jenkins The job of is to automate this whole thing So, if your code comes, it will be built.

That will be the test, that will be The monitor will be built back and tested. If it is deployed then the symbol of DevOps would be If you have looked carefully, a Infinity is infinity where your The code happens Build Test End One of the deploy build test and deploy. It is a very popular symbol and because such a There's a tool that lets you do that because There is a tool that gives you a CI/CD CI means Continuous Integration. That code is also being built, so Continuous Integration: If you have done it in Computer Science You must have seen integration tests and unit tests.

Continuous deployment or delivery, what is continuous delivery and this What is Continuous Integration then CI means Continuous Integration where Your test cases are run using the CD. It means Continuous Delivery Meaning of Delivery What is it and why is it not deployed here? Used delivery / deployed How can you do it when you So that thing is delivered to you right means You manually go and click on the order. When you want something Deployment should happen automatically where you are a Delivering that thing at the press of a button Ho That Time You Use Continuous Delivery.

It is being uploaded so basically all that If it is deployed then if you complete the entire process Automate it where you can do anything manually If you are not doing it then that is continuous. Deployed butt where you have a little button Have to press or trigger If you need that automation then it should be there I hope the delivery is You must have understood that because such a. Tool that allows you to use CI/CD in DevOps Helps and what is CI/CD I have explained that to you, so this much part So you can understand it very easily.

[Jenkins and CI/CD Pipelines] AWS (EC2, VPC, IAM, S3, ECR, EKS)

Everything will be practical together, come on, Jenkins. That I'm on the VM If I talk about VM then you can do it on Azure also. You can also do it on AWS. You can also do it on GC, you can do it on your virtual You can do it in Box as well as in Docker. For this we are going to use AW Pay, let's go now, I'll go to my Now I need a server here at AWS. If you want to make it, then making a server is very easy e2 I will click on it and on e2 I will click on it.

I'm going to create a server, so come down a little bit. Go here and go to Launch Instance Go to Launch Instance and click on End Go to the launch instance and name the server two like Jenkins master Jenkins Master name given now here dew imagine. Images Which images do I want to use? I want to use nuts because It looks great and the nut is very accurate and Out of that also your x64 bit x86 means Type this now if you are doing this tutorial.

Things to do: Create servers, agents There are a lot of things to be done, that's why I'm going to take t2 medium which is t2 medium He would have had a little more rum. I will tell you the whole story, now it's over. You will have to create a key pair so that you If you can connect to that brick instance I will go to Create New Key Pair. Save it by dot pay and create a pair do it So your Jenkins-CLI.jar If it happens then take 15GB to be on the safe side Can you say that work can be done even with 8GB?

There is no problem but for the future We are also going to look at Docker in the tutorial. I would carry 15GB to be on the safe side. Successfully Initiative UC try refreshing it a bit I made it quite big, maybe that's why it looked so good He is not running, yes yours Instance is running now, keep it incremented. Click and connect to the instance. Whatever is the method, through this you can directly You can connect it on your browser.

[Jenkins and CI/CD Pipelines] AWS (EC2, VPC, IAM, S3, ECR, EKS) (part 2)

Is the public IP address and this is your user If your name is Nut, then with this Nut username you can connect as you would connect A browser window will open and your EC2. Instance, which is a VM, is your virtual machine. This is one way, I will tell you another way also. Click on Connect and or SSH Select Client SSH Client Meaning Is this terminal so this terminal This is our SSH client, so that's why Go to the SSH terminal near you and look for it.

SSH, try writing that SSH sends such a message If it is coming or not then SSH client will be able to Pass must be installed to use this method. You can do EC2 Instance Connect with This method can also work, so SSH You can open the client with the complete steps given below. SSH client so I opened the terminal I have done it, open terminal, locate your The private file is yours, which is private. The file is I placed Jenkins.war in the download folder If so, you need to go to the download folder.

If you want to locate it, then it is very difficult to locate it. It is easy, you can either find it or ls filename what was the file name Commodities #pm yes it is visible, if it is correct then Download is complete, run this command if. No, it should be read only, so if you If you are on, then there is no problem if you Go and score, let me tell you this first. I am connecting so CMO 400 to this what happened it became read only and its After connecting to another instance using is Public DNS, so this is public DNS.

I will use the SSH command, SSH - Jenkins Key I means he is giving path Where is your key located? If I am inside the download folder then So I directly named it Key. What is ubuntu is my username and @ What is EC2? You have to go to DBN Ubuntu. Because we have selected Ubuntu, this is it Commands To install Jenk, this is the tool A GI based tool is what the tool is made of In the Java programming language, Java.

[Jenkins and CI/CD Pipelines] Dockerfile and image builds

Pay developers created Genki which allows you to Can automate CI/CD which builds Test was deployed to automate that This tool was created in Java, so that's why we Install Java before installing Jinx. If you have to do it then Jenk Requires Java to run so install it how will you install it First, run the sudo at update command. It happens, I will tell you its explanation I am sudo means as a super user, that is we People can't run as administrator in Windows.

Let's do it, we get permission, AT means Application Package Manager And what will be the next step after updating? Sudo att install funcconf jedi k17 GRE TO OPEN JDK MEANS JAVA The development kit is fine, install it. Java Run Time Environment will also do everything And this font Confide and go here and throw it here I went and threw it, it's done, now he says. This is going to be your complete 100% Java. Once Java is installed After

that you have to insure that Whether you have Java or not, it is for The command is java high fun version so copy this Do it and this progress bar will end.

Slammed Jawa Han Version of Open Jedi If the run time of version 17 OpenJD is correct This means you now have Java installed. If the chances are very high then long term What does support mean that once it is made He kept it aside, he is going away, he is going away. If it is good then it is very good So you should take the one with long term support. If yes then type the entire command in sudo dbo get Let me explain it to you: sudo w Get sudo means again super user permissions W Get means get something from the internet.

Funo means one path, on which path you Which path are you taking from the internet? Store To USR Sher Keyring I'm bringing one of the Jenkins so Jenkins' Whatever it is, it is an open source tool, so no put an installation of it anywhere and may harm your system. It is authentic and good, absolutely official Jenkins If there is a key to use it, then that You are installing the key, first of all Let's copy the key and go here.

[Jenkins and CI/CD Pipelines] Jenkins installation and setup

If you throw it then sad w gate this location Pay store it and get the key from this URL Come on, the dot has come, what should we do now? Take the key and print the echo meaning what dab said by is key to is key to Sign and install your packages There is a list of it,. There is a source list in that source list Add Jenkins with this command will take it and add Jenkins so here I'll run this command and it will work for me.

Add Jenkins to your sources list Now you can add it to the source list and do it. The same command as I mentioned earlier, which Han Gate is what happens to the internet Things come up so sudo at get Updates will kill my system. Updates: Check out the Jenkins package junkinwali now. Things are starting to get updated, now I will install Jenkins directly. Sudo etiquette install jks enter that's it That'll take a couple of minutes and your Jenkins' Will be setup so setup Jenkins It is not too difficult for you Initially, run a few commands and mands.

There will be no problem now Jenkins has been installed on your system. If you have to see it, how will you see the system? There is a controller that tells the status. Start can tell the status of Jenkins It can stop it, it can restart it All things so if you look at the status then Jenkins. Continuous Integration Server is active and running so let's get this going now we What we want Jenkins to do If my system is restarted then also sudo for automatic restart System CTL is enabled because doing this.

Your junking is your start The generator starts automatically as soon as the system starts. If it happens then I also want it to happen Enabled Jenkins now that is my Jenkins It will be running on port number 8080, so I'll go to my instance, here's mine. Instance and the IP address of this instance If I extract the IP address, this is it. My instance will go into details of this Go here to AW S2 and In details, this is the public IP address and It will be running on port number 8080 but.

[Jenkins and CI/CD Pipelines] Jenkins installation and setup (part 2)

As I drive, this wheel keeps turning He is roaming around and saying Kant Why does B. Reese know because Jenkins is What's up for now is his Joe Port The number is 808 and it is currently blocked by the firewall. They go to the firewall and we have to open it. Go to security if you have to. After going to security, you can go to security Go to the group, security group means fire There is a wall, and go into the fire wall.

By editing these rules, i.e., incoming traffic I've made two edits, now add a new rule. Add a new rule to port number 8080 Custom TCP on 8080 means custom Transmission Control Protocol Anywhere and IPV means anyone So if you want, someone can access. You can also give me access or give me my IP If possible, just your IP address That's where you'll have access to Jenkins, and so on. If you go back to port number 8080 You'll go and hit refresh, side.

Can't provide never mind again Hit refresh what's going on fix it Let's open a new one and see don't do http s do http Enter is run and your Jenkins is in front of you. Unlock Jenkins So first time Jenkins If it starts like this then it remains locked. If it is there then no one can take it out because it is locked. If the person cannot open it then the one who has Only those people will have access to the server.

Joe Jenkins initials The password remains here, where the lib Jenkins Secret initials kept in the admin password And this initial admin password remains Where does it live? What to do if sudo remains Cat and this path, this path from here Got it, cat it, cat means inside it Show the things and hit enter. Go here and throw the administrator Password is there and continue like If you continue, never save it now. Do it, yes, so the Jenkins tool is very All plugins exist because of those plugins Jenkins Gets More Powerful From Let's You have to

connect Git to GitHub.

If you want to connect then there is a plugin for git Docker has to connect to Docker's plugin Happens if AW is to be connected If there is a plugin for AW then it is like this.

[Jenkins and CI/CD Pipelines] Monitoring with Prometheus and Grafana

If there are plugins then we will use them now. This is the URL of your instance. If you go to this URL then your junking Once it opens, save and finish it. It's Jenkins' is ready so your Jenkins' If the setup on the VM is done then how much is it now 25 It's been a minute, we've just covered two points. I think while covering all these you Jenkins will take six to seven hours comfortably Reading will be covered completely, let's go, this Jenkins is Ready start using Jenkins so when in 2018.

I first saw Jenkins in my office I was shocked, what is this? What is this new item, what is build? What happens to history, I don't know How will I manage this view, who is this cue? What is this all about so this is Jenkins This is the UI of Jenkins where you If you can do all the work of Jenkins then this The UI runs in the browser so what I. This is the base system, now zoom it a little bit.

Let's do it so that you can enjoy it more. This is the page where your Jenkins' jobs will be Display all your jobs here So when you are looking for junk food then click on such If you do this, this dashboard opens up. Jenkins has a dashboard where you can view jobs. You can create an agent. If you can do things then you will understand one by one that Let us first understand what these things are. What is a job, because you understand it like this Suppose you want to automate anything like If you want to create a CI/CD pipeline, then this What happened to the pipeline, you got a job.

If you want to clone your code, what should it be? That one job has been automated and you have been done. If you want to run a Python script, then What if you have a job, so if you Look carefully at the Jenkins logo I mean, what's the best part about it? This is the butler, butler means this man is waiting. You just give us the order and we do the work. If we do junking then generally we do jobs Let's create, so we are the best First click on a job and then create it.

[Jenkins and CI/CD Pipelines] Git and version control

Click on A job, now we are here These items make the item with the song No item means something box type Think about the folder type where we. Your first job will be in DevOps, by the way Jenkins' first job is to do everything Now I have given the name of this job to do. You can do everything you have to do, one after the other. Steps have to be given in free style In the project pipeline you will find the decelerator I will also tell you how to build pipelines.

In multiconfine, like if you have multiple DevOps Staging Production in Environment If you want to do it then I will tell you that too and Folder means your simple folder. Multi branch means like here in the group Git has many branches. Master Branch DevOps Branch Production Branch, you will get some automation on that branch. If you want to do it, you can do it and if you Created an organization folder Inside the organization folder, you can find many If you can keep jobs, then many such items.

Click on the style project and I will do it simple free style We clicked on the project and clicked. This form opens what is this form This is your confession form that you are not eligible for the job. This is a simple job, simple job. Discard Old Build means what build It happens when Jenkins says that this is the man who Once the entire automation is done, a build If you do it a second time, then you get another build. It is possible that whether she fails or passes, it makes no difference.

If it has to be built, it will be built, so let's So I created a job in which I had to pay some. If you want to get it printed then it is printed for the first time. Create a job and build that job build what do I want now Do I need old builds that are my old If there are builds, it has to be discarded. If you want to save storage, you can do this. If you don't want to save, leave it as log rotation What happens, how many days should you keep the build?

[Jenkins and CI/CD Pipelines] Jenkins jobs and freestyle projects

Two more maximum builds to keep in two Write days, that means write two builds, so that Whatever builds happen in the last 3 days, It will delete everything and maximum two of yours. I will keep my build but I don't have that much brain. If you go and check, there's a notes app here. Django Notes app Ek This I Project I am going to give you and the other project that I am going to get it done, that is your three tier Application Blogging to Wanderlust Karke.

Project Wanderlust here's this mega project this is also me If I get you to do both these projects, you can fork, fork means make a copy. If you can make your own copy, here We are going to do the project right now. If we leave it as is, the GitHub project For now, keep it as it is, parameterized. Meaning you are passing any parameters right now Throttle build means you bike You drive it, you give it a ring throttle You might think that if you turn it around like this then it will be 100-150.

If you do it, it will still stop at one build. That is why it is called throttle Concurrent build means in parallel if you If you want to run the build then there is a source for it. Code management means if you need help with GitHub If you want to take up a project, use this. Build triggers means when to run this build For if you want to run it from a script So trigger build remote if you are using any one job want to run it after it ends.

Build after this project periodic Meaning you want this to happen every morning at 6:00 Once the job is running, you can create a cron job here. Can you also write what this cron is? I will tell you further about the GitHub hooks you want the developer to push to GitHub And if Jenkins job starts then Git for him You can also put hub hook and similarly pole. There is an environment in which if you Even if you add a secret environment variable You can put what is most important to you.

Build Steps Go Here for Free in the style project and add build step You have to do it and click on the execute shell here.

[Jenkins and CI/CD Pipelines] Jenkins jobs and freestyle projects (part 2)

Click on it and in the execute shell you will get a The box will open, inside which whatever you want to shell will appear. Why am I saying this because these videos would have been Isn't it the more people it reaches? Hopefully giving steps that my work I want it to be automated and such things should happen and I also want that What should be created in the last, it should become a folder. M kd i r - P DevOps So what is M KDR a creates a folder -p means If the DevOps folder is present, do not Make it if it's not present, then make it so.

Mkdir Han Pa DevOps and it's done DevOps and make it to the DevOps folder Given some steps to do automation To Jenkins and I told Jenkins Save So what does your job do? Gets to print, like and share begs for subscriptions then his What it does next is create a folder And it echoes that folder CreatedDocumentsFragment here What will you do if DCD Continuous Delivery? Two brothers, the build schedule will be done. The build has been scheduled, now that build is so It was so small that it was not visible here.

Click on the first job, you will find it there Look here it is here so what is this one this indicates that your build is successful. This tick means successful, now this The build is complete, you can see its output. That what actually happened, what we wanted to print Then we wanted the folder to be created so that He is making everything so it is started by users. Gives this location on your system then This is your system, if you Go to cd swer lib genqi workspace First job, here your folder is this path.

Printed by MK DI R Han P DevOps To He might have created a folder as well, so let's see. There is also a folder named lo DevOps by doing ls So this is automation that you did freestyle and des folder CreatedDocumentsFragment But how are things at the basic level? Only if you learn will you be able to move forward, now understand What are decelerator pipelines or. What are pipelines, pipelines like this understand that this is your tap this is the tap cool now this is the tap and this is the garden this what is this garden now in this garden tap water what will you do you a pipe.

[Jenkins and CI/CD Pipelines] CI/CD concepts / Jenkins jobs and freestyle projects

Maybe the water should become a little purer It can also get spoiled, so you are here in the middle If you apply a filter, what are you doing? A source You are going to a destination from using a In pipeline and b you are creating stages So in the same way, Jenkins also CI/CD Builds pipelines where you can. There are stages like stage one Stage Two, Stage Three, Stage Four, Such The pipeline is built, what could be stage one? What could be the code clone stage two What could be code build stage three What could be the stage four test?

Deploy build test deploy pipeline If you can make it, let me show you. How do you make it, so how do you make it free style? The project is done, let's leave it. Come to the dashboard and click on a new item If you click, click on New Item Now I want a CI/CD I named it Pipeline Construction. CI/CD is or demo CI/CD why I'm demoing it for you freestyle Then I went and that was it, command dhad dhar dhad Just write it down and you will get the pipeline, just click on it.

Same is build triggers, if you want then like this Build all that is completely the same as you What is important is this definition pipeline. Script Here you will find a script You have to write it which is very simple. Pipeline do world a world Pipeline is simple, he wrote pipeline I turned on these curly brackets, curly Brackets off, agent, any agent I'll tell you what happens next, just watch it. The topic of agent will also come, this agent I will tell you further and he will give me more stages.

Given stages means that in that pipeline what are the things that are going to happen then stage, there is a stage called J Steps. Will replace world with, I want to get it done,, name Now you have multi stage There could also be a pipeline, such as this one. There are stages, add another stage inside it. I put, then you put something else like Enter a stage named Create Folder and Like it is written echo, you can also say h can s means shell command and in that.

[Jenkins and CI/CD Pipelines] CI/CD concepts

You can see M K D I R Han P DevOps To A folder named DevOps will make this simple So what you were doing in freestyle You can do the same thing in pipelines also. What is the benefit to you stage by stage You will keep getting the output that the first stage has passed The second stage passed and the third stage failed. If it happens then he will not go to the fourth stage. The pipeline is if I minimize the stages If I do it, these are your stages and these two Your stage is a stage what does a Stage prints and another Stage creates folder if you want.

You can also create multiple stages with lettuce. When you build it, see how it turns out. This build will come out perfectly. It is running here build has run It's done, now click on it and build one. Folder is made in style, it is also made in, the devs folder has been printed. Meaning, like you did LS, first job I created the folder of DevOps in cd dot Will you see the demo CI/CD folder created CD Go to the demo CI/CD and also the DevOps in it.

If the folder is created then you are automating now Using decelerator pipelines but what We need to create such folders in the industry. There are no, there are no folders in the industry Actual projects in the developing industry If it is possible then let's move on to the actual projects. I work but one more thing in the industry What happens in the industry is this We wrote Agent Annie here so Jenkins It operates on master-slave architecture. One remains the master and the other nodes or Then there are other servers where If the work is actually done then it is the master's work.

Distributor to agents now here It's written Agent Annie, so it's also built. Jenkins is also running on the same server. How is this industry built on servers? It happens to be your If Jenkins is on a master server then it Jenkins server remains then your There are nodes, your nodes are these which Nodes reside here and are called agents. Such agents live and work as agents What remains is that Jenkins will give them work and Here the job will actually run so that Memory issues in Jenkins To be safe, I want to setup this agent.

[Jenkins and CI/CD Pipelines] Monitoring with Prometheus and Grafana

What I would do first is create an instance of I'll go, look, you'll go to Jenkins too,. Go to Dashboard and then Manage If you get junked, I will write you a warning. Will come Building on the built in node can be a Security Issues You Should Set Up Distributor builds then built in node Meaning where you installed Jenkins. That's what your built in node is, so if you are building a job there then it Not a good idea, so best is to set up N Agent then click on Set Up N Agent.

To create an agent you need a second You will need a server just like this demo. This is your server, become this one. If it is gone then let me create this server also. You can name the worker anything like agent I will do that which nut uses and instance I will keep the type t2 micro because I don't have to do a lot of work on this, just. Is very important and HTTP If you want, you can enable HTTPS. 8GB of storage is fine, so what should I do?

After I launch the instance, this What I'm doing I'm Jenkins' agent I am making an agent, what is it, you can do it with your Built-in nodes generally don't run builds. You place an order on the built-in node and Where does that order get executed? Agent Pay which is a separate server so a separate I created the server here. If you look at this, Jenkins here Master and Genie Agent both running Is Now Jenkins should be installed on the Jenkins

master.

The agent who is there, because No need to install but Jenkins is built on which tool? It is built on Java. You must have Java in the agent. If you want, let's connect the agent. Install Java in the agent So this Jenkins agent came connect Go to it and connect it with the SSH key. If you do it, I have used the same SSH. I'll go to another, my local And I'll go here and paste it, so I If I connect then why this what why I am doing this because I am from an agent.

I want to connect so this is what I've done.

[Jenkins and CI/CD Pipelines] AWS (EC2, VPC, IAM, S3, ECR, EKS)

The agent would have named it here. This is called agent oh agent give it some orange color Which color should I give it orange and this should be it gone is mine master What color will you give to Master and Master? I will give you the blue, give the blue to the master. I will give you this orange and I will give you this orange, so this I am done with my master and this is mine. What do I need in an agent?

If you want to install then install Java So here is my best command here You'll go down a little bit to Jenkins installation. With these two commands your Java will be installed If it happens then I will go and find an agent. I will install it Simple, now how will this agent connect? Because neither agent knows each other's name Do anything about Plus Agent? How to connect even if there is security If you do, I will give you a method for that.

Let me tell you because we connect through SSH. What will SSH do, you are a connect from computer to another computer You can connect using SSHS simply. You can do it, how will you do it using Public and Private key pair so that it remains private key He will be near the computer to which he is connected. To be done from some other server and which is to be done from some other server. The server you want to connect to He will have the public's first What we have to do is to tell our Master Have to connect to agent then master But what will I do?

Cd isse dar sd SSH to dot SSH name There is a folder where all your SSH keys are kept right there I key I will generate a public private key pair How will I generate SSH's population? I will generate the SSH key from the Jan and enter it. He is saying that please tell me the name of the file. I have made it, now I should keep it private. I want to keep more public on the master server. Private Master Server on your agent It is already there which is public, so that cat ID and pub, where do I have to put this?

[Jenkins and CI/CD Pipelines] Jenkins security, credentials and access control

If I want to put it on the agent then I will put it on the agent I will go where he is asking me to go. To install or not to install Yes, it's fine and I will go to the agent I'll add the public key. Being an agent is fine, someone can call it an internship or Permanent not during probation period Directly create two agents now Description This is an agent Ubuntu machine Agent became number of executor meaning. How many jobs will run in it at a time?

I am one job one executor at a time I would like to run the remote root directory meaning What is the name of the directory of the server? If so, I will write/home/ubuntu All my work is in the one-to folder. There will be a label, this is very important, label Whatever happens, all the work is done through this. The agent Annie that was written in the pipeline So we will put a label in front of the agent. Tell me whatever you think is funny in the comments.

Shout out to them for free like this now use This node as much as possible Make the node use the latch method by the latch agent Connecting it to the controller doesn't help. It has to be done through SSH, through SSH. Through host means the IP of your agent What is the address so go to your instances now But Go here go to instances and whatever your If he is an agent, find out his IP address. How to find it, here is the IP address, you got it.

Credentials What are the credentials of the agent who It is done to connect,. I will come to my master and this If it is private then whatever is my CAT ID This is private property, I have to give it to you. Why the Jenkins master because this Master of Jenkins with this host If you want to connect, I will go to Add. Click on Jenkins Now I don't have a username with password. What is SSH Username with Private Key If so, I enter the SSH username in the drop down.

I put this in Ubuntu's name, what is the username?

[Jenkins and CI/CD Pipelines] Jenkins security, credentials and access control (part 2)

My second instance of Ubuntu What is the username and private key? Enter Add it directly, then add it and Go over here and knock him down, this is not What is private, I told you like It was told in the explanation that Jenkins. The master needs to connect to the rest of the servers. Jenkins will come to the private key and the rest The public will come to that and then I will do this. I added a private key to Jenkins.

Because the poor guy failed, why look at the log Take a look at the logs and you will understand. It is saying that SSH authentication failed. Means Jenkins has what private I used it to do that SMS. The agent had the public's Otherwise let's go cat id dot pub it Who will they give this which belongs to the public? I would give it to the agent, so I went to the agent and CD went into the SSH folder which would be LS Kiya will be inside the SSH folder Authorized Keys is in VIM Authorized Keys.

I'll go and here I'll put my key I will do it, I will press it, no, what have you got? I pressed it and came to insert mode and here At the end, you will add this key. I don't know if it will be done in an hour, and What will I do in this, add my key I will give you authorization, once the authorization comes, then this. It goes that the edit is in progress and you close it. Given window then instance went to master node Went there and connected it, it worked It happens quickly, no problem connect it to Liya CD D SSH Cut Id pub and copied this.

Where did I go and hit my agent? Given that the public has access to the agent Went and what I did Agent naming it master of it and here was its color Blue Now re-establish this connection. You understand that built in how to create agent other than node If you understand the agent, tick next to the agent. Put it on Where did the go, Jenkins Agent, you. I have come how to make it by doing SSH, now this The decelerator pipeline is to be operated by this agent.

[Jenkins and CI/CD Pipelines] CI/CD concepts / Jenkins security, credentials and access control

If I want to run it, I'll go to my dashboard. Pay and demo CI/CD will go to Confil Pay I'll go now to this agent here. I have to label it, what kind of label? I again added everything to my repository If it is inserted then you can go to the junk file Yes, and this is such a label, you can see it completely. Remoting.jar folder, there is a folder will be created and a folder named DevOps will be created I need to know if my automation is correct.

If it is running from then I will click on build And here it should be built. Successful and if I do it here, can If the UC workspace is created, the CD Workspace Demo CI/CD Becomes Demo CI became CD and inside became DevOps so Deploy a job from the master to another server. That's great, now I'll give you this Declarative Pipeline Demo How do you create a decelerator pipeline? You can create a new pipeline. New Items named after Django Ci I will do CD Django in CI/CD I click on the pipeline and click.

There is one Decelerator A Django P for Django P I will make CI/CD whatever this project will be. If that is a Git Up project, then the Git Up The URL of the project is this URL. Django Notes P To Copy This URL All links are available in your description. I can use the Django CI/CD project URL I put this project URL on it, if your This private repository should be accessed in advance. Or go to credentials and select credentials You can also add a scene if not I will put it in the display name Django Notes P So this is just for display, now.

Build Trigger Poll SCM GitHub Hook I want a trigger for SCM polling. That when commit is done on GitHub then this pipeline I want it to be automatically built. I'll select this and go to Git. Hub Hook Trigger for HCM Polling so I will just do it, advance project. The option, display name hum Have you done this before? Let's run the pipeline script Right now in the pipeline you will get it completely well Let me teach you how to build a simple pipeline.

[Jenkins and CI/CD Pipelines] CI/CD concepts

How will you make it, then first you will write pipeline wrote now this The pipeline must run on an agent. Write down what the label will be, so what will the label be? There will be stages, you will have many stages inside the stages. All stages will come stage for example code This is the code stage. You will have a stage Is the build? It has reached some stage. Build then your stage will come After doing tests and tests, the stage will come What Deployed Build Test Deployed will also end now for every stage What are the steps inside?

Steps, so each stage in these steps puts it in Let's make it a little full screen. If I put the steps, what will it look like basically? Cloning the code then following its steps what will happen to This is Building the Code is why I'm doing it. I am doing this so that you can have a After we get the place holder, we Code bill test in actual also deployment

Palang The Code Diple The code is finished now it's mine. The pipeline has been built.

I will save this Saved this is a cicid for janggo p, now build it, build it. Named Pipeline View, yes, so here it is. A plugin called Pipeline Stage View If it remains, install it, it's. Pipeline stage view for you It is installing very nicely, it looks very beautiful Now restart junk when the Installation is complete now that you can restart your browser will. Reload automatic when Jenkins is ready to her Once you install a plugin Jenkins' once it starts again so when Till this restart is happening, we will also do a little gets refreshed.

You go to the dashboard and open your demo CI/CD was that click beyond and made two Now watch your code build tests Now I understand it is visible in deployed view. I learned how to install plugins and How to show it in pipeline view So your code became Build Test Deployed Now we have to put the actual masala in it. Enter the code in the stage with the code from Let's So what I'll do is this code of mine. This is the stage of this code, what should I do at this stage?

[Jenkins and CI/CD Pipelines] Git and version control / Dockerfile and image builds

I want my code cloned from GitHub I will give you the complete project. Let me explain my project What is Git? My code is stored on Http. Here is my code on git. I will build this code using Docker. I will build it through docker then i would also like to add that Docker Hub is where our The images repository is Docker Hub This code should also be pushed to Docker Hub. I would like to and then I would like to on your instance at If you want to git clone it from here git clone then docker build from here Docker push and deploy go here and.

I want all these things from Jenkins. All this stuff Jenkins' through If I want to, let's get to work. First I have to clone git, so I What you need is this URL. I need this URL and its branch name. I'll go here echo In This Is Cloning the Code I will write Get URL Now I will give the URL here, this My URL is done what is my URL. Where I want to git clone now After this what branch do I need?

Which directory of this GitHub repository do I need? If I want a branch, I want the main branch. The Gut URL is this URL and its Branch man, first of all, I just ran this much Check if my code is being cloned this is coding cloning the code one. I click on Build Now. Click on Build See what the code is doing by going to the logs Let's see, look, are you looking at the logs? Is cloning the code and cloning your code too If it's done then I'll go to my agent.

P So this is my agent and agent P is going to Later I went to my Django CI/CD folder. I go to Django CI/CD and do LS So look at my Dockerfile API Manage Pava All the code that was here was in the API Dockerfile. The code has been cloned, how did you do it? Go to Build and go to Console Output You can see it by going to the console output. Speaking this is cloning the code and See Cloning the Git Repository Ho And Its The commit message also came after code cloning If you were successful, your first stage was the code.

[Jenkins and CI/CD Pipelines] Dockerfile and image builds / Docker containers and commands

The clone thing is over, this is it. Building the Code Testing the Code Deployed If the code is successful then this is the remaining stages. Understanding how you can help someone You can also pick up the code from anywhere via GitHub. I have picked it up now, next build now. I will need Docker for this so I What will I do if I install Docker? Here I came to my agent and the agent I did sudo at get install dcker Come on, here I am installing Docker.

That I can use Docker which is my image If I can build it then I will use Docker I am installing it here, it is very important, because if there's no Docker,. Installing Docker The command was very simple, S AT Gaol docker.io done, try docker ps Permission denied means your user What do I do if I don't have permission to sudo. User mode han a capital g means user Modify it and add it to the group. The name of the group is Docker Who to add I have to add my current user and If you refresh the group then newgrp Docker, reset the Docker group.

Do two refreshes, do two basic ones if I Now I will do docker ps means list the running Containers so I did not get an error so that's it. Showing now here I am on my What I have is my Dockerfile So what makes a Dockerfile a Docker image If it is built then I am here this is building the. After the code I will put a shell command which will say I will speak docker build -t nati meaning tag de I am being him.

Notes p latest and where is this location Dockerfile that is in this place so docker build nutty notes. P latest so this is the name of the image this image is the tag and what is this dot that file is kept at the location, so simple this Why am I building this? I am building it because at the time of build, what The application will be built and you can

by testing Let's see the deployed kernel build stage. It's saved, click on Save and Build.

I clicked on the build button and the code is Build failed, let's check by logging in.

[Jenkins and CI/CD Pipelines] Dockerfile and image builds

Permission Denied to connect to Docker De man socket at this permission denied to I have given permission, still why? Denied is coming so let's see what our Which user is the pass user then need to show it to the user Ninja technique is a very simple technique. S h hum i so I see that Which user does not have permission? Save it and try building it and it will work The fourth one in build history Check its console output to see who I am.

I am Ubuntu so I have Ubuntu So added It was so what are you saying permission Denied or connecting to Docker socket Oh Nattu, it is not working here, whom am I talking to? I'm checking on the agent. We should go, so if this doesn't work, there may be another way. That may be restarting your Jenkins or restart the system We should go, no problem. Here's to Jenkins and the system. Try restarting and then sudo reboot You can restart the system by Let the system reboot normally.

If he takes it, he still understands what is happening. I'm going to do a Docker build and it says Who is denied permission to Ubuntu? Permission is being denied, it's. Let's try it, let's try it Are The server with this Jenkins is called a Jenkins worker. We connect to the node will connect the agent It's the same, maybe it's Now what am I supposed to do here? Docker ps, it's working fine. Run it again If you want to see, I will come here once.

I also hit restart sudo system ctl Restart chains just to make sure If there is no problem then why Hit restart once too, Where is the master, so I said to the master. I mean what it will be like is logging back in. Django CI/CD has failed on your end Let's try building it now because right now Once the permission is granted, click on Build. Tax Gave can you see this now it is very It is more likely that you have given permission but If that permission is not being reflected then a Restarting the bar is a good option.

Code cloning who am I step one come see It is possible that your Docker is being pulled.

[Jenkins and CI/CD Pipelines] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Dockerfile and image builds

Everything is happening, so this step git code clone this one step is yours The Docker build step is complete. I am here so this is the step of the code Build step if you have a developer So you can also get it tested here. There is also a push step which I will discuss further. Here deployment If the unlock was successful then we cried. Poor thing, it was a red pipeline. It's finally green and if I'm here I'll go to my agent and look for Docker By creating images, my image was also built.

Hogi entered, see notes app became 44 Seconds ago, now by running this application Let's see what I do next. I will go to Confill my pipeline I will keep doing this until I reach the final pipeline I don't make it, I go to the camp and end up here. I went and built it in testing if there are any test cases here, please write them. By the way, there are no test cases here, developer It depends on the pay, deployment Palaikompu What happens to h docker run command remains then docker run Hyin D means run in detached mode.

Give port 8000 to the container background Map it to 8000, why only 8000 wrote and why not wrote because here The Dockerfile of our project. Your own application runs on Rs 8000 runs on 8000 and which uses the code for its The image name is that is Django Notes Latest So here I click on Notes Latest. I will give it to you, it's done, now by doing this I Application And click on Build Now. The old one is showing that build here It is happening and Deployed Key IP Address and its Port If I open number 8000 then my P will start.

Need But Security Group Firewall 8000 The port is not open so I will go first. On the instance and on its security group Where will I go? I will go to security. I will go to the security group and here Which is bound by rules for incoming traffic I will add a rule named 8000 to it. Added a rule named 8000 and set it as Anyway I have given the IP because the application is any The rule should be saved and run from anywhere.

[Jenkins and CI/CD Pipelines] Docker containers and commands

I went here and waited, but I tried to go but it didn't work. Kill it and open it again on another. Do it and you can see your Application deployment Palaikompu Tell me something better, I'll tell you I am the most First, you can read the Docker Compose Dull here. You can also do this via Docker Compose To deploy Curr Compose, you simply need to enter your agent If you want to install it then type sudo at get Install Docker. Compose v2 This is Docker Compose v2 which you can use You can install it and great, you can install Docker Create your application through Compose as well To deploy, you will have to go to the confill.

Why Docker Compost is needed Because you will build it again, don't you see? I will show you now that I have fought I got the CI/CD and rebuilt it. I did a build again,? Why is it deployed because the port number Things are already running at 8000,. If your things fail, docker run We don't use this command in production. Go to config and where docker run command It is written, just do it instead. Docker Compose up -d what will happen if the container is running, it will do nothing.

If the container is not running, it will do so now. Save it and build it now. Do it now and see you er will not come look show two seconds The error will not come, you will be fine now. What is Docker Compose Docker Compose Basically there is a file where your Docker build and docker run commands are the same. They write it in the same place, it is Run the file on port number 8000. I have written it done, still running Because the port number is already 8000.

If it is allocated then docker ps and the port No. 8000 P for already running containers kill it then Docker stop container and docker rm meaning What will happen if you remove the container? The already running container will be removed. Will go and run the container with Docker Compose If I think so, these are all the errors that I am getting. If you are having problems then you can also get the solution.

[Jenkins and CI/CD Pipelines] Git and version control / Jenkins security, credentials and access control

It is good that you can do this What do they call this, this build? You can see this image of Notes Pop which has been made. You can push to Docker Hub, you can do this. You need to login to Docker and install Docker You need credentials to log in What do I do if this happens to Docker Hub? I'll create an account on Docker Hub. I need to push this image to Docker Hub. If I want to do it, what do I have to do here?

Will Deployed come in place of this test? Would write I am pushing Is push to Docker Hub? Push to docker up this is pushing The Image Two Docker Hub will then push the image to Docker Hub How will most First and Docker login What will he do then, with his image? The name and tag of the image will be changed. Docker Image tag which will be the old image like Notes p latest old image name wah Will replace your Docker Hub username But how is it when you are from your local.

If you push to Docker Hub, the image It contains your user account of Docker Hub. The name remains that username slash image The name has such a format, so that's why I re-tag the image. Accordingly, the name has been changed and this is what you should do. Why am I doing this because What happens when your code comes into production? It is built and then stored in a repository. Push is pushed to an image repository It could be Docker, it could be Nexus.

It could be a lot of things what also happens maybe what a Y Repository Query is one where You can put your images on it, so many ECS ECR Elastic Containers across the board. If the registry is also of AWS then You can also push there Docker Hub is the Best thing that works easily or something Noticed Docker login to log in. I need a username and password, how do I get it? That's why Credential binding is required What is credential binding?

Do not hide your credentials If it is in the environment variable then it is credential binding so I can put the password here I will not put the credential binder So let's do credential binding.

[Jenkins and CI/CD Pipelines] Jenkins security, credentials and access control

I'll teach you how to do it Save it save it fails will be gone because you are not logged in go to dashboard I went to Manage Jenkins now go to Manage Jenkins. What you have to do is enter the credentials I have to go to Credentials and Add one global credential Where did the credentials on Manage Jn Case go? Go to Pay and add a global credential Now here I enter username with password. I will add which one is my Docker Hub's so What is my username?

Once I create the token I can install it on my Docker Hub. I will create a personal access token and this I access my account through the token. I will token look here on token Went and generated new token and used two tokens. I don't need read, write, delete access You need full access, read and write if you want. You can also give it because we are just bridges. Pushing images Read and write is fine, generate it so Got your token Here is the Docker PAT Gone, copy it, I have copied it, now you Go and paste that token into the password.

Once done, enter your username and enter your token. Put fear in the given ID Hub Cred Docker Hub Cred means Docker Hub Enter the same in the credential description. Enter two Docker Hub credentials and click Create Do it, then Docker Hub is a software called cred Id is inside which this is my username And this is my password so username and. Password me Docker Hub cred name ID Now I can get this thing from my If I use it in the pipeline then I will Django CI/CD on the pipeline.

I will go into the camp and be very careful watch very carefully who push To Docker Hub, I need my username in it. If you want a very secure password then I'm going to do credential binding. How am I here with credentials I will use it with credentials If you need syntax, I have one. Repository created in Node to do CI CD here I have a complete CD of it I have given it completely If so with credentials, this is an inbuilt.

There is a function inside which you can enter the username and You can remove the password from the environment You just have to provide the credential ID and from that.

[Jenkins and CI/CD Pipelines] Jenkins security, credentials and access control (part 2)

The password variable will come and that password Your password will appear in the variable user The name variable will appear, in which your user will be present. Watch Your Syntax to Remember Syntax Don't Need It But Good To Have It With credential written with credential written now Inside this there will be an array pass so square bracket square bracket passed her. Inside there will be another function username password Which is the username from the credential ID If you remove the password, I will be here.

I will enter the username and password which is a It's a function, it's a function, what will it do? Credentials ID I Capital D Small Credential ID Now it will remove the username and password from what is my credential id my The credential ID is Docker Hub Cred so this. My password variable and username variable You can put it in your own way. What should I name the password variable? Docker Hub pass so this is my password variable and I have a username in the same way What happens to the variable is the username variable in which I will put.

Docker Hub User put this now this function is something like this It looks like this, it looks something like this. It's been, let's be with credentials username password Remove credential ID from Docker Hub Password from the ID named Cred and remove the variable and remove the password Docker Hub Pass Username Retrieve Docker. In the hub user and its scope will be Whatever scope remains, it will just be cancelled. That it is easy for you to read You're done with it's done echo with Credentials If yes then with credential username credential ID turned password variable.

I should not even expose my username now that There is an environment variable in it, this variable The values will come and the subset will be absolutely fine. Far everything seems fine Let me run this and see if Docker Logging in with my credentials from the user Can he tag the image using name and password? He is also pushing it, you should save it and see. Let's click on Build Now. Push to Docker fails now. It's done, build, build is done.

If we copy this then the spelling Don't mistake it's with credentials I may have misspelled it, so with Creedence done T I A L's Save it and click on Build Now.

[Jenkins and CI/CD Pipelines] AWS (EC2, VPC, IAM, S3, ECR, EKS)

It will come, now click on build now. Give me your 12th build and whatever it is, go ahead. Logs in or not Clone Successful Building the Code Gets Built Pushing To Image to Docker Hub Login Successful Very good docker image tag also done. Docker push o push also got pushed, what's the matter push success oh great success so basically Your build is done, Docker push is also done and I will go to my Docker Hub to deploy it. Pushed less than a minute ago means Jenkins ne Automating your code in such a way that Clone and build and push to Docker Hub It is happening and What do you want to deploy, this train?

Your code is this war, it is back Build and run, so here If the image comes, I will edit it. This build will run automatically Have I added this to the agenda? Agenda, you should understand it like this that whatever I told you I was reading more than that. I am fine Declarators give web Hook and bhuk knot and bhuk so knot and bhuk what If it happens, I would tell you, this is your git. Hup this is your GitHub and this is Yours Jenkins this is your

GitHub this is your Jenkins now wants you to commit to GitHub.

If so, Jenkins should know and Jenkins Complete automation by cloning code to GitHub Keep it running, meaning you also need to click No, how will you do that, you will do it on the web Hook. If you give then go to security groups My instances are like masters Why do I open the security? Because I need GitHub to be connected as well. Do it to him I go to the security group and My port number is 8080 where Jenkins is where is the port number 808 running But junk is going on, I would have done it anyway.

That GitHub can also connect to it I'll save the rules now, ideally. What should happen if anything is pushed on the girth A web hook should be triggered which Jenkins Pay GitHub Send some requests Git Now what is the url of the hub dot whatever its. The IP address is 8080 GitHub Jenkins has a hit on this URL If he kills then Jenkins will know something good.

[Jenkins and CI/CD Pipelines] Git and version control

Has happened and Jenkins then triggers further bill He will do it, now whoever hits it will be us. If people confiscate, here is our Jenkins URL. 8080 is your IP address and your URL. I'm going to go to my GitHub. After this is my repository of Django Nuts You fork it and the same thing with your To sideload, go to Settings now. Settings means you don't need to access the settings of GitHub. Go to your repository settings If you want to go then go to settings and here web Click on two webcams and click on add a Click on the web and add the book now.

This is getting printed Now what Application Payload URL Then Payload In the URL you put Your Jenkins URL is just ahead of it. Write a GitHub web hook like this, what will happen with it Jenkins will know this, which is at this location But my Jenkins is running right look at this I'm junking at this address. Jenkins will understand that from GitHub Web Hook has come, what is Web Hook online A Request Response for Communication They send request response and they eat it.

Jenkins should understand that He is hungry, that's why it is not SSL I want it, yes I understand, No SSL required, no secret There is nothing like this event you would like to. Trigger This Web Book Send Me Everything Joe This active ad webcam should be used for this. What I did was I went to the website and added Jenkins in the payload URL done in Webc Enter the URL of the GitHub web hook as Endpoint given content type A it kept disabled SSL why.

Last delivery was unsuccessful, invalid ATTP response 302 Edit this try it git up try webc and json and update try web browsing Take Update Way Go to Bhuk Recent Deliveries and check it out. It did not happen a Why is the last delivery not happening? Not Successful 302 so git now And Bhok, try something like this and see why not go It's been, just gut hit by the same issue It appears that you are using the URL here when. If such small things keep happening, Put a slash in front of V Bhuk and see Did Jason Wesson use that application?

[Jenkins and CI/CD Pipelines] Git and version control (part 2)

Is this what that yes application Jason yes this is right and update it by clicking on the link given below and recent Still not in deliveries Why is its HTTP not working? I am not using secure ATTP S I GitHub Web no problem This won't work, what do you do for now? Keep this in mind that in case your delivery fails If this is happening then you have to use these things Your URL Your Content Type x w form URL Encoded Disable Verification Send Me.

Everything Active Bus and It Got Worked Last delivery was successful, come on, this So it's going on, I mean, I can do anything on my git. My Jenkins will know Let's tell Jenkins right now that Git If something comes from the hub, you should take a look. What to do is go to Build Triggers and GitHub Hook Trigger for SCM Polling I have to keep it active, that's all I have to do. At the funny thing I'm going to start with CI/CD.

I will go to the last build now when is 843 I mean, it happened a while ago,? I will change to GitHub Django I am going to share the Docker GitHub of Notes. It is composed, I don't want to build it. I want it to go from one image to the next. I'll take it out here And if this is committed then this pipeline It should run automatically here. If the commit works or not. She left so can you see that I committed there Did And bill number 13 is automatically triggered, what a wonderful thing, Jenkins, what a wonderful thing.

It means it has been completely automated, this is called Continuous Deployed that I am not checking manually I pushed the code and built that code. Doing, testing, pushing Docker on Hup It's the right thing, if you think it's right then comment. Write so that When I tell you this, your What you wrote completely what stages code build how much It looks dirty now, I have filled it up so much. I'll have to write another pipeline tomorrow. I'll write all this

hard work back to you tomorrow.

Something to What is a deployed library Jenkins in that you have a repository There is a repository inside which You can reuse Groovys code in your Pipeline is so cool,.

[Jenkins and CI/CD Pipelines] Jenkins shared libraries / Git and version control

You can reuse the code in Let me show you the shade library, then understand it like this Take that this is your pipeline, this is what you have with credentials and so on are written in full. No, if you want to reuse it then Create a shared library, I'll show you. How to make it, go to your GitHub, go to git and create a new repository. Create New Repository Create it and write its name. Jenkins shared Libraries like shared libraries What is the meaning of libraries, such small Code Spats that you can reuse Jenkins Shared library has now been made public.

Keep the report, everyone can use it, you can also use it. Do it and just create it, now whatever you want Add a file to this repository. The shared libraries It has a folder named What is vers, vers means variable So create a folder called variables. I will give it, I wrote it down and it was done, a shed was built. Type in search while scrolling shared or Libraries So Global Trusted Pipeline libraries are the name of those Shared Library: We kept going like this Let's say his real name is Global.

Trusted Pipeline Libraries Sharable Libraries available to any pipeline running On the system this library will be Trusted means without any restrictions. Your branch name will appear So its branch name is main so I am here But I will write Now where can you find this library of retrieve method. If you are bringing it from then select Modern SCM I am fetching the project from GitHub. Get the repository here go here this Those who go into the code and clone the code,.

HTTPS or copy this and go and post it Now if you have a private repository then Here you add credentials like no Enter your GitHub username and password. Add the credentials of the public repository So let the behavior remain as it is. If you save it, then basically whatever you shared This is the library repository, Jenkins. Shared Library Add it to your Jenkins I have done it but I have not added it to your pipeline. Kiya to Django CI/CD me jao gaye cuffing Now what you have to do is go into the pipeline.

[Jenkins and CI/CD Pipelines] Jenkins shared libraries

I went in and then I got up and started on it The library should be written like this and in brackets put whatever your pip library name is. If it was there then I had put shared, so put shared Now, as you come forward, what will he say? Tell me he will say there is an error, it is, he will say The error is "Look, unexpected token pipeline". Shared Library So the stage code is what I'll do is stage I'll do a before the code.

Let's put a stage in which I say I will get it printed but actually printed I will not call the shared library. Let me do it, if I don't write the steps then Joe's wonder It is a lace project, yes, it has steps. Step script right let's go Stages to is the stage in which The script comes from a shared library. At the funny thing I built this If I do this, a new stage will be created which The name will be, now this is the log, the thing that is printed.

Go to Locks Par Printed But I said somewhere in the whole pipeline, I did not write how this got printed. Changing is disturbing your pipeline. Things you wrote here like this Script this is cloning the code And clone successful, this is what it is. What name will you put in it and file it? Name will be git or clone. I will pass the parameter as a string what URL so I will pass the URL and what if I pass the string branch I will pass in the branch and URL and this.

I will put a branch in the branch that is there. The advantage is that tomorrow when you clone the code Whatever URL you enter in it, He will clone it, whichever branch you put, it will be there. Will clone it now here I have my One of the repositories And shared is shared Jenkins shed lib yes this is it now you go in it If you look, I have typed code everywhere. If the url is in the git url The URL is entered in the branch, this is you As it is, you can use this also and whatever.

[Jenkins and CI/CD Pipelines] Jenkins shared libraries (part 2)

I have written that, you can do that too, just do it. Eliminate all this in your pipeline right now. Went What you have to do is write it here. Script inside script and script Remove this git URL from your git. Was In clone you passed the url first And then you passed the branch and enjoyed it. I passed a URL to the clone And passed a branch similarly i docker Can also create your own group for builds So here's a group of Docker builds as well.

If it is made then it If I remained, I went to Vers and created an add file. No need and users Name Image Tag Yes Project Name means whatever is the name of your image and The image tag is yes done commit Give me what I am doing, this is my gr. I'm reusing the Docker build. All this remove docker build what is this build of mine The script is Script in which I have no idea Tell me what to do, I have to pass my.

Image Name What was the name of the image Notes P This is my image name for Notes P What else had to be passed, image tag passed Had to do the right image tag. Push to Docker Hub if you can automate it Also see Jenkins Grue of Docker Push here it is So copy this, how is this all this. I will give you the links and it will be for you. Add to shared library Two Docker pushes so that was messy looking code You are cleaning what is in it Project name means the name of your image tag and Docker Hub username Docker Viewing Credential ID.

Hub Cred This must be the same as your ID I've passed it into the password variable I think I had something like this near Docker Hub Docker Hub user, this thing, once. Make sure you have installed Docker. Hub cred so Docker Hub cred is correct Docker Hub pass Docker Hub pass is true Docker Hub user docker Hub user is correctly logged in Docker now user and docker now closes did docker push. With the image tag to the user is correct and what is its name docker push very If you commit this, your y which is so big was written this cut hoke kitna ho.

[Jenkins and CI/CD Pipelines] Git and version control

Just this much script will go What docker push needs to docker Push your project name The name of the project is Notes app, what else do you want, tell me quickly. You can Docker Compose's simple Docker Even if you keep it composed down like this it's done, saved it, let's go Let's build that with Groovys. With shared libraries Decelerator two let's C it burst It doesn't matter if something explodes, you should No, if you are going to tell him again then he will be silenced.

You can also do that but if you realise that is T then If you accidentally tear something like this Can you fix it? It's completely done, done, very Great, now I want this to be my This is the pipeline, this pipeline is in GitHub itself If kept, I will copy the pipeline Copied it and go to my project. Where did my Project My Notes Read this project and here is the Jenkins file It's already fine, I'll update it. I will show you because you have to view the file again and again.

No need to go to the UI of Jenkins and make changes No, you can also change it on GitHub. Here is the pipeline script Do it by doing pipeline script from SCM Pressed two enter now what is SCM is git repository url so this is the repository. Please give me its URL, I have copied it and given it to you. Credential Public URL Public repository is not a problem Yes, give me the branch, see if the branch is here Do it here, what will happen with this.

This will cause the script to be named after the file. What is the name of the script here Just save the file and see it I know what will happen, I will show you this. You won't need your pipeline. It will automatically run from GitHub. I went to this GitHub and found this Let me make it short to show you a demo. Feels good no how do I show this happened now what should I do in the Jenkins file Should I tell you what changes should I make?

From the lettuce I want to Docker Compose down before compose up So what happens is my latest code I get that first the container stops and then.

[Jenkins and CI/CD Pipelines] Notifications and email / Jenkins security, credentials and access control

If it happens then I am like this then see what happens now This is what will happen now I changed the genki file I've done it, I'll commit the change. Committed changes as soon as I commit My pipeline gets triggered there You should know, look, this has been triggered now. Pipeline, it's this pipeline that We have seen it on Get Up, so I'll add it here if you like it Ho Ga Decelerator Checkout in Stages SCM then code then build then.

Push to Docker Hub again Deployed Action Level Declarator Check Out Docker Hub Push Deploy Library Sub you can do it now here you go your Application is still running where agent. I taught you something which is one of the Jenkins An intermediate should come at the basic level Shared libraries have also been increased at the level I told you, it's not much at the production level. I got him signed in, now, tomorrow's The boy who came gave him all the plumbing He has access to manage Jenkins as well.

After going to Jenkins you have to go to If yes, you can see the roll base here. While Jenkins i'm restarting do Lose weight, drink water and roam around at that time. Manage back on Jenkins' Manage You've just got on the bus and you'll see something new. Security At Security you A roll base will be visible, where did you see it? Here are the authorizations Yes, these users can do anything. Tell me, what will happen to the agent? Build the job not to create or delete it No to doing it, yes to reading the job and Discover karne ya mab wahi reed ho Got access to run but not to view.

I understand that the deployed is not showing up. How many people understood the control, if it goes, tell me quickly User Management In Jenkins role based now next segment There we have a CI/CD Will project with three tier app. Three Tier app Kh Jahan Kubernetes Argo Sea Sadie Prometheus will see all these things DevOps will create a pipeline of cops where OWASP TV SonarQube: Use these three things and you will also receive email notifications. If you want to do this, you can click on Join Now.

[Jenkins and CI/CD Pipelines] Notifications and email / CI/CD concepts

In the next segment, we are CI/CD Project DevSeps and Email Notifications We are going to see this, so this is today's project. It is going to happen, wherever there is one Developer pushes code to Git Up on Jenkins But that code is a bridge of that code CI/CD This from DevSecOps Practices where dependencies are checked. All these things happen and then that image is scanned. Scanning the code of that file system and then push to Docker Hub. Once the image is pushed to Docker Hub, then from there that itself will trigger a job that will cd The job will be continuous delivery and that thing when.

Your code will also change, change its image and upload it to GitHub and from GitOps Argo CD will pull that code and create an e on the KS cluster Once this. Deploy monitoring and logging of that app Things that will happen if your app fails during this time You will receive an email if it passes. If it goes then how many people will get the email In these shorts, I understood that till now we People used all the tools they read about.

This project has been created by How Many of you are excited for this project Super Amaze How many of you are really excited to Build Something Like This Where Everything Is getting. Automated Superb Very good, very good, now Let's see how excited I can make you. I'll give you some patience, like I say hype would have created This is your CI pipeline where you Whatever you do, you do it through parameters. Do you do it if you have to update something?

Git will run things through parameters Code check out here on DevSecOps If you are looking at these four things, Ho tivi OWASP SonarQube here Security aspect has been added Environment variables will be setup in Docker. The images will be built, pushed, and then There will be a CD pipeline build that will again Kubernetes will verify your Docker tax will update the manifest of Kubernetes and Also push your updated code to GitHub. After doing all this, it will email you. How many people will go to see how this page looks?

Application where you can click here like this By doing Let's Test Devs, can you pick any image like this?

[Jenkins and CI/CD Pipelines] Monitoring with Prometheus and Grafana / AWS (EC2, VPC, IAM, S3, ECR, EKS)

Bali is the image where I went on adventure I love your blog here, taste It will be created on August 31, you can get its back. You can also check the end where the back end is Accessibility and even more features of this application I have a CD where I can update it. If I do, then this application here EKS cluster gets updated automatically But whatever Kubernetes on that EKS cluster Whatever resources there are, they Added all targets via Prometheus and those targets are displayed on the dashboard.

It has been shown through where there What service is causing the problem? What is happening in which service every moment The thing is, let's give you the back end. If you want to see the deployed then back end key Deployed, you can see whatever you want to see. Which network is lagging the most? Everything in One Project Now How Many of You are excited for this project. Tell me quickly in chat because we are Gonna learn some new things like Argo CD GitOps can we do it Not side by side, you can't do it side by side.

By the way, I mean obviously if you are different from me If you go a little faster, you might even be able to do it. I would highly recommend not touching your Laptop and not touch your keyboard you can use a pen Take the paper and do as I say. I will tell you exactly what to do Many of you want to listen to that exactly Tell me what to do in this, look good. What to do in this, first of all First of all you have to tell yourself how many of you will be taking this challenge of Completing this project yes per git docker.

A lot of crazy things are going to happen, let's go. You should know that you can only This project is not like any other in the world. Through Project DevOps Practices Deployed the idea of today's session it's not Just someone gave me a mega project or No, no, I want you to understand. What is DevOps in this Why DevOps in this, when you go into the depths, then things will become clearer Let us first understand what this application is.

[Jenkins and CI/CD Pipelines] General DevOps discussion

What is it I hope my screen is showing All these things are visible First of all let us understand what is an application? This is a three tier application in which this The front end tier is in React I hope Do you all understand these people, this front? As it is written and which is its database That's MongoDB's database now. A lot of people will be making transactions in Input/output operations will continue to be performed, hence There is a caching mechanism, what is Caching is if you want to use a server for the same thing.

If you are repeatedly querying then that It gets messed up because there is load in it. If it increases then for the same thing it You can store it in a cache which can be saved in Redis Also called cache, Redis is a key value. Payer Cache Storage Service allows you to Get the data document you want Based on key, retrieve key value. This is the complete application which looks something like this something like this application is great you can do things on this right now this.

How did the application come, where did it come from, why did it come First of all, This is the first episode which will give you I want to understand exactly what this application does. Whenever you run any application of DevOps What Deploy does is only then you can find its errors. Issues can be further deep-dived into services, so this is our first component. Back end node means you have a lot here All JavaScript files can be found here If you go here But it was said that this is you will go here you.

Chat that it is an application that was developed by a developer The name of that developer is Krishna. Mast Krishna is an open source application. Let me give you a shout out so this is him The application is, it is a good application everyone fine now Next, we will get a machine that will take us through all the. Things will drive a cluster of Kubernetes And your Jenkins master will make all this If we make things then for this we need a We need a master machine, what do we need?

[Jenkins and CI/CD Pipelines] Terraform and infrastructure as code

If you want a master machine, this is the master machine. What will happen in this master machine i2 instance that will provide some security Some with pairs, some with groups With IM Rolls Confers that all or all of our output. If all this can be deployed correctly then master How can we make a machine to make it There are many ways to directly here go to Terraform go to e2f and a. Make a great C to instance how many of You think they can make an EC2 instance How many people are using Terraform?

This is what was in the Terraform session Terraform instance will be created in it Because I have to drive, because I have to drive now, two things can happen, either you. Master, run the entire Jenkins on this. Run the entire master on this or just one Create a worker node and do things on the worker node Let's go, this is also, so both things Possible is best practice j worker. Create a node but if you want it in a single machine only Everything has to be run because if we If you use it, it will be of no use.

If there is an under-utilized machine, then everything will be in it. You can operate it, but this machine Jenkins will be inside it and Docker will be from here If your case is about to become a cluster then this. The size of the machine and the type of instance can we keep this thing tad micro Because we have to build Zinc, build Docker There are a lot of things to be made, otherwise we This. T dot I will have to keep it large, in t2d large It will cost a lot, that's absolutely right.

There will be a lot of expenditure in KS cluster also. This Project and When Ever You Think You R You are ready for this Deployed Large is the instance size which You will need to pay for this project on a masternode. Meaning this will be installed on your master machine,? On this master machine, we're junking setup. From there all these things will be created, from there, all of this stuff. Banegi How Many of You Understood Point Number two, the second part of this project Where we need to configure a master machine I have to configure it, that means all the ports.

[Jenkins and CI/CD Pipelines] Notifications and email / Code quality and security scanning (SonarQube, OWASP, Trivy)

Whatever ports are required to be kept open I have to give permission whatever is required If we want to create this cluster from here then Awww a seal I need is understand. It is coming that whatever I have learnt till now is from All that relates to us in this master machine I have to do it, now master confirm it once Then we have to configure Jenkins. Which is your DevSecOps Jenkins needs to be confederate Jenkins needs to be confederate To do this we will have to use tv for image Scanning is the images we create or If our code is scanned then.

Tree will also be used for image scanning Then there is a dependency check, sometimes how many people know that in the middle there is a log named 4h There was a library, that library of Java. It will have to be done, then we will code. Quality check the quality of the code Is it good or bad or what it is? If we look through the SonarQube, then these three Here are the things you need to set on your master node: Have to do it up and then through Docker.

Things will be built, all the images will be pushed. If it is made then we will need Docker also,? I understand how many of you are Getting this, that you also need Docker, same He will run everything from here and then consider himself here. If you want to send an email through this then send an email Integration is also required and you can do all these things. Will do it through Jenkins on your master machine Now, you can do this on workers also.

Just to Save Cost Just to Save Time I am the one and only Master Deploy I'm going to show you worker deploy. You will do it because I did Jenkins session I have got this one done, I just and I will only show it on master. Time should not go forward or backward, you people should have it only once Everything will be fine in Now once you have your DevSecOps, I mean Bakan Once setup, we'll set it up again. ES Cluster Kubernetes cluster watch is your this is your third ch is Your Fourth, we call this cluster a particular will set up in the region whatever its Whatever things are required, they will provide everything.

[Jenkins and CI/CD Pipelines] Git and version control / AWS (EC2, VPC, IAM, S3, ECR, EKS)

Minimum 20 minutes for cluster formation should have been If it takes 20 minutes, then it takes 20 minutes. That we will keep the time with us until we People will take a break or whatever, we saw that. Your cluster will be ready with two notes. Again, these notes also have security group. It happens if you don't forget these notes too Security group how many of you Realize that these notes also have security There is a group and we have to port it also.

Fourth Point is coming The fourth point is understanding yourself How to create an EKS cluster and The cluster has to be managed and that manages. If the master machine is from the cube then the cube will be of Seattle. The commands will run on the master machine which will The cluster will now access this cluster In this we have to set up Argo CD. Setting us up Argo CD in a cluster Argo CD updates this cluster Will do will update this cluster how Many of you understood this well before Let me explain what an Argo CD is.

Argo CD is a Continuous Delivery Tool Which is specially for Kubernetes Designed where you can find Kubernetes You can sync whatever services there are. They Deploy to whatever your repository is You can connect so that directly Your Kubernetes from your repository The manifest files will come and they will be on your server. Sorry Cluster Pay What does DeployD do with a Git repository? Kubernetes takes the Nity's manifest file and Goes to your cluster and updates it How many of you get the understanding of Argo CD Simple Tool a simple tool.

What that does is get your from GitHub. Kubernetes manifest file like from let's Here if you go to Kubernetes So this will see Kubernetes here The manifest file is frontal back Andel MongoDB is the deployment of MongoDB. All these files are a service of MongoDB. On a cluster of Kubernetes Deployed CD Works How Many of You Group what were we doing just now we What were we doing kubectl Apply Online for Deployment Apply Online. F service, so that automation is not needed right now.

In the gig of the Using Argo CD, he Have a manifest file which we can update How many updates should arrive on your cluster?

[Jenkins and CI/CD Pipelines] Helm and charts / Terraform and infrastructure as code

Of you understood this, do you manually I don't want to write anything, I need five reps. Go to the Hub and repair two Argo CDs. Simple will make it five Understanding How Many of You Gut It Now the last part is monitoring, here we People will use Helm, here we will use Helm Helmet Files Remains persistent Volume R. Packages for bank or service accounts It's a package manager Simply through that package manager we Graphador is about to do it, not single click There will be four or five commands, but only four or five.

Will remain Here in connecting the observational I do the same thing with automation through Helm. That's why you should understand that Helm what happens Helm is a package manager Where you can see the manifesto of Kubernetes Installing files using Helm Package and deploy it directly to Right code would automate everything. How many people understand six Components of our project The project has six components, work hard six Today we will learn six components of Today we will learn all six components of Everyone.

Ready everybody ready during this some of the lettuce If a step is missed, some steps may happen next. What would I have done My Wunderlist Mega Project I make this repository my Why clone locally? Why, because I am sorry for Kubernetes via Terraform We have to create a cluster, not our own. Was the master machine to be made or before that? You have to listen to these things to understand them. On but still I love you I will explain it to you, do it a little.

I have said it enough, so Ganpati Bappa Morya Let's get in the name of Ganpati Bappa Started, let's just do what I did, local Pay Git clone your project CD Mega Any project on my wander list. I cloned this project of mine. I have received all my files What will I do now, I have to go I'll go to Terraform this is my e2f Terraform variable Everything is ready near me at AWS Pay. I AM I AM I will create a user I'll create a user how many this thing People know I'm a Terraform admin.

I will create a user named and give it admin access.

[Jenkins and CI/CD Pipelines] Terraform and infrastructure as code / AWS (EC2, VPC, IAM, S3, ECR, EKS)

I will then provide its security credentials. Many of Go to this and here in your local I will run AWS Confil, how many people to know That's how AWS configures It's fine locally or on any machine. Here I go and VIM. SSH jan a main bana di jisa I have named it Terra, I have named it Terra Dia and I have Terra's Dot Pub here It's here, Terra Kid Pub is here. This is the location where I can this is what I'm doing this is my this I am running the part that I made the master machine how many of you got this i am making a.

Master machine is VIM Whatever your location, this I had changed it I think we have people here But Terra had kept the pub as it is. Your location will change, you people You can do it through Terraform or you If you want, you can also do it directly on Etoo. No problem, no one will judge you The one right there is Terra Automate's default. VPC creates a security group where But let's allow port 22 and we MI ID and instance from the people variable How many people will understand that they will type it out?

Aaya what is MI ID and what is instance Type here I want to use Ubuntu machine I am going to have the MI ID of Ubuntu I'm going to use type t2 large so tad. You will get a large amount and the control will be done. M variable tf will go to us to mine This is the region, this is my ID, if you want, you can You can confirm that by visiting an instance. What is the MI ID of a Ubuntu instance?

Terraform's confession, what do you need? Don Clear now just Terraform in it so that I can create my own cluster and create servers that My master was the machine and I became, done. Fam Plan this thing again, you do this thing directly You can do it on Sattu also, it will save time but it will be complete. If you want to do it end to end then I'll just go through The entire process, so four things are going to happen: security.

[Jenkins and CI/CD Pipelines] Terraform and infrastructure as code / Notifications and email

Groups Keys and Your instances, one thing to note here. There's one thing that's missing in Terraform. Just to note You need to root the block device's storage I will have to keep those 30 GBs, keep 30 GBs. Why would it be necessary because our work is done in 8 GB If it is not going to be made then keep 30 G as well. Terraform Apply Let's Now he will apply and say that the we did you plan to execute it Yes, execute Is Come on, your security group is also being formed.

The instance is also being created Instance also I think around 30 seconds In it will be up and running, the instance has been created, so now we What will you do? You will open a only and only for your master machine so can you C this automate instance t2 large this is Ready Using Terraform What I Will Do Now This is my instance. I will do this connect so this instance directly this one I keep the open Yes, let's call him master, this I will open it in a, Master.

I will try to connect it So this is my key with whose help we People are going to connect that so here Terra's kept So with the help of that Terra, I. I need permission from Terra. Connect Done Your master machine is ready Yes and No your t2 large master machine is ready This is easy. Simple Fun Easy Now what am I now I'll go to my dummy And I will understand that on me and this master machine What things do you need to install?

The things you need to install on the master machine He has to do this Jenkins is Seattle Argo CD is Jenkins worker is For tutorials, we are all on Master. You are doing it, if you want you can work as a worker It's Again An Easy Step OWASP SonarQube Email Notification Monitoring and Clean Up These. All things are set to us through the master machine have to do up Now the ports that you want to open in the master machine The things to do first are listed here.

All the ports at once Open all that, I understand.

[Jenkins and CI/CD Pipelines] Notifications and email / Terraform and infrastructure as code

This is the master machine, this one in it There's a lot of things going on,? It's not just SSH and ATPS that work. There will be many more ports opening. 3 From Hajj to 32 Hajj 767 open it Why is this Kubernetes? Nits are for node ports generally, and I will add port number 80. Port number 80 i think already is port Number 22 and 443 are already there then I what will I do 6379 Can Avon Guess Why 6379 What is the port of Radish cured.

I will ask what is its port what is its port now another port remains 465 which stands for SMTPS simple mail Transfer protocol is. We will send the S of our application MTP S we will go back to our The application will have some ports like 3000 From Up to 10 Hajj, this is through our application. The ports will be P then the port number is Our 25 generally remains SMTP Simple Mail Transfer Protocol is fine and That was completely different, there was SMTP and now we.

People will keep 6443 Who is it for 6443 Generally It is fine for your test I got it, done save The rules security group of your instance has been updated. I have even done it now, on every stage I meet you I will expect something because again it's How many people does an end-to-end project take? That is the ports that I added manually Can you put those ports here? Ingress here in TF on Terraform can add to How Many of You Think They Can Do Can you how many of you get first Idea for open source Contribution: How many people got the idea?

That there is some project somewhere in it You should see Terraform or a port that Could be the first open source contribution to In this project I will tell you about open source. I will keep telling you about the contribution as well, Good Done, let's move on now. To our server which we have created. If you did, you would have named it are you master machine this is our master machine Now on this master machine first of all So sudo AT get update machine update hit.

[Jenkins and CI/CD Pipelines] Docker containers and commands

Hit update on the machine Now what should I do about this? Setting up the tools one by one My tools these are Docker done Jenkins' I had to set up all these tools one by one. If yes, then sudo at get install docker d Come on Docker Compose if you want to build this project with Docker Compose because many people will say. Forget about Kubernetes Bati, you should also use Docker Compose. If you get it done, you can do both of them.

When we do docker ps it If an issue arises then to solve this issue So this is our version run docker.sock If the thing has to be opened then sudo Seech M. The master machine is your own machine, Docker PS started working, its alternate method What is it, tell me its alternative method In chat sudo user mode - append to a group docker. Whoever my user is logged in to His/her and newgrp docker newgrp docker what happens to your Docker group It gets a little reset and refresh.

That it reflects in your group go What happens with the newgrp Docker This refreshes the Docker group. That you can restore the system completely. I don't have to do it, it's done, now move on. That's it, now the master machine

on Jenkins For Jenkins you have to confine What you need is Java installed. Whatever one wants is the end You can install Jenkins directly. How many people have seen Jenkins before? You have setup something for Jenkins Kuch Confession Chahiye and Jen Kins Mast Set gets up If so, paste the same confession and press enter.

Is multiple times right now with a different The relationship became Let's be good Yes, we have from 3000 to 10000 Ports were kept open, this is a security issue. It could also be because the port numbers are 80, 81, etc. Open port number 90 Open this is obviously a There could be a security issue but again a mistake If there is any port left here, then someone's port should be What if our time gets wasted there? I have given it to you so that time does not come, it is yours There will be intelligence which will tell that, each one Do it and open it but again it will take time.

[Jenkins and CI/CD Pipelines] Terraform and infrastructure as code / AWS (EC2, VPC, IAM, S3, ECR, EKS)

I hope it is fine for the tutorial part but in real life you The port specified in Terraform just as open You have to do it right in production You will not do production directly now. What should not be done got it we saw that security We have given ranges in groups, we have ranges again, it can be Security Issue But Just for the Sake of This. Tutorial repeatedly tells me port number 808 Port number 9000 is not on here and there Read so the entire range was given, how many of You got this that if I on my instance Let's go.

My instance right now was and go to its IP address like this So it is already open or why is it already open Because I have kept the range open. How many of you got to visit me again and again I understand that I don't have to update it, it's good. Samsam super what do I do from here Start setting up your server I will give you sudo Kat This and from here I entered the admin password has given Add yours to the master's group.

Stop all this stuff, done, install now, suggested The plugin which he is initially saying Initially the plugin he is talking about Let it install, now initially. He is talking about pipeline and Git Ya on OWASP. He is saying let it happen, we will do it later. I will do the complete setup properly Create your username enters email address Yes, this is the URL of your instance. Go save and finish your Jenkins now you are ready Can Start Using Jenkins' is fine Done, everyone understood this part.

How we junked the master machine You have set it up, now next to yourself Creating a cluster of EKS from the master machine, you have to master the machine and get the case. We have to make a cluster of this. How a cluster is formed Annie Idea how many of you have S CT L, so to create this cluster, first you need IAM user with access Keys and Secrets to Access Terraform. I am the admin of this one, so I will take an hour.

It has run before but this one ran first today.

[Jenkins and CI/CD Pipelines] AWS (EC2, VPC, IAM, S3, ECR, EKS)

For the server because our master I will create a new machine. Access's command line interface Understand its tag value if you want So you can keep it or let's keep it Wonder Lust The name of this project is Wonder Lust. If you get it then this secret access will be done. I will go and add CLI, this is the command As it is, I will copy this command. This command will enable AWS SSL IConfer. Go Is Done AWS Confido yes it's coming cha means it is working fine so I'll go now to my Access Key By copying this access key I will paste it here and.

Enter and copy this secret access key I will stop screen sharing. Copy the secret access key AW's Confine I will paste you in Here you can copy the secret access You have to go and paste it in the default region. Do not keep any name, any output format How many of how to Confinet If you do this, then the shortest command is I see that, yes it is mine. What I will do is I will now call the E case CTL.

I will start installing the case For CTL I need kubectl Because the one who manages this cluster The CEO who is the director is. kubectl so kubectl we do will take install then client version wv 1.9 kubectl is installed now. To create an EKS cluster that Pre Requisite A W Sal I Cube CTL, this CTL, we made it. Many of you are getting till here that puri became a thing Is Now I will make this cluster my master If I go to the machine, what will I do first?

I will create ES cluster whatever my The name of the cluster is whatever is given in the region I want whatever version of Kubernetes I want to give that first without note group. I will make it but before that I have to go here Our You will have to go and check the cloud formation whether the cluster named by mistake or that. It's not here, it's not here, so we People can make it, otherwise it creates problems. I will do this case create cluster name Wunderlist Region US E2 Version 1.30 without node The group got it, "I'll do this thing and I will press enter here

this is your one.

[Jenkins and CI/CD Pipelines] Code quality and security scanning (SonarQube, OWASP, Trivy)

Why are you doing it one by one because we Project only Fully deployed within an hour How many of you honestly seen Videos complete end to end in one or one and a half hours. Deployed how many of your actually I saw two videos, so I am telling you, nothing will happen in an hour. Jenkins will configure the machine. Configure Jenkins Let's run a program in Jenkins How to configure Jenkins Some of the things you need to do in Jenkins had to do This is if you keep only Master.

If you are, you need to create a new node. No, but if you are adding another agent If yes, you can create a new node and save that node How many people can you attach to Jenkins? I was in a fundamentals session where I had a master and an agent were created and both of them How many of you were There in that session, how many people are like this? Who was also in that session on Day One and Today is also a day when simply means two months of rigging.

If you want to do Docker, then we have to do it. Take your master, my master now It's stuck, no matter what's in it. I mean, I will create it on a new server. Created on a new session what will you take Let's go open a new terminal. There are a lot of things that can be done on the server. Even if you don't go, I will still go because You are doing it through Jenkins, right, without Even after that, but still the server is on one side Let's start a new session.

A new server session has been started here. The cluster is already being created on the master machine So I opened a new one and a. Which will fix your file system Scanning all these things will be fine Let's have a great TV, the TV is installed, now we are What will we do with this SonarQube? If you install the server then docker run in en Interact Terminal Detach Mode SonarQube Run the server SonarQube on port number 9000 LTS Community, copy this.

Here's the SonarQube in the background. Server, which is the server of SonarQube?

[Jenkins and CI/CD Pipelines] Notifications and email / Code quality and security scanning (SonarQube, OWASP, Trivy)

Running on port is any one who can read the docker run command and tell me which port But 9000 what do we Again port 9000 needs to be opened No, because we opened the range. I would have done it one by one, you would have said Again, security group, understand that I Why did you open the range? Things are smooth SonarQube is starting Let it start, give it some time is taking That SonarQube is up all systems are Operational Loading Login to SonarQube How many people are using SonarQube?

We will simply setup the Kube server using Docker. Through initially admin password remains Is Its what you are after I can change it in yes how many of you can Create an a suner cube server easily bun. Jenkins Done Tivy Done Sonar Done Docker is fine Is What's next now? How far has your cluster reached If the cluster is being formed then it is difficult to form the cluster. Started at 4:30, just 436, 45 minutes left It will take more time, until then we What can we do till then, we are doing well.

Email system does its own email How many of you want to do email then her There are some pre-requisites for this like LETS You have opened this email from I opened any email and now. P and here is app Password if you need Gmail SN application If I want to run it, I will have to verify it. I want to make a P and I will name it Jenkins I will give you the forest, you can name it. Who has to go to Jakus and after going to Jakus Manage Jax and then just give me a minute and Scroll down to Manage Menu and then click on you have to enter the credentials.

You can send emails, yes, I mean, if you got this, Gmail is a mail server that You have to enter his credentials here. Password which is p password how many of You Understood This is how Gmail lets you enter your password. I have to create a Gmail password and You have to put it here once you have Password created for you on ManageJson Then go to the system.

[Jenkins and CI/CD Pipelines] Notifications and email

After going to the system, you have to go to Email Kiss on the notification part Pay Email Notification Extended Email notification and here you will find SMTP. If you want to write the name of the server then whatever happens to Gmail his server name is s MTP gmail.com so this is the MTP server of g mail. What is SMTP, simple mail transfer protocol so if you get an email If you want to send it, then this is the server for that.

The SMTP port is generally 25. It happens but if you want to run yourself through P then 465 SMTPS is the protocol we You have to use it, after doing this you will get advance Go in and add your credentials. You have to give it simply and then use SSL. Here you can test email notifications. STP will be available for testing For testing the server STP gmail.com default User Advance go Keep it as it is and here you just User needs to use SMTP authentication.

Name you what is this to test it Just test whether my email arrives or not. SSL which is the same as the above confine used she was SMTP 4 65 and just test the confession by sending an Email me, to whom should I send my email? Tell me someone please email me Two ha, Dheeraj, if you don't get an email from me then Be respectful, say yes give but let's see let's taste it then Dhiru Yadav 5@gmail.com Test. Confession o sorry port number not 456 If it is 465 then you would have messed up such things.

That's why email was successfully sent. Still Send yourself an email and see if your Whether the confession is true or not, is the trust complete? I entered my own email incorrectly. Email or Send If you see test email to send From Jenkins, people understood that How to send email through Jenkins How many people understand it went fine Is Done, now save this confession. I accidentally entered the wrong email address. I had given nothing else, so this is the test Confession: No one should steal your password.

No problem, email test is just fine If it is there then you can keep it, which one was it, it was 465,.

[Jenkins and CI/CD Pipelines] Code quality and security scanning (SonarQube, OWASP, Trivy) / Notifications and email

Is done So here we are, we have Gmail setup. Docker is done, SonarQube is done, EV Done, let's see before we do the OWASP. Your cluster has been created, your email address case cluster is Ready your e case cluster is what Have you made Ready 463? Will go Email notification that's for testing let's do it now what do you have to do Add nodes to your cluster. It's to give cluster is ready if you do it kubectl get Why is Node No Resources Found?

Your cluster is ready Node in the cluster No, I did not add it, why did I not add it, because When you entered the command it was without node. There were groups, what do you have to do now? How do I add my node group? Many of you understood this yourself Before adding groups, O IDC Provider remains Open ID Connect Open ID Connect Open ID Connect US to Cluster Named Wanderlust Why does Open ID Connect happen? That your ads account can talk to this cluster Click Enter Open ID is connected Now look here, create CTL of E.

Node Group Cluster Wanderlust Region US You can change your region to East. Lena Name Wanderlust Nut Type Tad Large Why because I consider myself very Prometheus Grafana Argo CD running a lot of things I need two notes, minimum two. Maximum two-node volume size 29 GB 30 Put GB, put 50 GB, it's up to you. This is the important thing about the case node group. You must have an SSH public key. What is my name, whatever is written here what was written was e case node group how many of you understood this Step How Many of You Understood This Step E-Case Note Group should have its own.

Let it be made, till then we will do what is the Use of this key By using this key you can access your Inside the nodes of the EKS cluster. I am is it done I want to search OWASP, I searched OWASP. OWASP Dependency Check 5.1 or 5.5.1 this one What to do and what to do, do it at once Let's take a plug called SonarQube Scanner We have to do this so that we can run SonarQube Server.

[Jenkins and CI/CD Pipelines] Code quality and security scanning (SonarQube, OWASP, Trivy) / CI/CD concepts

Be able to attach it to yourself and what else you need I need a Docker plugin so that Docker If you can run the commands then Docker also has a plugin. What else do you need, pipeline? Why do you need stage view,? Pipeline stage view here To see the view so that you can understand yourself I should know which job is for how long. Build and so on such a complete stage View Jaye This Is Called Stage View Pipeline Stage How many people want to install four plugins?

They How many people think four plugins I will install it We will also take Blue Ocean Why not try this cool looking blue ocean? Just Click On Restart Jenkins N Installation is complete, it is junking will restart and drink water Take what is this second wall own Let's restart it Admin Let's get this done. Ours is fine whenever we restart everything is complete It will be installed, this thing will be fine Restart will happen after some time It will be done now, let's see your notes.

Groups were also created, two nodes in the cluster Everything is ready now if I make a cube CTL gate nodes then how many How many notes will be shown in my. Cluster two is visible now, become the correct cluster everybody able to set In the course of conversation this became this became It became I am having fun chatting with my. There should be an understanding of what is happening I'll go a little slowly I understand it clearly till here is now This will keep happening because it happens It will be fine, I will come to Jenkins.

Do it and install it from Gib.com So Joe OWASP' account of the original Git Gib There is a Gib repository, here are some things from there It will download the files which will take you 20 minutes. May occur when you run the pipeline Getting it because he'll download something. Is I need to install Jenkins and SonarQube If you want to integrate, Sonar was here. Go to administration and then You have to go to security again. The administrator user is visible for Click Create a Personal Access Token Someone please enter the name of the token here on tokens Give me whatever name you like, Jenkins The name of the token sounds good.

[Jenkins and CI/CD Pipelines] Jenkins security, credentials and access control / Code quality and security scanning (SonarQube, OWASP, Trivy)

Enter 30 days expiry to generate This is the token, copy it somewhere. Go and slam it so you can get it back I understand that you have a token. Made for integration of SonarQube Admin for Token Now that token as I have here Jenkins' by making a token of the google2 cube I will put it in the credentials then Global Pay Clicked. Went to add credentials then here now But how many people think that I am using the username I will use it while we only used tokens.

Only token was created, no username. It will be used here, secret will be used here Text Secret Text and whatever you The token was copied into that secret text Put it in and give it an ID of the SonarQube. Put in the Sonar Key or Sonar Token whichever Add whatever you like and create it. You have got the goldsmith's token too. What will we do with that goldsmith token? Integrate it with your Jenkins I will show you how Again go to tools and after going you will see Search Sonar like these are tools Maven's conflation is Java's git if you.

If you want to do it on a private Git repository then Here is the Gradle installation. Got your own SonarQube scanner meaning This server of SonarQube is interesting to me. I want to add it, enter the name of the server. Of Sonar Put Give me two more installs and automate SonarQube now its After that, all you have to do is save it. Things to do if you look at the diagram Also updates the GitHub repository Jenkins with updated Kubernetes tags Or update the Docker version tags So Jenkins has the password for GitHub How many of you think that.

Personal access to Jenkins and GitHub You can integrate through tokens What to do: Go to your account settings But Go to Developer Settings Go to Personal Access on tokens go create a new token named. You put its name here Wanderlust Because put something here Give access to the repository and save it. You can copy this token somewhere. Tell me the System Tools plugin nodes Security Credentials Credential Types I will add I will add in the credentials right why Because I don't have the git credentials yet.

[Jenkins and CI/CD Pipelines] Code quality and security scanning (SonarQube, OWASP, Trivy) / Jenkins security, credentials and access control

In the ID I will put GitHub cred or then GitHub credentials GitHub credentials Total Create so I through Jenkins which Credentials of the person doing things on the tool I'll give it to you first,. Is Now I have to go to the SonarQube too. If you want to integrate it then I need to You have to go to the system and go to the system I'll have to go to the SonarQube after Search here, yes here is the SonarQube. Went to installation d SonarQube d Went to the SonarQube now here I what I am doing this, I will add a Sonar server.

Can you tell me the URL of the Sonar server? Here goes the server information of Sonar. URL I copied this and here I am Paste how many of you got When Jenkins reaches this server, he Authentication token will be required, that token will be in Sonar Picking up from tokens. I will take how many of you understood that How to access SonarQube through Jenkins Integration First, create a SonarQube token. Add that token to the box And then you just have to enter its URL.

Tell me which one is this I have made an easy one So I wrote master here. Take Done, now I will give you a How Many of You Think Jenkins Pay TV OWASP Sonar Docker Email All these things we People have accepted all Confer is just a small step away when. Jenkins will hit SonarQube to give him a Web hooks will be needed to automatically hit So I will go to the Sonar server and After going there, I went to confession.

What is a web hook and how to create it? If I need to send something from Jenkins to Sonar So if I need a snack, I'll be here. I will go there and create and eat the food of which I will give it the name Jenkins Web Hooks. I will copy the URL of Jenkins and put this now this is a web hook so SonarQube web hook I will create this No secret, nothing Create, what will be the use of this and hunger?

What's the job of the book whenever in Jenkins If any job comes, send gold to the goldsmith.

[Jenkins and CI/CD Pipelines] Code quality and security scanning (SonarQube, OWASP, Trivy) / Notifications and email

Job completed then sonar Jenkins will send a reply that the job is completed. Of you got this, we say that When will your delivery partner arrive? Jenkins address near this SonarQube This should be how many of you get This should be the address of Jenkins and Because of this address he will tell that you are my. Is Done, everything is clear till here, Khushal Mangal All good how many of you have Completed to four and three Meaning I understood mentally u 4 Andy understood I came well, how many of you done good.

Let's fix Argo CDs Let's set up Let's Set Up Argo CD So Argo Again we're going to set up the CD. Go to your repository and after going to the repository, you see What have we done, we have done Ne Junk Master E Seatel OWASP Did SonarQube email Argo saved. Argo, do all this as it is flow You also have to implement it in This is the worker node one I skipped Tivi SonarQube yes let's Confil Argo CD What is Argo CD.

It's a CD tool that is specific Kube works for deployment ctl create namespace Argo CD this where will I run Tell Cube Seattle Create namespace Argo CD ye Argo CD me install kahan. I will do this, I will do this KS cluster but its All the work will be done on the master machine. kubectl get namespace so the default name here is space If the cube node is leased, it is public and it is system then I'm installing the Argo CD here.

I'll create a new namespace. Argo CD name then kubectl create namespace Argo CD Ho gone apply Argo CD manifest now argo The CD project is GitHupP on it His manifest is all written down. Just apply it I applied twice and made sure that Also, your Argo CD's server will back up Service Deployed Pods all created at once Easy now you kubectl get pods do. Argo CD's to see that the pods become are they becoming or are they not becoming So can you see the application of Argo CD?

The controller is running Server is running Notification control is running Redis repo server is running everything can U C This Don pull now what do i have to do install argo.

[Jenkins and CI/CD Pipelines] Argo CD installation and setup

CD cel came Argo CD seal came Where is this Argo CD of mine? This is my Argo CD, it is gone. It is a UI application, if you want the same thing If you want to do it on CLI then you can do Argo CI. If you install it then copy it and sudo curl silent location silent here What is it about whatever gets downloaded? Things don't print on the screen will be Enter silently, it got installed, what is this? Argo CD CLI And we will give permission to that CLI.

That we can give him the CL of Argo CD. Can run the Now take a look at the services of Argo CD. Bar so all these services are then Argo CD Server is a cluster IP server Repo Server Cluster If you want to run your IP on port then What type of service should it be? If you want to run it on a port then What should be the type of service, only two People answered, it is correct, the note should be put but here.

All cluster IP is so cluster It will be to have an IP, they have no problem with it. Cluster IP but this let me out as someone If you want access to the port then Joe Argo CD server Its type should be node port sorry what does RB COD X mean. Permission in X means executable permissions usr local bin Argo CD here But we have created a tool called Argo CD. Downloaded it and gave it permission like If I write Argo CD like this Made it executable, it worked fine Now what do I do with that cluster IP?

Whatever type you want, just Argo I will patch it for the CD server. Whenever you update something So you update the whole thing when you patch something and you take a particular thing. If you add a patch to it, I Only to Argo CD servers I want to put the note so I took the patch and If you go here and update it, then kubectl. Patch Service Argo CD Server in the name Space Argo CD Patch Spec Note Make Pot Enter Argo CD Server Patch Now Cube Seattle Gate Service Argo CD.

[Jenkins and CI/CD Pipelines] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Argo CD installation and setup

What will my Argo CD server be on the node port started walking Argo CD started running on the node port What NAT port is the server running on? Which one matches with port number 80? Make mess gives If it makes a big deal, then port number 8 306 with It is attached 30169 then I will go Your instances But this is my node,. Node Which One Wanderlust Node This is the Wanderlust node This is its security group This is Wonder Plus Nut, how many people understand this?

Coming that my master pay Argo CD Master is not running So I'm here I'll go to the Wanderlust node on Security I'll go to the security group. In its bound rules I will go what to add I will give you this port, which port is this? 30169 What port of Argo CD is this? I will write Argo in the port description Put the CD Anyway Save two taxes Now go back and go back one more time and back Go now what do it go into it.

The group, which I will name EK This will do, if you make it yellow in colour and its IP address This is the port number 30169 tell me what should be run on this. HTTP is what it should run on Tell me this Argo CD Advance Go Proceed because we have The certificate is placed on this Argo CD should be Running How Many of You Can Install Argo. How many people will buy Argo CD? Install Setup I will do it, Atul will be able to do it, what about Others Main Cluster I created Argo CD on the cluster Installed Argo CD's Note Port Open Did you go to security groups and.

I opened the port and I'm just running it. It's over five Steps The initial username is Who is admin and what is the password? If you are an admin, you will have to enter your password first. How to find the password You will get kubectl Argo CD name Argo CD in Secrets of Space Removing the admin secret password of Will have to remove admin like for Jenkins. It was just like that, Dheeraj, so cube Removed password from CTL Argo CD I got the password and this password goes here.

[Jenkins and CI/CD Pipelines] Git and version control / Notifications and email

In this case see this case in this case Enter your password and sign in. Taken Easy Now on User Info Go and update your password. Throw it away because that weird password I don't remember Will enter any password, put it in and save the new password. Here comes the password Just got my Argo CD setup too. If you want to do it then go to settings Go to Repositories Go to Argo CD What Argo CD is a simple CD. The server resides on which Kubernetes from GitHub Maff comes with a file and a Kubernetes of Kubernetes If you update the cluster, I What are the things required in the ARG CD?

Repository will be required and admin Did you change the password? What I need is a repository. If you need it and you want a cluster, then How I connected the repository There are two ways to use HTTPS from Asus You can do it through ATPS also,? The default project name will be repository. The URL of your repository is now The git clone URL is this one This is the HTTPS URL and go and post it here. If your repository is private, Provide your username and personal access token There's nothing else to do, just connect and see Low Connection Successful How Many of You Understood That How to set up Argo CD.

How to attach your GitHub repository have to do gating I understand it, yes, good, let's go. Just go, let's go Next, set up your I Think Argo CD as well. Went, Argo CD is also setup. All your setup email notifications done This is also done, SMTP is also done Let us go Take a break now, setup Observability is almost done The remaining is the obligation, I will tell you at the end. Mohit replied given that Superb what happened is good Let's go great Have many, the break was for 5 minutes,.

From 48 Lake 53 Till once cluster IP to note bud Case explain f u get time c This is a service cluster IP type. Cluster IP so this is the I of the cluster now I can't access this, expose me. The application has to be done So I can edit this type as well.

[Jenkins and CI/CD Pipelines] Jenkins shared libraries

I can tell him patching so I Patched how to patch kubectl Patch Service Argo CD This service is called Argo CD Server. Patch it inside the namespace its type note put two All these cluster IPs are different. You might be seeing it but it is on the cluster. Is Let's read this Jenkins file How many of you have written this kind of James How many people have this type of junk file? Written it is written what is it What you do in this stage is build a pipeline.

You make me an agent and create stages. Give that code build scan push Great, you make the stages. Inside this, steps inside stages You write steps within correct stages You write this, see this with credentials Do you write steps or write all these things? If you need some command on any shell run some command or any tool So you write S correctly. Tell me now if I need to docker push to two places, meaning Pushing to my account at this stage and at the second stage some other account.

I have to push on this stage of mine account has to be done in the second stage and do it on the account So how many times can I do this push command? I'll write this one with credential image. How many times should I type push y if I have to go to two places But if you want to do it then do it on two different accounts. How many times will it come generally, ideally how many should come twice have to do 50 So 50 times simply but what if I tell you that this I will write the code inside a function.

I will call this function So what is the function, these are my lines of code. Doing wrong so what are you If you do this, you will make a function of this thing. Thing Whatever this thing is, make a function of it. This def means in Python here The call happens, you'll create a function. If you want to push to Purve's account then loop Will something happen to you again and again? You will have to provide some more variables. He went there Let's call this thing in Jenkins Library Shared library that you can use with the same code can be shared across multiple pipelines You will loop the same thing over and over again.

[Jenkins and CI/CD Pipelines] Git and version control / Dockerfile and image builds

Repeatedly rotate the pipeline through your same code Can be shared to multiple libraries Yes, how many people think that this thing You must have written five at least five times. For the project This docker push and docker build five times It will be written differently for different projects. It will be written different times what you should do You create a repository Create a repository and share a library Or whatever you like, this is made by Madhup. The repository is lion where you can do whatever you want.

Create a folder named verse like this In which you want to write whatever code you want. Checking out your code via GitHub If so, check out the code, become your function name. Here are the GitHub branches and what's inside. The syntax of that gvi is git url b this bean is the same thing that you see here I will get how many of you get it that this is what I got. URL here URL sent here I have sent a branch, I can do the same thing.

By using a shared library in which I will write code checkout and code checkout I will give you this URL and this branch. Is like this understand this My GitHub Here's a project to do. P There is a project here, two tier P There is a project here Node app here is a project Notes Jenkins file for each project Write down how many people are there for each project. Write the zenki file There is a Genie file for each project you write yes, in how many projects have you used the same Docker The build command is the same Docker push command that is run.

Same Docker build command Same GitHub clone Sem Docker Push beans Same Butt Different same thing, now whatever you do is same You are letting the dry do not repeat. Yourself How Many of You Have Repeated Yourself in the Last Two I have prayed, put your hand on your heart and swear I have done it many times, I have corrected it What are we learning now? Learning the DevOps Mindset Through Projects so what are we gonna do People here, we write these common things.

Let's create another repository for this.

[Jenkins and CI/CD Pipelines] Jenkins shared libraries / Dockerfile and image builds

Let's name it Shared Library or Jenkins Shared Library whatever you want to call it Shared Library and in this Let's name the function let's docker push Docker Push Dot Why Groove? Declarator pipelines r groovys Inside this we would write a def call What are the def calls and whatever you have inside it There are parameters and you will pass them late. What parameters are needed for Docker push now? I would have needed a certificate Need credential ID named Docker It requires a user of Docker Hub and Password needs to be correct yes or no.

I don't want a username and password, so what do I need? Did one of that things This simple example created a shared library. In whichever project I have Docker push Am I going to write this entire code? What am I going to do Tell me I'm going to do it, look. I'm gonna do this I'm late from Docker push Look what I'm doing, can you see This What Is This Docker Push what is this docker push it's function it shed library where.

I will name my project sorry image I gave the tag that was to be given, my user gave it. Gave the name Oh, now I can do what I need in that function. Can write and I'll call that function here. Let's get what I need Docker build has to be done in which docker build command and the user is docker user project The name is the tag of the image,

just look at it. What I do I want to get Docker built, I want to get the back end built as well.

I want to get the front end built as well. If I want to get it done then what am I saying, what is this? Docker Hub username of Atul Your Docker Hub what is the user name remember this I want Gaud Atul, if I put it here. If I give it to you, it will be fine, yes or no, just me You will have to enter credentials and that is a different matter. You have to tell Jenkins. You would have to tell Jenkins that's it pick this code wait I will tell you how to do it will be right here for you, where is it, Jenkins?

[Jenkins and CI/CD Pipelines] Git and version control / Jenkins shared libraries

I have to go in and tell Jenkins that, because pick up from here do this Go to Manage Jenkins and Search for Global Trusted Pipeline Library Global Trusted Pipeline Libraries means shared libraries Global. The library to use the pipeline should like This is this shade of lip, what should I do here? I will go to ManageJans P I'll go back to the system I will go and search for that thing here. Ten libraries will be trusted Meaning they can run without any restrictions, I will add the name of the library now.

What is the default version now If you have a branch then it is your shared branch. This is a library, this shed is a library, who is this? From branch p y main branch so I am here I will go and write it and Bus Modern SCM means this shared library will be picked up From which repository will git be lifted from? I'll wake up here, I'm wanderlusting repository or shared library I will put the repository where all this information is available.

Grubhi Syntax I'll go to the code and copy it. Here in the project repository I will put it in the shared library or understand it. I am coming, I am not coming is coming Yes, I will do just this much, that is enough. Save it Take the same if your private repository If it were there, look there, you have the option If this was your private repository So you are on the system Go to Trusted Pipeline by going here. You had an option to use GitHub credentials.

Insert how many of you understood that Did not put in public repository would have been private So it was love, that's why I let it go private. Getting It How Many of You Are Understanding Shared Libraries, let's use the shared library. There will be a repository with that name in Jenkins. Added everything inside that repository shared that code into this pipeline attach it with Just like importing in Python How cool is this for a junkie advance? Session Mast Now tell me how many people have their Shared library at the office can implement Will use it in many people's offices They might not know much about it Not being used in everyone's office.

[Jenkins and CI/CD Pipelines] Git and version control

There was that, his meme had come which I used to say I am and I do not do what I do not say I don't do that either, I don't do anything. Pipeline Agent Label Node Tell Me Now If I'll run it through another agent. Let me insert the label node But if I run it on my own node So what to put here? Will have to what will have to be put here Annie Fine environment sonar home tool sonar now me Can anyone tell me the S capital of this tool Sonar ONR Why I wrote this.

Is the tool Sonar y why did I write Why is it written because if you if you go to your manage Jenkins and if you On tools. If you go and see, you will see the sonar Cube, you have named this tool Sonar, so I understand that Why did you use the name Sonar S Capital? Why did you use how many yet this is coming has come Right front end docker tags like I mentioned It was that in our project, this Front End This is the back end.

The back end image is also updated. There will be a front end image someday. 3.7 needs to be built on version 4.2 If you want to build then I can directly build it here. A version that can insert a tug will be built tomorrow interesting haya Mast, whatever version I put, that build will be will go and now stages of parameters See what the validate parameter is doing. The pipeline will penetrate Again tell me docker tag tell me what is docker tag what is a docker Tag What is a Docker Tag Look at this very carefully see this My repository is right.

If I clone this repository Lu and run inside this repository I have a front end and a back end. CD went to the back end me and if I did docker Build High Fun Name the image. Lust of the back end Son, whatever you write after putting a colon in front of it yes what is that what do you call it are tags speak again Tell that if I have a particular. If you want to use this version of the image then I will use the front I'll use its Docker tag for the end.

[Jenkins and CI/CD Pipelines] Dockerfile and image builds

Docker backend for images with backends I will use the tag if I understand it. I do it here version 2.1 Docker Build -T Wanderlust Version 2.1 and Dot What version of my build is it now? Is 2.1 so here I am going to put the back end docker tag what will I put 2.1 I did this manually I build this from the Jenkins side I will build it automatically. If you take it then no version is used here I am waiting, these are the parameters,.

I pipeline I will create the through parameters of the pipeline I will put it in right now, I will just give it to you explaining I am fine Right now I'm just explaining. What is this Docker tag to you my Docker what is a tag Now whenever you do a docker build, here But there is a tag for the image, that is a tag. That's, I understand how many of you get This means that if the front end image is built To do this, run on a particular version.

I forgot about you, the latest one So let's get the latest image build from me. If I have to, I should put the latest one there. I will give it to you, if you want to build it then put V2 Give Done, what are these parameters, we ask the user will take ParametersValidateParameterBrotherIf If the parameters are empty then your. The pipeline stops and the build fails if Why are my parameters empty? Because the name of the Docker build image is preceded by I have to add some tag, yes or No tag has to be entered if yes and no if no.

If you put it, what will happen, that latest You will pick it up but it will be a mess here. What am I doing if my tags If not, I would have caused the pipeline to fail. I'm fine if there's no tax on the pipeline. I've validated the parameters, how many of them you get. Keep telling me this quickly in chat so that So that I can go to the next stage. The validate parameter is just checking if the user parameters that were front end Docker tags back end Docker tug whether it is inserted or not Enter the workspace cleanup script.

[Jenkins and CI/CD Pipelines] Code quality and security scanning (SonarQube, OWASP, Trivy) / Git and version control

WS Is this shed library Is No Clean WS Jenkins one Has an inbuilt function Look how I ticked it,? He cleans it up just fine Clean WS deletes the workspace When Build Is Done Once your build is done go So clean it to save storage. I accidentally came across something that Jenkins had done. If something accidentally enters the directory, Clean win, Keaton, all that stuff. Stage git code checkout what is this, I have put this code inside my checkout. This URL and this branch only with this code Check out this shade How the code got Call git url and pass the url branch Pass Jingle Ala How Many Of You Understood Code Check.

That allows me to complete the git clone No need to write a complete git clone of I wrote that code in a shared library. DVD scanned and kept in the library where the DVD file system is scanned It does it right o Wasp Dependency Check o Wasp Dependency Check What it does OWASP Dependency What is Junk Shed Plumbing Shed Library. What is Ovaz Dependency where there is a keyword Dependency checks that your current OWASP in directoryentry installation Where did this come from?

Has gone beautiful Sundar dot sfer current directory yes I said the same how many of you understood this Stage S Well, if there is any problem with the current directory. No, he is doing great Dega Stage SonarQube Code Alice Where Shade named SonarQube Analysis Have you used the Sonar library? What is the name of the hat tool Wanderlust of the project Name is fine, now I have to read it further, Sonar Go to Cube Analysis's Shade Library And read that with Sonar.

Environment SonarQube API which is also You have also mentioned the name of that sonar. Scan with the sonar tool you had at home. For the project name and for the project key If you want, scan by the name Wanderlust Look how many of you are getting it that I. Which is hard work and I put it here and I just call that thing here doing It did not come Code with SonarQube environment means what you will scan it through the SonarQube.

For this you need SonarQube API what is SonarQube API It's Just a Variable Name If I had written Mask Ali here, he would also have Variable name becomes but what is pro main.

[Jenkins and CI/CD Pipelines] Code quality and security scanning (SonarQube, OWASP, Trivy)

What is being passed to this variable name? What is being passed to this variable name The sonar is passing and what is it? Sonar this S capital O N A R what was this There was a tool that I set up and gave it to you so that it is available on SonarQube server. He will use the sonar scanner to It will scan the project and run this thing. It seems and It's these things are meant to be that you every Checking quality at one stage Checking dependencies File system check How many of you are you doing it on every stage.

How to scan sonar [music] How do sonar scans do all this? Of T here is the mill ha how many of you have done this is such this big The script is whatever the script is. Such a quality get time out wait for Quality gate so this is pre defined The syntax I externalized I gave it and here I injected it. Copy and paste this from c this SonarQube quality is this ya wait For quality gate that you don't understand This is a junkie Syntax is Here it is with SonarQube environment.

You picked this up from Jenkins' documentation. I have to read one thing, Cha is Noon S DevSecOps With Jenkins How Many of You have completed How many people have done this? Back End Environment Setup Front End Element what is this setup what is it This Asu what is this Consider it gone that this is my project Lust is its back end. I'm going to show it to you via Docker Compose. Explains so VM Docker Compose look at this How many services are there in the project?

DB Back end front And how many services are there in Redis? Total four services in this project The name of the mongo DI container is mongo service Image Name Mango Latest Volumes Back It has picked up volume from the end data and expose this port number. How many people got this service cleared? That a MongoDB container from this service becoming Is Now back And container name back and build dot / back And this line has no meaning, this line has no meaning.

Everyone is bronze, what does its build mean?

[Jenkins and CI/CD Pipelines] Dockerfile and image builds

When do you build in docker Someone said it right, wait, wait, someone said it Khushboo is right, my Dockerfile is inside the back end folder this is the meaning of this file This Line. When you build the back end container Yes, the location of its Dockerfile is Let's read this big ARG CD now. What is an environment file? Back and DAV Dader what does it mean what does it mean it means back Finally, an environment file The IP address in that environment file is The port number is related to its environment.

There are specific things if I If I run it, my IP address will be different. How many people understand If it comes, then this back end folder has a dot in it. Invi docker is some variable like MongoDB Key URL: Redis Key URL: Port: front end url this cookie this something There are some environment variables that affect this written in the file So what is this Jenkins stage? Does Update back end What is the environment doing and updating it? The back end environment is designed to Tell me what is this IP address?

1822 4.72 54 what is this what is this IP Whose address is the public IP address of Etoo? IP address Now mine The node of this cluster What is an IP address 13.5 22417 where I want to run it IP address is this is this IP address and This IP address is matching Is. Otherwise, if I build it Let me docker build it So this It will not work, it will not work That's why there is something known where is the age That's why there is something known as a.

DirectoryEntry Updated front end New so if I have to understand the back end environment A script to update the variable If it is written then look at it, VIM update back end. New where you just need an instance ID I have to put it in, I mean let's give it to me, master. Node is mine what was the name of my master node It was automated, this is the master node, its IP The address is 18.10 2 39212 this IP address is on my side have to put.

Address Updated back and new I'll go to this instance ID.

[Jenkins and CI/CD Pipelines] Dockerfile and image builds (part 2)

This will copy the ID of this instance and copy the instance ID and I'll go here and paste it, so instance What will be obtained from the ID? The ID of the instance Get the IP address of that Situ instance from and I'll go to his back end folder I will go into the file containing the ghusuk and from sed to sed. How many people know what this is Said what is said Stream editor so you can edit any file If you can edit it, I will go in and edit it.

Going into the particular back end element Docker Replace my previous IP address. Tax I will give you how many of you got this Automation I entered the instance ID Instance Key ID will not change, IP address will change. The nun who used to be this, now she's become that. I will have to see why she became a nun. Instead of what it was before, your IP will be The address will come up Why is line number four in the set command?

Wrong instance ID in the set command Have you picked up the batch seven server? The server is down from the batch, poor thing. That's why when that problem occurred, I was VIM Updated back and new Go and update me on whatever IP address to do it like this Let's run this Wonder Less Node. Let's run it on Wonder Less node. This is the IP address of the Wanderlust Note 13.5 I copy its I instance ID did paste here escape call dub k dot / date back and new updated git Diff 13.5 Went Easy.

Easy Same Automation VIM I'll go into the updated front end DS and Update the instance Give No, you don't have to provide the instance ID. Watch Update Back & New Watch Very Attention listen from instance id is a variable where You need to provide the instance ID. What is an IP address? It's again a variable that In AWS to describe instances If you give the instance ID, you can You can query and get the public address. You can get the public IP address.

I got the IP address, now what do I do? If it is there then leave it, if it is not there then find the file and update with the new IP two what class are you in I mean, I'll teach you shell scripting again.

[Jenkins and CI/CD Pipelines] Dockerfile and image builds (part 3)

I'm feeling strange because Look here it is written which class do you study in At this level you Otherwise you are going right to DevOps you If you need to go, look carefully. Everyone understands what is The Use of This Script Environment File Is it to change the IP address? Escape Colon W K Enter How Many of You Group This is what this script does for you. Changing the IP address of the environment Let's get it simple, come back to your Jenkins pays how many of you think that back To set up the end environment and To set up the front end environment.

I have provided the script you need here. Parallel is written because you are doing two jobs You can also run these in parallel if you remember. GitLab I taught you if you have three If you want to run something parallelly in the environment You can write it in parallel to parallel The task is to execute both at the same time. You will be just to save time getting It will be running in two stages simultaneously. Is the Docker build images directory back end docker build This image name is whatever you want to use with docker.

You will enter the tag, this is your username front end docker build this image name whatever you tag How many people will you explain this to? I need what is Docker build what is docker build you me Tell me what is docker build If it is built, I also know that but. Is this thing this Govi syntax here came from docker build which executes the docker build command runs internally and whatever the username, whatever Project name, whatever is included takes it shared Library is shared code.

Push to docker h docker push again tus How many people understand this part of the shared code? Here comes Docker Build Damage and Docker push yes there is a thing docker to push what about yourself what do we need to push Let's get the credentials build so on Docker Hub Go to Docker Hub. After that go to my account settings and here on the personal access token Go and generate a new token here. Please name the token something Batch 7 Mega You also like read write delete Give access and generate Two, here comes your personal access token.

[Jenkins and CI/CD Pipelines] Jenkins security, credentials and access control / CI/CD concepts

Copy and paste it somewhere, it's done Tell me where you have to give the token. Where to give is absolutely in the credentials Enter correct credentials and go to Global On credentials Let me go and see what his name is. Copied the token ID, you do the description Docker, please give me this ID Docker Why did you give Why is it because in a shared library If you go and look at docker push, the username. Happened is Docker written I understand what he will do now.

Credentials wherever it is Once I get the credentials, I will enter my password. What am I What will come in the password Docker Hub The password will come. I checked my Docker credentials put it in Jenkins so that it can do this thing Correct How did you do all this manually, it was so easy,. That's why I named it Mega Project. That you have to do it all at once Make a project then you can use it understand I come in Gone is the push to Docker.

Success if this pipeline is successful. What If it happens then whatever artfed is there will be saved. A build job will be done Wander Less CD A new build will be triggered so how to trigger New Build You Just Write The Keyword Build and the job name and whatever your parameters are. That's fine how many of you understood This is how many of you understood this line pipe What percentage of Indians will get this pipeline? I understood it's a big pipeline Corporate pipeline butt understanding.

Is the group copy it, do the last build, last build I mean, I'll tell you there's another job. Joe CD Job There is a continuous delivery job because here But anywhere you see Kubernetes Anything is mentioned Is there any mention of Kubernetes or anyone else anywhere? No mention of deploy or docker run or anything. CI is just Continuous Integration. To Check If My Quality Checks Are Past My images are build CI job this is a The bar ends only if it is successful.

CD How Many of You Got This is if this success happens then only CD CE Happens Then Only CD Will Happen Understand, let's go Just copy this who do you go to?

[Jenkins and CI/CD Pipelines] Jenkins shared libraries

Create a new pipeline which are named Wanderlust CI because CI is pipelined I would have done are this description this is the ci pipeline of Wanderlust CE Pipeline is Old bits are discarded because. We are saving storage, keep it for one day only. Keep two builds to the max from it Don't keep too much or take some 36 Create the build and then upload the screenshots. Is this the hard work of how many of you Adding screenshot of scene t 50 60 build There is success in the last one so it Good, obviously it's good, but it's not storage.

Efficient so B storage Efficient now just want to in git project You can enter the URL of your project by entering yes I also entered the URL of this project here. If you want automated, you can put SCM but you can parameterize it. If you want to enter build parameters then you will have to manually insert atlist and tags If you fall then there will be no SCM will pipeline script this is pipeline. Copied the script and pasted it here How many people were given the entire pipeline?

I understood the script how many of you have understood this pipeline script and here just do any Two because we didn't use nodes. People should understand the entire script. Came Gone 50 No, how many lines should the script be? How many lines would this script have been? Please do a project with 500 lines Jenkins Pipeline Does You Think It Makes Sense, how many people are feeling that yes I was also in that category who said that. Make a 500 line junk How Many of You understood in your heart that yes I only I did not eat it, it is a different matter but someone else did.

I told you, so I am telling you that this Cheese does not need ina pasan The need for efficiency is correct you need efficiency in your Pipeline No Lines of Code Save have it done. Overall, we have to focus on quality and not The Quantity is Done CE Pipeline I also started looking at the CD pipeline. Hand CD Pipeline Take So let's go to Gittops to get into Git After that a junk file is kept again shared Library Agent, I'll change it.

[Jenkins and CI/CD Pipelines] CI/CD concepts

Parameters: How many people clear this stage? Have a clean workspace Clear Checkout Code Clearly, this is just to print the Docker tax. That I can see on the console that Which version are you running and that's it Docker image tags are done, this is also done. Clear Update Kubernetes Manifest what is it Now tell me what is the function of Argo CD? What is the work of Argo CD someone me Can you tell me how much I Explained according to Argo CD what is the work of.

Let me show you a diagram so that it is a little easier Easy Read Argo CD work case file update It is not to be done Colors are the work of Argo CD from GitHub. Taking the manifest file to the cluster Deploy your manifest files Updated is that how we Will Deploy Depends on the Manifest. File If We Do a Continuous Delivery CD of manifest files and Argo CD no manifest file to take I have been deployed, I have just been automated by someone.

You have to update it in the same way as this update Argo CD will pull its files He will do it and send it to my cluster. Kubernetes will tell me what I am doing here. Updating the Kubernetes manifest Inside That Said Wanderlust Back End Beta Change the tag to back end dull What is the back end? I tell you dull I mean, when you look at the Kubernetes inside it You will see that DIR Kubernetes means Kubernetes in a directory named nits Enter the directory named Kubernetes.

If you need this version update in the backend have to do So whatever your image name is in the back end is whatever tag you send into the CI pipeline That tag is back end dal update. If I have version two in CI Dala then Gittops Practices through Kubernetes Which version of the depl in the nits cluster Gotta tell me version two The deploy pipeline builds that version. She will push it and call the CD Kubernetes. The old version that was in the Nity's manifest Update it and put version 2 Argo CD will pick up that version two and deployment End Simple understood how many people.

[Jenkins and CI/CD Pipelines] Code quality and security scanning (SonarQube, OWASP, Trivy)

The Hum Log version came into the CI pipeline. Will put from lats to that version of tv through OWASP through Scan through sonar That version will be built and scanned How many people understand the CI pipeline? How many people understand this CI pipeline? Has arrived First Kapil, you understand the CI pipeline. I What is a CI pipeline and what does it do to you? Takes a version as a parameter Builds, scans, and builds on Docker Only the one who pushes his image builds, scans, and Docker Once pushed to Docker When I went here, I triggered the CI job.

Which will save the version of the Docker image on Git Argo will update that updated version The CD will have bridges and the Argo CD Kub On Kubernetes Why is this CD a deployment pipeline? The pipeline is because Kubernetes Where did this go? This Kubernetes The manifest file is back In that, the wanderlust back and beta, this The image is its version, that's what matters Will update its version which CI. The version you put in the pipeline and the CD The pipeline will update GitHub and if git updates to hup then Argo CD cluster Deployed in your own lines in your own.

Words Explain What CI is Doing And tell me what the CD is doing, explain it. What will it do after that and now git will push hop p whatever it is it will push GitHub Fifth note of musical scale Tell me, I didn't understand the version part. No, what you understood write a line I did not understand the part about Kunal's weight. Write whatever you understand and not what you don't understand. I will explain it later but what I understood right now So tell me what you understand Is What CI does right now is build images.

Everyone, it's significant B You what is the use of CI and CD tell pipeline ci build cd update end There are only two people in the class and the rest are. Let Me See How Many People Are There are more than 100 people but only two People Are Saying Ci Taking the Code CD Delivering CI Docker Push DevSecOps of CD Deployed CI git clone dependency check.

[Jenkins and CI/CD Pipelines] Git and version control

Code Quality CD Update Via Argo CD No I just used the CD pipeline asked in this pipeline Argo CD these words There is no What this pipeline does is just so much to me. Doing deployment Palay wrong this pipeline Rohin is updating the Deployed Hub. CD Replace environment variable incorrect Environment variables are replaced in the CD not only is the environment variable we use in CI were replacing Ojas Tejas Making Manifest File to be Used. In Argo CD is the answer I was Expecting Tejas is preparing the manifest files that the version received from CI Joe Tag from CI I got that version, we are getting it ready.

GitHub, so here we just update You are doing bad things even if you read and say I would be happy even if you don't give it to me It goes here and only updates are happening. Has manifest files What happens once the files are updated? What would happen then the code is updated and pushed to git. How to get user with credentials How many people remember the name and password? Is This GitHub credit how many of you remember what this git Hub He will bring the credit from me and take the token.

What will commit the element variable Will update and push to my repository But, push to my repository. He will finish me once he pushes me Kubernetes manifest file updated Then on to Argo CD. Handy git tool name normal git Normal this is now once this is done and My updated code is in git up Then came the Argo CD, bringing it with its manifest. Deployment is simple with GitHub Getting the manifest file onto the cluster deployment is not replicating because only Delivering the code, take this is not found anywhere in Deployed Agra.

It is written that Jenkins is going to eeks I wrote Argo CD is taking the git Repo and then going to ES Jenkins code and Update all versions in the GitHub repository will put and Git Operations. Argo will hen it is known as Wt group Ops now by making the mistakes of scolding the scolding How many of you understood the real definition Off Git Ops real definition of GitOps not yet.

[Jenkins and CI/CD Pipelines] Notifications and email

Will you forget what is GitOps I am In which the code is basically pulled from GitHub Repo and GitLab Repo and Thrive Repo Server Tools like Argo CD directly rely on. The Git Code Not Any Pipeline No One Else And the operation of that thing is just here and Only two people working with your git and Argo CD on the plate. This is the work of giving to us by putting it in our hands. Did they Chacha Give Big Round of Applause for Chacha Y What is their job?

Updated tags y I understand it How many of you I understood How many people have understood this far? Updates and the manifest file simply git will put it on hup and my Argo CD So connected to GitHub from here Deployed Wanderlust Application Has Been Updated End Deployed Wander Less Application. Nishan Wanderlust application has been updated And deploy when it will come it will come when My Pipeline Success It will happen, but what if my pipeline fails? I will also get an email from my about wanderlust.

Application Build Feld in understanding I asked what is the role of DevOps in the pipeline So she should go, she will go but if It has failed so what If it happens then it will go. No, how many of you understood, tell me Gave Come on, copy it, that's all there is and this By copying I create a new pipeline in Jenkins would have made new item. Pipeline Wanderlust CD Watch Now CD Capital, why did I write the name like this?

Wanderlust CD because the CI pipeline If this is mentioned in the build job name then this name Mention I did not understand the CI pipeline In this junk file, CI, and If this pipeline is successful then. A promoted build of the CD. To trigger it so it does Now it is simple in this also this is the CD of Discard Wanderlust Old Build Keep one day's build, keep two builds, max two. In Max and pipeline script this is my script And I have kept the agent on a single server.

How many of you understood How many people understand each stage? I What Steps Will Thorets Manifest? The script will run where the name Kubernetes is written.

[Jenkins and CI/CD Pipelines] Notifications and email (part 2)

Directory entry of [music] Gagi entered inside her, what will she do next? Will edit Wanderlust back and beta Whatever tag is inside, it is used in the back end. If you update it in DL then the back end will be DML. Updated the name of Wonder Less Back & Beta in Tax I will give it Done Save it Tell Lou or email guy if this pipeline Success is success if this pipeline is Success. Email text attached log what does this mean happens in your email which is part of the pipeline There are logos, they will get attached, how cool is it That the log file is being attached automatically.

If it is successful and whatever the build number in it Even if there is a failure, the mail will be sent. Build that Wanderlust application with logs Fay understands There is a template, it can be created quickly using GPT. Save now Look what has happened to you, Varal CE Vandal CD both become correct yes or no both become paid But this is the Argo CD deployment CD deployment Your Y in Argo CD application. This cluster, this one, in which this Argo CD is a two-node cluster.

If you have to connect it together then you can connect it to Argo CD. Go to Settings > Clusters Go to Kabar Niti Default near it's default Does it have a cluster of wanderlust? It will deploy if the cluster is yes or no. I had to create a cluster, cluster add I will have to tell you how they do the ad. Tells So first of all you I have to log in to Argo CD. If you have to login to Argo CD then This is the URL of Argo Sady 1358 This is the URL to log in through.

It happens with your username admin whatever Log in with your username Enter Certificate error because No HTTPS, will you proceed? It was for Argo CD. The server password is Enter admin logged in successfully context Updated login then my Argo CD The CL that has come is with my server I am logged in now what do I do? I will do it now Argo CD cluster list I would look at Which clusters do I have?

[Jenkins and CI/CD Pipelines] Terraform and infrastructure as code

I have the default cluster. If I want to put a cluster of Wanderlust then what should I do? I will name the cluster of Wanderlust first I will take it then kubectl Gate Context Whose Cube Seattle's Joe I will get the context from it. That I get the name of my cluster Current Context Astic Name Terraform Admin to Wanderlust to Ked Come How Many of how to know your cluster How to find out the name of. Do you understand how to find out the name of the cluster?

From kubectl conf Context Take will get the name of Take Taken Now Argo CD Cluster D Now this is my cluster I want to add him and give him his name. What wise cluster of lust whatever EKS whatever me I can enter the name you want. Of you understood This fire CD doesn't know where I am But I logged into Deploy and gave it my I am naming the cluster It is understood Aya enter do you want to create a service.

Account for Argo CD Manager Yes Now Look what that service account is doing. Argo CD Create Cluster Role Argo CD CreateDrawAnaNavigator CuttedCreatedDocumentsFragment How can a booty access the cluster? A role called Clear Argo CD Manager built into your kube-system and Gave him access so that he could deploy now if you go over here and Refresh If you do, you will see. Your cluster will be created, can you see it? This Your Wanderlust Cluster Is Now added how many of You Got to Know How to Add a Cluster in Argo CD How to do a cluster in Argo CD How many of you are getting value from.

This session is a heavy session, but once If I apply for this session then I will be here for all the years and months to come It's no year, invest your time in new things. Learning Implement Loonga How Many Of You're getting value and how many people still I am full of energy, it has been three hours. Who keep that in four hours of class He was sitting and he learned everything, Javed Khushboo Rohan Good job Rohan you have given all the answers.

No matter how many times you gave, three or four times you chatted would have done the right thing Gave Sapna join what a thing so it's good Ojas Super.

[Jenkins and CI/CD Pipelines] Argo CD applications and sync / Git and version control

Let's do application what is an application If you add the application you will get Argo CD's through pass deployment New Let's create a new app called this application Let's put wanderlust cool small name. Keep the Wanderlust and Project Name as default This is the sync policy, that's the whole game here. I have automated it so it will be yours syncs with git up How cool is that you tell Jenkins Get up updated Get up updated Argo CD has launched Kubernetes should prune resources meaning if something.

Delete old resources yourself The heel automatically picks up the old one Take a selfie and auto create name Space, if I have any application, its name Space created file. What kind of people are they now, I'm teaching them. Why not keep it because it is going on I understand that this is a Application created Wander List Project Name Default Sync Policy Automatically, automatically I'm syncing. I am feeling the sale of my code cluster if an application accidentally breaks If it happens then the image from the old cache will be played She is using it.

Selfie, a T-shirt, auto create here too namespace here if your namespace If there is no Kabj in the manifest file then this It should be auto created, now let it remain. One that I already kept connected so drop seen in down Revision means branch is mine, main this put the main Path Kubernetes's manifest file is on which path It is kept in this repository. Tell me in the repository where is the file of Kubernetes The name Kubernetes is kept in Kubernetes in the path from in the folder of so I will say in it Kubernetes Nits done.

Which cluster should I be on? Which namespace to deploy to? Wanderlust namespace but it's Now it's done for me Tell me my application whenever Which port is deployed to? When my application will run So on which port will the back end run this file What do you think when you see Kubernetes On which port will the application run? Chalega back and low everything Wrong I am so proud of you this happiness I might need Meditation I Might Need Uday by Uday I Don't meet Uday Bamal I mean control Uday.

[Jenkins and CI/CD Pipelines] AWS (EC2, VPC, IAM, S3, ECR, EKS)

I just said 8080 despite saying so much means next level Pay Legend Kubernetes Pay Node Port Pay You run something on service and if you The thing has to be accessed The Target. Port is what's inside the What is a Pod Node Port? Tell me outside how many people understood it How many people feel that this session is needed? Tell me which front end in front end The port must be running on the net. This is Who This namespace contains the selector front and is associated with deployment Which ports should be exposed?

Karegi Sabko T-Shirt Chahiye Good One Sab T-Shirt I am able to understand how many Of you can revise the Kubernetes end to end project session where we A deployment was created in which notes were kept. The port number was 8000 and now it has 50 pods. It could be port 8000 for 50 pods Where will we reach then all of the pod's Deployed with a service Binders' node strips are exposed So that exposed note pot will become 3100. This 3100 which is the note pot of the Wanderlust node I went to the security group and attached it.

In whom will I give the security of Wanderlust This is a group, whose node is this? This is my If there is a node of the head then this 31000 is this front end I will add it after doing P and what was the back end? Because I need you to be this cool This thing is to show that the back end is running so done so back end P Annie Where Save Rules Be Went Done. How many of you understood that Kubernetes If you have leftovers on Nits, now leave it, I will I am not explaining how many people understand I came to know that there is no one for Kubernetes.

Why use node port for service? Note ports are generally above 30000 If the port remains then all the people He would have said 8000 even by mistake in this answer. You were asked by the interviewer Kubernetes Which port does the application run on? Here Tak All Good Everyone Cool Great node port is use for external Pats Correct Yes, create it. Application should I create this application which I made one called Wanderlust, which I am putting the source here and its I deploy to this destination Argo CD only needs two things.

[Jenkins and CI/CD Pipelines] Monitoring with Prometheus and Grafana / Code quality and security scanning (SonarQube, OWASP, Trivy)

We need a source and a destination. It happens, it creates am yes have gone It's gone, that's it was Went Now let's start the CI/CD pipeline. How many of You Want Me to Run a Sea CD Pipeline for this will come bugs Will come too I should definitely come because for the first time When you build. You will do it when you build it for the first time. Or no, if not then what will come in the if condition? It was written in this condition that if there is an error then go here.

Wander less CI error dekh lo kaisa Looks now build with parameters Click on the Docker tag of the front end What is the front end Docker stack needed for? Needed Tell me what docker tack v2 back end is needed tell me put this also v2 put on build Two Now this OVASP is for dependency check downloads some files for which. It takes you 20 minutes only for the first Time Gating It OWASP takes 20 minutes for the first time I understand what that is I mean, let me pass 20 minutes of time.

Y now Don't throw the parcel box, we are with me Why is the box for this chair still there? I don't know it's Don't ask why, now watch How to Monitor Your ES Cluster and Your Kubernetes Components and Workloads Using Promg Fana, first of all understand this thing that. Helm is a package manager so how many of You remember when we were Grafna and We were installing Prometheus. Through Docker Compose and every time we People scrap in Prometheus Dame Confident because our Docker Using Sea Advisor for application.

Node Exporter for the node exporter were using it for someone else and used it for something else were doing but if I know that EKS What This ES is a Kubar Kubernetes Cluster Service by amazonbusiness.in [music]. Why edit again and again yes and no when I have to Know in a namespace of Kubernetes There is a kube-system and in the kube-system everything The system monitor is the API server. The API server handles all connections. System Proxy kube-proxy Network If he sees things going out If I know all this, then do I Prometheus YAML I'll put all this confine already, yes.

[Jenkins and CI/CD Pipelines] Helm and charts / Monitoring with Prometheus and Grafana

Or not if I know what that confine is Yes, I will already put it correct, so Helm In this Prometheus's confine already nugget Here is the promise for this case in Helm. The value of is already entered, so what should I do? I will go and first look at the Helm chart. The tool is Helm, install it first. I have already installed it, It's been get held as it is so get held S is installed if you cut it and get it If you go to Held S and see Helm.

Wrote the install script I haven't seen the script explained yet No one will ask why this How the script works is up to you. Run the script and install Helm yes It will be fine, you just need to run the script. Will do and install Dega how many of you can create such Script to Install Jenkins Docker Kubernetes Tell me how many people can make such a script. Which has shal script and only in that The command is written, thank you How many of you can do You will make this, I think it is Madhup's one Maybe someone from Madhup So a repository.

Gone is the Dev tool He did all the installation here. Script prepared so how many of You can do something like This is a good way to contribute to this Repository to install Helm Chart I did a butt here on Argo CD's. How big did you write for How many of you can do something like that You can make this project open source. Script Get Helm Installed Now Helm Repo d stable this stable a repository Where all the stable applications are there His list come fine stable Now in that I am Prometheus Community, this.

I have a Helm chart and am going to install it. With this Helm repo add Prometheus Community What's happening to me? LM Repo I have a list of Prometheus's helpers. The charts are here, now if you look at this If you go to the

Helm charts of Prometheus All the Prometheus codes, whatever code you have As Now I am not going to explain all this. I'm going to do it, there's no need to do it as much as There is also code, all that is written If you need to connect Prometheus to Redis.

[Jenkins and CI/CD Pipelines] Monitoring with Prometheus and Grafana / Helm and charts

If it is there then what will happen for this, all that is also written It happened that someone was asking me, isn't it node? Given Redis Exporter RabbitM What is Node Exporter? Node Exporter Everything is given in it, so it is open source. Contributions are what people need to do It should be automated and easy, so people If you contribute, what about yourself now? If all this stuff has arrived, then do I need it? To another node exporter and sea advisor and anything No, because everything is Helm Chart.

How many of you got this him what is a chart kube manifest file and all the values are co one Click Install I Got It Getting It. What do I have to do, I just have to Start installing the chart So first Prometheus creates the namespace The name Prometheus became the namespace of the name Key kubectl Get namespace A Name. I lift it into space to see Whether my promise was made or not, my promise was made. Did anyone notice that Wanderlust Also made is Did We Create Wanderlust manually Did we create kubectl get?

Who did not namespace who made it Jenkins did not make it, Argo CD was made. Jenkins doesn't have that much access. Helm install stable Prometheus install community Prometheus All installed now by tap Will go to hell install Prometheus Community In the Name Prometheus is big so it takes time. A little more similar to how you went to prom I installed it, you have to install it. If you want to do it then can you see Prometheus? Deployment Prometheus Bun If you're Prometheus's If you look at the parts of Prometheus Running Yes and No Prometheus Running How Much It was easy, why did we spend three hours in it?

In this you took three minutes Told you to understand this in three minutes So it took three hours,? How many of you are understanding this Promise I installed it so now I will see the service of Prometheus because I need to access it too Neither Prometheus's where is the server. All those cluster IPs on what I cluster I can access the thing on the IP What will I do if I can't do it?

[Jenkins and CI/CD Pipelines] Monitoring with Prometheus and Grafana / AWS (EC2, VPC, IAM, S3, ECR, EKS)

Stable Prometheus who is to his service edit it I'll give it to you, which is the stable Prometheus I will edit his service. I will give you a patch and I can also do it. By doing this here if you want load balancer you can do a load balancer here if you want. Port so you can do nut port Escape colon null key enter Prometheus service has been updated Similarly, you have to verify That Prometheus service update. The service Prometheus got updated 3245 This note is ported similar to yours Grafana service should also be updated.

Update the service of Grafana also Give it to me and I'll do it here. Gaffa Notep Updated now Grafana again See the service lo to not port not pot grafna so when you When you install Prom, Grafana is inside it. This Now 3094 and what is the port number here 3245 Let's go Which one should I go in? In the group I will go to the master node or I will In the security group of this cluster owner I will go Tell node cluster ones very good let'S3 You are coming right after an hour.

What was 3245 Prometheus then 325 Prometheus Anyway, what else will I add? 3949 No 309 49 Grafana Anywhere Save Rules Now add a new in this case. I will give where I will give this IP address Which port should I use on this IP address? I will drive Port of Tell Prometheus 3245 Enter and your Prometheus set Don Cool Prometheus setup on the cluster. Open a new like this and here you will see what to do of Grafana 3949 Grafana enter boom Grafana also installed i what did you tell email username will be admin but.

You will have to pick it up from the gate secret Grafana password will have to be lifted so prom The operator password is its prom operator password its never Save how many people have set up. I have taken it, I haven't done it yet, but how many people understood that an ekes cluster How is Prometheus and Graf set up? I haven't seen Prometheus and YAML anywhere. Edit What is a promised dull in which you haven't set up scrap targets yet, have you?

[Jenkins and CI/CD Pipelines] Code quality and security scanning (SonarQube, OWASP, Trivy)

You go to the status target Go can you see as many Kubernetes as you can The target is everyone has attached themselves. The pipeline is still 17 You could have extended it for a little more, three minutes. Any talk how many of You Are you finding this beneficial for me today? How many of you think for this you can do a Simple small task, first of all you have to do this thing to be implemented how many of you Think They Can Implement This Project.

Address I don't know this at first build run Went And, let's take as much time as it takes. Wait, let's start with the SonarQube code. Gone are his bugs onebill code smells all Things checked out Is Getting Docker Build Images What does it take to push to Docker Hub? If pushed What I have my phone yes can I get it on this phone Email I don't know whether he will come or not, but let's Hope for the Best, I am enjoying seeing this pipeline.

You are doing CI here and further Go and CD too You will see that it takes time to build. First time because it was completely from scratch If I am building then time in the first time takes It is the totally summer and most awaited event of batch. Sen, I told you, you people don't you believe it show the console output O Like who knows you are sleeping inside We need to see the console output, can you do that? What are you doing inside then console output I can see that whatever clean up happened earlier Then came the repository, then the TV file system.

There was a forest bill, it was detected, whatever way S has high vulnerability, now to whom will all this report go, developer? To All this will go Happened OWASP check files downloaded then Processed after download to be After dependency check Analysis has started, fine analysis has started Happened Collecting Artifact Passing File Dependency check report generated SonarQube Environment. Woke up again this is the SonarQube server 9000 Go to that and make this thing, this thing of wanderlust project and then all these things happen Python code quality gate seen in Go.

[Jenkins and CI/CD Pipelines] Dockerfile and image builds / Code quality and security scanning (SonarQube, OWASP, Trivy)

All the codes and quality check of all the codes happened again Whatever the report was what then became Every line, every file was checked. The analysis has been successful and the report has been sent. Created a SonarQube task in SonarQube Quality Gate came and passed Automation file ran in Wonder Lass CI Environment Update Happened [music], a dubs I did not confirm what I did. You can confer could able to locate credentials are check this Running in a very junk workspace Docker build happened what happened with docker build Well, these are test cases that written in the pipeline.

Integration Tests Get all featured posts here Update has been posted Oh my god, a lot has happened now with Docker build. Wonder Less V2 in successfully tag front end also tag Done, logged in as well, he did docker push also do There should also be a push on login Finish Success O! It means if I check my Gmail Karu Wonder Less Application Has Been Updated Deployed Success Two Minutes Ago it's cool it's cool But we checked the pipeline logs. While doing something You see, that thing should not be ignored.

What I Feel able to locate credentials in your card Running AIDS Confer Now what AIDS Confer is We will not do it in the pipeline. Where is he If you look at the step in the beginning then You need to attach an IAM role to Jenkins. At two ways you can do AWS You can confide in an AWS confider And secondly you can do So let's create an IAM role on this instance. I am on I Am I'm going to go over and get a roll made I will take it and make a roll that I.

I will make an ad service and I'll put two in it two I will create a specific role for the instance So that the AWS conf error. If you don't come then you did the next one now in this Grant administrator access I will give it, I have not mentioned the role name anywhere. Next is the role name EC2 Wander lust roll this I will make a roll so that The E2 instance will get access to The AWS Confer thing and just create the roll.

[Jenkins and CI/CD Pipelines] Jenkins security, credentials and access control

When I did it, I was told that my EC was two. Wanderlust Satu Wanderlust Roll Mil Gonna roll this over to me on my master node I will have to attach it to the instance I will go, I will come today after removing the bill. I have an automaton called my master. Road was his actions will go security. I was that role of EC2 Wanderlust. What was my etoo instance where it The thing was going AW S he is unable to confirm the credentials So what I did was roll that instance itself.

I gave it to Bhaya, did you understand what AIDS it is? To grant device access Who understood that AWS Confer did not work, enable to locate cut then What I did was I deleted that instance itself give a roll Dia how many of you got. How many people understood this when we AWS confers So the shell is the one that gives its command prompt The shell remains and it gets access but Jenkins user doesn't have this access. I will build with parameters in version 2.1 I will run the version 2.1 or whatever version it is, I will build Now look how much time it takes to build.

Have done all I understand the thing Now what I will do is now I will look at its logs I will check This scanning and vening completes it. Where that environment variable will now reside updates the Can you see running automation? Bash update there is no such error of able to Locate AIDS Credentials how many of you get it how to solve that How many people understand the error? No, otherwise you will do the same thing, this step. Would have skipped Then I'll tell you, skip that step.

If you do, it will be Let's version 2.1 pushd version 2.1 push can you see Scheduling Project Wanderlust CD Wanderlust CD Project is Now Scheduled means I will go to Wanderlust CD. Can you see automatically it is triggered Automatically, I did not even touch this CD. Did automatic ticker now give me this Tell me, have I done anything in the last hour or four? In an hour, anywhere on GitHub committed Some? No, but if I go to my project Jenkins updated environment variables Now f Means This is the environment variable that is updated.

[Jenkins and CI/CD Pipelines] Notifications and email

Argo CD updated variable a minute ago automatic lift Lee & Argo CD full deployment taxa How cool is that I did on the Argo CD I didn't say anything to anyone, I said everything to anyone. How cool is that if it doesn't happen then you can Anyways refresh and it syncs can you see this it is Let's start syncing, right now our application. Do you remember what the port of the application is? 5173 If you say so, I will come home and give you the T-shirt.

Tell me what is the port of the application front end is 3100 31000 This sign cannot provide secure Do not run the connection o HTTP s Run it over HTTP and then you've deployed it. Hours If you see the Yum tutorial in an hour remained If it is, it is very like it is very possible but this Four hours to explain things Lagte ts Correct Text Fore. Hours Show a Funny Thing Look at this Now this Dove Yes, I did it right,.

What I do I do something beyond this would change I am on the front end of it I'll go to the source of it and. I'll go to its pages and I'll go to its home page. I will update and write that batch sen is and I will commit to this, commit Now I've committed this, I'm just I will tell my Jenkins that Jenkins I've committed something His version 2. Back end What was the version of this, such that every tax would have come Yes son 2.1 2.1 let it be like this Build how many of you found DevOps fun that.

I took a decision two months ago that Should I do DevOps or not? In between there was de-motivation that job in devs It is not in the app, it has no influence. Didn't even touch the server anywhere has full power Automated How are you liking DevOps means as long as you won't love DevOps. Until then you won't understand why Whatever is happening to me is fine It's whatever it is, however it's happening Why is this happening, you don't understand it You will find it so loving DevOps ya pre.

I understood the request, how many of you I think I might have received an email here.

[Jenkins and CI/CD Pipelines] Monitoring with Prometheus and Grafana

Wanderlust Zero Minutes Ago Success Hoye yes Run Success Very Interesting Application to be seen 3100 Argo CDs Can You See sucked a minute Ago sucked a minute Ago how cool is. This was the mega project for you where But you can build a three-tier application with the help of DevOps. Through Practices Deployed this is how you do it I know it Might have been tough, how did you post all this? Blog Title Do You Wish More and More Students to Learn DevOps Blog Content You your thoughts Vim author name vas If I have been to only one place in this entire world, then I.

Select Adventure was posted Blog Post Successfully CreatedDocumentsFragment More and More How Many of You Love The Project How Many of You See Another Thing not now not now can you Grafana's Through Dashboard. In the namespace of my Kubernetes Show a dashboard of the number of pods yes, can you see this, can you see this can you see This is how cool it is This is how cool it is. This is fun, I enjoyed it, so all in one. You can build everything on the spot, see the pod.

You can have whatever pods you have on the back end. Deployed, we just did one thing Sorry, not the back end, you go to the front now. If you all used it too much Suddenly he grew up and said, "You can take it for the last." F

minutes increased or decreased in the last few minutes How much network did the transmitted packet cover? How much Wallace namespace oh a good one so here you go and you can See this back and pay last no second in cum It's done but now it's fine, can you just do it?

To do Yes, junk auto cut should definitely be done on commit. It is possible but you have entered the parameters The project will run if the parameter is removed. Monitoring of Application real time monitoring of your Application how many of you Enjoyed how many of you enjoyed today Session Today Session B Do who liked today's session the most? Go and after going to the end of the project you just Have to do one small thing which Is E's Ctl Delete Cluster Wanderlust In your particular region, let's say if you.

[Jenkins and CI/CD Pipelines] Notifications and email / Terraform and infrastructure as code

If you have created it in the US East Region, then US Should I delete it or give some screenshots to someone? Can you send the logs you received via email? If you want to see it, you can also see whatever logs are there. They all give you should not do it like that but if you go some It's like you're messing something up here You blew this script out of the script. Gave it and saved it What if you ran this pipeline?

Gotta know So what If so, you will receive an email Saying what If a mail comes with the name Build Fail, then I Only the build failed one was showing no one else not a scene. Here Done Featured Post Kaise Ha Us Pare Kaam It is running and is currently on in process, so if you go to the dashboard of Wonder Lust. The build has failed and if you want to know more about it please contact me via email. Fifth note of musical scale You will go, the email did not come but the build failed.

You should get an email, no problem on that. The fix will work fine Oh, it's not working at all This pipeline is not working, it has no error. Came Well, you can try it in the open blue ocean also. Can Ho Same Pipeline looks a little prettier look same pipeline your wanderlust It will look a little beautiful just like this It's raining and here's something else. There is no Blue Ocean Nothing much, you should have received an email. I Showy Feld it comes like this Wanderlust application Build failed I tested this last night It just came out that I did something by mistake.

If you haven't failed very well, That's why it has happened so I too will come and say Think of it from my side, I just want you to Let me delete a cluster that Wood B. Delete my East Wanderlust cluster, the cluster should be deleted. The next thing I want to do is It's a little emotional but we'll have to do it anyways how many of you are Ready So Batch Seven Server habitat Sus to be terminated Boat [music]. DevOps with you I enjoyed it a lot Some?

This is what I will say and what else, this is your We can use Terraform to install the server. Through At this server police station is ours Terraform We can destroy it through destroy. Can that is Yes, I'm going to be able to see you over here. Practically speaking, the work of DAPS was It's over here with your new things. Ho you all are just so thanks a lot For that this I messed you up a bit Master before cluster destruction Don't make such a mistake.

GitLab CI/CD

Source file: transcript_lFfqBxZf-6c.txt | 15,383 words kept of 20,308 | 39 sections

[GitLab CI/CD] GitLab runners and tags / GitHub Actions

Important topics that are going to be discussed in like variable secrets Artify includes self-hosted runners Or maybe it's based on the season runners. We are going to deploy the project. Let's start reading GitLab for now. DevOps is the first to understand that GitLab What is GitLab? It is a tool that helps you Provides some project management capabilities What do you mean, like you read GitHub You create a git repository and put You post your project and that project. You manage the source code of You version control all these facilities All these capabilities are also available in GitLab.

What's the specialty of GitLab? Integrated into it CI/CD features Integrated CI/CD The features are there, they are in GitHub Actions. It is also absolutely there, but GitHub Actions is a It came in a very later stage and is still there. It hasn't matured enough for people to use GitLab. Skip to GitHub Actions and go to Git Lab is hair since a long time, very time plus if we talk about the difference In short, you can use GitHub as your own. I can't run this thing on the server.

What does this mean if I should go www.GitHub.com But this opens the UI of GitHub.com for me. K so this is the UI of GitLab.com to me It opens, there is no difference between the two. You can keep many of your Git repositories You can do it securely. You can do anything but when it comes to yourself I took a look at the environmental letters of K Created an AWSP E2 instance p me git for your company. If you want to run a hub, I won't run it for free.

GitHub Enterprise I have to take the edition, meaning if I use GitHub Compare with Compare with GitHub I'd recommend using GitHub to create your own instances. You can't run GitHub forever It will have to be run on the cloud only, which is GitHub.com GitHub.com Or you can get the paid Enterprise Edition. It will happen when it comes to gut love gut love you can also run it on your own instance. You can run for teams, meaning train against Shum GitLab.com own instance self You can also make it hosted, how many people can this The most important difference I understood was that.

[GitLab CI/CD] Docker containers and commands

GitLab provides you with a self-hosted, on-the-go experience. Provides the ability to run on a server as well Meaning in short, if you do Docker run GitLab and you do docker run GitHub in it. The one with GitLab, you can install it in the inner Docker Can run the container but can't run GitHub Can you make GitLab open source? Open to use GitHub What is used by Microsoft, how many people I understood the difference which is the most major There is a difference, there is a huge difference Let us go.

Bucket Again is a product of Atla like Git Hub is the same as Bit Bucket is pretty much The differences aren't just in the bit bucket Integrations with Jira Confluent. Cumin is used in a very seamless manner Integration with GitLab and GitHub It can also be done with but as many features as bid If you can't get what they give in a bucket, I. Hope you all have got an idea that gut Lap Bitbucket GitHub All of these There are platforms where you can share Git versions You can use the control system but some The platform offers good features like.

GitLab for CI/CD for DevOps Features for DevSecOps Gives dev a great fit for sec ops Features also provide inbuilt capabilities So if you're a DevOps engineer And if you want to choose between GitHub. GitLab, where everything is up to you. Dependent is you are the soul owner of the Project Than GitLab Is The Better Option How many of you understood this? What is gut love its an open source Tools where a lot of people use them You are buying this thing if you can afford its pricing.

If you see it, it is for personal use Completely free obviously for team use They charge a premium Plus you can self-manage and dedicated servers You can also take very interesting things. It means a lot that you are your own You can also manage this option yourself, Git Up This option is not available for free, get up charge does for this thing let's have a. Got the overview GitLab, I've got a little overview. About GitLab Now let's see what gets out of this.

How is GitHub different from BitBucket?

[GitLab CI/CD] GitLab fundamentals

Someone has a particular characteristic called gut Lab is particular use for CI/CD DevOps & DevSecOps Bid Bucket is Mostly Done for project tracking and Managing GitHub Is Mostly Done for Open. Source Contributions as many as open source Contributions are becoming huge Projects are Majorly GitHub use thik Now how to use GitLab is as simple as that. If you want to see it directly on Google Lab Why GitLab why GitLab is very famous Basically you can see that it is very fast The amount of funding that the GitLab has received.

As many people are using it Very fast in comparison plus GitLab A tool that has enabled AI to do very well It is integrated but, that is git. The hub was also used in the pilot and Gith Copilot and came early is perfect But these people have started this under the name GitLab. If you have removed it, then GitLab He started his own business by the name of D Git Club D. An AI butt is removed that specifically Made for DevOps and development So they are focusing more on DevOps.

If you are a GitLab user, you can read this. That security deployed planning is very It's crazy, plus its source code is He has kept it open so that can see it. Mast in an open source model every time there Was a gap, this one, he completed it beautifully. If you are keeping it cool then you should definitely Try GitLab Now how to use GitLab, we directly Anyone in this session watching the practical No theory, no PPT. You will go directly to GitLab.com Pay and there you can go to sign in As soon as you click on Sign In, you will see that your username and pass are there.

That people do not misuse it, they This L has been kept with a very high level of security. This is important because gut labs have many All the features are free, so no matter how many Creating an account can misuse GitLab That's why they verify email and phone number. Within the first three days you This has to be verified, these people are Does not accept credit cards, no payments I only take your phone number and.

[GitLab CI/CD] Git and version control / CI/CD concepts

We take the email address so that it can be verified. Engineers, if you tell me then the group lab of CI/CD features remain those CI/CD Features include the Help Tool Kit. If yes then select role DevOps Engineer you will see Can you be a software developer or development team? Lead DevOps Engineer System Administrator Security Analyst Data Analyst Product Manager Product Designer Saw some only and only development and Deployed Teams is the focus when you group Do you join the lab? Healthcare domains around the world The domain this tool is specific for Developers and DevOps Engineers so you Select DevOps Engineer I Am Signing Up for GitLab Because I Want To.

Store my code I want to move my Repository from GitHub or Bit Bucket to Git Well, today we will also see that gut How to move a repository from a hub to a group lab. I want to learn the Basics of Gut I Want to Use Gut Labs I with my assisting repository general people For this we use GitLab ci or its There is a tool which is very powerful,? We want to learn base this Basics of Git Done Store My Code What do you want to use it for?

Tell me 1 2 3 4 5 6 which one would you choose Wood Go For Four ci who will be using got lap my The company and team Just Me Just Me for now. What would you like to do to create a new project and join a new join en Existing Project I Want to Create A New Project Explores Things Like That Email Updates I Like to Receive. Updates about GitLab yes I like it is to learn new things continue and this After doing it, you tell me a little bit.

To get to know yourself so that your Create and import to make your experience better Your First Project You can start your project You can import from GitHub via Bitbucket. The GitHub import is here. He used to come as an icon, but what did that achieve? The guy who first time about GitLab I know he used to see GitHub, then he These people used to run there and use the GitHub icon. Repository by URL if you select Repository by URL and The URL of the GitHub repository in it Whatever you put in, the project gets done.

[GitLab CI/CD] Git and version control

The Concept Of Groups the concept of groups like My team is where I work. It's like a group has formed DevOps is a group DevOps is a group Prepared is for The script is a group created for Scripts is a group created for docs. This is groups what is a group is A collection of team members and their Projects Theek Group is a collection of Team members and their projects now look like. There is a group called DevOps within that group.

There are multiple projects for example CI/CD A Manifests Or Kubernetes The NITY manifesto is such projects and these Who will have access to the projects? Everyone will have what is dead in the DevOps group Like I am added in the Deb group so I will have access to its projects. He is my, he is my colleague and he is also dead. He

will have access to it from the lets you have any There are people who do not have access to the DevOps group They only care about scripts.

If you can add them, then this Seg Gun team Management: This work is done by GitLab. I have groups called groups like You won't see the name GitLab anywhere. It will be visible only in and all these in GitLab things are called projects these Projects That Speak All Things GitHub The user name you have in the same group happens and is inside it. Repositories are your groups in GitLab. There are projects within that group So Just a Small Difference repository is called on GitHub The project is called GitLab Pay I Hope You understood the difference in an easy way.

It must be that, when we talk about GitLab Do So here we go, groups end projects so this is your group and its inside your project when you talk to GitHub. If you do this, here comes your username and Here comes how many people your repository I have understood the difference in how you get The Difference Pretty I hope you have come when you. It's a no brainer Gab So if you talk about GitHub Co, then here But I saw the difference and understood it was yours.

Domain name gab.co.uk it's yours group or your user Name This is your project or your This thing is called a repository project or this thing your user says name this thing called group let's go.

[GitLab CI/CD] SSH and remote access / GitLab runners and tags

Create group names Projects R Organize into groups late from us people T.W.S. Dev Create a group called Ops and Inside that, let's see the project Demo C I CD Demo There is a program called CI/CD. We will make the project, then this project in the group so our URL How will GitLab.com become WBS DevOps One? Why have I put this one because all over World someone and T W S DevOps name If he had made something of it, he would have destroyed it.

Ten Things Are Done include getting started read me Create a new project with a simple Create the project and you will see it now. Dega GitLab UI now let's say to me If you want to add someone, I can add them. I will go Right now if you are a beginner then if you select I have said that I am just learning, so you should All items will be visible in the Learn GitLab set. Up Your Code Confirm Your Project and things like that but how are we right now It's been a month since I became a pro.

Go directly to your project here it is Project Now if you need anything on GitLab push pull clone fraction things that you I used to do it on GitHub also, for that you need SSH. Or SSO Single Sign On Sale to you will have to use and if you want to use HTTPS If you want to use it through then you have to use personal. Access token will be the password in Group Lab If there is no such thing then either you have to go to SSH Keys will have to be used or personal Access token or single sign on.

As you mentioned It will be done, now I will do it completely. Today's tutorial should be done through SSH only. I'm going to log in to AWS. Create a new instance, let me create a new instance. We will quickly see Create an instance K My Authenticator, let's create an instance and We will play today's entire session on instances People Great Latch instances, let's call it GitLab. Server test where we get together If you are going to make a self-hosted runner also That's why we do it by batch server.

[GitLab CI/CD] SSH and remote access

Remember to get an Ubuntu instance Let's take the x86 architecture Let's buy a pair of T2 micros. Turn on HTTPS traffic in If you want a case then it is so much All you have to do is create a Latch instance, what's up, that's great, so yours. Instance is getting ready, well done Pending I hope you all understood would be how to create a group how to Manage Projects You Can See GitLab's Main Is. You will go to the home page and there you will see the project You will see now this demo CI/CD is a Project which is private is fine only those who are added can use it.

It's a private project, just like you can go to groups, multiple groups You can create a new group lately. Create a project and name it after it. DevOps Notes can create DevOps Notes Public can keep inila is repository With read me create project so now group Your DevOps notes have also become public. It is public, now if you want, add notes in it. You can do whatever you want Let's check the instance is corrected or not our instance is running this thing to do Connect Great Let this be your terminal.

Already happened on and here's what we're going to do Have fun accessing your GitLab server We will do it, so here is our GitLab. Server to SSH Ready Yes, very good, this is our server. Ready now this server and your

GitLab.com We have to connect all these pull push things have to be done randomly so that's why we People will do SSH gen SSH gen a. The key will be generated and will be of ED 25519. We don't need any names, we also need some passphrases.

I don't want the pass, I won't give it again. Your key is ready CD dot In SSH I'll go and IDD 255 this will be my private key and this will be my public key account or from this to instance to me. If you want to push or pull code from GitLab It will remain on the private instance and whatever my It belongs to the public, I will put it on my lips. Copy this Lenge GitLab will go And you can't push all pull repositories using SSH Ants Key to Your Profile Add SSH will do And here the SSH keys one will come here, this.

[GitLab CI/CD] SSH and remote access / Git and version control

You can go here very well and Your settings are here, it's done. Security Settings and here are your keys I can't see where she lives yes where User Settings Profile Account Billing Chat Access Token Email SSH Key And You Can. Just add a new SSH key and add the SSH key You copied from your instance You had copied the public, go here and He threw it down and picked up the title himself. Took authentication and signing and a Put an expiry date on the ad, just do it.

You will add the SSH of your SSH. This group is registered with a club account Now you are free to act as if you are the master of your own will. You used GitHub, you can use GitLab. Do it, push it, pull it, do it, whatever you want, pull it Let me show you this, your account is Authenticated with SSO means you g Let's see what is auto DevOps if you automatically do this. Creating a CID CI/CD for the project If you want, you can also enable auto DevOps.

If we build our own pipelines, then this will also be closed. If you do this, this auto DevOps feature is just Available only in GitLab, not on GitHub Come on, let's close this one too, now that's done. This is your project that you can code. Clone with SSH Pay Go Copied and Directly Here's a MKDI R GitLab, make one. Let's turn it a little orange and let it go. It's done GitLab cd GitLab and here But we can do git clone same command Same command as you used on GitHub Git has the same commands as git clone or GitLab.

How to clone your project's SSH Key command Enter are you sure you want to connect Yes I am pretty sure and Done, you have created a private repository. How many people have been cloned via SSH? Got it Setting Up a Project is on GitLab as simple as GitHub any It's not different, come on, it's the real thing. The real difference comes when You cannot create a demo CI/CD project. Do you If you import then on new project Go to Create a Blank Project What do you do instead, import the project.

[GitLab CI/CD] Git and version control / CI/CD concepts

Or any other open source platform. You have 4000 followers, come on anyone Pick up the repost, tell me which one to pick up. Notes P yesterday I did two tear flax up Before that I did Python Bash Blaze React Django to Do Wanderlust Micro Services Prem Graph Fumble Lots Of There are repositories you can pick up any that. Like the most used repository is the most There is also a more folk repository, so just its Grab the URL, grab the simple URL and Repository by URL and click on Git.

Go to the repository by URL and slam it down. Project name as it is no to do CI Everything rotten comes on its own, everything comes on its own It comes to you, you don't have to do anything but yes. Under which group did you get this project? You have to put it, you see that you have TWA Whether to put it in DevOps or notes So I put it in the DevOps, that's it. Visibility level makes it private Why are you picking up private keys from a repository?

If so, create a project now, this thing It takes a minimum of 5 minutes, but this It's a small project so it's quick. I went and imported 90 Days of DevOps. It was minimum, it took me 5 minutes When we read Terraform, we will use GitLab. We also integrate Terraform through I will do it, but let's close it for now. Your project was Here's why this no-do-CI/CD run ten Commands Note to do CI/CD Run this command As it is imported, it's as it is It has been imported, you can also do this.

If someone pushes to this repository So let's push here too for that thing. There is an option Mirror This Repository Mirror Meaning, you moved your hands like this in the mirror also. You If you have just imported it then here There will be changes that are not reflected here. It's over great Done, I hope you all have understood. How to create your own commits in GitLab How to manage projects How to Use GitLab How Many of You are finding GitLab very easy because The option for everything comes in writing plus you.

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If you have already read GitHub then GitHub Becomes very easy if someone gets lazy for the first time If he sees it, he will feel that he has understood something. I am not coming, I am on the side for example Git Der is a concept of pull The request here is a pull request. No, it's called a merge request here. What is called a merge request then pull request in GitHub Merge Request Chocolatey in GitLab Eat the Eclair one or the Cadbury one Chocolate is chocolate, so it is like that Correct.

How to use GitLab to get you to a good level I have understood how to create it well. Manage Projects and Groups You Viewed You can see the visibility of those groups. You Now let's see The Elephant in the Room How Many Pipelines to Create in Git? Of you are excited about creating a Pipeline in GitLab because for the same thing So let's see what GitLab is famous for. We are famous for the projects we have done I checked and also imported it from GitHub.

I even got it created myself through SSH. I cloned it too, I did everything, please get this done, then I will enjoy it again. It will come Now let us first see that how do you build the first pipeline so you're a Take on a project, any project. No to do don't take a demo CI/CD Take the project Demo CI/CD You have taken a great project, now you What Will you add a new one to this project? File is you add a new file i hope you People should branch me and tag me and this is the same group No need to teach things that are easy.

The branches are also the same as those in Git. There is a system in which what is tag release Tag version tag is also same in GitHub There is a scene here, go to New File Click here and always remember if you. Creating a CI/CD pipeline in the GitLab So you need to use a tool like Genki externally. No need to install Get all the features of Jenkins here Deta to GitLab is very famous for CI/CD and more than CI/CD CI then you don't need to install Jenkins separately.

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No need CI means build your code It is being tested and everything Integration testing is happening, you can see everything here. You can only do what you need to remember Pay close attention to your file name. To what your file name should be Dot group Lub Hyn Sadwa A There should be a as soon as you wrote this The option called Apply a Template is unlocked. The option called Apply a Template has been unlocked. Because GitLab only and Only understands this thing if you have GitLab If you want to create CI/CD pipelines then your file should be named dot meaning hidden Internal GitLab Next to - Next to.

CI means generally used for CI For code checks for testing Then dot emal for quality analysis Meaning you will get Python group or Java in it. You do not have to use the syntax of Directly Key Value Pair The value of this has to be entered in Pair definitely reminds you of Docker Compose. Can I use ML files in Docker Compose as well? If this happens then first of all you should understand this How should the file name be? How GitLab CI works So GitLab works with CI Focus on two very important things Stages End Jobs So whenever you have a CI/CD on GitLab build a pipeline Yes, you build CI/CD pipelines for that.

The pipeline consists of stages and Jobs come in stages, this is it Something seems familiar to me When you build a pipeline, inside the pipeline Comes in stages and comes within stages Are. The steps are the same as in GitLab when you create a CI generally only makes C pipelines but you can also do CD in this whenever you CI. When you build pipelines, you have stages in it. Mention has to be made and those stages If we have to mention jobs then let us understand, what could be the stages?

Let's look at yesterday's example, when we People had read Git Jenkins, so we were Pipeline Start Used to Pipeline Stop There used to be a start and an end on this. It does not start and The stage used to come in view of the end pipeline It was not visible in the view, then what about us?

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Used to come here and then a what would have been our There used to be a stage and then we used to go ahead. What would happen then we had another stage and then further We used to go, what would have happened, we would have had a stage and there would have been another stage here. You can create as many stages as you want. Let's say this is stage one, this is Stage two this is stage three this is stage This thing is called four end One Pipelines You create pipelines from a file Whose name is Dot Gut.

Lab Cole - ce dot wa ml means if your If the repository has a CI, you have a The pipeline will be simple pipeline now What are these Bhaya stages and what are the stages? What are these jobs inside, so understand it like this

Take whatever stages you have This is just a stage from Let's Build is the second stage. The third stage is the test. Push and the fourth stage is Defining the first in Deployed Lab If needed, go to GitLab and here But first of all Define the stages if I want and.

Let me zoom in on GitLab for you. Go in and first define the stages Do it now, there can be multiple stages So we will use a list, meaning a - like Stage is build, second stage is push, not Test 3rd stage Is. Push is the fourth stage. These stages define who is mine Which stages are going to happen already? I told you, I told you already, now this Joe Stage Tell me what will come under this stage, what will you get inside this stage?

Jobs: Your jobs will appear within this stage. You have a job from Let's, let's call. Build it job build job now which build job is this Which stage does this build job fall under? Comes inside the stage Inside the build stage Now and which one are you working on in this build job? Want to run the script in this build job I want to run in Echo This build is Don using docker build Haiti Using the dot command Please show the command something like docker build.

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Hight Dot or Docker Bee dot hight tag whatever you can understand I came in, many of you got it, how many people. It means that it has been clicked in such a way that what you what will you do will you do it Stages multiple stages us multiple. Which of the stages stage build and for that build these one Job, then we will call it a build job. Can we or should we do anything to this, you can do the job, you can definitely do it.

It's up to you, you can name it whatever you want. You can give it any name, it's just on stage. It should be referred to which stage it is. After doing this let's So I have to create a test job now, so you Create a test job Put the low stage, which test and for this put a different script, which script. Echo in a shell script That this is testing of our Docker build Have a small test for the Docker build It's, now you can do anything similarly.

You can also git push jobs done and What stage will come in the push job? Tell me your push stage will come, yes this if I Push Replaces Did a push to Docker Hub and this is what I did here But if it is not written, it will come. If you get an error, you will have to go through these stages. Keep the name and the stage the same. If it is required then the build will come here. There is also a build here, there is a test here, so this too test here is push so this is also push And you put the script what is the script this script is a.

Keyword is in which you can shell nicely You can run the script, like this also can you do that This is pushing To Docker Hub, yes you can do it like this. It is simple that you can also tell this directly Yes, multiple lines of scripts are used How clean are you doing it, how clean is it? Stage script is neat you do one more thing Can you get a cheat sheet now what's in it All the keywords you have will go into it.

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You will find all of them for example General build tests for stages deployed pre dot post dot prepost What happens before build? After that you can do all these things If you are creating a Docker build stage The script in it Rules from all over the world, so this is your cheat sheet. Which you can use anywhere I'm just telling you how to create a pipeline. It's to make and Also made for deployment palay Low deployment Palay Two ECT instances done now, that's so neat, Jenkins.

I had curly brackets and curly Open and close brackets all over the world There were things in it, there's nothing like that, yes Because this is YAML syntax. I don't have curly brackets there end values then this object ends its value object and its value is It happens in a certain way, you just do this much. After doing this you just have to copy this file to your Commit to the repository Commit messages put it Edit group Lub Ci dead GitLab ci commit do main me.

Commit it, no problem, do it in the main Given commit, now as soon as you commit Look, here the icon of GitLab CI has been created. I became an icon, I became an icon, that's all you need. Somehow you have to create this GitLab ci file to your repository as soon as you This wrong CI file in your repository This option will be unlocked for you will go. Build so the build option is unlocked for you where you can build pipelines What jobs are in your pipeline?

You can also see your pipeline editor yourself. You don't know the syntax of let's say YAML So the pipeline editor also gives that your I will also correct the syntax of YAML. You can schedule the pipeline artificially You can see, you can do everything related to So you don't need to install Jenkins tool separately. There is no need to put all those features here. If given, you go into the pipelines as soon as Adding a GitLab CI file to your repository Email

is created, then your pipelines are here It will start getting made, here.

Watch the pipeline being built, are you seeing what? Is this what is this is the thing Tooth so as soon as you create that mml file Automate Your Pipeline Is Running Automatically So if you go here, you go to the pipeline.

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You can see that all this is automatically added as well. If you pipeline this is the ID of the pipeline If it remains, then click on it. If you go then you can see build test push and Deployee Pay job is running How Many of You Are Enjoying Gut Labs, I did not install a single word in it. Did Yes, I am enjoying the same Jenkins ones. I am able to create pipelines and here All my things are also happening interestingly.

Like we used to create agents on Jenkins And Jenkins was said to be Jenkins server This is the Jenkins server. Master server on the master server of that Jenkins Vote Never run jobs on the deployed node Execute but in the GitLab, we If agents have not been added then this How is automatic deployed and where? Deploy when you unlock it a little below Go to Settings Go to Settings After leaving you have to go to CI/CD Click on CI/CD to give CI seed like Just click and you will see it here.

Runners like you had agents there Because there would be runners here anyway. Runners are the processes that pick up and Execute CI/CD Jobs for GitLab Simply Expand Now Project Runners Meaning your self-hosted runners like now I can create you instance and run my own runner on it. You will setup it and then there is an instance. Runners this instance runners all over the world all over the world GitLab ne some Servers are set up like this blue shared Galab Ogg Runners Manager GitLab Such Duniya.

I have told you many such things throughout the world, how many are there now? How much are 101 servers for the project? There are 101 servers where you can upgrade for free. The project deployment Palay Fukat has just one thing for you What does it mean to be careful? Runner's Tail Last contact three months ago, this thing told you Keep in mind, it's been three months since this runner No one has seen this, this is what green means Runner is active Runner is online last.

Contact one minute ago look so many of you There are so many active runners nearby that it's. There are so many active runners You can also make your own runners and GitLab You can also use free runners of People understood the concept of runners.

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How many of you get the concept of runners Put your hand on the pipeline a little more I'll teach you about runners. A little more power of love, a little more power Let's go to GitLab CI now whenever you This is a CI file, how can you edit it? You can also do this in the pipeline editor. You can do this in the pipeline editor using lets If you have written stages as stage then tell it there He will tell you to correct the spelling of and.

You can manage your entire project there. Here is your file, it's great Edit: Source code control is here Commit here, it's cool for Example, I went into this file, now I what to do. I have to keep parallel jobs have to keep parallel jobs, what is parallel jobs? Jobs means what should I do? Is the stage I am in the testing stage. What can I do for, put parallel stages here. Can I do parallel After the completion of Stage One of the stages Stage Two here sorry not parallel stages it's called Parallel Jobs S means this is the test stage.

Three jobs will run in parallel, try this. Create a job, any job. Create two Let's Say Dev Tests Job dev Test Job Done Now This dev The stage of the test job will be given to you in that. You have to put it in the stage where you have parallel I want to make jobs with let's test me I have to create parallel jobs and scripts for put that in. Eco tested for dev and similarly this thing What you can do is turn it off Give me a copy of this You can do that for production by doing Prod Stage Test Tested for Prod It.

You did it, now what should you do, two stage test this is your explorer on its y third go source control so what this will do is this can commit to your repository put it in the commit message. Edit Parallel Builds For test you did it and commit to main Click on the branch and do you want to Commit to branch continue and you will see Loading GitLab Web ID: See your chances.

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Bean Publish Now go to your project Demo CI SD Build on Go Pipelines Go automatically the pipeline is triggered too Look, the pipeline is triggered automatically. It is so true that you also care about the environment. GitLab is giving you GitLab Runner I am also giving you pipeline GitLab I am giving you the code also GitLab gives you everything It is running, click on this build. Let me show you how does parallel Build Looks Like Can You See This is a stage in which parallel bid is all three together can you see all three run together getting.

This is how powerful this is that you have a Production Dep Staging All Three at the Same Time I have to get the test done, runners too, just you. I have to build a logic and write the command It's up to you, so be it. How beautiful the push job looks. Not that you can do Multiple Jobs in a Single Stage, now you can go to any stage and click on it. You can and after clicking you will get that The output of the stage will also be visible.

Tested for dev tested for dev, you can see the output now. Your runner, see which one was used, three green Mother-in-law linux.com so you can see one of your A good runner is also ready, he has I have told you the thing, look it is running right now. Beautiful Funny everyone A CI/CD pipeline will be simple to create. At least create a demo pipeline and GitLab Run Dega I'm Pretty Surely all the people in this batch will do it. Many people may have done it too Because it is so easy, it would have been so easy So tell me something the whole world could have done.

Let's talk about variables that if I have to use such variables in it Passwords are secrets, so how can they be I will use it, which is very important. For Example, I went to my ID now I This build is done using command a but I need to enter my project name here. If there is then there are some inbuilt variables like what is dollar CI project Name: What does this mean?

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CI project name: This There are inbuilt variables, you can change it Galab CI project name is now predefined CI/CD variable Chat Channel Chat Input CI API. I want to see the author of the CI commit from Lets that who has committed this so this is testing out Build done by and here I am mast dollar. What can we know about the CI commit author by putting it? It will happen that this The testing that is being done is for this build Here is what this developer has committed For whoever committed that Testing is going on, how cool is that first.

The variable is defined from the commit branch itself. Testing for dev commit message There was a lot of tag title happening for this message. There are so many CI runners, which runner should I use? I was so cool, she's so beautiful, pre Defined variables means too many variables Many have only this much so you are limited You can use variables But I hope you understand that this. How to use it, it is inbuilt Variables then come your user defined variables that you Create an object with a variable name variable let you have a variable made.

Deployment A variable named Diploy et made it and gave it a name production then you Created GitHub Git Lob I used one of the user's keys and made this is my. Secret Key Simple Now Deployed Production You have to use it from here. I am pushing to Docker Hub for then how to use it You will do it directly dollar Deployed now here Pay Tested For Di using dollar now what your GitLab user. That using this GitLab user means this I am giving you a demo on how to change the variable.

You create your own variables You can also use GitLab's built-in variables. I thought that to create a variable you can do this Now, create a variable and put this dash in it. Why not put it because you are using the variable You are creating an object and a list of variables. If you are not making it then there is no dash on 'y'. It will be applied directly, the same principle as You do this for the environment in Docker Compose environment and variable names I have written it without dash, put it here.

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Aided Even variable for Pipeline Edit Some Variable for Pipeline Commit to Men Branch Success Your Chances Have Been Committed Done Now you will go here to the demo build in CI/CD. If you go to the pipelines, you'll see Automatically edit your commit. The variable is there before you even see it. The pipeline would have been checked that the You have also messed up the pipeline. Whether the changes are right for the project It runs automatically so if these shirts You are in 2024 and if you are looking for automation If you are not using GitLab then you are missing out.

You are doing a lot of things, I hope this Everyone must have understood the thing. How cool this platform is Build Job Pay I'll go through and find the variables I used. Pre defined they will appear here echo this CI project name

build is done using Locker Command This Demo CI/CD What is This demo CI/CD is of my project. The name is what I've created here from the variable Printed this demo CI/CD build Is done using command this is fine great Beautiful now beans I different for her I will see A pipelines fine.

From Let's I used it somewhere else as well Right, put your user-defined variable somewhere. Use that used for push So you can also see the variable for that push. Will give seen Deploy the value entered and push the hub It has gone but, I did not enjoy it, I will enjoy it later. When you're here in secret Will you show me my password for AWS? The account ID is Or my Docker Hub username and password that's what I want to use in this too How many people do you want to know without showing them to the user?

How to do it like you do in Jenkins You set up credentials and that username And the password comes out and you can do it without Show me how you can run the pipeline, let's do that. Let's take it, you have to go to settings If you want to go to CI/CD, click on it. You have to go here to the variable. Do it to him Expand so here there are three types of variables.

[GitLab CI/CD] CI/CD variables, secrets and artifacts / CI/CD concepts (part 2)

Are like protected masks expanded directly you a You can access another variable from a variable It is of no use, you look at this mask You are the most important, so let's go. Start creating environment variables CI/CD Variable Add Variable Now do we have to keep it for the entire environment or For whom should it be kept visible or. You have to keep the mast, whatever you want the meaning of mast to be. Do that and the value of the variable is not passed to anyone.

Let's say Docker Hub users, tomorrow you There must have been a lot of confusion, Docker. Docker Hub User E Protected Mask End Expand Now similarly you take a variable Take a mask for the entire environment. This is Docker Hub pass password / token or Then the personal access token is Docker Hub Get Personal Access Tokens now on Docker Hub But I hope everyone has understood this part. That on Docker Hub You can also create your own personal access token. Go to your account and go to your account settings Go ahead and log in if you need to.

Using token, you can create a personal access token. I'll create a new token if you can. To whom I will give it GitLab token and I will give it read, write and delete I will copy this token and use it in my Should I put its key in CI/CD? Docker Hub Password I will enter the Docker Hub password I pasted this value in the value field. As soon as I added the environment value Just cancel it, now this username, this password someone who is not visible to anyone You can also use it in pipelines.

No one will see it, it will always be mast The value will not always be visible Mast aayi let's done variable ho Now let me use this variable. You can use the variable by dollarizing it. How many people can understand variables? I understood that you can create variables like this. I came in variable and let this same variable So I have to use it here so I Directly go to the ID and edit it. Give That Deployed Dock hub user Docker Hub user and it docker Where will the hub user come from in your environment?

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Get up from It will neither be passed to a pre-defined variable nor to the user. Use from a defined variable or from your secret be Docker Hub username s environment Committing the variable It's done, I hope you find it all easy. It will happen because we are on the difficult part right now. Difficult Part Pk, there are logs from let's in every job. Logs those logs to me in artifacts But, we have to store it and then our learners They say that We were going to artificialize, we are artifacts.

To know in advance that artificial You ran test cases from that test Some log reports of cases were generated now this Logs and reports are downloaded to you in one file. To do this you need to download it You can share it with your team. Such things which you can collect from Your Pipeline So That It Can Be Shared With others are known as And you have to store that log somewhere. You have stored some photos in the drive and You can always download one J Frog is artificial, did you say that?

An artifact lives Let's from AWS Artifactory there If you put it in or you GitLab yourself You can use only that artifact I will show you Go to Settings Go to CI/CD Settings I went to CI/CD. Went Now what did you do here did it here You went to Artifacts and expanded Kiya keep artifacts from most Recent Successful Jobs You Ticked Meaning the artflute of the jobs you will have. It will remain simple, yours already Now what do I have to do next?

Next what i have to do Simple one second What do I have to do next about this thing? How to use it in your pipeline I went to the ID I will use in the pipeline. Simple now which stage can job Artifactory You Need Late Test job test job let Echo these are my taste The result is correct. What do I have to tell you about my pipeline? To know where its artifact is located Artsy Factory, where is the artifact factory?

Will you call Artifactory as Artifacts? Demo CD, inside it all your Consider these files as another folder.

[GitLab CI/CD] GitLab runners and tags / GitLab CI/CD pipelines (.gitlab-ci.yml)

You are building a build Or you can make logs anything you want. Above is this is just the name of the folder from let's I named it Logs Now this is my logs in this folder Of Your Pdflog will go inside What is Pdflog? Dlog is a file which I want to Preserve from My script and from my job simple now. This whole deployment of late has come to something now. I this too Our Echo this log e from dollar ci now watch How to name CI stage here stage ha ci job stage lets This is from CI job stage and this is for me.

I have to put logs / appends this to logs how many People know the difference between single and double what is the difference this double is used to. Append and Single is used to just create log a dlog and also of this. You can tell the artifact in this that the paths are the same above logs simple that I have a logs of the name I made Artfed where I keep all my Storing logs. I will not do much artifact I stored as much as I wanted.

Stored for that many jobs only, generally people What do you do with the entire pipeline? As with push jobs Even if you don't want logs, they will still come. Goes watch is not good that is not I want to get my rights, I don't want anything like that. With that being said eco tested for Prod GitLab Users It's Over Done commit this Two added artifacts commit to main Now watch as your success comes your way Change has been committed in your build.

Go where did your build go, go to the build go to pipelines you will see that your The pipeline is running but there is a lot of A cool thing has arrived, just a little bit. I will show you later, it will be very cool. I think I'll show you when the build is finished. How Much Time to Build a Pipeline Takes No, not 6 minutes, but 1 minute 54 seconds. 1 minute 54 takes two minutes to complete a pipeline It's taking about two minutes to build.

Is it fine for free a for free That's all I get We Did Create the Logos folder what is it like in that you Like if you give the path then that folder it auto.

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CreateDrawAnaNavigator If you look, you will see the download folder here. Where your two Artfed for Some Region Lets See Test Job is now felt why let's see the Region Log Dlog No Such File and Directory See this is testing out of Docker Build This. Why master, because whatever I wrote He's mastered it's Now yes correct if you get a pre Define a variable sorry not pre defined variable you have to create a folder whose. You are creating the name yourself then you have to create a folder but here is a Artificial, whatever things, whatever, whatever jobs It ran an automatic job dialog created in which you can download it.

You can see it nicely but let's see you I want to make something of my own name with Let's Something called logs or something has to be made then it. Do you want to create a folder named or create your own If you need name logs then you have to do it. Added User Defined Logging folder commit Done now what he will do is first this folder He will create it and then he will run your things fine. Will first create the folder then run things Come on let's go and check it out, look at this, it's added If the artificial pipeline fails then here I came as a cross if anything fails.

She goes here and the cross comes, otherwise she Tick comes now go to build in the pipeline Go, you can do one more thing late. If any pipeline fails, you click on it. By doing this you can run that thing that much pipeline go back that much pipeline go back you can run let's go now Let's see how our pipeline is right now. Performing Dev Prod Test Both of these things It's on, come on, let's C Test Job Failed Again Let's See what is y saying This lagad log no such directory why is it.

Giving No Such directory let's see that it's like this Why are you spelling log correctly? I felt Lock Lets See Let's D Bagsargi This is the thing that You should know before using the Logs folder. No, why a small mistake but very pic so it's a good thing that this thing is your He was detected before going into production.

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I will tell you and you can Directly Minor Fix To CI these small ones live I towards the End of the session my office pipeline too I will show you, I can't wait, show me now. I can't show you the code, but Let me just show you the pipeline. This is my pipeline, just to show you. That you will know that How does a pipeline look like? I might not be able to show the code for the pipeline to build Pipeline show ya how the pipeline of my Project Looks Like Can You See This Build Test Report.

Just did a simple build, lots of testing We have testing all over the world. There are all over the world and then there is his Artifi deployment Mo CI/CD Pay and see our recent pipeline Whether built or not, minor fixes to CI is currently in process late. The pipe the job has been built Come on, I have introduced you to my office. The pipeline is also shown. That many people ask that again Jenkins Why are you not using Jenkins also? I will show you that too, but if you CI has to do only one project related work GitLab many organizations r Moving to GitLab at the Same Time They Want.

People who have junking so that their It is if old projects are also supported. Let's create your entire pipeline once. Artifacts p now you're gone Artifacts p now You will see that Deploy or whatever the test job is to files Which files are artifacts, zip, and Meta data, what should you do with it now? Deployed Sorry Deployed job and wherever from to file You kill him with great pleasure download Now I'll show you a very That's right Here's my thing Artifact Test Result This Log Is From.

Deploy you see all of the logs I've captured From my pipeline A very good way to manage your logs has captured I hope you get artifacts You must have understood how to use it Artificial. How many people even understand GitLab? Coming up in I Can Create a Pipeline I can create my own variable I can Create My Own Secrets I Can Create My Own Artifacts Let's go, need a break or you want me too.

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Continuum need a break or need a break let's Take a small brake, let's see where we are. Are So We Saw How to Create Pipelines We Saw Variable Secret and Artifact V So just now next we will be talking about Self-Hosted Runners and Sus Runner Sus Runner, we have seen it, basically. Software as Service GitLab somewhere Already installed and running on your runners There are also known instance runners You get 100 plus instance nurses Where automatic things keep running Is Let's Talk.

About Self-Hosted Runners and Sus Runners So basically these runners are your runners which are also instance runners, meaning you Will you expand this instance runner as well? SS runners say that there is already a gap here. The lab's software is installed and it will pipelines If you run it, now you will see this and you will see some You will also lift any pipeline Here you will find a little about this runner. You will get a lot of information like this is a pipeline So where did this pipeline go?

Build Jaws Where did it run? It ran on this runner. Six green sus linux / y to it by default, git is the runtime of lib but you can You don't want this to run on GitLab's runner this is our own this is our own. You go to the runner page, go to the CI/CD. You go to runners now you to new Click on Project Runner and the three dots By creating with runners, creating runners With Runner Registration Token is disabled You will have to install it manually.

Click on New Project Runner Done One Click on New Project Runner and add it now. Tax to Specify Jobs That the Runner Can Run like Jenkins has labels that I put the dev label here from Let's And I said that as many DevOps. All those who care about the environment will be on this runner. Will you run on this agent or on the runner? If you go, wherever you have put it in the pipeline Def put that on this particular runner You will go to the labels like you used to put in Jenkins.

Dev would have been assigned to that particular agent Similarly, you can put tags here like these jobs Doge dev Prod Staging and your job is Assigned to the agent or this runner.

[GitLab CI/CD] GitLab runners and tags

I will go because in India it is called agent. In the GitLab, this is called the runner's fix. Don't put description, it's time out Fine, no problem, click on Create Runner. Just give the label and click on create runner Now do many different things for yourself. There are types of runners like you can use with Docker You can also run the container as a runner, Kubernetes You can also run the Kubernetes as a runner Run uxx1 so how do i install GitLab.

Click on Runner, now your architecture See which architecture you are working on. When the instance was created, the x86 version remains. AD x86 also remains MD64 so it Select Download the binary for your Give the system the permission to create a GitLab runner User Install Group Lab Runner as a Service Copy. Do it and go here sudo so user Become a super user, become the one who scripts I copied it, throw it here. Drop it and hit enter, now that cool install will make your GitLab runner he's your Install or download GitLabRunner Binary for Your System First he Binary Downloaded it, then gave it some permissions.

A user of a group club runner created by user d A user of GitLab Runner created and If you start one of his services then it will start Group Lab Runner. System ctl git Lob Runner Check the status, first check the command, GitLab Runner yes let's go so GitLab runner Status group Lab Runner Service is running group. Lab Runner Services Running So how do you do this thing You will install it on your system and you will go to CI/CD Settings Pay Register Runner Pay Operating System The operating system you.

I want to make the runner run How to install it, click here All its installation methods will be available here. Your GitLab Runner is installed. He is done with your project. To register means to understand this in such a way that this is Your Machine This is your machine it's This is your machine which one is it Nginx and its your GitLab. Galb.com what is your post on Galb.com Your project has been placed on your neck. Happened what is in this project what is in this project In my dot gut Lab CE Dull File Now this project is By doing this he has now become a runner here but this.

[GitLab CI/CD] GitLab runners and tags / GitLab fundamentals

The project was not registered with this runner So I want to register this project now. How will I copy and paste the following command into a command line and register to a runner then GitLab.com token This is a secret token, I went here and after going here I. Paste that command: GitLab runner register GitLab enter now he is asking Let me shorten it a little bit so that you have good visibility met now he's asking enter the gut. The LB instance URL is now your If it is running on GitLab.com then GitLab.com Leave it as is Enter Verifying Runner It is valid enter a name for the runner Let's put its name in a.

Jobs so I want that on my shell only May my job be executed You can also do this in Virtual Box. You can also do it in Docker, you can find it somewhere. You can also get it done, it has to be done in late shell So I'm going to put the shell here. Enter Done Runner Registered Successfully Fill Free to Start It But If It Is Running Already the confession should be automatically loaded, it's done now, what should I do now?

What do you have to do Have you registered a new runner as soon as You register and from there this project The connection is built because of its. You have the tool and you can register a new one. Runner However, if you need to run a runner manually If you want to do it then you run GitLab Runner Run. GitLab runner run you run but your either already registered successfully and Git Check love runner status Low services running, everything is going very well How many people think he's in just three?

Step two and step one and plus Installation in three steps Register a runner how many of you can do that For me I want to know now you Go to CI/CD settings in Runners. Get your assigned project running on Runners Def remained Now you can either send these instance runners Disable it, give it to me for free We don't want the things that happened on our own runner. You will either do this or you will Keep your ego aside and run it like this If it works then it is fine, if it does not work then it works on this.

[GitLab CI/CD] GitLab runners and tags / GitLab CI/CD pipelines (.gitlab-ci.yml)

What will we do, we are Indians I want it for free, so this is it. If I do a particular job If I want to run on this runner then I have to I will have to tag how many of you got this. It is written in stone that if I get any job If you want to play it somewhere, give me this tag. Let me run the build job from So what tax will I pay? Tell dev now this tax dev what does it take me You will find out that this particular job is dev only on instances with The push job will run on the dev instance.

Will run only on the deployed dev instance The dev test job will run on the dev instance. Let's do a production test job. I don't want to run it, let's run it for now. Of you group this Tags System Tags means that you On which executor is the particular build drive which executor to run it on Done went to commit Added a cuter Tags. For runner whatever execution on runner I have done the tagging required and committed it. Pay Runner's Tag have commit to me oh yes what are you went fine Now I'm not a CI/CD so I can use my I will go to the build, where did my build go?

I will go to the build, I will go to the pipeline, now you will see your pipeline after your commit. Whenever you need it, you can directly Go to the job and find out why you failed. Job Fail Prepare Environment status runner to mil usko den wai it Failed Preparing Environment This is Done Job Field is fine, you can go and check it out here. Ho GitLab Runner Run Done Run Time The platform is fine, the platform is also working come on let's See let's see if it is failed then let's Try to redeploy, let's see if there was.

Some Kind of Issue you can also retry failed jobs here you can do it, this can also happen. That your runner is about some particular thing Because of this he might be taking some action like I'm trying to force something to run So it is also possible that he may take some action.

[GitLab CI/CD] GitLab runners and tags / GitLab CI/CD pipelines (.gitlab-ci.yml) (part 2)

It must be going off like right now my build The job is done, build my job. Succedence Seeing Job Field Let's see how I prepare myself, this environment is failing every time. While our Joe build job was which build job was this one He was getting close, meaning there is something. In my particular instance that resets my environment If you go to build job here you will see that the runner ran very well and got Sourced and everything is going well everyone.

Agree to this that runner own this GN BA whatever personal our own runner was Everything went well but, that's the next stage. I am failing what does it mean that whenever the runner comes here, whenever When the runner comes up, it logs the old instance. Means once the runner starts If so, log out the old instance. What do you have to do to make it happen? I know there is a particular file Watch CD/Home/GitLab Runner and his You will go inside LS you will not get anything if you do LS han a will get a bash logout bash log out meaning.

GitLab Runner keeps logging out Why is GitLab Runner doing it, so we are will go em dot bash log out And there's something written here. That wherever you have to go through some particular process And ho ja the ye when leaving the Console Clear the Screen to Increase Privacy Clearing this screen And the runner of my instance is. If it's closing, tell me what's going on The thing needs to be logged out repeatedly Why is it needed? Executed by Bash when login shall exist when Leaving the Console to Clear the Screen increase Why do I need privacy?

I'll run my runner back After doing this I want to run my runner again. I'll do it now and see what happens Now I am at my jobs I will go to the pipeline and this I will execute the pipeline again. Sorry not download then execute I will do it, now see the one that is going to log out. The option was to log out after every build. I was getting logged out after every build. He was No option can you see this How many of you can a runner run from end to end?

[GitLab CI/CD] GitLab runners and tags / Dockerfile and image builds

How to set up with real-time troubleshooting shooting with a Pipeline Done Now What Do You Do Checking For Failed Jobs Post Done it's done Now what you will do. Else Sorry CD/Home GitLab runner builds CD Builds this Whatever your entire project is, Your instance has arrived TW S DevOps Your entire project Your instance Just the way GitLab on hum. People created a home directory home Stutoo / apps and a working one at that directory and your entire git directory was in it Cloned and running the project How many of you remember me?

How to do it, what to do Went Now, I understand the runner as well as the pipeline. I understood it and also understood the environment variables. I came and saved Tell me how many people understood the runner? What we should do We need to connect an instance to the Google Lab and on that instance we have to create a runner and that runner is your project. It will run so that you can be free of GitLab. If you can't rely on instances, we're covered.

Self-Hosted Runners Age Well, please show me one deployed project also. How many through CI/CD pipeline People want to see a project deployed. With a CI/CD pipeline, let's do this in One go is fine, let's do it in one go Let's do it in one go So this is your GitLab demo ci sadi similary. Similarly I cloned you a project made someone remember One of the TWSS DevOps We had the project cloned. This complete this Project You can develop this

project You will do this project?

You can take deployment, I have plenty of them. I will show you So I don't have it anyway, if you want to See I have some but this one will do it for you Dockerfile required or Dockerfile is there. It runs on port number 8000, let's clone it. The git clone it Now what do I have to do with it? The project with the CD node here gives me a dot git Lub C I'd email I need to get this file but from him First, let me tell you a little about this.

What kind of stages should a project have? We are going to tell you, we will build the project Then we will test this project.

[GitLab CI/CD] GitLab runners and tags / GitLab CI/CD pipelines (.gitlab-ci.yml)

Will we do it or yes then we It also gives a push to the project Will also use Docker Hub I will do it, I will push it to repo, yes. Is Docker and then we will build this project. What will remain deployment and that's it. If it remains, then the construction job will be there in What will be there in it, what will be the stage will remain Build everyone understands this. I used to use Vim and it was working fine too.

What do you need yourself Am Docker build -t then What notes not Node Plate Test dot And the tags that remain will remain yours. Dev because I want to build this on dev I want a test now similarly I have one Job whatever will remain will Some of it will work on stage test scripts will remain like. What script should I keep, I don't have anything for it You can add it if you want So Tags dev now here you are The important part comes Push.

Job now tell me what stage I am in I will write push I will write or what tell me so that I understand that you. I understood that I will write push no not push to Docker Hub whatever my stage name is Same stage name as defined above Push to Docker should also come down here. Come I should have done this now after I've done this What I did was yes, scripts. First of all what do I will docker login Now where will my username come from?

Dollar Docker Hub user jainki me to vid Credentials and things around the world are fine. IIN P DOLLAR Docker Hub PASS LOGGED IN What docker image to push again? Tag the old image name node Latest old image name then new image Of Name Dollar Docker Hub user slash your what node app Latest this will be yours and you will keep this thing What to do. Do you want to push to Docker? You will push this same thing to Docker. It is happening, meaning it has become so clear that he How many people can make this thing themselves?

It is understood that he himself is also a complete I can make it complete, what will I do now?

[GitLab CI/CD] GitLab runners and tags / Git and version control

Tag this too I will give you Def and [music] so and here Deploy as much of the GitLab as you have Put all the knowledge in this stage. Deployment what is the script Will it be Docker Run or Docker Compose and if you are using Docker Compose via You are running it and put some tags for it. I have You should check your Docker Compose YAML once. Where in this image node p batch six It is written where I consider myself only and Just keep it that way, Done, now you have to check this once.

Whether your runner has Docker on it or not Yes, that's fine, wait for the error to come. Git status on branch master modified docker Compose aided group lub group d dot ya group d Dakar Compose Group D Dot Galb Ce Git status two file edit git commit hynum Edit CI/CD and Docker. Chance Done now git push Origin master because it is the master branch is in Enter was also pushed Meaning you'll go here and hit refresh. You can see added CI/CD and Docker changes have also come here, which means You also learned how to use GitLab to create a Push pull through SSH with even instances.

You can also do a normal bit of a git There is work, the pipeline may fail It has already happened, let's see why it has failed. You will go to the pipeline in the build You will go and this has failed you, why did you fail? It has been seen There are jobs Build job conf should implement the script run and trigger keywords mean your Mamal Invalid, your YAML, you have done something wrong. Elite the script oh you have here messed up using scripts here Look, he did it, he said it directly.

Script is not allowed now here No quick fixes right now. Do the script This then the last one will also be messed up, yes now After doing this, just commit the same. Thing in the pipeline editor or in the ID could also do now checking pipeline Go to status Go to pipeline Now this Your pipeline is being checked is pending Stuck it, your pipeline is stuck it. When your runner is not available.

[GitLab CI/CD] GitLab runners and tags / GitLab CI/CD pipelines (.gitlab-ci.yml)

What happens when your runner is not available? Your stock comes so you Go to Settings in CI/CD Go Runners It is just available If it is not working then git Lap Run the runner and see, running in system. Mode Try to Build Again if Still Woe Struck comes We'll Find What's The Issue Let's Try to Run [music] again Running, let's see that he's getting a runner. Or if you are not able to run then it is coming [music] come on let's see what is the [music] Problem No to do CI/CD has problem why are you coming Let's Run Pipeline.

To Again Master if you want to change the parameters If you want to enter then you can enter parameters from here. You can also add something if you like it If you want to know the parameters, you can enter them from there also. This is happening by running the second pipeline Let's see if that one works or not. That you can know what if Something in itself, something in the pipeline The problem is, let's try to run this. TW's demo CI/CD pipeline here Try running it Go to Build Pipelines Run Pipeline Lets See build job done but then these are Pending means this pipeline is running Your runner is active.

Your runner has no problem, your runner is fine. Is doing a good job runner, your doing well There is a problem with that pipeline. The Note2D pipeline run try and see Let's see this pipeline is Gating Stuck let's bug that why is this getting Stuck This Job Is Stuck Because of One of the. Following problems there are no active Runners Online No Runner for the Protected Branch And no runners that match off of the tag c Runners look for protected branches It is also possible that a.

Runner No for Protected Branch yes we [music] Go to CI settings and see if this There is some problem with the particular pipeline. This project runner is dedicated to that Project Just Enables You to Know What to Do to do Now you can assign a runner to your project. Go to Build Pipelines and you Can o can you see the stuck part is gone now That's not stuck right now, your pipeline is Pending it's not stuck so go here and do that.

[GitLab CI/CD] GitLab runners and tags / CI/CD variables, secrets and artifacts

Runner Pay is connecting but it is not Stock It's being built right now, let's see. Go to the console log and see what it does in the build what's up lat C is still pending Yes, there could be one more thing, my. If you don't have Docker installed, Let's go and let's try to install Docker, to be on the safe side, we're going to Also installs Docker on the instance So that there is no scope that. Stk sudo a pt Gate Once the update is done, we will Also install Docker will take [music] Let this be it Gone Sudow AT get install docker and come and docker Compose.

What do I want to do, let's see what's right for me You need to change the command a little bit. Will have an error I will kill you, let's go C this will be yours install it in No problem, we go on our way line pipe Pay and run this pipeline back Are. Let's run the pipeline and let's see If our runner is, runner, I've done this job. Why Runner Picked Jobs Because We People have given runners for this project. Runner A, look, runner has been assigned.

Is and can you see successfully build and Tag Note P Job Success [music] what other jobs are doing come on Done, which push to docker gut failed? Pu Docker Hub feld y lets see Why it got failed because whenever you There will be a problem in the pipeline, can not perform An Interact Login from Non TT Y Device Let's. C Cannot perform active login Device Two Let's go We for all of you fighting with errors can't Perform interact login from non TT or Device solution was provided here Let's see the solution.

No solution found if your branch is not Protected than your variable in GitLab should also be not protective o let's go And this is your branch. Let's protect it from everyone First let's protect our master Branch What problem is happening in branches? Your branch is master This branch is unprotected. T GitLab link i down probed not Need the link problem was the variable was Not Presented for the Pipeline Oh, let's see what this thing of yours is Who is the pipeline?

[GitLab CI/CD] CI/CD variables, secrets and artifacts / CI/CD concepts

For C node CI/CD, whether that node Is that a variable for CI/CD? To Settings Let's go to CI/CD And let's try to get the variable [music] Come on God, why does this happen to us? Came Import Variable from another project GitLab Let's See If We Can Do That's what we call project level. CI/CD variable Project Level CI/CD Variable, let's see I hope each project. We have to define this is It won't be hectic, will it be anything? There's something very powerful about it will happen, we will find it, let's find it Looking for project level variables This is my project settings Variables store information that you.

Can use in job script j this that is fine Expand Tax ha so this is how I TW DevOps this Let me assign the variable to the project itself. I understood that in this project, whatever All of them will get these variables like this Let's try

something and see Let the variable environment be all master description this Docker Hub password Key Docker Hub Near. Lee and passed it here as the d variable so this How I'm doing it at my project level, now similarly I Let me do this too, this is Docker Hub user.

Project Gets This and We Will Go to Nuts to do CI/CD and Let's Try to Build the line pipe Again, let's See if your project variable can pick this up. Run Pipeline Le Trot Again let's build job how many people ya side by side If they're doing it side by side I have one more wonderful thing For. I have also created a project here on the jungles. Very interesting app with cool version Can you show me, I will make it public.

That everyone can see this, it's public Only See You Can Access this is another one I've kept you failed why fail ho ya pu job failed. Again incorrect username and pass something put it wrong I have tried using the Docker Hub username or the Docker Hub username. Honna Dajal Are Just Mistakes variable Docker Hub user is bean Updated with correct name now go and will not run the entire pipeline, only Retrying push jobs Push only. Retrying the job to help yourself take what is happening Permission denied I tret connect to docker D Man What happened to the socket?

[GitLab CI/CD] GitLab runners and tags / Dockerfile and image builds

I don't have permission to my Docker yet, what should I do? How did he build it, if this There is no permission Why did it build if it had permission? No Any thoughts on this that if it has If he doesn't have permissions then he ran the docker build command How Run Annie One Any One Who Can think sudo user mode - G.I.D. Uban to the group of Docker Let's c sudo reboot But I am having the same doubt that Think about how it would have built it.

Let's try to connect the instance. Again and Let's Try to Run the Pipeline Finally, and this time I have very Our pipeline of feeling Will sucked let's connect the instance There are instances running Connect GitLab runner. If you check the status then the services are running So come on, let's run our line pipe Again this time take the Lord's Naam Ganpati Bappa Morya Let's See If. Our Pipeline Gets All of the Permissions All of the Permissions The Password All of the Required Tools Her If the thing is near the pipeline, this should Work fine at the same time, lose say fail.

It's done what are you saying Why is permission denied? Has been [music] First of all, you can see that your Who is the user he should work Fine, let's see permission. Denied on GitLab Runner Docker Let's See Why Are Docker Is Tried for this Add the user mode gitlib runner to the service Restart Docker and let's see. T Works user mode docker git add lub runner done cat / tc / group lts c key your cut ssi group let's see near your docker Ubuntu and getline both have.

Let's try to run this again and see If my GitLab runner required permission Have you got it or not? Docker build is Happening So you don't have to give permission to either user. Why because your user is a group leader runner name is so that user you have to you will have to pay 4p for it. Lab Runner has to see Not Your Ubuntu User Let's test your pipeline now. In push to docker is done but deploy failed Let's See Why the Deployed Is What is the Docker Compose command saying failing?

[GitLab CI/CD] Git and version control

Not Found One reason for this is that, our The command of Docker Compose is something like this so like this cd GitLab node a here Docker Compose is such a command, what can we do? Were Docker command was coming from compose and hence If not found then what will you have to do? I am Dot Group Lob A bit about CE and Docker Compose commands You have to do this by modifying and make this change Git D DUT GALBE CE GROUP commit hynum Added Docker Compose Command Changes Git.

Push Origin Master what is happening This is something we did on GitLab. If it was pushed directly then it should have been pushed first. Bridge tax Take master and rebase your branch Take What happened, have you done it? It's over, this Git session should also be revised. Found that some conflicts occurred in GitLab CI Let's see what conflict has arisen. GitLab CI This generally happens when you will also change the master branch here. You will also change it, yes, look at this, something here So where did the complex come from?

Start scanning where Conflex No How many of you are getting that one? It's Not Very Much the Work of a DevOps Engineer As straight forward as you like youtube1 20 minutes we will be creating a If the pipeline was not like this then all this would have happened. Both Modified Gate D dot group lub CI/CD group commit group commit The commit message will be my Pass Git commit git push Origin Master, what happened to him? Let me git push this Why If You Want to Integrate Come I don't know why he is doing this There is no such scene, we

directly see it in this.

Not letting things happen so a small There is only change, there is not much drama so V can directly change it in my The file with GitLab has a small change. Too much time is being wasted because of let's see pipeline e running let's Go to a pipeline, what does it look like this time? You are watching Struggle that first YAML was wrong then your r I was stuck, then what about you? There was some error then your Docker There was a permission issue, then the project Could not find the variable then here But your bit in the command of Docker Compose.

[GitLab CI/CD] GitLab runners and tags

A mistake had been made, so with all these Mistakes you make when you completely abandon your job The happiness that comes when you build it No, she is different so don't go for it. Easy way, whenever you take any tool, always keep it Search end to end to see if you get any error. Resolving the error is occurring Resolving the Aya error and then When you go to your instance and then when you go to r security group and when you port number 8000 where your P is running.

When you open it You will tell the whole world that Access Tax Take So you'll sit on your Jenkins' runner this git Lob Your P Should Be Running Ideally your p should be. Running Done How Many of You Got to Learn GitLab in One Video that three hours sitting GitLab still Batch Six, you have put in a pull request. Let's understand, taught how many of you understood A project deployed via a CI/CD Pipeline in Your Git Lap how many of you got some Value out of this session and that's it.

Same old story if you don't get it from this session If you have got the value then definitely share about it. This session is if you get value from this session. Got it because you have u in the same video So what is Gut Love? How to Create a Project How to Import a Project from GitHub and upload that project to GitHub through Deployed do take a screenshot of this and Post on your Lincoln so that everyone. Let us know what is GitLab Now how many of you want end to end notes For this, whatever you study today, everything Notes, take it, take it Quick Excellent.

Will post a post where he This will tell you that this is the repository about Git. What are you doing for a month, three months, one month So post this right now, all of these Git Labs. That this report that you are sending is about this thing I want to create a PDF, I understand. Tool You can make a PDF from this, and that Share the PDF here So that everyone is able to know your efforts s well If you want to see you can directly go Two here.

[GitLab CI/CD] GitLab runners and tags (part 2)

Deb's GitLab Zero to Hero Every Day to You What to do is already covered Everything Project Everything Has Been Covered four kit have it as a pdf give him Shout-out to Lincoln. Docker's notes from the build I have yes Very notes, you must have seen the update Nut Index Engine Access Controller Multi Stage Docker Scout is a great note taking tool. Proper Screenshots So great work and all this notes all this pdf when Community gets Two Things Happen One They also learn and they always remember.

The Guy Who Taught Them So try to do that on your Don't limit your knowledge to yourself, but in the community Share it, it will do two things for you. Let us understand that the pipeline that remains, that's what the DevOps pipeline is for. Where your code Build Test We also saw four at the Deploy Lab. Stages Defined Code Build Test Understand this way after coding here But you do some kind of code Quality Analyze whether the code you have written That code is correct is there any flaw in that code.

There is some vulnerability to SQL injection There are some bugs and everything is about code quality. The scan is done and then your Docker build Is the build image created from this correct? If yes, then you can do it through TV or Docker. Scout's through Scanning an image Scanning an image Then your application test will be done. There is some compliance which is called Wasp Maybe open source or online web application Security protocol is something I care about. I don't know its full form but if you If you do it, then it needs to be known.

Wasp Open World Wide Application security project so this is one such project that sees to it that your All security checks done in the application All these things and codes have been done or not. Quality you can do so cube to SonarQube so that first code build test Deployed Code Quality Analysis Build Image Scan Test OWASP and Then Deployed Quality Analysis Security Image Scanning Security Web Security. Project security, so whatever your Getting into DevOps first Another layer has been added to what was happening.

Where you also see security and That is why it is known as dev Seck Ops that whenever your project is created in it Security parameters should also be checked and Thatchi yer is some how a DevSecOps. Engineer Age Well Because Whenever Your A pipeline will be built, you will have security in it. I will have to put cheques with that being said I Hope you enjoyed today's session here.

Argo CD and GitOps

Source file: transcript_B0bTEeM34DU.txt | 30,627 words kept of 72,259 | 79 sections

[Argo CD and GitOps] GitOps principles / Argo CD notifications and integrations

If reading Argo CD in the same video Want to become a DevOps Engineer? Where all the Argo CDs are Concepts covered. Introduction to gits.org CD of GitOps Principles. How Argo CD is Different How to Connect to Argo CD from Other Providers How to setup Kind Cluster on EKS How to run your first Argo CD on a cluster. You will create the application in all three ways from the UI Everything declaratively from the CLI Even e-commerce apps can be purchased through Argo CD.

You will learn to deploy this, look in front and I'll tell you extended things that Argo What are the features of a CD? Projects, App of Apps, Multiple How to run on clusters? Integrations, Automations, Slack Notifications Send it, send it by email. I need a well-secured HTTPDPS service. I will do the configuration on the EKS cluster. Argo That's all there is to the CD. Argo Project There are other projects as well, like Argo Rollouts. Outs, Argo Workflows, Argo Events If the entire CI/CD project stops so much If you have gone, do some interview preparation.

First, let's understand, what is this GitOps? We DevOps was doing DevOps. This was GitOps Where did you come from? This Why is GitOps necessary? Whenever I read anything Or I teach three things first. Why do you need GitOps? Do you need an Argo CD? How does an Argo CD come about? No need to register anywhere There is no need to send an email anywhere. You will get all the open source stuff for free. Let us understand what is GitOps? First of all understand that this is Argo CD Some things you need to know to read this tool What do you need to do first?

Where I have taught Linux. Kubinetis Docker, Jenkins, GitHub, Shellscripting, Teraframe Ansible, Git Labs, I have read and kept everything. You need to read GitHub. GitHub Because of this you will understand that these GitOps Why are things important in. You should read I have Docker because I will run it in containers. Docker is very useful for running the application. Argo CD as a concept It's built for Kubernetes. Cloud Built for native applications. What GitOps is all about. I'll tell you diagrammatically, GitOps What happens?

[Argo CD and GitOps] GitOps principles / Git and version control

First of all, understand that Git is What is the version control system here? Your source code Then you can share the source code as well as your Also write Kubernetes manifest files You can see the ymail files that are stored here. You can use your Terraform You can also keep infrastructure files. Meaning if you have to build infrastructure. If you want to build infrastructure, then The files are also on Git. If you Kubernetes Application Kubernetes The files of the application to be deployed Your source code is also on Git.

Which is your actual application There's a code that's also kept at the gate. If all the things are kept on the grid, then why Git became the single source of truth Give. Meaning, it should not be the case that if tomorrow I got a cluster of Kubernetes. This is a cluster of cubes in production. That the user can come here and do something in the cluster itself. That the user went to the cluster and started this application This did not happen tomorrow.

It will happen that the user comes and installs this application. If the source code is in single source of Truth Complete Your Kubernetes App Yamail The files are located here. Application code Yes, the infrastructure is all here. That's why Git is considered a single source of truth. This thing is called GitOps. Meaning, if You should use the application in Kubernetes. What to do if you want to increase the replica as well? What about Kubernetes in the GitHub? This is the y file of the application, update it.

End if update in git Only if this happens will it be applied to the cluster. You will not go here and apply manually. You will not apply in any other way. The first thing Git does is your version Control. You can see which one is currently The version of your app is running on my cluster Pay. Because merge requests, pull requests, Review of code, all this is already installed on Git Because of this, your security is also a It's visible in place.

Plus observability You can also observe who did what and when Committed, who changed what at what time When did you change it?

[Argo CD and GitOps] GitOps principles / Git and version control (part 2)

Git gives you a lot It gives all the benefits due to which Git Ops became so famous. Your Git repository should be a single Source of Truth for Your Deployment for Your operations. How can GitOps work like this? The updates of the Kubernetes app on the cluster How will they work? A controller is required per cluster. A controller is required that can handle changes from Git. Pull.git from your GitHub Bringing Changes to Bridges and Kubernetes I'll run it in your cluster.

Who will be the controller? This is a service There will be a tool that will help you manage your cluster It must be running in. The name of that tool is Argo CD. GitOps Helps to do on the cluster sid. Argo CD always in cluster Your code will always be on GitHub and You will pull your code from the Argo CD and You will deploy it on Kubernetes. It's a simple This is the explanation of GitOps, what is it? Declarative means manifest files Or whatever the configuration files are.

Someone going in the air like this Don't write the command. That is where the declarative comes into Picture: Version control git which version There is a control system that keeps its commits Track your version history by tracking branches And that's why version control works. Automated us What should happen as soon as any new change comes on your application or your On infrastructure or anywhere else. Be deployed in an automated manner. An automated controller Which is Argo CD. Automated methods Observable however many logs you have, however many you can see all of them in the matrix.

Low End Intro to Gates Argo CD Go away. Declarative that the email manifest version is Desired State Automated Argo CD Observe, everything is written. Understanding GitPos and Traditional CI/CD What's the difference? It is already ready to deploy the application Why are you teaching us a new tool? This is where your git lived. The source code was kept. It remained Your Qtys cluster. Here Your application was deployed. You used to do CI/CD using a tool called Jackins. Let's say these Jackins uncle what you Did you use it?

Meaning the code came, It was built, tested and deployed here.

[Argo CD and GitOps] Helm with Argo CD and production GitOps project

You can use DevOps on this. First you push through Jenkins yes. You push the changes through Jenkins. They write the code and push it to Git. What does the CI/CD say? End then if your application is correct then it will push or gets deployed. If I to a Jenkins that is outside the Kubes cluster One tool that is kept is the CI/CD tool Traditionally, he has permission for this cluster. If needed, should I provide its credentials? You'll need to provide credentials for this.

Of this Jenkins and this Clubis cluster There must be a connection. Where you have to provide credentials What comes up there? Availability means if Jenkins rolls over tomorrow So if Jenkins' server and your Connection between Kubernetes clusters lost When I went, I had all these problems. Its What is your CI? Continuous Integration is available on GitHub itself. Do it or do it through GitHub Actions Go through Jenkins. What should you do about the CD part? For this you can install Argo CD.

Argo CD Put it on for him, and what will he do? He will remain sitting on the cluster. He will continue to sit on the cluster. These changes He will do this bridge. As soon as the developer pushes the changes Did this manifest file or whatever code is there will go to GitP and from there whatever The configuration is that it is pulled from here. Argo CD will be installed on your cluster. If Argo CD is in your cluster No credentials required.

Role base You can give him access through access control. No connection is needed because He's sitting right there in your cluster. This overhead Jenkins has repeated overhead of pushing It disappears. That's why this wonderful Pipeline is written in the comparison chart Direction which is the traditional CI/CD Always pushes. Argo CD or GitPass Operates on pull mechanism. Which is source of Truth always lives here at Jenkins' lives nearby. Because the Jenkins of Joe Jenkins The entire game depends on the file. Source of Truth Your Git which is the repository of your manifest file Your Helm charts are yours The source of truth is Security CI/CD.

The tools that are there should run within the cluster itself. Drift Detection Yes This is asked a lot in interviews.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster / Deployments, ReplicaSets and rollouts

The question is, drift detection What happens? Drift means tomorrow your Application exploded. Jenkins is not Checking if your application is running Or not. That will happen when his next When the deployment happens, he will go and push No. The replicas of the application that are running and The manifest file in your GitHub is let It has been said in Gihab that two manifests should be given. Requires replicas and your Qubes cluster If I crashed in a car, immediately monitor it. If he does it and goes here and makes two pods then Healing Factor and Drift Detection Corresponding tells the thing to auto heal This is a very powerful feature.

You need to view the CI/CD logs to There's nothing to do here. Many people ask what is DevOps,? If What are the security risks in production? What are the availability and how many There are also huge risks if you such tools as Argo CD If you study tools, you get a job. What is the difference between GitPos and traditional CI/CD? Tell me, what is Argo CD? I did a little I gave the trailer of Argo CD here that he pulls. Think of it like an open source There is a project called Argo Proj.

Let's make it a little smaller so that the whole It should be done in this. Argo CD Argo Roll out. There was your Argo Work Floss So these four projects were made. Argo Praj yes so Argo Praj if you look So these initially they four big Projects were created by Argo Workflows Argo CD Argo Events and Argo Rollouts from that to make continuous delivery. The project that was for Kubernetes That was Argo CD, I mean, very, very This project is a more popular project It is written in Go language.

There are so many issues in the repository, There is so much focus and most importantly, so much They are stars. Too much, too much It's a widely adopted project. Let's do the Argo CD. Like you Heard all this in the Argo Project There were projects. Argo CD was one of them. Project that does what it basically does For your Kubernetes cluster Deployment does it. What does Argo CD do to your Kubernetes?

[Argo CD and GitOps] Git and version control / Kubernetes architecture (control plane, nodes)

Because it's part of GitOps. Runs on the principle of GitOps What does the principal say? All manifests The file, all the code, is on GitHub. Argo CD What will he do with it? Will he pull it, and where? Deploy your Kubernetes cluster Pay. This is what Argo CD does. It's a Continuous Delivery Tool For Kubernetes which uses GitOps Principles. Use GitOps Principles It does. The question arises, how is it Here you do how, why and what. What is Argo CD, you have told me but Answer to why we need it is in the gates because all the Security and Overhead and Observability and Logging so many constraints were Argo.

CD solves that thing, how? How does it work, first of all you To understand Argo CD, use Argo CD. You'll need to understand architecture. For Argo CD There are some components. Argo CD has some components which You will have to understand it very carefully. The first component is the API server,. Let me call this API server. The second one remains Here's your repository server. These three There are many important things for you In Argo CD. As many UI components as there are Any number of requests are to come, I have to know the responses, all those things.

The API server sees it. API means Application Programmable Interface. The API server that handles all requests whatever you're coming from the command line interface whichever UI is coming from somewhere else All these requests are also coming from sees even those coming from git. Requests coming from the web are also He looks and this is the front desk from here The repository takes the request and gives it to you. This is the repository of the server. Understand the work like this, that the repository It's a server,.

It's your Treat GitHub as a Git repository clones it and it uploads it to the Git repository as a recipe He understands the book and whatever recipe book there is Nor does he take it inside himself. Whatever manifest is on GitHub I have kept the file, I will take it inside me. Will understand all the manifest files and will Pass will make a copy of that repository. Talking to these people about these repositories Any requests to the server and API service Your Cuban team is coming to talk to them.

[Argo CD and GitOps] Deployments, ReplicaSets and rollouts / Git and version control

Will deploy things to the cluster. Sync things across your Kubernetes cluster I am in, I am running, I am running, I am not How are things going, everything here So from the API server you take requests. From the repository server, you map a repository with Git and in Application Controller Actual Kubernetes talks to the cluster. All of them These are the services running inside. Your Argo CD's CLI outside The command line interface also remains. Who will take all the requests? API server will take over.

Repository server will take over. Application Controller It will deploy it on Cuba. This way A great bit from Argo CD's Yours Architecture is made. We It has been seen that why Argo CD and How Argo CD Works. From GitHub He will bring the code. Whatever the request is, will process and deploy it on Kubernetes, this is similar to Jackins X It seems like it's a flux CD Which are the tools that work on GitPos and deploy to your cubes. Why am I reading the Argo CD?

Argo CD basically gives you a command line Allows deployment through the interface. Allows deployment through the user interface and also through declarative files It allows for great deployment. All three How you can do it in Argo CD Deployment with GitApps.FluxCD On the other hand, it doesn't give you everything. CLI will give.jkins.exe a different overload Because it's a part of Jenkins. Argo CD in short for your Kubernetes He is very helpful. I have also The whole thing is put here, how it all the components and how they work with the rest I'll show you here.

RGO The UI of CD is very much like that It has strength and a very easy UI. Jenkins So he is very strong but very complex. YAML and it doesn't fit into the entire ecosystem. You've got a little something for Jenkins X. There is more learning to be done and the adoption of it is very little compared to Argo. I hope you understand that Why Argo CD is so important and Argo CDs by Jenkins and Flux What is the difference between them?

No So you can pay AWS, no? Let's understand what Argo CD is. Argo CD on your Kubernetes cluster If it runs, a cluster of cubes is required.

[Argo CD and GitOps] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

First of all for that you Kubernetes and One Look at the shot, don't be afraid, I will explain it in this. You can also find a Kind of Kubernetes at your local store. You can create a mind cluster or on the cloud You can also create one. I'll walk you through AWS. I'm a I'll click on Launch Instance. There is no difference, so it has become one now. I will create a key pair named I will give Argo Let's create it, I don't have it.

If you are on Windows and using PuTTY If you do, use dot ppk. Create Here is Argo CD and One Your shot is great. My In the network settings I'm allowing SSH I'll leave the traffic on. HTTP It's to turn on HTTPS. Because I need to connect via SSH. I'll put it in storage. I take around 20 GB because there is a lot I will run all the applications and show you. Initiated launch of instance done. I created a test server.

All these steps of SSH client I will follow it as it is written. This requires an SSH client. Connect your instance using public DNS. By the way, to run this setup You need Docker for that. Let's run all these commands one by one. What does sudo apt get update do? You need to install Docker. Kind of a cluster of Kubernetes. Sudo ap install dockerio So I install Docker. That Docker is there when you install it for the first time. If yes, then he would not have permission.

I would have written after installing Docker ps so that permission denied He tells me, so what am I supposed to do? I need to give permission to my current user To run Docker, I ran this command I'll give you permission. If I If I run docker pi, it will work fine. To create the cluster I need kind for this. Cuban is in Docker This Cuban Is in One Short Video I've told you. Whatever kind You need a package for the cluster here I'll have it.

Chmod X means it's made executable. I kubectl also has to be inserted because Interaction with the Kubernetes cluster So kubectl is very important. Install kubectl from the Style Guide.

[Argo CD and GitOps] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / Kubernetes architecture (control plane, nodes)

I, you have to do it in Linux. If you have created the executable then kubectl version - afn running from the client version 1.34 is running Well done. You can also take Argo CD through Helm. You can install and normal kubectl You can also install it through. If you have Helm installed If you want to do it, open the Helm guide and By going here you will get three simple commands. The script has been downloaded. Helm is also taught by me.

We will form a cluster. There's also a script called Argo CD dot I have put in it whatever I have just manually all those things, fat I've put you in the script. Make me a kind of cluster. This is my kind of config file Kind of like configmail to vi Kind ConfigMail So that will create my Cubetis cluster. This is my way of creating a Kind Cluster. This is the manifest file. Kind cluster means you are looking for a cluster type Writing the manifest file.

API version Kind then you make for a Kind cluster You are using the API server for networking. This is basically the address of your instance. Internal API Server Port 33893 He will talk to me. Control Plane your main control The plane will remain of the Cubant cluster and two We'll put the worker node in. Wq enter this is your kind config created. Using this config What do we want to build? We want to build a cluster. The command is very simple.

Cluster name Argo CD cluster of. Kind config in config I'll use this command. I'm going to copy and go over here and hit enter. What I've configured provided, on that basis one Kubernetes A cluster of is being formed using Kind. Run it on Linux, run it on Mac OS, Run it on AWS, EC2, whatever. Because it runs on Docker. Kind Cuban is and Docker must It'll do. Let's create the control plane. Once this is done, you will get the Argo CD.

The project which is there, the tool should be installed You will have to do it. Here are two ways to install it: There are different ways. The method is Much easier way from the Helm repo and This is what is used in production.

[Argo CD and GitOps] Helm with Argo CD and production GitOps project / Helm and charts

The other way is your cube CTL method: I will tell you both methods No scene, no scene No, I'll show you quickly. Kind of created cluster, it's great. Production style which is the hell It is recommended, what should you do? In the Helm repo add the Argo project There's an Argo Helm repo, add that. Argo has been added to your Helm Repositories. Helm Repos Please update so that you get the latest news. The charts will come to you. The latest hand charts have arrived Yes.

Space will have to be created for Argo CD. kubectl create namespace Argo CD Created. You need to install the Argo CD via Help. What if Helm Argosy is as good as Manifest is all the charts and send that package to your Kubernetes Sara will apply within the cluster Will install all versions of Helm. If you manage it yourself then you can see it here. Sakta Ho Name Argo CD Deploy Status The deployment has been done well. If you like Argo CD's UI If you want to access it, run this command.

kubectl Port Forward Service Argo CD server namespace Argo CD 808043 So run this. Whenever I cube a ctrl If I forward the port then my The screen is there, the terminal is there, that's stock It happens because its logs come And so on. I'll open a new terminal. Open a new terminal, a new session of type I'll do it. Back to CD downloads I will go and my instance I connect to the instance from another terminal. I am doing this so that Forward as many ports as Argo CD allows I have to do that at this terminal.

I'm all about port forwarding on what is I will run the command and click on the green one. This command was kubectl Port Forward Argo CD's Here I Go Who will I give it to? Before that, he First if you want to see Kubernetes Argo CD pods are running in the cluster Whether or not kubectl Get All In The namespace Argo CD One Time Look at the things. Your The deployment is up to date. You will need to copy the lo command from here.

Forwarding ports, but what I want is I'm here, which is my Argo CD server,.

[Argo CD and GitOps] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / Argo CD installation and setup

I'll make a slight change to the command. I'll go to that instance. I'll go to the instance's security group. This is my security group,. Here I will add a rule that The port number which is 8080 is the port you Open it. If you want your own If it's for IP, enter my IP. This must open on your IP address. There are a lot of thoughts going through your mind Block them and this IP address Go ahead and type in 8080 and you'll see it here.

Insecure connection HTTPS I will tell you when you do it on EKS, then I will tell you. Click on Continue to Site and Go to Advanced and click on Proceed Your Argue CD server is ready for you. What a great thing, it was done within 1 hour. I even got Argo CD installed. Go into your notes and You have to pick up the initial admin secret. Initial admin secret with this command Where do you enter this command? By the way if you have this on in the background If you want to use it then control it and kill it.

If you want to do it with the kubectl If you have, you can do that too. If you If you want to do it with kubectl then I will give you a I will do the work and give it to you. Kind delete cluster Name What was the name of the Argo CD cluster? Argo CD, so the Argo CD cluster The name was Argo CD Cluster. Kind Delete Cluster Argo CD Cluster

You can blow up the entire cluster. You The entire Helm repo is also deleted.

Pay Kind Create Cluster lets you create a new Create a cluster. That is the installation of Argo CD. Using official manifest cube ctls It is easy, that too is the same method, cube By applying CTL you get a namespace You will have to create like you help What does everyone do to help. Keeps you already packaged here If I have to do things manually, I'll go here. Pay deta kubectl create namespace Argo CD Done Create Good I will bring it here now.

I just put the installation manifest I'll apply it. All the services that are available, He will create all the resources he has.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster / AWS (EC2, VPC, IAM, S3, ECR, EKS)

If you do kubectl get ports in The namespaces Are Go CD So you will see that the container is being created. Some init containers are running. This too Is this the ready status of the Argo CD server? There is no scene for it and the In this way you have got error lost connection here It had become two ports, the way you Port forward of Argo CD's service. You did it in the same way as you did in the manual Please forward the port of the installation as well.

Let's check if its port has been created or not. No, yes the running is done, let's do the same now. How to access Argo CD's servers You will do a good port forwarding and here Pay you will go and refresh then your Argo CD must be playing but it is fresh. Its password whatever you test @ 1 2 3 If you had entered it, it would show not logged in, so log in. If you click on it, this will be your manual method.

Current password and Very secure test @123 and tests I added it as @123. This We had a great time with the Argo CD cluster. I've shown you how to setup Argo CD inside I taught you. Argo CD on client cluster done. After installation of Argo CD on EKS I'll tell you. I have to show you that How to configure HTTPS in EKS With Argosy CD with Cert Manager I'll show you later. When I do this one Teacher then I show setup on EKS I'll give it to you.

I'll tell you Let me explain the components of Argo CD. It is important because your application You can use that application by using the same. Deploy sync on a Qubitis cluster Yes. There will be a project, that project name It will come here. If you start a new project If you give, you can name the project anything. I started the project with Let's. You can also use your GitLab wherever you are. One or the other repository is given whose name This is the Argo CD demo where I installed the app of apps command line declarative image Updater Applications All the UIs are in place.

[Argo CD and GitOps] Git and version control / Helm with Argo CD and production GitOps project

All the applications I have posted a demo of this. You can connect it to with your Argo CD cluster. You Want to access via SSH, http HTTPS want to go through. GitHub app, Google Which way do you want to go, Cloud? Select HTTPS There you go. You have your Git repository. Argo CD Demos you do that You can put the project here. What do you need to do with repositories like I did? I told you here in the diagram that your The repository server is what your repository is for.

Server, what will it do, it will upload it to your GitHub or GitLab will save whatever code is stored on it as A recipe book. Interpret the files and deploy them will do. This repository server It is very important here. If you need multiple clusters If you want to connect then you can click here You can also connect clusters. Cluster which remains clustered by default That's a cluster. If you need more clusters, If you want to add then you have to use Argo CD Add via CLI command line interface I have to do it.

First of all I have a cluster which is my I'll add the cluster. I have the Argo CD. Which is the command line interface from which I will talk to your cluster about that. I'll go to Argo CD's CLI. I would use a command line interface like Argo Well, it's not finished yet. Argo CD Argo CD Yes, done. If I Argo CD version client to here I should do it. From my command line I'll log in to the Argo CD server.

All the Nibb Pookies ones Don't take this as a bad sign, log in. Argo CD login is the command that lets you log in You will need to login to the public I need an IPP address. Once I'm logged in, The type of cluster running on your system that cluster is being linked to your Argo CD Let's go clubbing with the server. Argo CD Account Gate User info. Helm and Manifest Installation with manifest file. I like Helm I actually like it more.

If you If you are a beginner, you can do it through manifest also. Preregistration what-ifs I need a cluster, I need an Argo CD I need the CLI, I need to be logged in.

[Argo CD and GitOps] Kubernetes Services and networking / Deployments, ReplicaSets and rollouts

I'm trying to setup my Qubes cluster on Argo. There So, kubectl config gate Context What does that do? My What is the name of the cluster? I'll just do it Argo CD Cluster Add whatever is the name of my cluster. Give it to Argosy cluster add to which cluster? I Kind Argo CD cluster that I have I want to add it. I want to name it Argo CD cluster. I want to name it Now what I'm adding is HTTP. This will create a new service account.

What is a service account? Argo CD Manager A service account named will be created which will You will be able to authenticate using the role Binding and so on. Saw the role binding of the cluster It has been created. Argo CD Manager role Some access will be to deploy to get Resources in your Qubits cluster. Cluster Added Now if I go to Argo CD Cluster If I list it, I will see it here. This is my cluster whose name is Argo CD clusters have been added.

Someone inside that cluster So if I Go to UI, go to Settings and If I go to clusters, you can see I can see the Argo CD cluster. This is how you add a cluster. Your Kubernetes Argo CD cluster Added the cluster with the add command. Automated R Deploy from CD I want to do this on my Kubernetes cluster Pay Run. Project Name Project The default is 0. Sync policy if I want that Joe of Kubernetes There is a cluster and my Inject app is on it.

Automatically sync Manual sync If I go then this sync policy remains. I click on auto create namespace so that if I have any namespace If it is not created then unnecessary errors will not occur. This is the source of engineering. Where is the manifest file? I'll show you where it is and I'll tell you was right this is the Argo CDs demo It has a UI approach, engineering and I studied engineering in it. The deployment file is installed. Created an Engineering deployment file which has three replicas.

On this port It works. This is the manifest file that is stored on git. It's not on this server, it's on Git.

[Argo CD and GitOps] Kubernetes Services and networking / Git and version control

If I were to put my Kubernetes cluster on If I want to run Engineering then I can directly I'll connect the gate. Which branch is it kept in? Argo CDs Demos in that repository The UI approach is the injectix path. I will write it in the path, yes its UI The Underscore Approach There's some issue with the UI, please check it out. The Argo CD Project have put together some Not done. Inside his manifest The file is already there. Git Is the Single Source of Truth.

The cluster's URL is then Where from? This is 172.131 this is my own kind of cluster so in this I will put the namespace in default I'll give it to you. If you have a namespace Just put it here. As soon as you create, you will click on it, such a great thing Project application will be made in your Default project. If I Let me go to cluster and write cube here. I will put it in the sync Auto create name in this git repository Should I synchronize space?

As soon as I synchronized the inject service. Inject's deployment pulled Three replicas started being made. If you understand that from One UI How to access the application from the UI of Argo CD You can deploy it through Gits. As long as your eyes are open that browser And until the visual images are visible. Cube Whistle I'll do gate service. Here I look The engine is a service kubectl Port Forward whom service injects service port Forward it, forward the port on which one?

You map port number 80 to 8081 So this is the service port that I have here I will write the port of my system. I'll write it here and I'll address 0.0 Enter now my 8081 engine The map is done. 8081 To If you want to put it on my IP then my IP Put it on. Not HTTPS, HTTP Welcome to Inject. Engineers to use on your cluster. Remove the engines and take any image Put that app and your app will be deployed.

You can use the CLI You can also run it from the CLI method and It's easy. You first Deployment, which is this repository.

[Argo CD and GitOps] Git and version control / Deployments, ReplicaSets and rollouts

Here's how to switch to a CLI approach. All the commands are written for you. Argo CD App Create What will happen with this? This will create your application. What is the benefit of creating an application? Create an application gives you a small thing You will find out where to put the repository. Destination What will the server be like? What will the namespace be like? Self-feel to Joe What you did in the UI, these ones, these ones Quickly go to your Applications on the UI This one that was filled Which one will be yours?

Let's do it via the CLI. The Argo CD app Create is this command. It requires a lot of You will need to enter all the flags. CLI approach Apache or whatever your There are demos inside it which are Apache The app I'm going to show you is on this path The Apache CLI approach is provided here. Apache's deployment file is And So On So what will be the path CLA approach a Apache so What did they do here? These paths were entered, now the destination server The path of your cluster here Which private URL will come, then that How to get it?

Argo CD [music] Argo CD Cluster List And this path, this one, this one, that demand So copy this and this Go to the destination server here Slam it. What does sync policy automated mean? How did we create Nginx in the first place? Clicking the Sync button It was lying there. Cell phone means if you accidentally You deleted the post here. Your Replicas to in the manifest file on GitHub One for all that is written, sync it. Autoprun whenever you want and even destroy It can be done automatically.

This does not mean that he will blow up auto prune means next Whenever you reload at the time, the old By deleting it, he will make it cool. Argo CD app create apache app you app The name of the repository is this, the path to the repository is this The path is the destination server, see it is done. Glitch in destination server As I said, modern editors are like this Do not use modern editors I would say use Vim as well.

[Argo CD and GitOps] Kubernetes Services and networking / GitOps principles

Apache service Four replicas were shown on Git. Replicas for so I made four replicas here. If I will do this here cube whistle, get pods so look There are four Apache's running. Like I injected I had exposed the service of I'll expose that too. Here I am Cube Seat Port Forward Engine There was a service, similarly there is Apache service also. Apache service is here and I will put the port number at I'll send you to 82. It is running on port number 80.

If you What can you do if you want to confirm? kubectl can do gate service and you will see that the Apache service is on port number 80. Command line Interface Argo CD app create. You I told you from the command line as well. You can do something that I don't want to command. You don't have to go in line or go to UI. My app gets deployed on Argo CD Through Declarative Approach Meaning So I will create a file like this: I will create a YAML file like this The file in which Argo CD Application of.

What should be its repository? What should be its cluster? All these things are one It should be written in the manifest file. This I have the manifest file uploaded to my git. This way I can do the actual GitOps My Q is also my Argo CD setup. Gitp is in.YAML of my Argo CD is also in git. My Cuban's The manifest file is also in git. Git becomes the single Source of truth making it pure GitOps. In the declarative approach I will give you a I'm going to give you an example of an online shop.

This is an example of an online shop, how is it How is Docker built? Docker in One How are shot.ml files created? Its CI/CD how What do you want to do? This YAML of Argo CD is your Kubernetes The cluster will understand. Definition CRD So this Yammel has Argo CD in it CRD of the application is created which You are looking at the application,. A CRD is made for this thing and that CRD That's the only difference that applies here Or that's the only trick.

A CRD needs to be created for the application.

[Argo CD and GitOps] Namespaces / Argo CD projects, App of Apps and multi-cluster

Argo CD install Need, Need CLI and KubeCettle We have all four of these. The cluster is running fine. Create a CRD Creating a CRD is too much. To create a CRD, you have to create an application,. Kind of an application because you are a You will create the application in Argo CD. Its API version is rgo prog.io alpha v1 alpha this is what you need to enter. That you don't have to learn the whole thing, this Argo CD is This is a one-shot.

All these fields are written In your CLI, you have written this command I had put this Argo CD app created for this. Equivalent Yammel File This is Kind Application Meta Name Online Shop App What is this will be the name of your application. Why will the namespace be Argo CD? You'll also find Argo CD's namespace Kubernetes The

Argo CD namespace inside the cluster This CRD has to be custom made in it. This spec in These fields of yours were named The project name and all that will come here.

Spec Project Name Project Default Source repository HTTPS GitHub.com You can download your Whatever your username is, enter it and do it. Target Revis means Branch Main Path is whatever your path is. Where did Argo CD go in demos? A declarative approach here at Demos It is. Here I'm going to put it in the cluster list. This is the manifest for creating CRD She's in. This is my GH report, this is my Server URL Sync Policy Automated True If the cell field is true, it will remain automated.

Mast click-click sync I'll have to do it. I will apply this to my on the cluster. I would copy this I can either copy it or use Argo. I have CD demos on my server. Cloning into Argo CD demos is here. Gone are the CD Argo CD demos so I Whoever was doing it on the git, come here. Declarative In the approach and in the online shop I have what Online shop appl the online shop app. I will do this on my cluster.

kubectl Apply Hafen f Which? This is my YAML file, this is my cubit The cluster will be applied in the namespace Argo CD Enter Application Argo CD Praj Online Shop App Created means here if I will see.

[Argo CD and GitOps] Deployments, ReplicaSets and rollouts / Pods

Cancel then online shop application should be created It's gone. I've also shown you how to use the CLI. I showed it to you and I also gave you a declarative means YAML file So this thing is working. It's running, no problem, it's great cube CTL Gate Service A online shop service with port number 3000 Pay is running so port forward when What should I replace the Apache service with? Will go online The shop service will be available for those who are in the cluster.

The port inside is port 3000 And what I have to keep out is 8083 tax I'll give it to you and open the address. Enter Port forwarding is done. The online app is deployed. Your e-commerce The application has also been deployed. I am the other one, this is mine, yes so What do I do here? kubectl Gate pods This is the page of pods Get Kiya Online Shop. Ctl delete If this enter is deleted then delete it from here also. Should I If I refresh here, git What is he saying on the git pay which is the source of The truth is, it says three.

Here if I look at the cube CTL Gate pods There are three running. Becomes the source of truth. I will go to Git. Not Argo Project, not here. I'll go to Argo CD Demos, now I'm going to take a declarative approach. What will I do with the deployment of the online shop? I have three replicas in it, I'll increase it to five replicas,? Commit Changes committed Commit Now I delete it here. I gave this pod a cube,. Ctl delete pod this one got deleted but kubectl get pids if i do o this sync did not auto sync yes this issue Auto sync comes, yes this issue comes Because your HTTPS.

I am insecure, that's why I feel like I'm drowning, but no one else is. It's if you just put it in the sync Click and click on Synchronize What if two? You see, some pods and a few. This way Say my git becomes the source of truth. I can make as much noise as I want here. kubectl Gate Deployment kubectl Get Deployment Online Now Shop Five's kubectl Scale Deployment Online Shop Huff Replicas is equal to 10 letters from me Scaled it.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster / Deployments, ReplicaSets and rollouts

By breaching security by entering the cluster I'll sit there doing the scale. If it says five on the git, That's five. This is how I git Can I tell you what my truth is? How many replicas do I have running? If I Scale and Deployment I should make the replicas one that the five brothers Hit delete. This becomes a single source of Truth. Yourself Great application by applying the manifest You can create it. Such applications can create questions No. Tag Lindin on whether you Argo CD Are you enjoying these one-shots or not?

We have You deployed the application. I've seen the UI method, I've seen the CLI. Let's take a look at some of the core features of Argo CD. Let's talk about the core features of Argo CD. If I do, there are a few things you should keep in mind How to make projects? Multi-cluster How do you manage your application? You will learn it very well by then but I will tell you I want to teach in depth because if you understand how to write YAML, then you can do all these.

This is us People created such a project as online shop It was a new project. It's like Kubernetes A cluster has a namespace. The amount of access you want to give to the space The number of resources to be grouped Have to combine, same as in Argo There's a project. In a project, you What do you do? You create projects and You

add applications inside. You have a Created project Front End Project. In that you created an application online shop Applications.

You have access to all the You have to give the application to all those to whom you have to pay the fee. You have to give permission to this project. The spec should be What is the name of the project? Project app Which one is going to run? Which repository should it be? All of them You need to know how to write permissions. I Let me quickly teach you how to do a project How to create a YAML file and how to apply it happens.

Vim creates a folder I'm here. Some project folder Is it? Mkdir Projects, let's go to CD Projects.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster / Git and version control

A one here project.YAML this is a great write up I will give you this project Dotal, it is very There will be a simple file in which your API. What will be the version, will it be your Argo? Question dot In IO, your V1, all these things which I have put it here. If you go to projects then here you will You will get the project. V1 Alpha V1 which You can also use it from here. The important part of this is that it is kind What will happen is that this YAML file that you create What kind of person are you asking?

I will have the name and your name will be the space. Namespace generally whatever you want for the Argo CD. The namespace for the YAML files you create is Your Argo CD remains the same. The project What do you want to name it? Let's say Front End Project. For example, if Give you a description of this project So whenever you go here, new What happens when you go to the project? Project name and description It comes. This Project Holds a New Front End front end app.

Its source What is a repo source repository? What do I put in the repository? My repository was these Argo CD demos So this is the demo repository. You can add your repository. Whatever project is in your repository is kept, whatever application is kept, Drop his repository over here. Source report And here's the list. You may be in multiple repositories You can also take out applications from here. You give And your server lives there. Your Argo CD server Private URL remains and namespace if You need to enter a namespace from Let's If you want to put front end then you can put namespace You can put it in.

The server has your server's If the URL is there then quickly click on the server link here. If you want to get the server URL, What happened to him? Argo CD cluster What will happen to me? Take this URL and Put your project into the server in ML I'll give it to you. This is my URL where My application project is about to be created. Cluster Resource White List means who will tell you which ones from which groups Resources are needed.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster / Namespaces

Namespace service, all those things come Then deployment within apps, all that In groups, that group and its kind Everyone is allowed. This cluster resource I will whitelist here. All types of things in cluster resources All types of resources are allowed to me. Who are in the namespace whitelist? Are you allowed to use namespaces? I'll allow all the namespaces. Project name What is this? Name front end team so front end team and Front End Admins: This role is for this project. Applications All applications Allow the front end team to do everything.

I will give you my in the YAML file and put it here in my Rolls. All these things come in meta data and your spec and whatever your source and The destination becomes all of this. Just once Let's take a look at the CAT project. The repos in the source repos Yes, I think everything is looking good. kubectl Apply Hafn if Project Dottle in the namespace Argo CD will do this. Enter the project front-end team Created. As you refresh, you will see that the project named Front End Team It has also been created.

The front end team The project holds a new front end app. These things that you made here cluster resources and whatever These roles, events, all things are added here. It's interesting that If you learn how to make YAML then you will Put Yamel in the gate. As soon as your cluster is created, you will be able to do all this. All these things What is the default project? What are the project roles? How Want to make a project? Everything in Notes So that you can get it anywhere No problem.

From the project list, you can use this command You can see which ones you have. You can see everything here, it's great now Under the project, you have to create an app. From a declarative approach, meaning YAML If you want to make an app then here is an M I've written an email for an Nginx app. There's just one small thing you need to do What you have to keep in mind is that I did all this I have already taught you how to type applications.

This is an application type CRD.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster / Namespaces (part 2)

In Spec you must enter a project name. This way you just apply. With this you just apply on Your Kubernetes cluster. First of all, you need to know your cluster. If you want to change the server URL, you can change the cluster's For server url I do one thing. Download this application Where will you run it? In the Project Front End team I'll drive it. Is the project application running? If you Look at the project here carefully. Project default project default Project defaults, so whatever application is running It is running on the default project What I want the front end team to do is If I go, I'll just change it.

These inject into the UI approach in Argo CD Deployment and service of is written. Create namespace is set to true. I want to install this app on my Argo CD. I'll run it in a cluster. kubectl Apply -f Nginx App Dottle In The namespace Argo CD enter application created Now if I go here, you will see a new application was created called Enginix. Front end project front end team so who You created a new project. This destination came under Argo CD Cluster name: Whatever your cluster name was Space front end and this great healthy your The app is running fine.

I've seen the full sync from Git. How did you create a project? Created that application under that project It's easy. You also did it, you also studied the theory, you Did practicals also, you understood these short I realized that all these things are very easy,? Hard work too One time for the first time you saw YAML I have to learn to write that both of these things that I have written Lemmel banaye engineering app and. Only Some cluster URL namespace all these Things change.

The rest of the configuration It stays almost the same. There is an app for Nginx, there is an app for Apache and there is an app for This is my online shopping e-commerce application. Creating an Application Kind of Application It is. It's just that, your which is the process of creating a YAML file Your method should be very simple. Like I made an app, I made a y apac app In this app I wrote a y file Then what I did was I installed another app.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster

Created Nginx app in that app its I wrote the YAML file and created the third app. Which one did you give Online shop app wrote its ml in it All I need to insure is this It is to be ensured that whatever these apps YAML file, my configuration is correct in it. The URL where my Apache app is placed is I showed you the command line method earlier. My Apache app is placed on that path. This is the URL of the server, I have put it here.

On Git, use an application. This is how I will create the YAML file of another application. Created the YAML file of the Inject app. The destination URL of the server becomes My Nginx child app is gone Child app means this was the diagram like this Understand that this is the parent and this is the child. I will also change it and put it in so that it Download your manifest file from my repository Pick it up. Is there an app of apps here?

Vim or the folder containing the Apps app I'll go to the App Store. Root app.YAML What's in the root app.ml? My root app It will run in the default project and the code He will bring the app of apps from GitHub,. Which is the URL of the cluster. kubectl Apply -f root app.YAML in the namespace Argo CD Argo CD what will happen with this The root app will be applied and it will go to GitHub Enter will deploy those three apps like this.

Much fun and enjoyment that has already been created How fun! You will go inside the root app and you will You will see Apache child is an application. Engineering's child is an application. Online Shop Child is an application. All the applications are there and this is the root app. At how the route looks and in this All the things inside, if you branch it out If you want to do it, you can see that too. All you need to do is learn how to make YAML.

That, I have What the Argo CD is capable of. That's how Argo CD came about. Argo CD Which by default has one of the cubes It holds a cluster which is called Incluster What do these clusters say?

[Argo CD and GitOps] Kubernetes cluster setup (Minikube, kind, kubeadm, EKS) / Kubernetes architecture (control plane, nodes)

In cluster so if you on your u CLI You'll go and write Argo CD. You will see two cluster lists. Cluster is one in cluster is one Argo CD cluster so right now we have two If there is a cluster then it becomes incluster and Argo C Where did Qyburn go? One of our Argo CD clusters Correct. I'm the one whose name I'll put Broad cluster. Understand that these clusters are your devices. The Argo CD cluster is your staging environment and prod cluster is your Production environment.

The question arises, whether Argo CD is so capable that it Get DevOps going too, get DevOps going too, Run all the apps in production as well. Deploy it to a single cluster. This cluster of ours is Argo CD. Argo is running on the cluster. Even though it's running on this cluster, all these Cluster Prod Cluster It can deploy applications. DevOps App Production app and staging app. Production apps that There is production, we will run an online of the shop and which is our third which is We will run Apache on the staging app.

By booming three applications on three clusters It should be deployed. What about your Do you have a third cluster? Because if we Look at Argo CD Cluster One and these Cluster two. Why are you writing YAML? The server address is the private IP. I went to EC2 Went to Running Instances. Just enter your API server address. I'll just put in a control plane because I have to save my RAM, my poor T2 It is large and will do as it is but it saves RAM.

I'm doing a little escape WQ Enter now I'm going to do this Another one using kind config Prod I'm going to create a cluster. Cluster Name Prod kind config kind config prod datal so this kind config Prad A Kind Confiq I created another cluster using prdml. Cluster could not find a log line that Machete Reach Target Multi User System Detected C group is fine, no problem In the cat kind cluster client config which What was in the used one? 172 31 84 and in cat kind configured What have you used?

A Kind Creation Cluster Fraud kind config Kind Configm Let's try it.

[Argo CD and GitOps] Namespaces

Let's try creating another cluster Using a Config Let's See If This Works. The port is already allocated. I came and had a port issue, so we People will change the port. VIM kind config What will we do to change the port? Make it 94 so that the port people You should not face any issues. There are Argo CD demos and here we are. Let's move on to multi cluster. As I said, on multi-cluster Multiple clusters are needed. There's an in cluster and Argo CD is a cluster.

If you want a larger If you buy a machine, you can also run a production cluster. Let's create two clusters for you here. These clusters and Argo Both CD clusters will be visible. In your application Your The whole game happens here in the source destination. The destination server is your I put it here. Enter the URL of your production cluster or the URL of any cluster you own. We People go on their On the DevOps app. I'll try it on these clusters.

Usernames are a game of source and destination Complete and of your ML files. It's done DevOps your app. Whatever your server URL is, if Your cluster is formed Unfortunately, it could not be made because of lack of resources, put its URL here for you. Namespace default Everything looks fine except. kubectl Apply -f Develop And your prod app dot YAML in the namespace Argo CD apply enter both apps You have created it, now go here Applications You will see both the apps being created here.

I'll see what the difference will be one of your The Argo CD cluster will be on the star and a You can see it on Incluster here. DevOps which was DevOps app his destination That is in cluster and which was online shop Prod his destination what Argo CD Cluster Interesting Right That Online Shop The product we made for production. Thought the app was his destination Argo CD Destination Joe Deaf His Destination In Cluster destination you go to it afters You can also put where did the sort by go?

Where did anyone go in this, yes here it is on the side Now you can put it in clusters here. These clusters so in these clusters we have a I've already put it in.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster / Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

I need it from Let's Argo CD cluster. Deployment is a part of applications. I hope it is clear to you that You can deploy to multiple clusters You can. You just need to put that cluster on Argo. You should know how to add it to a CD and use it as an edge. These are your apps, now what do you get from here? Blow it up then Argo CD app delete this one enter are you sure you Want to Delete Argo CD App and All Its Yes, blow up the resources.

Route So look at this, you got the root app, so this root app blow up the Argo delete cd app a sorry Argo CD app delete root app whatever. It is a root app, remove it, I am pretty sure. Application Set Template Based Multi App Deployment. You are a You make an app of apps. Wrote its YAML, wrote its YAML and If tomorrow such If you had to make 50 more apps, you would have made 50 more. The kind of The application is set up.

Application Kind Saw it, saw the app project and set the application So you've seen these three kinds. Its name Space will remain Argo CD because Argo A CRD is going to be created inside the CD namespace. Application set.Generator now in its spec There are multiple types of generators. List is a generator, Git is a generator, so Git You can also give multiple paths to the hub and things You can generate it. The name is Nginx And its path is UI approach.

The app you have in CD Demos is Nginx. If you have an online shop app, this is it. This is the app, this is the variable name of this app Talking. Put the project in Spec 2. Whatever you want to add to the repository Put it, put a branch and you will be on the path Where do I need to enter the path variable and what should be in the path? We have to deploy that too. Zoom in too much so the vim gets a little messes up in the list generator What will happen to us?

Destination Servers Do you need to install them on these clusters?

[Argo CD and GitOps] Namespaces / Git and version control

Or your Argo CD cluster I have to drive. I want to see Argo CD on a cluster I can get the server URL Argo CD I'll put it in. You want the production to be If you run it on a cluster of then you can use the server URL that Put that in. Namespace The default is true. Git will use your above Path is in which repository it is kept Are there? Lates to Apache And what will its path be? Whatever your CLI approach These CLI approaches UI These folder names and Apache was inside it.

Once confirmed Let's check if the path to the Apache app is correct. We'll confirm with our repository. Argo CD Demos is our repository.CLI Approach is our path in which this Kubernetes' manifest file kept All set, so I just have to add this much. Escape:wq Enter Now apply kubectl -f list generator dotall in the namespace Argo CD Enter Application Set Demo List created now if you go here then here You will see that so many applications are created instantly. Argo CD Appset Application Set List.

If you look here, you will see that One called Argo CD Demo List The application is set up, within which a Parameters have been generated. CTL Get Service in the namespace. Port forward command That's all we have in life. It would be better to do it on EKS cluster. Run it on the ES cluster, man. I'll go to Security on my instance. The Git code is posted on GitHub. You see how I Quickly created the applications set and three In fact, there are four applications.

If I do I will give Argo CD Appset delete which one Demo List Inside Joe Argo CD Joe Demo List Inside Delete Argo CD Yes, delete it, enter, are you sure? Want to delete all of the applications If you want space specific If you want an application then you can do specific By doing A B C D I can open all the apps. Argo CD Asset Delete Argo Let's try it again. Argo CD appset delete these enter are you sure i want to delete yes I want to delete Deleted Now I will go here from the bottom I will be deleted.

[Argo CD and GitOps] Argo CD notifications and integrations / Deployments, ReplicaSets and rollouts

How interesting Argo CD is. Your entire deployment should be I've prepared butter and kept it ready. There is a project that will give you cool multiple Package applications by creating templates Help me do it. Configuration management is done from Let's I have one YAML file which is of inject. There should be two in DevOps. Docker Ansible Terraform Argo CD Sub One I read it in a shot. As much as possible to the people What should be reaching is not reaching. Very important for engineers When did your deployment fail?

If anything goes wrong, you An email notification or Slack notification You should come. Tell us whether you want email notifications or I need one with Slack. You have to understand that If I If you want to send an email, please send it to me on the email address You have to send it to me on my email from Lets. If you want to send it from any email to any email If you want to send a pay then you will need to use an SMTP It seems to be a server that we use for Gmail.

Verified all good now what should I do I will create an app password here. As soon as I created this app I get an app password. This is me I'll copy it and put it somewhere here. Off you go for your Argo CD. Here you will basically generate a notification. To do that a notification controller You will have to install it. I am a mkdir notification Is there a notification folder anywhere? Notification I'll create a folder named. Gotta keep it a secret.

I'm off to my Argo CD here There is a folder called Notifications. Important things you get in the notification The thing to keep in mind is notification Controllers, triggers, and templates. Subscription means where you Send it, email it, Slack it For this example, we will send an email So notification controller What happens? You now have a Kind cluster running. Let's check out our Argo CD. This means you have to return your Return the token to the cluster that is logged in.

Enter your public address, username, admin, and Our password is very secure. Test @ 1 2 3 enter and here is my login Successful Argo Cluster listing done.

[Argo CD and GitOps] Argo CD notifications and integrations / ConfigMaps and Secrets

My cluster is running Argo CD. The Argo CLI is installed. The email you receive will be It will be written which app has been deployed? Meaning when the email is sent or when the notification is sent I want my success on If the app is built, you will get a notification. That's it, run, enter config map. Argo CD notification configured to Now you have the notification template. What you have to do now is Secret Joe You kept this password hidden,. You create secrets like this too.

It'll Be Kind of Secret. Argo CD Notification Secret will be its name and The namespace will be on the Argo CD. Copy this and Go to your notifications folder in vim secret dot YAML by doing it here Throw it away. The Gmail account you used to install the app I created a password. Remember that this Email is the username, Email is the password, these This is a variable and this is its value,? The variable Isn't this a variable we can use in the future as well?

Your variable name is exactly Think carefully before putting it in. I thought Enter your email username and email password. Escape:wq Entered file save kubectl apply -f secret Kiss Me In The namespace Argo CD I will apply it and my secret which is mine The password of the app is and my email ID is. He has reached the cluster of Argo CD What will I do now? I am I will run the application but before that I also need to check my template to see if my What was in the template?

The template gave you a I'll show you the bar. This notification There is a config map,. Even then it will work, you will find it in the config map. Name for Argo CD notifications Space Argo CD will keep you going Just the URL of Argo CD in the context You have to give public IP, not private IP. The service that will remain inside it The username and password used in it will run That would be dollar email username and dollar Email password.

It looks like these variables. These are the same variables. That you used in secret, I have named the same variable here.

[Argo CD and GitOps] Argo CD notifications and integrations / ConfigMaps and Secrets (part 2)

Argo CD Notifications CM, we have given notifications and other Did you install the controller and those secrets? He will take the password and give it to me. If the protocol, its host name It is smtp.gmail.com And the port is 465, that's it, and here If you want to receive it from another email address, you can You can figure it out. That's it for your service email. Configured Password Username Host Port Username Is Basically Your Email Address Only this is done, the service is configured now you make templates for example subjects What do you have to write in the email that you.

The CD is saying that the health of my app is like this Has status successfully degraded a stuck or Whatever it is, what is in the message now? Application name, namespace, status, Reviews, last what-ifs? If I want that My app is deployed so it will come Email deployed, tick here in green Gone or you do this hooray application deployed

You could write something like this Successfully deployed or health is. Compromise by doing something you want now You can make as many types as you want.

Based on such triggers, you can create email templates. An email is of deployment and health is of degraded. Whenever my app's status is Health Degraded If so, this email will be sent. I have created the config map as follows: I'll copy that and put it here. Vim config map via email dot ymail I'll save it and go over here and drop it. First of all, Argo CD's I need to change the URL. We'll keep the rest secret. Here goes my URL remember public IP Doesn't matter to anyone I tell How do I remove it from here?

Just go over to your Argo CD and you're, whenever you feel like this, it is not possible To load data request be terminated like so Understand that this has not hung. Copy this public IP and Go over here and throw it. URL of the CD Enter. kubectl Apply Huffn F Config Map Email Template Apply What has happened to me now is a secret here. Config map both have been applied I've got the template and the triggers are here. If you look at the last three commands We have completed this notification catalogue.

[Argo CD and GitOps] Argo CD notifications and integrations / Git and version control

Deployed by secret deploy I have given the config map and deployed it. I deployed three things, I did them. Deploy which apps to deploy it for The notification should be on. Argo CD Notifications this Argo CD this complete you get all these things,. You should know how to make YAML. I'll enter my email here I'll give it to you in a bit. Here again the same application What is the repo URL for the work? My destination for Joe Q Argo CD His server is What is the private address, please give me that.

I will write here Argo CD Cluster You could have given me this one from the list end part. I have got it, now I will put it in this Pay this is my Argo CD cluster list Now your application set is gone. On my repository It is kept there, so I will go and put it here. Project you Do you want to work on a specific project? If you want to put it then change the project here. Saurabh Jain's or Rishikesh's or random people That you are my, that's correct but Enter your email and that too please enter the correct email address.

Because you will ask, did you receive the email? How is my status showing Argo CD's train won't feel this good Visham.com So I have put this that if health If it gets downgraded, please send it to this email. I can now deploy this application. Check here whether the app has been created or not. What should happen is an email send I'll quickly check if Here on deploy look here this app o app So it's torn, it doesn't matter why it's torn.

The email may not have even arrived when this degree You will receive an email when it is done, it is currently being processed. Isn't it, when it gets degraded then the email will come. What we What was the repository URL that you entered? The repository URL at GitHub.com. Shm 153 Argo CD demos let's go that one Go to the repository. I go to Docker Hub and Where is my image on Docker Hub? Child DevOps Latest Joe My Push Image It's right here, I'm going to pick it up.

[Argo CD and GitOps] Argo CD notifications and integrations / Kubernetes Services and networking

As soon as I picked up that image, I went here and I run the sync once. This is image not found or image. Will you check my email first on your phone? Alas Let's show you this on the big screen Where did it go Come here, come on the big screen, no is coming Here comes the boom Argo CD Tea App Successfully Deployed 3:34. Pay Application Tea App Nempase Argo CD Sync Status Sync Revision is here. Cluster URL in details It's here too.

Please see our cluster URL. What the cluster looks like. Email Send me an email. Wrong image number right now. Go to GitHub and upload the wrong image. We picked email instead of Slack. Image What is an updater,? What do you need that The scandal I just committed when I went here Opened GitHub. When you are a DevOps Engineer, you should have the image There should be updates that should be automated. That should follow GitOps principles. There is a service called Image Updater which What does it do?

If your latest image is Tagged. Any latest in your repository? If the image is tagged then the ID of that image is in GH It goes and updates itself and your It runs the application. Image Updater Is a Service in Argo CD that handles automatically Naming and versioning of your container images. If I use any Docker image Now there will be some version of this image, some Let's have tea This is my image of this name from the app. Whatever version of this will be, whatever tag it will be, let's is from V 1 This is the version from 0 Let's.

In your manifest file, you can add whatever you want. Image here in Deployment Manifest file I keep it in my possession. Let's say my developer I have requested a new version. I go back and forth and tweak the deployment a bit. I'm a I will create a path, a way I will give that automatically Whatever version changes I have made It gets detected and the latest version is It should get updated automatically in the gateway. That's what Argo CD's image updater does.

Argo Set CD image updater which keeps checking whether your Docker When is the tag of the container image Changed.

[Argo CD and GitOps] ConfigMaps and Secrets / Git and version control

Like you changed v1.01 I did something like this, let's immediately use this app. Detect will tell Argo CD that, I have detected it. I'm going to go to git and hit commit and here He goes and commits. What will Argo CD do? You won't have to update it as you used to. As soon as the new version is installed on Docker, docker push As soon as the new version was installed on Docker, it The image updater, the image updater, is Detects the version and this version.

Update it in your manifest file He does it. This thing on the principle of GitOps Automatically syncs to Argo CD Deploys your Kubernetes Cluster. You changed the docker here. Automatically Image Updater installed on Git. Git sync and Argo CD deployed to the cluster I did it. You don't have to change the YAML file again and again. Mode that it writes back to Git. That's a git remove commit creates commit In which the tags of your image are there Updates happen. You can use any on GitHub Pick up the project and see if it is open source general.

I have an image updater He's about to commit to GH. I want GitHub's I need a personal access token. I can use my I'll go to GitHub. I rubbed it a lot, I hope you also rubbed it a lot. Work hard, work harder and you can also get all the gadgets that fulfill your dreams. Ho Argo CD in One Short token this is a token so that I While deleting, remember that you people do this Don't be silly. Whoever has access to the repository It will.

Access to the repository Save the token somewhere here because That too disappears. There will be an access token. Access token so It's a secret. Do I need to create a secret file? Secret File But before that, Argo CD Image Updater from Project Lab You should install whatever service is there. It was a notification image Which is in the updater image updater whether now no vim no vim needed We have to apply directly to Cube Citytel The manifest image updater's manifest. Apply and he will do everything in a jiffy.

RBAC created ConfigMapDeployment You have also created it and your image updater is running.

[Argo CD and GitOps] ConfigMaps and Secrets / Kubernetes Services and networking

What is an image updater? The service is as I told you. Let us go Now what does he do, he watches the image updater that you did a pull/push of a Docker image Version: He sees it. For this I also need Docker credentials So let's log in to Docker. Docker Login Waiting for authentication. Go to Docker Hub and go to Docker Hub After that you should log in to your Docker Hub. You can call it Argo CD. You got one for the Argo CD.

Create a token and enter its expiry date. We are new I don't want to do mine like this. This is how you can change anyone's image. You can push it by tagging it. I don't have this image. The Docker tag V1.0.0 I tagged and did a docker push V1.0.0 I did that and pushed it in. Because that image The updater doesn't know which image to send. Detect which image to scan He has to do it. Update your image in your deployment.

This Argo CD has one in it There is a folder in Argo CD Demos named There is an image updater in it. You If you want to make it for production then This becomes your base image, base manifest files and from there you can create multiple replicas, You can make multiple copies. Image The updater needs customization. I want to use the git credentials which I have provided here But this is the personal access token of Git that I have I had saved it and I will keep it secret.

Let's put in the secret. The image Let's put in the updater's secret. The name Its will be Argo Image Updater Git Crates Copy this and here vim secret dot By YL Now put your GitHub username here. kubectl apply, you will say that, I can't see it because of your photo. kubectl apply -f secret dot yl in the namespace Argo CD Enter Secrets Argo CD Image Updater Git credentials updated Now this is Argo CD Image Updater Git What did he do that he was running.

This is a service, whose should I show? These Image updater, this service is running,. Running this with my geth credentials We've met. Give him the Gate credentials Now what will you do?

[Argo CD and GitOps] ConfigMaps and Secrets

My image updater will make the changes to git. Argo CD Image Updater Argo Project Image List Image List means you have your Docker Hub username And you have to enter the name of that image here. Will?'s image updater update this Docker image Will continue to scan regardless of the DevOps image Did you get it or not? As soon as he finds her, he will grab her. Git's secret method is Argo. CD Image Updater Git Credentials to He will also get the secret.

The versioning strategy we will use What we'll do is SemVer semantic versioning. Three things to remember were the Argo CD images You need to enter a list of images into the updater. You have to decide which images you want to see. I will put my username and tea DevOps is my I will put the name of the image in the right back method. What I want is I want that Argo GitHub's credentials were from the CD. I'll take and update strategy semver bus That's all you have to do, just do it once Keep everything auto-updated automatically in your app I will show you how it will happen.

If you want, you can use this application on these clusters. Here is my Argo CD cluster. The image The name is child DevOps. My Docker image is now It matches. Press CTL Apply hfan f cha app dot YAML in the name Space Are Go CD Enter Tea app configured so means here if I'll go to applications AR This repeatedly. This problem is only and only kind cluster comes in. If you run on EKS, it Such problems will not arise. Check your image updater logs now.

Check the locks on the image updater. Argo CD This image updater service is running,. Starting Image Update Cycle Considering Zero innovative applications for updates. Starting Image Update Sal Considered One Annotation Processing Result Application One The image is considered, he saw an image and held it. Lee Watch Now Starting Image Update One Image Cons Considered Processing Result One Applications An Image Considered Again do it An image considered will take some time It will take some time, what will happen with this. He initialized a commit on GH hit And basically successfully updated the live Application.

[Argo CD and GitOps] Monitoring with Prometheus and Grafana / Argo CD notifications and integrations

In the app If you look at Can you see what happened a few seconds ago? Author Argo CD Image Updater Comment Build Automatic update of tea app. Isn't it true that I only Pushed gate sorry pushed docker. Docker push like You see this version, I will show you now. I'll give that to my Docker Hub on the child In DevOps The versions that are kept are two, one and Zero is the latest. Docker push here I'll also push the third version.

Argo CD GitOps principles on This image will be updated here from Git. Argo CD will fetch the latest image and Qubitis Pay will deploy your app Updated. Went to the git And Argo went on to CD demos. You will click and you will see that here Your version three update on the live manifest It's done. I Nothing done, no CI/CD pipeline Nothing played. I want you to Prometheus Grafana integration also shown Let me tell you that notifications are also being sent.

The images are also being updated. Prometheus can also build Grafana dashboards I am sure you should put it all in. I wish I had a Grafana Dashboard visualization Prometheus server is the same as Argo CD Take all the metrics that you have and Adjust it in Gaffana and make it look good. Create a dashboard so that my teams Let them know that in the Argo CD How many applications were deployed and how did it happen? I need it here There will be a cube Prometheus Grafana stick.

I have already taught it, but I would tell it again. I am sure that if you have your If you want to monitor Argo CD, you need a Monitoring seems to be a problem. Consider that your The Argo CD is playing on which one? It's going on Kubernetes on. Kubernetes on the right is a cluster of cubes inside which your Argo CD is running. It is correct Argo CD is running. What you need now that these Argo CD dashboards visualizations Reports, logs, metrics, all of that.

Things you'll need if you're planning to buy this Kubernetes in the cluster put a monitoring stick which is Grafana and Prometheus It will be fun. This Grafana and What does Prometheus do?

[Argo CD and GitOps] Monitoring with Prometheus and Grafana / Helm with Argo CD and production GitOps project

CTL Get All in the namespace Argo CD So you'll see. kubectl Get pods in the namespace Argo CD To You'll see it has Argo CD There is a server, there is a repo server, there is Redis, There are many application controllers. It's running on this cluster,. If from here The Matrix is this Prometheus creeping in Come and send Grafana to become Prometheus Go to a data source And Grafana becomes a visualization dashboard It'll be fun,. You have to see some services which is basically This is the Argo CD Matrix and Server The Matrix and one more remains.

Server Matrix, all these notifications You can take all the controller matrixes. Prometheus has to pick up everything and throw it away In. It's easy to install Prometheus and Grafana. There is one thing that is called Cube Prometheus Stk. Krupp Prometheus Stuck is a graph Helm chart which will give you the complete Prometheus Grafana Install it on your Qubits cluster You will first need that Helm chart. Helm chart of Helm repo add Prometheus Add it. Create a namespace called Monitoring of the name.

Inside this namespace, your All apps will run. Helm install q Prometheus Stack in the namespace Monitoring. There is a hell chart of the community, why in it Prometheus is stuck in it with Cubitus Inside Prometheus Grafana Sub with Connections He will create it and give it to you. He will It will take time to deploy because again It's Your help. It takes a lot of time, Helm but Helm It will make your life a lot easier Add the Helm repo and then install Helm I have to give this ML, nothing is written.

It is written, look, it is written, look why it has become written Prometheus stuck dis deployed complete Deployed and admin user your Grafana username is admin and. If you can get the password from this thing then take it. Take this and throw it here and you will get your password. The Prime Minister operator will come out and then Prime Minister Operator is your password so let's check it now where is Grafana running so kube CTL Gate Service In the namespace Monitoring so your Prometheus.

Prometheus is running 1990 your gafana Can you see the alert manager anywhere here?

[Argo CD and GitOps] Kubernetes Services and networking / Monitoring with Prometheus and Grafana

Prometheus Stack Grafana Yes there you go Grafana is running at 80, so these are all the port numbers where Things are moving now we have to port it Port forwarding will have to be done. Show me the commands for port forwarding. By The Way You get all the services of Argo CD. Like Service Matrix, scrap all that If you want to bring it then you will need a service monitor. There is a CRD service. It's a very simple manifest. This Argo CD docs, isn't it, everything inside it Things came out.

Let's say I need monitoring. Here I put Monitoring These Inserted this. Trigger service from here, all this I've got my things. I want to see the Grafana dashboard. You have three services to choose from. The first service is Where did Argosy go? Argo CD Matrix then other service Argo CD Server Matrix and third The service will be your Argo CD repository Server Matrix. These three services have to be availed. These will be the names and the rest will be completed on the label.

Why the Promise Tag Label Everything else will remain the same. Matrix, Matrix and Repo Matrix. Escape wq try the cat matrix once See if anyone has come up with something wrong? API version monitoring kind server Service Monitor has to become three things. This is the first Which Is for Your Argo CD Matrix. This is the second one which is for your Argo CD Server Matrix. The third one This is the repository server matrix. kubectl apply -f Matrixal In the namespace Argo CD apply do Done Service Matrix Created Error Converting YAML to json YAML control characters are not allowed something My made a mistake while copying.

This, this, something has gone wrong, this, this Something's wrong, it's I'll delete it. Give it and apply it. As many as the Argo CD matrix Where will they go now that they are Matrix? Argo CD Server Where will the matrix of the matrix go? Because there's a There is a small thing called labels. Prometheus Stack Port of Grafana I want to forward it. I'll show you If that repository was not there, how could I Finds the port number of Grafana I go to kubectl get Service.

[Argo CD and GitOps] Monitoring with Prometheus and Grafana / Kubernetes Services and networking

In the namespace Monitoring I came like this. Service jani to cube Prometheus stock gfana is its service name. Port It's running at number 80. What was the name of the service? Cube Prometheus Stack Grafana In the namespace Monitoring Now I need to know which port my instance is on. I need to map the number to the port number How much should I do? With port number 80 first I see this Let me try running something on my EC2 at 3000. No one will say anything to me Grafana will run.

kubectl port forward service Cube Prometheus Tag Grafana in the name Space monitoring. This port number is done. Hafen Address 0.0.0 M percent so that it is in the background Keep going. I also want to see Prometheus so that I It turns out that all the data came in Prometheus or he hasn't come. Where does Prometheus come from? Cue Prometheus Stuck Prometheus here it is. Why Prometheus Take Prometheus This The name of the service is... That's what I did for Grafana what he did for Prometheus I'll get it done.

9090 Done enter now this 1990 promise is going on And Grafana is running at 3000. What was ours, not Grafana? Let's move on to our applications. You will go to status target If you look at the health of the picking up metrics from everywhere. Argo CD Application Controller Argo CD The matrix picked up Aago CD's repo Bringing server metrics up. He has all the metrics properly. We need to There's no need, because when the need arises We'll see to that. Whatever happens to us, Argo We had all the CDs we needed.

The Argo CD server is matrixed up. The repo server is up and Argo CD metrics I needed three. Let's move on to our Grafana. Grafana and Grafana http It's running on HTTPS. I'll tell you about HTTPS later. If it is there then what is going on, let's email There was admin and the password was a pram operator It comes from that command, login a what. Here's how it works in Grafana There are connections, so there are connections. Already your Prometheus in Sorceress The data source remains.

In that, Your entire data source will already be there.

[Argo CD and GitOps] Monitoring with Prometheus and Grafana / Argo CD projects, App of Apps and multi-cluster

It came like this I selected Prometheus as the data source. I want the Argo CD ones. Argo So I need Argo CD app info. Select Label Let's put this in. Let's put the project in. Argo CD App Info project default. The default How many apps are in the project? Search here On the dashboard, type Argo CD Enter. It Look at you, a very strong Argo CD. Name Argo CD Folder Dashboard Import. All your dashboards were created that from day one My Argo CD server is running.

There are two clusters inside it. One is our Argo cluster. CD cluster and the second one is our in cluster. One application There are five more applications in the front end team. The default project has a repository Connection is available. The sync status is synced here. Status of the controller, this is the controller The status of your cluster The cluster boomed a little while ago Was. It will be done automatically, it will be visible here Your tea app project looks healthy.

Sync Your sync status is slightly out of There is a sync, look, go and be a DevOps engineer. Go and see, this is the controller status. Your controller status is what's up What happened in the last 30 minutes? You can use the DevOps engine. This repository The URL of this repository has also been uploaded. How many times did we sync that whole thing here? You Argo The CD cluster was also monitored. You've also updated your image. Monitoring was also done, Prometheus was also done Gave it, also integrated Grafana.

A little about role base access control Please tell me that too. Access control is a very important This is a topic. Understand this way that when I am here Login to your Argo CD cluster I have logged in as an admin user. Please share your username and password with the entire team. These Argo There are two types of users for CD. These local users will either have to use the UI access to the console or Access to your command line interface (CLI) Or there will be access to tokens.

I teach you that in Argo CD How to create local users. First, Let's go to your in the same folder where our He's the security guy.

[Argo CD and GitOps] kubectl commands

Whenever You have to create a user, whatever the user's name, whatever the user's role is, whatever the user's You also have access to that in a config map You make it. Here I have the Argo CD. User CM ConfigMap has been created. Kind of Name Argo CD in Config Map Meta Data config map namespace Argo CD.now What's in the labels here? App Name: Argo CD CM Part of Argo CD.So What does this basically do? It is useful to identify all the resources that are a part of this particular name.

Here As I told you, there are two types of accounts. Local Config map for local users config map local user config map dot yummail or you can name it anything. kubectl Apply - fl user config maple enter now argo CD Config Map Configured Now Here You need to check if I have users. Its command is very simple. Argo CD Account Update Password Hashing I entered my account from here. Argo CD Account Update Password Account. Argo CD Cluster List Ah, whenever it gets stuck here My connection is lost.

I'll close this and do it But it shouldn't be like this. I'll go to CD downloads and go here. Own Log into the running EC2. Control-R Recursive search in Argo I'll run the CD service server. This I also started the server. While this There is everything on the cluster. Let me put a roll from admin, put a role DevOps Engineer, Put a roll of tester in. DevOps engineers probably need more policies. What can my DevOps engineer do? That is a roll that is read only All applications can be gated Apart from applications, another role is to read Only apps can be synced Deny objects that have a read-only role.

If so, it won't be able to sync. If your If the role is admin then it can manage the applications Star means action here get sync Star in the line of action what is action Get Delete Invoke Create Update Sync Everything. As an admin you can do everything on your Aloud with the application All policies are allowed to the admin. It only has a role for a few applications. Or Whatever your front end teams or default ones There was also a project, he can get it.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster / kubectl commands

There is a file here named Here's Argo CD RBAC. There is a local account or local User who can access the system either through UI or CLI These are accessed through tokens. P It's a policy role developer so a what is the policy for developers I have a It is the policy of the developer to My Gate can do it for applications. All the files in the project name Getting applications allowed for that Is. There is another policy that which is developer role What can he do for the application?

Sync allowed What's the policy for admins? End Go to your server and set VIM roles to Let's make it and then go and dump it here. Escape con wq Enter kubectl Apply -f Rolls YAML in the namespace Argo CD Enter Configured Now let's set Alice's password. That Alice can be given what roles You'll see that too. Account Update Password for Account Alice To Alice Your password is the admin password. Developer See how to write your own DevOps Developer Now what I will do is apply this thing I will give you escape call wq kubectl Apply Hafen F Rolls Dottle.

My project / all the applications in the namespace ROCD can this gate o My project is not a valid resource What was the name of our project? The developer can only gate my By syncing the project he is allowed to It is possible but there is no policy for deletion. A sync will come Why yes because he can get it to sync He can do it but he cannot delete it. Like the VIM roles I put in. The policy is that if your role is admin If so, you can do something to any application.

Escape:wq enter kubectl apply -f roles In the namespace Argo CD configured Gone. You can also use Argo CD. What would have been the SSO scene? You can also login through GitHub. Click on SSO Do it and complete it completely Read it and follow it just one by one I'll explain it all to you. Understand it like this is the Argo CD This is the Argo CD Argo CD is an identity within itself. Control open Identity Control Connect I have all I mixed it up because I was very late.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Argo CD security, RBAC, SSO and HTTPS

Open ID connect this one There is a tool that allows you to download files from third parties like GitHub Gmail Lots of different octas Auth Zero, all these services are there. It helps to connect with them Open ID Connect to OID and Argo CD It remains with the user automatically, meaning it remains inbuilt. This is what we call Dex so you have to dex and GitHub Basically, you have to get it connected. It's called Single Sign On (SSO). End here I need to create a new Auth app by going to whose name I will put Argo CD app.

By doing this I entered the public IP of the instance here I gave it. HTTPS in URLs as shown in above GitHub Only allow HTTPS callbox URL. We the people We still have to grow up a little. We'll have to work on the EKS cluster. Let's create an EKS cluster now because I need HTTPS,. I need HTTPS, If you want a production cluster, then for that I told you no in the beginning. That we have to go to the HTTPS one Part.

First of all, To build EKS, we need people from AWS. Create a user Yes, the user whose name is given EKS User or EKS Admin whatever you name If you want to enter it, the EKS admin will provide it and. This is called a part of the AWS Management Console. Next, now give it a Attach the policy, whatever policy you have basically your administrator because I want to create an EKS cluster. Requires VPC, lots of load balancers If I need things, I will I will enter administrator.

Next create administrator access The user is now my EKS admin. You've become an EKS admin. I want to enable it on security credentials. I You will find Argo CD inside an EKS cluster I will tell you so that we can use HTTPS completely. Configure the EKS cluster and then We were able to do SSO. Which is known as AWS CLI If you need the AWS CLI, I have it. Pay You Bonus HTTP Hosting Argo CD You'll get it. Go to this and select AWS.

CLI I will add to this that AWS How to install the CLI.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Helm with Argo CD and production GitOps project

AWS CLI Install, just write this nicely From the documentation you can learn Linux, Mac OS, Windows You can install it in any one,. My AWS CLI on my local Computer local laptop config ho It's gone. Vi is non as EKS ctl. What does the EKS CTL do? This will help create the cluster. This is the complete command that you can use: EKS CTL Will you install it? Enter password EKS CTL Try the version, it should work, if it doesn't work So that means that EKS CTL is yours not installed version has arrived, which means your EKS CT.

It is installed, which means you are using the EKS cluster. You will create kubectl in your local If you want the Cube CT version, I have it. Installed kubectl Helm need Helm I have it installed in my local Generally everything remains installed I want a great domain name. It's, even if there is no domain name, it will work. You have it, all you have to do is EKS ctl create cluster Argo CD cluster in the region whatever US West one and these like.

If you want to write the command then what should I do? I want the US instead of the EU West One West To me, generally I put all my resources So this is EKS CTL Cluster in EKS from Create Cluster A cluster named Arbo CD will be created. EKS It takes a lot of time to form a cluster. Not much to become of EKS test Meaning as an Amazon employee, don't say that. I told you about integration, I told you about image updater, I've specified the obligability via HTTP.

Argo Rollouts, Argo Workflows and Argo Events. Argo Praj Argo Project Joe Comes in that Argo CD roll out. Events are a different Story, WorkshopLow Is A Different Story. She'll be in a different video because It falls under the Argo Project. In Argo CD This is not the case. As for our full CI/CD Project with ROCD. I'll tell you I'll show you the project quickly. If you look carefully here So this is a multitier application in which Orders Checkout Card All of these There are applications.

As you imagined All the applications have come to mind now.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Helm with Argo CD and production GitOps project (part 2)

In the folder with its Argo CD. I have done the project here. This is its role base access control over access to these things Now what all is included in this project? In this The applications are numerous. Retail Code These retail store cards, catalogs, check out, orders, UI and the functionality of these applications The Git repository is here. This is our full CI/CD project. If you go to git there is a branch named afs. You will go to the branch named GitOps So you'll get workflows.

Here Work threads will be very strong and complete How is the entire CI/CD pipeline built? I have told you all those things that How is Everything is Mentioned Now made? The cluster is already forming. You will be able to see your cluster. An OI DC There is a provider. As I said Argo CD me oidc so that thing you I'll have to add one. I'll have to build it. Cluster access Get nodes and this will happen. Name Space kubectl will create Argo The CD will be installed, it will be done,.

I hope I am confident that the last 5 6 In an hour you have learned all this. Once our cluster is created, we All these things will apply. K Aya that successfully created I thought the cluster was created but That matrix server was created. Successfully created add on VPCCAI, I think it will happen in a few days. Seconds it will be done o became EKS cluster or CD cluster It's ready, it's 5:48 now, folks. How much is it 12 minutes 12 minutes is good 12 minutes Great, our cluster is now complete.

If you are an EKS CTL Gate Clusters run this command in the region US West to then you'll see that our Cluster Argo CD Cluster Should Be Live. The Argo CD cluster is live. If you If you want, you can also see this EKS here. EKS Elastic Cubes Service You can see the production here. Cluster God, two clusters are not working. No, there is no scene, everything is fair for you people. I'll set up a context config of. AWS EKS Updates Queue Config Region and Name from Argo CD Cluster You will.

E's CTL Utilities Associate From an OIDC provider, you basically get a Add the OIDC provider.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Kubernetes cluster setup (Minikube, kind, kubeadm, EKS)

Open ID Connect So that you can get information from different sources like GitHub You can connect to Gmail. OIDC Approve your RG CD cluster Are. Let's add a node group now. EKS CTL Create Node Group in the cluster Argo CD region What does a node group mean? A Let me add an extra node. Region EUS you Whatever region you use, according to your needs. Even creating a note group It will take some time because it is an EC2 Arrange all the things to create the instance If he does, he will do the building management note.

Kind Do it on the cluster. Whatever you If you want to do it on the cluster, do it. I am going to because there is something new in the market Not coming. DevOps in the market right now has reached a different level. Where Generative AI Agentic AI ML Ops New of ROCD Work Flos Kubernetes AI Applications on Versions Kubernetes Because the world is talking about AI. If it is left then it is enough for me for batch 10. Interesting topics have been found Generative AI for DevOps Your Interviews The preparations are too much I am excited for Batch 10.

Our entire cluster is ready. I want to create my local cube context config. AWS EKS Update kubeconfig in the Region Whatever my reason is, what is the reason after clearing it? US West to name Argosy Cluster Enter Now I have my context. If kubectl get nodes then I should have three nodes. Basic There should be two not three because the control plane is what EKS manages. The control plane doesn't work for you. You will see nodes here. The control plane that EKS manages?

Two nodes are visible and Your instances may have been created there as well. You go to AWS, here you go to EC2. You can also see your instances here. These two of yours Argo CD cluster nodes are also visible. Installation of Argo CD then kubectl First I use create namespace Argo CD A namespace in your EKS cluster I will create a CD named Argo. Either I install it through help or can I install it manually? Custom Resource Definition Argo CD Project created.

Deck Server It's all created.

[Argo CD and GitOps] Ingress and ingress controllers / Pods

It even includes the Redis server. If there's cash, The Argo CD that remains stateless. It doesn't have its own database. If it remains, it stores the cache in Redis. All the things so if you have a highly available Creating an Argo CD Server in Your EKS cluster so you can scale that as well So this happens in a very strong way. Argo CD is done now what do you All the pods in Argo CD I have to wait for him to get ready.

The command is kubectl wait for condition ready All ROCD time up means up to 30 seconds Will wait and see how all Argosy's Now! There are also pods, whether they are ready or not. Argo CD's application controller has been created. Application Set Controller became.dex became It's done, notifications are done, reds are done Yes. kubectl Get Pods in the namespace Argo CD Everyone will be running. All are running to your Argo. Argo CD hum log You were doing some small-time work here at this port.

We have an ingress controller. Let's install the ingress controller. Helm repo add engine ingress so an ingress from this Add controller to your Argo CD server I do. Ingress Engineering has Bean added to repositories. At the Helm Rambo Get the latest charts from the update. From help install ingress controller Ingress and enable SSS pass-through I need to do SSL pass through and by doing this I'll just press enter. Still This is your Argo CD from Let's. I need a load balancer.

His For this I am using Nginx's Ingress Controller. The ingress controller is putting it in. As long as all the pods of Cert Manager are active Wait until it happens. That tomorrow we will enter our email, application I'll put all these things in. We will wait until all the Cert Managers If the pods are not ready then the condition met all Are you ready? Project CD Projekt mkdir EKS Argo CD EKS Argo CD Now What do I have to do here, let me Encrypt your Cert Manager using If you want to put Sirt Ishar then Sirt Ishar.

Let's use Cluster Issuer here Just a small message to you and your email ID. It's just a small thing you have to put in which in If it is important then VI Let's Incript I will create it by coming into it.

[Argo CD and GitOps] Kubernetes Services and networking / Ingress and ingress controllers

All you have to remember is Escape wq Enter VIM Let's Encrypt. V1 will be the type of cluster. The issuer will be Let's Encrypt Metadata Prad, this is very important, now this name This is a URL, this is basically one such server. Where you enter your email, that's a If the certificate is issued to you then this The cluster issue does the same thing. You just need to enter your email address. I'll put my email here. If my cluster issuer has Told the city manager, let's Enter this email into the encrypted server.

This Let's Encrypt product is not this hum log We'll use it in our ingress. Argo CD ingresl This is important for you So all you have to do is see that you Making an ingress. You're creating all this on Argo CD. Namespace Argo CD ingress HTTPS You have to put annotation and here you have to put your Just write down the domain. Copy this and put it here Ingres dot y enter and paste this here now You have to do the thing rgo.

This It will only work if I have my own DNS. In Go Daddy or whatever my DNS is Updated the load balancer URL for this cluster It will work only if it happens. Let's enter the URL of the load balancer. How do you do it in this kubectl get? Service in the namespace Ingress Engines Hit enter. Here you will see a load balancer. You'll find the URL for "Which is this?" My load balancer's URL. Here I will go to DNS, good domain And I have to go to DNS and update it.

I'll put in Argo CD and I'll put in value I will put mine which is of load balancer The URL was what I got from here,? Let's make argocdin w shm.com I need a C name record for this. Argo CD should have its own load in the value I will save the URL of the balancer. Let's update the DNS records. I'm going to call it Argo CD, this is also here. CTL GATE Certificate but before that you Apply the ingress you created as well.

kubectl apply - F Ingress What will this do? Your Argo CD will know that I have my account on this domain.

[Argo CD and GitOps] Namespaces / Argo CD projects, App of Apps and multi-cluster

Argo CD Server is set to true. You have an HTTPS Connection Secure's Argo CD Server is ready which is on EKS scalable Now let's go to the next level and login to it. Here's my Argo CD running. If you still don't know how to make it, then come here Go away, oh God, he must have arrived by now. How many six seven how many hours have we been studying People have been reading for 6 hours, till now It must have come to make an app, still Argo CD Go to the Demos repository.

Take a declarative approach Whatever you like, whatever you like, go ahead. Give whatever you like, take it and get what you want from it. Two brothers great deploy our online Shop on any other app from Argo CD Demos You can use Path Multicluster online shop Let's go to the multi-cluster online shop. Where did you go Yes Will go to Multi Cluster Online Shop. Connect Repo Gone via HTTP Went. Default project name in project Optional. Multi Cluster Online Shop So Multi Cluster Online Shop So Application Name Online Shop Project Name Default Sync Policy automatic Auto create namespace source in Argo CD Demo Revision Main Branch Path Multi Cluster.

Online Shop This Is Our Path, this is our path, the destination URL. In whatever our cluster URL is there No need for namespace and just create Simple, all you have to do is this What happened to it now There are invalid namespaces, including namespaces. Take I was telling you Our own namespace is default The default namespace was too small Things keep happening,, it's done. Online shop deployed and running well All your Argo CD is playing perfectly Well, if I make a replica here There are three, if I make it four then you will get There will be no objection.

Deployment of a multi-cluster online shop I've changed it. As soon as you sync it You will do that the three pods have now become four. I had to do SSO. To do SSO we How much time did people spend on an EKS course? The authorization callback is rgo.com Register by doing this on API Dex Callback You apply. I need to I want to use it in my cluster.

[Argo CD and GitOps] ConfigMaps and Secrets / Argo CD security, RBAC, SSO and HTTPS

Argo CD tea has health is degraded to hours Ago. Stop I turned it off, turn it all off, man. We need to enter our client ID and that I had to paste it, so this is our Client ID and this is our client's Secret, so secret has come here and this. Our ID will take us to setup our To tell on the site of Dex It will be installed on your Argo CD cluster. If you still want to make it, you can You can build it using GitHub and Create.

Organization Name Argo CD Org put it just to do this demo contact Please put it in your email what happened? My email address is in Contacts. Configure Argo CD is now running the cluster. Install and run in the Argo CD namespace Both are fine, that's all. I need to cd GitHub secret this needs to be created where Client ID and Client Secret You have to put both in. I did this and here is the VIM secret. Escape and enter so I have a secret ban I'm done, now I have to make the next thing.

That's my config map where my Public IP of the instance and my gate If the name of the organization keeps all those things then These make it CM VIM Config dot Yammel. The client ID will take the name of the org. Let's now create both our secret and config. Inside my Argo CD that OIDC Connected to GitHub Now it has a roll base Access control is also created. Its role would have been to make base access control Yes. Dot Why this because which email has what access?

All the things I need are fine now Number of members in the organization Like what is the name of my organization? What is the role of a repo admin? I give Group Shubhamnath 5@gmail.com I give the role to rippo admin by the What do they do on the repo admin role? Escape:wq Enter kubectl Apply -f to what? Apply R backlink to namespace Argo CD Do it. Argo CD rollback access control Configured. The only thing left now is left, that is No, we did it Secret had applied Dex Configuration done, backed up Now we are deploying Argo CD.

[Argo CD and GitOps] Deployments, ReplicaSets and rollouts / Argo CD notifications and integrations

Will restart so that login via The Git button should be visible here. Am Argo Was it Argo? Not the Argo CD, man, here I am. Log out I should log in via GitHub And login via GitHub is there which is Saying shout-shout single sign on in Your production cluster is ready. Username and email address: This is my email address. This setup also controls ingress. Install Argo CD on ES. UI Declarative is all done. Argo Workflows, Argo Events, all of that. I'm going to tell you how to scale this Argo CD.

Because Argo CD is out tomorrow. Just Have To Scale the deployment that you have. The deployment of Argo CD Scale it up. As I mentioned Argo CD is stateless. Meaning it doesn't have its own database. You have an available pod. API servers, repo servers, Application Controller, Raid Is External If you have a database, you can do that. If you want high availability You can install the official High Availability HA Manifest. I can There are already a lot of things like this in Redis.

Joe Ha Proxy The server was not there. The HA proxy What is a proxy server? This happens to your load balancer. It's as I told you, it's simple, you just Have to scale which is kubectl argo CD Scale Deployment Argo CD

Server Replicas Three, so what happens to that? The single replica that was running is now running three The replica will run and your application will be Available for many people to access So like kubectl get pds in the name. I did the Space Argo CD.

Argo CD Scale Deployment Argo CD Server, make three replicas of this. This is the Argo CD server. At this pending This is being built right now, initializing everything right now. If it is being created then you can do this one repo If you have the same server than let's Create two replicas like the repo server now Oh because of H A he made it, all. Which is It would be if you could see it, You can verify the scaling. First of all, whenever you deploy RboCD Domain based EKS cluster So that you can work at scale.

Your login system will be it should Via SSO.

[Argo CD and GitOps] Kubernetes Services and networking / Argo CD projects, App of Apps and multi-cluster

Or Gmail or Whatever its button TLS Everywhere means Must have HTTPS secret management.role Based access control should be with Least privilege. Your logs are all monitored. Your scaling practices Having your Git as a single source of truth Monitoring alerting is available to you. For backups I used shell scripting I taught you how to take backups. Let's move on to our project and project I will need it to go to the Where did my Kubernetes go? Argo CD helps with that. An AWS Containers that are available on the official repository They are such great projects.

One of those retail stores The sample app is a project that can be downloaded from your dvs The has done a great fork and It is good to add more chilli and spices to it. Here you have Java, that's connected. The service you checked out It is connected to Redis. The service is connected to DynamoDB. The service with the catalog is connected my from the SQL. This whole multi-tier This is a microservices application. Monolith Why didn't you apply? The service that will take your orders tomorrow If there is some problem in it then other services Your UI must run.

You get a complete repository of everything I won't need to update it. Micro This would have been the biggest advantage of services and Kubernetes is a tool for microservices. It is considered best for if you have a micro Services need to scale. Introduction to the product project I hope you found that a cool It's a project. This is an e-commerce project. You have seen the benefits of microservices. You will need the full A monolith of a large repository You will not have to maintain it.

You can decompile it into small services. It happens, microservices and monolith Architecture. Why Use Amazon EKS Auto Mode in iOS? First of all, this whole To do the project you will need a You'll need to create an account on AWS. You can enjoy this project. You can run it on AWS. If it's Compare production Kubernetes to EKS Let me make it a little bigger. Zoom in I give you EKS Elastic Kubernetes Service provided by Amazon There is a production ready Kubernetes service.

Where you can create your own clusters and The control plane part is Amazon He looks at it.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Kubernetes architecture (control plane, nodes)

If you want to go into production, You have to use cloud services. You can make clusters production ready That's EKS. We'll switch to Auto mode. If we talk about EKS, it's simple Cubinate is cluster so if you have cluster How is it formed? Components of a cluster If what they are, I'll simply revise it. This is very There is an important thing which is called Control plane. The control plane is your entire cube. APIs to it Server Scheduler Your Controller Manager Everything resides in it.

Whatever other The thing is they are your worker nuds. These are your worker nodes. Your control plane is Amazon. You Only and only on these worker notes which There are applications,. You deploy the applications, you'll manage. The heavy lifting that happens The control plane is managed by Amazon. You just need to apply But to run these applications What do you need? First of all You need compute, meaning your EC2 Instances then require storage Meaning your EBS volumes would need to be reinstalled Is the network mean what your CNI container is Network interface drivers then VPC.

This, this heavy lifting with EKS this You also have to manage EC2 Then Manage EBS Manage VPC Manage ALB, you also have to do all this. I will manage the control plane. I will also manage your storage. I will do it and

will also manage your network. I'll even manage your load balancer. If you talk about security, then security We'll manage that too. AMI (Amazon Machine Image) It will be immutable. Data will be in the plane via Amazon EKS Auto mode and it uses it in the background Carpenter.

Carpenter for scaling you Let's use your lot load has arrived. Automatically your nads Are there EC2 instances that can scale? If the load is too low, it scales down. The cluster is so easy. Super Simply click on Create a Cluster. Quick Configuration with EKS AutoMod Tax Your cluster name will automatically appear. The Kubernetes version by the latest one Take it. 1.33 is now your IAM role. For security, you can create recommended If you click on the role then Joj will be able to see the roles.

You can select any field for the node.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Argo CD projects, App of Apps and multi-cluster

You need your own VPC. How to create an Amazon EKS cluster Auto mode enabled. Whenever this cluster is formed, it takes less than 10 minutes to make It is the fastest. Meaning if you are production ready Talking about clusters, it is less than 10 minutes. When you go to its computer, first how I had to manage my nodes myself,. That add this node add EC2 forget add Do it, it's not like that, there's a background in it I am using Carpenter which is EC2 Auto-scale instances to 10.

Wait a minute, this cluster will appear within 10 minutes. It will be ready nicely, I hope you understand it nicely. I must have come to know what is EKS and What is the Auto mode of EKS? Let's say that our Infrastructure is what we have today. The infrastructure to be built in the project What is it and which one should we get? First, you Creating an EKS cluster on AWS. You will need the AWS CLI. What is a command line interface?

All you have to do here is AWS configure Bus. This is the access ID that is coming, Secret Access Key is coming, default region Name is coming, default output is coming. All these options that are coming to you, this is coming because you have AWS You must have the CLI installed. This Installation of whatever the project requires I put everything in Read Me Meaning if you want to make this project locally What do you need to do to drive? CLI Terraform kubectl Docker Help How will you take it?

It has been pointed out here that AWS CLI If you want to download it, run this command. Download Terraform on Linux If you want to do it, do it. kubectl on Mac OS Yes. You have to do it in Linux. AWS How will we connect on CLI? First of all what do you There's nothing much to do, no hard work You have to go to Amazon or AWS. You have to search for IAM on the console and You will click here and it will be great on IAM.

Whatever you need access to For example, you need access to EKS. Full cluster admin access if needed Select that.

[Argo CD and GitOps] Terraform and infrastructure as code / Helm with Argo CD and production GitOps project

Go to the terminal and run a command. AWS Configure You can only do this if you have AWS CLI version two is installed. Copied the access ID from here Enter and Secret Access Key Here Copied it from there and posted it here. You just saw my secret access. You have to simply Try running the command. AWS S3 LS thats It. It will reveal It will be possible to have as many S3 servers as AWS has. That You Terraform in One Short Video You will understand that Terraform is a Infrastructure as Code is a tool that allows you to The entire infrastructure will help via code to automate It helps a lot.

Terraform Halfon version I have Terraform version 1.57 Terraform version is out of date with the latest Version is 12 lo br Update Terraform Done low blue upgrade Terraform. We will do as you say, what should we do? The Latest Version You Can Upgrade by downloading it from There is no scene, this one will work too. No problem, come back Let's say you have Terraform installed,? There is Terraform, there is also KubeCT. This is KubeC Why is it important for you Talking with a cluster of Kubernetes So let's try it out kubectl version Client version 1.33.2 so kubectl also We have it installed.

If you have If not, it's written in Read Me. Helm Let's also take a look at the Helm version. Let's install Helm Let's take it. Here Go and look here it is written Helm If so, just copy it. Boom so I have the Helm now The Helm file is downloaded. By doing a little chmod 700 If you make it executable then ls huffn I have to look at the gate Helm, I have permission Yes, read, write, execute permission I have given it because of 700.

Other users has no permission. Get Helm Where did he go? Initialize Terraform End Review infrastructure code structure. Meaning our project structure is We need to review that. From Terraform What do you want to make? We have to create it using Terraform. Is our EKS Elastic Cubinity Service and in this we People will enable auto mode so that this Our cluster should be formed from the pat and this Terraform will also install Argo CD.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Terraform and infrastructure as code

Helm inside our EKS cluster as well will install our EKS cluster What is it inside? Ingress of Engineering controller so that outside traffic inside, traffic outside, all this one If it can be connected, this ingress controller will also The rest of the retail stores will be as good as the app. Manifest files are like files that are used to create pids. Deployment has to be created, service has to be created Yes, Ingress has to be made. All these things via Helm charts will be taught to you.

First, I'll go to my repository. Because everything is in our repository. I'll move this to the main branch. CD documents are one Get into the folder nicely. Who What are the deployment strategies? I don't know my Git repository. You have to clone it here. This is The command is, just copy it and put it here. Let us go and clone the gate. If I slammed it, my repository is retail. Store sample app will be cloned I will clone you, phat, I am now a clone.

One The folder is Argo CD. There's Terraform and the rest of the branching. You will find all these files on Terraform. Addonsf Argo CD Locals Main Output Read Me Security Variables Version You'll get all these things. Infrastructure VPC And your EKS cluster creation. Both We use prebuilt modules for this You will create a block called module. Your The EKS cluster and your VPC are small. This victory A win for our entire DevOps community So here we go, folks. Click Done and if you want to install Terraform Now you will open the files, yes they are colourful Watch now talk about module VPC If you need to create a VPC, Create a VPC here.

This is VPC, what is it, a private There is a cloud inside which all your Resources will be there, they will talk to you. If you have a If you create VPC then this Name CIDR Range Tenancy Availability Zone CIDR subnets have to provide all of this. Source Terraform AWS Module VPC This One Line All the files your VPC contains Security Availability Zone gives you all that No need to pay attention, it works You will go. Just put in the version, you will get all this.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Terraform and infrastructure as code (part 2)

AWS Registry Terra Form VPC and that's all you have to do. Terraform on AWS VPC Module Click it. In the name you've entered the variable Created a VPC with cluster name. Where did the variable clusterName come from? Here I have created a variable tf where Pay Cluster Name Default Value Retail Store Environment Dev You can also put production You can. Your version of Kubernetes Take what you want. VPCT Take whatever CIDR block of VPC you want. What would be the namespace of Argo CD?

You can configure all the variables here. Availability Zones, Public subnets, NAT gateways that connect to the outside network Address translation helps you improve your public and I set that to true for private subnets. Internet gateway So, all these I've made things here. This By doing this your VPC module will be created. This is how I setup an EKS cluster. I have also provided the configuration here. Source Terraform AWS Module EKS AWS version picked the latest version I.e. Cluster name cluster name cluster K Public access to endpoints Do you want to pay?

Creator Admin Permission is very important so that If I can create, I have to give it to you. I told you about EKS auto mode,. Compute nodes that scale automatically The network automatically gives it. The load balancer does this automatically. The configuration appears to be this: cluster computeconfig enable true. You will do EKS Auto Mode of your cluster, and in that Enter the VPC ID and create a key so that Encryption is also required. If the cluster login If you want to keep it enabled then make it true.

If I'll go to Terraform here cd Terraform I'll go and write it here, Terraform In it, that it is me, as much as my Modules are as many as my back end has. All the pat starts downloading This means that in the background What's going on? This is the entire code, this Here is the complete code, it is from the internet. All the things I had No Hell, Argo, CD, everything through this. My Terraform installation What's wrong with the command? Terraform Initialized in the folder.

Here But my Terraform is done initializing. What should I consider in my architecture?

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Terraform and infrastructure as code (part 3)

I want it in my architecture. EKS Cluster, VPC, Argo CD Hale and Nginx's ingress controller. Until my VPC is created, my EKS cluster will not be created. Until my EKS cluster is created My Argo CD won't be made until then. Argo CD will not be created inside applications How will they be deployed? Step wise if you need to install anything Like this is my Terraform folder If we close this then it is Argo CD.tf In which I have written the configuration of rgo Everything.

Everything on the Argo CD It's written. If I do it here What will happen if you apply Terraform? This Argo CD Main VPC EKS will all be created and connected to each other It won't be possible. I'm going to start with I will create a VPC. Target equal to module which is my module, make that VPC and enter, what's the point of it? I will have to create my VPC first. Terraform apply target Module Target Equ Module VPC Now These Do what will be in all these files?

Will find where our VPC is located VPC is in and what is needed in it So if you need all these things then only that much It's going to work. This is because the VPC has to be completed. Availability Zones If you take all these then the total How many resources will you create in the VPC? VPC will start getting created. As long as the VPC is created. Let me tell you that this Terraform I've done this completely automated. How is ArboCD running in Terraform?

In this, your provider That would be a wonderful thing. If you've used AWS, If you have used the provider then you can use AWS Related components of its resources Cool, do it in a pinch through modules You can install it. I also took Helm's provider. Took the provider of cubiness. Time random string I used for everything I have taken the provider. I am telling you that, whoever helps me The provider is on my EKS cluster. Take all the commands you want to run.

My help should run on EKS cluster. Kubernetes is my end of the cluster Point whatever token is configure everything So I configured everything in versions.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Terraform and infrastructure as code (part 4)

I've given you the variables TF I had explained it. You If you talk about production then your load Which route should be in the balancer? What should be HTTPS traffic? Security Groups I need some security from myself. If the outputs Show me some output like cluster What is the name? What is the cluster suffix? Meaning what is written at the end of the cluster What is the endpoint? The VPC has been created. In the output you can see which VPC CIDR block and the VPC ID it became So try copying this VPC ID.

That is actually created in your account Whether it is made or not, because you can't tell. Your VPC ID search in US West 2 Do it, if you don't get it, it doesn't matter, go somewhere else It must have been made, yes these are two, click on this. Two, here's the retail store VPC, and here's your VPC ID paste 090 has been created one The bar has become a VPC, what can you do now? If you can also create a cluster then I would use Terraform Apply Target is equal to module dot now my module was in main What was the name?

VPC's name has become The name was Retail App EKS, so by copying it Go here and put it on the module dot retail app EKS My Enter That's all I have to do and my EKS. The cluster will obviously form after 10 minutes which We created it manually, please check it once. EKS Admin, make me the admin of EKS admin. I have given cluster access and I have reviewed it. Two complete and create so which cluster is this? EKS cluster this is that cool this How to access it?

I have also given the command here. I have just put it in, you don't have to do much There is a small command, just copy it. AWS, where did you go? AWS EKS Update kubeconfig just copy this and Go to any of your favorite command lines Just drop it. Whatever your cluster name is. Cannot list resource nodes in API Group cluster scope. Because this The user does not have this particular access. Whenever you face such access issue by the way Our AWS EKS cluster is built How many of it was through Terraform?

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Terraform and infrastructure as code (part 5)

If there are 40 resources to be added, then say yes. He is able to describe the cluster himself Your cluster is active and running. That will list the node groups and will add a policy. He will see the current access because my I only have access to the cluster. AWS Pay-Only Access I gave him access to the cluster. Nads are auto managed by EKS Auto Mode. That's when until it fixes this thing I will keep running it iteratively. That Is Where Our Industry Is Heading, look, it has come and gone, excellent.

Permissions have been applied successfully The cluster now has access to everything quit this Now if I do kubectl get nodes boom my node near me Ready. kubectl Get namespace My I have my namespaces ready. Interesting I understood that our industry is not AI is heading towards. There is a cluster of retail stores, that is us making it through. A whole day of great DevOps learning, GitOps You will get it. Which video from the playlist If you have seen it, tell me that too.

Like The one that got the maximum votes is Linux. Kubernetes Then Docker then Jenkins then GitUp. We've initialized Terraform has been given. The structure of the infrastructure code is Completed. Deploy Core Infrastructure VPC with Proper Networking and EKS cluster. EKS It takes around 10 minutes for the cluster to be created. EKS cluster takes around 10 minutes to become... Figure Cube Certification Access to the Newly Created cluster. I entered that command, do you remember? AWS EKS Update kubeconfig Just name your new cluster that will be created.

Put it here and your kubeconfig will be updated. Go there and search for Terraform is Terraform for DevOps Search Mast now here if you find one If I had to create a simple EC2 instance. At how much hard work it takes and what data it has. Key Pair Security Group Egress Ingress All things instance so many things You can use better modules than writing Yes. That's what I'm trying to achieve through this project. I'm trying to tell you that if you don't have to go into production.

Let's take what else into our code Things are in. Our Terraform will be created from Terraform How will the EKS cluster be created?

[Argo CD and GitOps] Helm with Argo CD and production GitOps project / Helm and charts

What did Terraform just do? First VPC Then I created an EKS cluster. EKS Cluster so we have full faith in you I hope you'll make it. This inner person These are the applications of Argo CD. Joe Helm is the engineer ingress How will all this happen? Here you will find all the micro services. Like UI, which is UI micro The service is written in Java and Spring boot html css all It is combined but to run it, The Dockerfile is written here, so Maven.

Through which this build happens, all this is written completely. It has happened, we also have to learn all this from Docker. If you want to run it locally then It can also be run through Docker Compose. The important part That's it, this is my retail store How will all this be done after the applications? This will be helped through charts. Helm charts will help you understand the entire terrain. Not writing the manifesto of Kubernetes You just need to dot the values in the charts.

YAML just insert your values in it The definition of what is paid is what is paid That's already an email template. For example, if you want to see deployment then deployment The template always remains the same. Which resource limits, auto scaling Whatever you want to enter will be in the values. Similarly, orders that There is a micro service and there is a chart inside it also. Because everything is happening through Helm. Helm What does he do? Hale says, ", separately." pod separately, deployment separately, service separately Do not make ingress from.

I have given those values here in different micro Enter the values in the chart in the service I have given it. If you want to change the image then Change this line. The values that Helm has, Helm charts, these are up to my cluster How will they arrive? They will arrive on Argo CD. CD stands for Continuous Delivery. I have also prepared the complete document of Argo CD. You will find a copy of the Argo CD. You will have to create the project.

Store project which is in source repo like this The repository will pick it up. Fork the repository and This will be your repository.

[Argo CD and GitOps] Git and version control / Terraform and infrastructure as code

What will be the applications in this? UI application, this Argo CD How to pick it up in the Helm charts Where to find the location, then SRC in the path The values written in dotted lines in the UI chart. It will pick it up and deploy it. I found that Argo CD is available on this repository He will go to this branch. This folder on the branch is SRC UI Go to the chart and see the values written in it. He will use the values and deploy it.

I give it to you on this particular repository Retail Store App SRC I will go into Go to UI Go to Charts Go to Charts It will go into the values and the image. The value of the repository is written to it Deploy will send its templates to you If you want to see the templates, they are here. This is what HPA has to be done, this is what ingress has to be done If so, the template remains the same for everyone.

Have fun Whatever happens, because this If you want to add this repository to the project You just clone and just images You change it here and there. Deploy the application from this repository You can. This is a plug and play Repository. If you go back to the code Go to Terraform Go to Argo CD then Here it is written that the whole thing is Argo How will CD apply QBC? What is the path module in the Argo CD namespace? Argo CD / Projects Meaning This Path Argosy in Project created in this path Argo CD Slash Applications The applications were created.

Once you've run Terraform Apply Done, your application will be up. Our EKS cluster is now created. 71H9 Go to AWS and look at the console. By refreshing So our old one got deleted which We created it manually and the ones that were created automatically It was created using Terraform and is up Retail Store 71H9 It's done. This will have to be done because all this is done through the CLI. I don't have to do anything manually in Terraform. The cluster that was created before this I had to do a lot of things manually in it.

[Argo CD and GitOps] kubectl commands / Pods

If I Let me run this simple C command AWS EKS update kubeconfig region US West to name retail store so what my Will the cluster connect? By the way, the notes won't be the first time. Because there's an empty cluster. Cube CTL Create NS Demo from Lets Nginx is a demo Nginx name Let's create a space and see. kubectl Run Nginx Look at the autocomplete, it's all coming up. This image injects me into the namespace Demo Nginx Enter Now this pod is created.

The cube CLI GATE Notes If I do then my You'll have a note ready. It will take time because kubectl get pods In the namespace demo Nginx a little bit It will take time. You can watch kubectl get pods Low Watch is not the same Tell me, install watch, see what a watch is. There is a small command which you have to No need to run a command repeatedly It will have to be done as a complete process. Watch Cube Citadel Gate pods If you start running then you have come to running.

kubectl get Your node has also been created in NS It will be. It has so much strength Security is there. Every 14 days, every 21 days These nodes keep refreshing for security For the purpose. In this which OS MI Amazon machine image use It is immutable, so it is the best in security. Nach if you want to read more about it If so, search Amazon EKS Auto Mode like this And this mast will give you its complete information. My Kubernetes cluster is created.

Configure Cube Ctil Access to the newly created cluster Gone. Deploy the Platform Ingress controller from services.ng Certificate Manager and Argo CD Platform services are what you call yourself How to deploy it. Simple Terraform apply I won't target this time. Because now if I apply the entire If I do it, then whatever is already created will be refreshed. Not as many as created Yes, they will be created. This is the best part of Terraform. That I just met you face by face, first Up the VPC then up the EKS cluster Now what I will do is to upgrade to Argo CD and ingress controller and my Helm whatever thing.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster / Namespaces

I will do all that I know it's been an hour since our project was seen I am not coming because it is architecture if so you get the framework. If you go, you can run any number of projects. Because your EKS cluster It's empty now. Argo CD will go into it. Node Key values will change. What extra do I get in the cluster? I need it if my cluster I need a certificate that my application SSL Use a certificate. If you use HTTPS, that's it.

If you also want to add monitoring Yes. Full stack of Prometheus and Grafana Your cluster will be enabled. I told you that you are told that This is Jenkins. Let's focus on the Cubitus cluster. Let's Helm release the Argo CD still. Argo CD is being created. Meaning Argo CD projects are being created. This file is Projects Argo CD. Retail store

Create a project named Sample Application Whose code is in this repository picking up from. In this namespace is putting in. The applications that are These are all the applications of Argo CD.

The chart of the retail store with the cart It's being made. Deployment is taking some time. Most public will get an application that is flaking Secret Shop. Infrastructure built using what teraphone Application Deployed on EKS Yes Continues Delivery Using Argo CD, great Deploying Argo CD Projekt and Application Baat Argo CD Project Created. The admin password of the CD is sensitive. Enter the namespace of Argo CD If you need to buy Argo CD locally It will work only on cluster but if you want local I want to access using port forward.

Its command has also been given. Give me the end point of the cluster. Cluster Name Suffix OI DC If you Connect to this cluster with Open ID Connect If you want to get a third party application done then that too I have given it to you. I have given it to the security group. Ingress Enginix Load Balancer Load The balancer URL is also displayed. If you want to see Argo CD's abs, check this out. If you want to see the cluster, check this out.

Cube CTL Gate pods All the namespaces All the namespaces There are some beds too, bring them all, let me also come.

[Argo CD and GitOps] Helm with Argo CD and production GitOps project / Terraform and infrastructure as code

How many trees there are, oh my god, so many All are, so what all have become of the pods. Our certificate manager has been formed so that we will be able to do SSL HTTPS Engineers, this is what we have demoed. Ingres Our retail store has also become the controller Cart has been created Catalogue has been created Checkout is complete and orders are created. Quickly, click on the load balancer. My load balancer supports my application Deploying. You Have Deployed the Most Public Secret Shop on an EKS cluster All automated.

Meaning, if tomorrow I You can also clone the repository here to any I'll play it on my laptop. Prerequisite EKS and AWS If the CLI is configured or Terraform, then this Your application will be deployed. You should change its public image. Order Place Your Order Arrived Your application is working Perfectly Using Terraform Argo CD help Argo CD has done this I don't know how I deployed myself. Let me show you kubectl get pods in The namespace Argo CD So Argo CD is running, everything is running Now I have to look at the Argo CD UI.

I will write this from Terraform output I will return whatever output there is to myself. You will see the port for Argo CD. At the forward command here, yes here it is. kubectl port forward copy this I did it and went here and threw this cube. Ctl port forward port our 8080 My cubes at 80-80 on localhost The cluster needs to be connected, let's try it. Take the here and go to Terraform in the folder and write the Terraform output. Secret Can you see the secret?

I must have put it in Read Me somewhere Find a secret somewhere Yes here is the port forward to Argo CD UI kubectl Argo CD Gate Secret Copied the initial admin password. I threw this one at my terminal. I copied this And go to your UI and select password. Went and slammed the sign in Hackarm so this is your Argo CD All How to use this application Sync All Healthy How did all these things come to be? What is Chat Me Aago CD in One Write down the UI and CLI of Argo CD Using YAML Files Completely Explained I'll explain this in more detail.

[Argo CD and GitOps] Argo CD projects, App of Apps and multi-cluster / Helm with Argo CD and production GitOps project

Let me give you this code here applications and projects. This The project is a retail store project. The things that are written, name namespace, You can also create all of these manually. Like you said here, project name. Put my new something to describe my New Engines installed from Lets Description put something create Once done, you can project through UI. If you are creating this, please refer to GitHub. If you want to add a source to the repository, you can edit and add the ad source here.

On the same thing I did through the code Is Name namespace Is Description Source Installed Repos Gate's rip repository. Destination: Enter the full address of your server. These are also similar applications. Everything in Kubernetes is a YAML file and A manifest. It's going on but, I need something like this that if I Here is my main app, this is its source I should change it to. Like in the UI I go to And go to UI and then Source and then Main.

Because right now The current images are in it using Argo CD What image is that? Which image is it now? Manifest, all these things are public,. If he uses the image, the public image is public. Our own images It will take a build. The Deploy Platform The service is the ingress controller. Phase Two of the Kubernetes Application

Understanding. We need every single micro Business purpose and technology of the service Understanding the stack is essential. As a DevOps Engineer only and only on Argo CD Deployed.

Quickly picking up code via Terraform Deployment done. That is not how a DevOps Engineer Works. One that is real In essence, a DevOps engineer His mindset is DevOps. DevOps means What happens to Development and Operations? Got the Argo CD folder, Terrafom folder I have understood what we have not understood. Cart's Read the Dockerfile to see what it uses. You also generally Ubu, Amazon, Linux, Redat, all these public You can use the images. This Maven, Java 21 is using this. If you read the files of your cart source files, you have to go somewhere The files will appear.

If you have Java files are visible it simply means the Docker should be doing maven build mvn install mvn clean will do all these things That's right, see, he'll do all these things.

[Argo CD and GitOps] Helm with Argo CD and production GitOps project / Kubernetes Services and networking

Multistage Docker builds are used here now what he has done is some user has been added for security purposes And basically a jar file has been created. If you want to build Java applications This is done. Your Cart Got It You Cart It uses Java. Here is main.go where your If the developer has used Google, then If you need to write a Dockerfile in golang So how is it made? First, you can download a base image of Amazon Linux. Your main is complete with go build The application will build.

Once that It has been built, you can access it from app.main So generally, creating Docker images, Creating Docker files is easy. If you Understand who wrote the code in the background. The question arises that We understood that the cart Functionality It's written in Java from Let's. Which is behind my The SQL is DynamoDB, which is behind Redis, how do we configure it? You of the catalog Go to the charts. Go to Charts and then go to Values. You will understand that the image being used If you build it, you can build the image from the Dockerfile.

If further down will you go So you will understand that MySQL A public image has been used for this is the version image. The service is running on this port. Cube starter What does a repository do? Deployment Strategies, Helm, Ingress, Role Based Access Control, Kind I have brought all these things here. Each module, the business of each microservice Perp, look at his technology stag. Review the Dockerfile you created. Helm Chart The values of the templates are That's very important. If from let's to me I have a card service.

Helm chart is what values does he use In the background? He uses Dynamo So I've used the public image DynamoDB It is also kept aside. If you like this If you don't want to use it, what can you do directly? This is very It is more important for you to understand that hell How are charts being used in this? How do you understand service? EKS Auto Mode lets you gives the data plane. Which is your node Security and your storage. By the way, if you If you want to see service to service then you can do simple Let me explain.

[Argo CD and GitOps] Kubernetes Services and networking / Helm with Argo CD and production GitOps project

Microservice Overview Catalog Service Uses Go and MySQL and Provides API for product catalogues That the product information is shown on your UI You can also use CartService Java Dynamic Uses customer data in the cart Stores. Check Out Service Temporary Customer data is provided for Redis caching Order Services manages my Order and Integer in Java with SQL Data The UI is your front end that lets you access the API. Apps are services that handle calls to each other. This is a typical E-commerce Microservice Architecture with Each Service handling special.

Service to service communication is now complete. Applications of Argo CD You've checked the definition and sync policies. How the service is being exposed is very It's important. All the services that are there if you go into it In service, they are exposed on port Number 80. If you look at all these ports. All these HTTP is exposed on port number 80. The ingress controller just redirecting. You Pick up any, pick up check out service and Go check out its service. This also port number 80 Pay is exposed.

Its ingress He manages everything. Ingress Joe He manages everything, doesn't he? You Go to UI And go to the UI ingress,? In Ingress you have to do everything The thing you will get is that this is the UI of the retail store. GitOps what happens is that I put something here I changed the UI in the application. SCM Source Code Management is yours. What is ECR, what is ECR, elastic Container registry means whatever containers Images will be at the Helm of ECR What it does is it fetches images.

What does Argo CD do on an EKS cluster? Deploy it, it's such a simple task The question arises regarding this ECR. How will this Docker image arrive? Write down your ECR here on AWS. Have you written the ECR? Fully Managed Elastic Container Registry. You can create the repository here. If you want to push me then demo private UI Image Tag mutabs mutability mutable Meaning change can happen Immutable meaning change If it is not possible then I want my. View Push Commands Click View Push Commands.

This Is How You Push an Image To Your Repository. AWS ECR Gate Login Password Username AWS This is it.

[Argo CD and GitOps] Helm with Argo CD and production GitOps project / Helm and charts

Anyone can use it in their terminal. I have to build the UI using let's. Here is where my Dockerfile is located. I'll just do it Docker. AWS ECR Login Password Copied it and posted it here. This should now be the Dockerfile here. What is the Docker build image? That much time It will take Docker to build and so on. It will take that much time and he will build it completely. Docker is also very easy to understand. They are masters of DevOps and now they want a project.

Your private image is yours. I'll tag and I'll push here I will then go and get it to Helm and he Argo CD will deploy the latest image. I don't want to I wait for hours to see when my will come It will build. When will he No dipl here I will tell you one simple thing I give which is known as CI/CD Continuous Integration You've seen the Continuous Deployment CD. I took that Argo CD will say that whenever your Source Repository Changed Those Things I'll deliver it.

CI How will you do it? CI Continuous Integration means that anything in the code Detect the changes that have occurred. All of this if it happened automatically then that is called Continuous Integration. CI/CD in Continuous Inti Integration: What does it take for you? Check out the code Then if your code is latest then its Detect Changes Then What Will Happen docker build then what will happen docker tag then What will happen then? Docker push to ECR then What about my latest image?

The tag is the update tag in Helm charts. This If all these steps are done automatically If you go, I will have the latest image that will be created. Its value will be shown in the Helm chart. From here From here In Continuous Deployment I used Argo You have already set up the CD. This becomes my complete CI/CD pipeline. The catch is if I didn't do this CI and I Directly updated in Helm Charts Go here manually So can my Argo CD deploy that thing?

Meaning this is the entire CI/CD pipeline It remains and not from GitHub This will be built on Docker, this will happen again There will be a push in ECR These.

[Argo CD and GitOps] GitOps principles / Helm with Argo CD and production GitOps project

It will be deployed on EKS, so this DevOps lives Code Build Test Deploy CI/CD now know what the problem is What's the problem? If I Go and change the values directly And this Argo CD deployed it Then your application must work. Are those values GitHub Does it match the code on the Is it? After a small error in DevOps Chances are that if your engineer DevOps engineer manually helped updated the charts and added Argo CD If he picks it up then check your code.

The code that will be on Git will be out. Those in deployment and those in production The match that's going on won't happen. That's why The concept came from Gitaps what is the difference between Gittips and DevOps? Speaks to your Git and your version control The system is the single source of truth. Meaning Whatever is written in your GitPap code The thing will go into your deployment. If your GitHub has a If you need a replica, you can use it in your EKS cluster.

There must be a replica. If it has 10 If it is written then it should be 10 here. The way you do something is to change the code in GitHub. Phase Three of this The project is the most important Face Till Now That here you have to understand GitOps a lot. Review GitOps Principles and Benefits Single Source of Truth Meaning I was quietly building this No, this is such an unpacking of Docker, this quietly I built it and I quickly got here.

Docker tagged And did the docker push. Because the docker push I am doing, my Helm chart will pick it up. Can I upload my source code to my GitHub? It will be on GitHub. New Folder Whose name will be GitHub Work Floss And you have to make work floss. What is work flow, this is what happens The workflow is what I just described. This only that changes in the code were detected in the code If changes are detected, docker build will happen,

Docker tag, Docker push, update Helm Charts Pay and Argo CD that thing Deploy it automatically.

Let's get started with GitHub Talking about actions. By the way you The image is in the private repository.

[Argo CD and GitOps] Helm with Argo CD and production GitOps project / Helm and charts

If you look at this image, Where do I put the value after picking it? Helm Chart Put it on. Let's understand GitOps right now. You can find all these workflows on GitHub. Defaults to static HTML Deploy the pages. Deploy static HTML pages then you just configure this Click on the button. Docker Container Publish Want to do it? It You can also schedule that in the 23rd minute Run it every day at 8:00. If I whether I push anything to the main branch or to the main I create a pull request on the branch So build this image on this registry.

It should be ready and whatever will be built should be local. They will not pay, they are the runners of GitHub. The one running Ubuntu and checked out Your code will do whatever the latest code you have Docker will take the code that has been pushed. What will the build say is Docker's metadata Docker build and push Docker image build and Push Joe's commands we have ECR's put that here and that's it Will push your image. That you don't need much You don't even need to write.

This 404 is coming because your There is no work flow in the main branch. No, but, if I work flow If you want to write it yourself then I will tell you. Kiro, I will tell Kiro, Kiro of Do me a favor, enter your idea I want to generate dot GitHub work flow for this Application which does three simple things three things What will it do? Detect any new changes Build and Push Docker Images to ECR And update the values in Helm charts Boom, enter.

General After understanding your source code well, one can AI can also create workflows for things. All this data is local cache. He's a Spec will create your There is a requirement, let's say you require Do you do Gib Actions Work Flow for Retail Store Application CI/CD Pipeline which includes I have a requirement CI/CD. The pipeline has to be automated whenever the main The code will be pushed to the branch and this will happen whenever If the microservice changes, this will happen.

If multiple microservices change then this What will be the second requirement as a DevOps Engineer I Want Docker Images Too B.

[Argo CD and GitOps] Kubernetes Services and networking / GitHub Actions

Automatically build and push to ECR. This is a project that I have created. If the requirement seems right then you can apply for it. You can also do it from here for example I want it to be the best DevOps and Gates Practices Or if you have a requirement then write like this. Told that Detect Changes, Docker Bill, Tag, Push, I'm showing you all this by design. CICCDL Trigger on push build then build service like so This will also create the entire YAML What a great move to implement.

Task 3: Send an email if my build fails so on. If you click on this task will you do it First create main CI/CD as a DevOps The engineer thought that first git work A folder with flows will be created then path based. Filtering will be the card catalog and so on Then the reusable work flow will be the build service Then AWS Authentication ECR Access. If you want to know how the authentication will be done then you will have to Giving your secret access keys to Geeb You will read it, yes they all give it like this.

Nothing, don't worry, docker build multi Platform, all these things, service Specific configuration configure go service If he doesn't configure all these things, he himself, you will just finalize the task. Let's take you to the last You don't need a task. All in one DevOps Engineer does it in day to day work. Boom Create Main CICCD Workflow File. Here Your GitHub wfls folder is created. If you want to make CI/CD then in GitHub wfls CI/CD. That too was told by the from DevOps.

Create me The CI/CD workflow file has been created. If you go to CICDC, whatever your Docker push and build is action, setup build exe Yes, need AWS account ID It must be, you need a reason. All these things, all these things here, update the chart Path made it all. I'll go to a new branch. The name is Git Check out GitOps. I created a new branch I have created this in the new branch. There is a simple pipeline called deploy which what do you do if you push my Path SRC Meaning in Branch GitApps If This is the GitPop branch in the GitPop branch.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / ConfigMaps and Secrets

In this in SRC file if my developer Look, the developer will push it here only,. If someone wants to change the UI then he can change it in the UI. Will change the catalogue then in the catalogue If he changes then I will change my work flow This work is done by a DevOps team. Here you will find your I have to make sure that I have the entire repository I don't want any change. My pipeline should not run. What do I need in the environment?

Go to your Gibe repos settings Secrets and Variables Actions New Repository Secret Access ID Your Access ID Now the user who has accessed your email address Case cluster created with same user ID. Use it and create a separate user for git. That because of that user this Changed Don't Mix Up Your Users So what I'm going to do now is I'm going to go to AM. This new There is a user who now has which-which I need access to GitHub. ECR has to be given.

ECR Me Re, give me admin for now, but you What do you basically need? You can buy Amazon Q You can ask what all You can ask from here what all permissions should i give to my gib actions what is this Zoom in a little bit, all permissions should. I give to my GitHub Actions user In IAM So That It Can Build Push Images to ECR Use the best security Practices. GitHub Actions ECR attached policy document. What's the point of Attach This Create Access that's the power of AI I this I'm not saying that AI will take your job.

Here in the middle there are agents who use tools are using use AWS. Keys GitHub User Key Let Me Have Access Keys will come right now I got the access keys,, so this is mine. Access ID Now here if I If I refresh, a user has been created. Gotta GitHub Actions ECR user aay now i What should I do in my retail store app? I'll go to settings I'll go to settings After going into Secrets and Variables I will go to actions and here Repository secrets are mine I've changed the repository secret I will configure it as written here.

[Argo CD and GitOps] ConfigMaps and Secrets / Helm with Argo CD and production GitOps project

Go To Your Get Rip Settings Secrets Variables Actions New Repository To Access ID Secret got its name and Secret Access What happened to the ID of that user? The access key is Create Add a Secret Access The ID has arrived, what else do you want? Need secret access one more Create a repository secret and enter its name. Secret Access What is its value? AWS Region See Read Me. What AWS region is the region? This is how you have access to GitHub.

I have given you the details of my AWS account. Not the entire AWS account Given. Cannot create cluster raster This. On means when will this job run? Push changes to this branch on this source It will run on AWS in this environment. You have the secret in your region. There are three jobs in the job market. Detect Changes, Deploy and Give Summary. The job runs on Ubu. Meaning, on the runner of GH. GH automatically on its runner Will run it. Output Change Service Steps Change Output Change service what is this means where But also if there are any changes in these steps The steps are this is a key and these are its.

Deploy this lot This matrix is the dot service matrix. Meaning of Matrix Parallel Jobs. Here the credentials are configured. You have accessed the ID secret. You have access to ECR. All the same commands are written here. Name of the repo Account ID Region. How did you find out the account ID region? Here if you go on ECR So whenever there is a push on ECR, if you go to your private registry whenever If it is pushed, what is the name of that image?

Your account ID then DKR ECR US West Region Amazon AWS your One way is to enter the name of the repository and then the tag. ID DKR ECR Region Amazon AWS then Retail Store Your Image Name Then Service Name of UI Cart And So On Then If the repo is not there then create it. Create will build and push docker build Same command docker build docker push The command will be pushed. Once the image is pushed If done, update the values of the Helm chart.

Updating Helm values for this So first you have to see that Helm chart where is it kept.

[Argo CD and GitOps] Git and version control / Helm with Argo CD and production GitOps project

Just doing this and that successfully Updated images. Commit those images to GH too. Because when you commit on GitHub What happens? If you commit to GitHub If you do, the work will start. It The jobs that were to detect changes, deploy, Summary. This was your CI Continuous Integration. Once you've committed I did the Argo CD work again on Git. I have mentioned it in the applications. That if in the GitOps branch of this repository If any changes occur then these values Deploy it.

Values The Deploy CI job has added values here. Changed it And Argo CD deployed that. Meaning, whatever goes to my gate Argo CD will deploy my cluster Pay. Git status now ER policy is ECR policy. This is a commit called test commit. That All my images build it. You So I said that you will change it on read me and Not good if the pipeline gets triggered He's showing you a demo. I'll do if git status So I have five files changed.

My status became five ready to commit. If I commit this get Commit hfan.m added minor changes to src Microservices I will put something like this for testing. The commit has been made. Because what does the pipeline say? Pipeline says on push to the branch gate offs. If I do gate Push Origin Get off. If you want to do it, you Take the diff of this branch, this repository and Try pushing in yours. Git Push Origin Gates Enter The push happened. As soon as the push happened.

I'll go to the repository and go to the GitHub header Recent Push. Pipeline is running added minor changes to Click on SRC Microservice Testing At the end of that you will see Detect Changes service is it getting some errors Oops, some errors came up, we'll fix them. I will do it, my job is to fix the error. What error is occurring in Deploy Cart Build and push card pay User is not authorized To perform ECR create repository we I gave him access to push but.

If you are not given access to create repository Now what I will do is create that user. I'll give you access to the repository. Amazon Q So if you write Amazon Q ECR Create Repository Access Was Not Granted Given to the GitHub now this user name is You will have to pay properly, otherwise he.

[Argo CD and GitOps] AWS (EC2, VPC, IAM, S3, ECR, EKS) / Git and version control

Get to the User Gateway Actions Using the Policies ECR Create Repository Access Was Not Granted Given to the GitHub Actions user. The ECR in this repository He will assess the policy that he has already kept. What he is saying is that the ECR policy is missing is That's so cool. List Create Read Update End All ECR Related Permissions On AWS, ask if this user Who has all the permissions? He will tell which permissions and Needed? Tags vj list karne Permission is also required.

It's not their job to do it themselves. Take decisions on your own, pipeline on your own Everything will be messed up. If you tell the AI That build this end to end makes garbage It's very bad. You need a human in The Loop. You may not know that ECR How to add a Create Repository policy Yes. The Permission Issue You Encountered Should Now Be Resolve Your CI/CD Pipeline Should Work Walk Smoothly Click on rerun jobs so that If you missed something, you will get it back.

Reinfeld doesn't have to pretend to be committed Click on Jobs and all these things fail. It's done, click on rerun jobs and let's go. Back to those Jobs metric tricks. I saw the whole build and push thing. This is the best part of DevOps that you We get to learn so many new things. You have to be mindful that you are not following the tutorial. Did you read anything about that project? Let's Deployment Summary My gave it to me too. This branch But it happened.

Build and push Docker images done. The Helm charts have been updated. Argo CD will sync it. It said that Argo CD will sync automatically. Which is this branch in which Have all these things happened? This branch name What is it? This branch is called GitOps. I am not taking it from the gate of the branch. Because when we Demonstrate the project in the beginning When we were doing it, we used the public image. The private image is in the GitOps branch I have kept all this branching.

Strategy explained in the.d file How is my branching architecture? If the developer changes the code, If it is in branch then manual deployment Public ECR images will be used and Umbrella Chart Complete Guide to RGCD But if the developer changes the code.

[Argo CD and GitOps] GitOps principles / Helm with Argo CD and production GitOps project

GitOps branch in production branch I will then trigger GitHub Actions which Buldan will build and push to hell charts Will upgrade and Argo CD sync. Private ECR will update the images and will update individual apps. It means that This is my Joe Argo CD. GitOps This is the configuration of Argo CD. I want to apply it in my cluster. Because right now what I running in the cluster, what is that running Right now the ROCD running in my cluster It is running on the main branch.

kubectl apply - F Argo CD In which applications of All the applications, whatever they are in these namespace Argo CD Enter. At my updated Argo CD Configured Updated Target Revisited Which Is Get Ops will use that. You can use your Argo CD if you look at the applications then all Look, they're processing. If you look at Target, you'll see GitOps. Branch and all are healthy. You coded on GitHub Actions changed that. Build Test Currently in testing stage I haven't added it.

I built it and Pushed to ECR. Helm Charts Updated and synced by Argo CD And every Argo CD is syncing now. Because he will have to complete Moving from a public image to a private image If it is there, it will be completed. Check if my application is running or not. If I want to see the UI of the applications So I can directly use the Terraform app. Terraform Output And here I get the URL of the retail store. Just run the echo command.

Nothing is going to happen at the Main Branch. Branch Is Your Branch with Public Images. Go to the GitOps branch. I'm going to go into the source of the UI. I'll commit to the GitHub branch. As soon as I What will my git gib actions commit? It's been pushed on the branch. If I go to my sample app So the push has happened to the GitHub branch. Why weren't the others deployed? Because I have written that when If the service changes, change it accordingly.

Build End Push is pushing the UI now. Helm Values Let's see what will update it. Build End Push UI is happening. Build and Push UI is underway. You must have Argo CD skills.

[Argo CD and GitOps] Monitoring with Prometheus and Grafana / Terraform and infrastructure as code

EKS Auto You should also know how to mode. Ticking off the agenda So where were we, you saw how Access to Argo CD as an admin You have entered the port with the credentials You also saw the forwarding of GitPos. The principal of your application. Deployment was also monitored and observed Improved Argo CD's sync processes Health Monitor just got declarative Configuration Management GitOps Getb Work You've seen Flos' self-healing capabilities. You have taken the URL of the load balancer. Test Different User Workflow Browse Add to Card Checkout I have taken it.

Monitor application health and Performance metrics have been done. Practice Trouble Shooting Common Deployment Let's practice the issues. There are some common deployment issues that you A changed image, changed secrets. Learn to identify Roll Sync Issues in Argo CD. These Sync Issues Occurs when your configurations are correct No. Infrastructure as Code vs Terraform Have you done Kubinets cluster management? Container orchestration done in EKS auto mode GitOps Methodology Viewed on AWS How to integrate cloud services Production Trouble Shooting End Monitoring View Declarative Infrastructure Management (See DevOps) You got the idea of the practice version.

How is monitoring and control done? You may also have seen Observability Practices For If you don't see me, do what you want to do to me more Do nothing I just have to go here and your Where in Terraform. Find Prometheus Do you see it somewhere Prometheus's Terraform It So what do you have to do in advance? This much part is for not commenting Just fill out the application form and that's it. Monitoring and Observability will also be enabled. That's the beautiful part of having infrastructure s code.

Because your The infrastructure is in place. Enabled Kubernetes Prometheus Tag and put it in the namespace. Package modules were available so he I turned it on. Modern Microservices You get the idea about architecture patterns. Deployment Strategies Video I'll link it to you. GitOps Edge A Deployment Methodology You Understand How does production work? End to End You've seen the DevOps workflow.

Linux for DevOps

Source file: transcript_e01GGTKmtpc.txt | 22,914 words kept of 44,723 | 59 sections

[Linux for DevOps] Linux networking commands

You will mostly have things that a DevOps to the engineer when he Boot loader, oh, so many things. Tell me this [music] Service provider like the one in Pune Broadband or broadband or airtelxtreme.in Devices Identification Is it a server or what is a server, see the server. No, it is a very simple thing, just understand it like this that there is a computer that has a specific job. To serve information to you Please bring information of any type Bring information to serve now this If I say bring me this email, then that What will become my email server if I.

Should I tell him that, this is my file, this is your file Give it to me on your server and I'll have it whenever I need it. Give it back, then it will be the file server. File server, now I'm saying that. I have my data and I will give you inside that database If I want to insert it, it will be database server now I have a server where. What about my web server that provides web www.htp.gov.in Whose job is to serve whom?

I will serve you, you will serve Client to serve client now The first question you will be asked in the interview The question asked is about server and client. Tell me the difference between the work of the server It is to serve information to the client Work is done to Request for Information What are the servers and what are the clients? I told you there is an application,. Where the server youtube4 1 This must be some IP address How are domains and IP addresses linked?

There is also a server for that work which Let's say DNS domain name server understood That's how the internet world works. If you write facebook.com [music] I have a web server that has some of my images. Some files are static data is static Data that needs some more work If not, then where will that data be stored? It will be stored on a web server. I have a And there is a server where an application is running. Where some computation needs to be done Calculations have to be done, logic has to be established Dynamic data is giving static If it is not dynamic then it will be an application.

[Linux for DevOps] General DevOps discussion

Server so if someone asks you web server and What is the difference between an application server and Web servers generally serve static data. Applications generally use dynamic data services. Acts like your web server has been Nginx So through Nginx you pages are static pages Application Server Lets So this happened to you because of some war or Nut Jas's application is running on a server. If it becomes an application server then this In this way you can also know the difference web How many applications of the server are there now?

There are types of applications, see. There are two types of applications, broadly If I tell you, stand alone and web Is the application a stand-alone application? Stand alone application you ever Have you been there, like to some airport? How was your feedback and Clicked and went away or sometimes It happens like this, put a coin and the work is done. That is your web application. The application runs on the Internet and Supporting email servers Database servers are applications There are servers, there are a lot of things.

That supports that web application And that's what a DevOps engineer needs to know What type of application should it work on? If he is doing it on a stand alone application He is working on databases and other things. There won't be much need for connections if he is working on web app then database Connection with the front end Connection with the back Connection with the end Connection with the cloud. This is going to be a very long series Take it easy, no problem, Now the question arises that what is Application Support and Maintenance What is it?

It happens that I had come to study DevOps and support. There are applications, aren't they all the same? Running on a particular operating system like nuke operating system windows operating system mac operating system so all these Applications running on the operating system They Are the applications running properly or not? Kisi Type You No Care Affection Love Whether support is needed or not Application Support and Application Maintenance If work is done on it then it is just. The basics are where you need to know that if the application is running, at any time Issues can arise at any time, it can happen that It has a connection to an email server.

[Linux for DevOps] SSH and remote access

If the connection is off, your emails will not be sent. Being a DevOps Engineer So true and tell me that What is Linux,? The reality is that 90 Q Applications do not work on Wiz, where do they work? The operating system which In 1991, a long time before I was born Linus Tor Valds created it even before He created an operating system that Basically, anyone who needs an antivirus No, you can do multitasking. Security is top notch, it is open source people You can contribute to that Can improve the operating system And the best part Unlike Windows, it will not happen that your.

The meeting is in two minutes and Windows is Updating in your production Issue has come Windows is updating your marriage It's about to happen and your wonderful email to you. I have to reply to Can't Windows This is updating so all these To solve problems, men Application security is the solution to this problem. To solve We use majority companies 90 More applications than It is a very prime example But if it is paid then one of it is free. Version or the open source version is free you are great open source you ever You can also use it, so that's it.

Introduction to nax that this is a The operating system is fine, operating The system is where the application runs. Windows operating system inside Install it and virtually [music] This is another way to install Linux The third method is AWS or Azure or else A virtual machine on GCP (any cloud). You need to buy the CD of Viz 10, Viz x, Viz 10 No, we used to buy CDs and bring them on that. Does matter of OS comes in Ubuntu Kali [music] Software Remote Location Server Tools Meaning If this is my I and here I am A remote VM or a server.

I want to connect to how do I do that There are many ways to do this There happens to be one RD P Remote Desktop Protocol that you Access any deck with the remote. You can, the second one is SSH Secure. You can also access the shell like this I will tell you both of these things in the upcoming series.

[Linux for DevOps] Linux boot process

Where I'm just telling you that How do you connect to a remote server? Should be because as a DevOps engineer You can also manage the systems from the comfort of your home. No trouble shooting so I need all these things, now, what is a kernel, a boot loader? What is a shell what is it let's go now I'll teach you in depth about Linux. It happens that its name is Linux kernel. Where does the name Linux come from? Its founder is Linus To He named it X, so it's Linux.

It's Linux itself, now something about this kernel. The job is to run processes, this is what But the colonel is like this. Programming Objective C like C The kernel is now written in programming. I need a program that does this Talk to the Colonel, we should find something like this. Need a program that talks to this kernel And then your shell arrives What is a shell? A shell is basically a platform is a terminal where You can talk to the Colonel through the Colonel's For example, I want Linux.

Kernel Linux, the kernel, has a program has been written to create a folder. It is written in C programming now I don't know C but I want to create a folder. I'll tell Shell to create a Folders now have shell commands for them What are commands? Shell commands There are many such things like mkdir All commands are in the upcoming videos I'll see, but we can use those commands. By doing this you can talk to the colonel and he Commands are called shell commands.

If you Shell should be well understood and Shell What happens is the shell remains a gateway to your and the gateway between the colonel and you So if you Understand what a boot loader is when. When your computer starts up, the first what is the first thing you remember When my father was not there in my childhood, he brought me here. I brought the CPU a long time ago. Papa had brought a CPU and we used that CPU What I used to do was very simple.

They would press a button on the CPU and that button What would happen to that CPU when it was pressed?

[Linux for DevOps] Linux boot process / Linux processes and services

The power was on and the hard disk I used to lose power, now that hard disk What is kept inside such a hard disk Inside, your operating system installed and what's inside it. The operating system which is the kernel has There are some processes, one of them process boo boot loader what is boot loader There is a boot loader. There is a process like this which is the files of your operating system which starts the operating system Those files are required to It runs, I understand that it is not a shell.

Shell what happens between you and me Sorry, just an interface between you and the Colonel. What is a boot loader? There is a process that runs first and Wakes up the whole thing at home how Mummy is the first one to go and wake everyone up. Wake up, a new day It has started, get to work as it boots The loader says get up, get up everyone Turn on the processes, our users have come. If you have already run Nginx then the boot loader is very It is important to know that You should know the full form of RUB I'll tell you G R U B to.

It means if I start this laptop If you have to do it, what will you do first? It will also work, its full form also Something global happens, it always remains There is no, there is something here No, maybe it's, there's no scene. We can write that it is completely Form Grand Unified Boot Loader Saw grub is a boot load so gr r ybi is a boot loader so if you. Ask in the interview, can you name any boot? Linux loader GR ybi simple much more in it inside it there are icons inside it there is a Open the terminal like the Start button You can do a lot of things like.

Butt Linux system architecture This is what remains, your shell remains in the center Sorry, no shell, Colonel, to talk to Colonel. Your utilities in and outside the Vala shell applications or whatever you want to run Whatever hardware applications you run If you want it to have an architecture like this. If you want to do this in detail then Linux System You can also see the architecture, you will find it interesting.

[Linux for DevOps] Linux disk management and LVM / Linux boot process

You'll notice that you have a shell that's the thing does it with the colonel and then the colonel basically who also have hardware that runs the thing So if you rebuild it, your. The Colonel who lives in Shell tells him that Colonel, do this and what would the Colonel have done? He does it with let's folder create or free up some disk space So that hardware is I'll do that and run the shell What you need is something Requires applications such as terminals.

Or Notepad or whatever So apps shell kernel hardware this is a flow You should have But it's not hardware its applications Shell, Colonel and Colonel will change what? To the hardware so if you need to Diagrammatic representation So this is the diagrammatic representation That center houses your hardware Became a printer, becamera, becamera Whatever happened to that camera, how did it come to me? Is connected to the system through kernel but if I need to turn that camera off and on. What I need is an application.

Will be the application shell will say that shell He would speak to the Colonel and the Colonel would drive the thing. Architecture resides in Information about the hardware lets From this you will get disk RAM CPU all these things If you need information about it, If there are some commands then the commands are as follows. I had kept it for day two but you can do something. You can extract information about the hardware like if you whenever, maybe I don't have it in me, but if you Free Han wrote in his Linux I think What happens for this remains free -.

There is something that remains, it doesn't matter If you don't know why No DF Fun A E for your hard disk no, it's free, but for Mac. If you write such commands also No command the free command is a common utility of If it is a little different then something Commands change, not much,? If you need information about the hardware You need to know how much CPU is being used. You can write top if you want that How much is the hard disk being used?

You can write DF - if you like rum If you want to see it, you can write it for free.

[Linux for DevOps] Linux processes and services / Linux basics and file system hierarchy

That Everything in Linux Starts from the root folder cd / this Whatever remains, this root folder remains, root folder means if I go to that. If I want to tell you diagrammatically, everything starts with a slash This is called the root folder, now inside it. You have many folders like Home and us r then bin remains binaries remains then ETC remains then your variable live and many things live if you want to know about all these things. All the things covered and all the notes will be given to you.

What is usr what is bin ETC Ver Ye cd / cd is a command to change DirectoryEntry ETC bin volume system in folder files private usr temporary files y Everything is files and folders. Linux keeps everything sane If my logs come, I will put them in Ver So CD / Ver See the / logs here logs, whatever logs are generated Let's go here, what do I need? You can see what's going on in the CPU. What is the command, top, so put top and this If all the processes are completed then what is the total processes running on every computer to It depends on the computer, but this computer.

There are so many total processes in it, out of which two Processes Bus running Five processes running It is happening and the rest are sleeping, great processes What happens in the background? That your system can run There should be help and the first process that happens Neither his PID is one, see this now P ID One P ID

stands for. Process ID So if any of your processes If the ID is one, then understand that the boot loader has First of all he launched the process that is Why its process ID is one.

That was it, now the status of the process From which seconds do they come, see here many You must have seen the states, look at the state sleeping. Running so there are states that a If the process is running then it is running if she is asleep or not working If it is sleeping then it is sleeping. If that process is killed If it has happened then he is killed or terminated.

[Linux for DevOps] Linux processes and services

If that process continues like crazy It is there but if you don't want to give it to me then That zombie process is so different There are types of processes and all of them. The status of your processes is that if your If the process is started then just open it. Running will appear if your process is stopped If so, then it has stopped coming. This is a 45 minute video where you can in the forest You have understood the basics, you have understood the basics in Day One.

It happens [music] F step so what happens in that DevOps Engineer said in the first step AWS Account Setup for Recruiters And Hiring Manager Receive More What Happened This, so what this is, you're a DevOps. You're an engineer and you basically have an AW If you want to create an account on S then how to create an account Let's create it, to create an AIDS account. Why am I saying this because, this is the way Right, either you can use WSL Use Virtual Box on AWS Create an account and use the virtual machine.

I am going to tell you about the virtual machine method. To create an AIDS account, go to AW S go on he should know you DevOps If you want to become an engineer, then do that. He should go to AWS and go to After this you will have to sign up for Visa Both credit and debit cards are accepted, you can create an account by giving an OTP. If you have an AWS account then my account I will go and after I go, maybe I You have to sign up and in sign up I What will I do?

I signed up and It's asking me for the code because I made a little security If I have kept it then this code will come on my phone. It goes and comes what have I done Created an AWSP account and I rent a house in AWS You have to search for e e2 stands for Elastic Compute The cloud where virtual servers reside. To create a system in that data center So on the latch instance I'm going to I'll go and Linux Let me create a system called for DevOps,?

[Linux for DevOps] Linux users, groups and permissions

I want which character, why character No problem, you can also take Ubuntu. I am telling you how to create an instance. Because what you have come to read is DevOps where you level up every day I have to go and learn something new every minute. That's why How Does Internet Work When we started in the first minute and Now at the 50th minute, how an instance. Makes you want to know all this stuff and that is why in the date that will be This is where we'll start that instance.

What this instance is a server what is a server is a web server Where In the parquet, what do we call this thing? This web Server un or Nginx is installed and I What will I do to this Nginx, this Ubuntu? I will connect, I will connect and with this I know what the benefit will be from this. One for Got an Ubuntu system Got a Linux Are you watching this Are you watching Welcome to Bantu? If the computer is your nuke, then whatever There will be instructions, we call it Commands, so if you want to get some particular work done is your.

To do this you need a shell did not create the instance if you are using virtual Box Ho Vegro Ho W Even if you are running on /, all these commands. It will remain the same for you no matter what so let's start so I go to my keeping away your [music] If you catch it, that is what the terminal is. This is the terminal from where we can shell If we catch it, it means here we People will enter shell commands like a simple C I will tell you the command if today's date.

Friday, December 15, 9:00 am It's 9:00, that's the time in UTC. The shell command date asked the colonel, "!" what is date kernal ne hardware meaning Understand that the date has been displayed on this display. There are many such commands coming in, the first command is LALS. Means list everything that is in a DirectoryEntry [music] So make a directory entry Now I want to see what a guy named DevOps is up to. DirectoryEntry The details tell us that this app is This user created it and it is his There is something Parmish, Parmish, we will see later.

[Linux for DevOps] Linux basics and file system hierarchy

What do I have to do DirectoryEntry Where you are present, where you work are you working DirectoryEntry If you are in a folder named Ubuntu inside Home Name Key Command to create directory entry and file. It is a directory, it gets denoted like that It's all so dirty, everything I have written that I do not have to clear it to everyone. If you want to do it then you can do it after clearing CLR. If you clean the terminal, like this There are some commands like ls PWD M KD I R what is this CD DVD so The times are gone but what is a CD?

For change directory so if I get the current Directly from someone else DirectoryEntry So he will tell me that Home Ubuntu Devs You are in it, which means you are in DevOps right now. I didn't know how to enter the directory. How to exit the directory, give cd As you take a step forward, take two steps back. Dot and came back PWD did Look where you were inside Home Ubuntu DevOps now Come home come to Ubuntu two steps back Now I have to go from Ubuntu to Home.

If I want to go back then cd dot now I Where have you come? Home I have to go further back CD dot now where have I come root DirectoryEntry Late, I have to go to the bin folder. Binaries are kept in the CD bin and now I went inside the bin, look how cool If you are feeling like a hacker then do such things Now I have to go back to my camps. Where was it in the folder cd/home Then it was in a folder named Ubuntu Then there was a folder named DevOps, I came here.

You can see that you can open any folder from the CD. I will remove the Fidti Acti I will go and try LS but I won't find it. What it is View the directory using the rm command Files are removed from directory entry Alphabets are written and spoken Flag so I am doing remove recursively Everything Chch is inside cloud enter now. How to remove and delete files How to remove folders How does another command remain -r md I R DevOps so what does that do to you can remove the folder directly.

[Linux for DevOps] Linux basics and file system hierarchy / Linux text tools (grep, awk, sed, cut, find)

New Fa I want to create another file demo Faith alone is touch demo file dti akti now by doing this I will make a demo file but this file is MT. I will do cat this is not cat but cat demo file.txt Enter returned nothing because the file is empty. That happens whenever a developer or Then one goes into coding which is the most The first program I write is. World should be printed here too To print World, we need We use echo and then double Whatever you do inside quotes but The world is created by those studious people.

I will redirect this angle I will redirect it from brackets Whatever output came, put it in your demo File in Dati Ekti Now if I do cat demo file dti XT you will see so. What cat does is read the contents of the file. Shows what Echo does on the screen Whatever you put in front of it, print it. If you want to write a file In which something like this is my file is written So echo this is my file and you can use it as new file.

No problem it will make it will make Try LS and see if my fat is formed then if you don't want to use touch, you How much can you do this directly? Meaning if you have a zip file, it is a zip file means four files It has been compressed by compressing it I have created a large file, large file. Is it a compressed file or You must have done it in zip format,. Unzip the files inside that zip file. To create a file then this head tail What is head tail, what happens like this Suppose I have a file called Inside which some lines are written like I wrote cat new file.txt now its.

There is nothing inside then cat ma file.txt Inside it is written 'This is my file'. This is my file, now this is total What is the number of lines if it is the same I like the top five or top 10 lines from the top. Keeps monitoring, for example, a real Let me tell you the life scenario, you are The DevOps engineer is checking the logs.

[Linux for DevOps] Linux basics and file system hierarchy

Daily Logs are coming to you again now You don't have to read from top to bottom. If you want to see the last line then what are you doing? Do tell - aapki log f to naye Logs will keep coming from below and they will be visible here will stay they will keep looking here watching Yes, your terminal has not opened. I don't have to wait for a minute, let's go a little more dangerous ones commands fun Now we understand that it is coming This file is my new file.

Home Direct No Where is PWD Yes, the home is in Ubuntu, here I have to give it a If you want to copy it to a particular folder What is a particular folder mkdir. DevOps, so there is a folder named DevOps in it. I want to copy

the new Fidgety act So I have to copy CP copy Whose source? I want to copy the source file to a new file. It is empty now where do I have to copy it Have to copy inside the folder see / come went inside the folder and by doing this I will press enter, now what happened.

The file has been copied, which means it is here as well. Show list is also inside this dev folder CD Des and Ls see new file copied So you can copy it now from the left. I have a Y here in the DevOps folder There is a file Touch Des Fidti Ekti DevOps file.txt What should I do now that I am fat? KD IR Cloud a new folder Made a cloud that's an empty CD If you try to cloud it, you will find it empty.

I have to stay in this folder, meaning home Ubuntu By staying in the file kept inside the DevOps Copy inside cloud file folder. If you want to do it then cp is kept in the DevOps folder This is what happened to the file which is the DevOps file Yours Source How Folder Name File Name This. Why did you do that because you are not in the folder If it is outside the folder then the path to the folder and copying it from there to the cloud.

[Linux for DevOps] Linux basics and file system hierarchy (part 2)

If I do Enter, then CD or LS Cloud then you will be listed What are the things inside the cloud? How interesting is the CP command that you can You can copy the thing also question I get this cloud folder Copying the entire file to the DevOps What I told you if you. If you want to copy some things inside then CP -R Recursively copy the cloud to two DevOps Enter inside now if you go to DevOps You'll find the cloud folder there.

Because I recursively saved the cloud folder Copied it into a folder named DevOps How cool is that, doesn't it mean you can do whatever you want? This file is new fati ekti this cloud I don't want to copy it inside What to do to move inside the cloud I have to move my CD cloud to this. I have a deb file inside which is new and torn. The action is to move inside the cloud Do what the meaning of move is in the cup?

Copy both in the source and in the destination The thing is created from the source in the move Moves from the source to the destination What will I do if it goes away? I will do CD DevOps here which I will get new I have a file and I want to move it from here to the new one. Let's rename me DevOps I want to make a nose for devs, how do I do it? Don't think I have one thing and he has another.

Get deleted in the source So from Deb So move DevOps move DevOps to linux For DevOps Rename Now look DevOps has been renamed How accurate is Linux for DevOps? The move command is very interesting. That's interesting, now this ws command. W S means welcome but linux.com file.txt It has a line There are four words, one, two, three. The file is of 16 bytes, bytes means storage is taking up so much storage then LS Han L. If you do, you'll see my The file is 16 bytes long Do you understand how much storage space it takes up?

Let's take Now that I have a Soft Link this is soft link and I want to write this as a DevOps file I wrote in the dev file.

[Linux for DevOps] Linux basics and file system hierarchy (part 3)

If I do what does cat DevOps file do to me You will get it this is soft link This is soft link PWD do this which. The file is located under Home in Ubuntu. Sluban 2 / en is the command to create a link h s is The Command to Create a Soft Link Soft Link means if the file placed in this path. Blue color LA N was seen AT this shows more details so look at this Arrow showing view soft link file Showing means this is a link to this file location its simple means if I.

What if I CAT soft link file? I know it will be printed, this is Soft Link: Now tell me what is the best part If I go CD What Was the Cloud and Then DevOps. This was it, no, no, no CD, yes, here And I will change this to this file was Change and my DevOps file I will change it and where is the original one? If reflected or not, then CAT Soft link this file was changed if Something kept in your original location If yes then you can use the LA command and the link command.

Shorten it into any one short link You can link it to a shortcut and that thing you can use Do what happens in a soft link? If I did RM remove I'm in the Linux for Dev folder DevOps file from cloud folder I removed it Look at the soft link file given, it has turned red. Softall soft link will delete if source is I understand what a deleted link would be. Echo where should I write this write to file I will write it in this file.

I will do it if I do CAT, it has come, now I am trying to make it hard. Alan now what is the path to Linux for deb a and put its file name link shortcut link I will give you the name, enter the hard link file. When the file is returned, it looks like this: hardlink see hardlink is here now if I will cat hardlink file It is printing but if I remove it I will give you

my original one.

RM I have removed this one now if I LS If you do this, the soft link file gets deleted again.

[Linux for DevOps] Linux text tools (grep, awk, sed, cut, find) / Linux basics and file system hierarchy

Hard link cat hardlink The source was deleted but the shortcut was not deleted. Acts like you have Cat my fidty acty and inside it this is My file is four such words, let's say one The word is two bytes long, I need to pick up the first byte. You have grown it, right Hitesh, work so tea is tea double e not tea and not tea so what does t whatever you command you will put echo from let's. That thing will also be printed on the screen And if you want to put it in a file, then that It will also be put in the file, show me now Echo.

This halo is called a haloed t ekti If you want to put it in a file then use pipe operator In t helod t ak t now see what will happen. Which is Echo and will also be printed and this load t will also move to a new file Enter came here now you are seeing a. Load t akti file has been created and if you Cut haloed t will do halo print inside it to tee is a command which is used to.

This is, some toy mail function has been done This happens because I have made a resolution I changed this if I reset it. He'll be right If this is what I did by resetting the VIM So look, it is showing correctly and it is showing correctly right now. He is not causing any problems This is from VIM and the EC2 in this browser If the issue is with the instance connecting then This editor is simple, just understand this much that. Open the Vi editor on your terminal.

If I open this file for you like e filename data activity whatever and open this file In this file you have to insert So if you press i then insert mode will be activated. Let me zoom this back in because You won't see this is done now if you cat You will make your demo file, whatever edits you make and everything will appear here Look what happened. Sort function then you can use vi editor or You can use both the vimters, both work the same.

[Linux for DevOps] SSH and remote access

Does VIM stay a little advanced looks a little better, now, let's come to the last section. Is where the real fun begins This above Linux was basic. I don't remember but Western Line Harbour Line Central line is something like this so there is a port near the Harbour Line where The ship comes and docks its ship in the port there. Park it just like a computer There are many ports nearby. There are ports where you can access things so if you Connect your computer to another computer If you need access to a port, you have SSH.

22 is the access to port number 22 on SSH. I don't know when you do the SSH. When you latch the instance Here neither one nor the other pair is asked and we The last time people created one Which was called Nginx for Depp's I Hope You will have that downloaded now. This key is created from which two parts There is one for the public and the other Stays private that of the public and that of the private It should be on the laptop or on that server.

Private Connect Private Key near him It should be something you are local to If you are using it on server then go to local Private should have public which is the server so whenever you. If you create a key, then in the background The command runs with the SSH key Generate SSH Key Generate two keys does public and private, then when You do this part,. Pair then one command behind running ssh Jan To Pair Naam Like if you pair the create key then the public key goes to the instance and the private key If it comes to you in downloads then.

Went So I am here, oh my god, here Yes, but you can see a lot of things. Was nq for dev i downloaded it which was for this instance now what should I do I want to do this nak for DevOps I want to do SSH using this machine. What will I do here, I will terminal shell I will, I'll open my terminal. I will take it and what will I do, I will take it. I'll go to the screen here Instances Pay and [music] Try putting it here ssh, enter this command to see in your command prompt if something comes up, it means you have an SSH client.

[Linux for DevOps] SSH and remote access / Linux disk management and LVM

You can pass a now locate your private key file now Where is your private file? Next dot pm found download folder I'm in the download folder Now what I will do is run this command if necessary, then to assure your Key is not publicly viewable what would it be. I will definitely tell you that, but you want this command. Go ahead or skip it if it If you run the command, you will get this command You will get to run it directly if you run this command.

If you don't run it, you will need to run a command called sudo. If I have to add it, I will tell you later. Go ahead and run this command for now. AOD 400 So what is the command SSH command -i -i means path to your private key file what is private key file nqfs key pm and your username is username ubuntu then What is the username Ubuntu? Rate your public DNS here to compute N Cloud Ye Ho SSH Path To The Key File username: Username of the user you are logged in with.

You have to do and f and this is your itu address Enter Are You Sure You Want to Continue Connecting yes I am sure and like you yes. You will see you soft link they were playing all those things they came I went here and understood how you From any computer to any computer You can connect provided and. You can see the disk usage of the disk, so how does this happen? How much storage is on your system? How much storage is left and occupied is all that Things you can see from DF So he is saying disk file system so this file system has so many blocks.

It is there and it is used so much, it is so available, meaning TempFS has been used for only 22 times. Temporary file system has so many blocks And not even a little bit has been used. If it is available then know about your storage like this If I tell you then what will happen if DF Han A He goes and tells it at a slightly higher level. Gives that 7.6gb storage was 1.7gb It is used and 6GB is available, the same thing is shown So if you have any concerns about disk usage If I want to know then you can do another DF.

[Linux for DevOps] Linux processes and services / Linux basics and file system hierarchy

It happens that I don't use DU most of the time. It's like Disk Utility You need to know some particular Want to know about folders from Let's me dnv dot means of this current folder Want to know which quickc file. If you ask where it is stored, he tells you that he has not come. What's left dot do yes then see DU.com means which section is in this directory There is a hidden folder called cache. The hidden folders are generally dot So if you LS Han A If you do this, you will see all the hidden folders.

The cash has appeared, now whatever It's a cloud folder, that's just a few bytes. Eating cash for debt eating so many bites The cloud inside Li De is so If you are eating bites then some information Shows the About folder, now I want to see the processes. What top processes have a PS name? PS tells you that you have Total Processes Which Second to Total Processors No, the total processes tells you the top. The top command shows the processes PIDB ID to P this is running all these Processes have system d kth read cc C RCU NET NS Too many processes running You can quit by pressing Q and P.S.

Let me tell you that particular bash On which process ID is this command running? It's running fine, so some processes Commands live ps top done now f What is a user? F User in general so I don't use it but it is for a particular file This means that your home Ubuntu The file system process is running on this file system. System what happens if you are a network If it is in the file system, it says a nas What do you say, SSS means network attached?

Storage Meaning if you are driving a two-wheeler, there is a lot of All file systems can be attached, so F The user tells about it. If you want to kill any process If yes, then you can kill me with let's If there is any process then I have topped it up now. If you want to kill a process from it, don't do it generally. To kill a process but if you want to kill If you have to do it then kill him and then whatever The process ID is something like 162.

[Linux for DevOps] Linux processes and services / Linux text tools (grep, awk, sed, cut, find)

I don't know if there will be some process I will kill. He was trying to do Operation Not If he says permitted then you don't have permission. It's basically to kill this thing. Kill any process if you want to kill it If so, the free command will be on your disk. Space tells me free if I write I have a total of this much rum in it. Used to it, so free, that's what I need If you want to see it a little better, free here H so me 949 means around 1gb rum Received 174 ABs used 430 ABs are free.

No hoop what is it noop generally Neither do we use DevOps in the application Do you have any application from Engineer Lets? The logs for that application are coming Logs lets you store those logs in a file. If you want to do anything, what can you do? Command is like this is free han h its which I also have output that I need to store in a file. Cat noped out look put that output Now what the benefit is. F A something remains here n5 or something yes -a means number of lines to head number of lines five node dot out so means above five lines this is what if i did tell him n fnop dot out so five lines below.

He will tell you, are you looking at these five below? Line or the top five lines then tell such work does head and tail do things that The last command is vm st which is virtual Your memory remains about him. If I had shown you everything then M So this is so much of your total memory. It is free, there is so much cash in this, that, if VM If you do this then the active and He is active and he shows all this.

If he is telling you about your rum then as a The DevOps engineer will tell you this sas command somewhere. Otherwise it would have been used somewhere, how many fathers We have covered the commands 10 I covered 30 commands, oh my god, how long did it take? In an hour, I hope you A little bit of meaning in Nginx commands You must have gained confidence that right now, Linux commands have started to be written and.

[Linux for DevOps] SSH and remote access / Linux users, groups and permissions

Started with LS, learned to do SSH VM status is showing disk The usage is telling me that the rum is telling me so much I am doing all the things so I hope you like this. First of all, I will install it on my instance. I'll go and do the instance. Connect, what is this connect, what is SSH, I have told you in day two that How do you access an instance of an SSH? If you haven't seen it, then here it is Let me copy the SSH command and paste it here.

I am my terminal means I am local to connect this EC2 to I'm going to try it so this is a little I can enlarge it for your visibility. I will give it to you so that you can see everything clearly. Give, it has grown quite big, no, it is quite too much. I have grown up, let's do it, it's, someone said Done, let us first understand that I have already mentioned in the first part system level commands whenever a DevOps The engineer opens his system and.

That is why we will use all these commands one by one. Load average is a little bit of load in this There is not much load on the pay and only one user is active. Activities to see when they logged in I will write down everything that was there and I will write down both of them. The command is kind of similar but who am I to There is only one user return from whom you A list is available of which user and when All of them were logged in.

Things now Chach ye wich command ye The journey of DevOps engineers has been very The work comes to me from Lets pythonanywhere.com Take it to me python's installed is bash. It is installed but which batch is bash If it is installed then I can write which bash like this If I give it to him, he will tell me the location of that bash. Bin in your/usr folder It's bash, that's the bash I'm using. You can say which A or anything CP is so CP It will tell the location of the command and similarly if You.

[Linux for DevOps] Linux users, groups and permissions

There is some fault in the system and that fault is from a particular version of Java now Your developer said that whatever you have The version is that is version 1.7 from lates and which. Which Java is installed on your system? There are small things in the system level that a DevOps engineer should know need it now ID is a very important thing. The UID of the Ubuntu user is 1000 and that is the group ID of Ubuntu. This is it and from which second do you have groups?

There is a group in the system called Ubuntu. There is a group called ADM with a dial out name The group is CD floppy sudo audio dip There are so many video groups in your system. How many users are there in it, it is the same as Ubuntu and if you add a user, it will reflect here. If done then it will be in system level command Why am I telling you this because tomorrow if your Too many users have entered the system.

How will you check that all the groups have arrived? You will check it like this, now I will give you a I want to tell you about Brahmastra. Brahmastra or a VVIP man or whatever it is sudo or sudo very It's a dangerous thing, it's a very important thing. Today we will understand what is sudo. This is the house, you go to your house I am here now you are here this is fine you are there Ubuntu you are Ubuntu what is Ubuntu.

You both live in the same house, how much longer can you earn a package, what a good person you are. If you want to adopt Ubuntu then I can't do it unless I get my father's permission. If Ubuntu wants to do this, it He uses a Brahmastra that Papa do to do means super User Do If you need anything If you are not getting permission then you can use sudo. If you use the command, what is sudo? Sudo is basically a group within which this System administrator or root user Added Show How ID wrote groups look now sudo name group of.

Got it, sudo is not user sudo is a group and in that group I have permission to do a crazy amount of Thing for example I have to do in this system.

[Linux for DevOps] Linux users, groups and permissions / Linux basics and file system hierarchy

If I want to see all the users then I what will i do cat/etc What is /, not GRP, one second One second, what do we do? Etsy List Users I find a user named Ubuntu. Here, this is the Ubuntu user's path. There are some folders which You are not allowed to go inside like Root I rooted the CD, permission denied. Permission denied, now what should I do to So it is not letting me enter the route. This is what I am giving you in the ETC folder.

I was going inside, CD etc., only he knows. I was giving it but why not let it go in the route He is in the route, that's why he is not letting me go. Because it is Papa's room, it is Papa's Can't enter the room without permission so what do I have to do what will I have to do you have to use sudo or. I must have explained it well but when It will be better when the user parts come.

If you understand then what is sudo basically Does sudo give you super user permissions? Shutdown if you have installed the system you have to shut down if you If you want to shut down your system, type Two shut down butts You can't just shut down like that. What should you do if you need your father's permission? Get added to sudo Papa's group and then shut down sudo shutdown done The system is going Down for power of eight Monday ho understanding coming in Is your system shutting down at this time?

If you go, you will understand that you are trying to shutdown first The as a normal user fails to set fail. You could shutdown it, but if you said, sudo shutdown with permission from the father He said, the broadcast message has been done. 1053 UTC in 1ST Indian Time How much is 4 by mit Monday means 4 If I think the system has been shut down then Look, all these commands have been completed, so you. I tried it but it didn't work out for Papa but he took that sudo The shutdown is understood to have happened in such a way that sudo means super user do super user key.

[Linux for DevOps] Linux users, groups and permissions / Linux networking commands

You have done the work with permission. By the way, we can turn on this instance again. If the system needs to be restarted These are all the commands, generally you will see All these commands are covered in the rest of the tutorial. Kill them all, it should be like this It doesn't matter if there are problems like that. If I connect this system, The system will be my connection because it I have turned it back on from shutdown. This is the IP address you will get to face No, this changes sometimes, so your The old SSH command may not work.

Sudo paste it's gone, yes, now it's gone, why do? Because his IP address had changed. Because it is AWS, they change it. Let's live, it's done, now we understand that What is the next command? The next command See our sudo shutdown reboot Do that shutdown means turning off the instance? Turn it off, reboot means restart the system So if I do sudo reboot good If I just reboot then no It's, I will fail, because Papa Father reboot the system is restarting now.

You have already taken up so much of our time. You have only your one and only Not able to acquire lock permission denied Still asking are you root, are you dad? I will do sudo at install docker d come now see package docker come is not available package is not Why is it available? I know because AT Application Package Manager in your system This Come see now Joke what happened at get Installed has no installation candidate Now this simply means that whoever uses Docker.

Libraries that need to be installed No, those who need simple packages can get them from this system. It can also mean that I am not If your system is not up to date, you what can you do sudo clear first to the screen sudo at get update you can update your system. Requires some security patches If yes, then all those things will be installed for you. You will install Docker, which is your The old command was well I'm going back to the command Instead of writing, let me give you a hint.

Search means all the commands you have done till now If you have written it, can you search it back?

[Linux for DevOps] Linux package management / Linux users, groups and permissions

Lets I learned some Docker related commands Look, the entire command has appeared and you just need to enter I want to press it, look at me, he got the docker right now. Following Additional Package Will Be The following new packages will be installed Do You Want to Continue Now Got Docker now to install things If he found a way, it means whenever You are installing something new. Internet Than you have to use get and your system I want to keep it up to date so this is Docker will be installed Now the question arises, where is Docker?

Folders named Docker usr bin docker This is how you can install a DevOps Engineer Docker Kubernetes Junkie Terraform You can install anything and with it You can play, so you understand AT. If you need to remove Docker, run sudo AT remove Docker d so what happens with this will go 113 disc pace will be freed meaning Docker will be removed so if I do. I want to uninstall Docker so I can do this I can uninstall it this way,? Docker is not found because It has been installed, we are not getting used to it.

I went to Windows and said, program files Go to the control panel and that thing install uninstall it this is so cool Isn't it simple now? What is APTM, what is DNF, Pacman what is portage what is this joe m dnf Packam portage application package Managers are just that, this application There are different types of package managers. On which platform will Yum run? If you had it in this place a red hat red hat so for that It probably happens RPM is fine if it is RPM then fifth But write that too on the 15th.

The name Gentoo is so cool, do you understand Gentoo? Application Package Manager is gone, so this is something There are system level commands which are DevOps Engineers should understand you now The Big Thing is Users and Files Management said, well before that this. In this house so if you drink W If you do PWD then this Ubuntu knows where it is, it's in the house. You are denied without Papa's permission. You will bring anyone into your home, strange love Have you seen this story, but don't do this.

[Linux for DevOps] Linux users, groups and permissions / Linux basics and file system hierarchy

How can you do it if you take permission from your father? Where did this user become where did the user become user made in house so if I cd dot If I go then now I have come home. It should be there, then I will use it, where did it go? P A Dal SWD so these are all the commands I think You have come sudo user add has come who am I The password has arrived, pod SDDD,? What I'll do is run sudo PASS.

Have to do Me I don't want to log in with Ubuntu user huh? I am showing it because when you instance from the Ubuntu user when I connected to connected and won to user your If the primary user was Show information like this. How to change user password How do you log in with that user or Then how do you switch users, it is simple. I can't delete it, I can delete it using sudo. The name has been added, now it's Ayer Tappu And the crowd is these three are by nature testers When he was hired in G, this tester were tester and Ubuntu and.

These testers will also have roles and permission. Give permission Can't give it to individual users google2 Hajj must be working a lot of people, a lot of people work, all of them What will you do if you can't give a little? You will create a group named DevOps You will create a group and include Ubuntu in that group. You will make a group with Ayer and Tappu and You will add the crowd, let's make it quickly. I have a group now what do I have to do If I want to make something called DevOps, what can I do?

I will give you sudo group d DevOps sudo group d DevOps enter now a group named DevOps has been formed, now. I told you, Cat Street Spa AASW D just for user P A double SWD happens here for the group GPA will be double SWD enter lo Sorry, no, the RP of the group is fine. The group was there, I was writing GRP a little more. If you had used your brain then this would have happened with GRP.

[Linux for DevOps] Linux users, groups and permissions / Linux basics and file system hierarchy (part 2)

I could have also written enter, the list has come, now see Whenever a user is added, his/her A group is formed, see a group named Ubuntu When Docker was installed, it also had a Docker. A group was formed, a crowd was formed, but what we Originally created the group it was DevOps Now what if I don't use sudo user? RO UP Group D tester enter now this There is a tester group in that list over there. If it comes in the end then see CAT ETC group The tester looks Tester Devs, these two groups are.

I am similarly my Ubuntu user I'll put that in as well, Ubuntu. I will also put the user in the DevOps group. They would have put Tappu and the crowd in it. It's, but before that check that Whether you have been added to the group or not How do you do that, that was the command, see the control R search which command named group. It is true that there is a group called DevOps in it. If there are 25,000 employees, then the command will be given 25,000 times.

I will not run it, don't do this, sudo p Edel Eddie pod SWD means you group to add to Who has multiple users? Multiple users were users from which second this one copy paste comma copy paste comma copy paste to ayar tappu bhidhi Which of these three should be added to the group? Should be what is sudo switch What is a user

and where are all these files? Should know right now if I ID If I do the ID, what do you think of my lug?

These users are Ubuntu will he have DevOps in groups Yes Look UID Ubuntu Groups Ubuntu to Ye Where did this go? Again Group cat / atc / It's a group, so be it, but let me know from this I want to delete the group,? What to do Give sudo group del as if user del was group del which group delete I will do the tester one, let's do the des It's important, sudo group del. Tester entered now if I group check Let's do it.

This was done because those users Proper permissions regarding files Proper permissions regarding folders This was the main reason users could get access Butt to manage Why does the file need permission?

[Linux for DevOps] Linux users, groups and permissions

Permission must be obtained to access that file. Because you can also keep sensitive data Yes, we have a lot of importance. It's to give permission to the files and that Today we will learn how to give it. The most common way to do this is to use the file permissions command. First, if I have a folder from Let's Who linux for dev cd L. If you have permission, your user will have read access to this file. Execute permission is there for other groups What permissions does it have, read, write?

Execute permission is available to other users. Read what permissions users have It is not right, it is executed, I understand is that you go into a house said went to my father's house you are at a house. Ubuntu has made the door, so what about Ubuntu? If it is not there then it cannot go inside, right Can't do it so enter your permission user Group and other users or file permissions, how will you write all this, Aad Ek Aad Ek. I will tell you a simple technique for the next 10 stay with me for a minute very meditative look to hear from Now file permissions in.

Throw it away, throw it away, now we will understand this image what happened to the file permissions uploading yes let's understand this read write execute where did read write come from. Let's change it to 0 1 2 3 4 5 6 7 now The first line is Alternate DXD x d x d EKS second line dd Dash dubbed the third line Dash. Means read no write only execute means what is this f Yes, it has got odd X meaning F O Odd X means tell me Odd X in chat Meaning Sen so its permissions of this folder.

I need permission to use its cloud based on 7 I need to change it to a 7 and a 5 I have to change the permission to two odd one odd one odd. All open for all, anyone can use it Rest I understand what file permissions are. Was Now the DevOps file.txt You will tell me the permission for this RWD. RWDRRDD look at this in front of you chart what was rw copying copy it and put it here take two texts and put them here now its.

[Linux for DevOps] Linux users, groups and permissions / Linux basics and file system hierarchy

I will drink water only if you tell me your permission. I take Look, this first one is blank, I mean Is this directory is not a DirectoryEntry Permission is 6 64 meaning Users can read and write this file. What is execute, what does execute mean As you execute the program Neither do you execute shell scripts If the one is exec then execute permission It happens Now what should I give by changing its permissions? What is it Give the user read, write, execute permissions But the group should not have any permission and other How will you do it if the users do not have any permission?

Meaning users should have read write execution The rest of the group should not have any permission If users do not have any permissions, then see read write execute where is seven is seven. Dash where is zero is zero dash Dust is zero if you write CHM Odi 700 files Sorry, the name was hidden in my photo. Chmod 00 file name then your work will be done So C MO 700 this and your file name is dep file.txt Enter now look l a han l aad a dash.

Dash the permissions you needed You have given it and as soon as you have given it Granted executable permissions and see if the file is green. It's so cool, isn't it, as soon as it's executed Permission granted, file turned green, green means executable i understand File permissions seemed easy, a little bit in it Maths is Maths because it is. That memorize odd one 7 zero means dash I could tell that too but Who knows, you might lose your memory. If you don't remember what you will do, then you quickly You will make such a diagram and quickly Will you give me permission, let's move on to the next topic.

U mask What happens if [music] Mask 002 What new number have you brought now? Mass what happens is that whenever you add a file Do Or do you do any work by him? This is the group's read, write, and execute and read and execute like this and what is the y mask of this computer Mask 002 so basically by default permissions. If you want to change it, then how to do it CAT.BR.S Maybe it doesn't happen in this CD cut, data, RC, yes it is kept

somewhere in it.

[Linux for DevOps] Linux users, groups and permissions / Linux basics and file system hierarchy (part 2)

What is this bash rc, your command Prem is looking bash about him The complete profile is available here. Using this mask number let's go chmod ho lus funnel ho seed Now see what happens Like this file is Demo FIDTI XT What is its owner, L.S. I am a demo fiduciary and this is my owner and the group is Utu but now I am the owner of the demo file I want to make it and I want to make it. A command you can use is CGRP is the same thing that if you If you want to change the permission of any group Let's say Han l This was the demo file.

I was in the Ubuntu group earlier, now what am I doing? Sudo seize your app to the DevOps group I make it the owner, meaning the demo The file is txt, that is DevOps instead of nutu. I want me if I do it I will be able to do it. I gave it to DevOps, I got it. Change Group Command is a very easy thing, this file permission User group management is very easy. He is saying that there is no zip at all.

You will need to install the Zip utility. How to install sudo att install Zip, take this command, copy and paste Now you have a lot of files. If you want to send all the files to someone So it will take a lot of time to get all those files. Zip it up, make it a file, send it and Then unzip it, I'll tell you, now like I have this cloud, this is so many what do I do if things are Give Zip me up.

If yes then apply it then recursive zip will happen will enter and look at the ad Compressed them all and put them in LD Zip I put it in, it's so cool, now when I. CD goes to the cloud folder instead What's inside? Because you have the unzip when you zip If you install it, it gets uninstalled. Un zip el FD Dot Zip I can see LD Dup in front of me. Why is this happening LS Han L Permission Doesn't it have Yes, I can see it in front of me, but this Why can't you recognize me?

[Linux for DevOps] Linux basics and file system hierarchy

Am what is this unzip l fd dt zip dt zip to No, why are you doing this by doing LFD? Were trying to copy were trying In which cloud wall is LDF DZIP Navigate to unzipped files in a folder I tried it, it was a spelling mistake. Is cd cloud unzip files enter now what I will unzip whom to LDF dump I will unzip sh Done unzipped ls Did that the CD Linux is in front of you? For DevOps all the files in front of you with Permission is so cool isn't it and unzip so if You want to compress any file.

Decompress means enlarged, similarly Your gun zip all these tools remain which The beans work, a lot of waste. Which is better than collecting goods from here and there It is a particular use case, do only as much as you like What does zip do to compress? You can use both zip and unzip. I will go but see what Gun Zip does. Take Gun zip same gunp command is use to Compress and Expand File in List of Files It's done, just that zip is it Look, he will make you log in now, leave it for later.

There is a zip, it is called dot zip. Extension gun zip me dodge j naam it happens because there is nothing else, yes There is one thing that is very important for you. That is tar, what is tar, tar also It only compresses, it is fine, tar is also less tar I will show you If it is with you then the tar command remains. Let's take me to the clouds tar -x in z f, what have you written, what have you written This is what aqui z af means files jad means.

Zip V means verbose and you remember all this You don't have to keep it, be afraid, just write it down. Flags Linux and that tells it all right He will tell you everything that if you x If you want to write then x means to extract If you want to compress the files then. Meaning, the file that will be created, its file name The extension will become tar, whatever file is created, it will be created.

[Linux for DevOps] Linux basics and file system hierarchy (part 2)

Archive Cloud is MT What cloud and then I got cloud d tar.gz I will write this What are you entering? You have Claude A compressed file named tar.gz has arrived which you have compressed using tar There is a utility to compress the files properly What is C? Verbose what is the root of this front showing Using Gun Zip Compression Gunp Meaning and Why dot g h f means see files what does A mean F means yes. Specify the file name this file It's done, now if I do something then I will get cloud I will get it now what should I do.

If I want to extract the clout jeep then I what will I do tar Extract Verbose Meaning Show Me While extracting, not here. We will extract the cloud folder Will extract in Linux for Des Seep Cloud Does to Linux for Dev So now going to Linux for DevOps, I. Cloud Folder Extractor h so tar actac ex verboso that's enough Yes, that's enough XSV

JAAD means you just have to use gun zip This is enough Whom. Cloud is not running dar dj jd if you like this If you want to put something in a folder, you can also use file enter You have to use the file name Enter Yes.

LS man it came in the folder of the same name So if you go to CD Cloud, you will DevOps and unzip file new file all got If so, did you. It's been so long, one and a half hours, oh my god. What tar does is either Zip or unzip it for you will mean either you compress will do or extract If you want to zip or compress it If so, whatever command you have, it will become yours. Tar will be formed because you can compress it.

If you need to unzip or extract If you want to do it, then tar ha one x Extract Now if you see the screen in front. If you want to show things then we are verbose if you have a gun. Zip, this gun zip, use this zip. If you want to do it then it is if you have to file If the name is given then it is FF,.

[Linux for DevOps] SSH and remote access / Linux basics and file system hierarchy

It has come, tar has also been done, meaning atar This is Attar, now I don't need any file from local. If there is a panchani on the server then how from Panchanga Let's This is my server This is my server. This is here PWD this home Ubuntu this my server is and this is my local, it looks a bit small, please enlarge it a bit. Two this is my local now in I create a folder in Documents. Am kdir SCP Secure Let me create a folder called copy test So CD Secure Copy Test now this is my [music] There is a secret text written inside it.

This file is a secret file in my local I want to put it on the server here. How does the PWD want to put it here or there? How to insert copy command Demo Fidget Action Linux for Demo I can copy it with the copy command. I will use it, I will just use secure copy Secure Copy What Secure Copy does SSH, that you can go from local to local Using pem file on any computer. You can go but if you need any local Use SCP to copy the file.

If needed, how to find the location of the pem file, where was my PM file, look, look. To go to the instance, the y SSH command is SSH, this is the command, copy it Do it, copy this command. Copied Now paste that SSH command here. Write scp and SSH commands Paste it, that's it, what do you need? Have to do SSH Don't have to do it SSH If you don't want to do SS then you have already done it. You need to SCP I want to SCP using this file map For Debs, now this CD into the Downloads folder then PWD.

Use the pem file and my file which was what is my file name The secret is Copy file.txt to any Camel machine The address of this Ubuntu machine is given in the colon. Whatever location you want to copy it to Let's say I'm PWD Home in Ubuntu What should I do if I want to copy it? Where did I go, yes this is my Ubuntu here In this instance, give a colon.

[Linux for DevOps] SSH and remote access

This is your command what is this is your command this is your command what is this is the private key. What is a secret file? Source file is source Meaning the thing to be copied and this is Ubuntu what is this thing this is your destination So what am I saying scp copy Command Copy the server using this private key. This item will need to be delivered to this destination. Copy this command Enter he's saying something so yes and no ha do You want to connect, yes I want to connect Yes, now see what happens in the secret file.

Anything copied, drop it anywhere If it hasn't happened then it's so cool to me now. This is the Linux for Debians folder. Have to bring it from server to local How to move DevOps folder from server to local I will use the simple sc p command -i same thing same thing. Same thing scp han i copy this whole thing Copy it completely. Not the SCP command to copy it after I After this you have to give the path of your pem file You can find that path in your system.

I will be in and what do I have to copy I am placed at this location Linux for I have to copy the DevOps folder, which I mean You have to copy what it is. It should be a folder that needs to be copied recursively. The -r seems to be copied recursively Where from this server to this server location What is the location named Home Ubuntu Folder In the folder named Linux for DevOps. In this directory, I will write dot Read to copy it to this directory back relax read scp copy Command -i ye path to private that is, I copy the file recursively.

I am because the folder is there, that's fine What is the source is the Ubuntu machine and this folder what is the destination dot enter now Look, go ahead, a lot of people are watching. Waiting to go home What's done, first download the zip file Gone is 100% Download Cloud Tar Also 100% Done Demo If the file is also 100% downloaded then If I do LS. R Sync comes here and changes the whole story.

[Linux for DevOps] SSH and remote access / Linux basics and file system hierarchy

What happens if you have sync than I have added I am in the remote You have to do SSH first, so this I'll show you the SSH You are syncing with. SSH is fine, yes if it has to be done like this It is simple, it doesn't cost you much. R, I will write it and I have it in me, come on If it is right then what should I do Yes, RC yes what do i do. This is my source and this is my destination Now I have to use SSH in between.

If it happens then I will just pick up the SSH one. Where are you going to be It's gone, you won't get it like this, stop. We do it fun, so what should I do wants I want to I want to write or sync Command Which Uses SSH. A pem file what was dot pay, what was it about nak phar DevOps -e means execute a command SSH do this AV JD or what AV JD What does A stand for Archive V.

To the local folder and to the server let's go Let's see by doing So R sync what to do han e SSH has to be done Whom I pass to file where I path to. I did write my path somewhere, yes this If so, this is my path This was the path I needed, I did it did a r sync SSH done. I said yes and sync this then what to do with myself have to do What is AV and Root AV Root Basically what you do is create an archive.

I mean, compress it and show me what you are compressing it and send it after zipping it. Send the item to your local folder So what is the name of the local folder, you So take this Now look how cool it looks. Use Sync R Sync SSH file and Linux for deb which is in my folder local and which is kept in my remote sync it Enter, see what he does now. If I do this here LS did not come what should have come yes it came right Look, didn't it come?

The CD has arrived Can you sync from both sides?

[Linux for DevOps] SSH and remote access

Maybe some files are missing from your remote There is more in the local area, just get it synced. Change the source and destination to your local I have more in remote and less in sync Get it done, change the source and destination. I think this might be a bit heavy for you but Really remember one thing if you practice If you do it then what is SCP or sync What is this, you will understand it all. Things that are not asked in the interview If you practice well then SCP and R, you have got all the sinks done for day three.

Is covered in X for DevOps A Simple About Networking Commands I will tell you a scenario, like you be a DevOps engineer now a DevOps What does an engineer do in day to day life? What DevOps Engineers Do in Day to Day Life So what is their job generally? Systems Pay Applications Deployment lets you run that application If it hasn't been there, then first of all, a DevOps what does an engineer do like this from Its. Is anyone running this application,?

Is any application running from let's Here and this application for some region stop If he has gone then what will he do, that eat server or. Whatever server it is, it will pin it and see that Is the application running properly or not? Whether the network reached there or not If you are getting it or not, then you can use it for that. If you can, how do you ping it? It has happened because I am taking this class A little testing before we start If I was doing it then SSH is already there but, what did I have to do?

This is a simple pink command, this work Verifies that your connection is working properly Or else he will tell you everything, then. Net St is Net St means network Statistics about your network Tell me all the things that I have If it gives then net state enter now net Statto went or a trace route happened or. NS lookup done Telnet done IP It's all a particular thing,. What this is saying is that.NET tools You have to download the package named Then you can run all these commands.

[Linux for DevOps] Linux processes and services / AWS (EC2, VPC, IAM, S3, ECR, EKS)

Like net st net st now net st very import command too much Why is it important to know if you pay attention? What NetSt did top p look top This is a very long list, he topped it. Transmission control protocol, so whatever data It is being transferred and all the movies I see, I use the internet, it's all TCP If it comes under the protocol then the protocol Which is the TCP receiver? No one, sender, this local address means My WiFi connection is local.

Address Foreign Address NS and Craze technology and its port and then tells State Establish means everything is fine now. I'm running Linux for Dev on my instance sudo now I can reconnect I'll see and I'll run the net st command again. Watch Net St Saw this TCP Established But Foreign Address 10666 28 which one is this is my phone Is there any WiFi, this is a hotspot, check it immediately. It is kept active in the cache because My previous instance was this one It was a little disconnected, so in that case I am he was showing that.

What are the protocols now that you If you use EC2 then many Sockets are sockets, understand the meaning of sockets like this that if I have to connect one system to another system. System D Done, all these sockets are there, so its So there are many connections to this Leave it, but what is important is TCP. Confine Fig I mean Internet a sorry interface now when I do If Conferral comes, look, I know Docker 0 A0 now what is this I don't know anything.

Instance pay day two or day three we People had Docker installed, so Docker What he does is he creates his own network. Whatever network interfaces there are They will appear here if Confine interface Well now Docker itself has one The interface brings the network interface so He has shown what is Ethernet. Just think of it as if you have an EC2 instance taken you have an S2 instance Now that instance, the internet is All it takes is an Ethernet cable. If you have it, put in an Ethernet cable, not a cable, but a wire.

Something, yes the WiFi is here, something So it takes an Ethernet cable now this There is an Ethernet cable, what is it made of?

[Linux for DevOps] Linux networking commands

It will be connected, it will be connected NIC means Network Interface Chip or cable or something still remains Let us see the full form of NEC. Network interface card sorry is the network interface card now this is basically a network interface card. If it is used to connect the network When you run this command if A Any number of Confinite Work Interface Cards A complete detail of what is connected comes up. A0 means a network interface that has Private IP address is IP v6.

This is how your work is done with Docker's private The IP address is this and the broadcast IP This is the address, now, understand the Docker one. I came across that Docker when you install one If you do, it creates its own network. A0 also understood that EC2 has a NIC Or a network interface card is attached. If it is there then it tells an interface that this AO what is AO means loop back means your system so let's say this is your server This is your server.

Let's do this NC something on that server Not connected is a stand alone server Still you can find that server You can use it as an internal server like. You have your laptop, in that laptop you Created an application using ReactJS or Node. Any application made using JS If you create it, you can access it locally. Cheese knows what the local host says local what kind of spelling have you written local host so what is a local host a local Server which is a loop back server.

It remains and the IP of the local server We all know the address. 127.0 So today you must have understood that this Where did 12701 come from? It's an interface that The loop back interface is running internally on your server so this was your conferencing This is information, it is not like this, how it would be. With your Internet Service Provider It is on a server somewhere. Sends the request to either DNS or The domain name server resolves that request. The server then goes to that location and like this Hopping means from one server to another Server hop from second server to third server While doing your request.

[Linux for DevOps] Linux networking commands / Linux text tools (grep, awk, sed, cut, find)

If you are doing this then you will also get the same error. Do sudo att install traceroute do this, after doing this you will get this new The Colonel is giving you nothing else. First, what does the trace route do? This IP address to us to means your Went to this address from the system then went here Then it went here, then it went here, let's see this. The one that went here is this, so this 10066 or this Let's see where the IP is located address finder let's see He does it a little bit like CID.

What do you say, I don't know, if you do something here Pay enter IP address and get IP Details of where this IP is from Where are you from, oh, I am not from anywhere, no. I didn't see any, I'm looking for another IP. Concentrate on your studies, yes, so IP The address we found last is here. This was If it comes in response then who tells it This trace shows the route, so if you If asked in the interview, tell me that If I have a source and a destination.

I need to go all the way between the source and the destination. How many times internet or where IP If you want to know where the hop occurred, then trace the route. Answer: But, I don't like the IP. There was a server, then the IP went from there to here. Reply means packet loss and some time It takes time to connect, that happened, then this Went to the IP address and then took some time took this to connect because. After connecting for a

while, he can now see its IP address.

Will go to where the DNS of c will be yes it Look, let's find it now and see what it is. Let's find out what this IP is. IP address lookup You see what is this IP address? He will drive it and say too many hops, it is like this, so this is basically the command that you Must know Trace Route and Trace Path, see what MTR is. I have used the MTR command like this I learned what the pin command is and what trace is.

[Linux for DevOps] Linux networking commands

The path command lets you ping and trace paths if you want to run them together, you don't have to run them separately. MTR Let's Sen Vishumangal It will also tell you to enter, see the ping done and the path also. I told you to look, the data ping is also sending packets. Is also sending and giving you IP addresses He is also telling the path, look it is done, it is done. For commands with paths, it does this 30 times. I understood what would be best, MTR Stand For My Trace Route my trace route or trace route Whatever you say, MTR is very cool,.

MTR is a very simple command. Particularly good no port number in this How did you give yes I Think like this, no, port number NS lookup for port number fails telnet is there for it, so telnet ones. I will tell you the port number, so basically What happens is that on every port number I have a NS lookup if the domain is mapped You understood that,. Which IP address is on a particular website? The one with 80 is public, that is this IP It is given so that you get such answers Same if you will gossip NL up google.com you will find something similar to this that which.

It is an IP but its port number is 443. This is a very interesting command, similarly you can use Telnet like Telnet I told you about the port number. Host name helps in connecting Let me tell you that for example, if I Enter the host name and this computer will tell you what is the host name of then this IP address. What is the name of local host, so you can do it like this You can run any of your applications with Lets.

If there are people living there then they have IP If you want to show the address then show the IP address You do this and get the complete IP address He will show you everything. I have it in the loop back address This is my address eth 01 or whatever My Ethernet cable is this and his is this If he shows all the details like this then. Crazy command, this one is also ip Ed Kong, what is Ed Kong?

[Linux for DevOps] Linux networking commands (part 2)

Got and id got IP means internet Protocol I Interface and Id Internet of Wireless kind of network so if I will eed config so me good if you wireless. If you want to do this, you need wireless tools You will have to install it, you are watching today's How deep is the session going? Tell me about net internet networking If I wanted to know, I had to install Net Tools. Could it be that what the commands differ from You can fill in the commands used in networking.

You can write net stop ping mtr trace out trace path ss remembered from dig If you want to know the IP address of any domain I have to dig out the complete information. Enter so I'm basically running an Ubuntu server. Come to the command that if who If it's gone then what is NC net cut oh this I did not see. Address routing protocol or something remains A RP protocol remains that Address resolution is very crazy Meaning, look at the address of your device.

The host requests a MAC address from a router. That router assigns a MAC address to that thing. This is the type of internet tool that you can know it by rp command, which is a rp If the command is, then my address is this And the Ethernet cable that is attached. If the MAC address is available then all this networking There are commands that will be useful to you as an RP. Networking by These servers do not have tools by default.

You need to install extras, so if plug status then link B Detected Link B Detected unplugged so I uninstalled docker So docker uninstall if I write But the interface of Docker was. There is no butt plug left, which means it won't work. Running an Ethernet cable loop back that If that is also running then everything is running then if Plug status indicates whether your interface is running. It remains now these are the commands The reason I wrote this separately is because it This is a command that you generally don't use.

Get curls that It is like curl, which is the curl command. What is your URL command? To call API endpoints, understand what API is like this that you Yes, this is your server where you have some data.

[Linux for DevOps] Linux networking commands / AWS (EC2, VPC, IAM, S3, ECR, EKS)

You don't talk directly to the server. You will tell the API that, take this server. The server will also tell the API that, take this and give it to your client. Two, so this is what the client requests. The server that responds does Why do we use curl to get the Response from a server to any API call If we use curl to do this, what do we do? Let's take a sample API for testing Let's take a simple API. Let's take it for testing, where did it go, take this.

H x gate means if I take any data ana curl han x get and ye API end Point enter and see, the entire data has been brought. Put the pipe in front of that curl and put Jay Ky two and that thing becomes beautiful will go oh he is saying that JQ Not found, I did not install JQ. JQ wasn't even covered, yours was just Curl covered curl with I did J Q I also told you so that you can do better If you bring me cute, beautiful, lovely.

Doon a m kd i r downloads Is A spelling, I am downloading this correcting the spelling I'll be fine CD downloads now this I need to download that file in the download folder. ATTP Request Send Awaiting Response 200 saving to file and you If you do ls your file is done Download it if you have it on the internet Anything that is public and downloadable. What does curl and w get mean Understanding Curl API Calls in General Downloading W Get Things To do this, you can interchange both the keys.

Use it trace route i told The trace route given to you remains the same IP hops through the Internet Neither does it remain the same. This is very important so IP tables Completely identical to your VPC. I read the chapter when you were not Vishumangal Made 11 days ago by AWS VPC And VPC peering in that I have explained the concept I told you about route tables, so route. Tables what remains that if your y server You have given internet access to this If so, what is his address?

Protocol is everything that has routes It shows the same in a table format.

[Linux for DevOps] Linux processes and services / Linux users, groups and permissions

The thing is IP Tables Ltd IP Tables are not good IP tables, those tables Is Maybe IP No Tables No IP Tables look good IP Tables Help Let's see what happens with IP Tables. In the IP tables here up to chain insert This is a list of all the rules printed in the chain. Rules IP Tables - - List Looking at the rules have could not fetch rule permission denied you Must Be Root o sudo ip Tables List rules Oh, so this one with IP tables.

You cannot run without root permissions. These are all tables for Docker. The more you go into depth, the more you Crazy things will get these W Gate we had it covered Well, what is a watch, watch which There is a command,. This is if you have to run any command and I have to watch its output again and again. If you want to see the output every two seconds from You can do it from watchlets if you have the command drive Is MTR my trace route did I.

Keep stirring this thing every two seconds Keep running, keep running this thing every two seconds so this is what he does so Watch top, this is what I generally do, so every two. Seconds later a top thing keeps playing That's all that remains Yes, so you can change that watch also. Can you watch n = 5 Top to every second will run you then you Whatever you want to loop, now watch five seconds later and it will restart That five yes look updated.

This thing is M top so every 5 seconds Later, if you want to run something repeatedly, If you want to run it then you can do this thing when. Network Map Lives N Map Network Map Remains to install this thing again If it falls, what remains of the N map is complete A network map shows where the network is located. Kind of like the route where it's going from The route table remains the same If the situation remains the same then the networking There are commands, they are endless, there are many of them.

Commands are important, know what they are What is important is that you understand We should know that we use all these commands.

[Linux for DevOps] Linux networking commands / AWS (EC2, VPC, IAM, S3, ECR, EKS)

Why are they doing it, how are they doing it? Because as a dev engineer you have a lot You will have to ping a server once. The IP address of the server will have to be understood as the packet Whether there are any losses or not, how much latency is there. You will have to understand all these things, that is why use it. Its port number 80 is open and please tell me. A map, what are you doing, that's all I have kept port number 80 open because If I don't maintain a website, how will you maintain it?

You want to scan a website. If a port is accidentally left open, then Attack further but don't do this generally people Like a map - And whom should I let go of? Try visiting google.com [Appreciation] google3 80, very good google's edit your routes change I have to do this system with let's If you want to add public IP then route. It shows the entire gateway. Route table containing routes which I IP table What he was saying was basically If the explanation was about the route, then the route table Now I will show it in this when.

When you go to VPC, there is a route in it. There is a table that tells you where the internet is located. Routing is happening from the subnet which is Where is that routing happening so route One thing happens by doing tables, see the whole thing. The route table shows this or that route. Table yes look at this is what happens 0.0.0 This is for internet access If it is Internet Gateway then look at the Col IP Routing Table Destination This Gateway This Then.

This is what's going on through the Internet gateway. Ith zero interface ya i face meaning The interface is, it is great, isn't it, everything you Go to linux.com for free and here View all and go here and you will understand. Whatever in end to end networking I must have told you everything in it That's how the internet works. How is the TraceRoute OSI Layer OAI What is the model? What is the TCP IP model? Real time doubt clearing is everything in it, everything related to was covered.

[Linux for DevOps] Linux text tools (grep, awk, sed, cut, find)

Networking, so you must also see how We use commands like awk sed grep If you can master it then first understand it. If you want to become a pro at something, You have a little bit Programming Pro You should know programming, a little bit. It is not that it comes in large quantities or completely Learn how to code as a DevOps engineer It should come, this question remains,. Take out the for loop and run all these things you should come How do you program in Linux now?

Shell scripting is used for that. Shell scripting but shell script You don't have to write the commands in full. If you want to play the game then how to do it then for awk AW so AW There is a tool called a command that itself. There is a programming in it, so what do you mean? If you have a file in that file You need particular The line has to be drawn to the range of lines Do you want to take it out, tell the count or anything?

Programming has to be done for that particular data So you can use awk with so today I Showing you a real-life demo of Oak So let's go to our instance. For example, you have a file, you have a file called log. File logs and system logs come here,. Logs something is there something Logs have come x y root xy root The logs have come, now from these logs you I want to extract some data from LETS. Or how to extract line number 50 If you do it, it will help to do all these things.

If it does, then today I will tell you in actual I'm going to show you how to create a log file. DevOps meets the engineer and inside it By using this command awk sed grep He shows magic, let's show him magic again Connect to your running instance. Let's take it through SSH, this is my system. Let's learn KOAK as well, so MK DIR Let's create a folder called a day file. In which we will discuss all the important things of today.

Will do cd de f what will i do now I'll bring a log file, then we'll go.

[Linux for DevOps] Linux text tools (grep, awk, sed, cut, find) / Linux networking commands

If you do programming, you will use awk now. Let me see, it's very simple now, DevOps You are studying everything from Bhale,. If you want to pay then the application is locked now. This command was mine awk info print 1 2 3 45 lalg let me What do I do if I want to write to a new file? Give This command greater than means redirect the Output to new file means only if dty Ekti and only if dlog what now I know I only have information.

If you go then CAT only info only and only info You got it, it's so right, whatever you mean You want a small one from a big log file,. The problem with this particular IP address If I had an IP address from Let's If you want to find any of these, let's go here. Do you have an IP address? Instead of addresses and info, I just put the number I will write down the IP address that was there. Of IP is 13 so this IP address total 13 Now you have learned how to use the awk command.

After doing this, he told me this thing with so much love. It's cool isn't it, but, the real fun will come then. I understand that particular incident or Particular issue came at 8:30 or 853 I only need the logs from 853 hrs now. There is a requirement from you in the interview also It is asked which is the better choice between oak and sed

What is the difference, there is a requirement. Isn't CSV comma separated values or so It should be CSV or separated.

Values T SV Ho Mean If you want formatted data then works but What to do if you don't have formatted data? What to do, after reading Oak, you should Read what is Sed, see Sed also It is a tool similar to awk, the only difference is sed. Full form stream editor so whatever Your data is coming and is displayed on the terminal. Or streaming data is coming from that You can also edit things in real time. Let's understand how sed works.

Hit like because awk generally gets You don't actually have to work There are two see oak people in the industry I am an example in all these industries.

[Linux for DevOps] Linux text tools (grep, awk, sed, cut, find)

Write what instead of ', write sed, that's the only difference So much so that in awk you have curly brackets I had to write things inside here You have to give expressions with lets. Lines to print info which line If it appears on Sed Info On which line do I appear? However many lines there were, 15 lines or however many There were also 15 lines, only those 15 lines He quit after giving So Sed Same Works Same As Oc Just That.

The difference is just this much, I will explain it to you. Works in column by column sed Structured Semi-Structured Unstructured works on anyone Line by line, then, the third command comes. Grep but before moving on to the third command Many of my learners told me that, you are teaching so much about all these things. You are teaching me in the same way as Jenkins too. I also want to read Docker in this way, I comment I read, don't everyone want me to read like I did?

I have studied Linux and I have also studied Docker. I should also read Jenkins, I should read Kubernetes I'll also read Terraform, I'll also read Embel If I also teach you Grafana, then, this I have brought the solution to this problem on the train. Vi Shubh am.com yes website It has changed a bit, a new website has been created. What does Grape do if it is If you understand the full form then it is Global regular expression pattern meaning if You need to select from a file or the entire system.

Some regular expression or something If you want to find it by matching the pattern then you If you use grapes, then it is great. Interesting thing, I mean, like you in sed Just like you were extracting information from If I write info in the grep command And if I give my file name then all the information will come. It will go, yes it will come, enter, see all the info will come, Ogg is also doing the same work, Sed is also doing the same.

Grep is working also What is the benefit of all the use cases? They are different from lets inform me I have to find and I have put info in capital I did not write it, I wrote such grape I A Ao and looking in the application logs.

[Linux for DevOps] Linux text tools (grep, awk, sed, cut, find) / Linux processes and services

I did not get it but grep which command He has a power which is called are case in sensitive - -i meaning Case in sensitive you If you want to write in small capitals, write in small. If you want to write, write in any case, you write it. Enter, he will find it and bring it and the best Tell me the part, there is more in it, so many If the grep is not finished You mean -i in case sensitive. I found the info in the application log and found this info Do you want to see the count of how many times it came?

For whom to file and you will get the answer You will get 16 for doing all this. I had to write this many lines and in grep it was enough If this is enough then use grep. The case is different, the use case of awk is different, files The structure data is different for sed. Line by line I hope you feel Clarity must have been achieved that oak said grape this All three are very next level Is There are next level commands but their use.

The case is different, let's start one process. I wrote PS AY x PS AY All the processes have been completed, everything is fine. I have come to know all the processes. We need those processes from this which are our What do I do if the Ubuntu user is running PSYX I will grep through all the processes Enter Ubuntu, now only and only Ubuntu. The process has come, it is so interesting. This also happens in real life, you are someone You write a command in the output of that command.

You don't want to miss some things from grep It works quickly, it works quickly Let's give me just more of this I only want this column, what should I do? What are you doing Everything What is LVM EBS What is Dynamic Storage Management? What is a physical and logical volume? That's what your volume is called now. Understand what external volumes are If this happens, you will get an AW2 instance. Lets create server in the

same way as AW SEBS or Elastic Block Store Lets you create volumes from lettuce today In the tutorial we have three volumes will create volume one volume two and volume.

[Linux for DevOps] SSH and remote access

Three Volume One Honey Singh I remembered but I did not get lost anywhere. Create a volume, one volume will be 10GB There will be one volume of 12GB and one volume It will remain 14GB. The default aids give 8GB and If you fall short of 8GB, you can use this extra You can also add volume, this is basically it. The entire fund remains You can, but if there is no EBS in Azore then This is the complete tutorial on EC2. Let me create an instance here.

I create mostly things I am going to explain the theory practically. If I am not going to do that work then its name Somebody put on TWA volume You can name the instance anything, it's yours. Here I will select Ubuntu. I will take you on a Uban to machine I'm going to demo the volumes in full. Sikhi hu wa instance type t2 Why do we take the micro because it is free? Tier is mu will read for free a pair of We create it and write its name.

TWA Volume Volumes TWA Volumes key and create it by pairing it Look, the pair has been downloaded now. We are going to do SSH using payer. Network settings to SSH Allowing you SSH traffic in You just have to do this much to stay at TPS. Give it this time, now confill storage, now What is storage? Space which is also your volume so when By default, you create an instance when If you create an instance, the minimum AW He gives 8GB and that is your root.

If there is volume then you should take minimum 8GB. You can take less than this, you cannot take seven. That's enough now let's do this After you launch the instance, that's it. Do this much, what have you done with this Created an instance of an AW Which has 8GB storage i.e. Disk size How much is 8GB, which is also called the root volume? What is the root volume, it is 8GB. Just understand that, now this is your instance. Launch done means in your instance you can store up to 8GB of data on it.

I installed an SSH in the download folder. Client is open now this is my Terminal is basically the same as SSH Client speaking, locate your private key I downloaded the file tod vy key.pem I will look in the folder ls td volume key.

[Linux for DevOps] Linux disk management and LVM / Linux basics and file system hierarchy

It has been located, now I have permission to do so. If you do not want to keep it publicly viewable then use this command I'll drive it to chmod 400, so this It won't be publicly viewable,? This is also gone, now what will I do, SSH tax I will give you on my instance so that I can give you If I can teach you all about volumes Let's do SSH quickly and I. It will connect Interesting Now what has opened up here?

Interesting now here I am going to put one of my AWS instance is open. Understand that there are two commands which you should pay close attention to. One of them is LS of LSBL. BL means List the Blocks List the Blocks means from which second, disk or You have attached volumes of that If it displays then I will have X VDI The 8GB disk is attached. What is Equi, Ex Vadi means your The root volume is named Aqui de you in the extended version when If you write then / DevOps S X.

If you also write vadi, it means device xvda1 14 15 16 these are all its partitions It is well partitioned and now it is I will tell you the second in command of LSBAA. I am going to tell you that he is Dae Han h means your How much disk space is free then Joe DevOps Root dev root means your root volume. Out of that total that is 6.8 gb.com that's mount slash mount I'll tell you what happens, it's yours. [music] Once you mount this thing you will get a You have to provide the mount point like a slash what is the slash where is your linux from Whatever you do, it starts with the CD.

Slash now your complete Linux from here If it is started then it is mounted at this location is your file system so that's why mount means Binder with one location for your volumes. Give it, it is called mounting and your Attaching a volume to an instance He is told that attaching is, so if You attached a volume to your instance. If it is there, it does not mean that it is a mount.

[Linux for DevOps] Linux disk management and LVM / AWS (EC2, VPC, IAM, S3, ECR, EKS)

At some location you have to pay extra I have to mount it, so this is me I will show you first so that you understand I have come here, two commands which are the most. It is important that LSBL gives you will show all the blocks and DF Han which will give you. All mount points and your storage will show simple now let me create and volumes so I can do my I will go to AW instances and. Elastic block in the side bar in the side bar store what is this it is a store lives where there are a lot of volumes If I find it, I click on Volumes.

Here I will click on Create Volume I'll click and I'm a general purpose I will create an SSD. Hard disk is basically an external drive I'll put it in size 10GB,? What am I going to do, where is my place here? My diagram is gone, yes this is going to be my diagram. That on my AW e2 instance I am going to make three volumes, Volume One Volume to and Volume three, let's make it So, one will be of 10GB and one of 12GB.

14gb now what is io ops in it The size has been changed to 10GB ibus means input How fast should the output operations be? Much throughput so much now Availability Zone Whichever Availability Zone I swear right now the availability zone is inside I am in the US way to and the availability. You can access your instance in the zone you are in. Create your own volume snap7 ID don't create volume from snap7 What happens if you have a volume have you backed it up is called a snapshot now can you.

Do you want to create a volume from the backup? No, I If you want to create fresh volume, don't Create a volume from snapshot now this After doing Create Volume I did So you can see the volume is 10GB. This volume is created as soon as You get created here. It will appear in such a state as if you If you refresh, that state becomes available. This means your volume is now available. To be attached, I will give two more volumes.

[Linux for DevOps] Linux basics and file system hierarchy

Create volume of 12GB, everything else will remain the same And one more I will make the rest of the 14gb Create volume will remain the same, so now I have sorting according to size. If I give you 8GB, 10GB, 12GB and 14GB, these four Volumes have been created, which one is 8GB? By default it is associated with that instance. I have four volumes of 10GB, 12GB and 14GB. Made just now and all three are available These three volumes that I created are available Now see what I will do, understand this like this I made all three volumes, like this.

I'll teach you what to do, so what do you have to do? To this 10 GB one and Select it, selected it, go to sh And click on Attach Volume as soon as You clicked Attach Volume. Instance you will have to see which If I want to attach it to the instance then my The instance name is TWA Volume This is the instance that I just created. Select the real game here If it starts then what is the name of the device It should be that which you b1 Eddie CDs are all generally root volumes.

The names remain but if you need data volumes If you want, the name remains in his name SD F SD H I J K L A O P So you who. I took the b and I added the attached volume I went to do it, now you will get an error that sdb For Yax Device Attachment Point Is Already In use, this is your already used. Why because it is the root volume, so For this you will have to take SDF and whenever you SDF will take that His name became X-Vadi, so whenever we.

I will create ADF and its name will be XD F will become like this, so ADF is installed. I have attached the volume now which is our 10GB What will be the volume in use? It will be done means it has been attached Similarly, yes, as soon as it becomes in use So try it here with LS BL. Block list and see that look at this Aqui df1 If he brings the GB1 and GB14 ones as well What will I do?

[Linux for DevOps] Linux disk management and LVM / Linux users, groups and permissions

I will go to actions and attach volume. In the instance I'm going to put the volume here. Device name I will attach the instance to There is a '' after 'f', I will put that in. I have attached the volume now that my third I picked up the one with the remaining available volume. I will take action and attach the volume and run my own in the instance I will take away whatever was mine and If the device name contains G followed by H.

I will put it in SDM, you have three Why did the volumes even come together in one volume? Anyways We see that those two new Attached volumes, can you see them here? We will try LS and BL and Equi df2 Butt is attached to your instance If it is not mounted then I will also mount it. I will teach you, stay tuned, let's move on to

the next point. I hope you have a great time You must have received an introduction So understand this that when you have these volumes made these volumes made these To make volumes usable, it.

Must be converted to physical volume, what is physical volume? It's like you made Volume One This made Volume Two and this Volume Three This volume one that was made was yours This volume two was 10GB and yours is 12GB. Volume your 14gb now i What can I do, combine all of these? Combining all these, I am a volume I can create a group which we will call I have a volume group and I can use it as a volume group. I can create a group, what will happen with this?

10gb and 12gb how much it became 22gb and 14gb became 23 and this 36gb The total volume group of will become mine How interesting and this volume group. I can take out small slices from of logical volumes Logical volumes are the siblings of logical volumes. Logical volumes There are partitions whose sizes you can choose. Can you according to your usage meaning I have three volumes from Let's The total volume of the group is so many times What is the size of 36GB, I took 10GB out of it.

I can remove a logical volume of I named it C Drive.

[Linux for DevOps] Linux disk management and LVM / Linux users, groups and permissions (part 2)

It is that out of 36GB, 10GB was given to it. Created a second logical volume of 10GB And whenever needed, I increased it by 2GB. Reduced 5GB and so on, this is what happens Your physical volume is your volume group and this is your logical Volume manages it all LVM means logical. Volume Manager or Logical Volume Manager so let's start learning LVM If you give it, it goes away quickly. Wrote warning running as a non-route User functionality may be unavailable when also you have to run LVM or whatever You have to do things related to volume.

You have to run it as a root user or every Put sudo before the command, which is very boring. If yes, then do sudo scu and become the root user. You have become the root user, so go to AV A now. You need LVM which means Logical Volume Manager. Who I am all these things, these discs or volumes or blocks If there is no physical volume then right now I. If you have to create physical volume then let's let's create and what then I will go How to create a physical volume I will go to LV M and now I will create PV.

Meaning physical volume created is PV. I will now create a physical volume. For whom will I create S DevOps / Aqui D A XD F what is this my First device means first volume then / DevOps / exsvi d h e fg yes. XSV DG and what is the third day / xsv d a so what is this, this is my first volume, honey. Sing Volume One this Volume Two and this Volume Three These three were my blocks what am I going to make with those blocks I'm going to create the physical volume of the box.

As soon as I press enter, physical VolumeCreatedDrawerNavigator It is of 10GB, 10GB is free, this is of 12GB It has 12GB free and so on great bun Now I have gone, these two are equal df2. I'm gonna name it TWA Volume Group Tied Volume Group Name I'm going to create a volume group. In which I run xvfb-run The volume group named Waji has been created How much will the total space be?

[Linux for DevOps] Linux disk management and LVM / Linux users, groups and permissions (part 3)

I will show v to a volume group List it in the Teads Volume Group. Two physical volumes and one logical volume No volume and its total size 21.99 GB means this size is almost 22 GB. You are free now, how did I teach you physical You create a volume group from a volume. Create a logical volume from the volume group If you can, what will I do here? LV CREATE LV CREATE means logical Create Volume I Create Volume A logical volume with a size of I will give 10gb so I create a volume of 10gb.

I'm going to name the one I'm going to give I said a LV means logical volume and this I will create a logical volume. From Volume Group to Volume Group What is the name Teed AVG is so cool, haven't you already? Physical volumes created a volume group from those volumes from the group You created a logical volume, but. The question arises whether you have this third one Why did you leave the third one alone? I left it alone because I'm going to make some more, some more mounting.

I will do it, you will know it in the end but In one place I have given you the information about LVM Physical Volume Group and Logical If you have been told to create a volume. LVM has finished its work. There is another

thing in LVM called PV. Display So when you display PV So the entirety of your physical volume All the information is visible here. If I displayed PV then PV The name of the volume group of this PV This PV's size looks so complete.

It goes well and similarly you will You can also display and volume groups You can put all the things about Similarly, lv displays logical volumes. How much LVM can you display? I hope you have understood that What are physical volumes and what are logical volumes? What are volumes and volume groups? What are ticks in Linux now? Please teach me how to mount also because If we are doing LSBL then I can see that XSV DF is Disk is one of the XSV DG LVM.

This one is out and Aqui Da is now It's just a disc, you haven't even touched it.

[Linux for DevOps] Linux disk management and LVM / Linux basics and file system hierarchy

When we are doing DF half A in There is no sign of people. Aqui dá aqui dez aqui dá these people So, I can't see anything here, that's why It is not visible because it is not mounted. Let's mount it than mount it First a question will come that aqui df2 then eqvi df2 now let's go to me If the volume is a logical volume, then the LV. S This is my logical volume TW SLV I want to mount this B Mount what does it mean look this is your phone Now this is your device, you can put USB in it.

If you take it and attach it, then now you have When you have attached this to your instance This USB actually gives you files in it. You will have to mount it or Then I will have to unmount it and I will do that. First, I will teach you a DirectoryEntry LV mount means logical volume which is my I will mount it here and press enter. First of all, my LV is created which is a logical volume I will format it, how will I do it?

I show the format AKA Ed ext4 now your path is / DevOps steed s Volume group from it A logical volume, this is the path you need to. You will have to format it first with this command. Creating a file system to format it This means creating a file system. In which it will remain blank, now this format will be Now that it is done, how do you mount it? Will you do it Mount Aoy NT Mount now this which Your location is: Consider this as a source.

You will write it and where you want to mount it. Ho let's say I want to mount the / M AT TID'S LV MOUNT YAHAN I This is my mount location so my logical volume I want a mount or a binder for it. Enter with this location mounted now The second command I told you about was ls. These blocks of yours would have been visible from BLK. The second such command mentioned was df -H now look df han a in finaly DevOps / Mapper TWA Waji means among the Volume Group This is a logical volume of 99.8 GB.

[Linux for DevOps] Linux disk management and LVM / Linux basics and file system hierarchy (part 2)

She is mounted on MNT TWA LV Mount now this place is Riu Jabal You understood how you were before When you attach a volume, its logical Create a volume and then mount it. Let me show you something really cool, CD. / M AT / whatever your TWA LV mount was Now I'm in this MKD I R DevOps whatever I want I can because this storage It has become my extra storage Extra USB storage for me. The instance has never exhausted its storage.

Why would it be because I know it How to attach and mount a volume It's so cool, I found it interesting and liked it. Let me now do the two videos and I will do the CD DevOps. I will give you and here is a VIM. Let's get out of here, now if I will do it / M NT TW S LV Mount DevOps and inside which is now I can read that file from anywhere. Can you see that I have that which is my mount If the location was there then it has been made useful then if someone asks you what is the difference Between Volume Attach and Volume Mount.

Attach lets you add a block to your disk. You give it to me but you can put that block on the mount. You make a, you make a like this If you need to make the block a ub then mount Now let me show you one thing if I Should I mount or unmount someone? NT TW S LV Mount Co I If I unmount it, it is unmounted now. If I want to access my file Effort I will no such file r DirectoryEntry Simple Mount Had its source and destination then Mount The source and destination are mounted.

Run the cat command and access it. Never use, reuse, or reuse your volumes. Point which was mounting volumes in I am telling the truth, I have not taken it anywhere. AWSEBS and Eat Instances In How do you mount both of them,? Here we were taught LVM. In LVM, physical and then groups and then Logical then by doing that you read Mount But directly if someone is made of EBS How to mount a volume That's why I did the last in LSLK.

[Linux for DevOps] Linux disk management and LVM / Linux basics and file system hierarchy (part 3)

Gonna say that's what I've got in volume The group is not created, even the logical volume I haven't made it, how do you mount it? I have the same method, just a little command It's here and there, I'll show you, look at me If you have to create a mount location again M KD I R / M AT. / a tw s disc mount disk mount means cd smnt I have two types of mounts, one The logical volume is mounted and a disk I have to teach you both.

My slash DevOps / acqui d a is what rah went to aqui da as a third one to this I will format it but now this is a disk. Formatting a logical volume that is not The method is a little different, even a lot No, it's just a little different, mkfs.xfs With ext4 you can directly create and use logical volumes. You can use enter for this so what do you say Dev XD Contains LVM Two Member If you remember to run the file system, we Did you create a physical volume for X Vadi?

That's what you still have to mount it. I told you one with logical and this one let's go I'll tell you the second one, then come out. If we go, its file system is created, meaning This XSV DH has been formatted, now it I will mount then mount / DevOps s. Ekdi A this is my source turned destination what will I put in / m nt steed s Disk mount is interesting and enter, try DF Hyn H. MNT With the disk mount and DevOps Mapper Ted a Joe It is a logical volume that is mapped MNT TWA LV Mount Co Drop A.

People have given AWSEBS to ETU How do they manage with instances? You also told me that I understood LVM also. Given that Logical Volume Manager is Look, let me tell you one more thing that happens when We run LVM so one of the LVMs When the promotion opens, LV S this The command displays the logical volume. Get the LV display done, this is the command. It runs inside LVM but it is a bean You can also run the command outside of LVM.

[Linux for DevOps] Linux disk management and LVM / Linux users, groups and permissions

Earlier, LVM was just a tool that Canonical created it. The people who created Ubuntu He gave the same commands to the tool and If you give the same command in your shell then Both of you can do whatever you want but. Interest comes when you are dynamic, do you do storage management now? Let me explain what dynamic storage is. You should understand that you have understood both of these. A volume using physical volumes Let's create a group of new diagrams Makes balls So this was physical volume one and this physical Volume two was, using this you had a great experience.

This volume group Correct your logical volume from One of your logical files from this volume group The volume turned out to be great, and logical. The volume has been exhausted from this, now if You might remember this was 10 GB, this was 12 GB. GB so the total volume group that was created It was 22 GB, out of which we used 10 GB. One of Logical volume named TWA LV What was its name when it was made? I can extend it by 5GB more, oh that's a lot You told me the right thing, how can I do it?

I'll show you, look, right now, DA -H DO M NTTS TWA LV Mount Key How much storage is there, it is 99.3 GB, now look at me I'll go into LVM and I'll go outside of LVM. I can also go into LVM and I will write LV Extend LV Extend means logical Extend the volume a little bit -l 5gb add do pa gb add. Logical Volume Path o So now you have a If I want to give the logical volume path then my What is a logical volume path and what is a dav / DevOps Stead.

SVs and then Steed LV this is my Logical Volume Path is Enter Physical Volume TW SVG Not Found In Volume Group TW SVG So My Y The command is a bit confusing. I'm so messed up I'm doing it to you like this You don't have to write directly to your LV Enter this in Extend 5g and see now Why did I do this? I did not put it because the TD LV is already associated with a volume group So if the volume group has extra If it is 5GB then it will be allocated, that is why.

[Linux for DevOps] Linux disk management and LVM

What I did was lv extend a 5gb path What did DevOps say in the Tead Volume Group? In twd lv means logical volume now Size of logical volume change from 10gb Two 15GB logical volumes successfully reset Now exit and do df5 and you can c Good to TWA this mount volume so. If you do, what is not visible in the first tree Wasn't it x I couldn't see it in the tree anymore. I can see how the beans appear You get more storage on the mounted volume is over and if you do df5 then you will be seen adjusting somewhere else.

It will be reflected but if it is not there If it is happening then you can see it in LS BL. That a little bit of 5gb from aqui df he Picked up and 10GB of Aqui DG He picked it up, it's so interesting.

Shell Scripting for DevOps

Source file: transcript_9Xl1ZTk3BQw.txt | 16,414 words kept of 30,251 | 42 sections

[Shell Scripting for DevOps] Backups and log processing scripts / Cron jobs and automation

Pay you will get all the information related to beginners to beginners in one video. He asks me, how much shell How much should one know about scripting? How to Write Shell Script Loops What is user input? What is a variable if you Programming Shell Scripting Nothing You can still learn the basics once them. Intermediate shell after covering How scripting changes ownership How cron jobs and automation do all that How to extract viewer logs Gonna see Advance Shell again We will go into scripting how we Networking How We Do Backups.

That you've come this far, some projects Also take Real World Applications and projects are going to happen how You can call AWAKE Cloud How to automate your own shell scripts. How can you backup and restore? Ho using cron job using log rotation g Yes, this is very important. The project is how will you parse the logs How will you do log analysis, everything is included, if you want to learn in one video If so, please like it and leave me a comment.

Things are going to be figured out and move on. Wally in his DevOps Engineering career Let us understand what a shell is and what a shell is. Why are we learning scripting? First of all, you should understand that Shell Scripting remains Linux-based. Based then you have to understand Linux first What is the architecture of Linux? If you want to understand the Linux architecture I will tell you a simple formula. You should write ask write ask Now understand the core of Naxals.

The heart is what the heart is for Colonel has such a Many such programs are written Programs are written so that Linux operating system works The program that runs inside the kernel. You can network the hardware You can manage it by installing software You can do a lot of things that talks directly to the hardware, which talks directly to the hardware. Your kernel is from hardware talks and the hardware is basically its Above is the kernel but the kernel is then C programming.

C programming to user What will happen to the user if he has access to the kernel? He needs a terminal that is A door is needed through which he can access The name of that door is Shell, that door.

[Shell Scripting for DevOps] Linux processes and services / Cron jobs and automation

The name is shell, so it is the layer above the kernel. Shell done and on top of that shell Utilities are applications based. Bases live that talk to shell and Your apps talk to Shell Kerner This top layer comes on so if you look at the ruffle So this is your I know circle, it is very bad. Made it for you but I hope you got the idea. You might have thought that this was the architecture of Nax. It happens and what you are reading is this shell You are trying to read shell Why, when you read Shell, you read Tap can handle and once you have to tell the Colonel.

Once how to handle it, you can automate the task. You can do a lot of things if how to automate the kernel, the shell I understand now, the Linux that lives there. Nor is there any way to talk to the shell that remains. There are commands for the thing which we people Shell commands speak What does the shell say? Commands and the shell commands that exist She is very cool and very simple For example, if I have to print something If I want to get it done then the command is echo.

If you want to create a folder, the command is m KDIRM If I To list the files DirectoryEntry If you look at the sequence from top to bottom. If you write it in a file, it becomes shell script goes so I How do you read shell scripts easily? Before that, a little about Shell. Let's understand the history of Linus Tor Walds created Linux but the shell There was a shell, he needed help for that. There was a guy whose name was Steve I think the name Steve is born There was a guy named Steve Bourne, he Help line to create Store Word Completely written in C programming.

Dangerous thing he then helped make the shell So the thing that Born created was called SH which was the shell and its location Slash in your Linux bin means sls h is in binary folder. That shell was kept at the location, that shell The file was kept there and from here the shell The scripting starts without / SH This is your shell now which is formed.

[Shell Scripting for DevOps] Shell scripting control flow (if, loops, case)

He brought more improvements to the community also brought further improvements so it became Born [Appreciation] again Born Again Shell means B.A.S.H. Bash is the Bourne Again shell that he created above which made the entire shell scripting possible. Shell was running in the same and improvements and Aya Bash, his location was also like this/bin slash bash means So you don't have to learn everything that is most popular nowadays. The most important thing going on is bash, so you need bash. I want to learn bash so this was a little bit Brief History About Shell Just understand what kernel hardware is.

Manages everything but the Colonel I have written a C program like this I need some language to talk to the colonel So that's why we need shell commands And if you run shell commands in sequence. Write it from top to bottom, a little programmatic If you write it in that then it will be a shell script. Before that, you need an environment. To run that shell script The environment will give us AWS. We have an 8 instance on AWS. We will create it and on that instance we will complete it.

Shell Scripting for DevOps course underway I will be here for a while and Here I am shell There's one called Scripting for DevOps. I will create an instance of shell scripting for instance named DevOps which is Ubuntu's The instance will be Ubuntu instance And this teat micro will remain free tier Eligible, free, eligible now. Many people will say, you have a crush on us. If you can't see the micro then don't region from here Select it in AWS and if you If you are not an AIDS account holder, you can Locally via Virtual Box or.

Can be run through WSL or any other way You will need a pair of keys to create one. They do it and name it Shell Scripting for DevOps's Shell Scripting for DevOps key There is a name called RSA Prem and you can do this. You don't have to change anything in this don't want to change don't want to change anything you have to launch the instance so if you What is bash second to yourself I need a Linux machine.

[Shell Scripting for DevOps] Linux users, groups and permissions / SSH and remote access

Must be able to connect to a Linux machine You should know how to connect, that means it should work. How will we do SSH one thing I have made videos on that before. Click next to see if the instance is running. Click on the check box and Go to Connect as soon as you have Connected It is written SSH Client there. SSH command is written this SSH and your This key name and instance name are used to do this. Pay and Instructions AWS Ne I have given it to you, do it, he is saying open it.

An SSH Client SSH Client means your I opened the terminal, then what is it saying? Locate Your Private Key I Have a Private Key Where is it kept here in the download folder I have kept it in CD downloads The private file is kept, run. This command if necessary to insure your key is not publicly available so viewable to Whatever permissions are there, they should close it. If you want chmod 400 then it means this It should only be readable for its own sake.

For your user, other groups and others There are groups, there is no extra permission for that. Two chmod Shell Scripting for DevOps Done now LS Shell Scripting for DevOps If you look at the permission of the key then L S I N L Whenever you want to see the permissions, how many. Permission is there, which means this pem file is correct. As easy as a minute What are commands? Let's look at shell commands. Inside echo and double codes and look at this Printed Print Done If I want to create a folder What I do is M KD I.

If there are shell commands, then remember that diagram. The big diagram that I had made Remember that you have to do this in hardware. I had to create a folder on my hard disk. He must have had a colonel's program, something But what shell command did you enter A KDI R Now if you need to do the same thing in shell script. If you want to create it, you will have to create one file at a time. How will that script be made?

For this you need to setup an environment One more thing will be needed, one more thing will be needed and that is A text editor text editor text editor now that text editor.

[Shell Scripting for DevOps] Linux basics and file system hierarchy

Your choice might be Vim Could be nano and could be G Edit You can do whatever you like, however you like. I like Vim, Vim is very cool So what I would do is I would first open Vim I will see how you run it, how will it work Let's run it and write vim, then you. The action is to create a text file which Dotty acti this extension happens that I want to create a text file so dot t. If I want to make a python then what will be the dot Tell me quickly in the comments about the Python file.

Vim t akt enter now this has opened my vim Editor, what should I do here now? Mode is not blank, it means command Now that mode is taking your command, you Two commands to remember: one is colon db Meaning write and quit means what you wrote. Escape colon double key enter came out And if I file cat then new The line has also been added, so in this way you Learn to use Vim or any other editor. Let's write our first shell.

Script is, there is a lot of excitement. Create a folder called mkdir or You can do one more thing, it is very simple. I can do as many scripts as I can They will be maintained in a Git repository The name of that git repository is shell Scripting for DevOps now you can clone it Can you do it? How will you clone HTTPS? Clicked on this copy and here Go write it down. Clone paste and here you can clone, now here comes a folder called Shell.

Scripting for DevOps is now called go inside the folder this folder Go inside and create a folder M K D I R Day 0 Day 1 Day G and more. All the scripts that remain today will be will put it in the forest and so that you Now learn what I have to do. I want to write a script, how to write a shell script Now I will write it as I told you. If you want to create a text file then end it with dotty.

Acti if you need to create a shell script So ends with Dr.

[Shell Scripting for DevOps] Shell scripting basics (shebang, variables, input) / Shell scripting control flow (if, loops, case)

A so I make a shell script named a first or Then Halod a load a what is this load h.a your The first is a shell script, how do this shell? The script is Because It Ends with a Dot S The dot ends with AH, so this shell Now let's go inside the script, let's go inside now. Will cause Linux to run this file as a shell Know how to execute the script If it works then how it is whenever you write any file about it.

I should tell you some information, then Shell What is an information about the script? It must be that, who is this script? I told you so There are not many different types of shells like bin slash s h bin slash bash like There are different types of shells, so that thing what is called shabang what is called shabang. What is shebang, I will tell you shebang what happens Hash Exclamation Mark / Bin Slash Bash means like we watch Marvel or Then let's watch the XYZ movie.

Let's watch its movie from here If you keep watching movies like this then it will There are production houses that say this movie is brought to you by this production house to. This shell script is brought to you by this Shell, for this you have to tell this first line It appears that, the first line is hash exclamation mark/bin/bash than Interpreter who knows what happens. It happens that what the interpreter does is Ba reads the line and executes it. He understands which shell Which shell to run?

There is a method, this hash is like this Let's call a thing a comment, this thing Let's say comment means your shell script I won't use this thing anywhere but this. There is a kind of indication, there would be a comment That, this is your script on this thing It is based on this thing, so this would have happened So let's comment and shebang. Let's do it, we're in insert mode hash exclamation mark a little bigger Let's make the screen have an exclamation mark/bin/bash slash bin Where did you get the/Bash from?

Colon double k enter I've saved it I saved the file and now I can see it.

[Shell Scripting for DevOps] Linux users, groups and permissions / Shell scripting basics (shebang, variables, input)

Where is this bash kept? I will Ch bash chch bash so he is saying us r bin bash lookup ch bash command gets to know What shell is running on your system? You are placed at this location / SR There is no need to put it but it is just for you I was telling you for knowledge, now. Vim went to AH, we went to Bash and then Insert Now we went into a mode, what did we do, we did one Echo printed it right now.

You have to do the same thing in a shell script The only difference is that whenever you shell You write scripts like movie scripts It happens that a man speaks a dialogue and then. The second man speaks the dialogue, then the

third When a person speaks a dialogue, it becomes a script. You have to write a script as it goes. Our They are learners, they are learners, what do they say? What do the learners say, from DevOps, we will comment, you have done this.

Please like also Give me one of these boys for sure There will be a little way to ask for likes and comments It's casual, someone will definitely write it, let's go. What did you write, you shell Commands Shell Command This is the shell command echo Shell commands are written one after the other and they become What a shell script has become so easy. It was so easy, now I have written so much escape colon double k entered but you We have to understand this much, let's go back once again on yourself as to what you want.

The shell script was created using a text editor. You execute the shell script If you execute it, you will get the shell The script needs to be executable. Whenever we do it, we He tells me what permissions the file has. Can do it other group members People in the group can read it and write it. The rest of the children may be some other Users can read it anywhere No executable permissions Executable means X, write X here should have come and if there is no permission then.

I need to send my file to HelloDigital If you want to give executable permission then I will have to do chmod what He changes the permission now.

[Shell Scripting for DevOps] Linux users, groups and permissions / Shell scripting basics (shebang, variables, input) (part 2)

This first dash tells us that this Is the object a file or DirectoryEntry If it's not cute then I need to read the current user. If I have to give right execution also then what should I do I will write seven means read right Execute now this next one is this What permissions does a group have? If I don't have read write execute then I If you don't want to give execute then read right Let's call the permission fi then the next one The group got permission for five, now that.

The last one others that is read write read just That's all, if you just want to read it So write 754, only read can write I can't do it, I've done so much, now this permission. Execute the executable file How to do it, it is simple, dot written dot the directory where your file is located go in heload a is dot / heload a means. Your shell script dot s h complete Shul script and enter watch the full picture Shall script be run TWA speaks, how are you learners?

I think within half an hour I had my first If you have learned to write shell scripts then this All the things you get throughout the course We need an environment, we need a shell. We need to setup the environment and we did it today. Let's start the day two, but before that, welcome to Train V Shivam on the channel and you are watching Shell Day to Day One of Scripting for DevOps People have finished it, which is very. It was interesting where we Introduction to Shell Environment I had set up everything that was necessary so that A shell script can be written so that today we Basic cell scripting Skills that a DevOps engineer needs.

First of all, he has to understand What are variables? What are variables once If you understand then you will understand the argument. I should say that if I had any one from Lets If the argument is passed then what does it mean? There are two things, for loop and while loop. DevOps are great, all these things I will teach you how to understand variables better. This shell here is basically Our instance is here, we have reached Day One.

[Shell Scripting for DevOps] Shell scripting basics (shebang, variables, input) / Linux users, groups and permissions

If you did, you would know gotta how to make this so it's here Now what I will do here is I will shell it In a folder named Scripting for DevOps. I will go and today is day two so I will I will do MKD IR day to name me so I created a new folder called Day Two. If so, exclamation It will look like hash here and exclamation/bin/bash it's done Now I need to put it in this if file I want to write a comment, what does any comment mean?

It is very important if you have a shell I've written the script and you've got to give it a lot of thought. Variable is variable its value is very She can be very capable now, just like the firstborn is tomorrow. I will not write what I will do Write the variable and put its name's value here. He will feel that just like this name, so is this Both are names and strings, so how can Explain to the shell that this is a variable, then its.

If you invest your dollar further then dollar that variable identification by applying If we make it executable then C H M Odi 755 What is this? If I give it, it won't work but date is this command Right, if I were to buy dollars and this was for pennies put inside then it will run so if you type any command. I don't want to change, I want you to be like

this Ask the user what their name is and the user You will get to know that the name is If yes, you can write echo here.

Will go and you will have to write at that time which any variable you need and that value user will go into a variable called name and you can print that variable. What is the use, know at the time of creating user Its use is how VIM creates your data one I created the file create your and put it in slash is exclamation bin bash and here. I put the read username and if you want, it Proms Ho Han P enter username enter username like this I have put it in, now try running it, Echo You.

[Shell Scripting for DevOps] Shell scripting basics (shebang, variables, input) / Linux users, groups and permissions (part 2)

Entered and here dollar username is escape colon double k enter escape colon double k Enter good two ways you can bash create user. Sudo user add han m and here dollar user I can name and echo that User dead or new user dead, something like that I can speak New user dead means I'm a. I will add the user by taking input from the user and I will show him that new user has been added. What I'm going to do here is file Make it a little executable like chmod 77 Sorry 755 Create User Ds Enter and dot / create user ds enter the.

The firstborn is done, the home directory is also added. The blank space after this, These are your arguments and it's all yours. The argument is this is yours Argument number zero is whatever comes after it. That is your argument number one. What happens after that is argument number two. Meaning, when you write yours on the shell You go to the terminal, look at this terminal is open dollar terminal is open It happens when you write dot / firstborn. The world is like that, dollar zero means zero eth argument let's you put it before that wrote that Iyer wrote Iyer, so that would be dollar one If I show the first argument, it's like.

At the first argument, a year, this is yours. The first argument lets you get the second If you want to give an argument, then this is your thing. The second argument is these arguments And you can enter these arguments in the code or in the script. Those things happened enter the name put anything I gave it to you entered now here But Ayar and Bhade are not visible anywhere. Can you do the argument and the characters? Of these firstborn How do you see the world or the world of the firstborn?

Argument and dollar one it's the first If it is an argument, what can you do about it? If you create it, it will work, then you will get the argument I understand how cool it is Understand what you understand I'm here, you understand. Variable you understand the arguments Now let's take a look at what if else that's what a conditional is look at condition It is very simple sometimes we.

[Shell Scripting for DevOps] Shell scripting control flow (if, loops, case) / Shell scripting basics (shebang, variables, input)

Let's start the script first. I will make an exclamation mark hash this one Bin Bai without bash so what does this script check does Disclaimer: Now what is this disclaimer? You are getting things done, I had come to study DevOps. If you do this then the condition which is yours if The block that happens is like this I Disclaimer comment block was closed,. Our conditional cover is also gone, now understand what are loops see what is loop I know I'll tell you the loop What is one thing that keeps happening over and over again?

Name is USA What is Bhupendra Jogi loop, are you thinking it is a loop? I am going on, I am going on, so I consider myself a Let's make a loop, now many people will say That, he is copying Easy Spot. Give me these funny examples I have been teaching for nine years, let's come back. To our place v I'm loop dot Let's create this name, for loop dot s now what is a for loop Look, if you have any one condition,?

If you have to do something repeatedly So you apply a loop, let's tell you the loop. Shh exclamation mark/bin/bash now this is for end While loops are fine, both for and while I'll show you what you'll remember,? Something from let's again and again Now I have to get an echo done from Let's, are you doing Echo, something good? Put it whatever you want to repeat Yes, keep it as it is and outside of that you can Put a loop, put a for loop now.

Which is nam this is a variable this is a variable whose value If you have put one then it means if you have any. If its value comes there then understand it like this that the first time the loop runs, demo one The second time the loop will run demo to the third time. Demo Th 4 5 Ideally it should be like this Lets C Let's C Let's give it a try Syntax errors may occur, no problem. Will do it so chmod 755 permission Let's give the for loop enter dot / for Lpad A Enter Syntax Error Unexpected Token: Now understand this like this.

[Shell Scripting for DevOps] Shell scripting basics (shebang, variables, input) / Shell scripting control flow (if, loops, case)

Which is a bit like the syntax of a for loop It remains different, it remains a little different So what we're gonna do is I'm for loop that s Now this thing is for such and such. If I do ls I will do it, you can see demo one demo two demo Three demo four demo f Five folders have been created. We're in furlough now I want to I want to name the folder and the folder and that How can one understand it through arguments?

Whatever the name of the folder will be, it will be dollar one dollar one means the first argument I show I So here I put I will give dollar one dollar one means first. Please write the argument somewhere here,? Colon double k enter now watch now watch funny Which is the repository of 90 Days of DevOps,. Check out this 90 Days of DevOps repository. Boom By writing this, LS Saab, all the days became day one. LS did see if all of them moved away then you are like this.

You can also create and delete it, this was it for loop what is while loop so v m while loop d a while loop funda. The concept of Vile Lup is very simple, what does it say? That as long as any condition is any condition The for loop continues until it is satisfied. Here I have put the condition that dollar no less than equal to less than equal to means less is dollar num e less than equal to 5 so do if this is so do echo print something and this.

Equal to Nam Psav you edited it like this How simple, how cool it is, now here per done escape colon double k enter bash while loop zero 0 1 2 3 4 5 six times printed if you want some one. What happened here, the echo of Namaste is heard Value Plv Pv Something is happening Let's fix this, let's get some divided Batu and Nam Lesson This is happening. A Simple You Can Do How To [music] Increment Value In this shell, it remains simple, perhaps mine.

Bash while loop enter then 0 1 2 3 4 5 6, this has been printed in everything, let's go to our Maybe something went wrong with your condition.

[Shell Scripting for DevOps] Shell functions, arrays and error handling / Shell scripting control flow (if, loops, case)

This is called a operator because it is one of your The variable is done, you have to put the multiplier inside it. Bash while loop 0 1 2 3 o this Why is setting the condition true? You kept doing hit and trial experiments Do it a little, you don't have to know everything For the first time, you would experiment like this stay put let's see what will happen let's try this. If you get zero, what do you think of yourself? Came that if nam matu dub e 0 now what will you do it Mode Operator Mud operator in shell script mode operator in shell script now you can see that How to use modules, beach.

There is a space in it, let's put a space in between. In this A lot of things depend on the syntax. If it goes then I have shell here Scripting and back project notes If you keep it kept then if you go to the notes So whatever I have taught today is all about this. What's left now is your functions for loop. While loop done function remaining function What happens, see the function, you should understand it. It's very easy function very It is simple, what is the function?

I will do it or I will do something again and again What I'll do is I'll create a function I will create a function and name it Give is. Loyal is loyal and inside this function I'll tie things up, I'll add things up I will make a club and group inside this. If I want to check anything then I can simply use this function I will call this function I will take that I wrote function is loyal and loyalty I have written all the steps here.

When that function is called, it runs. Function is Loyal is like this is loyal this is a function And whatever our code of loyalty is I'll tie it up in curly braces, I'll tie it in curly braces. The function is loyal to these curly braces I have put it inside so whenever I If I call Loyal, it will work now. Let me show you one thing this code was running I will not call this function anymore.

[Shell Scripting for DevOps] Shell functions, arrays and error handling / Shell scripting basics (shebang, variables, input)

What happens escape colon double k enter Dot / check if zeta loyal nothing happened Nothing happened, I know why Function created, function is loyal is From a function named loyal, the function is like this. Parens are identified by the y of the function I wrote down everything inside and called him. I will use it, it is simple to use Write to Loyal here and call him, this is. Non-S calling, so I would write it here am this is function call this is a function call and above What is this is this Function Definition, this is the function definition.

Where the function is created and here Escape colon double if function is being called Now what I will do is simple run. Why would I write dollar one, dollar one, because I will write it because I am an argument guy I'll pass the name of Look, right now this dollar one I am now. I'll pass the argument how cool it is Not if you put dollar one dollar inside the function If you are writing this then what all this is. All this is going to be passed to your function.

If the argument is, I am loyal to where my The function call is followed by the name of the argument I will write it down, I can put anything in it. Can also run for and similarly you can basically do anything in shell scripting. Scripting for DevOps' repository I just checked if it was forked or not. A beginner who is How do they write shell scripts? If an error occurs in that function then How to handle it gracefully I don't know if backing up a server is a good idea.

You can now try shell scripting. Let's give it to you, we have come to our instance We have shelled out this instance on Dedicated to scripting for DevOps So let's do a MK DI R. It is called Day Three where we discuss today's If you are going to keep all the things then it should be Day Three I am here now to tell you the basic fundamentals. I'm going to explain how error handling works and Why is there a VIM error?

Handle dts Error Handle dts here now But I will do it / ya shab bin bash and Here I will give you a simple function I will make Create DirectoryEntry.

[Shell Scripting for DevOps] Shell functions, arrays and error handling / Shell scripting control flow (if, loops, case)

I enter a simple command mkdir demo is now a function that returns demo now this Let's call the function create DirectoryEntry The second time the function is called So this. If there are no directories then the code will not run further. Should the code stop then see what happens if I echo with Let's This Should Not Work Because the code is interrupted, right in the code If something goes wrong, this line will not work. It happens escape colon double k entered now Let's give this file some permission.

AOD 700 and error handling done now I dr error handle a Enter This Should Not Work Because the Code Is Interrupted It's for the first time Because if you do LS for the first time then. It kept moving because he kept moving from top to bottom. What I do is I'm handling errors on Now let me see if create directory if in create directory It gets messed up, I don't create directory if in create directory What if something goes wrong? Be it to do the code is being Assitate Already The Agist code is exiting because The directory already exists and exists So I took Exit One.

It doesn't happen but if you want to do error handling then you have to play with if conditions Exit one means exit code one, come out go come out with a code one. This is the only thing that needs to be exited, a project. It's as simple as that, I go escape code a Enter creates a file e m deployment A one I have a Django app Which we follow from start to end. Deployed that error handling through shell through scripting Be deployed and excited and want That I should bring more such videos and If you bring me a project, let me know immediately.

That I can tell you in the comments so that me and hurriedly day one day two day th day Four Day F makes a lot of videos like this. First of all my work will be to clone the code so First I will clone the code. From where will I clone the code? I will do it with git so this is my GitHub Isn't it there, I will search it here.

[Shell Scripting for DevOps] Backups and log processing scripts / Shell scripting basics (shebang, variables, input)

Any one of the Django projects in the repositories So take what is above Django Notes P Ye removed Django notes p these notes app I want to run Shell Scripting. I'm going to go here and copy this as well. I'll do it, that's it, now I'll do git clone. Echo cloning the Django app cloning the Django app here I will do it like this and here I will write on Git Clone done, now understand git clone forever. If it happens once, then it will work for the first time.

Will it go a second time and give an error a second time? Required libraries like for this code Just look at what you need Look at this, look at this, look at this in DM, right I have Django, I need Python, I need Richt, I need Docker. I want it, it is simple, I want so many things now. The second function I'm going to write is It is very simple to install I'm going to install the requirements I am going to do the requirement as per.

Docker is done, the engine is done, so here I am. Installing Echo Mar Dependencies Installing dependencies This is done now sudo at get install What a duck and Engine and if you get any promotion then say yes. Y this is done now if there is any An error occurred, but there is no error in this. Required Restart is fine if you don't see any functions. To any service to anything If you want to restart, you do it right. Lettuce helps restart your engine if so sudo System CTL Enable Docker is fine by the way Hido System CTL Enable Engine Enable What happens the next time you.

Will go fine required R K R a required tart required What will you do after the restart? You your What will you do in deployment? It is a simple Docker project, so Docker what will you build, what will you do with docker I am not talking about build or this repository. If it is written in it then copy it from here And put it here, put it now. What's next is docker run copy from here Take this and put it in, once it's done you will get this much.

[Shell Scripting for DevOps] Shell functions, arrays and error handling / Linux users, groups and permissions

After doing this you will get a Calling functions one after the other like what was the first function you wrote People clone code clone no code. I made a clone call, then what did I do, install it? Did the required restarts and at last You people have deployment like this like this by doing this deployment You can do something like this, you started Write, beautiful, beautiful and this. You can end the same line here that Deployed Done Deployed Done Now Ideally this code should run because What a DevOps engineer does is what a developer does The installation dummy is given as.

The Cod 77 means every user, every group, other to the user but we will only give read to the owner Right Acute Permission to Di Paly Now watch the day to day live of the engineer. I dotted Degree Now watch the entry deployment started Everything is going well, deployment is taking place. I installed the watch Neel is running code d also clone You must have taken it, now see it is going well. Something good is happening, Deit done but dement done speaking meaning Ideally, what has been deployed, but it has happened what if I do docker ps docker only It's not working and it's saying.

The one before you has been deployed Why error handling came into play? If it is important then let's make an error Let's see what the error handling is. The error is that it is unable to prepare for the contest. That thing is Dockerfile not found most First, where is it looking for the Dockerfile? If there is no Dockerfile then how to handle it you will do it whenever Deployed DirectoryEntry before that or whenever you git clone Or will you handle the git clone now?

Git cloned escape colon double enter Escape Colon Db Enter Now I can run this code I'll run it again and see what happens Cloning into the repository will be fatal. Destination Django Notes Already Assisted and This is not an M directory o meaning understand I am thinking that it is already existing, so it is not What will I do if I can walk? Deployment if I got an error while cloning, so I cd two, I will do this much, then Deployed Started Jahan Code How to handle errors in clone?

[Shell Scripting for DevOps] Linux users, groups and permissions / Backups and log processing scripts

If code clone encountered problem if code Clone problem occurred first. What could be the problem if your code If there is already assist then which CD should I do? Two what thing Django Notes P I Hope That's the Name Django Notes P Yes, this is the correct name and CD it in Django Notes. Took it and also got an Echo done before this The Code DirectoryEntry The fatal one had come, whoever it was already existed. We have handled it and fixed it Try it, let's try it, Dipl be the court gone look what he said the code Already exist means you have a handle.

Your code is handled now, next time He must have seen that look you saw that the Dockerfile not found error was coming,. It's gone Docker found file not found error Gone, new error has come Permission Denied while try connect to docker deem this docker Do why the file not found error occurred? I had made CD Notes app in the clone due to this What happened that DirectoryEntry and he drove he now next If you see the issue then install I

have no issues with the requirements.

You can fix any permission issues that arise with Docker. If you can fix it then you have permission. What was it that Docker's D man This var run docker sock this is the file He is having problem in this, he is giving permission to. If you cannot find it then VIM deploy what do I do I will give permission to this particular file I will give you CAO DNB whatever my current The user is. He should get permission for this file and do this work.

I'll do it with sudo do this I'll give you this now if restarts are required restart if not What will happen if it is required? There is a problem, it means there is an issue in your system. System Fault Identification Fide system fault identification and you Get an exit f come let's say to me What should I do if I need to restart Docker? I will restart Docker if from let's If I have to do it, I will write it here.

I will give you the system ctl docker a no system ctl restart docker this restart asap I did it too, now look here if any issue arises, the code will exit.

[Shell Scripting for DevOps] Linux users, groups and permissions / Shell scripting control flow (if, loops, case)

I won't go any further because I have taken the handle I have done it where here handle it here I have taken the escape column and double enter it. This happened, ran, ran, ran, well, ran, oh Look, Docker is asking for authentication. Two, you go there and put sudo. Because your Ubuntu user who is using AWS When you create a CCTV, you generally have a password. Put sudo and it should run now. Run vim deploy and enter deploy Executing Enablement Engine and Your Docker The build has been started, it's so cool.

How cool is it that the error is in the same script are also handled if not git clone If it happens, he handles it. The library is not installed and it is handled The system cannot restart I would handle it and finally deploy it. Is complete deployment palay is applying let's see And what happens next is a warning if Ideally, it should stop if an error occurs. Deployed Successfully Tag Notes Latest Oy Yes, yes, I have already been deployed. If I docker ps then that is on port number 8000 What should I do if this code is running?

I will go to the brick and its port Where did the 8000 go in the security groups? I will go and open its port at 8000. Do a rule custom TCP 8000. Two, if you need security, my IP Simply put it and you will be saved on your IP only. Rules have been made, now go to your EC instance le its public IP Go here and run it on port number 8000. What's going on you can see your application deployed what's the matter with error handling But, if I run this script again If I do, just see what happens.

Built node p latest and running perfectly fine everything is going well so it means your There is complete awe at this time, there is complete awe at this time. You can write it in your vim file. It was said that we can write in its file which Is it deployed or not? Docker run or its also giving error. Can you handle that If there is any problem in deploying then Echo what Echo what is deployment field Failed Mail the Admin instead of Mail the Admin.

[Shell Scripting for DevOps] Backups and log processing scripts

Mailing the admin you can do and so forth You can use a send mail utility or this one There is a utility that allows you to send emails You can do it if your deploy fails. This is one escape colon double k Enter Now try running it again and see deployed P, this went well, it went well, let's see that from deployed started to. Everything is working fine till deployed ended or don't see, what is this, deployed done, everything is fine.

Everything is going well, everything is fine Try CPS and see, yes everything is going well. Sen second ago mast he is becoming but one I can see something wrong, what's the problem? I can see that every time your Docker is restarting every time Restarting and it's running I don't want that, what do I want? If I don't restart Docker every What happens at times because it's a simple thing Isn't it true if you have written it here? If you have written here docker run and this Code This code is repeatedly entered on port number 8000 If I try, I will make an error, so let me do it.

Docker would not have been enabled either If something like this had happened to Let's, look. Error what comes escape colon double c and ter now See what error will come, error will come, enter did this happen deploy failed mailing the Admin, if deployed fails then. The life of a DevOps engineer is not so easy. Everything doesn't look green, look it's failing Why do it's been deployed? Because 8000 It is already allocated, in this case What will you do in this case?

Docker Compose so you write in the file here Docker Compose in Doge Install Requirements You will also install this and this of Docker Compose The good thing is that if you. If you want to restart then you can do it by up and down. You can allocate the pot, it is very easy So I need Docker Compose and where I made an error where I made an error So before that send it to the admin and here. What will I use instead of run, simple Docker Compose up is in Docker Compose up here it goes here it's done here I am a.

[Shell Scripting for DevOps] Linux basics and file system hierarchy / Linux processes and services

I will simply run this command, I will not run this command. I will simply run a command Docker Compose up through Docker Compose This project should run. Half an hour video I hope half an hour You will get one hour video of this No objections because my purpose is You have to teach here, that too, look at this. Error something came can not start driver Fail port already allocated so what should I do I'll give you this port by doing docker ps for now If I de-allocate it, then docker kill but see this.

You are thinking that deploy is failing but Deployed Done after Deployed Fails Why is it coming in writing, its simple meaning is That whenever you have mailed the admin If the code is not exited then here. You have to go and write exit one from this What if this is not a deployed done message? This will not give you an error because it The file which is cat deploy app this is the complete error Handled, the entire file is an error.

Handles everything from code clones to deployments You can operate this thing completely comfortably. I commit to Kit Status, I commit to that day. Why has it gone, what has happened because Day Three Inside I wrote a Django Notes P a direct Cloned a GitHub repository What is the reason for the problem Scripting for Des already a Git. There is a repository and inside that I have another one I have made a git repository which is very wrong If it's a matter, I'll do the work.

CD Day I will remove it, this is for you generally. It will not come but this is a common issue if you are a Clone the repository within the repository So this comes day three p go status so here this Our CD Death. RM Hafer Django Note yes done git Status git d These gits status git Commit ain am added Death and doing this I push the gate. I will do the origin main and here for all of you here where ya which shell scripting Is it a repository or a shell scripting repository?

Your Day Three account is in the repository. How to clone using commit How do you do it, what are its errors?

[Shell Scripting for DevOps] AWS CLI automation in shell

We have shelled as much as it takes Scripting for DevOps in this free course We have covered it, we saw it in Day One How do you setup your entire. Learn the basics of Environment Day to Day Learned intermediate shell scripting in Thr and a project Deployed Vance Shell Scripting and about debugging because I've done things like. I told you that as much as you can keep your shell I have read scripting and that is enough. What should I do for a DevOps engineer?

My heart just doesn't agree and I want more If you want to learn something new, then today I am going to tell you I have brought it for you AWS CLI AWS Apartment Won't you do ours? Where we are a Linking with AWS scripting Gonna see all these things Look, this is a very interesting thing, I'll tell you. What is the script about? Let's take that AWS is basically your cloud. It is a service platform where you can You can create all the resources You can build a lot of infrastructure.

Where can you do things like shell scripting? Shell scripting comes in handy here. This entire process is AWM Automating the way things are done It comes but, you can also Terraform that thing. I can, I can definitely do it and in this region Because of Terraform, Embedded made things happen. New tools were created because shell scripts We cannot do everything alone, but we It is very important to know that shell How does the script do all that? In today's tutorial, I will explain to you I'm going to see what things are How integration works for AWS So the first one that AWSIM is up, this is your laptop.

Where is your shell what is your Shell Now run this shell from AWS account You want to connect, how to do it G Shell Always run on the command line where the yi. If there wasn't a GI user interface, you would have had a CLI gets you and AWS gives you a AWS CLI gives aw a What is the first thing you should do if you get sick? Open a new and download AWACLI v2 Look at this, Edl v2 means Version two, the latest one, is fine.

[Shell Scripting for DevOps] Shell scripting control flow (if, loops, case)

Open it and here something basically There are commands that you can use to run AW CI You can install it, see when I told you I was teaching shell scripting here. Basic Scripting: Basically Shell Commands When you write the conditions in sequence You write with loops, you write the same There is a shell script so if I have a AW CLI through shell script. Do it and today whatever I have done in the last three days Based on what you have taught, ask yourself whether I can write a shell script and Read the commands of CLI from here and.

Install by writing in shell script Can I write a shell script that does this? If your answer is yes then leave me a quick comment. It is very important to install so your The shell script will be your shell The script will be something like this Let's say this is your shell script. In that, first of all you have installation Off AW SCL I but I install it every time I won't do it every time Install AW CAI only then install.

I would if I had a WCL If it is not installed then install AWS / ii only If it is not present, now I have to Tell me how I can do this thing. You can check if AWS seal is available installed if not installed then this The pipe that remains is the fallback if it is not installed then the installation one Run the script, if not, install. If so, run the installation script. You will have to configure your shell to do this.

I'll give it so that this shell AWS This is what you have to do to get linked with it. Next, our task is to create a T instance. We have to create a T instance. People can also create it through Terraform. Why are you bothering Ed Say in this and Why bother with a shell script? If you can write a script where you are running the AWS instance You can create S3 with the same script You can also create an instance and create an S3 bucket.

You can create RDS instances of anything. You can at the same time send that script to your You will be in control of that script.

[Shell Scripting for DevOps] Shell scripting control flow (if, loops, case) (part 2)

You can modify the script yourself. You can use it for personal use and your Shell scripting is a journey for him and you Can you sharpen me at the end of the session? I'll show you how to prepare your script. Let me give you a little challenge whenever we People What is an AWSium instance? You take a server on rent. This happens whenever you create an instance. The instance gets created but The instance's state is not running It takes four to five seconds, sometimes up to 10 seconds.

It may take seconds, so you can use a shell script How will you see through that this Whether the instance is in the running state What you will need for this is these seconds. In point you t EC2 Create You will do it, you will create and third point I will see you will wait you will wait that if Whether your instance is created or not. How to check if the instance is running or not You will do it, it is very simple whenever you take a t.

You create an instance and its state what happens to a state if that state If it is running then it means that the state is running. That the instance was created correctly else running if state is not running This means that everything is not right yet and when Till this state is not running, check me. I understood the script, didn't I? What to do is make a shell script do AW How to connect with AW Second AW from CAI BEAT YOU FROM CAI instance is to be created right became the third that.

Checking the state of the EC2 He is in running, he is pending, he You have to check all the things in which state. You have to do this thing, I am giving you this as a challenge. I want to give a shell scripting challenge For you Shell and AWS what do you need to do? I will give you a script for it. You have to understand and after understanding it, your I want to create any service as per ads S3 May be Lada May be RDS It could be anyone, so let's go.

If you understand the script, click on it.

[Shell Scripting for DevOps] SSH and remote access / Linux text tools (grep, awk, sed, cut, find)

The shell script is here create to I have created this instance which is quite strong. It's a C shell script, so you won't read it like this. Let's launch an instance and see. If we go, I'll go with my AW S here my shell scripting for Connect to the instance running with DevOps I will do it and connect through SSH If I take it here, SSH. Instance on where we are shell I'm going to run the script, now I'm coming. Go to my Shell Scripting for DevOps Now it is day one, day two, day three.

Shell script for DevOps a git The repository is, this is a git repository. Your branch is up to date with Origin Main But I am not up to date on Day Four. I can do a git pull. I'll take git pull as soon as I'm done. At my Day Four is here and here I am on Day Four. I'm going to go CD Day Four now its Inside is the script for create e2 a What I'm going to explain to you is E.M.

Create EC2 a so first of all this first There is a sentence, it is called Shabang second Sentence set eu pipe fail meaning if you error If you are handling it, then it. This advanced shell shows beautifully There is a flag of scripting, what do I do? I am checking AWS CLI this I'm checking if AWS If I can't find the command, I'll say it. AWS CLI is not installed Please install it first and exit. I'll do it and then I have a complete AW CI installed.

Method where I am installing AWS S. Next function Let's see Waiting for ECT Instance I will come here a little later Let's see how to create an EC2 Understand how it creates this thing. Once you have CI then that seal came From the command line interface you can use AW If you can talk directly then add e2 run instances This is a command that allows you to create e2. You can create an instance, just give you a You will have to provide image ID from Lets.

To make a kit and Now if the instance is not created then the instance ID will not be generated, see here how am I doing here what am I doing Instance ID is equal to this thing.

[Shell Scripting for DevOps] Shell scripting basics (shebang, variables, input) / Shell scripting control flow (if, loops, case)

Where does it get its instance ID from? Whatever query you get, that query instance whichever lets say this command gave 10 Create an instance or create a The first instance of will pick up the instance ID and the instance ID. I will get the output done, see I will get the output done So if you don't get the instance ID then It is simple that your instance creation Simple has failed and so you can exit. Go and if the instance ID is found then There will be no exit, the instance will come here You have to put it in the created list until it is running.

Wait for it to not reach the state I would call this method here. Pass the instance ID inside it I give it and by the way I have given it here The instance ID, AE ID, all these things Isn't it, I have created this as a local variable. Whichever way you pass it from the command line can you let's understand that wait for What does the instance wait for instance do? An AW SEAL runs the I's command Whose name is EC2 Describe Instances And in that instances I want to set the instance ID I pass it and put it in the query.

I wonder what the state name of that instance is? If the state name is Running then the instance and inside this while loop of yours Take a break from it if you are not running. Take a short break of 10 seconds and then Run this command and see that the state Is there running or not, if there is running then Break and come out done and as soon as. You will come out your waiting for instance These things will end now All the variables that are named are I passed it inside a main function.

I need the instance type, I will fill it here. I will fill in the subnet name here. I will fill in the ID and do security groups. I will give the instance name and all that All the variables are parameters, I will pass the function and Thatchi here you must have seen this here Check the line AWS CLI and. Or fall back to install AWS ci or I mean, he'll check first.

[Shell Scripting for DevOps] Linux basics and file system hierarchy / Shell scripting control flow (if, loops, case)

If adult cell has come then it is a good thing, then further It will increase, if not, then this fallback. Is it or not because what things are there in it There are four things it will check whether a CI is there or not No, if it is not there, there will be exit if. If exit is done then there is a fall back in fall back What is the installation of Ed CI once Ed Say installed, we have a brick instance.

We will create the instance once it is created then we What we will do is check the state of the instance. Escape call now yes main function I have made the call here and this thing that follows It is written means in the main function You can use any number of arguments. Escape colon double k enter now what do I will see you soon 700 Create to Instance to Whom I'll give enter now if I run dot s Create to instance so what first.

Maza Enter Ads is not installed please Install it first, let's install it now Let's go We'll create it Inside the instance, basically your check AWS ci install it first this as soon as. What is he saying creating EC2 Instance line 58 DS command not form, so he basically can't get the command. This returns one let's go then return one is done

escape column enter dub what is this If this command is not found, return one. Two, if there is no one return, then basically Your command will not run and AWS SSL will be will install enter no such file off DirectoryEntry something is messing up.

It's a CD, the CD was left in it by mistake, that's why There was a slight error while copying and pasting. I should use my brain a little bit Maybe the CD was left there, yes this is the CD Don't keep yourself if aids don't come. Run it and install it and just press the escape column w k enter run done now see the installation ad v2 on Done yes look at you installing cia Can Now Run Usr Local Bin AWS Version.

Meaning the AWS seal you received should be installed. It is already but when that EC2 instance I went to create it but I did not have any credentials.

[Shell Scripting for DevOps] Shell scripting control flow (if, loops, case)

If you get it, you have Edal CI, then you have it. The part has come to you but AW Credentials needed to connect to Didn't find the same, let's setup the credentials If yes then do not go to Connect to Instance You have to go to IM,? IAM and here's what you need to do. Shell admin put shell admin this is it He no longer needs access to the console. He needs CI access, so here Click next on it, now I will put this in it.

Let me put an admin policy in place Attach policies directly to Or you can say no to him basically give full access to EC2 full access to e2. Grant access so that it can create an S2 instance Got it, it's, I've given full access to e2. Next, go to your shell administrator. I am done now and I will go to shell admin. I'll go to security credentials and I will create access. What is a command line interface? He gets it and that command line It will generate access keys from the interface I understand next create like this Two done, this access key came and this The secret access key has arrived, copy it.

I will run the command aw a, AWS, Confer Done, here now. It's two times the secret access that was there. Giving this AWS seal eye as if it AIDS Seal IEC Two Instance Created Enter the default region name now it's up to you what I do generally I do generally a. US East 2 where I'm at All instances remain as they are in the US East to so copy this from here and here Go to the US East and write it down.

Output Format: Do not write anything for now It's completely fine not to write anything Enter a is configured now what to do Now you have Ed CI as well. It is also installed conf dot / create two instance co Run it now and see what problem comes up. Creating E2 Instance Error and Error Acad Missing Parameter When Calling Run Instances the Request Must Contain parameter image id so if you create easy In Two Instances Jaoge Mi ID is blank you instance.

You are building it and there is no operating system in it.

[Shell Scripting for DevOps] Linux users, groups and permissions / Shell scripting control flow (if, loops, case)

You are not uploading the image, how wrong it is. Go to instances Go to instances And here this MI ID has no You should also pick me up from the toilet. Two slammed Done Now instance type t2 Micro because you have a free tier It won't cost any money, now your name is Which key should be used to access this instance? Can you see that I No, yes, go to the pair of pair now. Pick up anything from here, I will pick it up.

Shell scripting for DevOps key is This one is the key to shell scripting for DevOps. I pick it up and go here and throw it down. What did I do, I went here and threw it subnet id what is subnet when Whenever you create an AWS instance It has to be created under a VPC and. It has subnets so let's say mine Shell scripting for DevOps this is what I have You can see which subnet the server is in. You can go into networking and here Look, the subnet ID is written in this subnet You can copy the ID if you need it from here.

Not copying what you can do VPC You can go into and go into VPC and your A subnet ID in the default VPC If it happens, you have to copy it,? Go to subnets and look here. There are so many of them, copy one of them. I copied it and went there and threw it at me. What to do: Add a security group So I either create a new security group Let me see, security is present here. Said the group or you can search Security Group Security Groups You can either create a new security group You can create it or assisting security.

If you can use the group then I will tell you this. Using an Assisting Security Group I'll take it, I've copied it and I've I wandered over here, it's done. My security group name is Saree Whatever details there were, I got them from AWS. Take it and fill it in and double escape colon Enter bus, this is the moment where we are Run this script and

Is Thorates C Practically Works? Enter Creating T Instance But waiting for this instance to be in Running state means it has been created but If it is not running then it will loop again and again.

[Shell Scripting for DevOps] Backups and log processing scripts

It is running, as soon as it starts running it will be like this I will exit the thing and boom no boom If it doesn't happen, you will get it in chat as soon as it happens. Boom if this thing works dum thing has gone away instance I this is now Running EC2 Instance Creation Completed You have an end to end AW An EC2 with Esc scripting. Created an instance and saw that it was an instance whether it comes in running state or not End Sub After this you exit this code.

You can go here and click on instances. Check if your instance is running You can see the shell scripting demo This is the instance it's running see it Your Initiation is going great, you can See it's now running yes when I start. I was here at the gate Status: This thing that was gate def this thing that was this return one and this part is giving me an error. You also have to solve this thing and This thing, what is its name, Shell?

Scripting for DevOps is a bridge to this. Create a pull request After creating it, I review it. Hope you can learn advanced level from a shell script. Interacting with Pay AWS If how to do it, then tell me quickly. Are enough for your interview so if you shell If you are preparing scripting then this much It is much more than enough but if you need more Looking at real world examples and. I need to practice a little, so I We have kept the days ahead, so today we Those who are going to see the actual shell Not scripting but shell commands is an integration of.

I will show you one such There is a special type of file where you have to run make commands. What are you saying, understand what make is like this what is the name of your file make file Now what do you want to do inside it? If you want to build something from it, you have a wrote the command build now what to do on this command You can write that command like. You want to write a shell command that docker build highty dot once this is done then you want all what should happen on this one.

[Shell Scripting for DevOps] Shell scripting control flow (if, loops, case)

Jaye docker tag image koi image tag ho jaye You want something like this, then you want that whenever I speak run so docker run command should run docker run whatever. That was your image and whenever I say up So what should happen in that, it should be built and run If something like this happens, it should happen, I mean basically. I am giving instructions using shell commands I am giving instructions for a make file Where the commands are very simple, build He's running, he's up to you, let's say You want as many containers as you can.

What will happen to the Docker system when it's pruned? However many containers you have If all those things are removed then simplicity will come. How elegantly you made one that you just Enter the command "Let's make build now" Run specific commands only for you If you have to do it then only run this much part. How cool it must be now, its real use What is the case that you have no connection with the letters? Whether to make it environment specific or Even if it has to be operating system specific Make files are very useful, for example You said that if my operating system.

All these things will run only if there is Nginx How much fun it would be if my operation System And here I will give you a demonstration. I'm going to show you how to use a make file. If you do, you would have taken a small application Here is a two tier flask application. Let's take a look at the two-tier application. If you want to create a file then how will you create it? Here is the code of the application which I would do I am a clone and I will go here and write git clone and this is my application's code.

If we go then CD two tier application now this I need to make a make for the application file now what I'll do is look at the make file so. This is already an expert, if we remove it then R You have deleted the m make file, now I am creating a new one. Let me make a file in the make file If I want to create a variable, how do I create it?

[Shell Scripting for DevOps] Shell scripting basics (shebang, variables, input)

Docker I found one called Compose, this Docker Compose. If I want to create a variable, how do I create it? Simple is variable name colon equal to now Which is your actual command for this variable Enter the value that should be there, so let's From me the value of Docker Compose in this. If you want to add it, I have added it simply. Let me know which operating system I want to know what it is, so I wrote O colon How to Equalize Now Operating System If you recognize it, you can write it inside the dollar.

Write the name of the shell, this will give you the name of the shell Operating system will also come now like I explained it to you in the tutorial here. Here you have to write the name of the command. Here you have to write the actual command. Let me create a command from Let's make a command There is build and what should be done in build? I want it to happen that my gets scared Echo Running in now here on our what is dollar Something like this should be printed that I can [music] So what about Docker?

Compose Up gets Docker in a detached mode What does Compose become from now on? Application Now I want to be able to do it whenever If I say down, what happens to Simple Docker? Composed down should be a simple and whenever What if I say clean? Docker system prune docker system prune - wi What does this mean, all the containers All the stagnant volumes are there If the things are images, they should be removed, then I have done this. I have made a small instruction set of my own.

Escape colon double k enter now what should I do I will do whatever is simple I ran the commands in my make file. If I give it to you, then if I have to do make and build, then make build enter now he is saying that command Make not found to can be installed to make which. Make and paste this so let's make It will be installed in two minutes now. What will make build make build from me What do I need to get my application built?

[Shell Scripting for DevOps] Linux users, groups and permissions / Linux processes and services

Go enter he is saying this no true fa directory Docker Compose no such file and directory make file line number eight at one If there is an error, let's see what the error is. If there is an error at line number in the make file T 3 4 5 6 7 8 there is an error one minute. Let's remove this line for that if this line is causing some error or not, let's do this and see, yes, one more thing. You have Docker to run Docker Must also install Docker command install that should install Docker Engine If that is not there then let's try that too,?

He is saying no such file of directory If he doesn't get this, then one work around is Docker Compose. Let's install Docker and see. If yes then sudo at install what docker d and Try installing Duck Compose. What is happening oh sh what sud at get install docker.io has no Installation Candidate because you have your If you haven't updated your system, AT Gate update the system first and. Try it out now, this is basically Docker Compose and Docker. I am installing this also see this once Check if your Docker is running.

Because this system requires access If it is Docker then try docker ps. If permission is coming denied then this is it. Run Docker sack, change this ownership then sudo sd a which is my current user I give ownership of the particular path So that Docker will not throw errors, give. Di docker ps no errors hit let's go now We see that in this file Docker There is compose, yes there is make file, let's go now what will I do, I will run make build, enter.

What he's doing look he started the build I've started building with just one command small command he built I have started, let's see what's next. It happens that I only built I have asked him to put up the containers I haven't said anything, you see how I Running the execution in a controlled way Let's just see what happens next. This is the complete build steps being run MySQL client all those things so if You must have seen successfully build means if I do docker ps the containers are still running.

[Shell Scripting for DevOps] Shell scripting control flow (if, loops, case)

If you haven't been there, you've seen how right it is. This is when I told the make file that In the build you only need to use Docker Compose If he wanted to build, he just built it. If you say you want to build it, then build it. It will make it up, which means it will also make a continuous run. I am doing a small issue, make no such file Why is this directory coming up? From the compilation command the error and the Source line given this is that thing nos file GCC Can't Find include, this is a very strange command.

Never mind, yes, let's come back to the Point Creating Two Tear Floss Cap Done Creating a Two-Tier Floss Cap Done If I'll do a docker ps next so you can see. Both my applications are two-tier. Both Tier 1 and Tier 2 of the application are running. Fixed with just one command from Make Up So whatever commands I want I can run there, let's see. Whether the app is running or not then I can I will go to the instances and after going to the instances Go to Security Group I see that port number 5000 here 5000.

In details go and pay 5000 Pay running then your two tier application It's going great, it's going great and this If you need to stop the application within a minute. Into this and a little bit of level of VIM Make File Adds Industry Touch Now what I want is that if my Operating System So how about this build command? Escape column double k enter let's make down First of all, download Docker Compose. What I'm going to do is build my make only if it's

Linux so what do I will do make up, not makeup Build is running fine in Linux build.

I noticed that only if I have Linux, it If I have Windows then the thing should be built. Set off command toi m make file here if I will put that If right here in the build itself if equals what is my operating system Windows NTT does something about Windows by doing this. Windows Anti-Corruption two so this small So if I have Windows then you simply Tell me Echo Please Add Windows commands and you can do this in Windows Try it, now make it small, yes.

[Shell Scripting for DevOps] Backups and log processing scripts / Shell scripting basics (shebang, variables, input)

Please add Windows command and after doing this You just have to do the end if end if done so this is how you create make files Can you escape colon double k now if I. I will do that make Down is complete, now make Build saw chala but Windows The command that was there was not run, I saw it running in. Here's what you can test on The operating system can determine this You can specify all the variables and build it.

You can do a lot of things, Now let me clean this up and show you simple make Clean now he's saying that unknown command System Prune Y Docker System Prune. Unknown flag y so basically that This poor guy did not get it, this did not happen I am getting it, what can we do is this Let's delete it for now. You can, such an Echo is very simple. If you liked the Make File tutorial Or not and I definitely need this app with make files Today we will learn how to make a backing up the server and most.

Challenge is where your task is on Day Nine Shell Scripting Challenge Directory Backup with rotation so this is your challenge In which I am helping you where. You are getting backup and rotation of the server or a directory entry and you did the backup of p1 then. You did a backup the next day and another one the next day backup of p2 then backup of p3 do something like this backups are being made backups are being made. You have 30 days of such a project.

30 Which is a very wrong way of speaking what do you do compress cups Do you use zip? You can zip it or tear it. By compressing this backup using You store it but still if it is 30 days old If you look at the entire year, How many backups will be made and how many. It will take up storage, what is its solution? It happens that only the last 5 days backup is done. Only last three days backup is required.

You have to keep the latest one with you and the rest you can You can delete it or change it to another one If you can move it to storage then today's.

[Shell Scripting for DevOps] Shell scripting basics (shebang, variables, input) / Backups and log processing scripts

Let's start, we're here The data folder on this server on your server Let's try CD data and see something here. We have to take backup of these, so let's go back. Create a script to take backup Yes, I am backing up DA. This script will be very beautiful. The script will use functions and Proper syntactical errors occur as many times as possible I will resolve the issues before you. We will learn a lot, first of all we Shabang writes the shh exclamation Marx/bin/bash now I'm a small I'll put a comment here, multiline.

Comment I will name him Dummy and tell him I'll give you dummy open and dummy close and I'll do this I will say this is a Script for backup with file system Rotation Day. Rotation means only the last five days will take backup and some usage of this script If you want to use it, then how You have to use dot s backup dsa this. The name of the script and the first one you need to enter Path to your source to be backed up and.

In the second argument, you have to enter the path to backup folder so that means you two If you have to pass arguments, one at a time Let's take a small check that the argument. There is a question to determine if you passed or not. Let's create a function that Display usage is fine, let's make it a function What will this do to display usage? Echo will print and this thing inside it Bus usage will print display usage It will print only if your conditions are met.

If the length of my argument is this The argument that happens if the arguments are equal to 0 if my argument is equal to 0 Does this mean that I have just given the argument? Call up Display Usage and set your if Turn off the condition, it was as simple as that What will display usage tell you? When will it tell you if you haven't passed the argument? How many arguments there are in total, if that total The argument is zero, meaning you haven't done it correctly.

If you run it, run it, press escape colon and delete.

[Shell Scripting for DevOps] Backups and log processing scripts / Shell scripting basics (shebang, variables, input)

Enter now what I will do dot / back upd I will try this, permission denied Why because LS Han L Back Upd It has read permission but If I don't have executable permissions then I can c. MO 700 Backup AH Karke Ise I will give executable permission, now it is green The file has turned green, a complete green signal, good to go. Go.backup.de is showing the usage. Saw that function run back up a path to your source path to your backup folder then I am yes this is correct so path to your What I will provide in the source is a backup folder.

Let's make it too, so M K D I R backups so this is ta folder this is my backup There is a folder and this is where I keep my files. If I give then PWD present working directory this will be my path so dot / backup s in this the first one is mine. The argument will be what will be my data folder and the second argument will be What if my backup folder is fine? Why because I have used it according to the usage I have scored, so far it is fine, now I What will I do, I will actually take a backup BackupDS is actually used to take backup.

To the backup function here. Let's write the function Create Backup Function named Create Backup I'll make it, which isn't too difficult. It will take three things: one source DirectoryEntry My backup directory will come from One Argument to is simple, the only difference is that I have to take backup every day,. I also need a time stamp, so I'll add another one. I will create a variable and name it Time stamp now how do I get this time stamp You will get a look It's not good what should I do.

Copy and I'm on my The time stamp variable in DSA is I will assign this to this. If you have a Assigning a shell command to a variable So the method is dollar pence and If you paste it inside then I will get time. I got the step, now what will I do, create? In my function, backup, I say I will tell you to zip it, zip is a tool.

[Shell Scripting for DevOps] Backups and log processing scripts / Shell scripting basics (shebang, variables, input) (part 2)

Directory and source directory meaning The source directory will be this source This directory is my dollar source. You will use the directory like this, this is what The source directory will remain, backup it and What should I name the backup file that will be created? BackupDirectoryEntry My time stamp so this dollar time stamp, I don't understand what this is. My backup directory is home.stu.s Backups will create a file inside it called Backup This time stamp will appear next to the name.

It will end with dot zip dot zip It will end and that's all I had to do. This whole thing will remain yours, this one The string will be the item you are passing It is in a string, so you can You will put it inside the double codes, see now. If you don't want it then it's up to you and this After doing this, you just have to send a message to Echo backup Generated successfully for now enter time stamp here Two of this time stamp.

My backup was generated and I closed it. Do it, it's done, now I will use this function. I'll call this create backup function I will call the create backup function. Escape Colon Wk enter dot / recursively dot / this goes I will give it and enter, see what it is saying. Could not create output file home Ubuntu back Up backup o why because this backup which was written Isn't it called something else? Backup S Yes, but you saw this even though the zip error came.

He got it printed and generated the backup. Successfully which is misly ding then you A condition will have to be added if the above If the command error occurs, generate a backup. First, let me check the backup. This Echo Backup in Generate Successfully Only when will it come if my higher one The command is How is the output of the above command dollar question mark will be known if its. Echo Backup generated successfully and if look Just turn it off, if this is what I have on top this command dollar question mark its What does it mean that the above command is a.

If the above command gives an error, an error occurs.

[Shell Scripting for DevOps] Backups and log processing scripts / Shell functions, arrays and error handling

If it comes then the value will be zero so if error If it does not come then generate backup successfully, simple Escape column double enter and here I am I will run my command back backups s. I will put enter this time, see how cool it is Backup Generated Successfully Not Error This is a warning message, so if you If you want to hide the warning message then. There is also a way to tell your backup DAA The output of the zip command is He redirects where a black hole is What is a black hole, such a device.

SWEE A device that is a tap whose output If it is not stored anywhere, redirect to it. Do it discard do it escape colon delete Now if you run that command, you will see That error did not come and it came that backup was generated Successfully for This Now Once CD. Go to backups and see, backup is also being generated and is also being generated well. It is happening but there is a small problem if backup it if it's backup script every The day went on successfully, every day went on, every day went on What will happen if it runs every day?

Your backup There is a folder, it will be filled, so here you will see You will have to perform rotation How will you do this, understand this simple funda. At this 39 15 34 35 36 you don't know Not which file is the latest, then a command There is L.S. That one of the latest five catch and Delete the rest of the people, this is what we need. Have to do it in shell scripting and same thing What is called rotation then I am back up as I will go and I will have a one here.

I will create a function and name it Function performRotation Function to create a rotation I will create this function what it will do is First of all, restore all the backups. Was the command that would bring up all the backups It was simple, dollar pence, what could be done with it? You can run any shell command. Iss Time's Sorts according to who sorts I have to sort this thing of mine to the backup directory here If I want to sort then I will do the same here.

[Shell Scripting for DevOps] Backups and log processing scripts / Shell functions, arrays and error handling (part 2)

Whoever comes, backup up, whoever comes, he I want to print this thing now. I need an array of backups so I can use this I'll create an array here so what happens I will get the list of file names. I'll go and here I am Echo and Dollar print backups inside it I'll get it done and this is an array so I'm going to put it here rate inside square brackets. Sorted backup according to the direct tree in whichever backup restic means whatever file Come from something that starts with the name Backup Now you can change one more small thing.

You have to call perform rotation that you Where will you create the backup and then recover it? After creating the backup, you can perform the rotation. You will call rotation the spelling of rotation Yes, perform call rotation Escape colon and enter now your last Will you run the command where did our last Command this enter can not access home Ubuntu. Backups Backup App No Such File Directory So what does this mean? It's taking the variable name, which is wrong. Em back up thats gonna do this and this The double quotes that I have put in He has put it a little wrong, this is till here only It will remain and this zip is yours.

Extras will be appended and its output Where can you redirect to 2 Grs DevOps Snal what does this mean It means whatever the standard You are getting errors, please discard it. This will do the job for you, let's go. Let's Run This Enter let's go to our Home Backup 41 and then 10 38 You will find all the things in a sorted manner. In this array, first we have to see that if my If the length of the array of backups is five If the length of this backer is more than five.

This will give you your length And if it is greater than what you are If you perform rotation then here what to give Echo Performing. If you want backups, then these are Echo backups,. Will need list slicing or array slicing Meaning if your array is backed up like this and you If you have written this then what does it mean?

[Shell Scripting for DevOps] Backups and log processing scripts / Shell functions, arrays and error handling (part 3)

Don't print the entire array in the array. Fifth Print ahead of the location, meaning if This is your array, it contains t 3 4 5 6 7 8. If there are such elements then its indexes would be is 0 1 2 3 4 5 if I borrow that array Only one and only one fifth of the entire array. If you want to show everything from the index then just And only this much part will be printed, so if you look carefully, our array is sorted.

According to time, so many parts are in your and delete this much part You have to give it, you have to keep the first five,. I follow this logic in addition to the first five Bring the last extras If I go, what will I do as a backup? A cool backup tool called Remove I will make a cool array named array how Let's make it like this, let's make it like this here I will create this array and inside it. I'll go to backup DSA and I'll go to another one.

That will run a for loop for backup in backups to remove backups to remove mine is an array so here I will put it in double quotes. Same as written above, backups to rim such and its array this like this and here's what I'll do double You have entered the code,. Forcefully remove any dollar backups I have to give you a simple dollar I need to remove the backup and this After removing it, oh and after removing this After I'm Done Performing.

You will see the backup generated successfully performing Rotation for five days and what happens with it The backup is also being generated and your You have only five backups. I made a video about cr cron Jobs so what do you do crontab -e Two crontab -e and here it will come that You can take scheduled backups through cron. I chose Vim from Lets. Put Pay To here and choose Vim editor. Here you have a simple cron Can you create a job for cron job?

I would like to give you a tip, you can go On a website called crontab Guru and here Whatever you get from cron what is cron There is a schedule that you need to do something.

[Shell Scripting for DevOps] Backups and log processing scripts / Cron jobs and automation

If you want to run it automatically at any time, then its This is a schedule, so you can do anything. I will give that every minute, every hour, every day of the month, every Month, every day of the week, now I have command here What is the command I have to enter? This is my A command. The backup DAA kept in Uban is You have to run it and I will give the source path in it. That is my/home st Ubuntu / data folder and my backups The folder is / Home/Uban So you can find your path computer-wise and here Backups and what I do after I'm done.

I am living every minute, every hour, every day of the month, every Run a bash command every day of the week that backups This is the source path to run the DSA script. It will give the backups path escape colon Double Enter See here crontab Installing New crontab now look what I'm going to do This is 41 46 07 52 46 meaning if I. If I change this to LS hight, it will be 46. The latest one is 52 days, 07 days, 46 days, this is the meaning What I mean by that is the 46 It was a minute, 45th minute, 44th minute.

The 41st minute and now this latest one will come backup because this is backup 2024 39 41 This should be deleted Why automatic because I have also made the rotation Backup has been performed. When I can run it back See 50501 This is due to the latest backup that came in. The backup of 39 was deleted. Automatic rotation has happened and, this The way you have done it so beautifully Wrote a backup and rotation script What does it do, it also tells about usage and backup.

Computer Networking and System Design

Source file: transcript_xN3BKHji12I.txt | 6,953 words kept of 30,878 | 18 sections

[Computer Networking and System Design] Networking fundamentals (OSI, TCP/IP) / DNS, HTTP and TLS

The entire flow of this session We will do that with a live application. Think of it like this application Dave's Board Which Is a Project tracking application. Inside this if you Look, there are these types of features and if you Talk about AI assistants, let's say Summarize This Project So it works perfectly. The project summary of the project is here. Wire up TLS let's say someone is tasked So may I ask what is TLS? Task done wire up TLS at the gateway so AI Integrated Project is now Inside the project there is a front end and a back end.

There is HTTPS, everything is there, how does it work How its system works To understand this, first understand yourself. You will have to learn computer networking. With that comes Your OSI TCP IP model. OSI TCP IP model This will also be covered in the same thing. People, you will say that OSI model TCP IP What is the model? After this, we Will move towards security For the Internet. To see how important security is Once you check the security, What is a VPC,?

Because every Anyone understand what Kubernetes is? The Internet is an interconnected Network for Devices. That Elon Musk There is a company called Starlink And there must be such a cool satellite and that in satellite The network signal must be going from Pune and that The satellite sends this network signal here. Even if you go by land route, then Too many geographical boundaries on the route It comes in. These poor people, how can our network Will you go? Appeared in Someone's Question That What is the source of the Internet?

The source is These cables connect one device to another. Understanding I came to know what is the source of internet? Which your ISP Provides internet to the internet service provider and this Internet Service Provider is your Local Area Network as WiFi It gives you internet. You're out to become a DevOps engineer. Packets What is a packet? A packet is basically a logical A packet, literally a pouch type unit did not send it like this by filling the data inside it A logical unit exists within which Some address remains, some details remain.

The data is encoded in that packet and is transmitted.

[Computer Networking and System Design] System design and scaling / Networking fundamentals (OSI, TCP/IP)

Packets say Now let's take this packet from Pune. That's as much as Time taken from source to destination That thing is called latency. This is very different from India's Got Latency So don't be confused. Latency is the Time taken for a packet to reach from A source to a destination. What size of packet can be sent in latency Can you? Bandwidth, he says that we have 300 Mbps unlimited data at Mbps There is a plan. 300 Mbps unlimited data plan We have.

Bandwidth You say it's my bandwidth 4 Mbps then data of that size can be transferred If you have a 500 MB Mbps There is a plan, but It's taking more time. If you are, you are not going to become a hero. Choose a 10 Mbps plan or a 10 Mbps plan If your latency is reduced, it The data isn't going away. To become this, you should have taken a 1 Gbps plan. Is it latency or bandwidth? The latency is 46.6 milliseconds.

Meaning Round Trip Time Latency Is the Time It Takes for a packet to be sent from your Computer to the network and back. The lower your latency, The lower your latency, the better Do what that means? Low Latency It's taking less time. If my latency You can see the download of 64.8 It is. The packet has been sent, but due to the congestion In networks due to the high latency, high Jitter The Packet Was Lost. Three things are very The first thing that is important router second thing switch And the third thing is the hub.

What does a router do? He says the router Restart. Your which is the router that the Internet Service Provider is the incoming network from the ISP and that thing that it is broadcasting to your network The router is routing the packets. Router's The work is not that you install the router The internet will start working. It is necessary to know whether the packets or data is transmitted between different network end If that packet has to reach your home you need a Router.

I will enter the Wi-Fi password and all these Connected in a local area network. They were connected through the router. No, because the router directs the packet.

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You play the router's band. Or Then you put the switch on. You can connect them all with a switch. Switch these data Center, look at such a photo. At this Data Center Switches Are High Performing Networking Devices You Need to Manage Extreme traffic with enterprise and Cloud data centers that provide high Speed, Low Latency, Connectivity Remains for connectivity. Images if you want to see it, it looks something like this. To connect it, you need a switch like this It is. Grandfather the times of switch of This is a switch from Grandpa's time.

Inside that port You connect your networking devices And things went from there. Information means Packet means packet in which your Information is traveling via Internet to data centers are the heart Off Your Internet. Once your site has an AWS.com Pay Just create your own instance. One second yes 4747 Now what is this, I have an account. EWS has arrived here, now whenever I create an EC2 instance. When I am also creating an instance I went to Launch Instance from Lets.Web Server Name Less Networking I created a web server called Practice.

I went into the network settings and Basically SSH allowed in security group I did it. By doing that, I created a 10 GB Took storage also. I took the S3 files. To our Internet Service Provider The money went in. I'll go to the instance and connect here. Click on connect After doing this copy it with SSH client I'll do it and call my terminal. As soon as I called the terminal it We've reached the terminal. CD into the Downloads folder. You have already understood that an instance You need to use the SSH client to connect to the This client has to be opened.

Chmod 400 I hope you all understand all this now. CMOD 400 We reduced the permissions from there. This The SSH command is We connected with this. It simply means that this system It's not up to date. Sudo apt By updating the gate You can update to the latest version by doing sudo apt get You can bring the packages to yourself. In One of the latest packages There is also a package called traceroute that can be downloaded.

[Computer Networking and System Design] Networking fundamentals (OSI, TCP/IP) / Linux networking commands

To install You should run sudo apt install trade out. This, that The IP address of the server is PayTrain is hosted on shm.com. Where is this Ubu IP 172 server? EC2 US West to compute AWS.com these IP Address Now copy this IPP address And go here and write it down. IP Location Finder IP Location Finder and this Go to and enter this IPP address and Find it. If you understand My server must have been in Oregon. Is Oregon That source is Oregon.

Someone from Oregon A switch that connects a particular network It connects. Route Google.com So Google Went here from this place then to this server Then went to this server 99.83.117 Let's see what 99.83.117 Where is his location? United States Washington Seattle Amazon Technologies then he went here On one of Amazon Technologies' servers Went and then brought this data from there 142 dots from this IP lookup fetched data from here From Google. It is understood that from a source He has a lot of trouble to go to another place.

Every hop There may be packet loss. If this data Are they images or static data? Images or If the data is static then from a particular server Comes what is called C D N CDN means content. Delivery Network CDM Because I want the data that is there The data that is stored in one place I want to take static data that does not change. If you open A Global Infrastructure AWS. If you're using AWS's global Infrastructure When you open it, you will see a huge map.

That's why one region comes under Availability Jones. Availability zone comes under region. Region It falls under the availability zone. If you go back to Global You will get this in infrastructure It's the Hyderabad region. Simple Don't worry, DNS IP is here. Let the others do the brainstorming Our job is to improve our mind so that walked longer than the others now I need you what do I do to reach. This address There is a particular protocol for this. Protocol Internet Protocol what is this Purushottam what is internet protocol Internet Protocol is basically a rule set of rule that says that if you.

The people who are coming here are saying IP Address IP Address It's Basically IPV4 address Now when this IPV4 address came because of this.

[Computer Networking and System Design] Networking fundamentals (OSI, TCP/IP) / Linux networking commands (part 2)

There is a protocol called Internet Protocol and the addresses in it is an IPV4 address. Thinking the same, these people got the IP address created IPP v4 and they thought we People 4.3 Billion We will create IP addresses and give everyone a unique address. One colon two colon three colon four kept like this if you go here what's my IP will go here So you will get my IP address like this 1 2 3 4 5 6. This thing is called subnet.

Subnet means small internet, so what about small internet? Even on the internet, according to the rules, Internet Protocol (IP) Yes or No If there is internet then internet according to the rules If the protocol remains, will it also have IP? What is a subnet of the larger Internet? CIDR Now everyone listens to my classes. As soon as I said CIDR, moms and dads Like apple CIDR vinegar. Drink water So CIDR basically Classless Inter-domain Range inter domain range that your subnet is an internal It's going on.

You can have so many IP addresses if your CIDR Notation Is / One what did you say, Aai Bagan, what is this So basically, whenever you want to make CIDR Is CIDR. Anytime you need to make CIDR. Visualize and Calculate Network Ranges with IP v4 sider blocks. That if you if you Create a subnet. Within that subnet Assigning devices a unique IP address So the way of doing that is known as CIDR. CIDR Classless inter-domain routing is a routing Mechanism that will give you A network range, range means range means lets 0.0.0 16 0.016 So write it here 0.016 Total number of IPP addresses How many.

65000 500 36 Now If I keep my CIDR notation like this I got 65000 65 IP addresses are available. It is just these two, these many The network address that will remain and the broadcast The IP address remains, but it will no longer be usable. Let's say Its IP address is 15.33.42 This is some IP address. This data It is serving you computer network He is teaching today. They Who is this IP address that is giving you this IP address? In a proper way Your domain name, such as google.com train w sham.com The IP address mapping remains a DNS Server Pay.

[Computer Networking and System Design] DNS, HTTP and TLS / Networking fundamentals (OSI, TCP/IP)

The DNS server that lives No, he does a lot of work inside. Your domain In the background of tnwsam.com, there is a The IP address is mapped. From there I'll teach you how DNS works. You enter a DNS query. First, that DNS query is sent to the root DNS server. Root DNS servers Where do DNS servers live? For example, if your route The DNS server for google.com is gadd If it is there somewhere, I would go there. From there it goes to the ask.com server as So what is ask.com?

The first thing about DNS is You should understand Your local system does the caching. Let's say if I write NS here look up ns look up name server look up Train Visham.com Now enter The IP in the background of the train w shm.com She's here. Let's get past that local cache. Went to the root DNS server. Root What is a DNS server? There is a telephone directory. TLD servers are very This is an important thing. The top level domain is What's the name?

The TLD server is the top domain server. Component of DNS Responsible Managing a Highest Level of Domain. Google's top-level domain servers are located here. Google as a top level domain operator Google Registry So Google will have a server like this. Which will contain all these mappings And then your root directory is Let's go to GD and these people go to Google. Google, Public, DNS, all these are Google's. These are particular DNS servers. From there your His IP address comes out and that IP address An authoritative DNS server Rolev says that, yes this is the IP That's right, this 2.7, whatever your IP is, and.

This will be your domain, both of these are correct. When this authoritative DNS server gives it He says that it is correct, only then that thing will be yours If the pass is loaded then the meaning of the story What does that mean? That's not how DNS works. DNS is very It goes through routes all over the place. For example, root DNS servers, TLDs Server, Authoritative DNY DNS Server. It occurred to me that if I were to When I write my name, something is in the background.

[Computer Networking and System Design] DNS, HTTP and TLS / Networking fundamentals (OSI, TCP/IP) (part 2)

This TLD server, Authoritative server, DNS server, these are some I didn't understand. Understand this way that every The server's job is different. One's job is to check Ensure that the response is from the correct authority. This is me No one is giving a master class on Jamtara. I am explaining that top level domain Why does this work? That's why You do not get an IP address directly. Its route from the first root DNS server level domain It gets verified. The top level domain such as dot now root What level is included?

Example.com now This comes in the top level. Dot dotb.in gov.in gov.in.gov.in Who will operate the TLD server? Computer Networking Master Class There are notes that you can download for free. The DNS is a database of domain names and IPP address information that enables computers To locate the correct IPP address for a URL. I've put it on Cloud Front Cloud Front has a DNS server. Domain Sorry DNS Server Not CDA Server Content delivery network through which I can Digital Products Directly Hosted There I can do it and give it to you.

It is understood that DNS is a database of The domain name and IPP address are in. First, From your computer to the DNS resolver Will go. The first thing it does is check your cache. DNS The route will go from the resolver to the server. From the TLD server the top level domain.com. From the authoritative server, he The item will be validated and then the item The website opens. How many processes work together to perform DNS tasks? DNS means Domain Name System.

S means Will it be the server or the system? What does the OSI model say? Application Layer, Presentation Layer, Session layer, transport layer, network layer, the data link layer and the physical layer. Connect your devices with the cable and those devices link data Then those devices are connected to the network. Connection is made and the IP address Data is transmitted through Transport layer A session is maintained so that the data The transfer should be done without any data breakage and then Your application from the session layer.

His security is checked to ensure that everything is in order. It goes to IPP address then it transports via The network then a session is maintained then its Security encryption happens and finally It goes to your application.

[Computer Networking and System Design] Networking fundamentals (OSI, TCP/IP) / DNS, HTTP and TLS

Browser inserted in www.tran sh.com So it went to this DNS and from here it gave me I brought the data. Internet Protocol at the Internet level The address it gives is IP Address. Media Access Control Sorry Media Access Control Is a Unique 12 Character Physical Identifier Permanently Assigned to a Device network interface chip controller Even then this device would have a MAC address Yes. You are in the network You can send the request in the network. If you send a request to the end network had been.

Data is transmitted via Packet. Packets are transferred in the network. Through the control protocol or the user Datagram Through the protocol. TCP If you transfer packets through the protocol Guaranteed if you do Guaranteed data loss If yes, the data will arrive. In UDP there is no guarantee that the data Went and arrived. There is a little bit of network latency in between. These are the notes that I have for you What type of network is this? These notes that I just gave you Sent over chat, it arrived over TCP.

This TCP IP moved forward and I got hold of a particular address. Take this data and this Send it to whoever requested it. Layer 1: TCP or UDP Whatever type of request it is and then the packet through the network It carries it. The packet is sent to my device Identifies the data link. At which layer does authentication take place in this? Tell me what you need for authentication HTTPS You will need it in the application. Here If it happens here then V layer does authentication Work in OSI session.

V layer does encryption Work in OSI, tell me fast, fast, fast quick Data encryption primarily happens in layer Six presentation layers now which layer does Application. Work in OSI layer seven application layer so if Please look carefully, your application is here. The application layer, presentation layer, session layer, then your transport layer, then your Network layer, Then your data link layer, then your physical Layer. OSI model which is ISO international standard Organization or International Organization Theoretical Its concept is practical Explain what the implementation is.

This is known as TCP. IP Model TCpAPI model which has four layers.

[Computer Networking and System Design] Networking fundamentals (OSI, TCP/IP) / DNS, HTTP and TLS (part 2)

Application layer, Transport layer, Internet Layer, Network Access Layer. The OSi model has seven layers. TCP The IP model has four layers. The OSI model works on a completely theoretical basis. How many People understood TCP IP And how does data move in the Internet? If my http I am unable to pass the certificate, no The protocol is not running, the application itself If it's not running, I'll try to use the Network Layer. I am going to create the application and presentation layers.

I'll check the top layer. I won't check the network. At HTTPS is the primary operator at layer seven because http adds encryption standard to HTTPS so and so Both the application and presentation layers keep working on that thing and. That HTTPS Is it the application layer, presentation layer, or session? In the TCP IP model, that entire He solved it completely and said That you go, these three are the same layer Whose name is application layer. How many people understood that there is a lot here There was confusion in the OSI model, so those People created a new model called TCPIP In which this application was told to these three Layer.

One is TCP And there is another one called UD P. TCp works client Server Works on client server model So the client sends a request to the server that a Server, join me. Meaning client syncs first The server will say, "I acknowledge that sync. There's a pipeline that's collapsed. SSH in the background Uses TCP. SSH in the background using TCP. SSH Secure Shell Always Uses TCP At the transport layer, weather you are running It's interactive and sending to the background Because TCP guarantees packet delivery orders.

It ensures remote commands and Data streams are executed reliably. On the other hand, by the way, TCP, this It also uses HTTP. There is a protocol The text to transfer from one place to another returns. HTTP is a protocol that transfers text from one place to another Allows transfer of transactions to another location It is. Protocol for Transfer of Data to the U.S. What does the TCP handshake say? Whenever I Did cl HTTPS colon www.google.com I did this, I hit enter and the data came in.

[Computer Networking and System Design] Networking fundamentals (OSI, TCP/IP) / DNS, HTTP and TLS (part 3)

HTTP uses TCP What is http s? You can send text Transferring from one server to another Up to the server. Someone came in between and took such a text and put it in his pocket. Encrypt HTTPS also uses TCP SSH uses TCP. If it weren't for TCP, If there was no three-way handshake, then this would also happen. TCP's Because of this, your mails and emails are received reliably. UDP User Datagram Protocol what does it contain Is it you or your server?

You tell the server to server data The server keeps publishing the data. Work How does Internet Security work? How does security work in the Internet? Security works, very cute In this concept, it is called fire. Ingress and Egress These are two brothers. If this traffic contains data, the request yes, there are packets so if there is a packet If it has a packet, packet means data information to because of network Layer It will have a unique address. If he has packet then it has an IPP address That would be the IP address.

Let's use the IP address. X and Z IP Address 12.2.3.4 This is the IPP address from Let's. I have installed it on my Ubuntu server. I added it to my Ubuntu server. Apt get install engineeringix I installed my Installed Inginix on Ubuntu server. This is one of my applications. An engginx The application is there and I have installed engginx and I started it. I have this IPP address, if I send this IPP here Type in the address and hit enter. I should have looked at Nginx.

Because This is my instance, this is my instance The security part of this is security If I go to that Its inbound rules, inbound means Ingress Inbound Rules is what it says. TCP protocol for port number 22 TCP is allowed for port number 22 Allowed. TCP on port number 22 General SSH TCP on port number 22 That's SSH. I want to set the port number to 80. Either I give Anywhere IPV4 If I give it, then see what will become Anywhere IP V4 If you gave it, look what it became.

/ Zero means / Zero means 4.3 Billions of people can access my server.

[Computer Networking and System Design] DNS, HTTP and TLS / Networking fundamentals (OSI, TCP/IP)

Ideally 4.3 billion people can access my server You can. Security Wise As a DevOps Engineer, you should do this What? I went to my EC2. I Went to the instance and got its public IP And went here and pasted enter. While I have not added any security to my server. If that thing is not put then I will give S I will delete it and press enter. Welcome to Engineering Welcome to Engineers I understood I understood Now If I add a HTTPS Certificate Two Ubuntu Server Engineers If I were to will come If I sit down to do all this.

For example, sudo apt update Then install all Python 3 or whatever. These steps are being given by the people from Google. If I sit down to do this So can I send my certificate to my fire I will be able to tell Wall that My Engineering is now HTTP as well as HTTP. Let's say Secure Socket Layer.SSL Secure Sockets Layer which is out dated. The latest thing that is used is called Transport Layer Security No TLS Yes Transport Layer Security TLS then TLS certificate It is what you call SSL.

Video Laike Aao SSL TLS SSL certificate gets outdated I understand it's over Yes, now Purushottam, now look who has come and the Prodigal Son Returns. What Then SSL So does SSL. TLS is the new thing. TLS has been called SSL. TLS has become a term used to refer to SSL. Read about SSL vs TLS. TLS Transport Layer Security is the Upgraded modern successor to SSL.Both Are the same thing but all original SSL versions Are Now Depreciated and Contains Critical Security Venables.

When he realized that, do something good with your career. Your future is a little TLS right now Secure with security. With outdated SSL security Not secure. Sudo universal firewall allow injectix Full. I used Nginx's rule Updated. If I do HTTPS and Go to my security group and Call the security group do one thing whatever remains do this, go to security and The port number of HTTP is different. HTTPS has a different port number. Or if you're here Go to HTTP You select HTTPS So the 443 is automatically locked.

Do this You give me my IP and save it.

[Computer Networking and System Design] Networking fundamentals (OSI, TCP/IP) / Linux networking commands

Your job Security and the concept of security Like the uncoupled firewall in my I speak in the language of Universal Firewall Allow Except means you are allowing a particular rule Uncomplicated firewall delete. Go here to Security and then Inbound Editing rules. Security is very It's important. A firewall is basically your ingress Incoming traffic and egress outgoing to monitor traffic to prevent It puts up a barricade for public facing It is very important for the internet. Packets come over the internet so IPP addresses also come through the network layer.

The port with the IP address. For proper security You think three things happen. One happens protocol, Then there's the IP address, and then there's Port. For example, if the protocol is HTTPS If yes, the port number becomes 4 3 If protocol SSH If so, what becomes the port number? 22 If your protocol is SMTP So the port number becomes tell 45 Something remains 465 yes If you want to see that, it's here. Go to Google and type in a network. Protocols End Port Cheat Sheet Network Ports and Protocols Cheat sheet: Here you will find the complete cheat sheet.

You will get the sheet Port number seven remains that of Echo. FTP The port number for this is 21. There is also a designated service for each service. There are small places where Your applications may be blocked. Like I have MySQL database. MySQL's database port Number 336 He's dedicated to going there. I If you want to do SSH, it is dedicated so that it will run only on 22. Me 27107 if you want to run MongoDB By doing this, something remains of MongoDB If he remains attached to her then like this Locations stay put, that's fine.

Went Now why did I tell you this because it All three things seem to give security If you look here, if you go to the instance. Go to Security Group and go to Edit security groups so if you carefully Look at the port here. Protocol and IP It looks like three things. For security you need protocol, IP And the port requires all three of these things. VPC or Virtual Private Cloud own internet own internet now You can have multiple subnets on your own internet.

[Computer Networking and System Design] Networking fundamentals (OSI, TCP/IP) / Load balancing, proxies and gateways

Can you have multiple subnets? That you can do that one I Public subnets And I kept one as private subnet. Public subnets and private subnets. What is the difference between a subnet and a private subnet? Resources that will reside in the public subnet Like the servers that will be with me late There is a server This is the server from Let's. Should I keep it in a subnet or a private subnet? Interconnected to worldwide networks gateway to access. I would have kept it in a private subnet.

Private subnets are now private There can be multiple subnets. Maybe there is a server like this There is a server that I don't make public. I would like to see the code for this as well. I should also make the server public, anyone can do it with Campering can plus anyone tomorrow. I can come and say, I am the one DevOps speakers in India We will keep it but you all know the reality. I'll put that on a private subnet as well.

I can put it in a private subnet. This is also a device in this private subnet So, it also has an IP address. Or what is the process by which these IP addresses can get a unique IP. Let's say to me In this public subnet only and only and only If you want to give IP to two devices then what is slash Will you come? 30 and this gives IP addresses to four devices. At the same time, I want my The server is placed in the public subnet The public subnet is now like this train This is the UI of shm.com This is the UI that is running.

The UI that is running is running in the public subnet. For this, there is a public subnet and a private subnet. There is another gateway in between which is called It is called Net Gateway. What does a NAT gateway do? Network Address Translation Gateway Translating your private IP address It gives a public face so that it The poor public people were able to access it. A NAT gateway is a managed cloud service that Allow resources like virtual machines in a private subnet to connect to the Internet and Other external networks.

[Computer Networking and System Design] Networking fundamentals (OSI, TCP/IP) / Load balancing, proxies and gateways (part 2)

It Translates private IP addresses into a Single Public Address Along Outgoing request. This private IPP address, its IP taken, its IP and it has a single IPP address. You go to AWS And here in the search write VPC Clicked Virtual Private Cloud. Go to Create a VPC. If it is written then how many IP addresses can be created Look, it is written in brackets 65000 If I make this 32, an IP, if this. If I make it 25 then 128 IP and so on.

Availability zone means within a region How many different places is your application located? You can create three subnets. Net Gateway, why is Net Gateway used? Does?Net Gateway Allow Private Resources To access internet from any availability Zone with in a VPC. VPC end points if you need that If you can access the VPC from outside, a Please provide endpoint. I have to provide DNS because I need it, not If you want train vs shm.com then it is like that Create VPC How many people can create a VPC Will you take it?

If I were to teach you VPC directly I would sit and not understand anything for an hour. At this VPC why subnet six because Three private subnets and three public subnets How many people understood that there are six subnets? What are these six subnet route tables? If you have any resources on that Remains for network routing. A gateway for network connections to the Internet Gateway IGW means Internet Gateway Finish Just Click on Create VPC Don. If that VPC has to be terminated So.

This VPC has been created. Go and delete the VPC. If your VPC is deleted then The Internet gateway will also be deleted. Routing Tables will also be deleted. Subnets will also be deleted. Wake up like you have a hangover reading DevOps for. You with full dedication Read up on DevOps. You can turn up the volume. Here I also need an authentication service. If yes, then a separate service is required for notification. Does he look any bigger now that he There is also a database inside, video upload service There is a notification service, authentication service.

That's it now as an application. The system is basically The Cost Off Not Thinking before You Build N Application.

[Computer Networking and System Design] Load balancing, proxies and gateways / System design and scaling

System Dis is the Cost of Not Thinking Before you build an application. Such a Netflix clone, a Netflix clone Using React. Whenever a request comes, it will load Go through the balancer. Load balancer so that tomorrow many people If you suddenly start watching Netflix then their The load should be distributed so that the traffic is distributed. Its data Stay tuned for MySQL. If You Don't Do That Then System Design This is the cost of not thinking before you build. I did not think that this application would be How do people access it?

Agar Static Files Notes PDF If I want to give it then I will put CD N here. I have to teach DevOps. Much A payment gateway will be required for that. A payment gateway will be required for this. Payment Gateway This payment gateway is my It is happening in a particular country. My country is great So for this payment gateway I need a GST I'll also have to get my marks. Submitting it to a particular place Then this payment gateway What will it cost to razor?

The best razor Pay provides services at just 2% For all transactions. My antilia is slowly getting smaller And my server costs are increasing. It Is the Cost of Not Thinking Before You Build an application. I'm a DevOps engineer, man. Designing that system can be a CI/CD pipeline. You too should monitor the infrastructure. As a DevOps Engineer System Dis is Variant. System design is very varied as a DevOps Engineer. I put that photo in the database and it took 3 MB. Will it load or will it load slowly?

It will load slowly because you are using 3 MB of data. Requires notification service, authentication need database, need front end, The back end is needed. UDP I will take it through ZM or in some storage I will put it obviously through ZM or Then UDP I will take it through so that there is fast connectivity. This Dev Board Is Your Project This is a project that you have created yourself. That you first studied Linux. After studying Linux you have established the basis of EC2 instance How do you make all those things?

You Read up on networking so you can learn about the Internet You understand how it works.

[Computer Networking and System Design] System design and scaling

You Study system design so that you can Actually, you will design the system for a Task tracking with application. You will create CI/CD of this application, You will create cubitus, you will monitor it, Everything Like an Eco System. The back end is built in Go Language.It has notification service. Email based authentication system You authenticate. Your job is to systemize this application. I have to create the UI of this application. The developer will build it. As a DevOps Engineer I should know that UI I have to make it.

I have this task, let's do it This build notification service to done Moved. You can also write in Python but I teach Python. I Artificial Intelligence in this application It is also necessary to summarize this project. I need Python We need a service that can talk to AI. Can you explain some applications of AI? Are you an AI application such as a model? It happens, chat GPT olama In the hail So these ulama One thing that will remain for me is AI The models will provide And these AI models My will call Python's service.

Ai Service will call Python's service. Python's service has some functions will remain which will call this AI and my back end will call this You can call Python. Or application programmable interface through. GRPC Remote Procedure Call So if I have to call a single I need to call a function from the server side. Use terminologies such as gRPC Remote Procedure Call Server to Server Communication gRPC Presentation Tier to Logical Tier Front End to back end communication API. GRPC So gRPC Is an open source framework created by Google That's why they named it GRPC.

This Tell me there are users in this application of mine. I have an email address in this application. To store the rest of the task I will use another database. Here I'm going to But this password that I am entering is a Authentication must go through the server. Because tomorrow if my authentication server fails or the app fails Or some other service failure. I don't Want entire application to crash. I'm here at JWT I'm here at a JWT Token Based JWT Java Web Token based a I will install the authentication service.

[Computer Networking and System Design] System design and scaling / Load balancing, proxies and gateways

I will install an authentication service. This from the Authentication Service UI and from my back end and my just and Will only talk to the users table. JavaScript Object Notation Web Token There is a different type of token which is quite The secret remains the encrypted token. I want my UI to be The notification service should come on. You are seeing this notification. Notification The notification should come in. I have a phone, this UI on the phone send a notification What if notifications for notifications It drops, it doesn't make any difference Does matter Missed any notification from Let's It.

I can use socket programming to That my notification service should continue running separately. My notification service should continue running separately. One Two Three four five Six seven eight I have eight services running that There are small services. In multiple What do I call services? I call multiple services micro. I call these multiple services Microservices. That one service database service, one service for another database Service, a Service Authentication Service, a back end service, one front end service, one AI Service, a notification service.

Microservice Architecture Microservice Architecture that from different independent services I have a complete system in place. I will create another layer and name it I will give Gateway layer. Gateway layer and I will tell the gateway layer that one lb Elastic Load Balancer Elastic Load Balancer Elastic Load Balancer's first user will be mine The internet will be here. My user will be here, he will first be at this gate gateway This gateway layer will load on it. There will be balance And only then will I proceed further.

This Important for the load balancer gateway layer Is. This is my application layer,. Application layer where my micro Services architecture is used. Application that uses microservices The architecture uses. I need to scale this application. If all the people come then the scale will increase immediately. I'm also using Kubernetes here. I can use Kubernetes Because Kubernetes Is Best For Microservices. Kubernetes is Best for Microservices And Kubernetes gives you horizontal pod Auto Scaler And also provides vertical pod auto scaler. If you have a server that has If you have 4GB RAM and you need to use 8GB of RAM, Either you install two servers of 4GB or 8GB.

[Computer Networking and System Design] System design and scaling

Install a large server, 8GB large. Two 4GB servers The server is horizontally scaling. For that you used Kubernetes. This application is now running. If you want to monitor that thing You used it for monitoring Prometheus Which is Will collect data from all these places Prometheus is collecting this data. To display this data You need Grafana. You Need Grafana to Display so that it is great How's your data doing? Logged in with the authentication service. Updates to the database whenever the task is updated So that whenever I refresh it Stay updated.

AI is powered by AI services. If you want to monitor it, Monitoring will also take place here. Grafana for CanUC monitoring Is. Monitoring the Dev Board Entire monitoring is also done. This is your DevOps is the board. You get something like this I put a while loop on it and I keep hitting it again and again So here is my Grafana dashboard. Spike is happening There has been a slight spike in requests. Kubernetes best for microservice Yes. System design three There are several types of applications.

Microservices , I knew about micro services and monolith. You Created multiple small services And a complete application is running from it. How many people are clear about microservices? It is a large application. Let's say it You ran a large application using Docker. The code of that entire application is given in one Housed within a larger application. Monolith Understand that I am currently live streaming. This is an application So that one application is a monolith Application. You must have seen this application in your office.

Inside which the code of micro services is kept It has happened. Applications typed where your small applications are running. A large, heavy object Application. A single repository inside Microservice code kept in one place It has happened. For example An online version of Google Cloud Platform Batik This is an online boutique application. It's Inside Multiple Service Front End Go In The card is in C. It All code in a single repository So this application is a Microservices Application But This Code Is What?

AI Agents for DevOps

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[AI Agents for DevOps] AI agents and LLMs for DevOps

First, let's understand why AI agents are becoming so popular. Why do companies want you to build AI agents? Why does the company want you to learn AI agents? AI agents are primarily software. AI agents are basically code applications that you can write. Which programming language do you use to write applications? How many of you have built an application? You can build AI agents in Python too. You can build AI agents in Java as well. You can build them in Ruby as well.

You can build them in many languages, even Rust. You can build them in different programming languages. Today, we will talk about Python. AI agents are basically Python code. That Python code remains, that can talk to LLMs. Compute means servers, servers, GPUs, CPUs, all these things were available in abundance, and due to COVID in 2019, everything became digital. Gemini nano, Gemini Pro are being made to generate images. Learn about AI Agents today. Can you explain what AI Agents are again? We take our problems to ChatGPT, saying, "Man, I'm working so hard.""Yet, I'm not getting a job." Forget about the job.

That is where you need you need an agent. What is so special about an agent? What is so special about an agent that you need one? Because an agent is a piece of Python code that has tools. It has tools, and these tools are compatible with the LLM. Meaning, if I tell the agent, "Tell me the current time." Then there will be a tool named "current time" that will pull the time from the system and give it to you. These are agents that have tools, and you can develop these tools too.

Zoom out a bit, that if you can develop tools. You have an LLM that is very strong, very powerful, and can generate anything. Can you build a tool that can run Docker commands and ask, " Can you tell me how many containers are running?""Yes, can you see why my pipeline failed?" Yes. The DevOps landscape is evolving, and this is what I presented at re: Invent. The problem was that generative AI has limitations, and that is why you need agentic AI. DevOps and SRE are seeing a new evolution with AI agents, and this can increase your productivity and production readiness.

[AI Agents for DevOps] AI agents and LLMs for DevOps (part 2)

How many of you go into DevOps daily? You log in to the server. You check by running docker ps. How many containers have died? How many containers are running, and do they need a restart, and all that. How many of you still go to the server to run commands like docker ps or docker logs, and later say, ", I can't remember the commands, " and do all that stuff? How many people have pipelines that fail, and they don't even know what happened?

You have an incident response and incident analysis. There is incident response and incident analysis. I don't know how to make a pipeline. The best part is that DevOps is evolving at the same speed as AI. You will have to learn that evolving DevOps. Let's understand how I create an AI agent. How will I build an AI agent? How many of you want to learn how to create an AI agent? HITL, Human In The Loop. Human In The Loop, you are important in this whole game, so you will request the LLM, and the LLM won't give an old response anymore; it will talk to the agent.

The agent has tools, and the tools, along with the LLM, will do crazy things to give you new responses and new productivity. AI Agents for DevOps Live, I will create this folder. I created a folder named AI Agents for DevOps Live and opened it. I will open a terminal inside this here. The new terminal has opened up here. This terminal of mine has opened inside the AI Agents for DevOps Live folder. How many people know Python? Python for DevOps was a part of the Josh Batch.

How do we create a virtual environment in Python? All the people who were in the Python batch. Python 3, now what version do I have? This is the latest version of Python running. Python 3.14, so python 3.14-m make a virtual environment. Create a virtual environment, create a folder that will contain the Python virtual environment. Why this, why are you making us create a virtual environment? Because you will install packages, and it's best if those packages are installed in a virtual environment!

For those practicing side-by-side or following along with me, if I want to activate the virtual environment here, what command should I type?

[AI Agents for DevOps] AI agents and LLMs for DevOps (part 3)

Linux based source bin activate. My documentation says, "How do I create an AI agent for DevOps use cases?" It says to use an agentic AI framework? We will talk about Strands Agents, which is open source. Today we will build and use Strands Agent. Pip means Python Package Installer. Python Package Installer means it is an installer where you just write pip install and the package name, and it gets installed. Share all the commands with us. I will write 'history' and I will share all the commands with you.

That's the Strands agent, but what is special about an agent that lets it perform so well, unlike an LLM? What do agents have, like tools, or maybe some answer tools, or maybe some great tools? That's why we will pip install strands-agents-tools as well. Where did you find this command? This is the official Strands Agents website, Strands Agent SDK. Go to examples here; the prerequisite says Python, and if you look here, where did it go? You can also go to the user guide; it says 'pip install strands', you can see it right there.

It says ' pip install strands-agents'. That same command, python-m pip install strands-agents, pip install strands-agents-tools, that's what I did, nothing else. Let's create a new agent. That is the extension for Python. In Python, if you want to import any package, what do you do? If you want to import any package, you use an import statement. From strands import Agent, from strands import Agent, so I am importing the Agent. Agent = Agent brackets are done. Create an instance of Agent. Like this by creating an instance, correct.

Amazon Bedrock is a service through which you can build generative AI applications and work with LLMs in the background. You will do pip install strands agents ollama. If you run this command, you will be able to run it with Ollama. I will run this command here. Pip install strands agents ollama. Amazon Bedrock is an AWS service that gives you permission to run large language models on the cloud. Some models might work, but if some don't, you can run Ollama locally too, so pip install strands as in Ollama.

By the way, on which port does MySQL run by default?

[AI Agents for DevOps] AI agents and LLMs for DevOps (part 4)

Can anyone tell me which port PostgreSQL runs on by default? The agent, what is this agent doing? It is using the Ollama model to create an agent. I mean, what is an agent? This agent is using an LLM. Did you understand this diagram, that your agent, which is a piece of code, is using the Ollama model, and your agent is created. I asked it, '', how are you?" It says, "I am doing well.""Thank you for asking.""Is there anything on your mind that you would like to talk about?""I am here to listen and provide information for assistance." How many of you learned how to create an agent in five lines of code?

How many of you, how many of you can say, "I am confident that I can install VS Code, install Ollama, download a model, install Swarms agents, write this code, and create an agent." Are you confident? Let's understand a bit: if I ask this agent- agent, let's have some fun with the agent. Let's have some fun with the agent. Let's write code for the agent such that the agent asks us...Input enter or you user input equals you-how many people understand what I have written here as user input?

You can write a system prompt for the agent. System prompt, system prompt -that you are a respectful agent, you are a respectful agent. You can train the agent a bit, saying you are a respectful agent. Similarly, you can import HTTP requests into tools. HTTP request, and you can add HTTP request as a tool; tools is a list, so you can add multiple tools too. To whom can you give the HTTP request as a tool? You can give it to your agent.

What does the agent have? What does your agent have? Your agent has an LLM, a system prompt, and tools as well. If I tell my agent now What is the current time and weather in Pune? What a lousy agent I have! If your agent, let's say, is not performing well, then 'what you can do', what can you do? This model, instead of using the Ollama model, 'what is the best cloud service provider' that can give you the best model access? We did HTTP, but how many people think this agent is production-ready?

[AI Agents for DevOps] AI agents and LLMs for DevOps (part 5)

This is a nonsense agent. What if I told you that by the end of this session, I will use Strands and Temporal to show you how to build AI agents that can monitor Docker. We can build agents for CI/CD pipelines. We can build observability agents. We can build Kubernetes agents. You can see a repository that says " Durable Execution for AI Agents.", what is Temporal? This repository must have opened for you. Docker monitor, simple agent, all these opened. Go here and clone it.

I will go into strands-temporal-agents and clone it. Strands-temporal-agent has opened for me. First, I will run a Temporal server on my system, which is temporal-server-start-dev. Here, I will run a Temporal server. Here, a Temporal server is running on localhost: 8233. Zero workflows, no workflows running in this space. I ran the Temporal server in one. I will go into strands-temporal-agents. I have a Docker agent. The Docker agent is very simple. What does the Docker agent do? The Docker agent uses Bedrock. Tell me, what command do you use for container status?

Anyone from batches 1, 2, 3, 4, 5, 6, 7, 8, or those who have read about how if you want Docker containers...Aditya says, "docker ps create". If you want to do the same thing in Python, what would you do? If you had to do that same thing in Python, what would you do? It's very simple, you use the Docker client- "docker client get containers". What is the Docker client? When you write Docker Python, when you write Docker Python SDK (Software Development Kit), you can run Docker through Python.

How many people knew that if you do pip install docker, import docker, you can run it like this? You will have to install Docker as a package. It will just install everything, one by one, very quickly. Whenever you clone that repository, you should install it like this. In the meantime, let me tell you what the Docker agent does. The Docker agent doesn't do much. I have imported Docker here and created a few functions. Like, container_info; if I need information about a container, give me the ID, image, and status.

How many people were in the Python for DevOps batch? How many of you were in Python for DevOps?

[AI Agents for DevOps] AI agents and LLMs for DevOps (part 6)

Temporal says that you have to run a workflow, you have to run your application. To run a workflow, I need a worker, so what will I do here? I will go into docker monitor, cd docker monitor and python docker worker. A worker is a process that keeps reading your code. Worker heartbeating config run time. It will keep monitoring the workflow and activity. I started a worker in one place. In the second, I will run my agent. My agent is in Python docker.

My agent is in client.py. I created this just now, I mean before this demo, it's a docker container health monitor agent that is monitoring workflows here. Your temporal is monitoring workflows here. I haven't run the agent yet. How many of you want me to run this agent? How many of you want me to run this agent and see that it never fails? An agent that will never fail. Can you tell me how many docker containers are running on my system? If I actually wanted to run this via command, what would I write?

Anyone, if I want to see how many containers are running, what should the command be? Docker ps, docker ps, cannot connect to the docker daemon, is the daemon running? Let's start the docker daemon. Anyone who can tell me how to start the docker daemon? Systemctl start docker and dockerd systemctl start docker correct Abdullah absolutely right, so I have started it now, let's do docker ps. How many containers are running? I have three containers running. Will my AI agent be able to tell this?

Can you tell me how many Docker containers are running? This entire workflow ran in Temporal. How many of you learned something new, that you can even monitor your agents with Temporal, so that your input and output remain monitored and are also being read. Dangerous, now can you show the logs for container demo-logger? Here, back to workflows, a second workflow was created. Can you show the logs for demo last 100 lines from container demo-logger? Can you restart demo-redis container? It would, because if you go back to workflows.

Can you restart demo-redis, successfully restarted container demo. If I go here and go over here, a demo-redis 5 days ago, docker ps demo-redis status up about a minute, up about a minute, meaning, meaning just 1 minute ago! For job-ready skills, you need your AI agents to not fail. Your pipelines should not fail. How the latest DevOps things work. How you can use DevOps in Agentic AI.